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# CAUSALITY AND DISPERSION RELATIONS

Volume 95

H. M. Nussenzveig

# CAUSALITY AND DISPERSION RELATIONS

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# CAUSALITY AND DISPERSION RELATIONS

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# PREFACE

*When a new particle or new fact is discovered, I notice that all the theorists do one of two things: they either form a group or disperse.*

R. P. FEYNMAN[1]

This book is about dispersing. It had its inception in a set of lectures[2] given at the Latin-American School of Physics nine years ago. The aim was to provide an introduction to dispersion relations by explaining their physical and mathematical basis in the simplest possible context, beginning with classical fields and then going over to nonrelativistic quantum scattering. While this aim has been preserved, the original lecture notes have grown into a monograph.

Dispersion relations in relativistic quantum field theory, which has been their principal domain of application, are not discussed at all. However, the mathematical techniques and many of the physical ideas involved are quite similar, and they are much easier to grasp in the setting of a more thoroughly developed and understood, albeit less fundamental, physical theory. In fact, several ideas have been lifted from the nonrelativistic domain to be applied in the relativistic case, sometimes with little justification.

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[1] *Proc. Aix-En-Provence Int. Conf. Elementary Particles, Saclay*, Vol. 2, p. 206, 1961.

[2] H. M. Nussenzveig, "Analytic Properties of Non-Relativistic Scattering Amplitudes." Universidad de México, México City, 1962.

The reader is assumed to be familiar with the elements of the quantum theory of scattering and the theory of analytic functions. Theorems that may be less familiar are extensively quoted, in the form found most suitable for the applications to dispersion theory. A long appendix provides an introduction to distribution theory.

In the seven years that have elapsed since this project was undertaken, I became aware of even more pitfalls than I had initially believed to lie concealed within the subject (cf. the quotation by R. Jost at the beginning of Chapter 6), and there are probably others into which I have fallen. It is also clear that many gaps still remain to be filled. Several topics in which the gaps might have been even more glaring, such as many-channel and inverse scattering problems and the three-body problem, have been omitted. The choice of topics reflects my own interests and limitations.

I am indebted to Professor Laurent Schwartz for an introductory course in distribution theory from which much of the material in Appendix A and in Section 1.8 was derived. I wish to express my appreciation to the publishers for their admirable patience during the long years it took to complete this project. It is a pleasure to thank Mrs. Margaret Barres and Mrs. Shirley G. McDonnell for their excellent work in preparing the typescript. I am especially grateful to my wife for her constant help and encouragement.

PART

# I

# CAUSALITY AND ANALYTICITY

... *"There's the King's Messenger. He's in prison now, being punished: and the trial doesn't even begin till next Wednesday: and of course the crime comes last of all."*

*"Suppose he never commits the crime?"* said Alice.

LEWIS CARROLL

"Through the Looking Glass," Chapter V

CHAPTER

I

# CAUSALITY AND DISPERSION RELATIONS

*There was a Young Lady of Wight,*
*Who could travel much faster than light;*
*She departed one day,*
*In a relative way,*
*And came back on the previous night.*

ANON.

## 1.1. Introduction

Dispersion relations have been employed in physics to a considerable extent over the past two decades. They have been used in many branches of physics, but most of the applications have been in the high-energy domain, in connection with strongly interacting particles.

As is well known, conventional relativistic quantum field theory, where the interaction is specified by means of a Lagrangian, has met with great difficulties in the treatment of strong interactions. The perturbation techniques that, together with renormalization, have led to successful quantitative predictions in quantum electrodynamics become completely useless for strong interactions, owing to the large value of the coupling constant.

Dispersion relations have provided a new tool for dealing with this situation. The development of this technique has been closely related to a program that was first formulated by Heisenberg [1]. The basic idea is that the interaction can be completely described in terms of the behavior of the particles when they are far apart from one another.

The interaction is usually observed by allowing the particles to collide, giving rise to several possible reactions. The collision process can be conceptually divided into three stages:

I. *Initial stage:* the incoming particles are moving towards one another, but they are still so far apart that their interaction can be neglected.

II. *Intermediate stage:* the particles get closer together and interact among themselves.

III. *Final stage:* the outgoing particles that result from the collision have moved sufficiently far away from the interaction domain that they may again be treated as noninteracting. The initial and final stages correspond to asymptotic limits in the remote past or future, respectively, and they can be described entirely in terms of free particles.

The operator that transforms the initial stage into the final stage is called the $S$-matrix (scattering matrix):

$$\text{initial stage} \xrightarrow{S\text{-matrix}} \text{final stage}.$$

It was first introduced by Wheeler [2], and it plays a central role in Heisenberg's program. Heisenberg's suggestion was to describe the interaction solely in terms of the $S$-matrix, omitting completely the description of the intermediate stage. The $S$-matrix was expected to contain all the information required to compute any observable quantity, including the cross sections for all possible reactions and the energies of bound states. This point of view may be compared with the "black box" idea employed in electric circuit theory or in the theory of nuclear reactions.

Instead of specifying an interaction Lagrangian, one would start from general physical principles believed to be satisfied by the interaction, and from them one would derive the corresponding properties of the $S$-matrix. These would then be expressed as relations between observable quantities, and, by comparing them with observation, one might presumably test the validity of the underlying general principles.

Among the basic principles usually taken as starting points are the *relativistic covariance* of the theory and the *unitarity condition*, which states that the sum of the probabilities for all possible processes must be equal to unity. Both conditions can be expressed directly in terms of properties of the $S$-matrix elements.

The fundamental assumption that will engage most of our attention is known as the *causality condition*. Actually, several different conditions are known by this name. The most primitive and probably also the most intuitive one can be formulated as follows:

*Primitive causality: The effect cannot precede the cause.* "Cause" and "effect" must be suitably defined in each case in order to apply this condition.

In the special theory of relativity, as is well known, the above condition is closely related with the following requirement:[1]

*Relativistic causality: No signal can propagate with velocity greater than c,* where $c$ is the speed of light in vacuum. Sometimes, this is also called the *macroscopic causality condition.* It is important to realize that primitive causality is more general than relativistic causality, as it does not depend on the existence of a limiting velocity for the propagation of signals.

In quantum field theory, a *microscopic causality condition* is introduced. This condition, also known as *local commutativity*, expresses the vanishing of the commutator of field operators taken at two spacelike separated points.[2] It is related to the idea that measurements made at two such points should not interfere. However, the microscopic causality condition is less intuitive and more remote from experiment than macroscopic causality, and the relation between these two conditions has not yet been fully clarified.

The problem of expressing the causality condition in terms of equivalent properties of the $S$-matrix elements is the basic problem that leads to dispersion relations. Causality usually implies that some function (such as a signal or a commutator) identically vanishes over a range of values of its argument. It will be seen later that the Fourier transform of such a function can be analytically continued into some portion of the complex plane. This leads to analytic properties of the $S$-matrix elements when the variables on which they depend, i.e., the energies and momenta of the particles, are extended to complex values. Finally, the analytic properties are expressed in terms of integral relations between different matrix elements, for real values of the variables, which are the dispersion relations.

The use of analytic functions leads to some seemingly unphysical features. Even though the dispersion relations are expressed solely in terms of physical values of the variables, it must be remembered that physical measurements have finite precision, and two functions that coincide within the experimental error are equally compatible with the measurements. They may, however, have entirely different analytic continuations. The analyticity is introduced by the assumption that some quantity is *exactly* equal to zero over some range. This brings up the problem of the stability of analytic continuation (with respect to small changes in the data). In the dispersion relations, which result from the application of Cauchy's integral theorem, not just values on the real axis are involved but usually also conditions on the asymptotic behavior

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[1] See, for example, Landau and Lifshitz [3, Sec. 1-2].

[2] For a more precise formulation, see Jost [4, p. 54].

at infinity in the complex plane, and it will be seen that they are stable in the above sense.

Dispersion relations were initially regarded as broad restrictions on physical theories (like the principle of conservation of energy). Comparing them with experiment would, hopefully, allow one to test the validity of the underlying basic assumptions, specially causality.

Later developments in high-energy physics led to an increasingly more ambitious program, in which it was conjectured (and in a few instances proved) that the $S$-matrix has wider analytic properties as a function of all its variables. It was furthermore conjectured that the dispersion relations expressing this broader analyticity, when combined with unitarity, might provide the basis for a complete dynamical scheme, allowing one, in principle, to compute the $S$-matrix in terms of a small set of parameters (masses, coupling constants). This extension of Heisenberg's program is mainly due to Mandelstam [5].

Even if one does not subscribe entirely to Heisenberg's original or extended program (as Heisenberg himself no longer does [6, p. 16]), the $S$-matrix, in view of its close relationship with measurable quantities, is likely to play a major role in the treatment of strong interactions. The study of the analytic properties of the $S$-matrix and of their connection with general physical assumptions should provide valuable insight into its structure.

What has the method of dispersion relations accomplished so far? One advantage of this method is that it deals only with renormalized quantities, so that infinite renormalization constants never appear. The quantities with which it works are closely related with experiment, and it has provided a convenient language for a semiphenomenological description of experimental results. It has also yielded some quantitative results, e.g., in pion–nucleon interactions. Finally, it has led to a new approximation scheme, different from perturbation theory, that may be more suitable for dealing with strong interactions.

The investigation of Mandelstam's program in high-energy physics is an extremely difficult task. This is chiefly due to the possibility of creation and annihilation of particles which, through unitarity, links together $S$-matrix elements corresponding to all possible processes. The investigation of analytic properties has been carried out thus far only in the simplest cases, and few rigorous results have been obtained; in particular, very little is known about the properties of production amplitudes. It is conceivable that, even if the dynamical scheme is essentially correct in principle, it may be so complicated that it does not provide a useful practical approach.

Partly for this reason, several investigations have been carried out in connection with simpler models, such as classical fields or nonrelativistic quantum scattering. Besides providing the simplest introduction to the subject, this approach allows one to test several conjectures in a far more familiar and better understood domain. The results thus obtained may also serve as a guide for the extension to high-energy physics. Even when a direct extension is difficult, it is often conjectured that analogous results are valid, as was done, for instance, with the Regge pole hypothesis. Some justification for such conjectures is provided by the fact that nonrelativistic potential scattering may be regarded, in some sense, as a low-energy limit of quantum field theory. However, there are important differences between the two domains, and such extrapolations may turn out to be unjustified. The results obtained in the classical and nonrelativistic domains also have intrinsic interest, because of their possible application to other branches of physics, such as nuclear physics (in particular, nuclear reactions) and many-body problems.

In the present work we confine our attention exclusively to classical fields and to nonrelativistic quantum scattering. We do not discuss at all the derivation and applications of dispersion relations in relativistic quantum field theory. An excellent review article, describing the main developments up to 1961, has been written by Mandelstam [7]. Progress in the derivation of rigorous analytic properties and some of the applications are described in more recent surveys.[3]

We divide the subject into two main parts. In Part I we consider the derivation of dispersion relations from general physical assumptions, in a spirit similar to what is known as axiomatic field theory. The main object is to clarify the relation between causality and the analytic properties of the $S$-matrix.

In the present chapter, starting with the simplest examples, we explain the connection between causality and analyticity at the macroscopic level. The mathematical foundations of dispersion relations, contained in Titchmarsh's theorem and its extension to distributions, are presented. The physical basis of dispersion relations is analyzed with reference to the problem of light propagation in a dielectric medium, leading to the Kramers–Kronig relations which, historically, were the starting point of the whole subject. A heuristic discussion of the connection between the macroscopic properties of the medium and the microscopic scattering by a single atom or molecule leads us to investigate the implications of causality for individual scattering processes.

The remainder of Part I is devoted to this problem. The nature of the scatterer need not be specified beyond assuming that some general physical properties, including causality, are satisfied. There is one important restriction, however: we treat only cutoff interactions, i.e., those that vanish identically beyond a certain range, which defines the radius of the scatterer. In

[3] Some of these surveys are listed in Ref. [8].

view of the short range of nuclear forces, a model of this kind has been commonly adopted in nuclear physics. However, it is well known that the behavior of strong interactions at large distances is better simulated by an interaction with an exponential tail, such as the Yukawa potential. Interactions of this kind are dealt with in Part II.

The simplest dispersion relations are those for fixed angular momentum, corresponding to the scattering of spherical multipole waves. Their derivation is given in Chapter 2. We first treat the case of a classical field. The structure of the $S$-function ($S$-matrix element for a given partial wave) is investigated; it is found that the $S$-function is essentially determined by its poles in the complex momentum plane. The treatment is then extended to nonrelativistic quantum scattering. The difficulty here lies in the formulation of a causality condition. Two different approaches have been proposed, one due to Schützer and Tiomno and the other one due to Van Kampen and Wigner. Both of them are presented and their relative merits are discussed. Wigner's formulation is in terms of a different function, known as the $R$-function. The relation between the properties of the $R$-function and those of the $S$-function is discussed. The concept of time delay in scattering processes and the role played by the completeness of the set of stationary states are also analyzed.

Chapter 3 deals with dispersion relations for the total scattering amplitude, both for a classical field and for nonrelativistic quantum scattering. The scattering amplitude is expressed as a function of the energy and the magnitude of the momentum transferred in the collision. It is shown that the dispersion relations for fixed momentum transfer follow from causality. Both in the classical and in the quantum case, primitive causality is sufficient: one need not employ the relativistic causality condition. It is also found that analyticity in the momentum transfer follows directly from the cutoff character of the interaction.

The important role played by the poles in the characterization of the $S$-function leads one to investigate their physical interpretation. An account of the results is given in Chapter 4. The basic problem is that of finding a satisfactory description of unstable states of a system, relating its transient time behavior with the poles of the corresponding $S$-function. It is found that one can associate a "transient-mode propagator" with each pole of the $S$-function, and for several explicit models it is shown that the response of the system to an arbitrary initial excitation can be expanded in terms of these transient-mode propagators. This allows one to discuss the limits of validity of the exponential decay law and to give a time-dependent description of resonance scattering.

In Part II, we consider a specific model, namely, nonrelativistic potential scattering, in which the interaction is explicitly given in the form of a central potential, rather than just being restricted by general physical assumptions.

Thus, the $S$-matrix can in principle be computed from Schrödinger's equation, and its analytic properties can be directly investigated. This leads to more detailed information, but the physical origin of the results is somewhat less apparent than in the former approach. Besides providing a concrete example to illustrate the general results of Part I, the potential model allows us to go further, treating also interactions with tails extending to infinity (e.g., of Yukawa type).

In Chapter 5, the results due to Jost, Levinson, Humblet, Regge, and others on the analytic properties of partial-wave amplitudes are described. In the particular case of cutoff potentials, the results of Part I are verified; the transient-mode expansion is also extended to this case.

The analytic properties of the total scattering amplitude are considered next, in Chapter 6. The Khuri dispersion relation for fixed momentum transfer as well as an analyticity domain in the momentum transfer plane are obtained, following Hunziker, by functional analysis techniques. For cutoff potentials, the causality condition of Chapter 3 is verified, so that analyticity in the energy follows from causality. In the spirit of Part I, it is also shown that analyticity in the momentum transfer, for interactions of (at least) exponential decrease at large distances, follows from a general condition expressing the finite range of the interaction, due to Omnès.

In order to proceed beyond the results of Chapters 5 and 6, it is necessary to restrict still further the class of potentials under consideration. Throughout most of Chapters 7 and 8, we consider only Yukawa-type potentials. These potentials have a special "smoothness" property, enabling them to be analytically continued in coordinate space, that allows one to extend the analyticity domain in momentum transfer. This is done, following Regge, by extending the definition of the $S$-function to complex angular momentum and applying the Watson transformation. The results, as well as the properties of Regge poles, which have found increasing application in high-energy physics, are presented in Chapter 7.

The extended analyticity domain in the momentum transfer finally enables one to derive the Mandelstam representation for Yukawa-type potentials, by applying techniques due to Martin and Bessis. By combining the Mandelstam representation with unitarity, as is shown in Chapter 8, it is possible, in principle, to implement Mandelstam's program in this case by constructing the scattering amplitude without going back to Schrödinger's equation; one gets in this way a "pure $S$-matrix theory." Finally, we also discuss Mandelstam's program and the physical interpretation of Regge poles in the case of cutoff potentials.

An exposition of some of the mathematical tools employed throughout the book will be found in the appendixes, which include a fairly lengthy one on distribution theory.

## 1.2. The Damped Harmonic Oscillator

To illustrate the connection between causality and analyticity, we shall start by considering a very simple example: the motion of a damped harmonic oscillator of mass $m$ subject to an external driving force $\mathscr{F}(t)$. The equation of motion is

$$\ddot{x} + 2\gamma\dot{x} + \omega_0{}^2 x = \mathscr{F}(t)/m = f(t), \tag{1.2.1}$$

where dots denote time derivatives, $\omega_0$ is the natural frequency, and $\gamma$ is the damping constant ($\gamma > 0$!).

### (a) Free Oscillations

If $\mathscr{F}(t) = 0$, the general solution of (1.2.1) is

$$x_0(t) = a\exp(-i\omega_1 t) + b\exp(-i\omega_2 t), \tag{1.2.2}$$

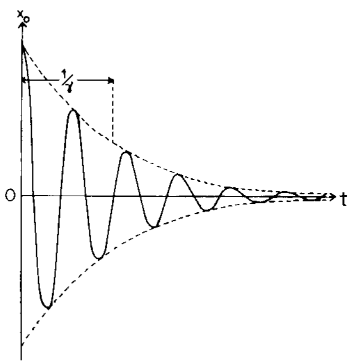


*FIG. 1.1.* Free oscillations of a damped harmonic oscillator.

where $a$ and $b$ are arbitrary constants and $\omega_1$ and $\omega_2$ are the roots of

$$\omega^2 + 2i\gamma\omega - \omega_0{}^2 = 0;$$

i.e.,

$$\omega_{1,2} = \pm(\omega_0{}^2 - \gamma^2)^{1/2} - i\gamma. \tag{1.2.3}$$

Thus,

$$x_0(t) = [a\exp(-i(\omega_0{}^2-\gamma^2)^{1/2}t) + b\exp(i(\omega_0{}^2-\gamma^2)^{1/2}t)]\,e^{-\gamma t}. \tag{1.2.4}$$

This represents, in general, damped oscillations with "lifetime" $1/\gamma$ (Fig. 1.1).

### (b) Stationary Solution

The response to a harmonic driving force,

$$f(t) = F_\omega e^{-i\omega t}, \tag{1.2.5}$$

is the stationary solution

$$x(t) = X_\omega e^{-i\omega t}, \tag{1.2.6}$$

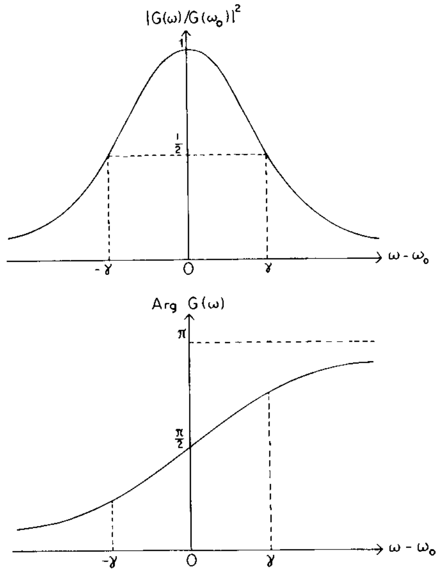


*FIG. 1.2.* Modulus and phase of the response function near resonance.

where, according to (1.2.1),

$$X_\omega = -\frac{F_\omega}{\omega^2+2i\gamma\omega-\omega_0^{\,2}} = -\frac{F_\omega}{(\omega-\omega_1)(\omega-\omega_2)}$$

$$= G(\omega)F_\omega, \tag{1.2.7}$$

$$G(\omega) = -1/(\omega-\omega_1)(\omega-\omega_2). \tag{1.2.8}$$

For weak damping ($2\gamma \ll \omega_0$), $G(\omega)$ goes through a sharp resonance at $\omega \approx \omega_0$. As shown in Fig. 1.2, $|G(\omega)|^2$ has a resonance peak of half-width $2\gamma$, while the phase of $G(\omega)$ varies rapidly across the resonance, changing by $\pi$ from one side to the other.

### *(c) General Solution*

Let us now consider an arbitrary driving force that can be represented by a Fourier integral,

$$f(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) e^{-i\omega t} \, d\omega, \tag{1.2.9}$$

where

$$F(\omega) = \int_{-\infty}^{\infty} f(t') e^{i\omega t'} \, dt'. \tag{1.2.10}$$

The corresponding solution of (1.2.1) follows from (1.2.5) to (1.2.7), by taking into account the linear nature of the problem:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{-i\omega t} \, d\omega = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) F(\omega) e^{-i\omega t} \, d\omega. \tag{1.2.11}$$

Substituting $F(\omega)$ by (1.2.10) and inverting the order of integration (assuming that all functions are sufficiently well behaved for this to be allowed), we get

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} dt' f(t') \int_{-\infty}^{\infty} G(\omega) e^{-i\omega(t-t')} \, d\omega,$$

or

$$x(t) = \int_{-\infty}^{\infty} g(t-t') f(t') \, dt', \tag{1.2.12}$$

where

$$g(\tau) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(\omega) e^{-i\omega\tau} \, d\omega. \tag{1.2.13}$$

The integral in (1.2.12) is usually called the *convolution product* of $g$ and $f$, and is denoted by $g(t) * f(t)$. The relation between (1.2.11) and (1.2.12) corresponds to the well-known theorem on the Fourier transform of a convolution product.[4]

The general solution of (1.2.1) is obtained by adding to (1.2.12) the general solution (1.2.4) of the homogeneous equation. The solution (1.2.12) describes the displacement of the oscillator due exclusively to the action of the driving force.

---

[4] See, for example, Titchmarsh [9, p. 51].

### *(d) Green's Function*

According to (1.2.12), $g(t)$ is the solution of (1.2.1) corresponding to $f(t) = \delta(t)$ (Dirac's delta function); i.e., it is *Green's function.*

Substituting $G(\omega)$ by (1.2.8) in (1.2.13), we see that the integral can be computed by residues. The paths described by the poles $\omega_{1,2}$ in the $\omega$-plane as $\gamma$ increases from 0 to $\infty$, according to (1.2.3), are shown in Fig. 1.3.

For $\gamma \ll \omega_0$ (weak damping), the poles $\omega_{1,2}$ are very close to the real axis, below the points $\pm\omega_0$, respectively. This corresponds to the narrow resonance situation pictured in Fig. 1.2. As $\gamma$ increases from 0 to $\omega_0$, the poles approach the negative imaginary axis, moving along the semicircle of radius $\omega_0$ centered at the origin. They join each other at the point $-i\omega_0$ for $\gamma = \omega_0$ (critical damping). For $\gamma > \omega_0$, the poles move in opposite directions along the negative imaginary axis: one of them approaches the origin, while the other tends to $-i\infty$ as $\gamma \to \infty$.

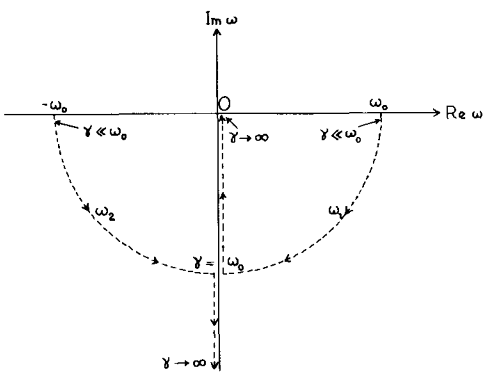


*FIG. 1.3.* Paths described by the poles $\omega_1$ and $\omega_2$ in the $\omega$-plane as a function of $\gamma$.

The important point is that *the poles always lie in the lower half-plane.* Let us now introduce the following notation: $I_+(\omega)$ denotes the upper half of the $\omega$-plane ($\operatorname{Im}\omega > 0$), $I_-(\omega)$ the lower half-plane, and $I_0(\omega)$ the real $\omega$-axis. When there is no possibility of confusion, we shall write just $I_+$, $I_-$, and $I_0$.

If $\tau < 0$ in (1.2.13), we can close the path of integration in $I_+$, where $G(\omega)$ is regular, so that

$$g(\tau) = 0 \qquad (\tau < 0). \tag{1.2.14}$$

If $\tau > 0$, the path of integration can be closed in $I_-$ and, according to (1.2.8), we get

$$g(\tau) = -2\pi i \sum \operatorname{res}\left[\frac{G(\omega)}{2\pi} e^{-i\omega\tau}\right]$$
$$= [i/(\omega_1-\omega_2)]\,[\exp(-i\omega_1\tau)-\exp(-i\omega_2\tau)],$$

or, by (1.2.3),

$$g(\tau) = e^{-\gamma\tau}[\sin((\omega_0{}^2-\gamma^2)^{1/2}\tau)]/(\omega_0{}^2-\gamma^2)^{1/2} \qquad (\tau > 0). \qquad (1.2.15)$$

Comparing (1.2.14) and (1.2.15) with (1.2.4), and recalling the definition of Green's function, we find the physical interpretation of the above results: an instantaneous impulse transmitted to the oscillator excites its free modes of oscillation.

Substituting (1.2.14) and (1.2.15) in (1.2.12), we get the solution

$$x(t) = \int_{-\infty}^{t} e^{-\gamma(t-t')}\,\frac{\sin[(\omega_0{}^2-\gamma^2)^{1/2}(t-t')]}{(\omega_0{}^2-\gamma^2)^{1/2}}\,f(t')\,dt'. \qquad (1.2.16)$$

Because of (1.2.14), the upper limit of integration is $t$ instead of $\infty$, which means that the displacement at time $t$ depends only on the value of the force at *earlier* times. Since the force is considered as the cause of the displacement, this is in agreement with the *causality condition* (cf. Section 1.1): *the effect cannot precede the cause.*

On the other hand, (1.2.14) followed from the fact that $G(\omega)$, the Fourier transform of $g(\tau)$, has a regular analytic continuation in $I_+(\omega)$, which tends to zero sufficiently fast for $|\omega| \to \infty$, so that the path of integration can be closed at infinity in (1.2.13).

This example suggests that causality is closely connected with analyticity. A more general discussion of this connection will be given in Section 1.3.

### *(e) Relation to Other Concepts*

The causal character of the solution of (1.2.1) is also directly related to the condition $\gamma > 0$, which means that the term $2\gamma\dot{x}$ gives rise to energy dissipation, rather than to the production of energy, as would happen for $\gamma < 0$. It is because of this condition that the poles (1.2.3) are located in $I_-(\omega)$, rather than in $I_+(\omega)$.

The damped harmonic oscillator is an example of a *passive linear system.* The causal behavior of such a system is closely related to its passive character. This point will be discussed in greater generality in Appendix B.

Another condition has often been employed (e.g., in electric circuit theory) to restrict the location of the poles to a half-plane. It is the requirement that the transient solution, i.e., the free oscillations of the system in the absence of a driving force, must tend to zero as $t \to \infty$ (or at least remain bounded, in the absence of dissipation). In the present case, (1.2.4) would increase exponentially with time if $\gamma < 0$.

For a linear differential equation such as (1.2.1), the boundedness of the solution is also related with its *stability* [10], i.e., with the effect of perturbations of the initial conditions on the behavior of the solution.

We also see in the present example that the exponential decay of the free oscillations (Fig. 1.1) and the resonance behavior of the stationary solution (Fig. 1.2) are directly related to each other and to the positions of the poles of $G(\omega)$. This relation is also quite general, as will be seen in Chapter 4.

## 1.3. Causality and Analyticity

The connection between causality and analyticity that was found in Section 1.2 can be immediately generalized [11, 11a].[5] Let us consider an arbitrary physical system subject to a time-dependent excitation or "input" $f(t)$, to which it responds by an "output" $x(t)$. Let us assume that the system satisfies the following conditions:

I. *Linearity (superposition principle): The output is a linear functional of the input. Thus,*

$$x(t) = \int_{-\infty}^{\infty} g(t,t')f(t')\,dt'. \tag{1.3.1}$$

Here, $x$, $g$, and $f$ may be distributions, rather than ordinary functions (cf. Section 1.8), but we shall not concern ourselves with their mathematical nature for the time being. The treatment in the following few sections will be heuristic: we shall discuss the physical origin of dispersion relations, postponing until later a more rigorous treatment.

II. *Time-translation invariance: The system is time-translation invariant*, so that, if the input is shifted (advanced or delayed) by some time interval $\tau$, the output is merely shifted by the same interval: $x(t+\tau)$ corresponds to $f(t+\tau)$. It follows that $g(t,t')$ can depend only on the difference between the arguments: $g(t,t') = g(t-t')$, so that (1.3.1) becomes

$$x(t) = \int_{-\infty}^{\infty} g(t-t')f(t')\,dt' = g(t) * f(t). \tag{1.3.2}$$

[5] Cf. the review article by Van Kampen [12]; also see Hilgevoord [13].

Let $F(\omega)$, $X(\omega)$, and $G(\omega)$ be the Fourier transforms of $f(t)$, $x(t)$, and $g(t)$, respectively, defined as in (1.2.10); in particular,

$$G(\omega) = \int_{-\infty}^{\infty} g(\tau)\, e^{i\omega\tau}\, d\tau. \tag{1.3.3}$$

For the moment, we simply assume that these transforms exist. Conditions for their existence, as well as a discussion of what happens when they do not exist, will be given later.

Applying the convolution theorem for Fourier transforms, we find, as in (1.2.11),

$$X(\omega) = G(\omega)F(\omega). \tag{1.3.4}$$

III. *Primitive causality condition: The output cannot precede the input*, so that, if $f(t)$ vanishes for $t < T$, the same must be true for $x(t)$. As in (1.2.14), this implies

$$g(\tau) = 0 \qquad (\tau < 0), \tag{1.3.5}$$

so that (1.3.3) becomes

$$G(\omega) = \int_{0}^{\infty} g(\tau)\, e^{i\omega\tau}\, d\tau. \tag{1.3.6}$$

The fact that the above integral is extended only over a half-axis has far-reaching consequences on its analytic behavior: $G(\omega)$ *has a regular analytic continuation in* $I_+$.

In fact, let $\omega = u + iv$ $(v > 0)$. Then,

$$G(\omega) = \int_{0}^{\infty} g(\tau)\, e^{iu\tau} e^{-v\tau}\, d\tau, \tag{1.3.7}$$

and the factor $e^{-v\tau}$ can only improve the convergence (note that this would not be true if the lower limit were $-\infty$). Thus, provided that $g(\tau)$ is sufficiently well behaved (more precise conditions will be given later), (1.3.6) defines a function $G(\omega)$ holomorphic in $I_+$.

There are many examples of physical systems satisfying the above conditions. We have seen one in Section 1.2. Other examples are: (a) an electric network, with $f$ the input voltage, $x$ the output current, and $G(\omega)$ the admittance of the network; (b) a dielectric medium, with $f$ the applied electric field, $x$ the dielectric polarization, and $G(\omega)$ the dielectric susceptibility of the medium [cf. Eq. (1.4.14)].

Before deriving further consequences from the analyticity of $G(\omega)$, let us turn our attention to the example of a dielectric medium, to consider the implications of causality in a problem of wave propagation.

## 1.4. Light Propagation in a Dielectric Medium

The propagation factor of a monochromatic plane wave in a homogeneous dielectric medium is

$$\exp\{-i\omega[t-(n'/c)z]\}, \tag{1.4.1}$$

where

$$n' = n + i\kappa = n + i(c\beta/2\omega) \tag{1.4.2}$$

is the *complex refractive index*. The imaginary part of $n'$ is the damping constant associated with absorption and scattering out of the beam; in fact,

$$e^{in'\omega z/c} = e^{-\beta z/2}\, e^{in\omega z/c},$$

so that $\beta$ is called the *extinction coefficient* (the intensity is attenuated like $e^{-\beta z}$) and $n$, which determines the phase velocity of the wave going through the medium, is the *real refractive index*.[6]

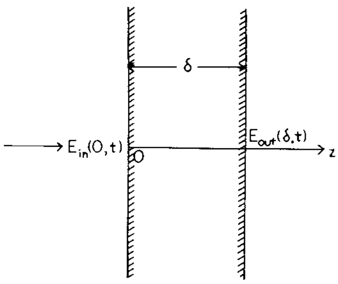


*FIG. 1.4.* Plane wave normally incident on a dielectric slab of thickness $\delta$.

Let us now consider a dielectric slab of very small thickness $\delta$, and let $E_{\mathrm{in},\omega}(z,t)$ be some component of the electric field of a monochromatic plane wave normally incident upon the slab (Fig. 1.4):

$$E_{\mathrm{in},\omega}(z,t) = E_{\mathrm{in},\omega}(0)\, e^{-i\omega[t-(z/c)]}. \tag{1.4.3}$$

The corresponding component on the exit face of the slab is[7]

$$\begin{aligned} E_{\mathrm{out},\omega}(\delta,t) &= E_{\mathrm{out},\omega}(\delta)\exp(-i\omega t) \\ &= \exp[in'(\omega/c)\delta]\, E_{\mathrm{in},\omega}(0,t) \\ &= E_{\mathrm{in},\omega}(0)\exp\{-i\omega[t-(n'/c)\delta]\}, \end{aligned} \tag{1.4.4}$$

[6] See, for example, Born and Wolf [14, p. 610].

[7] This is not strictly true, because the transmission and reflection coefficients of the slab must be taken into account. However, the correction is negligible for sufficiently small thickness $\delta$ and not too dense materials.

so that

$$E_{\text{out},\omega}(\delta) = \exp[in'(\omega/c)\delta]\, E_{\text{in},\omega}(0). \tag{1.4.5}$$

Comparing this result with (1.3.4), we see that $\exp[i(\omega/c)n'(\omega)\delta]$ corresponds to $G(\omega)$, so that, if we put

$$g(\delta,\tau) = (1/2\pi)\int_{-\infty}^{\infty} \exp[i(\omega/c)n'(\omega)\delta]\exp(-i\omega\tau)\,d\omega, \tag{1.4.6}$$

it follows from (1.3.2) that the transmitted wave on the other side of the slab, corresponding to an arbitrary incident wave $E_{\text{in}}(0,t)$ that can be represented by a Fourier integral, is

$$E_{\text{out}}(\delta,t) = \int_{-\infty}^{\infty} g(\delta,t-t')\,E_{\text{in}}(0,t')\,dt'. \tag{1.4.7}$$

In the integral (1.4.6), $\omega$ runs from $-\infty$ to $\infty$, although $n'(\omega)$ is usually defined only for $\omega \geqslant 0$. However, in (1.4.7), $E_{\text{in}}$ and $E_{\text{out}}$ are both real, so that $g$ must also be real. According to (1.4.6), this implies the symmetry relation

$$n'(-\omega) = n'^{*}(\omega) \qquad (\text{real } \omega), \tag{1.4.8}$$

which extends the definition of $n'(\omega)$ to $\omega < 0$. According to (1.4.2), it follows that

$$n(-\omega) = n(\omega), \qquad \beta(-\omega) = \beta(\omega). \tag{1.4.9}$$

The causality condition in the present case must be formulated somewhat differently from that of Section 1.3, because of the spatial separation $\delta$ between input and output. We must employ the relativistic form of the causality condition (cf. Section 1.1):

III′. *Relativistic causality condition: No signal can propagate with velocity greater than c.* This implies that $E_{\text{out}}(\delta,t)$ can only depend on the values taken by $E_{\text{in}}(0,t')$ for $t' < t-\delta/c$, so that, in (1.4.7),

$$g(\delta,\tau) = 0 \qquad (\tau < \delta/c). \tag{1.4.10}$$

According to (1.4.6), this implies

$$\exp[i(\omega/c)n'(\omega)\delta] = \int_{\delta/c}^{\infty} g(\delta,\tau)\,e^{i\omega\tau}\,d\tau,$$

or

$$\exp\{i(\omega/c)[n'(\omega)-1]\delta\} = \int_{0}^{\infty} g(\delta,t+(\delta/c))\,e^{i\omega t}\,dt. \tag{1.4.11}$$

This is again a relationship of the type (1.3.6), which implies that the function $\exp\{i(\omega/c)[n'(\omega)-1]\delta\}$ has a regular analytic continuation in $I_+(\omega)$. Since $\delta$ is small but otherwise arbitrary, $\omega n'(\omega)$ must have the same property. In fact, the only thing we have to exclude is the possibility that the exponential might have zeros in $I_+(\omega)$. Such zeros would give rise to logarithmic singularities in $\omega n'(\omega)$. If $\omega_0$ were such a zero, however, we would have, in the neighborhood of $\omega_0$,

$$\exp\{i(\omega/c)[n'(\omega)-1]\} = (\omega-\omega_0)^\rho f(\omega) \qquad (\rho > 0,\quad f(\omega_0) \neq 0).$$

Then for suitably chosen $\delta$, the function

$$\exp\{i(\omega/c)[n'(\omega)-1]\delta\} = (\omega-\omega_0)^{\rho\delta}[f(\omega)]^\delta$$

would have a branch point at $\omega_0 \in I_+$, which is impossible, according to (1.4.11). Thus, $n'(\omega)$ is holomorphic in $I_+$.

This result does not exclude possible singularities of $n'(\omega)$ on the real axis. In the case of a conducting medium, for instance, there would be a singularity at $\omega = 0$, due to the term $4\pi i\sigma/\omega$ in the complex dielectric constant [15, pp. 250, 260] ($\sigma$ is the conductivity of the medium). It will usually be assumed here that we are dealing with a nonconducting medium.

Since $g$ is real, it follows from (1.4.11), for complex $\omega$, that

$$n'(-\omega^*) = n'^*(\omega), \tag{1.4.12}$$

which extends the symmetry relation (1.4.8) to complex values of $\omega$. Thus, $n'$ takes on complex conjugate values at points symmetrically placed with respect to the imaginary axis. In particular, $n'$ is real on the imaginary axis.

The complex dielectric constant $\varepsilon'(\omega)$ is related to $n'$ by Maxwell's relation

$$\varepsilon'(\omega) = n'^2(\omega), \tag{1.4.13}$$

where we have taken the magnetic permeability $\mu = 1$. Thus, according to the above results, $\varepsilon'(\omega)$ also has a regular analytic continuation in $I_+$.

The last result can also be obtained directly [15, p. 256] by applying the primitive causality condition of Section 1.3 to the dielectric polarization **P** produced by the applied electric field **E**. For a given monochromatic component, we have

$$\mathbf{P}_\omega = \chi'(\omega)\mathbf{E}_\omega = \frac{[\varepsilon'(\omega)-1]}{4\pi}\mathbf{E}_\omega, \tag{1.4.14}$$

where $\chi'(\omega)$ is the dielectric susceptibility, which plays the role of $G(\omega)$ in (1.3.4). The causality condition (no polarization before the electric field is applied) implies that $\varepsilon'(\omega)-1 = n'^2(\omega)-1$ has a representation of the form (1.3.6), and is therefore holomorphic in $I_+$.

Notice, however, that this reasoning cannot be inverted to conclude from the analyticity of $\varepsilon'(\omega)$ that $n'(\omega) = (\varepsilon'(\omega))^{1/2}$ is also regular, because branch points would not be excluded. That is the reason why it was necessary to apply (1.4.11).

## 1.5. Physical Origin of Dispersion Relations

Can the real refractive index $n(\omega)$ and the extinction coefficient $\beta(\omega) = (2\omega/c)\kappa(\omega)$ of a dielectric medium be arbitrary functions of the frequency? According to Section 1.4, $n' = n + i\kappa$ is the boundary value of an analytic function that is holomorphic in $I_+(\omega)$, so that one might expect the existence of some relation between its real and imaginary parts. It will now be shown directly,[8] by a physical argument, that the causality condition indeed gives rise to a connection between the real and the imaginary part of $n'(\omega)$.

In fact, if the behavior of $n$ and $\beta$ as functions of the frequency could be specified completely independently of each other, it would be conceivable that a medium might act as a perfect filter for some frequency $\omega_0$. Such a medium would completely absorb this particular frequency, without at all affecting any other frequency. It would have $n = 1$ and $\beta$ sharply peaked at $\omega_0$ and zero otherwise.

Let us consider a slab of this hypothetical medium, as in Fig. 1.4, and an incident wave train $E_{\text{in}}$ with a sharp front, which arrives at the slab at $t = 0$, as shown in Fig. 1.5(a). This wave train is a superposition of plane monochromatic waves $E_{\text{in},\omega}$ of all frequencies, each of which of course extends from $t \to -\infty$ to $t \to \infty$. The component $E_{\text{in},\omega_0}$ is shown in Fig. 1.5(b). The vanishing of $E_{\text{in}}$ for $t < 0$ is due to the phase relations among the Fourier components, which give rise to destructive interference for all negative times.

The corresponding output field would be $E_{\text{out}} = E_{\text{in}} - E_{\text{in},\omega_0}$. Thus, as shown in Fig. 1.5(c), $E_{\text{out}} = -E_{\text{in},\omega_0} \neq 0$ for $t < 0$, so that we would have an output field before the arrival of the input field, thus violating the causality condition.

In order not to have a violation of causality, the absorption of one frequency must be accompanied by a compensating phase shift of all other frequencies, in such a way that they still interfere destructively for $t < 0$. Thus, the real refractive index $n$ at any frequency must be related to the values taken by the extinction coefficient $\beta$ for all frequencies. These relations between $\operatorname{Re} n'$ and $\operatorname{Im} n'$, the explicit form of which will be derived in Section 1.9, are called *dispersion relations*. Similar relations follow from causality in the general case of Section 1.3, which we shall now consider.

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[8] Cf. Toll [11a].

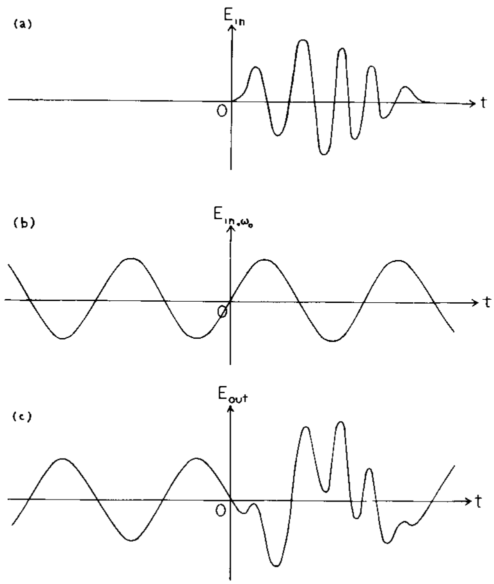


*FIG. 1.5.* A hypothetical medium acting as a perfect filter would completely absorb the component $E_{in,\omega_0}$ from the incident wave train $E_{in}$. The corresponding output field $E_{out} = E_{in} - E_{in,\omega_0}$ would violate the causality condition [after J. Toll, *Phys. Rev.* **104**, 1760 (1956)].

## 1.6. Titchmarsh's Theorem

### *(a) The Plemelj Formulas*

Let us consider a function $G(\omega)$ of the form (1.3.6). As we have seen, such a function has a regular analytic continuation in $I_+$. This is still not sufficient, however, for the derivation of dispersion relations: we must have additional information about the behavior of $G(\omega)$ as $|\omega| \to \infty$. Such information has to be obtained from the particular physical problem under consideration. In order to obtain a relation between $\operatorname{Re} G$ and $\operatorname{Im} G$, we need, besides the regularity of $G$ in $I_+$, some condition ensuring that it decreases sufficiently rapidly at infinity; otherwise, $\operatorname{Re} G$ and $\operatorname{Im} G$ can be completely unrelated, as shown by the example: $G(\omega) = a + ib$ (complex constant).

To begin with, we shall assume that $G(\omega)$ is square integrable, i.e.,

$$\int_{-\infty}^{\infty} |G(\omega)|^2 \, d\omega < C, \tag{1.6.1}$$

where $C$ is a constant. This assumption is often related with the requirement that the total energy must be finite. However, this requirement may lead only to a weaker restriction, e.g., that $G(\omega)$ is bounded. We shall see later how to proceed in cases in which (1.6.1) is not fulfilled (cf. Section 1.7).

According to Parseval's theorem, if $F(\omega)$ and $G(\omega)$ are the Fourier transforms of $f(t)$ and $g(t)$, defined as in (1.2.10), we have

$$\int_{-\infty}^{\infty} f(t) g^*(t)\, dt = \frac{1}{2\pi} \int_{-\infty}^{\infty} F(\omega) G^*(\omega)\, d\omega. \tag{1.6.2}$$

Thus, taking into account (1.3.5), (1.6.1) implies

$$\int_0^{\infty} |g(t)|^2\, dt < \frac{C}{2\pi}. \tag{1.6.3}$$

On the other hand, according to (1.3.7), $G(u+iv)$ is the Fourier transform of $e^{-vt} g(t)$, so that, again by Parseval's theorem,

$$\begin{aligned}\int_{-\infty}^{\infty} |G(u+iv)|^2\, du &= 2\pi \int_0^{\infty} e^{-2vt} |g(t)|^2\, dt \\ &< 2\pi \int_0^{\infty} |g(t)|^2\, dt \qquad (v > 0);\end{aligned}$$

i.e., according to (1.6.3),

$$\int_{-\infty}^{\infty} |G(u+iv)|^2\, du < C \qquad (v > 0). \tag{1.6.4}$$

Thus, *a function of the form (1.3.6) which is square integrable on the real axis is also square integrable along any straight line parallel to the real axis*[9] *in* $I_+$. In particular, it follows that

$$\lim_{u \to \pm\infty} G(u+iv) = 0 \qquad (v \geqslant 0). \tag{1.6.5}$$

Now let $\omega = u_0 + iv_0$ be a point in $I_+$ and let us consider a rectangular contour of integration $\Gamma$ with corners at the points $\pm U$, $\pm U + iV$, where $U$ and $V$ will eventually $\to \infty$. According to Cauchy's theorem,

$$G(\omega) = \frac{1}{2\pi i} \int_{\Gamma} \frac{G(\omega')}{\omega' - \omega}\, d\omega'.$$

[9] The left-hand side of (1.6.4) is the $L^2$ norm of $G(u+iv)$ (for fixed $v$). This norm is uniformly bounded in $I_+$ (actually, it is a monotonically decreasing function of $v$). A function $G(\omega)$ regular in $I_+$ and having this property is said to *belong to Hardy class* $H^2$ (cf. Hoffman [16] and Duren [17]).

Let us first fix $V$ and consider the integral along the right side; we have

$$\left|\int_0^V \frac{G(U+iv)\,dv}{U+iv-u_0-iv_0}\right| \leqslant \max_{0\leqslant v\leqslant V} |G(U+iv)| \int_0^V \frac{dv}{[(U-u_0)^2+(v-v_0)^2]^{1/2}}$$

$$= \max_{0\leqslant v\leqslant V} |G(U+iv)|$$

$$\times \ln \left| \frac{V-v_0+[(U-u_0)^2+(V-v_0)^2]^{1/2}}{[(U-u_0)^2+v_0{}^2]^{1/2}-v_0} \right| \to 0$$

for $U\to\infty$ [cf. (1.6.5)]. Similarly, the integral along the left side of the rectangle $\to 0$ for $U\to\infty$. Thus,

$$G(\omega) = \frac{1}{2\pi i}\left[\int_{-\infty}^{\infty} \frac{G(\omega')}{\omega'-\omega}\,d\omega' - \int_{-\infty}^{\infty} \frac{G(u+iV)}{u+iV-u_0-iv_0}\,du\right].$$

According to Schwarz's inequality,

$$\left|\int_{-\infty}^{\infty} \frac{G(u+iV)\,du}{u+iV-u_0-iv_0}\right|^2 \leqslant \int_{-\infty}^{\infty} |G(u+iV)|^2\,du \int_{-\infty}^{\infty} \frac{du}{(u-u_0)^2+(V-v_0)^2}$$

$$< \frac{\pi C}{V-v_0},$$

where we have employed (1.6.4). The last member also $\to 0$ as $V\to\infty$. Thus, we finally get

$$G(\omega) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{G(\omega')}{\omega'-\omega}\,d\omega' \qquad (\operatorname{Im}\omega > 0), \qquad (1.6.6)$$

which expresses the value of $G(\omega)$ at any point in the upper half-plane in terms of its values on the real axis.

Now let $\omega$ be a point on the real axis. In this case, the contour $\Gamma$ must be taken as in Fig. 1.6, avoiding the point $\omega$ by a semicircle of radius $\varepsilon$ in $I_+$.

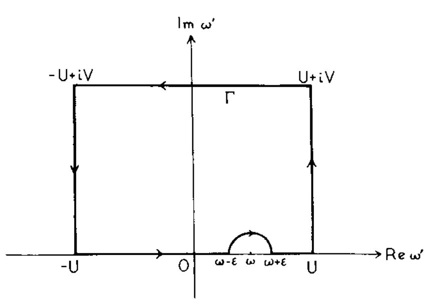


*FIG. 1.6.* The contour Γ.

In the limit as $\varepsilon \to 0$, the semicircle contributes a half-residue at $\omega$, so that we get, for $U \to \infty$,

$$0 = \frac{1}{2\pi i}\int_\Gamma \frac{G(\omega')\,d\omega'}{\omega'-\omega}$$
$$= \frac{1}{2\pi i}\lim_{\varepsilon\to 0}\left(\int_{-\infty}^{\omega-\varepsilon} + \int_{\omega+\varepsilon}^{\infty}\right) - \frac{G(\omega)}{2},$$

or

$$G(\omega) = \frac{1}{\pi i}\,⨍_{-\infty}^{\infty} \frac{G(\omega')}{\omega'-\omega}\,d\omega' \qquad (\mathrm{real}\,\omega), \tag{1.6.7}$$

where

$$⨍_{-\infty}^{\infty} = P\int_{-\infty}^{\infty} = \lim_{\varepsilon\to 0}\left(\int_{-\infty}^{\omega-\varepsilon} + \int_{\omega+\varepsilon}^{\infty}\right) \tag{1.6.8}$$

denotes Cauchy's principal value.

Comparing (1.6.7) with the limit of (1.6.6) as $\omega$ tends to the real axis from above, we see that it corresponds to the well-known formula in distribution theory [cf. Appendix A, Eq. (A4.24)]

$$\lim_{\varepsilon\to 0+}\left(\frac{1}{\omega'-\omega-i\varepsilon}\right) = P\left(\frac{1}{\omega'-\omega}\right) + i\pi\delta(\omega'-\omega), \tag{1.6.9}$$

where $P$ denotes Cauchy's principal value.

Taking the real and imaginary parts of (1.6.7), we get

$$\mathrm{Re}\,G(\omega) = \frac{1}{\pi}\,⨍_{-\infty}^{\infty} \frac{\mathrm{Im}\,G(\omega')}{\omega'-\omega}\,d\omega', \tag{1.6.10}$$

$$\mathrm{Im}\,G(\omega) = -\frac{1}{\pi}\,⨍_{-\infty}^{\infty} \frac{\mathrm{Re}\,G(\omega')}{\omega'-\omega}\,d\omega'. \tag{1.6.11}$$

These relations between $\mathrm{Re}\,G$ and $\mathrm{Im}\,G$ are the *dispersion relations* mentioned above. They are also known as the *Plemelj formulas*.

In agreement with the considerations of Section 1.5, $\mathrm{Im}\,G$ at one frequency is related to $\mathrm{Re}\,G$ for all frequencies, and vice versa. Either $\mathrm{Re}\,G$ or $\mathrm{Im}\,G$ can be chosen as an arbitrary square integrable function, but once one of them is given the other one is completely determined by causality.

*Examples.*

i. Let

$$\mathrm{Im}\,G(\omega) = -1/(1+\omega^2), \tag{1.6.12}$$

which is obviously square integrable. Substituting this expression on the right-hand side of (1.6.10), we find (e.g., by contour integration)

$$\operatorname{Re} G(\omega) = \omega/(1+\omega^2) \tag{1.6.13}$$

so that

$$G(\omega) = (\omega - i)/(1+\omega^2) = 1/(\omega+i), \tag{1.6.14}$$

which is regular in $I_+$.

ii. Similarly, if we take

$$\operatorname{Im} G(\omega) = \frac{2\gamma\omega}{(\omega_0{}^2-\omega^2)^2+4\gamma^2\omega^2}, \tag{1.6.15}$$

we find

$$\operatorname{Re} G(\omega) = \frac{\omega_0{}^2-\omega^2}{(\omega_0{}^2-\omega^2)^2+4\gamma^2\omega^2}, \tag{1.6.16}$$

and

$$G(\omega) = 1/(\omega_0{}^2-\omega^2-2i\gamma\omega), \tag{1.6.17}$$

in agreement with (1.2.7).

iii. An example in which $G(\omega)$ is not square integrable is provided by the formula [cf. Eq. (A11.4)]

$$\varepsilon(t)\,e^{i\omega t} = (1/\pi i)\,⨍_{-\infty}^{\infty} [e^{i\omega' t}/(\omega'-\omega)]\,d\omega', \tag{1.6.18}$$

where $\varepsilon(t) = -1$ for $t<0$, $\varepsilon(t)=1$ for $t>0$. This formula can readily be proved by contour integration. It follows that, if we take

$$G(\omega) = e^{i\omega t} \qquad (t>0), \tag{1.6.19}$$

its real and imaginary parts $\cos(\omega t)$ and $\sin(\omega t)$ verify the Plemelj formulas (1.6.10) and (1.6.11), even though $G(\omega)$ is not square integrable.

### *(b) Hilbert Transforms*

It will now be shown that each of the Plemelj formulas (1.6.10) and (1.6.11) implies the other one, so that it suffices to keep only one of them.

To see how this arises, let us start from (1.3.6):

$$G(\omega) = \int_0^{\infty} g(t)e^{i\omega t}\,dt.$$

It follows that

$$\operatorname{Re} G(\omega) = \frac{1}{2}\int_0^{\infty} [g(t)e^{i\omega t} + g^*(t)e^{-i\omega t}]\, dt,$$

or, changing $t \to -t$ in the second term,

$$\operatorname{Re} G(\omega) = \frac{1}{2}\int_{-\infty}^{\infty} \tilde{g}(t)e^{i\omega t}\, dt, \tag{1.6.20}$$

where

$$\tilde{g}(t) = \begin{cases} g(t), & t > 0, \\ g^*(-t), & t < 0. \end{cases} \tag{1.6.21}$$

Similarly,

$$\operatorname{Im} G(\omega) = -(i/2)\int_{-\infty}^{\infty} \varepsilon(t)\tilde{g}(t)e^{i\omega t}\, dt. \tag{1.6.22}$$

Comparing (1.6.20) with (1.6.22), we see that the inverse Fourier transforms of $\operatorname{Re} G(\omega)$ and $\operatorname{Im} G(\omega)$ differ only by a factor $-i\varepsilon(t)$.

Thus, let

$$p(t) = (1/2\pi)\int_{-\infty}^{\infty} \operatorname{Im} G(\omega)e^{-i\omega t}\, d\omega, \tag{1.6.23}$$

and let us *define*

$$q(t) = i\varepsilon(t)p(t) = (1/2\pi)\int_{-\infty}^{\infty} Q(\omega)e^{-i\omega t}\, d\omega. \tag{1.6.24}$$

Then, employing the identity (1.6.18), we get

$$\begin{aligned} Q(\omega) &= i\int_{-\infty}^{\infty} \varepsilon(t)p(t)e^{i\omega t}\, dt \\ &= \frac{1}{\pi} \mathop{⨍}_{-\infty}^{\infty} \frac{d\omega'}{\omega'-\omega}\int_{-\infty}^{\infty} p(t)e^{i\omega' t}\, dt \\ &= \frac{1}{\pi} \mathop{⨍}_{-\infty}^{\infty} \frac{\operatorname{Im} G(\omega')}{\omega'-\omega}\, d\omega' \\ &= \operatorname{Re} G(\omega), \end{aligned} \tag{1.6.25}$$

where we have employed (1.6.23) and (1.6.10), which was assumed valid. The interchange of the order of integration in the double integral is legitimate for square integrable $p(t)$.

We can now invert (1.6.24) to get

$$p(t) = -i\varepsilon(t)q(t). \tag{1.6.26}$$

This differs from (1.6.24) only by the interchange: $q \to p$, $p \to -q$. This is equivalent to the substitution $\operatorname{Re} G \to \operatorname{Im} G$, $\operatorname{Im} G \to -\operatorname{Re} G$, which transforms (1.6.10) into (1.6.11). Thus, for square integrable functions, (1.6.11) follows from (1.6.10), and vice versa.

On account of the reciprocity relation between $\operatorname{Re} G$ and $\operatorname{Im} G$, these two functions are called *Hilbert transforms* of each other.

The above results also imply

$$\begin{aligned} G(\omega) &= \operatorname{Re} G(\omega) + i \operatorname{Im} G(\omega) \\ &= i \int_{-\infty}^{\infty} [1+\varepsilon(t)]\, p(t)\, e^{i\omega t}\, dt \\ &= 2i \int_{0}^{\infty} p(t)\, e^{i\omega t}\, dt, \end{aligned} \tag{1.6.27}$$

which leads us back to (1.3.6).

The right-hand sides of (1.6.20) and (1.6.22) are sometimes called *dispersive part* and *absorptive part*, respectively, on account of the analogy with the complex refractive index.

### (c) Titchmarsh's Theorem

This theorem[10] is a precise formulation of the preceding results.

THEOREM 1.6.1 (Titchmarsh's theorem). *If a square integrable function* $G(\omega)$ *fulfills one of the four conditions below, then it fulfills all four of them:*

(i) *The inverse Fourier transform* $g(t)$ *of* $G(\omega)$ *vanishes for* $t < 0$:

$$g(t) = 0 \qquad (t < 0).$$

(ii) $G(u)$ *is, for almost all* $u$, *the limit as* $v \to 0+$ *of an analytic function* $G(u+iv)$ *that is holomorphic in the upper half-plane and square integrable over any line parallel to the real axis:*

$$\int_{-\infty}^{\infty} |G(u+iv)|^2\, du < C \qquad (v > 0).$$

(iii) $\operatorname{Re} G$ *and* $\operatorname{Im} G$ *verify the first Plemelj formula:*

$$\operatorname{Re} G(\omega) = \frac{1}{\pi} \fint_{-\infty}^{\infty} \frac{\operatorname{Im} G(\omega')}{\omega' - \omega}\, d\omega'.$$

(iv) $\operatorname{Re} G$ *and* $\operatorname{Im} G$ *verify the second Plemelj formula:*

$$\operatorname{Im} G(\omega) = -\frac{1}{\pi} \fint_{-\infty}^{\infty} \frac{\operatorname{Re} G(\omega')}{\omega' - \omega}\, d\omega'.$$

---

[10] Parts (i) and (ii) of Theorem 1.6.1 are related with the Paley–Wiener theorem; by (ii), $G(\omega) \in H^2$ in $I_+$ [cf. footnote after (1.6.4)].

The restriction to "almost all $u$" corresponds to the fact that, if we change a function over a set of measure zero, its Fourier transform does not change.

For a rigorous proof of the theorem, we refer the reader to Titchmarsh's work [9, pp. 125–129]. An (not very rigorous!) argument is contained in the above discussion. In fact, according to Section 1.3 and (1.6.4), (i) $\Rightarrow$ (ii). We have also seen that (ii) $\Rightarrow$ (iii) and (iv), and each of these two implies the other one. Finally, according to (1.6.18), (iii) or (iv) imply (i). Thus we have a complete cycle, showing that the four conditions are indeed equivalent. In particular, we see that, *for square integrable functions, the dispersion relations are not only necessary but also sufficient for the validity of the causality condition.*

A function $G(\omega)$ verifying one of the conditions of Titchmarsh's theorem (and consequently all four of them) will be called a *causal transform.* A simple example is the function

$$G(\omega) = 1/(\omega-\zeta), \qquad \operatorname{Im}\zeta < 0, \tag{1.6.28}$$

where $\zeta$ is any complex number in $I_-$. Another example is the function $G(\omega)$ defined by (1.2.7), which has two poles in $I_-$. Note also that any linear combination of (a finite number of) causal transforms with constant coefficients is a causal transform.

It should be emphasized that a square integrable function that is the boundary value of an analytic function regular in the upper half-plane need *not* be a causal transform. It is necessary, in addition, that the analytic function be square integrable on any straight line parallel to the real axis, as stated in part (ii) of Titchmarsh's theorem. This is clearly shown by the example

$$G(\omega) = e^{-ia\omega}/(\omega-\zeta) \qquad (\operatorname{Im}\zeta < 0, \quad a > 0), \tag{1.6.29}$$

where $e^{-ia\omega}$ is just a phase factor on the real axis, but blows up exponentially in $I_+(\omega)$, so that (1.6.4) is not satisfied. Thus, (1.6.29) is not a causal transform. As may readily be verified, its inverse Fourier transform vanishes only for $t < -a$, rather than for $t < 0$.

## 1.7. Subtractions

### *(a) Dispersion Relations with One Subtraction*

In practice, the function $G(\omega)$ may not verify the assumption (1.6.1) of square integrability. A common situation, for instance, is that in which a square integrable output $x(t)$ corresponds to a square integrable input $f(t)$

in (1.3.2), and there exists a constant $A$ such that

$$\int_{-\infty}^{\infty} |x(t)|^2 \, dt \leqslant A \int_{-\infty}^{\infty} |f(t)|^2 \, dt. \tag{1.7.1}$$

This usually corresponds to the requirement that the total output energy is at most equal to the total input energy (it may be smaller because of absorption).

It then follows from (1.3.4) and Parseval's theorem (1.6.2) that

$$\begin{aligned} \int_{-\infty}^{\infty} |X(\omega)|^2 \, d\omega &= \int_{-\infty}^{\infty} |G(\omega)|^2 \, |F(\omega)|^2 \, d\omega \\ &\leqslant A \int_{-\infty}^{\infty} |F(\omega)|^2 \, d\omega \end{aligned} \tag{1.7.2}$$

for any square integrable $F(\omega)$, which is only possible if

$$|G(\omega)|^2 \leqslant A. \tag{1.7.3}$$

Thus, $G(\omega)$ is bounded, a weaker restriction than square integrability. According to the causality condition, it still has a regular analytic continuation in $I_+$. This follows from the fact that, for any input $F(\omega)$ that is a causal transform, the corresponding output $X(\omega)$ in (1.3.4) must also be a causal transform. However, because of (1.7.3), $G(\omega)$ itself need not be a causal transform.

It is readily seen that dispersion relations of the form (1.6.10)–(1.6.11) are in general not valid in this case. For instance, $G(\omega) = B$, where $B$ is a constant such that $|B|^2 \leqslant A$, satisfies the above conditions; on the other hand, $\operatorname{Re} B$ and $\operatorname{Im} B$ are totally unrelated and (1.6.10)–(1.6.11) would be meaningless for this choice of $G(\omega)$. Clearly, (1.7.3) is not sufficient to guarantee the convergence of the integrals in these relations.

This example also shows that, given $\operatorname{Im} G(\omega)$ in the present situation, $\operatorname{Re} G(\omega)$ is not completely determined by causality, for if $\operatorname{Re} G$ is a solution then $\operatorname{Re} G + C$ is also a possible solution, where $C$ is an arbitrary real constant [compatible with (1.7.3)]. In fact, if $x(t)$ is a causal output corresponding to the input $f(t)$, the function $x(t) + Cf(t)$ is also a causal output. Thus, there is an arbitrary additive constant that must be specified before we can relate $\operatorname{Re} G$ and $\operatorname{Im} G$.

This constant is usually specified by giving the value of $G(\omega)$ for some (real) frequency $\omega = \omega_0$. We shall also assume that $G(\omega)$ is differentiable at this point (this is usually true, although weaker assumptions may be sufficient). Then the function

$$D(\omega) = [G(\omega) - G(\omega_0)]/[\omega - \omega_0] = \Delta(\omega)/(\omega - \omega_0) \tag{1.7.4}$$

is bounded for $\omega \to \omega_0$, regular in $I_+$ and square integrable on the real axis, according to (1.7.3). We now show that it is also a causal transform. Note that this does not follow directly from the above properties, as was emphasized at the end of Section 1.6.

For this purpose, let us take the input function

$$F(\omega) = 1/(\omega-\zeta), \qquad \operatorname{Im}\zeta < 0,$$

which is a causal transform. The corresponding output function $X(\omega) = G(\omega)/(\omega-\zeta)$ is therefore also a causal transform, and so is the function $\Delta(\omega)/(\omega-\xi)$, so that this function verifies the relation (1.6.7):

$$\Delta(\omega) = \frac{\omega-\zeta}{\pi i} \int_{-\infty}^{\infty} \frac{\Delta(\omega')\, d\omega'}{(\omega'-\zeta)(\omega'-\omega)}. \tag{1.7.5}$$

For $\omega = \omega_0$, this becomes

$$0 = \frac{\omega_0-\zeta}{\pi i} \int_{-\infty}^{\infty} \frac{\Delta(\omega')\, d\omega'}{(\omega'-\zeta)(\omega'-\omega_0)}. \tag{1.7.6}$$

Subtracting (1.7.6) from (1.7.5), and employing the identity

$$\frac{\omega-\zeta}{\omega'-\omega} - \frac{\omega_0-\zeta}{\omega'-\omega_0} = \frac{(\omega-\omega_0)(\omega'-\zeta)}{(\omega'-\omega_0)(\omega'-\omega)}, \tag{1.7.7}$$

we get

$$G(\omega) = G(\omega_0) + \frac{\omega-\omega_0}{\pi i} \int_{-\infty}^{\infty} \frac{[G(\omega')-G(\omega_0)]}{\omega'-\omega_0} \frac{d\omega'}{\omega'-\omega}. \tag{1.7.8}$$

This is the dispersion relation (1.6.7) [equivalent to (1.6.10) and (1.6.11)] for the function $D(\omega)$ defined in (1.7.4), thus proving the assertion that $D(\omega)$ is a causal transform.

The relation (1.7.8) is the typical dispersion relation for a function $G(\omega)$ satisfying the boundedness condition (1.7.3). As it requires the subtraction of a constant from the original function, it is called a *dispersion relation with one subtraction*, and the constant $G(\omega_0)$ is called the *subtraction constant*. Taking the real and imaginary parts of (1.7.8), we see that $\operatorname{Re} G$ and $\operatorname{Im} G$ determine each other once the subtraction constant is given.

The reference frequency $\omega_0$ is often chosen as $\omega_0 = 0$, so that the subtraction constant is the static value of $G(\omega)$. It may also be possible to take $\omega_0 \to \infty$, so that (1.7.8) takes the limiting form

$$G(\omega) = G(\infty) + \frac{1}{\pi i} \int_{-\infty}^{\infty} [G(\omega')-G(\infty)] \frac{d\omega'}{\omega'-\omega}. \tag{1.7.9}$$

### *(b) Dispersion Relations with More than One Subtraction*

Let us consider next the case in which

$$G(\omega) = O(\omega) \qquad (|\omega| \to \infty). \tag{1.7.10}$$

An example of this situation will be discussed later on (Section 3.2).

In this case, we need two subtractions in order to get a square integrable function:

$$D_1(\omega) = [G(\omega) - G(\omega_0) - (\omega-\omega_0) G'(\omega_0)]/(\omega-\omega_0)^2 = \Delta_1(\omega)/(\omega-\omega_0)^2, \tag{1.7.11}$$

where we have assumed that $G(\omega)$ is twice differentiable at $\omega = \omega_0$. We can now show that $D_1(\omega)$ is a causal transform, by a procedure entirely similar to that employed for (1.7.4).

The function $\Delta_1(\omega)/(\omega-\zeta)^2$ ($\operatorname{Im}\zeta < 0$) is a causal transform:

$$\frac{\Delta_1(\omega)}{\omega-\zeta} = \frac{\omega-\zeta}{\pi i} \int_{-\infty}^{\infty} \frac{\Delta_1(\omega')}{(\omega'-\zeta)^2} \frac{d\omega'}{(\omega'-\omega)}. \tag{1.7.12}$$

Taking $\omega = \omega_0$ and noting that $\Delta_1(\omega_0) = 0$, we can write

$$0 = \frac{\omega_0-\zeta}{\pi i} \int_{-\infty}^{\infty} \frac{\Delta_1(\omega')}{(\omega'-\zeta)^2} \frac{d\omega'}{(\omega'-\omega_0)}. \tag{1.7.13}$$

Subtracting (1.7.13) from (1.7.12) and employing the identity (1.7.7), we get

$$\Delta_1(\omega) = \frac{(\omega-\omega_0)(\omega-\zeta)}{\pi i} \int_{-\infty}^{\infty} \frac{\Delta_1(\omega')}{(\omega'-\omega_0)(\omega'-\zeta)} \frac{d\omega'}{(\omega'-\omega)}. \tag{1.7.14}$$

Taking again $\omega = \omega_0$ in this relation, and noting that $\Delta_1(\omega)/(\omega-\omega_0) \to 0$ for $\omega \to \omega_0$, we find

$$0 = \frac{(\omega-\omega_0)(\omega_0-\zeta)}{\pi i} \int_{-\infty}^{\infty} \frac{\Delta_1(\omega')}{(\omega'-\zeta)(\omega'-\omega_0)^2} d\omega'. \tag{1.7.15}$$

Subtracting this result from (1.7.14) and again employing (1.7.7), we finally get

$$G(\omega) = G(\omega_0) + (\omega-\omega_0) G'(\omega_0) + \frac{(\omega-\omega_0)^2}{\pi i} \int_{-\infty}^{\infty} \frac{[G(\omega') - G(\omega_0) - (\omega'-\omega_0) G'(\omega_0)]}{(\omega'-\omega_0)^2} \frac{d\omega'}{(\omega'-\omega)}, \tag{1.7.16}$$

showing that $D_1(\omega)$ is a causal transform. This is a *dispersion relation with two subtractions* for $G(\omega)$, the subtraction constants being $G(\omega_0)$ and $G'(\omega_0)$.

It is now clear that, by repeated application of the same procedure, we can derive a dispersion relation with $n+1$ subtractions, for the case in which

$$G(\omega) = O(\omega^n) \qquad (|\omega| \to \infty). \tag{1.7.17}$$

The dispersion relation is then of the form

$$G(\omega) = G(\omega_0) + (\omega-\omega_0)G'(\omega_0) + \cdots + \frac{(\omega-\omega_0)^n}{n!}G^{(n)}(\omega_0)$$
$$+ \frac{(\omega-\omega_0)^{n+1}}{\pi i} \int_{-\infty}^{\infty} \frac{\left[G(\omega') - G(\omega_0) - \cdots - \dfrac{(\omega'-\omega_0)^n}{n!}G^{(n)}(\omega_0)\right]}{(\omega'-\omega_0)^{n+1}}$$
$$\times \frac{d\omega'}{(\omega'-\omega)}, \tag{1.7.18}$$

where we have assumed that $G(\omega)$ is differentiable up to the order $n+1$ at $\omega = \omega_0$.

Additional subtractions can also be made even when dispersion relations with fewer subtractions are valid. If (1.7.18) is valid for a given $n$, it is valid a fortiori for $n' > n$, provided that the corresponding higher-order derivatives exist at $\omega = \omega_0$.

The purpose of writing down a dispersion relation with more than the minimum required number of subtractions usually is to improve the convergence at high frequencies, due to the increase in the power of the frequency appearing in the denominator of (1.7.18). This also makes the integral less strongly dependent on the high-frequency behavior of $G(\omega)$, which may not be known with sufficient accuracy.

This improvement is made at the expense of additional knowledge required about the behavior of $G(\omega)$ in the neighborhood of $\omega_0$. Instead of determining higher-order derivatives at a single point, however, which may be difficult to do in practice, one can also determine the value of the function at different points.

Thus, for instance, if we write down a dispersion relation of the form (1.7.8) at two different points, $\omega_0$ and $\omega_1$, and subtract term by term, we finally get

$$G(\omega) = \left(\frac{\omega-\omega_1}{\omega_0-\omega_1}\right)G(\omega_0) - \left(\frac{\omega-\omega_0}{\omega_0-\omega_1}\right)G(\omega_1)$$
$$+ \frac{(\omega-\omega_0)(\omega-\omega_1)}{\pi i}$$
$$\times \int_{-\infty}^{\infty} \frac{\left[G(\omega') - \dfrac{\omega'-\omega_1}{\omega_0-\omega_1}G(\omega_0) + \dfrac{\omega'-\omega_0}{\omega_0-\omega_1}G(\omega_1)\right]}{(\omega'-\omega_0)(\omega'-\omega_1)(\omega'-\omega)}\,d\omega', \tag{1.7.19}$$

which contains an additional power of $\omega'$ in the denominator for $\omega' \to \infty$.

In the dispersion relations employed in high-energy physics, subtraction constants are usually related with the coupling constants that characterize the interactions.

## 1.8. Dispersion Relations and Distributions

### (a) Distributions as Boundary Values of Analytic Functions

As has already been mentioned in Section 1.3, the assumption that the input, the output, and Green's function are ordinary functions is too restrictive for many physical applications, in which we have to deal with distributions. We shall now extend the preceding discussion of the relation between causality and analyticity to this more general case. The main concepts and results from distribution theory that will be required for the extension are given in Appendix A.

In order to extend (1.3.2), let us assume that the input $f_t$ and the output $x_t$ are distributions in the time variable $t$, belonging to the space $\mathscr{D}'$. To formulate the causality condition, we need also the space $\mathscr{D}_+{}'$ of all distributions in $\mathscr{D}'$ with support in the semiaxis $[0, \infty)$ (cf. Section A8).

The analog of the discussion given in Section 1.3 is contained in the following theorem:

THEOREM 1.8.1. *Let $\mathscr{G}$ be an operator mapping every distribution $f_t \in \mathscr{D}'$ into another distribution $x_t \in \mathscr{D}'$, and having the following properties:*

(a) *$\mathscr{G}$ is linear*;

(b) *$\mathscr{G}$ commutes with time translations* (i.e., $\mathscr{G}f_{t+\tau} = x_{t+\tau}$, or, if $\tau$ denotes the translation operator, $\mathscr{G}\tau = \tau\mathscr{G}$);

(c) *$\mathscr{G}$ is causal,* i.e.,

$$f_t \in \mathscr{D}_+{}' \Rightarrow x_t \in \mathscr{D}_+{}'; \tag{1.8.1}$$

(d) *$\mathscr{G}$ is continuous in $\mathscr{D}_+{}'$*; i.e., if we have a sequence $\{f_{t,j}\}$ of inputs tending to $f_t$ in $\mathscr{D}_+{}'$, the corresponding sequence of outputs $\{x_{t,j}\}$ tends to the corresponding output $x_t$ in $\mathscr{D}_+{}'$ as $j \to \infty$;

*Then $\mathscr{G}$ is a convolution operator; i.e.,*

$$\mathscr{G}f_t = x_t = g_t * f_t, \tag{1.8.2}$$

*where $g_t$ is a distribution belonging to $\mathscr{D}_+{}'$.*

In fact, let $g_t$ be the output corresponding to the input $x_t = \delta_t \in \mathscr{D}_+{}'$ (Green's function). Then, $g_t \in \mathscr{D}_+{}'$ by (c). By (b), we have $\mathscr{G}\delta_{t-\tau} = g_{t-\tau}$ and, by (a),

$$\mathscr{G}\left(\sum_n c_n \delta_{t-\tau_n}\right) = \sum_n c_n g_{t-\tau_n} = g_t * \sum_n c_n \delta_{t-\tau_n}$$

[cf. (A8.8)]. But any distribution $f_t \in \mathscr{D}_+'$ can be expressed as a limit of such linear combinations of $\delta_{t-\tau_n}$; i.e., they are dense in $\mathscr{D}_+'$ [for a function, one would write $f(t) = \int f(\tau)\delta(t-\tau)\,d\tau$]. Then, by using (d), one can prove (1.8.2) for all $f_t \in \mathscr{D}_+'$. Note that the convolution product (1.8.2) always exists, because $g_t \in \mathscr{D}_+'$, $f_t \in \mathscr{D}_+'$ (cf. Section A7).

Let us now make the additional assumption that *$f_t$, $x_t$, and $g_t$ are temperate distributions*, so that they have Fourier transforms (cf. Section A10)

$$F_\omega = \mathscr{F}f_t, \qquad X_\omega = \mathscr{F}x_t, \qquad G_\omega = \mathscr{F}g_t. \tag{1.8.3}$$

Then, if the convolution theorem (A10.18) applies, the Fourier transform of (1.8.2) yields

$$X_\omega = G_\omega F_\omega. \tag{1.8.4}$$

Some sufficient conditions for the validity of the convolution theorem have been given in Section A10. In particular, if we restrict the input $f_t$ to be a rapidly decreasing distribution, $f_t \in \mathscr{O}_C'$, Green's function $g_t$ can be any temperate distribution: $g_t \in \mathscr{S}'$. In terms of the Fourier transform, it is sufficient that $F_\omega \in \mathscr{O}_M$ (cf. Section A10); i.e., the Fourier transform of the input can be any $C^\infty$ function which, together with all its derivatives, increases no faster than some power of $\omega$ as $\omega \to \infty$.

We now consider the problem of characterizing the effect of the causality condition, namely, under what conditions the distribution $G_\omega \in \mathscr{S}'$ is the Fourier transform of a distribution $g_t \in \mathscr{D}_+'$. The analog of parts (i) and (ii) of Titchmarsh's Theorem 1.6.1 is contained in the following result:

THEOREM 1.8.2. *Let $G_\omega = \mathscr{F}g_t \in \mathscr{S}'$. Then, $g_t \in \mathscr{D}_+'$ if and only if:*

(a) *$G_\omega$ is the boundary value of an analytic function $G(u+iv)$, regular for $v > 0$.*

(b) *For any fixed $v > 0$, $G(u+iv)$, regarded as a distribution in $u$, is temperate, and $G(u+iv) \to G_u$ (in the sense of $\mathscr{S}_u'$) as $v \to 0+$.*

(c) *For* $\operatorname{Im}\omega \geqslant \varepsilon > 0$, *$|G(\omega)|$ is bounded by a power of $|\omega|$:*

$$|G(\omega)| = O(|\omega|^n) \qquad (\operatorname{Im}\omega \geqslant \varepsilon > 0), \tag{1.8.5}$$

where the power $n$ may depend on $\varepsilon$.

These conditions are not mutually independent. The above theorem arises in the theory of the Laplace transformation of distributions [18]. The analyticity of the Laplace transform in the half-plane of convergence is well known for functions; the polynomial boundedness (1.8.5) arises from having a temperate distribution.

*Example.* According to (A11.5),

$$1/(\omega+i0) = P(1/\omega) - i\pi\delta_\omega = -2i\pi\delta_\omega^{+}$$

is the boundary value of $G(\omega) = 1/\omega$, which satisfies conditions (a), (b), and (c). On the other hand, according to (A11.9),

$$\mathscr{F}^{-1}\delta_\omega^{\ +} = \theta(t)/2\pi \in \mathscr{D}_+'.$$

It is possible [19] to use Theorem 1.8.2 for deriving dispersion relations by taking as contour of integration the straight line $\operatorname{Im}\omega = \varepsilon > 0$ closed by a half-circle at infinity, and then letting $\varepsilon \to 0$. However, we shall employ a different method, due to Schwartz [20], that does not make any use of analytic continuation. This method has the advantage of making clear what is the broadest class of distributions that may satisfy each type of dispersion relation.

Let us consider first a dispersion relation without subtractions, such as (1.6.7). The right-hand side of (1.6.7) can be rewritten as a convolution product:

$$G_\omega = -(1/\pi i)\, G_\omega * P(1/\omega). \tag{1.8.6}$$

In order that a dispersion relation of this type be meaningful for a distribution $G_\omega$, it is necessary (at least!) that the convolution product on the right be well defined. Our next problem will be to characterize the class of distributions that are convolutionable with $P(1/\omega)$. For this purpose, we must introduce some new distribution spaces.

### *(b) Convolution of Distributions with $P(1/\omega)$*

We shall first introduce the space $\mathscr{D}_L'$ of *summable distributions*. A *summable function* is a function $F(\omega) \in L$, i.e., such that

$$\int_{-\infty}^{\infty} |F(\omega)|\, d\omega < \infty .$$

*A distribution $T_\omega$ is summable if it is a finite sum of derivatives (in the sense of distributions) of summable functions:*

$$T_\omega = \sum_{p=0}^{n} D^p F_p(\omega), \qquad F_p(\omega) \in L. \tag{1.8.7}$$

One can then define $\int T_\omega\, d\omega$ by

$$\int T_\omega\, d\omega = \int F_0(\omega)\, d\omega, \tag{1.8.8}$$

where the right-hand side is an ordinary integral. It can be shown that the result does not depend on the choice of the decomposition (1.8.7), which is not unique. $\mathscr{D}_L'$ is the space of all summable distributions.

*Examples.*

i. All summable functions $\in \mathscr{D}_L'$. In particular, any continuous function that is $O(\omega^{-1-\alpha})$ or $O(\omega^{-1}[\ln|\omega|]^{-1-\alpha})$ $(\alpha > 0)$ as $|\omega| \to \infty$ is in $\mathscr{D}_L'$.

ii. Any distribution with compact support is in $\mathscr{D}_L'$; i.e., $\mathscr{E}' \subset \mathscr{D}_L'$ (cf. Section A6). In fact, it can be shown that any such distribution can be expressed in the form (1.8.7), with all $F_p(\omega)$ having compact support. In particular, $\delta_\omega \in \mathscr{D}_L'$ and

$$\int \delta_\omega \, d\omega = (\delta_\omega, 1) = 1. \tag{1.8.9}$$

If $F(\omega)$ is a summable function, so is $F(\omega)\,\varphi(\omega)$, whenever $\varphi(\omega)$ is bounded. Similarly, *let $\varphi(\omega)$ be a $C^\infty$ function that is bounded, as well as all its derivatives. The space of all such test functions is called $\mathscr{B}$.* Then it can be shown that, *if $T_\omega \in \mathscr{D}_L'$ and $\varphi(\omega) \in \mathscr{B}$, $(T_\omega, \varphi)$ exists,* although neither $T_\omega$ nor $\varphi$ need have compact support. In particular, $\varphi(\omega) = 1 \in \mathscr{B}$, and we have

$$(T_\omega, 1) = \int T_\omega \, d\omega. \tag{1.8.10}$$

To characterize the distributions $G_\omega$ that are convolutionable with $P(1/\omega)$, we might require that $\omega^{-1}G_\omega$ be summable. However, the denominator $\omega$ might introduce a singularity at the origin, so that we shall replace it by $(1+\omega^2)^{1/2}$, which has the same behavior as $|\omega| \to \infty$ (other choices, such as $1+i\omega$, would also be possible). This leads us to introduce *the space $\mathscr{D}_L'^{(1)}$ of all distributions $G_\omega$ such that*

$$(1+\omega^2)^{-1/2} G_\omega \in \mathscr{D}_L'. \tag{1.8.11}$$

*Examples.*

i. Any square integrable function $G(\omega) \in L^2$ is in $\mathscr{D}_L'^{(1)}$. In fact,

$$2\left|\frac{G(\omega)}{(1+\omega^2)^{1/2}}\right| \leqslant |G(\omega)|^2 + \frac{1}{1+\omega^2},$$

and both terms on the right are summable, by the assumption that $G(\omega) \in L^2$.

ii. Any function $G(\omega)$ such that $(1+\omega^2)^{-1/2} G(\omega) \in L$ is in $\mathscr{D}_L'^{(1)}$. In particular, any continuous function $G(\omega)$ such that

$$\begin{gathered} G(\omega) = O(\omega^{-\alpha}) \qquad \text{or} \qquad G(\omega) = O([\ln|\omega|]^{-1-\alpha}) \\ (\alpha > 0) \qquad \text{as} \quad |\omega| \to \infty . \end{gathered} \tag{1.8.12}$$

iii. Any distribution with compact support $\in \mathscr{D}_L'^{(1)}$. Thus, $\mathscr{E}' \subset \mathscr{D}_L'^{(1)}$. In particular, $\delta_\omega$ and its derivatives belong to $\mathscr{D}_L'^{(1)}$.

iv. Any rapidly decreasing distribution is in $\mathscr{D}_L'^{(1)}$: $\mathscr{O}_C' \subset \mathscr{D}_L'^{(1)}$ (cf. Section A10).

v. $P(1/\omega)$ belongs to $\mathscr{D}_L'^{(1)}$. In fact,

$$P\int_{-\infty}^{\infty}\frac{d\omega}{\omega(1+\omega^2)^{1/2}} = 0,$$

so that $(1+\omega^2)^{-1/2}P(1/\omega)$ is summable.

vi. $G(\omega) = C$ (constant) does *not* belong to $\mathscr{D}_L'^{(1)}$.

We now show that $\mathscr{D}_L'^{(1)}$ is the natural space for defining the convolution product with $P(1/\omega)$. Let us consider first a test function $\varphi(\omega) \in \mathscr{S}$ (cf. Section A9). Then, according to (A8.4),

$$h(\omega) = P\frac{1}{\omega} * \varphi(\omega) = P\int_{-\infty}^{\infty}\varphi(\omega-\omega')\frac{d\omega'}{\omega'} \tag{1.8.13}$$

is a $C^\infty$ function. We have:

LEMMA 1.8.1. *If $h(\omega)$ is defined by (1.8.13), where $\varphi \in \mathscr{S}$, then* $(1+\omega^2)^{1/2} \times h(\omega) \in \mathscr{B}$ (i.e., it is bounded, together with all its derivatives).

*Proof.* We start from the inequality

$$\frac{1}{|\omega'|} \leqslant A\,\frac{[1+(\omega-\omega')^2]^{1/2}}{(1+\omega^2)^{1/2}} \qquad (|\omega'| \geqslant 1), \tag{1.8.14}$$

where $A$ is a constant. This inequality follows from the fact that

$$F(\omega,\omega') = \frac{|\omega'|[1+(\omega-\omega')^2]^{1/2}}{(1+\omega^2)^{1/2}}$$

has a lower bound for $|\omega'| \geqslant 1$. In fact, if $|\omega| \leqslant 1$, $F \geqslant 1/\sqrt{2}$. If $|\omega| \geqslant 1$ and $|\omega'| \geqslant |\omega|/2$, we have $F \geqslant \frac{1}{2}|\omega|(1+\omega^2)^{-1/2}$, which has a lower bound for $|\omega| \geqslant 1$. Finally, if $|\omega| \geqslant 1$ and $1 \leqslant |\omega'| \leqslant |\omega|/2$, we have $F \geqslant (1+\omega^2)^{-1/2} \times [1+(\omega/2)^2]^{1/2}$, which also has a lower bound.

It follows from (1.8.13) and (1.8.14) that

$$\begin{aligned}|h(\omega)| &\leqslant \left(P\int_{-1}^{1}\varphi(\omega-\omega')\frac{d\omega'}{\omega'}\right) + \frac{A}{(1+\omega^2)^{1/2}} \\ &\quad\times \int_{|\omega'|\geqslant 1}[1+(\omega-\omega')^2]^{1/2}|\varphi(\omega-\omega')|\,d\omega' \\ &\leqslant \left(\int_{-1}^{1}[\varphi(\omega-\omega')-\varphi(\omega)]\frac{d\omega'}{\omega'}\right) + \frac{A}{(1+\omega^2)^{1/2}}\int_{-\infty}^{\infty}(1+\xi^2)^{1/2}|\varphi(\xi)|\,d\xi.\end{aligned}$$

By the theorem of the mean applied to the interval $(-1, 1)$,

$$\left|\frac{\varphi(\omega-\omega')-\varphi(\omega')}{\omega'}\right| \leqslant \max_{|\omega'|\leqslant 1}|\varphi'(\omega-\omega')| \leqslant \frac{B}{(1+\omega^2)^{1/2}},$$

because $\varphi \in \mathscr{S}$, so that $\varphi$ and its derivatives decrease at infinity faster than any inverse power. Thus,

$$|h(\omega)| \leqslant C(1+\omega^2)^{-1/2},$$

so that $(1+\omega^2)^{1/2}h(\omega)$ is bounded. The same is true for all its derivatives. In fact, by (A8.11), in order to differentiate $h(\omega)$, it suffices to differentiate $\varphi(\omega)$ in (1.8.13), and $\varphi \in \mathscr{S} \Rightarrow \varphi^{(m)} \in \mathscr{S}$. This completes the proof of the lemma.

We can now define $G(\omega) * P(1/\omega)$, *for any* $G_\omega \in \mathscr{D}_L'^{(1)}$, *as a temperate distribution, given (for any* $\varphi(\omega) \in \mathscr{S}$*) by*

$$(G_\omega * P(1/\omega), \varphi) = -((1+\omega^2)^{-1/2}G_\omega, (1+\omega^2)^{1/2}h(\omega)), \qquad (1.8.15)$$

where $h(\omega)$ is defined by (1.8.13). In fact, by assumption, $(1+\omega^2)^{-1/2}G_\omega \in \mathscr{D}_L'$, and, by Lemma 1.8.1, $(1+\omega^2)^{1/2}h(\omega) \in \mathscr{B}$, so that the right-hand side exists, as we have seen above. It is readily shown to be a continuous linear functional on $\mathscr{S}$, so that $G_\omega * P(1/\omega) \in \mathscr{S}'$.

Finally, if $G_\omega * P(1/\omega)$ exists by the ordinary definition of the convolution product (e.g., for $G_\omega \in \mathscr{E}'$; cf. Section A8), we can suppress the factors $(1+\omega^2)^{\pm 1/2}$, and (1.8.15) reduces to

$$(G_\omega * P(1/\omega), \varphi) = -(G_\omega, P(1/\omega) * \varphi), \qquad (1.8.16)$$

which agrees with (A8.10), because $P(1/\omega)$ is an odd distribution.

The space $\mathscr{D}_L'^{(1)}$ is the natural space for defining the convolution product with $P(1/\omega)$. In particular, since $P(1/\omega) \in \mathscr{D}_L'^{(1)}$ (Example v above), we conclude that $P(1/\omega) * P(1/\omega)$ is a temperate distribution. Its value will be determined below [cf. (1.8.24)].

### (c) Distributions Belonging to $\mathscr{D}_+'$

A locally integrable function $g(t)$ belongs to $\mathscr{D}_+'$ if and only if $g(t) = 0$ for $t < 0$. This can also be expressed as follows:

$$g(t) = \theta(t)g(t) \qquad \text{or} \qquad \check{\theta}(t)g(t) = 0, \qquad (1.8.17)$$

where $\theta(t)$ is the Heaviside step function and $\check{\theta}(t)$ is defined by (A3.8).

In order to extend (1.8.17) to a general characterization of distributions belonging to $\mathscr{D}_+'$, we have to define the product of a distribution by $\theta(t)$ [we cannot employ (A5.1) because $\theta \notin C^\infty$!]. The natural space for this purpose is the space $(\mathscr{E}_0)'$, which is defined as follows:

$(\mathscr{E}_0)'$ *is the space of all distributions that are first-order derivatives (in the sense of distributions) of continuous functions;* i.e., $g_t \in (\mathscr{E}_0)'$ when there exists a continuous function $f(t)$ such that

$$g_t = f'(t), \qquad f(t) \in C^0. \qquad (1.8.18)$$

Clearly, $f(t)$ is only defined up to an arbitrary additive constant.

*Examples.*

i. Any locally integrable function $g(t)$ belongs to $(\mathscr{E}_0)'$ because

$$\int_{t_0}^{t} g(t')\,dt' \in C^0.$$

ii. $\delta \notin (\mathscr{E}_0)'$, but $\theta(t) \in (\mathscr{E}_0)'$; in fact, $\theta(t) = f'(t)$, where $f(t) = 0$ $(t \leqslant 0)$, $f(t) = t$ $(t \geqslant 0)$.

If $g_t \in (\mathscr{E}_0)'$ and is given by (1.8.18), we define $\theta(t) g_t$ by

$$\theta(t) g_t = \theta(t) f'(t) = (\theta f)' - f\theta' = (\theta f)' - f(0)\,\delta. \tag{1.8.19}$$

The right-hand side is well defined. In fact, $\theta f$ is a locally integrable function, so that its derivative (in the sense of distributions) exists. Furthermore, $f(t)\,\delta = f(0)\,\delta$ is well defined for $f \in C^0$ [cf. (A5.4)]. Although $f(t)$ is determined only up to an arbitrary additive constant, the right-hand side is independent of the choice of this constant. Finally, (1.8.19) is in agreement with (A5.12). In particular, we have $\theta^2(t) = \theta(t)$.

THEOREM 1.8.3. *In order that a distribution $g_t \in (\mathscr{E}_0)'$ shall belong to $\mathscr{D}_+'$, it is necessary and sufficient that*

$$g_t = \theta(t) g_t \qquad \text{or (equivalently)} \qquad \check{\theta}(t) g_t = 0. \tag{1.8.20}$$

The condition is obviously sufficient. To show that it is also necessary, let $g_t \in [\mathscr{D}_+' \cap (\mathscr{E}_0)']$. Then, $g_t = 0$ for $t < 0$, so that we can also choose $f(t) = 0$ for $t < 0$ in (1.8.18). Since $f(t) \in C^0$, this implies $f(0) = 0$, so that (1.8.19) becomes [note that (1.8.17) is valid for $f(t)$]

$$\theta(t) g_t = (\theta f)' = f'(t) = g_t;$$

i.e.,

$$\check{\theta}(t) g_t = [1 - \theta(t)]\, g_t = 0.$$

### *(d) Fourier Transform in $\mathscr{D}_L'^{(1)}$ and $(\mathscr{E}_0)'$*

We now show that the spaces $\mathscr{D}_L'^{(1)}$ and $(\mathscr{E}_0)'$ are carried into each other by the Fourier transformation.

THEOREM 1.8.4. *If $G_\omega \in \mathscr{D}_L'^{(1)}$ and $G_\omega = \mathscr{F} g_t$, then $g_t \in (\mathscr{E}_0)'$.*

In fact, by (1.8.11) and (1.8.7), there exist functions $F_p(\omega) \in L$ such that

$$(1+\omega^2)^{-1/2} G_\omega = \sum_{p=0}^{n} D^p F_p(\omega)$$

or, what amounts to the same,

$$(1+i\omega)^{-1}G_\omega = \sum_{p=0}^{n} D^p H_p(\omega), \qquad H_p \in L.$$

Let $h_p(t) = \mathscr{F}^{-1}H_p(\omega)$. Then, since $H_p$ is absolutely integrable, $h_p(t) \in C^0$ (it is even uniformly continuous and bounded [9, p. 11]). According to (A10.8) and (A10.9), it then follows from the above relation that

$$g_t = \mathscr{F}^{-1}G_\omega = \left(1 - \frac{d}{dt}\right)\sum_{p=0}^{n}(it)^p h_p(t). \tag{1.8.21}$$

Since the sum on the right-hand side is a continuous function, $g_t \in (\mathscr{E}_0)'$.

THEOREM 1.8.5. *If* $G_\omega = \mathscr{F}g_t \in \mathscr{D}_L'^{(1)}$, *we have*

$$\mathscr{F}^{-1}(G_\omega * P(1/\omega)) = -i\pi\varepsilon(t)g_t, \tag{1.8.22}$$

$$\mathscr{F}^{-1}(G_\omega * \delta_\omega{}^+) = \theta(t)g_t, \tag{1.8.23}$$

where $\delta^+$ is defined in (A11.5).

According to (A10.19), (A11.8), and (A11.9), it suffices to establish that the convolution theorem may be applied under the above assumptions. Since it may certainly be applied to $G_\omega * \delta_\omega$, because $\delta_\omega \in \mathscr{E}'$ (cf. Section A10), it suffices to consider (1.8.22). We shall only show that both sides are well defined.

According to (1.8.15), $G_\omega * P(1/\omega) \in \mathscr{S}'$, so that the left-hand side exists. The right-hand side is also well defined, because $\varepsilon(t) = \theta(t) - \check{\theta}(t)$, and $\theta(t)g_t$ exists by Theorem 1.8.4 and (1.8.19). The proof that the convolution theorem is valid may be found in Schwartz's paper [20].

In particular, since $P(1/\omega) \in \mathscr{D}_L'^{(1)}$, it follows from (1.8.22) and (A11.8) that

$$\mathscr{F}^{-1}(P(1/\omega) * P(1/\omega)) = -\pi/2,$$

so that, by (A10.16),

$$P(1/\omega) * P(1/\omega) = -\pi^2\delta_\omega. \tag{1.8.24}$$

This result is related with the Poincaré–Bertrand transformation formula [21, p. 56].

### *(e) The Fundamental Theorem*

The fundamental theorem on dispersion relations, in the no-subtraction case, is an immediate consequence of Theorems 1.8.3–1.8.5:

THEOREM 1.8.6. *Let* $G_\omega = \mathscr{F} g_t \in \mathscr{D}_L'^{(1)}$. *Then* $g_t \in \mathscr{D}_+'$ *if and only if*

$$G_\omega = G_\omega * \delta_\omega^{\ +} \qquad \text{or (equivalently)} \qquad G_\omega * \delta_\omega^{\ -} = 0. \qquad (1.8.25)$$

It suffices to take the Fourier transform of (1.8.20) and to employ (1.8.23).

Taking into account (A11.5) and (A8.5), we can rewrite (1.8.25) as

$$G_\omega = (1/2)\, G_\omega - (1/2i\pi)\, G_\omega * P(1/\omega),$$

or

$$G_\omega = -(1/\pi i)\, G_\omega * P(1/\omega), \qquad (1.8.26)$$

which is the basic dispersion relation without subtractions [cf. (1.8.6)].

Taking the real and imaginary parts of both sides of (1.8.26), we find

$$\operatorname{Re} G_\omega = -(1/\pi)\operatorname{Im} G_\omega * P(1/\omega), \qquad (1.8.27)$$

$$\operatorname{Im} G_\omega = (1/\pi)\operatorname{Re} G_\omega * P(1/\omega). \qquad (1.8.28)$$

These are the Plemelj formulas for distributions.

As in the case of (1.6.10) and (1.6.11), each of these formulas implies the other one. To show this, it suffices to take the convolution product of both sides with $P(1/\omega)$ and to apply (1.8.24) (the product is associative in this case).

We have therefore proved the analogs of parts (iii) and (iv) of Titchmarsh's Theorem 1.6.1 for distributions (without employing analytic continuation!). Parts (i) and (ii) are contained in Theorem 1.8.2, as we have already seen. We see also that the dispersion relations are simply the Fourier transform of (1.8.20), so that, apart from the distribution-theoretic considerations, the present method of proving them is quite straightforward. Since the dispersion relations make sense only for distributions that are convolutionable with $P(1/\omega)$, we also have the most general conditions under which the result applies.

According to Example i following (1.8.11), Titchmarsh's theorem for square integrable functions is a particular case of the above results. The gain in generality that has now been achieved can be appreciated by considering Examples ii–v, to all of which Theorem 1.8.6 can be applied.

*Example.* Since $\delta_\omega \in \mathscr{E}'$ and $P(1/\omega) \in \mathscr{D}_L'^{(1)}$, we see that $G_\omega = \delta_\omega^{\ +} \in \mathscr{D}_L'^{(1)}$ and $g_t = \mathscr{F}^{-1}\delta_\omega^{\ +} = \theta(t)/2\pi \in \mathscr{D}_+'$ (cf. the example following Theorem 1.8.2). Thus, according to Theorem 1.8.6,

$$\delta_\omega^{\ +} * \delta_\omega^{\ -} = 0, \qquad \text{or} \qquad \delta_\omega^{\ +} = -(1/\pi i)\,\delta_\omega^{\ +} * P(1/\omega). \qquad (1.8.29)$$

The imaginary part of (1.8.29) is a trivial consequence from (A8.5). The real part coincides with (1.8.24). Thus $\delta_\omega$ and $-(1/\pi)\, P(1/\omega)$ form a Hilbert transform pair.

### (f) Subtractions

Let us now extend (1.8.11) by considering the space $\mathscr{D}_L'^{(n+1)}$ of all distributions $G_\omega$ such that

$$(1+\omega^2)^{-(n+1)/2} G_\omega \in \mathscr{D}_L'. \tag{1.8.30}$$

Since $\mathscr{D}_L'^{(n+1)} \subset \mathscr{S}'$, we can define the Fourier transform. Let

$$g_t = \mathscr{F}^{-1} G_\omega \in \mathscr{D}_+'. \tag{1.8.31}$$

What dispersion relations can then be derived for $G_\omega$?

In order to reduce the problem to that already discussed, let

$$g_t = (d^n/dt^n)\gamma_t \tag{1.8.32}$$

so that

$$G_\omega = (-i\omega)^n \Gamma_\omega, \tag{1.8.33}$$

where $\Gamma_\omega = \mathscr{F}\gamma_t$. Then, according to (1.8.30) and (1.8.33),

$$\Gamma_\omega \in \mathscr{D}_L'^{(1)}. \tag{1.8.34}$$

It can be shown that, with $g_t \in \mathscr{D}_+'$, the differential equation (1.8.32) always has one and only one solution $\gamma_t \in \mathscr{D}_+'$; i.e., the condition $\operatorname{supp}\gamma_t \in [0, \infty)$ uniquely determines $\gamma_t$:

$$\gamma_t = \mathscr{F}^{-1}\Gamma_\omega \in \mathscr{D}_+'. \tag{1.8.35}$$

According to (1.8.34) and (1.8.35), we can now apply to $\Gamma_\omega$ the fundamental theorem 1.8.6 and we find, according to (1.8.26),

$$\Gamma_\omega = -(1/\pi i)\Gamma_\omega * P(1/\omega). \tag{1.8.36}$$

In order to determine $\Gamma_\omega$, we have to solve (1.8.33). According to (A5.11), the general solution of

$$\omega^n T_\omega = 1 \tag{1.8.37}$$

is

$$T_\omega = Pf(1/\omega^n) + \sum_{k=0}^{n-1} c_k \delta^{(k)}. \tag{1.8.38}$$

More generally, it can be shown [22, p. 123] that the general solution of the "division problem" (1.8.33) is

$$\Gamma_\omega = [G_\omega/(-i\omega)^n] + \sum_{k=0}^{n-1} c_k \delta^{(k)}, \tag{1.8.39}$$

where the first term denotes an arbitrary particular solution [like $Pf(\omega^{-n})$ in (1.8.38)] and $c_k$ are arbitrary constants. Once we choose the particular solution, the condition $\gamma_t \in \mathscr{D}_+'$ fixes the value of these constants.

Substituting (1.8.39) in (1.8.36), we get

$$G_\omega = (-i\omega)^n(-1/\pi i)\left\{P(1/\omega) * [G_\omega/(-i\omega)^n] + \sum_{k=0}^{n-1} c_k P(1/\omega) * \delta^{(k)}\right\}.$$

According to (A8.7) and (A4.13),

$$\begin{aligned}\omega^n[P(1/\omega) * \delta^{(k)}] &= \omega^n D^k P(1/\omega)\\ &= (-1)^k k!\,\omega^{n-k-1} \qquad (0 \leqslant k \leqslant n-1),\end{aligned}$$

so that, finally,

$$G_\omega = -(\omega^n/\pi i)[P(1/\omega) * (G_\omega/\omega^n)] + \mathscr{P}_{n-1}(\omega), \tag{1.8.40}$$

where $\mathscr{P}_{n-1}(\omega)$ denotes a polynomial of degree $\leqslant n-1$ in $\omega$. The coefficients of this polynomial are the subtraction constants, and (1.8.40) is the general form of a *dispersion relation with n subtractions.*

In particular, if $G_\omega$ is an ordinary function $G(\omega)$, and if we choose $Pf(G(\omega)/\omega^n)$ as the particular solution in (1.8.39), we get, according to (A4.14),

$$\begin{aligned}&-\frac{\omega^n}{\pi i}\left[P\frac{1}{\omega} * \left(\frac{G_\omega}{\omega^n}\right)\right]\\ &= \frac{\omega^n}{\pi i}P\int_{-\infty}^{\infty}\frac{d\omega'}{\omega'^n(\omega'-\omega)}\left[G(\omega') - G(0) - \cdots - \frac{\omega'^{n-2}}{(n-2)!}G^{(n-2)}(0)\right],\end{aligned} \tag{1.8.41}$$

which is equivalent to the result obtained in (1.7.18).

## 1.9. The Kramers–Kronig Relation

Let us now return to the discussion of light propagation in a dielectric medium and let us find out the implications of the above results. We have seen in Section 1.4 that, as a consequence of causality applied to the dielectric polarization at one point of the medium, the complex dielectric susceptibility

$$\chi'(\omega) = [\varepsilon'(\omega)-1]/4\pi = [n'^2(\omega)-1]/4\pi$$

is holomorphic in $I_+$. Furthermore, by applying causality to the propagation of a signal through the medium, we concluded that $n'(\omega)-1$ is also holomorphic in $I_+$. In order to derive dispersion relations for these quantities, however, we need additional information about their behavior on the real axis, in particular as $|\omega| \to \infty$.

At finite frequencies, it may usually be assumed that $n'(\omega)$ is bounded. An exception occurs in the case of conducting media at zero frequency. In fact, as we have already mentioned in Section 1.4, the complex dielectric constant contains in this case a term $4\pi i\sigma/\omega$, where $\sigma$ is the conductivity, so that we have a singularity at $\omega = 0$. Let us exclude this possibility by assuming that the medium is an insulator.[10a] It suffices, then, to consider the behavior of $n'(\omega)$ as $\omega \to \infty$.

If $\omega$ is much larger than the frequencies associated with the binding energies of all the electrons, they must behave as if they were free, so that the equation of motion of an electron will be

$$m\ddot{\mathbf{x}} = e\mathbf{E} = e\mathbf{E}_\omega e^{-i\omega t},$$

where $\mathbf{E}$ is the incident field (the displacements are much smaller than the wavelength, so that the field may be regarded as uniform). Thus all electrons oscillate in phase with the field: $\mathbf{x} = \mathbf{x}_\omega e^{-i\omega t}$ and, substituting in the equation of motion, we find

$$\mathbf{x} = -(e/m\omega^2)\mathbf{E},$$

so that the polarization is

$$\mathbf{P} = Ne\mathbf{x} = -(Ne^2/m\omega^2)\mathbf{E}, \tag{1.9.1}$$

where $N$ is the total number of electrons per unit volume.

Comparing this result with (1.4.14), we conclude that

$$\chi'(\omega) = [n'^2(\omega)-1]/4\pi \approx -Ne^2/m\omega^2 \qquad (\omega \to \infty); \tag{1.9.2}$$

i.e., the refractive index must approach unity at high frequencies, in such a way that

$$n'^2(\omega) - 1 \in L^2.$$

On the other hand, according to Section 1.4, the inverse Fourier transform of $\chi'(\omega)$ vanishes for $t < 0$, so that $n'^2(\omega)-1$ satisfies condition (i) of Titchmarsh's Theorem 1.6.1. We conclude that

$$n'^2(\omega) - 1 = \frac{1}{\pi i} \int_{-\infty}^{\infty} \frac{[n'^2(\omega')-1]}{\omega'-\omega}\, d\omega' \tag{1.9.3}$$

and, furthermore,

$$\int_{-\infty}^{\infty} |n'^2(u+iv) - 1|^2\, du < C \qquad (v \geqslant 0). \tag{1.9.4}$$

[10a] For dispersion relations in more general media, see [15, 22a, 22b, 22c].

It follows from (1.9.4) that $n'(u+iv)+1$, which is regular in $I_+$, must approach 2 as $u \to \pm\infty$, $v \geqslant 0$, so that (1.9.4) implies

$$\int_{-\infty}^{\infty} |n'(u+iv) - 1|^2 \, du < B \qquad (v \geqslant 0). \tag{1.9.5}$$

Thus $n'(\omega)-1$ satisfies condition (ii) of Titchmarsh's theorem and it is therefore also a causal transform[11]:

$$n'(\omega) - 1 = \frac{1}{\pi i} ⨍_{-\infty}^{\infty} \frac{[n'(\omega')-1]}{\omega'-\omega} \, d\omega'. \tag{1.9.6}$$

Taking the real part of both sides, we get, by (1.4.2),

$$n(\omega) - 1 = \frac{c}{2\pi} ⨍_{-\infty}^{\infty} \frac{\beta(\omega')}{\omega'(\omega'-\omega)} \, d\omega'.$$

Taking into account (1.4.9), this may be rewritten as

$$n(\omega) - 1 = \frac{c}{\pi} ⨍_{0}^{\infty} \frac{\beta(\omega')}{\omega'^2-\omega^2} \, d\omega'. \tag{1.9.7}$$

Thus, the real refractive index at any frequency is completely determined by the values taken by the extinction coefficient for all frequencies, as had been surmised in Section 1.5. In particular, if $\beta(\omega)$ is strongly peaked at $\omega = \omega_0$, as in the example discussed in Section 1.5, the above relation determines the corresponding value of $n(\omega)$ in such a way that no violation of causality can arise.

Historically, (1.9.7) was the first-known dispersion relation. It is known as the *Kramers–Kronig relation*, after Kronig [23] and Kramers [24], who were the first to derive it. It must be satisfied by any causal model of a dispersive medium.

The classical theory of dispersion, due to Lorentz, describes the medium in terms of electrons harmonically bound to atoms. The equation of motion for the displacement $\mathbf{x}_j$ of the $j$th electron under the influence of the incident field $\mathbf{E} = \mathbf{E}_\omega e^{-i\omega t}$ is taken to be[12]

$$m(\ddot{\mathbf{x}}_j + 2\gamma_j \dot{\mathbf{x}}_j + \omega_j^2 \mathbf{x}_j) = e\mathbf{E}_\omega e^{-i\omega t}, \tag{1.9.8}$$

---

[11] Note that the regularity of $n'(\omega)-1$ in $I_+$ and its square integrability on the real axis only would not suffice to conclude that it is a causal transform, as we have seen at the end of Section 1.6. On the other hand, the inverse Fourier transform of $n'(\omega)-1$ does not have a direct physical significance, like that of $n'^2(\omega)-1$, so that we could not verify condition (i) of Titchmarsh's theorem directly. This is the reason for the somewhat roundabout derivation, in which we had to employ (1.9.4) to show that $n'(\omega)-1$ verifies condition (ii).

[12] Actually, this equation of motion applies to a gas at low density. At higher densities, the field acting on an electron differs from the incident field. We shall consider only the low-density case.

where $2\gamma_j \dot{\mathbf{x}}_j$ ($\gamma_j > 0$) represents the damping due to collisions and radiation, and $\omega_j$ is the natural frequency.

The solution of (1.9.8) is

$$\mathbf{x}_j = \frac{e\mathbf{E}_\omega e^{-i\omega t}}{m(\omega_j^2 - 2i\gamma_j\omega - \omega^2)}. \tag{1.9.9}$$

If there are $N_j$ electrons of the $j$th type per unit volume, the polarization becomes

$$\begin{aligned}\mathbf{P} &= \sum_j N_j e\mathbf{x}_j = \sum_j \frac{N_j e^2 \mathbf{E}_\omega}{m(\omega_j^2 - 2i\gamma_j\omega - \omega^2)} e^{-i\omega t} \\ &= \mathbf{P}_\omega e^{-i\omega t} = \frac{n'^2 - 1}{4\pi} \mathbf{E}_\omega e^{-i\omega t}.\end{aligned}$$

Since $n'^2 - 1 \approx 2(n' - 1)$ ($n'$ is very close to 1 for a gas at low density), we get

$$n'(\omega) - 1 = 2\pi \sum_j \frac{N_j e^2}{m(\omega_j^2 - 2i\gamma_j\omega - \omega^2)}, \tag{1.9.10}$$

which is the Lorentz dispersion formula.[13] For large $\omega$, it approaches (1.9.2).

The dispersion formula for the susceptibility in the Lorentz theory is necessarily causal, because it was derived from a causal model. In fact, (1.9.8) is the equation of motion of a damped harmonic oscillator, already discussed in Section 1.2. Just like (1.2.8), the right-hand side of (1.9.10) is holomorphic in $I_+$ (its poles are located in $I_-$). Its real and imaginary parts satisfy the Kramers–Kronig relation, as can also be directly verified [cf. (1.6.17)].

The detailed propagation of a wave train in a dispersive medium for which the Lorentz dispersion formula is assumed to hold was investigated in 1914 by Sommerfeld and Brillouin.[14] The object of their work was to resolve an apparent contradiction with the theory of relativity: for a medium with anomalous dispersion, there can exist a region near the absorption line where the phase and the group velocity of light both become greater than $c$. They showed that, in spite of this, the front of a signal cannot propagate with velocity greater than $c$. In our present treatment, this is an immediate consequence of the analytic properties of the Lorentz dispersion formula.

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[13] A *dispersion formula*, such as (1.9.10), should not be confused with a *dispersion relation*, such as (1.9.7). The dispersion formula gives the explicit behavior of $n'$ as a function of $\omega$, whereas a dispersion relation is only a restriction, due to causality, on the analytic behavior of $n'(\omega)$. Any causal model must lead to such an $n'$, so that there is a broad class of dispersion formulas that satisfy the Kramers–Kronig dispersion relation.

[14] Cf. Brillouin [25].

If the only damping effect acting on the electron were that due to radiation, then, according to Lorentz, the term $2\gamma_j \dot{x}_j$ in (1.9.8) would have to be replaced by the radiation reaction $-\frac{2}{3}(r_0/c)\dddot{x}_j$, where $r_0 = e^2/mc^2 \approx 2.8 \times 10^{-13}$ cm is the classical electron radius. It is amusing to note that the corresponding "more accurate" dispersion formula would violate the causality condition.

In fact, the above substitution would replace the denominators in (1.9.10) by $\omega_j{}^2 - i\tau_r\omega^3 - \omega^2$, where $\tau_r = \frac{2}{3}r_0/c$. In addition to two poles in $I_-$, as in (1.9.10), this would yield a third pole located in $I_+$, at $\omega \approx i/\tau_r$ (note that $\tau_r^{-1} \gg \omega_j$).

It can actually be verified[15] that, if the damping term in (1.9.8) is replaced by the Lorentz radiation reaction, the corresponding equation of motion for an electron submitted to a sudden force predicts that the electron starts to accelerate appreciably within a time interval of the order of $r_0/c \sim 10^{-23}$ sec *before* the force is applied. This preacceleration violates causality, although the violation is confined to very small time intervals. It can be argued that such difficulties arise from an unjustified application of the Lorentz radiation reaction up to very high frequencies ($\omega \sim c/r_0$).

## 1.10. The Optical Theorem

The complex refractive index has appeared in the above discussion in two quite different ways. It has been linked through Maxwell's relation to the local polarization of the medium [cf. (1.4.13) and (1.4.14)], and it has also been linked [cf. (1.4.1) and (1.4.5)] to the phase velocity and the extinction coefficient of a wave traveling through the medium. We shall now consider the relation between these two definitions of $n'(\omega)$, by discussing the mechanism whereby the change in phase velocity and the extinction of the incident wave arise, in terms of the microscopic processes associated with the polarization of the medium.

When the incident wave falls upon the medium, it gives rise to oscillating dipole moments in each atom (or molecule). These oscillating dipoles reemit radiation in all directions, corresponding to a scattering of the incident wave. The transmitted wave results from the interference between the incident wave and all scattered waves. Thus, there must exist a relation between the complex refractive index and the elementary scattering process by a single atom of the medium.

[15] Cf. Rohrlich [26], especially Chapter 6 and the references quoted there.

To illustrate the nature of this relation, we shall again consider only the simple situation in which the medium is a very thin layer of not very dense material (e.g., a gas). The refractive index is then very close to unity, which allows us to make several simplifications. We consider a plane wave perpendicularly incident upon the medium. We want to evaluate the transmitted wave at a point $P$ on the other side.

The coordinate system is shown in Fig. 1.7. The $z$-axis is taken through $P$ in the direction of propagation of the incident wave, and the origin is taken at the entrance face of the layer, which coincides with the $(x, y)$-plane. The incident wave is assumed to be linearly polarized, with the electric field in the $x$ direction, so that

$$\mathbf{E}_i(z, t) = E_0 e^{i(kz-\omega t)}\hat{\mathbf{x}}, \tag{1.10.1}$$

where $\hat{\mathbf{x}}$ denotes the unit vector along the $x$ direction.

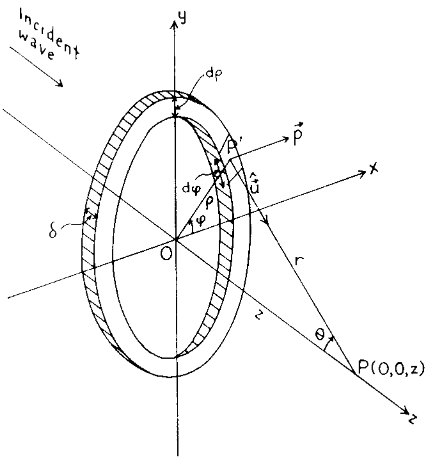


*FIG. 1.7.* Coordinate system for evaluating the contribution from a volume element $\rho\, d\rho\, d\varphi \cdot \delta$ to the scattered field at the point $P$.

The electric dipole moment induced in an atom by the incident field is given by

$$\mathbf{p}(t) = \alpha\mathbf{E}_i(0, t) = \alpha E_0 e^{-i\omega t}\hat{\mathbf{x}}, \tag{1.10.2}$$

where $\alpha$ is the polarizability of the atom. Let $OP = z$ be the distance from $P$ to the layer. We assume that this distance is much larger than the wavelength, so that

$$kz \gg 1. \tag{1.10.3}$$

The scattered electric field $\mathbf{e}_s$ at $P$ produced by an oscillating dipole at a point $P'$ of the medium (Fig. 1.7) is given by the well-known expression for dipole radiation[16]

$$\begin{aligned}\mathbf{e}_s &= (1/c^2 r)\hat{\mathbf{u}} \times [\hat{\mathbf{u}} \times \ddot{\mathbf{p}}(t-r/c)] \\ &= -k^2 \alpha E_0 (e^{i(kr-\omega t)}/r)\hat{\mathbf{u}} \times (\hat{\mathbf{u}} \times \hat{\mathbf{x}}),\end{aligned} \tag{1.10.4}$$

where $r = P'P$ and $\hat{\mathbf{u}}$ is the unit vector along the direction $P'P$.

Let $(\rho, \varphi)$ be the polar coordinates of $P'$ in the $(x, y)$-plane, and $\theta$ the angle between $P'P$ and $0z$. Then,

$$\begin{aligned}\hat{\mathbf{u}} \times (\hat{\mathbf{u}} \times \hat{\mathbf{x}}) &= \left(\frac{\rho^2}{r^2}\cos^2\varphi - 1\right)\hat{\mathbf{x}} \\ &\quad + \frac{\rho^2}{r^2}\sin\varphi\cos\varphi\,\hat{\mathbf{y}} - \frac{\rho}{r}\cos\varphi\cos\theta\,\hat{\mathbf{z}}.\end{aligned} \tag{1.10.5}$$

The contribution to the scattered field at $P$ from a volume element $\rho\, d\rho\, d\varphi \cdot \delta$ around $P'$ is $\mathbf{dE}_s = N\mathbf{e}_s \rho\, d\rho\, d\varphi \cdot \delta$, where $N$ is the number of dipoles per unit volume and $\delta$ is the thickness of the layer. If we neglect multiple scattering, which is allowed because of the small thickness and low density, the total scattered field at $P$ is obtained by integrating $\mathbf{dE}_s$ over the whole layer. The contribution from the last two terms of (1.10.5) vanishes after integration over $\varphi$, so that we get

$$\mathbf{E}_s = 2\pi N \delta k^2 \alpha E_0 e^{-i\omega t} \int_0^\infty [1 - (\rho^2/2r^2)](e^{ikr}/r)\rho\, d\rho\, \hat{\mathbf{x}}.$$

Since $\rho^2 = r^2 - z^2$, we find, by partial integration,

$$\begin{aligned}\int_0^\infty [1-(\rho^2/2r^2)](e^{ikr}/r)\rho\, d\rho &= \frac{1}{2}\int_z^\infty e^{ikr}\, dr + (z^2/2)\int_z^\infty (e^{ikr}/r^2)\, dr \\ &= (e^{ikr}/2ik)\Big|_z^\infty + (z^2/2ik)(e^{ikr}/r^2)\Big|_z^\infty \\ &\quad + (z^2/ik)\int_z^\infty (e^{ikr}/r^3)\, dr \\ &= -(e^{ikz}/2ik) - (e^{ikz}/2ik)[1 + O(1/kz)],\end{aligned} \tag{1.10.6}$$

where we have taken

$$\lim_{r\to\infty}(e^{ikr}) = 0.$$

---

[16] See, for example, Landau and Lifshitz [3, p. 191].

This is justified by the fact that the above limit corresponds to the radiation from infinitely distant dipoles. Such radiation must traverse an infinite thickness of the medium, so that it is completely extinguished, no matter how small the extinction coefficient may be: multiple scattering cannot be neglected for this contribution!

The final result for $kz \gg 1$ is therefore

$$\mathbf{E}_s \approx 2\pi i N\delta k\alpha E_0 e^{i(kz-\omega t)}\hat{\mathbf{x}};$$

i.e., by (1.10.1),

$$\mathbf{E}_s \approx 2\pi i N\delta k\alpha \mathbf{E}_i. \tag{1.10.7}$$

The total electric field at $P$ is

$$\mathbf{E} = \mathbf{E}_i + \mathbf{E}_s \approx (1+2\pi ik\delta N\alpha)\mathbf{E}_i. \tag{1.10.8}$$

On the other hand, according to (1.4.5), we have

$$\mathbf{E} = e^{i(n'-1)k\delta}\mathbf{E}_i \approx [1+i(n'-1)k\delta]\mathbf{E}_i \tag{1.10.9}$$

for small enough $\delta$. Comparing (1.10.8) with (1.10.9), we find

$$n'-1 = 2\pi N\alpha. \tag{1.10.10}$$

The only relevant contributions to the integral in (1.10.6) arose from the lower limit $r = z$, which corresponds to forward scattering, as we see from Fig. 1.7. Thus (1.10.10) must be related to the forward scattering amplitude. Indeed, according to (1.10.4), the field scattered by a dipole in the forward direction $\hat{\mathbf{u}} = \hat{\mathbf{z}}$ is

$$\mathbf{e}_s = k^2\alpha(e^{i(kr-\omega t)}/r)E_0\hat{\mathbf{x}} = f(k,0)(e^{ikr}/r)\mathbf{E}_i(0,t), \tag{1.10.11}$$

where $f(k,0)$ is the forward scattering amplitude.[17] It follows that

$$f(k,0) = k^2\alpha, \tag{1.10.12}$$

so that (1.10.10) becomes

$$n'-1 = (2\pi/k^2)Nf(k,0). \tag{1.10.13}$$

This is the relation we were looking for, between the complex refractive index and the forward scattering amplitude by each individual atom of the medium.

[17] Actually, the scattering amplitude for the electromagnetic field is not a scalar, so that this definition can be applied only in the forward direction; cf., e.g., Van de Hulst [27, Chapters 4, 5].

Taking real and imaginary parts of (1.10.13), we get, with the help of (1.4.2),

$$n - 1 = (2\pi/k^2) N \operatorname{Re} f(k,0), \tag{1.10.14}$$

$$\beta = (4\pi/k) N \operatorname{Im} f(k,0). \tag{1.10.15}$$

Let $\sigma_t$ be the total cross section for extinction (i.e., scattering and absorption) of the incident wave by an atom. The intensity of the wave within the medium then decays according to the law $I(z) = I(0)\exp(-N\sigma_t z)$, so that, by the definition of the extinction coefficient (Section 1.4), we get

$$\beta = N\sigma_t. \tag{1.10.16}$$

Comparing this with (1.10.15), we find

$$\sigma_t = (4\pi/k) \operatorname{Im} f(k,0). \tag{1.10.17}$$

This relation between the total cross section and the imaginary part of the forward scattering amplitude is known as the *optical theorem*. While (1.10.14) is only an approximate relation, the optical theorem is an exact result, valid not only in the present case but also under much more general conditions.[18] It is simply an expression of the conservation of energy (or of probability, in the case of quantum-mechanical scattering).

Substituting (1.10.14) and (1.10.15) in the Kramers–Kronig relation (1.9.7), we find

$$\operatorname{Re} f(k,0) = (2k^2/\pi) \,⨍_0^\infty [\operatorname{Im} f(k',0)]/k'(k'^2 - k^2)\, dk', \tag{1.10.18}$$

or, with the help of (1.10.17),

$$\operatorname{Re} f(k,0) = (k^2/2\pi^2) \,⨍_0^\infty \sigma_t(k')/(k'^2 - k^2)\, dk', \tag{1.10.19}$$

which expresses $\operatorname{Re} f(k,0)$ in terms of a directly observable quantity, the total cross section.

The result (1.10.18) no longer contains any reference to the medium, but only to the microscopic scattering process by each individual atom: it is a dispersion relation for the total scattering amplitude in the forward direction. However, the above derivation of this relation is far from satisfactory, since it relies on the validity of (1.10.14) and (1.10.15) for all values of $k$, whereas these results were derived under the assumption (1.10.3), which is certainly violated as $k \to 0$. We should therefore expect to find violations of (1.10.19) as $k \to 0$.

[18] See, for example, Messiah [28, p. 866]. For a derivation and a more precise formulation of the optical theorem in the electromagnetic case, see Born and Wolf [14, p. 665].

In fact, according to (1.10.19), we ought to have

$$\lim_{k \to 0} f(k, 0) = 0.$$

While this would be true for the scattering by electrons bound to an atom [cf. (1.10.13) and (1.9.10)], it would not be true for free electrons, for which the amplitude must approach the classical Thomson value in the low-frequency limit[19]:

$$\lim_{k \to 0} f(k, 0) = -r_0 = -e^2/mc^2. \tag{1.10.20}$$

This suggests that (1.10.18) should be modified by subtracting $f(0,0)$ from the left-hand side, which would make it compatible with (1.10.20). It will be seen later (cf. Section 3.2) that this is indeed what happens.

Having thus established, heuristically, the existence of dispersion relations for the scattering amplitude in the microscopic scattering by a single scatterer, which provide the underlying basis for the macroscopic dispersion relations, we should clearly consider next the implications of causality at this more fundamental level, namely, for individual scattering processes. We now turn our attention to this problem.

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[19] Cf. Bjorken and Drell [29, p. 357].

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CHAPTER

2

# PARTIAL-WAVE DISPERSION RELATIONS

*Probably most physicists will agree that we should have an S-matrix, that an S-matrix is a good instrument by which we can describe the results of experiments .... The S-matrix ... by its analytical properties ... must somehow express causality. This "somehow" of course involves already a lot of difficulties ....*

W. HEISENBERG[1]

## 2.1. Introduction

In Chapter 1 we discussed the connection between causality and dispersion relations for macroscopic quantities, such as the refractive index of a medium. We also saw that such a quantity is related, at the microscopic level, to the scattering of the incident wave by each atom of the medium. This suggests considering directly what are the implications of causality in the individual scattering process; macroscopic properties would then follow from this more fundamental treatment.

Let us therefore consider such a process, in which the incident wave falls upon a single scatterer. In a stationary situation, the process can be described in terms of the amplitude of the scattered wave at large distances. In the problems to be considered, the scattering amplitude is a function of two

[1] Proc. Seminar Unified Theories Elementary Particles, p. 8. Max-Planck Inst. Phys. Astrophys., Munich, 1965.

variables. One of them is usually chosen to be the magnitude of the momentum (or the energy). The other one can be taken as the scattering angle, but several other choices can be made and turn out to be convenient for different purposes: one can take, for instance, the magnitude of the momentum transfer or the angular momentum, as will be seen later.

We shall first be concerned with the analytic properties of the scattering amplitude as a function of the magnitude of the momentum (or the energy), when the second variable is held fixed. In the present chapter, we deal with fixed angular momentum; in the next one, with fixed scattering angle or fixed momentum transfer.

We shall not specify the nature of the scatterer beyond assuming certain general properties of the interaction, among which causality will play a dominant role. This approach has the advantage of enabling one to see quite clearly which physical assumptions about the interaction are relevant to the validity of specific analytic properties of the scattering amplitude. However, the generality of this approach is restricted: we shall only be able to apply it to scatterers of finite radius $a$, so that the interaction vanishes identically for $r > a$. This restriction will be necessary for the formulation of the causality condition. It will be lifted in Part II, but only by introducing a specific model for the interaction, namely, potential scattering.

We shall deal first with a classical field, and later with nonrelativistic quantum scattering. We restrict ourselves to *spherically symmetric* scatterers. In terms of the well-known partial-wave expansion, this allows us to fix the angular momentum and to deal separately with each partial wave.

## 2.2. Classical Field: *s*-Wave Scattering

The connection between causality and the analytic properties of partial-wave amplitudes for the scattering of a classical electromagnetic field was investigated by Van Kampen [1]. For simplicity, we shall consider, instead of the electromagnetic field, a real scalar field $\psi(\mathbf{r}, t)$, which satisfies the wave equation (with velocity of propagation $c$) outside of the scatterer:

$$\Delta\psi - (1/c^2)\,\partial^2\psi/\partial t^2 = 0 \qquad (r > a). \tag{2.2.1}$$

Actually, it turns out that the electromagnetic field can also be described in terms of two scalar functions of this type [2], $\psi^e(\mathbf{r}, t)$ and $\psi^m(\mathbf{r}, t)$, known as *Debye potentials*, where the superscripts $e$ and $m$ refer to electric and magnetic multipole fields, respectively.[2]

[2] For electric multipoles, we have $\mathbf{E}(\mathbf{r}, t) = \operatorname{curl}\operatorname{curl}(\mathbf{r}\psi^e)$, $\mathbf{H}(\mathbf{r}, t) = (\partial/\partial t)\operatorname{curl}(\mathbf{r}\psi^e)$, and the corresponding expression for magnetic multipoles is obtained by the substitutions $\psi^e \to \psi^m$, $\mathbf{E} \to \mathbf{H}$, $\mathbf{H} \to -\mathbf{E}$. Both $\psi^e$ and $\psi^m$ satisfy (2.2.1).

If we multiply (2.2.1) by $\partial\psi/\partial t$, the result can be written in the form of a continuity equation,

$$\partial w/\partial t + \operatorname{div} s = 0, \tag{2.2.2}$$

where

$$w = \tfrac{1}{2}[(1/c^2)(\partial\psi/\partial t)^2 + (\operatorname{grad}\psi)^2], \tag{2.2.3}$$

$$\mathbf{s} = -(\partial\psi/\partial t)\operatorname{grad}\psi. \tag{2.2.4}$$

The positive definite quantity $w$ may be interpreted (up to a factor having the appropriate dimensions) as the energy density outside of the scatterer, so that $\mathbf{s}$ represents the energy current density, and (2.2.2) expresses energy conservation in the outside region. We can also interpret $\psi$ as the velocity potential of sound waves, provided that $c$ is taken to be the velocity of sound.[3]

For $s$-waves, which are spherically symmetric, the stationary solutions of (2.2.1) are of the form[4]

$$\psi(k,r,t) = [A(k)(e^{-ikr}/r) + B(k)(e^{ikr}/r)]\,e^{-ikct} \qquad (r > a), \tag{2.2.5}$$

where the first term represents an incoming wave and the second one an outgoing wave.

According to the general definition of the $S$-matrix as the operator that transforms the initial stage into the final stage (cf. Section 1.1), we can define the $S$-function, in the present case, by

$$S(k) = -B(k)/A(k), \tag{2.2.6}$$

where the minus sign is introduced in order to have $S(k) = 1$ in the absence of a scatterer [in fact, in this case Eq. (2.2.5) holds down to $r = 0$ and $\psi$ must be regular there].

We assume that *the interaction is linear*, so that the general solution is a superposition of stationary solutions,

$$\begin{aligned}\psi(r,t) &= \int_{-\infty}^{\infty} \psi(k,r,t)\,dk \\ &= \psi_{\text{in}}(r,t) + \psi_{\text{out}}(r,t) \qquad (r > a),\end{aligned} \tag{2.2.7}$$

where, by (2.2.5) and (2.2.6),

$$\psi_{\text{in}}(r,t) = \frac{1}{r}\int_{-\infty}^{\infty} A(k)\,e^{-ik(r+ct)}\,dk, \tag{2.2.8}$$

$$\psi_{\text{out}}(r,t) = -\frac{1}{r}\int_{-\infty}^{\infty} S(k)\,A(k)\,e^{ik(r-ct)}\,dk, \tag{2.2.9}$$

[3] See, for example, Landau and Lifshitz [3, Secs. 63, 64].

[4] As usual, when we employ complex notation in connection with a real wave function, it is implicitly assumed that the real part must be taken at the end.

which represent incoming and outgoing wave packets, respectively. Apart from the amplitude decrease due to the factor $1/r$, both of them propagate with velocity $c$, without changing their shape.

The fact that we are dealing with a real field implies that $\psi_{\text{in}}$ and $\psi_{\text{out}}$ must be real, so that, in (2.2.8), $A(k)$ must satisfy

$$A(-k) = A^*(k), \tag{2.2.10}$$

and (2.2.9) gives

$$S(-k) = S^*(k). \tag{2.2.11}$$

This relation is called the *symmetry relation* for the $S$-function.

The next assumption we make about the interaction is that it is *energy conserving*: there is no absorption or emission of energy by the scatterer. Thus, the total energy flux through a sphere of large radius $r$, integrated over all time, must vanish: all the energy coming in must eventually go out again. According to (2.2.4), this implies

$$\int_{-\infty}^{\infty} dt \oint s_r r^2\, d\Omega = -4\pi r^2 \int_{-\infty}^{\infty} \frac{\partial\psi}{\partial t}\frac{\partial\psi}{\partial r}\, dt = 0. \tag{2.2.12}$$

It follows from (2.2.7) to (2.2.9) that, for sufficiently large $r$,

$$r\frac{\partial\psi}{\partial t} \approx -ic\int_{-\infty}^{\infty} kA(k)[e^{-ikr} - S(k)e^{ikr}]\, e^{-ikct}\, dk,$$

$$r\frac{\partial\psi}{\partial r} \approx -i\int_{-\infty}^{\infty} kA(k)[e^{-ikr} + S(k)e^{ikr}]\, e^{-ikct}\, dk.$$

Applying Parseval's theorem (1.6.2), we then find that (2.2.12) becomes

$$-8\pi^2\int_{-\infty}^{\infty} k^2\,|A(k)|^2\, dk + 8\pi^2\int_{-\infty}^{\infty} k^2\,|S(k)A(k)|^2\, dk = 0, \tag{2.2.13}$$

where the first and second terms represent the total incoming energy and the total outgoing energy, respectively. If we restrict ourselves to wave packets with finite total energy, we have to assume that $kA(k)$ is square integrable.

In order that (2.2.13) may be valid for every admissible choice of $A(k)$, we must have

$$|S(k)|^2 = S(k)S^*(k) = 1. \tag{2.2.14}$$

This is the *unitarity condition*, a very important property of the $S$-function, which here expresses the energy-conserving nature of the scatterer. It follows that $S(k)$ is just a phase factor, so that the only effect of the scattering is to change the phase of the outgoing waves:

$$S(k) = \mathrm{e}^{2i\eta(k)}. \tag{2.2.15}$$

The reason why a factor 2 was introduced in the exponent is that (2.2.5) becomes

$$\psi(k,r,t) = -(2i/r)A(k)e^{i\eta(k)}\sin[kr+\eta(k)]\,e^{-ikct}, \tag{2.2.16}$$

so that $\eta(k)$ coincides with the usual definition of the *phase shift*.[5]

Comparing (2.2.14) with (2.2.11), we find

$$S(k)S(-k) = 1, \tag{2.2.17}$$

and from (2.2.15) it follows that the phase shift is an odd function of $k$:

$$\eta(-k) = -\eta(k). \tag{2.2.18}$$

To relate the phase shift with the scattering cross section, we consider an incident plane wave of unit amplitude,

$$\psi_0 = e^{ik(z-ct)}, \tag{2.2.19}$$

and we apply the well-known partial-wave expansion

$$e^{ikz} = e^{ikr\cos\theta} = \sum_{l=0}^{\infty}(2l+1)\,i^l j_l(kr)P_l(\cos\theta), \tag{2.2.20}$$

where $j_l$ is the spherical Bessel function of order $l$ and $P_l$ is the $l$th Legendre polynomial. The $s$-wave term in this expansion is

$$j_0(kr) = \frac{\sin(kr)}{kr} = \frac{e^{ikr}-e^{-ikr}}{2ikr},$$

corresponding to the normalization

$$A(k) = -B(k) = -1/2ik \tag{2.2.21}$$

in (2.2.5). The contribution to the scattered wave from (2.2.5) is

$$\psi_{sc}(k,r,t) = A(k)[1-S(k)]\,e^{ik(r-ct)}/r. \tag{2.2.22}$$

The $s$-wave scattering cross section is given, as usual, by the ratio of the corresponding scattered energy per unit time to the incident energy current density associated with (2.2.19). With the help of (2.2.4), (2.2.21), and (2.2.22), the $s$-wave cross section is readily found to be given by the well-known expression

$$\sigma_0(k) = (4\pi/k^2)\sin^2\eta = (\pi/k^2)\,|1 - S(k)|^2. \tag{2.2.23}$$

[5] See, for example, Schiff [4, Sec. 19].

## 2.3. The Causality Condition

So far, we have assumed that the interaction is spherically symmetric, linear, vanishing for $r > a$, and energy conserving. Now we come to the crucial assumption that will be responsible for the analytic properties of the $S$-function: the causality condition. In the present case, it can be formulated as follows: *the outgoing wave cannot appear before the incoming wave has reached the scatterer.* In fact, it is sufficient to apply this condition at the surface of the scatterer ($r = a$) in order to guarantee causal propagation in the whole outside region, because we already know that the incoming and outgoing wave packets travel with velocity $c$ in this region [cf. (2.2.8) and (2.2.9)].

To find out the implications of the causality condition, let us build up an incoming wave packet having a sharp front, which reaches the surface at time $t = t_0$. Without loss of generality, we can take $t_0 = -a/c$. According to (2.2.8), we then have, for $r = a$,

$$\begin{aligned}\psi_{\text{in}}(a,t) &= (1/a)\int_{-\infty}^{\infty} A(k)\exp[-ikc(t-t_0)]\,dk \\ &= 0 \qquad (t-t_0<0).\end{aligned} \tag{2.3.1}$$

If $A(k)$ is taken to be square integrable, it follows from Titchmarsh's theorem that (2.3.1) will be satisfied if and only if $A(k)$ is a causal transform (cf. Section 1.6).

According to the causality condition, the outgoing wave at the surface must also vanish for $t < t_0$; i.e., (2.3.1) implies

$$\begin{aligned}\psi_{\text{out}}(a,t) &= -\frac{1}{a}\int_{-\infty}^{\infty} S_a(k)A(k)\exp[-ikc(t-t_0)]\,dk \\ &= 0 \qquad (t-t_0<0),\end{aligned} \tag{2.3.2}$$

where

$$S_a(k) = e^{2ika}S(k). \tag{2.3.3}$$

Thus, whenever $A(k)$ is a causal transform, $C(k) = S_a(k)A(k)$ must also be a causal transform. This is only possible if $S_a(k) = C(k)/A(k)$ is regular in $I_+$, as we see, for instance, by choosing $A(k) = b/(k+iK)$ $(K > 0)$, where $b$ is a constant [cf. (1.6.28)]. Therefore, it follows from the causality condition that

$$S(k) \quad \text{is holomorphic in} \quad I_+. \tag{2.3.4}$$

In addition, $S_a(k)$ must be bounded in $I_+$, since $S_a(k)A(k)$ shares with $A(k)$ the property of square integrability over any line parallel to the real axis in $I_+$.

To determine the bound on $S_a(k)$ in $I_+$, let us note that, by Titchmarsh's theorem, $S_a(k)A(k)$ verifies a dispersion relation of the form (1.6.6) for any causal transform $A(k)$ and any $k \in I_+$:

$$S_a(k)A(k) = \frac{1}{2\pi i}\int_{-\infty}^{\infty} \frac{S_a(k')A(k')}{k'-k}\,dk' \qquad (\operatorname{Im} k > 0). \tag{2.3.5}$$

Since $|S_a(k')| = 1$ on the real axis, it follows that

$$|S_a(k)| \leqslant \frac{1}{2\pi\,|A(k)|}\int_{-\infty}^{\infty} \frac{|A(k')|}{|k'-k|}\,dk' \qquad (\operatorname{Im} k > 0).$$

Let $k = k_1 + iK$ $(K > 0)$, and let us choose $A(k')$ to be the particular causal transform

$$A(k') = 1/(k'-k^*) = 1/(k'-k_1+iK).$$

The above inequality then becomes

$$|S_a(k)| \leqslant \frac{K}{\pi}\int_{-\infty}^{\infty} \frac{dk'}{(k'-k_1)^2 + K^2} = 1 \qquad (K > 0),$$

so that we finally get the bound

$$|S_a(k)| \leqslant 1 \qquad (\operatorname{Im} k \geqslant 0). \tag{2.3.6}$$

Since $S(k) = e^{-2ika}S_a(k)$, this implies that $S(k)$ may blow up exponentially, like $e^{-2ika}$, as $|k| \to \infty$ in $I_+$. That this can actually happen, and that the bound (2.3.6) cannot be improved, is shown by the following example. Let the scatterer be an impenetrable sphere, defined by the boundary condition

$$\psi(r,t) = 0 \qquad (r = a). \tag{2.3.7}$$

Then, it follows from (2.2.5) and (2.2.6) that

$$S(k) = e^{-2ika}. \tag{2.3.8}$$

Physically, the exponential factor represents the phase advancement, corresponding to a path difference $2a$ (from the surface to the center and back), suffered by a wave reflected at the surface of the sphere, as compared with one going through the center. Correspondingly, an outgoing signal can appear at a time $2a/c$ earlier than would have been possible in the absence of a scatterer.

The presence of the exponential factor leads to an essential singularity for $S(k)$ as $|k| \to \infty$, so that one cannot derive dispersion relations for $S(k)$ itself, but only for $S_a(k)$. According to (2.3.6), we need one subtraction in the dispersion relation for $S_a(k)$. We shall make the subtraction at the origin. According to (1.7.8), the result is

$$S_a(k) = S(0) + \frac{k}{\pi i}\int_{-\infty}^{\infty} \frac{[S_a(k')-S_a(0)]}{k'(k'-k)}\,dk'. \tag{2.3.9}$$

It follows from (2.2.17) that

$$S(0) = \pm 1. \tag{2.3.10}$$

The possibility that $S(0) = -1$, which, according to (2.2.23), would imply an infinite cross section for $k \to 0$, cannot be ruled out on the basis of causality, but usually one will have $S(0) = 1$. It then follows from (2.3.9) that

$$\mathrm{Re}[e^{2ika} S(k)] = 1 + \frac{k}{\pi} \fint_{-\infty}^{\infty} \frac{\mathrm{Im}[e^{2ik'a} S(k')]}{k'(k'-k)}\, dk', \tag{2.3.11}$$

which is the partial-wave dispersion relation for $s$-waves.

This type of dispersion relation is less fundamental than (1.10.18), because of its explicit dependence on the radius of the scatterer. It is also less useful, because $\mathrm{Re}\, S(k)$ and $\mathrm{Im}\, S(k)$ become mixed through the presence of the exponential factor.

The analog of (1.6.6) in the present case is

$$S_a(k) = S(0) + \frac{k}{2\pi i} \int_{-\infty}^{\infty} \frac{[S_a(k') - S(0)]}{k'(k'-k)}\, dk' \qquad (\mathrm{Im}\, k > 0), \tag{2.3.12}$$

which expresses the analytic continuation of $S(k)$ in $I_+$ in terms of its values on the real axis. Taking the complex conjugate of both sides, and applying (2.2.11) within the integral, we find

$$S^*(k) = S(-k^*) \tag{2.3.13}$$

as the extension of the symmetry relation to $I_+$. Thus the $S$-function takes on complex conjugate values at points symmetrically placed with respect to the imaginary axis. In particular, $S(k)$ is real on the imaginary axis.

## 2.4. Analytic Continuation to $I_-$

According to (2.2.17), we have

$$S(k') = 1/S(-k') \tag{2.4.1}$$

at any point $k'$ on the real axis. We now *define* $S(k)$ at any point $k = k' + iK$ in $I_-$ by extending this relation to complex values of $k$:

$$S(k) = S(k' + iK) = 1/S(-k' - iK) = 1/S(-k) \qquad (K < 0). \tag{2.4.2}$$

According to Section 2.3, since $-k' - iK \in I_+$, the relation (2.4.2) defines an analytic function in $I_-$, regular except for poles corresponding to the points where $-k'-iK$ coincides with a zero of $S(k)$ in $I_+$. According to (2.4.1), this function tends to $S(k')$ when $K \to 0$. It then follows from the theory of analytic continuation [5, p. 157] that (2.4.2) defines the analytic continuation of $S(k)$ in $I_-$. In this way, $S(k)$ *is extended as an analytic function in the whole plane, regular except for possible poles in* $I_-$, *i.e., as a meromorphic function.*

It also follows from (2.4.2) that the relation (2.3.13) remains valid in $I_-$. Combining this relation with (2.4.2), we get

$$S(k)S^*(k^*) = 1, \tag{2.4.3}$$

which is the extension of the unitarity condition (2.2.14) to complex values of $k$. We can also conclude from (2.3.6) and (2.4.2) that

$$|S_a(k)| \geqslant 1 \qquad (\operatorname{Im} k \leqslant 0). \tag{2.4.4}$$

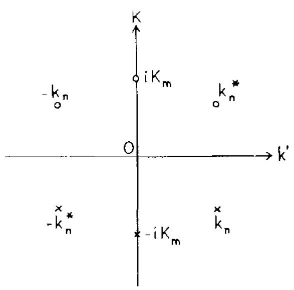


*FIG. 2.1.* Symmetry properties of the distribution of poles and zeros of $S(k)$ (×, poles; O, zeros).

The relations (2.3.13), (2.4.2), and (2.4.3) (only two of which are independent) imply several symmetry properties of the distribution of poles and zeros of $S(k)$ (cf. Fig. 2.1). Let $k_n$ be a pole of $S(k)$ in the fourth quadrant of the $k$-plane. Then, according to (2.3.13), $-k_n{}^*$ is also a pole, and, according to (2.4.2) and (2.4.3), $-k_n$ and $k_n{}^*$ are zeros of $S(k)$: the mirror image of a pole with respect to the imaginary axis is also a pole, whereas the mirror images with respect to the real axis or the origin are zeros. There may also exist unpaired poles $-iK_m$ $(K_m > 0)$ on the negative imaginary axis, with corresponding zeros $iK_m$ on the positive imaginary axis. The distributions of poles and of zeros are both symmetrical with respect to the imaginary axis.

The number of poles, which is the same as the number of zeros, may be either finite or infinite. However, if it is infinite, they cannot have an accumulation point at finite values of $k$, so that their absolute values must approach infinity. This is because $S(k)$ is holomorphic in $I_+$ (for all finite $k$), and the zeros of a holomorphic function are isolated points: they cannot have an accumulation point within the domain of holomorphy [5, p. 88].

## 2.5. Product Expansion

We now derive a representation of $S(k)$ in terms of its poles and zeros in which all the analytic properties derived above are explicitly exhibited. For this purpose, let us arrange all the poles $k_n$ of $S(k)$ in order of *increasing modulus*:

$$|k_1| \leqslant |k_2| \leqslant |k_3| \leqslant \cdots \leqslant |k_n| \leqslant \cdots, \tag{2.5.1}$$

so that $k_n$ and $-k_n{}^*$ are paired together. Then, as we have just seen, if the number of poles is not finite, we must have $|k_n| \to \infty$ as $n \to \infty$.

Let us consider the product

$$B(k) = \prod_n B_n(k) = \prod_n \frac{[1-(k/k_n{}^*)]}{[1-(k/k_n)]}, \tag{2.5.2}$$

where all the poles of $S(k)$ appear in the denominator and all the zeros in the numerator (multiple poles or zeros are repeated a number of times equal to their order). Explicitly, each complex pole pair $(k_n, -k_n{}^*)$ gives a contribution

$$\frac{[1-(k/k_n{}^*)][1+(k/k_n)]}{[1-(k/k_n)][1+(k/k_n{}^*)]} = \frac{(k_n{}^*-k)(k_n+k)}{(k_n-k)(k_n{}^*+k)}, \tag{2.5.3}$$

and each pole $-iK_m$ on the negative imaginary axis contributes

$$\frac{1-(k/iK_m)}{1+(k/iK_m)} = \frac{iK_m-k}{iK_m+k}. \tag{2.5.4}$$

A product of the form (2.5.2) is known as a *Blaschke product*, and each factor $B_n(k)$ is called a *Blaschke factor*.

If the number of poles is infinite, we must make sure that (2.5.2) converges. For this purpose, we use the following well-known [5, p. 15] convergence criterion:

THEOREM 2.5.1. *Necessary and sufficient condition for an infinite product*

$$\Pi = \prod_{n=1}^{\infty} (1+u_n) \tag{2.5.5}$$

*to be absolutely convergent is the convergence of the series*

$$\Sigma = \sum_{n=1}^{\infty} |u_n|. \tag{2.5.6}$$

In the case of (2.5.2), we have

$$\Sigma(k) = 2|k| \sum_n |(\operatorname{Im} k_n)/k_n{}^*(k_n-k)|. \tag{2.5.7}$$

Since $|k_n| \to \infty$, the convergence or divergence of this series (if we assume that $k$ does not coincide with any pole, so that each term is bounded) depends only on that of the series

$$\sigma = \sum_n (\operatorname{Im} k_n)/|k_n|^2 = -\sum_n \operatorname{Im}(1/k_n). \tag{2.5.8}$$

The convergence of (2.5.8) follows from the following theorem [5, p. 131]:

THEOREM 2.5.2. *Let $f(z)$ be an analytic function, regular and bounded in $I_+$, and let $z_1, z_2, \ldots, z_n, \ldots$ be its zeros in $I_+$, arranged in order of increasing modulus. Then the series*

$$\sigma = \sum_{n=1}^{\infty} (\operatorname{Im} z_n)/|z_n|^2$$

*is convergent.*

We can apply this theorem to the function $S_a(k)$, which is regular and bounded in $I_+$ [cf. (2.3.6)]. Since (2.5.8) is independent of $k$, we conclude that the Blaschke product (2.5.2) is absolutely and uniformly convergent in any finite region of the $k$-plane that does not contain a pole, and it therefore represents a regular analytic function $B(k)$ in such a region. It is readily seen that it also shares with $S(k)$ the generalized symmetry and unitarity properties (2.3.13) and (2.4.3).

Finally, by drawing the corresponding representative vectors in the complex plane, one can verify immediately that

$$|B_n(k)| = |(k_n{}^* - k)/(k_n - k)|$$

is $<1$ if $k \in I_+$ and it is $>1$ if $k \in I_-$, so that

$$|B(k)| \leqslant 1 \quad \text{in} \quad I_+, \qquad |B(k)| \geqslant 1 \quad \text{in} \quad I_-. \tag{2.5.9}$$

In conclusion, we have:

THEOREM 2.5.3. *The Blaschke product $B(k)$ is a meromorphic function of $k$, having the same poles and zeros as $S_a(k)$, and sharing with it the generalized symmetry and unitarity properties, in addition to having the same bounds as $S_a(k)$ in $I_+$ and $I_-$.*

Let us now consider the function

$$F^{(N)}(k) = S_a(k)/B^{(N)}(k), \tag{2.5.10}$$

where

$$B^{(N)}(k) = \prod_{n=1}^{N} B_n(k) \tag{2.5.11}$$

is the Blaschke product (2.5.2) restricted to its first $N$ factors. Clearly, since the poles and zeros contained in $B^{(N)}(k)$ are canceled out in the ratio (2.5.10),

$F^{(N)}(k)$ is regular in $I_+(k)$; furthermore, $|F^{(N)}(k)| = 1$ on the real axis. On the other hand, since

$$\lim_{|k|\to\infty} B^{(N)}(k) = 1, \tag{2.5.12}$$

it follows from (2.3.6) that $|F^{(N)}(k)|$ must become $\leqslant 1$ as $|k| \to \infty$. We might therefore try to apply the maximum-modulus theorem [5, p. 165] and conclude that

$$|F^{(N)}(k)| \leqslant 1 \quad \text{in} \quad I_+. \tag{2.5.13}$$

Actually, we have to be somewhat more careful, since there may be an essential singularity at $|k| \to \infty$, so that the usual form of the maximum-modulus theorem cannot be applied. However, we can employ a very powerful theorem due to Phragmén and Lindelöf, in a modified version due to Nevanlinna[6]:

THEOREM 2.5.4 (Phragmén–Lindelöf theorem). *Let $f(z)$ be an analytic function, regular in $I_+$ and such that*

$$|f(z) \leqslant 1 \quad \text{on the real axis,} \tag{2.5.14}$$

*and let*

$$M(R) = \max_{0 \leqslant \theta \leqslant \pi} |f(Re^{i\theta})| \tag{2.5.15}$$

*be the maximum modulus of $f(z)$ over the half-circle $|z| = R$ in $I_+$. Then there are only two possibilities: either (i) $|f(z)|$ increases so rapidly for $|z| \to \infty$ that*

$$\lim_{R\to\infty} \inf[(1/R)\ln M(R)] = \delta > 0, \tag{2.5.16}$$

*or else (ii) we have $\delta = 0$, and*

$$|f(z)| \leqslant 1 \qquad \text{for all} \quad z \in I_+. \tag{2.5.17}$$

Note that, if (2.5.16) holds, $M(R)$ can behave roughly like $\exp(\delta R)$ as $R \to \infty$. Thus the intuitive content of the theorem is the following: if a function is bounded on the real axis and regular in $I_+$, the only way in which it can become unbounded at infinity in $I_+$ is by containing some factor behaving like $e^{-i\delta z}$, i.e., like a phase factor along the real axis, but increasing exponentially in $I_+$. Such a possibility had already been mentioned in connection with the example (1.6.29).

For the function $F^{(N)}(k)$, according to the above discussion, $M(R)$ is bounded, so that we have possibility (ii), thus proving the validity of (2.5.13).

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[6] See Nevanlinna [6, p. 43]. Cf. also Hille [7, p. 412].

In the limit as $N \to \infty$, we get

$$\lim_{N\to\infty} F^{(N)}(k) = F(k) = S_a(k)/B(k), \tag{2.5.18}$$

and (2.5.13) implies

$$|F(k)| \leqslant 1 \qquad \text{in} \quad I_+. \tag{2.5.19}$$

Since all the zeros and poles are canceled out in (2.5.18), the function $F(k)$ is an entire function of $k$ without any zeros. Therefore,

$$f(k) = -i \ln F(k) \tag{2.5.20}$$

is also an entire function. This function is real on the real axis, where $|F(k)| = 1$. Furthermore, by (2.5.19),

$$\operatorname{Im} f(k) \geqslant 0 \qquad (\operatorname{Im} k > 0). \tag{2.5.21}$$

It will now be shown that the above properties of $f(k)$ imply

$$f(k) = 2\varepsilon k + \beta \qquad (\varepsilon \geqslant 0, \beta \, \text{real}). \tag{2.5.22}$$

This is a consequence of the following result[7]:

THEOREM 2.5.5 (Čebotarev's theorem). *If an entire function $f(z)$ is real for real $z$ and satisfies the condition*

$$\operatorname{Im} f(z) \geqslant 0 \qquad (\operatorname{Im} z > 0), \tag{2.5.23}$$

*then $f(z)$ is a linear function of the form*

$$f(z) = \alpha z + \beta \qquad (\alpha \geqslant 0, \beta \, \text{real}). \tag{2.5.24}$$

To prove this result, let us first apply Schwarz's reflection principle [5, p. 155], according to which an analytic function $f(z)$, regular in a region $D$ intersected by the real axis and real on the real axis, takes complex conjugate values for complex conjugate values of $z$:

$$f(z^*) = f^*(z). \tag{2.5.25}$$

Incidentally, for the case of (2.5.20), this property follows directly from (2.5.18) and Theorem 2.5.3, according to which $F(k)$ satisfies the generalized unitarity relation.

It follows from (2.5.23) and (2.5.25) that

$$\operatorname{Im} f(z) \leqslant 0 \qquad (\operatorname{Im} z < 0), \tag{2.5.26}$$

so that

$$\begin{aligned} 0 &\leqslant \arg f(z) \leqslant \pi \qquad (z \in I_+), \\ -\pi &\leqslant \arg f(z) \leqslant 0 \qquad (z \in I_-). \end{aligned} \tag{2.5.27}$$

[7] See Čebotarev [8]. Cf. also Levin [9, p. 230].

If $\Delta_C \arg f(z)$ denotes the variation of the argument of $f(z)$ round a closed contour $C$, this implies

$$|\Delta_C \arg f(z)| < 2\pi \tag{2.5.28}$$

for any contour $C$ contained entirely in $I_+$ or $I_-$, and

$$|\Delta_C \arg f(z)| \leqslant 2\pi \tag{2.5.29}$$

for any contour $C$ intersecting the real axis. We can now apply the well-known theorem [5, p. 115] according to which

$$\Delta_C \arg f(z) = 2\pi(N-P), \tag{2.5.30}$$

where $N$ and $P$ are, respectively, the number of zeros and the number of poles of the analytic function $f(z)$ contained within the contour $C$ (each zero or pole being counted a number of times equal to its multiplicity).

Since $f(z)$ is an entire function, $P = 0$, and it follows from (2.5.28) and (2.5.29) that $f(z)$ can have only one zero, that must be located on the real axis, say at $z = x_0$. Then

$$\varphi(z) = f(z)/(z-x_0) \tag{2.5.31}$$

is an entire function without any zeros. Furthermore,

$$\begin{aligned} |\arg \varphi(z)| &= |\arg f(z) - \arg(z-x_0)| \\ &\leqslant |\arg f(z)| + |\arg(z-x_0)| \leqslant 2\pi, \end{aligned}$$

so that

$$\psi(z) = \exp[i \ln \varphi(z)] = \exp[i \ln|\varphi(z)| - \arg \varphi(z)]$$

is an entire function with bounded modulus, and thus, by Liouville's theorem [5, p. 85], it is a constant. Therefore, $\varphi(z)$ is also a constant, and by (2.5.31) this implies that $f(z)$ is of the form (2.5.24), where the restrictions on $\alpha$ and $\beta$ follow from the reality of $f(z)$ for real $z$ and from (2.5.23). In the case of (2.5.20), letting $\alpha = 2\varepsilon$, we are led to (2.5.22).

Substituting (2.5.22) in (2.5.20), we find

$$F(k) = e^{2i\varepsilon k + i\beta}. \tag{2.5.32}$$

Since $B(0) = 1$ according to (2.5.2) and $S_a(0) = \pm 1$ according to (2.3.10), it follows from (2.5.18) that

$$e^{i\beta} = \pm 1. \tag{2.5.33}$$

Substituting (2.5.32), (2.5.33), and (2.5.2) in (2.5.18), we finally obtain the following result:

THEOREM 2.5.6. *The most general function $S(k)$ having all the analytic properties derived in Sections 2.2–2.4 is of the form*

$$S(k) = \pm e^{-2ik\alpha} B(k) = \pm e^{-2ik\alpha} \prod_n \frac{[1-(k/k_n{}^*)]}{[1-(k/k_n)]}, \tag{2.5.34}$$

*where*

$$\alpha = a - \varepsilon \leqslant a \tag{2.5.35}$$

*and the poles $k_n$ are all located in $I_-$, the pole distribution being symmetric with respect to the imaginary axis and such that*

$$\sigma = \sum_n (\operatorname{Im} k_n)/|k_n|^2 < \infty. \tag{2.5.36}$$

The expansion (2.5.34) is known as *the canonical product expansion of the S-function*. It was first obtained by Hu [10], and the proof of its validity under the present conditions was given by Van Kampen [1]. As was already noted following (2.3.10), the plus sign usually holds in (2.5.34).

The appearance of the constant $\varepsilon$ is related to the fact that the radius of the scatterer appears in the causality condition only as a bound: if $a-\varepsilon$ is the actual radius, the causality condition is verified a fortiori by defining a larger "radius" $a$. This introduces an additional time delay $2\varepsilon/c$ between the instant at which an incoming wave front strikes the surface $r = a$ and the appearance of the outgoing wave front, but such a delay is obviously compatible with causality.

It can be shown that the possible exponential growth of $S(k)$ as $|k| \to \infty$ in $I_+$, which has been physically related to the phase advance on reflection at the surface (cf. Section 2.3), arises solely from the exponential factor in (2.5.34), and not from the Blaschke product. It would suffice to show that

$$\lim_{|k|\to\infty} \frac{\ln|B(k)|}{|k|} = 0 \tag{2.5.37}$$

as $|k| \to \infty$ in $I_+$, so that $B(k)$ must be of slower than exponential growth.

However, we have to be very careful in defining a limit such as (2.5.37). In fact, when the number of zeros is infinite, they may be expected to cluster around certain directions, and such directions must clearly be excluded. The convergence of (2.5.36) actually indicates that most zeros must lie near the real axis. Nevertheless, it can be shown that the excluded directions form a set of measure zero. This is expressed in the following theorem [11, p. 115][8]:

[8] A more precise characterization of the exceptional set of directions is given in this reference.

THEOREM 2.5.7 (Ahlfors–Heins theorem). *Let* $B(k)$ *be defined by (2.5.2), with* $k_n \in I_-$ *and such that (2.5.36) is convergent. Then, for all* $\theta$ *in* $(0, \pi)$, *except at most a set of measure zero,*

$$\lim_{K \to \infty} \frac{\ln |B(Ke^{i\theta})|}{K} = 0. \tag{2.5.38}$$

*In particular, (2.5.38) holds without exception in any sector* $0 < \theta_1 < \theta < \theta_2 < \pi$ *which contains no zeros.*

This clarifies the physical meaning of the parameter $\alpha$ in (2.5.34): it plays the role of an effective radius of the scatterer, in the sense that the outgoing wave front starts to appear only when the incoming wave front strikes the surface $r = \alpha$. Thus, if $a$ is the actual radius of the scatterer, i.e., the lower bound of the values of $r$ such that the interaction vanishes outside a sphere of radius $r$, one would expect to have

$$\alpha = a. \tag{2.5.39}$$

Although $\alpha < a$ is compatible with causality, as we have seen, it would imply such a "soft" surface that no reflected wave would arise before penetration to a finite depth beyond it, which would hardly be expected to occur for any physical scatterer. In all specific models to be considered later, condition (2.5.39) is fulfilled. Thus, we shall usually assume that the equality sign holds in (2.5.35) (with the proviso that $a$ is the actual radius), and that the plus sign holds in (2.5.34). It then follows that *the S-function is completely determined by its poles.*

If we were to think of the dispersion relation (2.3.11), combined with the unitarity condition [which also relates $\operatorname{Re} S_a(k)$ with $\operatorname{Im} S_a(k)$], as some sort of "integral equation" (cf. Section 1.1), we might say that (2.5.34) is the general solution of this "equation." Thus there exists an infinite class of solutions, characterized by a denumerable set of parameters, the positions of the poles. Given any particular solution, one can obtain other solutions by introducing additional poles (together with the corresponding zeros). This is sometimes referred to as the *CDD ambiguity*, after Castillejo, Dalitz, and Dyson, who found a similar ambiguity in a study of meson–nucleon scattering [12].

It is clear from the above discussion that all the information about the internal structure of the scatterer must be contained in the poles, so that they play a very important role. The physical interpretation of the poles and their relation with observable effects will be discussed in Chapter 4. In addition to their location in $I_-$, they are restricted only by the requirement that the series (2.5.36) must be convergent. The sum of this series is related to a *sum rule*, which will now be derived.

Taking the logarithmic derivative of both sides of (2.5.34), we find

$$\frac{S'(k)}{S(k)} = -2i\alpha + \sum_n \left(\frac{1}{k-k_n^*} - \frac{1}{k-k_n}\right)$$

$$= -2i\left(\alpha + \sum_n \frac{\operatorname{Im} k_n}{|k-k_n|^2}\right). \tag{2.5.40}$$

In particular, taking $k=0$, we find the sum rule

$$S'(0)/S(0) = -2i(\alpha+\sigma), \tag{2.5.41}$$

where $\sigma$ is defined by (2.5.36).

Since the poles $k_n$ and $-k_n^*$ are paired together in (2.5.36), according to (2.5.1), we may rewrite (2.5.41) as

$$\sum_n 1/k_n = -i\sigma = i\alpha + \tfrac{1}{2}[S'(0)/S(0)]. \tag{2.5.42}$$

According to (2.2.15), we may also rewrite (2.5.40) in terms of the phase shift $\eta(k)$, as

$$\eta'(k) = -\alpha - \sum_n (\operatorname{Im} k_n)/|k-k_n|^2. \tag{2.5.43}$$

Since $\operatorname{Im} k_n < 0$, the last term is nonnegative, so that we get the inequality

$$d\eta/dk \geqslant -\alpha \geqslant -a. \tag{2.5.44}$$

As will be seen later (Section 2.11), this result is related to another formulation of the causality condition, due to Wigner (Wigner's inequality).

## 2.6. Extension to Higher Angular Momenta

For the $l$th partial wave, the stationary solutions of the wave equation (2.2.1) are of the form

$$\psi_l(k,r,\theta,t) = u_l(k,r,t)\,P_l(\cos\theta) \qquad (r>a), \tag{2.6.1}$$

where the radial wave function $u_l(k,r,t)$ is given by

$$u_l(k,r,t) = a_l(k)[h_l^{(2)}(kr) + S_l(k)\,h_l^{(1)}(kr)]\,e^{-ikct}. \tag{2.6.2}$$

Here $h_l^{(1,2)}$ are the spherical Hankel functions, given by

$$h_l^{(1,2)}(z) = \frac{e^{\pm iz}}{\pm iz}\sum_{n=0}^{l} \frac{(\pm i)^{n-l}}{n!}\frac{(l+n)!}{(l-n)!}(2z)^{-n}. \tag{2.6.3}$$

Asymptotically, for $|z|\to\infty$, $h_l^{(1)}$ and $h_l^{(2)}$ represent outgoing and incoming waves, respectively:

$$h_l^{(1,2)}(z) \approx (\mp i)^{l+1}\, e^{\pm iz}/z \qquad (|z|\to\infty), \tag{2.6.4}$$

so that, for $r\to\infty$, (2.6.2) becomes (with $A_l = i^{l+1}\, a_l/k$)

$$u_l(k,r,t) \approx [A_l(k)/r]\,[e^{-ikr} - (-1)^l\, S_l(k)\, e^{ikr}]\, e^{-ikct} \qquad (r\to\infty), \tag{2.6.5}$$

which reduces to (2.2.5) for $l=0$.

The function $S_l(k)$ is the $S$-function for the $l$th partial wave. In the absence of a scatterer, (2.6.2) is valid up to $r=0$, where it must be regular [like $j_l(kr)$], so that $S_l = 1$ in this case.

The asymptotic behavior of an arbitrary incoming wave packet at large distances is

$$u_{\text{in}}(r,t) \approx \frac{1}{r}\int_{-\infty}^{\infty} A_l(k)\, e^{-ik(r+ct)}\, dk \qquad (r\to\infty), \tag{2.6.6}$$

and the reality of the field again leads to

$$A_l(-k) = A_l^*(k). \tag{2.6.7}$$

The corresponding outgoing wave packet is

$$u_{\text{out}}(r,t) \approx \frac{(-1)^{l+1}}{r}\int_{-\infty}^{\infty} S_l(k)\, A_l(k)\, e^{ik(r-ct)}\, dk \qquad (r\to\infty), \tag{2.6.8}$$

and the reality condition again leads to the symmetry relation

$$S_l(-k) = S_l^*(k). \tag{2.6.9}$$

It is only asymptotically, now, that $ru_{\text{in}}$ and $ru_{\text{out}}$ propagate without change of shape: at finite distances they are deformed as they propagate, as a consequence of centrifugal forces.

Energy conservation again leads to the unitarity condition

$$S_l(k)\, S_l^*(k) = 1, \tag{2.6.10}$$

because the derivation, as in Section 2.2, depends only on the energy flux over a large sphere, so that we can employ (2.6.6) and (2.6.8). The only difference is that $\int d\Omega = 4\pi$ in (2.2.12) is replaced by $\int P_l^{\,2}(\cos\theta)\, d\Omega = 4\pi/(2l+1)$. Correspondingly, the cross section becomes

$$\sigma_l(k) = (\pi/k^2)(2l+1)\,|1-S_l(k)|^2 = (4\pi/k^2)(2l+1)\sin^2\eta_l(k), \tag{2.6.11}$$

where the phase-shift $\eta_l$ is related to $S_l$ by

$$S_l(k) = \exp[2i\eta_l(k)]. \tag{2.6.12}$$

Now let us consider the causality condition. Since the distinction between incoming and outgoing waves is now being made only asymptotically, at large distances from the scatterer, we can no longer apply the causality condition at the surface, as in Section 2.3. We must reformulate it as follows: *if the incoming wave vanishes for $t < t_0$ at a large distance $r$, the outgoing wave must also vanish for $t < t_0 + 2(r-a)/c$.* The time delay $2(r-a)/c$ corresponds to the time required for a wave front to travel from the sphere of radius $r$ to the surface of the scatterer and back.

By rewriting the integrand of (2.6.6) as

$$A_l(k)\exp[-ik(r+ct_0)]\exp[-ikc(t-t_0)],$$

it is readily seen that $u_{\text{in}}(r,t)$ will vanish for $t-t_0<0$ if we choose

$$A_l(k)\exp[-ik(r+ct_0)]$$

to be a causal transform. Similarly, by rewriting the integrand of (2.6.8) as

$$S_{la}(k)\,A_l(k)\exp[-ik(r+ct_0)]\exp\{-ikc[t-t_0-(2r/c)+(2a/c)]\},$$

where

$$S_{la}(k) = S_l(k)\,e^{2ika}, \tag{2.6.13}$$

we conclude from Titchmarsh's theorem that $u_{\text{out}}(r,t)$ will vanish for $t-t_0-2(r-a)/c<0$ if and only if $S_{la}(k)A_l(k)\exp[-ik(r+ct_0)]$ is also a causal transform. This is only possible for any choice of the causal transform $A_l(k)\exp[-ik(r+ct_0)]$ if (cf. Section 2.3)

$$S_{la}(k) \quad \text{is holomorphic in} \quad I_+. \tag{2.6.14}$$

The derivation of the boundedness condition

$$|S_{la}(k)| \leqslant 1 \qquad (\operatorname{Im} k \geqslant 0) \tag{2.6.15}$$

now proceeds in exactly the same way as in Section 2.3. Thus all the analytic properties that were derived in the case of $s$-waves remain valid for arbitrary values of $l$. In particular, $S_l(k)$ verifies the dispersion relation (2.3.11); it can be analytically continued to $I_-$, defining a meromorphic function with all its poles restricted to $I_-$, and the product expansion and sum rule derived in Section 2.5 remain valid for $S_l(k)$.

## 2.7. Nonrelativistic Quantum Scattering

We now want to carry out a program similar to that which we have gone through for a classical field in the case of quantum-mechanical scattering of nonrelativistic particles[9] by a spherically symmetric scatterer of radius $a$.

[9] For an introduction to the quantum theory of scattering, see, for example, Messiah [13].

Outside of the scatterer, Schrödinger's free particle equation is valid:

$$i\,\partial\psi/\partial t = -\tfrac{1}{2}\Delta\psi \qquad (r > a), \tag{2.7.1}$$

in units such that $\hbar = m = 1$, where $m$ is the mass of the particle.

For $s$-waves, the stationary solutions of (2.7.1) are of the form

$$\begin{aligned}\psi(E,r,t) &= (1/r)[A(E)e^{-ikr}+B(E)e^{ikr}]\,e^{-iEt}\\ &= \varphi(E,r,t)/r,\end{aligned} \tag{2.7.2}$$

where

$$E = k^2/2 \tag{2.7.3}$$

is the energy of the particle, and we *define*

$$k = +(2E)^{1/2} \tag{2.7.4}$$

to be the *positive* square root.

The $S$-function is again defined by [cf. (2.2.6)]

$$S(E) = -B(E)/A(E). \tag{2.7.5}$$

We assume that the interaction is *linear* (superposition principle), so that the general $s$-wave solution outside of the scatterer is given by

$$r\psi(r,t) = \varphi(r,t) = \varphi_{\text{in}}(r,t) + \varphi_{\text{out}}(r,t), \tag{2.7.6}$$

where

$$\varphi_{\text{in}}(r,t) = \int_0^\infty A(E)\,e^{-ikr-iEt}\,dE, \tag{2.7.7}$$

$$\varphi_{\text{out}}(r,t) = -\int_0^\infty S(E)\,A(E)\,e^{ikr-iEt}\,dE \tag{2.7.8}$$

correspond to incoming and outgoing wave packets, respectively. The scattered wave is given by

$$\varphi_{\text{sc}}(r,t) = \int_0^\infty [S(E)-1]\,A(E)\,e^{ikr-iEt}\,dE. \tag{2.7.9}$$

A crucial difference between the present case and that of a classical field treated above arises from the fact that all the integrations run only from 0 to $\infty$, rather than from $-\infty$ to $\infty$, because the energy $E$ is nonnegative [cf. (2.7.3)]. We shall see that this leads to far-reaching differences in the whole treatment.

The analog of (2.2.2) is the continuity equation for the probability flow:

$$\operatorname{div}\mathbf{j} + (\partial\rho/\partial t) = 0, \tag{2.7.10}$$

where

$$\rho = \psi^* \psi \tag{2.7.11}$$

is the probability density, and

$$\mathbf{j} = (1/2i)(\psi^* \operatorname{grad} \psi - \psi \operatorname{grad} \psi^*) \tag{2.7.12}$$

is the probability current density.

The total instantaneous inward probability flux through a sphere of radius $r \geqslant a$ is

$$\begin{aligned}\Phi(r,t) &= -\oint j_r(r,t)\, r^2\, d\Omega \\ &= 2\pi i(\varphi^*\, \partial\varphi/\partial r - \varphi\, \partial\varphi^*/\partial r),\end{aligned} \tag{2.7.13}$$

and it follows from (2.7.6) to (2.7.8) that

$$\begin{aligned}\Phi(r,t) = 2\pi \int_0^\infty dE' \int_0^\infty dE\, A^*(E')\, A(E)\, e^{i(E'-E)t} \\ \times [(k+k')\, e^{-i(k-k')r} - (k+k')\, S^*(E')\, S(E)\, e^{i(k-k')r} \\ + (k-k')\, S(E)\, e^{i(k+k')r} - (k-k')\, S^*(E')\, e^{-i(k+k')r}].\end{aligned} \tag{2.7.14}$$

We shall assume that the interaction is *probability conserving*; i.e., there is no emission or absorption of probability. The flux $\Phi(r,t)$ measures the instantaneous rate of increase of the probability $P_i(r,t)$ to find the particle inside the sphere of radius $r$,

$$\Phi(r,t) = (\partial/\partial t) P_i(r,t). \tag{2.7.15}$$

This probability can be defined by

$$P_i(r,t) = 1 - P_e(r,t), \tag{2.7.16}$$

where

$$P_e(r,t) = \int_{\geqslant r} |\psi|^2\, d^3x = 4\pi \int_r^\infty |\varphi|^2\, dr \tag{2.7.17}$$

is the probability that the particle may be found *outside* the sphere of radius $r\,(r \geqslant a)$ at time $t$. Since the scatterer does not emit or absorb probability, we must have

$$\int_{-\infty}^{\infty} \Phi(r,t)\, dt = P_i(r,\infty) - P_i(r,-\infty) = 0. \tag{2.7.18}$$

Since [cf. (A10.16)]

$$\int_{-\infty}^{\infty} e^{i(E'-E)t}\, dt = 2\pi\delta(E'-E),$$

it follows from (2.7.14) that

$$\int_{-\infty}^{\infty} \Phi(r,t)\, dt = 8\pi^2 \int_0^{\infty} k\, |A(E)|^2 \,[1-|S(E)|^2]\, dE, \qquad (2.7.19)$$

where the first term on the right represents the incoming contribution and the second term the outgoing one. This relation can also be derived directly from (2.7.13) and (2.7.6) to (2.7.8) by applying Parseval's theorem (1.6.2).

We restrict ourselves to wave packets such that $\sqrt{k}\,A(E)$ is square integrable. It then follows from (2.7.18) and (2.7.19) that

$$|S(E)|^2 = S^*(E)\, S(E) = 1. \qquad (2.7.20)$$

This is the *unitarity condition*, which here expresses the conservation of probability.

It follows that

$$S(E) = e^{2i\eta(E)}, \qquad (2.7.21)$$

where the phase shift $\eta(E)$ is connected with the $s$-wave scattering cross section in the usual way,

$$\sigma = (\pi/k^2)\,|1-S(E)|^2 = (4\pi/k^2)\sin^2\eta. \qquad (2.7.22)$$

Thus far, we have assumed that the interaction is spherically symmetric, vanishing beyond $r = a$, linear, and probability conserving. Let us now discuss the problem of how to formulate a causality condition.

## 2.8. The Schützer–Tiomno Causality Condition

### *(a) Formulation*

In attempting to formulate a causality condition for the scattering of nonrelativistic particles, we are confronted at once with several difficulties:

i. *There is no limiting velocity.* This by itself does not prevent us from discussing the effect of causality, provided that cause and effect are located at the same point in space, as in the primitive causality condition of Section 1.3 and in its application to dielectric polarization in Section 1.4.

ii. *No wave packets can be built which propagate with sharp wave fronts for any finite length of time, no matter how small.*

Physically, this corresponds to the well-known fact that, if one builds a Schrödinger wave packet having a sharp front at some instant, it will instantaneously spread over all space [4, Sec. 12]. This results from *i*, because, in order to get a sharp front, one requires infinitely large energies, which imply infinite velocities and instantaneous spreading.

Mathematically, property *ii* follows from the "one-sidedness" of the Fourier integrals (2.7.7) and (2.7.8), which represent "positive-frequency functions," of the form

$$f(t) = \int_0^\infty c(E)\, e^{-iEt}\, dE. \tag{2.8.1}$$

Such a function cannot vanish over any finite time interval, no matter how small.

In fact, according to (2.8.1), $f(t)$ has a regular analytic continuation in $I_-(t)$. It then follows from a theorem on analytic continuation [14, p. 177] that the boundary values of this analytic function over an arbitrarily small portion of the boundary of its analyticity domain, i.e., the real axis, cannot vanish, unless $f(t) \equiv 0$. Thus property *ii* is a consequence of the positive semi-definiteness of the energy spectrum (2.7.3).

It follows from these considerations that a causality condition of the type formulated in Section 2.3 cannot be applied here. However, there is another formulation of causality, closely related to that of Section 1.3, which does not require the existence of signals with sharp fronts, and in which cause and effect are located at the same point, so that no propagation problem arises. This formulation was employed by Schützer and Tiomno[10] in the earliest treatment of the problem. It can be stated as follows:

SCHÜTZER–TIOMNO CAUSALITY CONDITION. *The scattered wave packet at a distance r at time t cannot depend on the behavior of the incoming wave packet at r at later times $t' > t$.*

In terms of the wave packets (2.7.7) and (2.7.9), this means that

$$\delta\varphi_{\text{sc}}(r,t)/\delta\varphi_{\text{in}}(r,t') = 0 \qquad (t < t'), \tag{2.8.2}$$

where $\delta\varphi_{\text{sc}}/\delta\varphi_{\text{in}}$ denotes the functional derivative[11] of $\varphi_{\text{sc}}$ with respect to $\varphi_{\text{in}}$.

Let us find out the consequences of this condition, first in a purely heuristic manner, without worrying about convergence problems, which we leave for later discussion. Let us introduce a Green function $g(r,t)$ by

$$\varphi_{\text{sc}}(r,t) = \int_{-\infty}^{\infty} g(r, t-t')\, \varphi_{\text{in}}(r,t')\, dt' \qquad (r > a). \tag{2.8.3}$$

[10] See Schützer and Tiomno [15]. Unfortunately, although the authors *employed* the correct form of their condition in the analysis given in this paper, the condition they *stated* in the text was that of Section 2.3. This led to a criticism by Van Kampen [16], which has often been repeated in the literature. Schützer and Tiomno later gave the correct formulation of their condition, in a paper presented at a meeting in 1952 [17, p. 281]. In this connection, see also Giambiagi and Saavedra [18].

[11] Cf. Bogoliubov and Shirkov [19, Sec. 17.5].

The Schützer–Tiomno condition (2.8.2) then leads to

$$g(r,\tau) = 0 \qquad (\tau < 0). \tag{2.8.4}$$

Let us assume that $g(r,\tau)$ has a Fourier expansion of the form[12]

$$g(r,\tau) = \int_{-\infty}^{\infty} G(r,E)\, e^{-iE\tau}\, dE. \tag{2.8.5}$$

According to (2.7.7) and (2.7.9), the Fourier transforms of $\varphi_{\text{in}}$ and $\varphi_{\text{sc}}$ are, respectively,

$$\mathscr{F}\varphi_{\text{in}} = \theta(E)A(E)e^{-ikr}, \tag{2.8.6}$$

$$\mathscr{F}\varphi_{\text{sc}} = \theta(E)A(E)[S(E)-1]\,e^{ikr}, \tag{2.8.7}$$

where $\theta(E)$ is the Heaviside step function.

It follows from (2.8.3) to (2.8.7) and the convolution theorem for Fourier transforms that

$$\theta(E)A(E)[S(E)-1]\,e^{ikr} = 2\pi G(r,E)\,\theta(E)A(E)\,e^{-ikr}, \tag{2.8.8}$$

so that

$$G(r,E) = (1/2\pi)[S(E)-1]\,e^{2ikr} \qquad (E > 0). \tag{2.8.9}$$

On the other hand, according to (2.8.4) and (2.8.5), we have

$$G(r,E) = \frac{1}{2\pi}\int_{0}^{\infty} g(r,\tau)\, e^{iE\tau}\, d\tau. \tag{2.8.10}$$

Since the integral is extended only over $\tau \geqslant 0$, it follows that $G(r,E)$ has a regular analytic continuation in $I_+(E)$. According to (2.8.9), this implies that *$S(E)$ is holomorphic in $I_+(E)$*. Thus, the Schützer–Tiomno causality condition implies regularity of the $S$-function in the upper half of the complex energy plane.

### *(b) Analytic Continuation in the k-Plane*

Let $k = (2E)^{1/2}$, where the square root is defined to be positive for $E > 0$, as in (2.7.4). Then $I_+(E)$ is mapped into the first quadrant of the $k$-plane, and it follows from the above results that, if we regard $S$ as a function of $k$, it is holomorphic in the first quadrant of the $k$-plane,

$$0 < \arg k < \pi/2. \tag{2.8.11}$$

[12] The lower limit of integration in (2.8.5) has to be $-\infty$ rather than 0 as in (2.7.7)–(2.7.9). Otherwise, according to the theorem quoted following (2.8.1), it would be impossible to satisfy (2.8.4), since $g(r,\tau)$ could not possibly vanish over any finite time interval.

However, in contrast with the case of a classical field, causality does not imply anything about the possibility of continuing $S(k)$ beyond the first quadrant of the $k$-plane [corresponding to the continuation of $S(E)$ in $I_-(E)$]. In fact, it does not allow us to make any assertion concerning the behavior of $S$ on the positive imaginary $k$-axis (negative real $E$-axis), where the occurrence of singularities is not excluded.

Actually, it is well known that such singularities do occur in potential scattering, whenever a potential can give rise to bound states. A bound state corresponds to a normalizable eigenfunction, of the form

$$\varphi_p = r\psi_p = B(E_p)\exp(-K_p r - iE_p t) \qquad (K_p > 0), \tag{2.8.12}$$

where

$$E_p = (iK_p)^2/2 < 0. \tag{2.8.13}$$

Comparing (2.8.12) with (2.7.2), we see that it corresponds to taking $k = iK_p$, $A(E_p) = 0$. According to (2.7.5), this suggests that *a bound state with energy $E_p$ corresponds to a pole of $S(E)$ [$S(k)$] on the negative real axis (positive imaginary axis) at $E = E_p$ ($k = iK_p$).* In the case of potential scattering, this will be proved in Section 5.4.

It is not known at present what general properties of the interaction would imply the possibility of analytically continuing $S(E)$ to $I_-(E)$, with poles on the negative real axis, corresponding to bound states, as the only possible singularities. Some partial results will be discussed in Section 2.9. Here we have to adopt the (unsatisfactory) point of view of taking this as an additional, purely mathematical assumption.

Assuming that a single-valued analytic continuation of $S(E)$ to $I_-(E)$ across $E > 0$ is possible, what happens when we reach again the positive real axis, from below? We cannot expect that $S(E)$ will be single valued, with $S(E - i0) = S(E + i0)$ $(E > 0)$, because the former corresponds to $S(-k)$, while the latter corresponds to $S(k)$. For a classical field, we had the symmetry relation [cf. (2.2.11)]

$$S(-k) = S^*(k) \qquad (k \text{ real}), \tag{2.8.14}$$

which followed from the reality of the field.

In the present case, it can be shown[13] that (2.8.14) is valid when the interaction can be described by a self-adjoint Hamiltonian operator (this is analogous to the reality condition for a classical field). Another general condition leading to (2.8.14) will be described in Section 2.9. Here we shall again take (2.8.14) as an extra assumption.

[13] Cf. Wigner [20].

In terms of $S(E)$, (2.8.14) may be rewritten as

$$S(E-i0) = S^*(E+i0) \qquad (E>0). \tag{2.8.15}$$

The single-valued analytic continuation of $S(E)$ in $I_-$ which satisfies (2.8.15) is given by Schwarz's reflection principle [14, p. 177]:

$$S(E) = S^*(E^*). \tag{2.8.16}$$

Since $I_-(E)$ corresponds to the second quadrant of the $k$-plane, $E^*$ corresponds to $-k^*$, and (2.8.16) leads to

$$S(k) = S^*(-k^*), \tag{2.8.17}$$

which is the same as (2.3.13). In particular, $S(k)$ is real on the positive imaginary axis, wherever it is regular.

The analytic continuation of $S(k)$ can now be extended to $I_-(k)$, precisely in the same way as in Section 2.4, by means of the relation

$$S(k) = 1/S(-k). \tag{2.8.18}$$

It follows that $S(k)$ is also a meromorphic function in $I_-(k)$, where it has poles corresponding to the zeros in $I_+(k)$. The symmetry properties of poles and zeros are identical to those described in Section 2.4.

If $k-i\varepsilon$ is a point just below the positive real axis, (2.8.17) and (2.8.18) imply

$$S(k-i\varepsilon) = 1/S(-k+i\varepsilon) = 1/S^*(k+i\varepsilon). \tag{2.8.19}$$

Letting $\varepsilon \to 0$, the first and third members tend to a common limit $S(k)$ by the unitarity condition (2.7.20). Thus $S$ is a single-valued function of $k$.

To sum up, $S(k)$ *is a meromorphic function of* $k$, *with poles restricted to the positive imaginary axis* (*bound states*) *and the lower half-plane.*

If we regard $S$ as a function of $E$, however, we have a two-sheeted Riemann surface, corresponding to the kinematic relation $k=(2E)^{1/2}$. There is a cut along the positive real axis and, according to (2.8.15), the discontinuity across the cut is

$$S(E+i0) - S(E-i0) = 2i\,\mathrm{Im}\,S(E+i0). \tag{2.8.20}$$

The first sheet, which corresponds to $I_+(k)$, is known as the *physical sheet.* The $S$-function is regular on the physical sheet, except for possible poles on the negative real axis, corresponding to the energies of bound states. On the second (unphysical) sheet, which corresponds to $I_-(k)$, $S(E)$ can have complex poles, which would always arise in complex conjugate pairs (corresponding to $k$ and $-k^*$), as well as poles on the negative real axis, not associated with bound states [corresponding to poles of $S(k)$ on the negative imaginary axis].

### *(c) Discussion*

The derivation of the analytic properties of $S(E)$ given above is still deficient in several respects.

In the first place, we have not discussed in what sense (if any) the integrals (2.8.5) and (2.8.10) can be expected to converge. It is clear that the integrands are not, in general, square integrable functions, so that we cannot apply Titchmarsh's theorem 1.6.1. This is shown by the example of an impenetrable (hard) sphere, defined by the boundary condition (2.3.7), for which $S(k)$ is given by (2.3.8): clearly, the corresponding function (2.8.9) is not square integrable.

The most powerful result available to us is Theorem 1.8.2. In order to apply it, we have to assume that, in (2.8.5),

$$G(r,E) \in \mathscr{S}'; \tag{2.8.21}$$

i.e., that we are dealing with *temperate distributions*. This assumption leads to several consequences. By condition (a) of Theorem 1.8.2, the value of $G(r,E)$ for $E<0$ in (2.8.5), which was so far undetermined [cf. (2.8.9)], becomes uniquely specified by the analytic continuation of (2.8.9):

$$\begin{aligned} G(r,E) &= \lim_{\varepsilon\to 0+} G(r,E+i\varepsilon) \\ &= \frac{e^{-2|k|r}}{2\pi} \lim_{\varepsilon\to 0+} [S(E+i\varepsilon)-1] \qquad (E<0). \end{aligned} \tag{2.8.22}$$

Our assumption that the only singularities of $S(E)$ for $E<0$ are poles (corresponding to bound states) is consistent with (2.8.21). In fact, if $E=E_1<0$ is a simple pole of $S$ with residue $r_1$, we have, in the neighborhood of $E_1$,

$$\begin{aligned} \lim_{\varepsilon\to 0+} S(E+i\varepsilon) &= \lim_{\varepsilon\to 0+}\left(\frac{r_1}{E-E_1+i\varepsilon}\right) \\ &= r_1\left[\frac{P}{E-E_1} - i\pi\delta(E-E_1)\right], \end{aligned} \tag{2.8.23}$$

which is a temperate distribution [cf. (A4.24)]. The same would be true for an $n$th-order pole [cf. (A4.25)]. This is also equivalent to saying that the integral (2.8.5) should be interpreted as the boundary value of an integral taken above the real axis, avoiding the poles.[14]

According to condition (c) of Theorem 1.8.2, it also follows that

$$|S_r(E)| = |S(E)e^{2ikr}| = O(E^n) \tag{2.8.24}$$

[14] Cf. Schützer and Tiomno [15].

as $|E| \to \infty$ in $I_+$. The value of $r$ at which the Schützer–Tiomno condition is applied must be $\geqslant a$. It can be argued that it would only be legitimate to apply it asymptotically, as $r \to \infty$, since incoming and outgoing (or scattered) waves can only be sharply distinguished at large distances. In such a case, (2.8.24) would yield practically no bound on the behavior of $S(E)$ as $|E| \to \infty$ in $I_+$. However, if we accept the applicability of the causality condition even for $r = a$, (2.8.24) does yield an important bound on the asymptotic behavior of $S(E)$ in $I_+$. We shall see in Section 2.9 that such a bound (in improved form) indeed follows from Van Kampen's analysis. We see, therefore, that (2.8.21) is a consistent, but by no means harmless, assumption: it already implies several of the results.

We must also make sure that the convolution theorem, which led to (2.8.8), can be applied. According to the discussion at the end of Section A10, a sufficient condition for this purpose is that

$$\theta(E)\,A(E) \in \mathcal{O}_M,$$

i.e., that $\theta(E)\,A(E)$ be an infinitely differentiable function, bounded by a polynomial, the same being true for all its derivatives. In order to be able to cancel $A(E)$ in (2.8.8) to get (2.8.9), we must also have $A(E) \neq 0$. These requirements can certainly be satisfied by appropriate choice[15] of $A(E)$.

Another problem that arises in connection with the definition of Green's function $g(r, t)$ is that this function is by no means uniquely defined by (2.8.3). In fact, according to (2.8.9), the condition (2.8.3) determines the Fourier transform $G(r, E)$ of Green's function only for $E > 0$. Thus, if $g(r, \tau)$ is a possible Green function, verifying (2.8.3), so is $g + \Delta g$, where

$$\Delta g(r, \tau) = \int_{-\infty}^{0} \Delta G(r, E)\, e^{-iE\tau}\, dE, \tag{2.8.25}$$

and $\Delta G$ is an arbitrary temperate distribution. Due to the factor $\theta(E)$ in (2.8.6), $\Delta g$ gives no contribution to the scattered wave:

$$\Delta g(r, t) * \varphi_{\text{in}}(r, t) = 0. \tag{2.8.26}$$

Another way of saying this is that positive and negative frequency functions are orthogonal to each other with respect to the convolution product.

It follows that there exists an infinite class of Green functions, all equivalent with respect to (2.8.3). However, at most one of them can be *causal* in the sense of (2.8.4). In fact, if $g$ has this property, $g + \Delta g$ cannot have it since, according to (2.8.25) and the theorem quoted in connection with (2.8.1), $\Delta g(r, \tau)$ cannot vanish over any time interval, no matter how small. We have

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[15] To get (2.8.9) at any particular point $E$, it suffices to choose $A(E) \in \mathcal{D}$, with supp $A \subset (0, \infty)$ and $E \in$ supp $A$ (cf. Section A2).

actually seen that the causal Green function is uniquely determined [cf. (2.8.22)].

In view of the nonuniqueness of Green's function, we cannot express the Schützer–Tiomno causality condition as "the Green function must be causal," but rather as "there exists a causal Green function." Conversely, if a causal Green function exists, it is unique, and its existence implies the validity of the Schützer–Tiomno condition. In fact, according to (2.8.26), any difference $\Delta g$ with respect to the causal function does not contribute to the scattered wave, so that $\varphi_{sc}(r,t)$ effectively depends only on the behavior of $\varphi_{in}(r,t')$ for $t' < t$, in agreement with the Schützer–Tiomno condition.[16]

## 2.9. Van Kampen's Causality Condition

### *(a) Formulation*

A different formulation of the causality condition in nonrelativistic scattering was proposed by Van Kampen [16]. As we shall see, his condition is actually related to the conservation of probability.

Since the causality condition of Section 2.3 cannot be applied, in view of the nonexistence of incoming or outgoing wave packets with sharp fronts, one might try to employ an integrated form of this condition, by requiring that the total probability of finding an outgoing particle at a distance[17] $r$ up to the time $t$ cannot exceed the corresponding probability for an incoming particle.

However, as shown by (2.7.14), the probability current at finite distances cannot be decomposed into an incoming and an outgoing part. It also contains a rapidly oscillating interference term. On the other hand, if we integrate both sides of (2.7.14) with respect to the time, from $-\infty$ to $t$, we get, according to (2.7.15), the probability that the particle is inside the sphere of radius $r$ at the time $t$. The minimum requirement for the probability interpretation to be possible is that it must be nonnegative; i.e.,

$$P_i(r,t) \geqslant 0 \qquad (r \geqslant a). \tag{2.9.1}$$

According to (2.7.16), this is the same as

$$P_e(r,t) \leqslant 1 \qquad (r \geqslant a).$$

We are thus led to

VAN KAMPEN'S CAUSALITY CONDITION. *Let the incoming wave packet be so normalized as to represent one incident particle for $t \to -\infty$. Then the total*

[16] A similar point of view has been expressed by Rañada [21].

[17] More precisely, between $r$ and $r+dr$.

*probability of finding a particle outside of the scatterer at any given moment cannot exceed unity.*

The strongest form of this condition is obtained by choosing $r = a$ in (2.9.1), and we can take $t = 0$ without loss of generality, so that, by (2.7.15), (2.9.1) becomes

$$P_i(a,0) = \int_{-\infty}^{0} \Phi(a,t)\,dt \geqslant 0. \tag{2.9.2}$$

Substituting $\Phi(a,t)$ by (2.7.14), inverting the order of integration (which is allowed for a normalized wave packet), and employing the relation [cf. (A11.5)–(A11.7)]

$$\int_{-\infty}^{0} e^{i(E'-E)t}\,dt = 2\pi\delta^{-}(E'-E) = \frac{P}{i(E'-E)} + \pi\delta(E'-E), \tag{2.9.3}$$

the inequality (2.9.2) becomes

$$\begin{aligned} &\int_0^\infty \int_0^\infty \frac{B_a{}^*(k')\,B_a(k) - A_a{}^*(k')\,A_a(k)}{i(k-k')}\,dk\,dk' \\ &\qquad \geqslant \int_0^\infty \int_0^\infty \frac{B_a{}^*(k')\,A_a(k) - A_a{}^*(k')\,B_a(k)}{i(k+k')}\,dk\,dk', \end{aligned} \tag{2.9.4}$$

where

$$\begin{gathered} A_a(k) = ke^{-ika}A(E), \qquad B_a(k) = -S_a(k)\,A_a(k), \\ S_a(k) = S(k)\,e^{2ika}. \end{gathered} \tag{2.9.5}$$

The contributions from the term containing the delta function in (2.9.3) cancel out because of the unitarity condition (2.7.20). The right-hand side of (2.9.4) corresponds to the interference term referred to above.

Van Kampen's causality condition requires that the inequality (2.9.4) be satisfied whenever $A_a(k) \in L^2(0,\infty)$ (i.e., is square integrable over the positive real axis).

As has already been mentioned, the main difficulty in handling the present problem is that, in contrast with the case of a classical field, all the integrals in (2.9.4) are extended only over $(0,\infty)$. One would like to deal with integrals over $(-\infty,\infty)$. This can be accomplished by means of a Mellin transformation [22, p. 7]

$$\begin{aligned} A_a(k) &= \int_{-\infty}^{\infty} \mathscr{A}(\tau)\,k^{i\tau-(1/2)}\,d\tau, \\ \mathscr{A}(\tau) &= \frac{1}{2\pi}\int_0^\infty A_a(k)\,k^{-i\tau-(1/2)}\,dk. \end{aligned} \tag{2.9.6}$$

This is just another way of writing a Fourier transformation, as may be seen immediately by substituting $k = e^x$. Parseval's theorem becomes

$$\int_0^{\infty} |A_a(k)|^2 \, dk = 2\pi \int_{-\infty}^{\infty} |\mathscr{A}(\tau)|^2 \, d\tau, \tag{2.9.7}$$

so that the transformation establishes a one-to-one correspondence between functions $A_a(k) \in L^2(0, \infty)$ and $\mathscr{A}(\tau) \in L^2(-\infty, \infty)$.

By rewriting the inequality (2.9.4) in terms of Mellin transforms, Van Kampen then derives from it the following result:

i. *$S(k)$ has a regular analytic continuation in the first quadrant of the $k$-plane.* This is the same conclusion as was found in Section 2.8 by applying the Schützer–Tiomno causality condition. For the detailed derivation of this result and of those quoted below, the reader is referred to Van Kampen's paper.

However, (2.9.4) yields important additional information, leading to the following bounds on $\operatorname{Im} S_a(k)$:

ii. *In the first quadrant of the $k$-plane, i.e., for*

$$0 \leqslant \arg k < \pi/2, \tag{2.9.8}$$

*we have*

$$\operatorname{Im} S_a(k) \leqslant \frac{\operatorname{Re} k}{\operatorname{Im} k}, \tag{2.9.9}$$

$$\operatorname{Im} S_a(k) \leqslant 1. \tag{2.9.10}$$

The inequality (2.9.10) follows from (2.9.9) by applying Phragmén–Lindelöf's theorem to the function $\exp[-iS_a(k)]$.

These bounds should be compared with (2.3.6). Clearly, we cannot expect to obtain a bound on $|S_a(k)|$ valid up to the positive imaginary axis, because we already know that there may be singularities on it. However, if we stay away from the imaginary axis, (2.9.4) does imply the boundedness of $|S_a(k)|$.

In fact, Van Kampen's causality condition (2.9.1) can also be applied with $r = a + \varepsilon$, $\varepsilon > 0$, so that (2.9.10) can be extended to

$$\operatorname{Im}[e^{2ik\varepsilon} S_a(k)] \leqslant 1 \qquad (\varepsilon \geqslant 0),$$

or, with

$$k = k' + iK, \qquad S_a(k) = |S_a| \, e^{i\varphi},$$

$$\exp(-2\varepsilon K) \, |S_a(k' + iK)| \sin(2\varepsilon k' + \varphi) \leqslant 1.$$

For any value of $k$, we can choose $\varepsilon$ between 0 and $\pi/k'$ so that $\sin(2k'\varepsilon + \varphi) = 1$.

Taking the most unfavorable case, $\varepsilon = \pi/k'$, we get

$$|S_a(k'+iK)| \leqslant \exp(2\pi K/k'). \tag{2.9.11}$$

It follows that

iii. $|S_a(k)|$ *is bounded in*

$$0 \leqslant \arg k \leqslant \pi/2 - \delta \qquad (\delta > 0).$$

It follows from property *i* that, as a function of the energy $E = k^2/2$, $S_a(E)$ has a regular analytic continuation in $I_+(E)$. Let

$$F(E) = i - S_a(E). \tag{2.9.12}$$

Then $F(E)$ is also regular in $I_+$ and, by (2.9.10),

$$\operatorname{Im} F(E) \geqslant 0 \qquad \text{in} \quad I_+. \tag{2.9.13}$$

A function with these properties is called a *Herglotz function.* A general representation for this class of functions is given in Appendix C. According to (C12), we can write

$$\begin{aligned} i - S_a(E) = {} & AE + C + \sum_n \beta_n \left( \frac{1+E_n E}{E_n - E} \right) \\ & + \int_{-\infty}^{\infty} \frac{1+E'E}{E'-E}\, d\tilde{\beta}(E') \qquad (\operatorname{Im} E > 0). \end{aligned} \tag{2.9.14}$$

where $A$, $C$, $\beta_n$, and $E_n$ are real constants, $A \geqslant 0$, $\beta_n > 0$, and $\tilde{\beta}(E')$ is a bounded, nondecreasing continuous function. Also, by (C4),

$$A = -\lim_{|E|\to\infty} [S_a(E)/E] \qquad (0 < \varepsilon \leqslant \arg E \leqslant \pi - \varepsilon),$$

so that, by property *iii* above,

$$A = 0 \tag{2.9.15}$$

in the present case.

At any point where $S_a(E)$ is regular, and so, in particular, for $E > 0$, we have, by (C9),

$$1 - \operatorname{Im} S_a(E+i0) = \pi(1+E^2)\tilde{\beta}'(E). \tag{2.9.16}$$

The energies $E_n$ form a finite or denumerable set; they must all be negative, since $S_a(E)$ is regular for $E > 0$; also, according to (C6), $\sum_n \beta_n$ is convergent. Thus we can rewrite the sum in (2.9.14) as

$$\sum_n \beta_n \left( \frac{1+E_n E}{E_n - E} \right) = \sum_n b_n \left( \frac{1}{E_n - E} - \frac{1}{E_n} \right) + \sum_n \frac{\beta_n}{E_n}, \tag{2.9.17}$$

where

$$b_n = \beta_n(1+E_n^2) > 0. \tag{2.9.18}$$

Substituting (2.9.15)–(2.9.17) in (2.9.14), we find

$$\begin{aligned} S_a(E) = i + D + \sum_n b_n\left(\frac{1}{E-E_n} + \frac{1}{E_n}\right) \\ + \frac{1}{\pi}\int_0^\infty \left(\frac{1+E'E}{E'-E}\right)\frac{[\operatorname{Im} S_a(E'+i0)-1]}{(1+E'^2)}\,dE' \\ - \int_{-\infty}^0 \left(\frac{1+E'E}{E'-E}\right) d\tilde{\beta}(E') \qquad (\operatorname{Im} E > 0), \end{aligned} \tag{2.9.19}$$

where $D = C + \sum_n \beta_n/E_n$ is again a real constant.

The sum on the right of (2.9.19) has the form of a Mittag–Leffler expansion [5, p. 110] with respect to the poles $E_n$ on the negative real axis. The last term of (2.9.19) can also give rise to singularities on the negative real axis (and it can modify the singularity at $E_n$ so that it no longer corresponds just to a pole). However, it can be shown, due to the continuity of $\tilde{\beta}(E')$, that such singularities would give rise to a slower increase than poles. We conclude, therefore, that $S_a(E)$ *cannot have poles of higher order than the first on the negative real axis, and the residues* $b_n$ *at such poles are always positive.*

Although Van Kampen's causality condition leads to more detailed information about the analytic behavior of $S(E)$ than Schützer and Tiomno's condition, it still does not exclude singularities other than poles from the negative real axis, nor does it ensure the possibility of analytic continuation to the lower half-plane. As was mentioned in Section 2.8, extra assumptions are required for this purpose. No completely satisfactory solution has been proposed, but we shall now discuss some partial results that have been obtained.

### *(b) The Symmetry Relation*

As was mentioned in Section 2.8, the poles of $S(E)$ on the negative real axis should correspond to the binding energies of bound states. Thus, for a purely repulsive interaction, it might be expected that no singularities would occur and, for an attractive one, they should not extend beyond the maximum binding strength of the interaction. The problem then arises of how to characterize the attractive or repulsive character of the interaction in terms only of quantities defined in the external region. The following formulation has been proposed by Van Kampen [23].

For free particles, the expectation value of the energy for the $s$-state wave function $\psi(r,t)$ can be written as

$$\begin{aligned}\langle E\rangle &= -\tfrac{1}{2}\int \psi^* \Delta\psi \, d^3x \\ &= \tfrac{1}{2}\int \nabla\psi^* \cdot \nabla\psi \, d^3x \\ &= \frac{4\pi}{2}\int_0^\infty \left|\frac{\partial\varphi}{\partial r}\right|^2 dr,\end{aligned} \tag{2.9.20}$$

where $\varphi = r\psi$, as in (2.7.6). Thus we may interpret $\frac{1}{2}|\nabla\psi|^2$ as the (kinetic) energy density[18].

In the presence of an interaction, Van Kampen defines as the "energy in the external region" the quantity

$$E_e(t) = \tfrac{1}{2}\int_{r\geqslant a} |\nabla\psi|^2 \, d^3x = 2\pi\int_a^\infty \left|\frac{\partial\varphi}{\partial r}\right|^2 dr. \tag{2.9.21}$$

For $t \to -\infty$, when the wave packet is far away, this must coincide with the total energy $E_{\text{tot}}$:

$$E_e(-\infty) = E_{\text{tot}}. \tag{2.9.22}$$

The "energy in the internal region" is then defined by

$$E_i(t) = E_{\text{tot}} - E_e(t), \tag{2.9.23}$$

which should be compared with (2.7.16).

A repulsive interaction is defined by the condition

$$E_i(t) \geqslant 0. \tag{2.9.24}$$

For an interaction described by a Hamiltonian $H$, it is usually assumed that there is a lower bound to the energy; i.e., there exists a constant $B$ such that $H+B$ is a positive definite operator. This implies

$$E_i = \int_{r\leqslant a} \psi^* H\psi \, d^3x \geqslant -B\int_{r\leqslant a} \psi^*\psi \, d^3x.$$

The integral on the right can be identified in this case with the probability $P_i$ to find the particle in the internal region. This suggests extending (2.9.24) by postulating that

$$E_i(t) \geqslant -BP_i(a,t), \tag{2.9.25}$$

[18] Actually, such an interpretation arises when we consider the Schrödinger equation as a field equation, and it is not well justified in a one-particle interpretation. Cf. Schiff [4, p. 338].

where $P_i(a,t)$ is defined by (2.7.16). We have $B=0$ for a repulsive interaction. Let us consider this case first and investigate the consequences of (2.9.24). If we compare (2.9.21)–(2.9.24) with (2.7.16), (2.7.17), and (2.9.1), we see that the only significant difference is in the substitution of $\varphi(r,t)$ in (2.7.17) by

$$\frac{\partial\varphi}{\partial r}(r,t) = \int_0^\infty [-ikA(E)\,e^{-ikr}+ikB(E)\,e^{ikr}]\,e^{-iEt}\,dE \tag{2.9.26}$$

in (2.9.21) [cf. (2.7.5)–(2.7.8)]. This corresponds to the replacements

$$A(E)\to -ikA(E), \qquad B(E)\to ikB(E). \tag{2.9.27}$$

Thus, as may also be verified by direct substitution in (2.9.24), the inequality (2.9.4) is replaced by

$$\int_0^\infty\int_0^\infty \frac{\tilde{B}_a{}^*(k')\tilde{B}_a(k)-\tilde{A}_a{}^*(k')\tilde{A}_a(k)}{i(k-k')}\,dk\,dk'$$
$$\geqslant -\int_0^\infty\int_0^\infty \frac{\tilde{B}_a{}^*(k')\tilde{A}_a(k)-\tilde{A}_a{}^*(k')\tilde{B}_a(k)}{i(k+k')}\,dk\,dk', \tag{2.9.28}$$

where

$$\tilde{A}_a(k) = kA_a(k), \qquad \tilde{B}_a(k) = kB_a(k) = -S_a(k)\tilde{A}_a(k), \tag{2.9.29}$$

and $A_a(k)$, $B_a(k)$, and $S_a(k)$ are defined by (2.9.5).

Since (2.9.28) differs from (2.9.4) only by the sign of $\tilde{B}_a(k)$ as compared with $B_a(k)$, it yields the same information about $-S_a(k)$ that (2.9.4) yielded about $S_a(k)$. In particular we find, corresponding to (2.9.9),

$$\operatorname{Im} S_a(k) \geqslant -(\operatorname{Re} k)/(\operatorname{Im} k) \tag{2.9.30}$$

in the first quadrant of the $k$-plane.

It follows from (2.9.9) and (2.9.30) that

$$\operatorname{Im} S_a(k)\to 0 \quad \text{as} \quad \operatorname{Re} k\to 0+\,; \tag{2.9.31}$$

that is, $S(k)$ *is real on the positive imaginary axis*. It then follows from the Schwarz reflection principle, as in (2.8.16), that $S(k)$ has a regular analytic continuation in the second quadrant of the $k$-plane, given by (2.8.17):

$$S(k) = S^*(-k^*), \tag{2.9.32}$$

which reduces to the symmetry relation (2.8.14) on the real axis. Thus, *for a repulsive interaction, $S(k)$ is regular in $I_+$ and has exactly the same properties that were found for $S(k)$ in the case of a classical field.* In particular, it verifies the dispersion relation (2.3.11), it can be analytically continued to $I_-$, as in Section 2.4, and the canonical product expansion (2.5.34) is valid for it.

In the general case, in which $B > 0$ in (2.9.25), it is found [23] that $S(k)$ is real on the positive imaginary axis, except on the segment from 0 to $iK$, where

$$B = K^2/2. \tag{2.9.33}$$

The reality of $S(k)$ beyond $k = iK$ may again be used to continue it into the second quadrant, with the help of (2.9.32), so that we again obtain the symmetry relation (2.8.14) on the real axis. We can then extend the analytic continuation to $I_-(k)$, just as in Section 2.4, with the help of (2.8.18). In this way we find that *$S(k)$ is meromorphic in the whole k-plane and regular in $I_+$, except on the interval $(-iK, iK)$ of the imaginary axis.* The above assumptions do not allow us to say anything about the behavior of $S(k)$ in the neighborhood of this interval. However, as we see from (2.9.33), our expectation that singularities cannot extend beyond the maximum binding energy of the interaction is confirmed.

To proceed further, we need an assumption about the nature of the possible singularities on the interval $(0, iK)$ [those in $(-iK, 0)$ then follow from (2.8.18)]. We shall *assume*, as in Section 2.8, that *the only possible singularities are poles.*[19] We have already seen that they must then be simple poles. Finally, as in Section 2.8, we conclude that *$S(k)$ is meromorphic in the whole k-plane, its poles being located either in $I_+$ or on the positive imaginary axis.* According to (2.9.32), $S(k)$ is real in between the poles on the imaginary axis.

## *(c) Dispersion Relation*

Let us now proceed from these results and find out their effect on the representation (2.9.19) for $S_a(E)$. Since all the singularities of $S_a(E)$ for $E < 0$ are contained in the sum over the poles $E_n$, the last term of (2.9.19) is regular for $E < 0$, and, since $S_a(E)$ is real between the poles, it follows from (2.9.16) that

$$1 = \pi(1+E'^2)\,\tilde{\beta}'(E') \qquad (E' < 0). \tag{2.9.34}$$

Substituting this in the last integral of (2.9.19), we find that it may be combined with the contribution from the term $-1$ in the first integral, yielding

$$-\frac{1}{\pi}\int_{-\infty}^{\infty} \frac{(1+E'E)}{(1+E'^2)(E'-E)}\,dE' = -i \qquad (\operatorname{Im} E > 0), \tag{2.9.35}$$

as may readily be verified by contour integration. This cancels the term $i$ on the right-hand side of (2.9.19). In the remaining integral, we can use the identity

$$\frac{(1+E'E)}{(1+E'^2)(E'-E)} = \left(\frac{1}{E'-E} - \frac{1}{E'}\right) + \frac{1}{E'(1+E'^2)}. \tag{2.9.36}$$

[19] It suffices to assume that there can be only isolated singularities (cf. Van Kampen [16]).

The last term contributes a constant, which can be added to $D$. [Note that $\operatorname{Im} S_a(0) = 0$, so that the integral converges.] Finally, (2.9.19) becomes

$$S_a(E) = S(0) - \frac{1}{\pi}\int_0^\infty \left(\frac{1}{E-E'} + \frac{1}{E'}\right) \operatorname{Im} S_a(E'+i0)\, dE'$$

$$+ \sum_n b_n\left(\frac{1}{E-E_n} + \frac{1}{E_n}\right), \tag{2.9.37}$$

where we have substituted the constant by its value $S(0)$. By analytic continuation, this dispersion relation is valid on the whole first sheet of the energy plane.

Since (2.9.37) was obtained by simplifying a considerably more complicated expression, it is instructive to give also a direct derivation of it. We shall restrict ourselves, for simplicity, to the case in which *the total number of poles on the positive imaginary axis is finite.*

It follows from this assumption that $S_a(k)$ is bounded as $k \to i\infty$, so that, by (2.9.11), it is bounded as $|k| \to \infty$ in $0 \leqslant \arg k \leqslant \pi/2$ and, by analytic continuation [cf. (2.9.32)], in $0 \leqslant \arg k \leqslant \pi$.

Let us then consider the function

$$F(E) = \frac{S_a(E) - S(0)}{E} \tag{2.9.38}$$

on the first (physical) sheet of its Riemann surface (cf. Section 2.8). On this sheet, it is a regular analytic function of $E$, except for a finite number of simple poles $E_1, E_2, \ldots, E_N$ on the negative real axis. Moreover, since $S_a(E)$ is bounded, we have

$$F(E) = O(E^{-1}) \qquad \text{as} \qquad |E| \to \infty. \tag{2.9.39}$$

It follows that, for any point $E$ outside of the real axis, we have

$$\int_\Gamma \frac{F(E')}{E'-E}\, dE' = 2\pi i \sum_{n=1}^N \frac{\operatorname{res} S_a(E_n)}{E_n(E_n - E)} + 2\pi i F(E), \tag{2.9.40}$$

where $\Gamma$ is the contour shown in Fig. 2.2 and $\operatorname{res} S_a(E_n)$ is the residue of $S_a$ at $E_n$.

If we now let the radius of the circle tend to infinity, it follows from (2.9.39) that it gives no contribution in the limit, so that (2.9.40) becomes

$$\int_0^\infty \frac{[S_a(E'+i0) - S_a(E'-i0)]}{E'(E'-E)}\, dE'$$

$$= 2\pi i\left[\frac{S_a(E) - S(0)}{E}\right] + 2\pi i \sum_{n=1}^N \frac{\operatorname{res} S_a(E_n)}{E_n(E_n - E)},$$

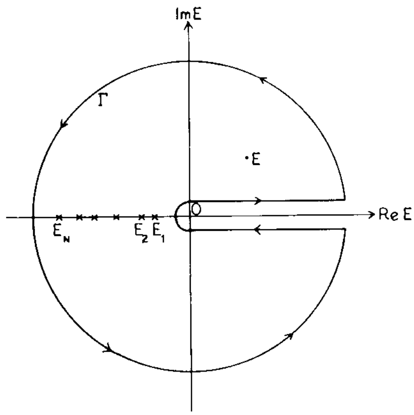


*FIG. 2.2.* The contour $\Gamma$ ($\times$, poles).

or, taking into account (2.8.20),

$$S_a(E) = S(0) + \frac{E}{\pi}\int_0^\infty \frac{\operatorname{Im} S_a(E'+i0)}{E'(E'-E)}\,dE' + E\sum_n \frac{\operatorname{res} S_a(E_n)}{E_n(E-E_n)}, \tag{2.9.41}$$

which is equivalent to (2.9.37). This is a dispersion relation for $S_a(E)$ at any point $E$ of the physical sheet.

By means of (2.7.3), we can translate (2.9.37) into a dispersion relation for the single-valued function $S_a(k)$, for $k \in I_+$:

$$S_a(k) = S(0) + \frac{2k^2}{\pi}\int_0^\infty \frac{\operatorname{Im} S_a(k')}{k'(k'^2-k^2)}\,dk' - 2k^2\sum_n \frac{b_n}{K_n{}^2(k^2+K_n{}^2)} \qquad (\operatorname{Im} k > 0), \tag{2.9.42}$$

where

$$K_n{}^2 = -2E_n. \tag{2.9.43}$$

Again, as in (2.3.10), we must have $S(0) = \pm 1$, and usually $S(0) = 1$.

If we now let $k$ approach the real axis in (2.9.42), we find, with the help of (1.6.9),

$$\operatorname{Re} S_a(k) = S(0) + \frac{2k^2}{\pi}\,P\!\!\int_0^\infty \frac{\operatorname{Im} S_a(k')}{k'(k'^2-k^2)}\,dk' - 2k^2\sum_n \frac{b_n}{K_n{}^2(k^2+K_n{}^2)}. \tag{2.9.44}$$

This is *Van Kampen's dispersion relation for nonrelativistic particles*. In the absence of poles (bound states), e.g., for a repulsive interaction, it coincides with (2.3.11), and similar comments about the dependence on the radius of the scatterer may be applied to it.

### (d) Product Expansion

It follows from (2.9.42), (2.9.5), and (2.9.18) that the residue of $S(k)$ at a pole $k = iK_n$ on the positive imaginary axis is given by

$$\operatorname{res} S(k)\bigg|_{k=iK_n} = -i(b_n/K_n)\exp(2K_n a) = -ic_n \qquad (c_n > 0). \quad (2.9.45)$$

Thus, in the neighborhood of the pole, we have

$$S(k) \approx -ic_n/(k-iK_n)$$

and, in particular, for $k = iK$ on the imaginary axis,

$$S(iK) \approx c_n/(K_n - K) \qquad (c_n > 0), \qquad (2.9.46)$$

so that the real function $S(iK)$ changes from $+\infty$ to $-\infty$ across each pole, all the residues having the same sign.

It follows that there must be at least one zero $\bar{K}_n$ of $S(iK)$ between two consecutive poles $K_n$ and $K_{n+1}$. There may be others [cf. Section 2.10(*b*)], but let us single out one zero $\bar{K}_n$ between each pair of poles and let us consider the (possibly infinite) product

$$\mathscr{P}(k) = \prod_n \frac{[1+(k/iK_n)]}{[1-(k/iK_n)]}\frac{[1-(k/i\bar{K}_n)]}{[1+(k/i\bar{K}_n)]} = \prod_n (1+u_n), \qquad (2.9.47)$$

extended over all the poles on the positive imaginary axis, arranged in order of increasing modulus.

We have

$$\begin{aligned}\sum_n |u_n| &= 2|k| \sum_n \frac{|(1/K_n)-(1/\bar{K}_n)|}{|1-(k/iK_n)|\,|1+(k/i\bar{K}_n)|} \\ &\leqslant \frac{2|k|}{\min[|1-(k/iK_n)|\,|1+(k/i\bar{K}_n)|]}\sum_n\left(\frac{1}{K_n}-\frac{1}{K_{n+1}}\right) \\ &\leqslant \frac{2|k|}{\min[|1-(k/iK_n)|\,|1+(k/i\bar{K}_n)|]}\cdot\frac{1}{K_1}, \end{aligned} \qquad (2.9.48)$$

where min[...] denotes the minimum value of the quantity within the brackets.

Since the last member of (2.9.48) is bounded in any closed set not containing the poles, it follows from Theorem 2.5.1 that the product is absolutely and uniformly convergent in any such set, and it represents a regular analytic function in it. Thus $\mathscr{P}(k)$ is a meromorphic function, with the poles $iK_n$, $-i\overline{K}_n$. We have

$$\mathscr{P}(k) = \lim_{N\to\infty} \mathscr{P}_N(k) = \lim_{N\to\infty} \prod_{n=1}^{N} \frac{[1+(k/iK_n)][1-(k/i\overline{K}_n)]}{[1-(k/iK_n)][1+(k/i\overline{K}_n)]}. \quad (2.9.49)$$

Now let

$$\tilde{S}_a^{\,N}(k) = S_a(k)/\mathscr{P}_N(k) \quad (2.9.50)$$

and

$$f_N(k) = \exp[-i\tilde{S}_a^{\,N}(k)], \quad (2.9.51)$$

so that

$$|f_N(k)| = \exp[\operatorname{Im}\tilde{S}_a^{\,N}(k)]. \quad (2.9.52)$$

Clearly, $\tilde{S}_a^{\,N}(k)$ is regular in the first quadrant of the $k$-plane; it is just a phase factor on the real axis, so that $\operatorname{Im}\tilde{S}_a^{\,N}(k) \leqslant 1$ and $|f_N(k)| \leqslant e$ there; along the imaginary axis, $|f_N(k)| = 1$. Furthermore, as $|k| \to \infty$, we have $\mathscr{P}_N(k) \to 1$, so that $\operatorname{Im}\tilde{S}_a^{\,N}(k)$ and $|f_N(k)|$ remain bounded by (2.9.10). Thus it follows from Phragmén–Lindelöf's theorem applied to the first quadrant [5, p. 177] that

$$|f_N(k)| \leqslant e, \qquad \operatorname{Im}\tilde{S}_a^{\,N}(k) \leqslant 1 \qquad (0 \leqslant \arg k < \pi/2). \quad (2.9.53)$$

If we now let $N \to \infty$, it follows that the limiting function

$$\tilde{S}_a(k) = S_a(k)/\mathscr{P}(k), \quad (2.9.54)$$

which is regular in $I_+(k)$ (because the poles have been divided out), has $\operatorname{Im}\tilde{S}_a(k) \leqslant 1$ in the first quadrant. Therefore, by repeating the above derivation, we must find that it satisfies a dispersion relation equivalent[20] to (2.9.42), without the term involving the poles:

$$\frac{\tilde{S}_a(k)-S(0)}{k} = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\operatorname{Im}\tilde{S}_a(k')}{k'(k'-k)}\,dk' \qquad (\operatorname{Im} k > 0). \quad (2.9.55)$$

Since the right-hand side of (2.9.55) is bounded as $|k| \to \infty$, it follows that $\tilde{S}_a(k) = O(|k|)$. On the other hand, $|\tilde{S}_a(k)| = 1$ on the real axis. We can therefore apply the Phragmén–Lindelöf theorem 2.5.4 and conclude that

$$|\tilde{S}_a(k)| \leqslant 1 \qquad \text{in} \quad I_+, \quad (2.9.56)$$

which is to be compared with (2.3.6).

---

[20] The equivalence of the integrals in (2.9.42) and (2.9.52) follows from the symmetry relation, which implies $\operatorname{Im}\tilde{S}_a(-k') = -\operatorname{Im}\tilde{S}_a(k')$.

It follows that $\tilde{S}_a(k)$ has exactly the same analytic properties verified by $S_a(k)$ in the case of a classical field (Sections 2.2–2.4), so that we may apply Theorem 2.5.6 and conclude that

$$\begin{aligned}\tilde{S}(k) &= \tilde{S}_a(k)\,e^{-2ika}\\ &= \pm e^{-2ik\alpha}\prod_m \frac{[1-(k/iK_m)]}{[1+(k/iK_m)]}\\ &\quad\times \prod_n \frac{[1-(k/k_n{}^*)][1+(k/k_n)]}{[1-(k/k_n)][1+(k/k_n{}^*)]} \qquad (\alpha \leqslant a),\end{aligned} \tag{2.9.57}$$

where we have written the Blaschke factors in the product (2.5.34) in their explicit form (2.5.3), (2.5.4). In (2.9.57), $iK_m(K_m > 0)$ are the zeros of $S(k)$ (if any) not yet included in the zeros $i\bar{K}_n$ of (2.9.47).

Substituting (2.9.57) and (2.9.47) in (2.9.54), we finally obtain

$$\begin{aligned}S(k) &= \pm e^{-2ik\alpha}\prod_n \frac{[1-(k/k_n{}^*)][1+(k/k_n)]}{[1-(k/k_n)][1+(k/k_n{}^*)]}\\ &\quad\times\prod_p \frac{[1+(k/iK_p)]}{[1-(k/iK_p)]} \qquad (\alpha \leqslant a),\end{aligned} \tag{2.9.58}$$

where we have joined together all Blaschke factors corresponding to purely imaginary zeros and poles in the last product. This product runs over all the poles $iK_p$ on the imaginary axis, taken in order of increasing modulus: $K_p > 0$ corresponds to a pole on the positive imaginary axis, $K_p < 0$ to a negative imaginary pole. The first product runs over all complex poles $k_n$ in the *fourth quadrant*, in order of increasing modulus. As in (2.5.34), the plus sign will usually hold in (2.9.58), and it may be expected that $\alpha = a$ (for potential scattering, this will be proved in Section 5.5).

The expansion (2.9.58) is the *canonical product expansion of the S-function for nonrelativistic quantum scattering*. It differs from (2.5.34) only by the possible presence of poles on the positive imaginary axis.

Taking the logarithmic derivative of both sides of (2.9.58), we find, just as in (2.5.43),

$$\eta'(k) = -\alpha - \sum_n \operatorname{Im} k_n\left(\frac{1}{|k-k_n|^2} + \frac{1}{|k+k_n|^2}\right) - \sum_p \frac{K_p}{k^2+K_p{}^2}, \tag{2.9.59}$$

where we must note that the first sum is extended only over complex poles in the fourth quadrant. Although $\operatorname{Im} k_n < 0$, we can no longer conclude, as in (2.5.44), that $\eta'(k) \geqslant -a$, because we may have terms with $K_p > 0$ in the last sum if there are poles on the positive imaginary axis [cf. Section 2.11(b)].

### (e) Analytic Continuation and Stability

Van Kampen's dispersion relation (2.9.42) shows that, in contrast with the corresponding relation (2.3.12) for a classical field, the values of $S(k)$ in $I_+$ are not completely determined by its values on the real axis: one must know in addition the bound-state energies $-K_n{}^2/2$, as well as the residues of the $S$-function at the corresponding poles $iK_n$ (as will be seen in Section 2.12, these residues are related with the normalization factors for the bound-state wave functions).

This result may seem strange at first, because an analytic function is completely determined by its values on an arbitrarily small interval. In fact, Heisenberg had originally suggested [24] that it might be possible to determine the energies and normalization factors of bound states from the scattering data (phase shifts) by analytic continuation of the $S$-function.

However, even though the analytic continuation is determined by the values taken by an analytic function on the real axis (or even on an arbitrarily small interval), it is not determined in a constructive way; i.e., it is, in general, impossible to give an explicit formula for computing the function in terms of its boundary values on the real axis. This is because the analytic continuation does not depend on the boundary values in a continuous way: arbitrarily small changes in the boundary values can produce arbitrarily large changes in the analytic continuation.[21] For instance, one may add to the function a term of the form $\varepsilon \exp(-ik/\varepsilon)$ $(\varepsilon > 0)$, which may be made arbitrarily small on the real axis, but can become arbitrarily large in $I_+(k)$. The analytic continuation of a function in terms of its boundary values is *unstable* with respect to small changes in these values.

On the other hand, if an analytic function is regular in a region, its value at any interior point can be expressed explicitly in terms of its values along the *whole* boundary of that region (closed contour), by means of the Cauchy integral. This is precisely what happened in the case of (2.3.12): it was not sufficient to know that $S_a(k)$ is regular in $I_+$; we had to know in addition that it is bounded at infinity, so that the "half-circle at infinity" does not contribute to Cauchy's integral. Otherwise, one could not have excluded for example, an exponential factor of the kind mentioned above [cf. (1.6.29)].

Similarly, if the function has poles in $I_+$, one must know the poles and their residues in order to subtract them and apply Cauchy's integral theorem to the remaining (regular) function. This is essentially what was done in the derivation of (2.9.42).

The stability of analytic continuation is a crucial problem, because all measurements have finite precision so that, in practice, any set of data are only

[21] Cf. Van Kampen [16, 25]; also Allcock [26].

known to within some error bounds, and we must make sure that any changes in the boundary values within the allowed bounds do not lead to drastic changes in the analytic continuation (cf. Section 1.1).

This stability requirement is the same one that appears in the formulation of boundary value problems for partial differential equations.[22] It must be satisfied in order that the problem be "properly posed" or "reasonable." Thus, for an elliptic partial differential equation such as Laplace's equation, a boundary value problem in a given domain can only be properly posed if the boundary data are prescribed on the *whole* boundary. In that case, if two solutions differ by less than $\varepsilon$ in absolute value at all points of the boundary, they must also differ by less than $\varepsilon$ at all interior points, because the difference is also a potential function, and, therefore, takes on its maximum and minimum values at the boundary.

## 2.10. The $R$-Function

### (a) Wigner's Method

Wigner's approach [20, 29] to the problem of causality in nonrelativistic scattering is formulated in terms of a matrix first introduced by Eisenbud and Wigner [30] in the theory of nuclear reactions, the $R$-matrix. For $s$-wave scattering, to which we have to confine ourselves here, we have to deal only with one matrix element, which will be called the $R$-function.

Let us consider again the stationary solution (2.7.2) of Schrödinger's equation for $s$-wave scattering by a scatterer of radius $a$:

$$\psi(E,r,t) = \varphi(E,r,t)/r \qquad (r > a). \tag{2.10.1}$$

The $R$-function is defined as the inverse of the logarithmic derivative of $r\psi$, evaluated at $r = a$:

$$R(E) = \left[\varphi(E,r,t)\Big/\frac{\partial\varphi}{\partial r}(E,r,t)\right]_{r=a}. \tag{2.10.2}$$

It follows from (2.7.2) and (2.7.5) that

$$R(E) = \frac{1}{ik}\left[\frac{S_a(k)-1}{S_a(k)+1}\right], \tag{2.10.3}$$

where $S_a(k) = S(k)e^{2ika}$. The unitarity condition (2.7.20) implies

$$R(E) = R^*(E) \qquad (E > 0). \tag{2.10.4}$$

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[22] Cf. Courant and Hilbert [27, p. 227] and Petrovsky [28, p. 62].

Furthermore, by linearity (superposition principle), if

$$\left[\frac{\partial\varphi}{\partial r}(r,t)\right]_{r=a} = \int_0^\infty w(E)\,e^{-iEt}\,dE \qquad (2.10.5)$$

is the radial derivative of an arbitrary wave packet $\varphi(r,t)$ taken at $r=a$, we have, by (2.10.2),

$$\varphi(a,t) = \int_0^\infty R(E)\,w(E)\,e^{-iEt}\,dE. \qquad (2.10.6)$$

Note that the integrals are extended only over $E \geqslant 0$, as in (2.7.7)–(2.7.8).

Now let us again apply Van Kampen's causality condition (2.9.2). Taking into account (2.7.13), we can express this condition as follows:

$$i\int_{-\infty}^{0}\left[\varphi^*(a,t')\frac{\partial\varphi}{\partial r}(a,t') - \varphi(a,t')\frac{\partial\varphi^*}{\partial r}(a,t')\right]dt' \geqslant 0. \qquad (2.10.7)$$

Substituting $\partial\varphi/\partial r$ and $\varphi$ by (2.10.5) and (2.10.6), respectively, this becomes

$$i\int_{-\infty}^{0} dt' \int_0^\infty dE \int_0^\infty dE'\, e^{i(E-E')t'}\, w^*(E)\,[R^*(E)-R(E')]\,w(E') \geqslant 0.$$

Inverting the order of integration and integrating with respect to $t'$ by means of (2.9.3), we finally obtain, with the help of (2.10.4),

$$\int_0^\infty dE \int_0^\infty dE'\, w^*(E)\left[\frac{R(E)-R(E')}{E-E'}\right]w(E') \geqslant 0. \qquad (2.10.8)$$

This inequality is the expression of Van Kampen's causality condition in terms of the $R$-function. Here a principal value sign has been omitted, because, according to (2.10.3),

$$\lim_{E'\to E}\left[\frac{R(E)-R(E')}{E-E'}\right] = R'(E) \qquad (E>0); \qquad (2.10.9)$$

i.e., $R(E)$ is a real differentiable function for $E>0$, except possibly at isolated singular points where the denominator of (2.10.3) vanishes.

Now let $E_1, E_2, \ldots, E_n$ be any points of the positive real axis where $R(E)$ is regular, $n$ being any integer. Let us consider $n$ neighborhoods of these points,

$$|E-E_i| \leqslant \delta \qquad (i=1,2,\ldots,n), \qquad (2.10.10)$$

where $\delta$ is chosen to be so small that $R(E)\approx R(E_i)$ in the $i$th neighborhood. Let us choose for $w(E)$ a function that vanishes outside of the neighborhoods (2.10.10) and is given by $w(E)=w_i$ in the $i$th neighborhood, where $w_1, w_2, \ldots, w_n$ is any set of $n$ complex constants. Then, substituting these particular choices in (2.10.8) and dividing the resulting inequality by $4\delta^2$, we find, as a particular case of (2.10.8), that

$$\sum_{i=1}^{n}\sum_{j=1}^{n} w_i^* \left[\frac{R(E_i)-R(E_j)}{E_i-E_j}\right] w_j \geqslant 0, \tag{2.10.11}$$

*for any selection of the regular points* $E_i > 0$, *the complex constants* $w_i$, *and the integer* $n$. The terms with $i=j$ in this sum are given by (2.10.9).

Let us introduce the $n \times n$ matrix $M$ with off-diagonal elements

$$M_{ij} = \frac{R(E_i)-R(E_j)}{E_i-E_j} = M_{ji} \qquad (i \neq j), \tag{2.10.12}$$

and with diagonal elements

$$M_{ii} = R'(E_i). \tag{2.10.13}$$

If we think of $(w_1, w_2, \ldots, w_n)$ as components of a column vector $w$, condition (2.10.11) may be rewritten as

$$(w, Mw) = \sum_{i=1}^{n}\sum_{j=1}^{n} w_i^* M_{ij} w_j \geqslant 0; \tag{2.10.14}$$

i.e., *the matrix* $M$ *is positive semidefinite.*

In particular, taking $n = 1$, this implies

$$R'(E) \geqslant 0 \qquad (E > 0). \tag{2.10.15}$$

Similarly, taking $n = 2$, we see that it implies[23]

$$R'(E_1)\,R'(E_2) \geqslant \left[\frac{R(E_1)-R(E_2)}{E_1-E_2}\right]^2 \qquad (E_1 > 0, E_2 > 0). \tag{2.10.16}$$

According to a modified version of a theorem due to Loewner [32],[24] *the positive semidefiniteness of the* $n \times n$ *matrix* $M$ *defined by* (2.10.12), (2.10.13) *for any value of the integer* $n$ *and for any choice of the regular points* $E_i > 0$ *implies that* $R(E)$ *can be extended over the whole* $E$*-plane, except the negative real axis, as an analytic function, and that*

$$\operatorname{Im} R(E) \operatorname{Im} E > 0 \qquad (\operatorname{Im} E \neq 0). \tag{2.10.17}$$

*The singularities of* $R(E)$ *can only be located on the real axis, and the only possible singularities on the positive real axis are poles.* It then follows, as in Theorem D3 (Appendix D), that the poles must be simple and have negative residues.

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[23] More generally, for any order $n$, all principal minors of the real symmetric matrix $M$ must be nonnegative. Cf. Beckenbach and Bellman [31, p. 59].

[24] The modified version is due to Wigner and von Neumann [33].

However, the above assumptions do not allow us to draw any conclusions about the behavior of $R(E)$ on the negative real axis. If (2.10.14) were known to be valid also for points $E_i < 0\,(i = 1, 2, \ldots, n)$ in (2.10.12), it would be possible to conclude that $R(E)$ is meromorphic in the whole $E$-plane. However, due to the fact that the integrals (2.10.5), (2.10.6) are extended only over $E \geqslant 0$, the negative real axis has to be excluded.

This is the analog of the difficulty already met with in connection with Schützer and Tiomno's and Van Kampen's treatments. In order to overcome it, we again have to introduce an extra assumption. In view of the connection (2.10.3) between $R$ and $S$, it suffices to assume, as before, that $S(E)$ has a unique analytic continuation across $E < 0$, the only possible singularities being poles. It then follows from (2.10.3) that $R(E)$ *is a meromorphic function.* Together with (2.10.17), this implies that $R(E)$ *is an R-function,* as defined in Appendix D, where the general properties of this class of functions are derived. It follows that $R(E)$ has all these properties and, in particular, according to Theorem D5, it can be expanded in the form

$$R(E) = \alpha E + \beta + \sum_{n=-\infty}^{\infty} \gamma_n{}^2 \left( \frac{1}{\varepsilon_n - E} - \frac{1}{\varepsilon_n} \right) \qquad (\alpha \geqslant 0; \beta, \gamma_n, \varepsilon_n \text{ real}), \tag{2.10.18}$$

where $\varepsilon_n$ are the poles on the real axis, and

$$\sum_{n=-\infty}^{\infty} \gamma_n{}^2/\varepsilon_n{}^2 < \infty. \tag{2.10.19}$$

Actually, with a slight additional restriction on the definition of $R(E)$, the above expansion can be further simplified, by making use of the relation (2.10.3) between the $R$-function and the $S$-function. Let us assume that the actual radius of the scatterer is $a - \varepsilon$, $\varepsilon > 0$, and that the $R$-function is defined at the slightly larger radius $a$. The causality condition clearly remains valid a fortiori for this larger radius, but now we have more than just boundedness for $|S_a(k)|$ within the first quadrant:

$$|S_a(k)| \to 0 \qquad \text{as} \qquad |k| \to \infty \qquad (0 < \delta \leqslant \arg k \leqslant \pi/2 - \delta), \tag{2.10.20}$$

because $|S_{a-\varepsilon}(k)|$ is bounded (Section 2.9, property *iii*) and the extra factor $e^{2ik\varepsilon}$ tends to zero.

In particular, let us apply (2.10.20) along the direction $k = e^{i\pi/4} K$, corresponding to the positive imaginary axis of the $E$-plane: $E = i\varepsilon$. It then follows from (2.10.3) that

$$R(i\varepsilon) = O(\varepsilon^{-1/2}) \qquad (\varepsilon \to \infty). \tag{2.10.21}$$

On the other hand, by (2.10.18) and (2.10.19),

$$\frac{\operatorname{Im} R(i\varepsilon)}{\varepsilon} = \alpha + \sum_{n=-\infty}^{\infty} \frac{\gamma_n^{\,2}}{\varepsilon_n^{\,2}+\varepsilon^2} \to \alpha \qquad \text{as} \qquad \varepsilon \to \infty, \tag{2.10.22}$$

so that (2.10.21) implies

$$\alpha = 0. \tag{2.10.23}$$

It also follows that

$$\int_0^\infty \frac{\operatorname{Im} R(i\varepsilon)}{\varepsilon}\, d\varepsilon \geqslant \sum_{n=-N}^{M} \gamma_n^{\,2} \int_0^\infty \frac{d\varepsilon}{\varepsilon^2+\varepsilon_n^{\,2}} = \frac{\pi}{2} \sum_{n=-N}^{M} \frac{\gamma_n^{\,2}}{|\varepsilon_n|}$$

is finite, so that the series obtained by letting $N$ and $M$ tend to infinity is convergent. Thus we can rewrite (2.10.18) as

$$R(E) = \beta - \sum_{n=-\infty}^{\infty} \gamma_n^{\,2}/\varepsilon_n + \sum_{n=-\infty}^{\infty} \gamma_n^{\,2}/(\varepsilon_n - E)$$

and, since the left side as well as the last term on the right tend to zero as $E \to i\infty$, the constant term must be identically zero:

$$\sum_{n=-\infty}^{\infty} \gamma_n^{\,2}/\varepsilon_n = \beta. \tag{2.10.24}$$

Taking into account (2.10.23) and (2.10.24), the expansion (2.10.18) is simplified to

$$R(E) = \sum_{n=-\infty}^{\infty} \gamma_n^{\,2}/(\varepsilon_n - E). \tag{2.10.25}$$

For the validity of this expansion, it is sufficient to exclude the possibility that $S_a(e^{i\pi/4}K) \to -1$ as $K \to \infty$, for (2.10.3) then leads to (2.10.21). This is why we had to choose $a$ slightly larger than its minimum possible value. Otherwise, the excluded possibility might arise, as shown by the example

$$S(k) = (i-k)/(i+k), \tag{2.10.26}$$

which satisfies the causality condition with $a = 0$, and corresponds to scattering by a zero-range potential.[25] If we define $R(E)$ by (2.10.3) with $a = 0$, we find $R(E) = 1$, which agrees with (2.10.18), but not with (2.10.25).

### *(b) The Connection Between the R-Function and the S-Function*

The $R$-function and the $S$-function are related by (2.10.3), which may be inverted to give

$$S_a(k) = \frac{1 + ikR(E)}{1 - ikR(E)}. \tag{2.10.27}$$

[25] Cf. Newton [34].

On the other hand, we have seen that Wigner's condition (2.10.8), from which the analytic properties of $R$ have been derived, is equivalent to Van Kampen's causality condition (2.9.4). The equivalence may be verified directly by substituting in (2.10.8)

$$w(E) = -ikA_a(E)[1+S_a(E)]. \tag{2.10.28}$$

Since the same physical assumptions have been employed in the derivation of the analytic properties of $R$ and of $S$, the resulting properties should be equivalent. This indeed is true [35], as will now be shown.

According to Section 2.9, $S(k)$ has the following analytic properties:

i. $S(k)$ is a meromorphic function of $k$.

ii. We have

$$[S(k)]^{-1} = S(-k) = S^*(k^*). \tag{2.10.29}$$

iii. The poles of $S(k)$ lie either on the positive imaginary axis or in the lower half-plane. At a pole $k = iK_n (K_n > 0)$, we have

$$\operatorname{res} S(k)\Big|_{k=iK_n} = -ic_n \qquad (c_n > 0). \tag{2.10.30}$$

iv. In the first quadrant of the $k$-plane,

$$\operatorname{Im} S_a(k) \leqslant 1 \qquad (0 \leqslant \arg k < \pi/2). \tag{2.10.31}$$

v. $|S_a(k)|$ is bounded in

$$0 \leqslant \arg k \leqslant \pi/2 - \delta, \qquad \delta > 0.$$

Similarly, according to Section 2.10(a) and Appendix D, $R(E)$ has the following properties:

i′. $R(E)$ is a meromorphic function of $E$.

ii′. $R(E)$ is real on the real axis.

iii′. All the poles of $R(E)$ lie on the real axis.

iv′. We have

$$\operatorname{Im} R(E) \operatorname{Im} E > 0 \qquad (\operatorname{Im} E \neq 0). \tag{2.10.32}$$

v′. We have

$$R(E) = \sum_{n=-\infty}^{\infty} \gamma_n^{\,2}/(\varepsilon_n - E). \tag{2.10.33}$$

In v′, we have assumed that $R(E)$ is defined at a value of $a$ slightly larger than the actual radius of the scatterer, as has been discussed in connection with (2.10.25).

Properties i–v are not all independent, and neither are i′–v′. It will now be shown, with the help of (2.10.3) and (2.10.27), that *properties i–v for S are equivalent to properties i′–v′ for R.*

(A) THE PROPERTIES OF $S$ IMPLY THOSE OF $R$

It follows from i and (2.10.3) that $R$ is a meromorphic function of $k$. However, according to ii, $S_a(-k) = [S_a(k)]^{-1}$, so that $R(-k) = R(k)$. Thus $R$ is a single-valued function of $k^2$, which proves i′.

It follows from ii and (2.10.3) that

$$[R(E)]^* = R(E^*), \tag{2.10.34}$$

and ii′ is a particular case of this result.

To prove iii′, it is sufficient, by (2.10.3), to show that $S_a(k)+1$ has no zeros outside of the real and imaginary axes of the $k$-plane.

Let us consider the behavior of $S$ on the positive imaginary axis, i.e., $S(iK)$ as a function of $K$ for $K \geqslant 0$. We have already seen in Section 2.9(d) that there must be at least one zero $\bar{K}_p$ of $S(iK)$ between two consecutive poles $K_p$ and $K_{p+1}$, because $S(iK)$ goes from $+\infty$ to $-\infty$ as $K$ crosses a pole (for increasing $K$). However, this condition can also be met by having an *odd* number of zeros between two consecutive poles. Since $S$ starts out from 1 at $K = 0$, we see also that it either has no zeros or else has an *even* number of zeros before reaching the first pole. These various possibilities are illustrated in Fig. 2.3.

Since $S(-iK) = [S(iK)]^{-1}$, each positive imaginary zero $i\bar{K}_p$ is associated with a negative imaginary pole $-i\bar{K}_p$. Thus, if we order all purely imaginary poles of $S$ by increasing absolute value, we find first an even number of (or zero) negative imaginary poles, then one positive imaginary pole, then an odd

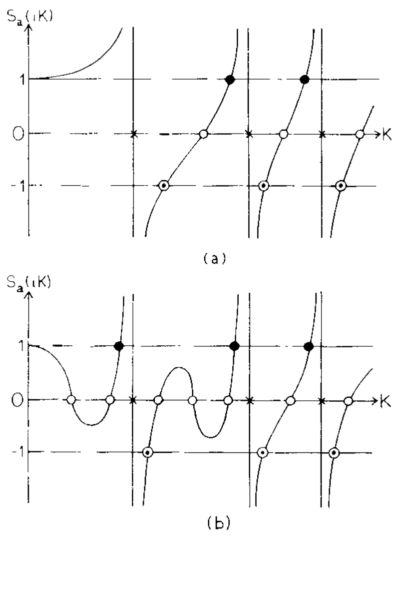


*FIG. 2.3.* Two possibilities for the behavior of $S_a(iK)$ as a function of $K$ ($\times$, poles of $S$; O, zeros of $S$; $\odot$, poles of $R$; $\bullet$, zeros of $R$).

number of negative imaginary poles, and from then on single positive imaginary poles and odd numbers of negative imaginary ones continue to alternate.[26]

According to (2.9.49), (2.9.54), and (2.9.57), we can write

$$S_a(k) = e^{2i\varepsilon k}\mathscr{P}\mathscr{M}\mathscr{N} \qquad (\varepsilon \geqslant 0), \tag{2.10.35}$$

where

$$\mathscr{P}(k) = \prod_p \frac{[1+(k/iK_p)]}{[1-(k/iK_p)]}\frac{[1-(k/i\bar{K}_p)]}{[1+(k/i\bar{K}_p)]} \qquad (K_p > 0, \bar{K}_p > 0) \tag{2.10.36}$$

is extended over all positive imaginary poles $iK_p$, with $K_p < \bar{K}_p < K_{p+1}$; if there is more than one zero between $iK_p$ and $iK_{p+1}$, $i\bar{K}_p$ is arbitrarily chosen as one of them, and the remaining zeros $iK_m$ (of which there is an even number between $iK_p$ and $iK_{p+1}$) are lumped together in the product

$$\mathscr{M}(k) = \prod_m \frac{[1-(k/iK_m)]}{[1+(k/iK_m)]}. \tag{2.10.37}$$

The remaining product

$$\mathscr{N}(k) = \prod_n \frac{[1-(k/k_n{}^*)]}{[1-(k/k_n)]}\frac{[1+(k/k_n)]}{[1+(k/k_n{}^*)]} \tag{2.10.38}$$

is extended over all remaining (complex) poles in $I_-(k)$. We have also assumed, for definiteness, that $S(0) = 1$.

We can now prove that all the zeros of $S_a(k)+1$ lie on the real and imaginary axes. For this purpose, we shall consider three cases:

*Case a. Finite number of poles*, $\varepsilon = 0$. Let $P$, $M$, and $N$ be the number of terms of $\mathscr{P}$, $\mathscr{M}$, and $\mathscr{N}$, respectively, so that $2P$, $M$, and $2N$ are the numbers of Blaschke factors they contain. According to the above discussion, $M$ is necessarily even. Thus $S_a(k)+1$ is a rational function and approaches 2 as $|k| \to \infty$. Therefore, by (2.5.30), it has the same number of poles and zeros; i.e., it has exactly $2P+M+2N$ zeros.

At least $2P$ of these zeros are on the imaginary axis. In fact, since $S_a(k)$ changes from $\infty$ to $-\infty$ across each positive imaginary pole, it must take the value $-1$ at least once between two consecutive positive imaginary poles, i.e., at least $P$ times on the positive imaginary axis (cf. Fig. 2.3). Since $S_a(-iK) = [S_a(iK)]^{-1}$, there are at least $P$ corresponding negative imaginary zeros of $S_a(k)+1$.

On the real axis, $S_a(k)$ is a phase factor. The phase of each Blaschke factor $[1-(k/\kappa^*)]/[1-(k/\kappa)]$ changes by $2\pi$ along the real axis if $\kappa \in I_-$, and by

[26] Cf. Wigner [36].

$-2\pi$ if $\kappa \in I_+$. Thus the phase of $S_a(k)$ changes by $(-P+P+M+2N)2\pi$ along the real axis, so that $S_a(k)$ must take the value $-1$ at least $M+2N$ times on the real axis. This exhausts the total number of roots, so that there can be none outside of the real and imaginary axes.

We also conclude in this case that $S_a(k)+1$ has exactly $2P$ zeros on the imaginary axis, one and only one between each pair of consecutive poles. Also, on the real axis, with $S_a(k) = \exp[2i\eta_a(k)]$, the phase $2\eta_a(k)$ goes through each multiple of $\pi$ between $-[N+(M/2)]\pi$ and $[N+(M/2)]\pi$ once and only once, and $S_a(k)+1$ has exactly $M+2N$ real zeros.

*Case b. Finite number of poles, $\varepsilon > 0$.* In this case, let us consider a rectangle, centered at the origin of the $k$-plane, and with sides much larger than the maximum modulus of all the poles, so that all poles fall inside it. Let the vertical sides intersect the real axis at $\pm\pi h/\varepsilon$, where $h$ is some large integer. For a sufficiently large rectangle, $\mathscr{P}\mathscr{M}\mathscr{N} \approx 1$ in (2.10.35), so that

$$S_a(k) + 1 \approx e^{2i\varepsilon k} + 1 = 2e^{i\varepsilon k}\cos(\varepsilon k) \tag{2.10.39}$$

on the boundary.

The last member of (2.10.39) has $2h$ zeros within the rectangle, all located on the real axis; thus $\Delta \arg[S_a(k)+1] = 2h\cdot 2\pi$ around the rectangle [cf. (2.5.30)], implying the existence of $2h$ new zeros of $S_a(k)+1$ within the rectangle. However, according to (2.10.35) and the above discussion of Case a, the phase of $S_a(k)$ now changes by $(M+2N+2h)2\pi$ from $k = -\pi h/\varepsilon$ to $k = \pi h/\varepsilon$, so that the $2h$ new points where it takes the value $-1$ all lie on the real axis.

*Case c. Infinite number of poles, $\varepsilon > 0$.* In this case we can apply the following theorem [5, p. 119]:

HURWITZ'S THEOREM. *Let $\{f_n(z)\}$ be a sequence of functions, each of them analytic in a region $D$ bounded by a simple closed contour, and let $f_n(z) \to f(z)$ uniformly in $D$, where $f(z)$ is not identically zero. Then an interior point $z_0$ of $D$ is a zero of $f(z)$ if, and only if, it is a limit point of the set of zeros of the functions $f_n(z)$ (points which are zeros for an infinity of values of $n$ are counted as limit points).*

Since the product representation (2.10.35) converges uniformly in any bounded region not containing a pole, it follows from Hurwitz's theorem and Case b that the zeros of the limiting function $S_a(k)+1$ must also lie on the real or imaginary axis. This completes the proof of property iii′.

To prove iv′, let us show first that the poles of $R(E)$ are all simple and have negative residues. According to (2.10.3), the residue of $R(E)$ at a pole $E = E_0 = k_0{}^2/2$ is given by

$$r_0 = \operatorname{res} R(E)\Big|_{E=E_0} = 2i[(dS_a/dk)_{k=k_0}]^{-1}. \tag{2.10.40}$$

Let us begin again by considering Case a above [$\varepsilon = 0$ and finite number of poles in (2.10.35)]. Let $k = iK_0$ be a pole of $R(E)$ on the positive imaginary axis, i.e., a root of $S_a(iK_0) = -1$. Then, as is clear from Fig. 2.3, we must have

$$[dS_a(iK)/dK]_{K=K_0} > 0, \tag{2.10.41}$$

because otherwise there would be at least three zeros of $S_a(iK)+1$ between two consecutive poles of $S_a$, whereas, as we have seen in the proof of iii′, only one should exist. It follows from (2.10.40) and (2.10.41) that $r_0 < 0$ in this case.

Similarly, at a point $k_0$ on the real axis where $S_a(k_0)+1 = 0$, we must have

$$\begin{aligned} i\left(\frac{dS_a}{dk}\right)_{k=k_0} &= i\left(\frac{d}{dk}\exp 2i\eta_a(k)\right)_{k=k_0} \\ &= -2\,S_a(k_0)\left(\frac{d\eta_a}{dk}\right)_{k=k_0} = 2\left(\frac{d\eta_a}{dk}\right)_{k=k_0} > 0, \end{aligned} \tag{2.10.42}$$

because otherwise $\eta_a(k)$ would go through $n\pi$ (where $\eta_a(k_0) = n\pi$) more than once, contrary to the result obtained in the proof of iii′. Thus $r_0 < 0$ also in this case.

Near a pole $E = E_0$, with $E - E_0 = \rho e^{i\varphi}$, we have [cf. (D8)]

$$\operatorname{Im} R(E) \approx -r_0\rho^{-1}\sin\varphi. \tag{2.10.43}$$

Thus, if we consider a small semicircle in $I_+(E)$ above each pole, it follows from (2.10.43) and the fact that $r_0 < 0$ that $\operatorname{Im} R \geqslant 0$ on each semicircle.

On the other hand, in Case a, it follows from (2.10.35) that $S_a(k) - 1 = O(k^{-1})$ for $|k| \to \infty$. By (2.10.3), this implies $R(E) = O(E^{-1})$, so that $\operatorname{Im} R \to 0$ at infinity. Taking into account property ii′, it follows that $\operatorname{Im} R \geqslant 0$ along a contour consisting of the real axis indented by semicircles in $I_+$ at the poles of $R(E)$ and closed by a semicircle at infinity in $I_+$. Since $R(E)$ is holomorphic within the above contour, $\operatorname{Im} R > 0$ for $\operatorname{Im} E > 0$.

In Case b, with $\varepsilon > 0$, there is an infinite number of poles of $R(E)$ on the real axis, but again, by considering a large rectangle, the same result follows. Finally, if the number of poles of $S_a(k)$ tends to infinity, as in Case c, $\operatorname{Im} R$ cannot become negative, and it cannot vanish for $\operatorname{Im} E > 0$, because this would correspond to a minimum. This completes the proof of property iv′.

It follows from properties i′, ii′, and iv′ that $R(E)$ is an $R$-function, so that, according to Appendix D, the expansion (2.10.18) is valid. However, as we have seen in (2.10.25), if we define $R(E)$ at a radius slightly larger than the actual radius of the scatterer, property v′ is valid.

### (B) The Properties of $R$ Imply Those of $S$

Properties i and ii follow immediately from (2.10.27) and i′–v′ [together with their consequence (2.10.34)]. To prove [15] property iii, let $k = k' + iK$.

Then, by (2.10.27), the poles of $S$ are the roots of

$$1 = i(k' + iK)(\operatorname{Re} R + i \operatorname{Im} R);$$

i.e.,

$$k' \operatorname{Re} R = K \operatorname{Im} R, \tag{2.10.44}$$

$$1 = -(K \operatorname{Re} R + k' \operatorname{Im} R). \tag{2.10.45}$$

If $k' \neq 0$, we can eliminate $\operatorname{Re} R$ and obtain

$$K \operatorname{Im} R = -k' K/|k|^2 = -\operatorname{Im} E/|k|^2. \tag{2.10.46}$$

Since $\operatorname{Im} R$ and $\operatorname{Im} E$ have the same sign, it follows that $K < 0$ if $k' \neq 0$, which is the desired result.

To prove property iv, let us assume, to begin with, that the partial-fraction expansion (2.10.33) of $R(E)$ contains only a finite number of terms. Then $R$ and $S_a$ are rational fractions and $R(E) = O(E^{-1})$ for $|E| \to \infty$, so that

$$S_a(k) \to 1, \qquad \operatorname{Im} S_a(k) \to 0 \qquad \text{as} \quad |k| \to \infty. \tag{2.10.47}$$

On the other hand, by property ii, $\operatorname{Im} S_a(k) = 0$ at any regular point of the imaginary axis. At a pole $k_0 = iK_0$ of $S_a(k)$, we have, by (2.10.27),

$$1 - ik_0 R(E_0) = 0,$$

where $E_0 = -K_0{}^2/2$, and

$$-i \operatorname{res} S_a(k) = [E_0 R'(E_0)]^{-1} = d_0 < 0, \tag{2.10.48}$$

because, by (2.10.33) [cf. also (D5)]

$$R'(E_0) > 0. \tag{2.10.49}$$

Thus, near the pole, with $k - iK_0 = \rho e^{i\varphi}$, we have

$$\operatorname{Im} S_a(k) \approx d_0 \rho^{-1} \cos \varphi,$$

so that we can draw a semicircle around the pole, in the first quadrant of the $k$-plane, on which $\operatorname{Im} S_a(k) \leqslant 0$.

Consider now a contour consisting of the positive imaginary axis, indented by semicircles around the poles, the positive real axis and the first quadrant of a circle at infinity. Then, as we have just seen, $\operatorname{Im} S_a(k) \leqslant 0$ on the contour, except along the real axis, where

$$\operatorname{Im} S_a(k) \leqslant |S_a(k)|^2 = 1. \tag{2.10.50}$$

Since $S_a(k)$ is holomorphic within the contour, it follows that $\operatorname{Im} S_a(k) \leqslant 1$ in the whole first quadrant.

To prove property v, let us note first that, as we have seen in connection with (2.9.11), it is equivalent to the condition

$$\operatorname{Im}[S_{a+\varepsilon}(k)] = \operatorname{Im}[e^{2ik\varepsilon} S_a(k)] \leqslant 1$$
$$\text{for any} \quad \varepsilon \geqslant 0, \quad 0 \leqslant \arg k < \pi/2, \tag{2.10.51}$$

so that it suffices to prove (2.10.51).

It follows from properties i–iv that

$$\frac{S_a(k) - S_a(0)}{k} + 2k \sum_n \frac{b_n}{K_n^2(k^2+K_n^2)}$$
$$= -\frac{A}{2}k + \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\operatorname{Im} S_a(k')}{k'(k'-k)}\, dk' \qquad (\operatorname{Im} k > 0). \tag{2.10.52}$$

In fact, in virtue of the symmetry relation $\operatorname{Im} S_a(-k') = -\operatorname{Im} S_a(k')$, which follows from property ii, (2.10.51) is equivalent to (2.9.42), except for the term $-Ak/2$. As we have seen in Section 2.9, only properties i–iv were employed in the derivation of (2.9.42), except for the omission of the term $AE$ in the original representation (2.9.14), which followed from property v [cf. (2.9.15)]; this corresponds to the term $-Ak/2$, which therefore has been kept in (2.10.52).

The right-hand side of (2.10.52) is regular in $I_+(k)$ and is $O(k)$ as $|k| \to \infty$. On the real axis, it is of order less than $|k|$, because this is clearly true for the left-hand side. Therefore, by Phragmén–Lindelöf's theorem 2.5.4, it is of order less than $|k|$ in the whole upper half-plane. It follows that

$$S_a(k) = o(|k|^2) \qquad \text{as} \quad |k| \to \infty, \quad 0 \leqslant \arg k \leqslant \pi/2 - \delta, \tag{2.10.53}$$

because the sum on the left-hand side tends to zero as $|k| \to \infty$ with $0 \leqslant \arg k \leqslant \pi/2 - \delta$. Thus we must have $A = 0$ in (2.10.52).

Consider now the function

$$S_{a+\varepsilon}(k) = e^{2ik\varepsilon} S_a(k) \qquad (\varepsilon \geqslant 0). \tag{2.10.54}$$

Along the real axis, just as in (2.10.50), we have $\operatorname{Im} S_{a+\varepsilon}(k) \leqslant 1$. Along the positive imaginary axis, with $k = iK$, we have, by (2.10.31),

$$\operatorname{Im} S_{a+\varepsilon}(iK) = e^{-2K\varepsilon} \operatorname{Im} S_a(iK) \leqslant \operatorname{Im} S_a(iK) \leqslant 1.$$

On the other hand, since

$$|S_{a+\varepsilon}(k)| \leqslant |S_a(k)| \qquad (k \in I_+),$$

it follows from (2.10.53) that

$$S_{a+\varepsilon}(k) = o(k^2) \qquad \text{as} \quad |k| \to \infty, \quad 0 \leqslant \arg k \leqslant \pi/2 - \delta. \tag{2.10.55}$$

The function

$$F(k) = \exp[-iS_{a+\varepsilon}(k)]$$

is regular in the first quadrant of the $k$-plane and, according to the above result for $\operatorname{Im} S_{a+\varepsilon}$, we have $|F(k)| \leqslant e$ along its boundaries. It then follows from (2.10.55) and Phragmén–Lindelöf's theorem applied to the first quadrant, as in (2.9.53), that $|F(k)| \leqslant e$ in the whole first quadrant, which is equivalent to (2.10.51). This completes the proof of property v and of the complete equivalence between the properties of the $R$-function and those of the $S$-function.

## 2.11. Wigner's Causal Inequality

### *(a) Derivation*

As we have noted following (2.9.59), the inequality (2.5.44) need not be verified for nonrelativistic quantum scattering. However, as was shown by Wigner [37], one can still derive a related inequality for the derivative of the phase shift with respect to $k$. According to Wigner, this inequality has an intuitive interpretation in terms of causality. We shall first give the derivation, postponing till later a discussion of the physical interpretation.

The starting point is the inequality[27]

$$dR/dE \geqslant 0 \qquad (\text{real } E), \tag{2.11.1}$$

which follows from the properties of the $R$-function [cf. (D5)]. Let us employ (2.10.3) to find the implications of this inequality for $S_a(k)$, where $k$ is taken to be real, so that

$$S_a(k) = \exp[2i\eta_a(k)], \tag{2.11.2}$$

where [cf. (2.2.15)]

$$\eta_a(k) = \eta(k) + ka. \tag{2.11.3}$$

Differentiating (2.10.3) with respect to $k$, we find

$$\eta_a{}'(k) = \sin(2\eta_a)/2k + E\cos^2\eta_a\, dR/dE, \tag{2.11.4}$$

so that, by (2.11.1) and (2.11.3), we have (with $k > 0$)

$$\eta'(k) \geqslant -a + \sin[2(\eta + ka)]/2k \geqslant -[a + (1/2k)]. \tag{2.11.5}$$

This is *Wigner's causal inequality* for $s$-waves.

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[27] We have to write $\geqslant 0$ rather than $>0$ as in (D5), because $R(E)$ may be a constant (this was excluded in Appendix D). An example is an impenetrable sphere, for which $R(E) = 0$ [cf. (2.3.8) and (2.10.3)].

### (b) The Inequality of Goebel, Karplus, and Ruderman

An inequality that is closely related to the above one was derived by Goebel *et al.* [38]. It is based upon the canonical product expansion (2.9.58). Let us distinguish[28] between positive imaginary (bound state) poles $iK_b (K_b > 0)$ and negative imaginary ones $iK_m (K_m < 0)$. Then, again taking $k > 0$, we can write (2.9.58) in the form

$$\eta_\alpha(k) = \sum_n \eta_n(k) + \sum_m \eta_m(k) + \sum_b \eta_b(k), \tag{2.11.6}$$

where $\eta_\alpha(k) = \eta(k) + k\alpha$ [cf. (2.11.3)],

$$\eta_n = \arg[(k+k_n)(k-k_n^*)], \tag{2.11.7}$$

$$\eta_m = \arg(k+iK_m) \qquad (K_m < 0), \tag{2.11.8}$$

$$\eta_b = \arg(k+iK_b) \qquad (K_b > 0), \tag{2.11.9}$$

and the first sum in (2.11.6) is extended over all complex poles $k_n$ in the fourth quadrant.

In terms of these phase angles, (2.9.59) can be rewritten as

$$\begin{aligned}\eta_\alpha'(k) + \sum_b \frac{K_b}{k^2+K_b^2} &= \frac{1}{2k}\left[\sum_n \left|\frac{k^2+|k_n|^2}{k^2-|k_n|^2}\right| |\sin(2\eta_n)| + \sum_m |\sin(2\eta_m)|\right] \\ &\geqslant \frac{1}{2k}\left[\sum_n |\sin(2\eta_n)| + \sum_m |\sin(2\eta_m)|\right].\end{aligned} \tag{2.11.10}$$

Since $|\sin(x+y)| \leqslant |\sin x| + |\sin y|$, it follows that

$$\sum_n |\sin(2\eta_n)| + \sum_m |\sin(2\eta_m)| \geqslant \left|\sin\left[2\left(\eta_\alpha - \sum_b \eta_b\right)\right]\right|, \tag{2.11.11}$$

where we have made use of (2.11.6).

Combining (2.11.10) and (2.11.11), we get (note that $\alpha \leqslant a$)

$$\eta'(k) \geqslant -a + \left[\left|\sin\left(2\eta_\alpha - 2\sum_b \eta_b\right)\right| \Big/ 2k\right] - \sum_b K_b/(k^2+K_b^2), \tag{2.11.12}$$

which is the inequality of Goebel, Karplus, and Ruderman.

In particular, if there are no bound states, i.e., no poles $iK_b$ with $K_b > 0$, this becomes

$$\eta'(k) \geqslant -a + |\sin(2\eta_\alpha)|/2k \geqslant -a, \tag{2.11.13}$$

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[28] It is assumed that the corresponding Blaschke products are separately convergent, as would happen, e.g., if the total number of bound states is finite.

which is a refinement of Wigner's inequality (2.11.5) for this case, since the sinusoidal term always appears with positive sign. The last member of (2.11.13) corresponds to the result already obtained in (2.5.44).

The result (2.11.12) also makes it clear that what spoils the lower bound $\eta'(k) \geqslant -a$ is the presence of bound states. The corresponding contribution [last term of (2.11.12)] is negligible for $k \to \infty$ [as is the term $1/(2k)$ in (2.11.5)], but it can become quite large as $k \to 0$, especially if there are bound states with binding energy very close to zero.

For instance, if there is only one bound state, associated with the pole $k = iK_b$, (2.11.12) yields

$$\eta'(k) \geqslant -(a+1/K_b), \tag{2.11.14}$$

which can attain large negative values as $K_b \to 0$.

### *(c) Physical Interpretation: Time Delay*

According to Eisenbud [39], the derivative of the phase shift with respect to the energy represents the time delay that the incident wave packet undergoes in the scattering process.

To see this, let us once again consider the incoming and outgoing wave packets, given by (2.7.7) and (2.7.8), respectively,

$$\varphi_{\text{in}}(r,t) = \int_0^\infty A(E)\, e^{-ikr-iEt}\, dE, \tag{2.11.15}$$

$$\begin{aligned} \varphi_{\text{out}}(r,t) &= -\int_0^\infty S(E)\, A(E)\, e^{ikr-iEt}\, dE \\ &= -\int_0^\infty A(E)\, e^{i(kr+2\eta-Et)}\, dE. \end{aligned} \tag{2.11.16}$$

Let us assume that $A(E)$ corresponds to a narrow energy spectrum, centered upon some energy $E_0$, so that it only takes appreciable values for

$$|E-E_0| \lesssim \Delta E \qquad (\Delta E \ll E_0). \tag{2.11.17}$$

For large $|t|$, the integrand in (2.11.15) or (2.11.16) will be a rapidly oscillating function. According to the principle of stationary phase,[29] the dominant contribution to the integral arises from the vicinity of those points at which the phase $\chi(E)$ of the integrand is stationary, i.e., $\chi'(E) = 0$. This corresponds to the condition of "most constructive" interference among neighboring $E$ contributions; at other points, contributions tend to cancel out by destructive interference.

[29] See, for example, Erdélyi [40, p. 51].

Let

$$A(E) = |A(E)|\, e^{i\alpha(E)}. \tag{2.11.18}$$

Then, the stationary-phase point in (2.11.15) is given by

$$\alpha'(E) - r\, dk/dE - t = 0;$$

i.e., $k = r/[\alpha'(E) - t]$. The "center" $r_{\text{in}}$ of the incoming wave packet will be at the point where the corresponding value of $E$ coincides with $E_0$, the energy at which $|A(E)|$ is maximum:

$$r_{\text{in}} = -k_0[t - \alpha'(E_0)], \tag{2.11.19}$$

where $E_0 = k_0{}^2/2$. Thus, as expected, the center of the incoming wave packet moves inwards with "velocity" $k_0$.

Similarly, from (2.11.16), we find, for the center of the outgoing wave packet,

$$r_{\text{out}} = k_0[t - \alpha'(E_0) - 2\eta'(E_0)]. \tag{2.11.20}$$

Thus the presence of the scatterer gives rise to a time delay (or advance, depending on the sign of the derivative) given by[30]

$$\Delta t = 2\eta'(E_0). \tag{2.11.21}$$

The center of the outgoing wave packet is delayed by this amount relative to its time of passage at a given point in the absence of the scatterer. The corresponding spacial retardation is

$$\Delta r = k_0 \Delta t = 2(d\eta/dk)_{k_0}. \tag{2.11.22}$$

Wigner [37] has employed this result to give a physical interpretation of the inequality (2.11.5) in terms of causality. The incident wave packet can be captured by the scatterer and retained for an arbitrarily long time, so that there is no upper bound for the time delay. However, causality does not allow an arbitrarily large *negative* delay (time advance). Classically, the maximum allowed time advance is $-2a/k_0$ ($\Delta r \geqslant -2a$), in agreement with (2.5.44). The additional term $1/k$ in (2.11.5), which is of the order of a wavelength, would arise from the wave nature of matter.

The relation (2.11.22) would also lead to a simple physical interpretation of the qualitative energy dependence of $\eta$. At energies for which the incident particle hardly enters the scatterer, the "retardation" will be close to $-2a$, whereas it will assume large positive values close to resonances, where the incident particle is captured and retained for some time by the scatterer. A more detailed discussion of this point will be given in Chapter 4.

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[30] In conventional units, $\Delta t = 2\hbar\, d\eta/dE$ (here we have taken $\hbar = 1$).

It should be emphasized that the above discussion is mainly qualitative, and the concept of time delay should be used with considerable care. In the first place, the shape of a quantum-mechanical wave packet can change considerably during its propagation, and the concept of "center" may accordingly lose much of its significance. Secondly, in order that the expression (2.11.21) may be applied, it is necessary that the energy spectrum $|A(E)|^2$ shall vanish sufficiently rapidly outside of its width, given by (2.11.17), and that $\eta'(E)$ remains sufficiently constant within this width. This requirement becomes particularly stringent near sharp resonances, where $\eta(E)$ is rapidly varying (cf. Chapter 4).

An alternative definition of the time delay has been proposed by Smith [41], who applied it to stationary states. It has been extended by Goldberger and Watson [42; 43, p. 485] to the scattering of wave packets, with the special assumption, however, that the Schrödinger equation is valid also within the interaction region. The following treatment, in keeping with our approach so far (cf. Section 2.1), does not depend on this assumption.[30a]

Let the incoming wave packet (2.11.15) be so normalized as to represent one incident particle for $t \to -\infty$, as in Section 2.9:

$$\lim_{t\to-\infty}(\psi_{\text{in}}(r,t),\psi_{\text{in}}(r,t)) = \lim_{t\to-\infty}\left(4\pi\int_r^\infty |\varphi_{\text{in}}(r',t)|^2\,dr'\right) = 1. \tag{2.11.23}$$

Since the left-hand side also represents the total incoming probability flux, integrated over all times, it follows from (2.7.19) that the normalization condition is

$$8\pi^2\int_0^\infty k|A(E)|^2\,dE = 1. \tag{2.11.24}$$

This can also be verified directly from (2.11.23) and (2.11.15), which yield

$$(\psi_{\text{in}},\psi_{\text{in}}) = 4\pi\int_r^\infty dr'\int_0^\infty dE'\int_0^\infty dE\,A^*(E')\,A(E)\,e^{i(k'-k)r'}e^{i(E'-E)t}.$$

Performing the integration with respect to $r'$, with the help of (A11.5)–(A11.7), we find

$$(\psi_{\text{in}},\psi_{\text{in}}) = 4\pi i\int_0^\infty dE'\int_0^\infty dE\,A^*(E')\,A(E)\frac{(k+k')}{2}e^{i(k'-k)r} \\ \times\frac{e^{i(E'-E)t}}{E'-E+i0}. \tag{2.11.25}$$

[30a] For an extension of this treatment to the scattering of plane wave packets, see [43a].

On the other hand, (A11.5)–(A11.7) also give

$$\lim_{t\to-\infty}\left(\frac{e^{ixt}}{x+i0}\right) = -i\lim_{t\to-\infty}\int_t^\infty e^{iux}\,du = -2\pi i\,\delta(x).$$

Substituting in (2.11.25) and performing the integration with respect to $E'$, we are again led to (2.11.24). Note also that the result is independent of $r$, as it should be.

With this normalization, as we have seen in Section 2.7,

$$P_i(r,t) = 1 - P_e(r,t) = 1 - 4\pi\int_r^\infty |\varphi(r',t)|^2\,dr' \qquad (2.11.26)$$

represents the probability, at time $t$, to find the particle inside the sphere of radius $r$ centered at the origin. The total (average) time spent by the particle inside this sphere is then given by

$$T(r) = \int_{-\infty}^\infty P_i(r,t)\,dt. \qquad (2.11.27)$$

The time delay $\Delta t$ due to the interaction is now defined as the difference between the time $T(r)$ spent within a sphere of radius $r \geqslant a$ and the corresponding time $T_0(r)$ in the absence of interaction:

$$\begin{aligned}\Delta t = T(r) - T_0(r) &= \int_{-\infty}^\infty [P_i(r,t) - P_{i,0}(r,t)]\,dt\\ &= \int_{-\infty}^\infty [P_{e,0}(r,t) - P_e(r,t)]\,dt\\ &= 4\pi\int_{-\infty}^\infty dt\int_r^\infty [|\varphi_0(r',t)|^2 - |\varphi(r,t)|^2]\,dr' \qquad (r\geqslant a),\end{aligned} \qquad (2.11.28)$$

where the index 0 denotes quantities in the absence of interaction. We see that no extra assumptions about the interaction beyond those already made in Section 2.7 (in particular, that it conserves probability) are necessary for this definition, and that it only involves quantities defined outside of the scatterer.

Substituting $\varphi = \varphi_{\text{in}} + \varphi_{\text{out}}$, where $\varphi_{\text{in}}$ and $\varphi_{\text{out}}$ are given by (2.11.15)–(2.11.16), and $\varphi_0$ by the same expressions taking $S(E) = 1$, we find that (2.11.28) becomes

$$\begin{aligned}\Delta t = 4\pi\int_{-\infty}^\infty dt\int_r^\infty dr'\int_0^\infty dE'\int_0^\infty dE\,A^*(E')\,A(E)\,e^{i(E'-E)t}\\ \times\{[1-S^*(E')S(E)]\,e^{-i(k'-k)r'} - [1-S(E)]\,e^{i(k'+k)r'}\\ -[1-S^*(E')]\,e^{-i(k'+k)r'}\}.\end{aligned}$$

Performing the integration with respect to $r'$, we find, just as in (2.11.25),

$$\Delta t = 4\pi i \int_{-\infty}^{\infty} dt \int_0^{\infty} dE' \int_0^{\infty} dE\, A^*(E')A(E)\,e^{i(E'-E)t}$$
$$\times \left\{ \frac{[1-S^*(E')S(E)]}{k-k'+i0} e^{i(k-k')r} - \frac{[1-S(E)]}{k+k'+i0} e^{i(k+k')r} \right.$$
$$\left. + \frac{[1-S^*(E')]}{k+k'-i0} e^{-i(k+k')r} \right\}. \tag{2.11.29}$$

According to (A11.5), we may replace

$$\frac{1}{k \pm k' \pm i0} \to P\frac{1}{k \pm k'} \tag{2.11.30}$$

because the $\delta$-function contributions all vanish, due either to (2.7.20) or to the fact that $k, k'$ are nonnegative.

The integral over $t$ is going to bring in another $\delta$-function, $\delta(E-E')$, so that it suffices to consider the behavior of the integrand as $E \to E'$:

$$[1-S^*(E')S(E)]\,P\left(\frac{1}{k-k'}\right)$$
$$= \frac{k+k'}{2}\{1-e^{2i[\eta(E)-\eta(E')]}\}\,P\left(\frac{1}{E-E'}\right)$$
$$= \frac{k+k'}{2}\left\{-2i(E-E')\frac{d\eta}{dE} + O[(E-E')^2]\right\}P\left(\frac{1}{E-E'}\right)$$
$$= -2ik\, d\eta/dE + O(E-E'). \tag{2.11.31}$$

We may now proceed to integrate over $t$ and to integrate the resulting $\delta$-function over $E'$. This leads to

$$\Delta t = 8\pi^2 \int_0^{\infty} dE |A(E)|^2 \left\{ 2k\frac{d\eta}{dE} + \frac{\sin(2kr) - \sin[2(kr+\eta)]}{k} \right\}. \tag{2.11.32}$$

The last term in the expression within curly brackets is an oscillating function of $r$. The average time delay due to the interaction should be independent of $r$, provided that $r \geqslant a$. Thus, averaging over $r$ to eliminate the oscillating term, we get

$$\langle \Delta t \rangle = 8\pi^2 \int_0^{\infty} k|A(E)|^2 \cdot 2\frac{d\eta}{dE}\, dE. \tag{2.11.33}$$

Comparing this result with (2.11.23)–(2.11.24), we see that it can be rewritten as

$$\langle \Delta t \rangle = \langle 2\eta'(E) \rangle_{\text{in}}, \tag{2.11.34}$$

where the right-hand side denotes *the expectation value of* $2\eta'(E)$ *in the initial state* (incoming wave packet); *this expectation value gives the average time delay due to the interaction.* In particular, if the energy spectrum of the initial state is centered about $E = E_0$ and is sufficiently narrow (as discussed above), we recover (2.11.21).

The average time $T_0(r)$ spent within a sphere of radius $r$ in the absence of interaction can easily be computed:

$$\begin{aligned}
T_0(r) &= \int_{-\infty}^{\infty} P_{i,0}(r,t)\,dt \\
&= 4\pi \int_{-\infty}^{\infty} dt \int_0^r |\varphi_0(r',t)|^2\,dr' \\
&= 16\pi \int_{-\infty}^{\infty} dt \int_0^{\infty} dE' \int_0^{\infty} dE\, A^*(E')\,A(E)\,e^{i(E'-E)t} \\
&\quad \times \int_0^r \sin(kr')\sin(k'r')\,dr' \\
&= 16\pi^2 \int_0^{\infty} dE' \int_0^{\infty} dE\, A^*(E')A(E)\,\delta(E-E') \\
&\quad \times \left\{ \frac{\sin[(k-k')r]}{k-k'} - \frac{\sin[(k+k')r]}{k+k'} \right\}.
\end{aligned}$$

Thus, finally,

$$\begin{aligned}
T_0(r) &= 8\pi^2 \int_0^{\infty} |A(E)|^2 \left[ \frac{2r}{k} - \frac{\sin(2kr)}{k^2} \right] k\,dE \\
&= \left\langle \frac{2r}{k} - \frac{\sin(2kr)}{k^2} \right\rangle_{\text{in}}.
\end{aligned} \tag{2.11.35}$$

Classically, only the first term, which represents the free time of flight across the sphere, would be present. The oscillating term represents a quantum effect, due to the wave nature of matter. In fact, this term is negligible for sufficiently fast particles ($kr \gg 1$). On the other hand, if we build up a wave packet of very slow particles, with $kr \ll 1$, the second term in (2.11.35) will practically cancel the first one, so that $T_0(r) \to 0$. This arises from the fact that the particles cannot be localized within dimensions smaller than a wavelength:

the uncertainty in position for such a wave packet is much larger than the dimensions of the sphere,

$$\Delta r \sim 1/\Delta k \gg r. \tag{2.11.36}$$

This clarifies the physical meaning of the sinusoidal term in Wigner's inequality.

The relation to bound states that is apparent in the inequality of Goebel, Karplus, and Ruderman (2.11.12) can also be understood, in terms of the range of the bound states. The bound-state wave function associated with $k = iK_b$ is proportional to $\exp(-K_b r)$, so that its range is given by $K_b^{-1}$. At low energies, $k^2 \ll K_b{}^2$, the scatterer may act like an object with radius $a + K_b^{-1}$ instead of $a$, thus giving rise to (2.11.14).

Combining (2.11.28) and (2.11.32) with (2.11.35), with $r = a$, we find that the time spent within the scatterer in the presence of interaction is given by

$$T(a) = \int_{-\infty}^{\infty} P_i(a,t)\,dt = \left\langle 2\eta'(E) + \frac{2a}{k} - \frac{\sin[2(\eta + ka)]}{k^2} \right\rangle. \tag{2.11.37}$$

Van Kampen's causality condition (2.9.1), which implies $T(a) \geqslant 0$, together with (2.11.37), thus leads immediately to Wigner's inequality (2.11.5), without having to prove (2.11.1) as an intermediate step.

## 2.12. Completeness

In the absence of a scatterer, the stationary normalized $s$-wave eigenstates, defined by

$$\psi_0(k,r) = (1/\sqrt{2}\,\pi)\sin(kr)/r = \varphi_0(k,r)/r \qquad (0 \leqslant k < \infty), \tag{2.12.1}$$

form a complete orthonormal basis in Hilbert space for spherically symmetric functions, i.e., any square integrable function $f(r)$ of the radial coordinate only can be expanded in the form

$$f(r) = \int_0^{\infty} c(k)\,\psi_0(k,r)\,dk, \tag{2.12.2}$$

where the expansion coefficients $c(k)$ are given by

$$\begin{aligned} c(k) &= (\psi_0(k,r), f(r)) \\ &= \int \psi_0{}^*(k,r) f(r)\, d^3 r \\ &= 4\pi \int_0^{\infty} \psi_0{}^*(k,r) f(r) r^2\, dr. \end{aligned} \tag{2.12.3}$$

The orthonormality condition may be expressed as

$$(\psi_0(k',r),\psi_0(k,r)) = \int \psi_0{}^*(k',r)\psi_0(k,r)\,d^3r = \frac{2}{\pi}\int_0^\infty \sin(k'r)\sin(kr)\,dr = \delta(k-k') \qquad (2.12.4)$$

and the completeness of the orthonormal set $|\psi_0\rangle$ (using Dirac's bra and ket notation) may be expressed symbolically as

$$\sum |\psi_0\rangle\langle\psi_0| = 1, \qquad (2.12.5)$$

or, more explicitly, as

$$\int_0^\infty \psi_0(k,r)\psi_0{}^*(k,r')\,dk = \delta(r-r')/4\pi r^2 \qquad (r>0, r'>0), \qquad (2.12.6)$$

where the right-hand side acts as the unit operator with respect to integration over all space of a spherically symmetric function.

Substituting $\psi_0(k,r)$ by its expression (2.12.1), we find that (2.12.6) reduces to

$$\frac{2}{\pi}\int_0^\infty \sin(kr)\sin(kr')\,dk = \delta(r-r') \qquad (r \geq 0, r'>0), \qquad (2.12.7)$$

which should be compared with (2.12.4). The expansion of the function $g(r)=rf(r)$ in terms of the set $\varphi_0(k,r)$ simply amounts to a statement of the Fourier integral theorem for Fourier sine transforms. The relation (2.12.5) is known as a "resolution of the identity."

Now let us consider what happens when the scatterer is present. Strictly speaking, the orthonormality requirement that would correspond to (2.12.4) in this case is only meaningful when the definition of the wave function can be extended also to the "internal region" $0<r<a$. However, we can argue that the infinity in (2.12.4) as $k \to k'$ arises from the infinite volume of integration, so that the normalization factor for the wave functions of the continuous spectrum $\psi(k,r)$ depends only on their asymptotic behavior[31] as $r\to\infty$. It is then readily seen that the correct normalization factor must be the same as in (2.12.1), namely,

$$\varphi(k,r) = r\psi(k,r) = (1/\sqrt{2}\,\pi)\sin[kr+\eta(k)] = (ie^{-i\eta}/2^{3/2}\pi)[e^{-ikr}-S(k)e^{ikr}] \qquad (r>a). \qquad (2.12.8)$$

[31] Cf. Landau and Lifshitz [44, p. 62].

In fact, by evaluating the integral

$$4\pi \int_r^\infty \varphi^*(k',r')\varphi(k,r')\,dr' \qquad (r > a) \tag{2.12.9}$$

with the help of the result

$$\int_x^\infty \cos(\sigma\xi+\tau)\,d\xi = \pi\cos\tau\,\delta(\xi) - \sin(\sigma x+\tau)P(1/\sigma), \tag{2.12.10}$$

which follows from (A11.5)–(A11.7), we find that (2.12.9) yields $\delta(k-k')$ plus terms that remain finite as $k \to k'$.

By analogy with (2.12.7), we are then led to consider the integral

$$\begin{aligned} I(r,r') &= (2/\pi)\int_0^\infty \sin[kr+\eta(k)]\sin[kr'+\eta(k)]\,dk \\ &= 4\pi\int_0^\infty \varphi(k,r)\varphi^*(k,r')\,dk \qquad (r > a, r' > a). \end{aligned} \tag{2.12.11}$$

We find

$$\begin{aligned} I(r,r') &= (1/\pi)\int_0^\infty \{\cos[k(r-r')] - \cos[k(r+r') + 2\eta(k)]\}\,dk \\ &= \delta(r-r') - (1/2\pi)\int_{-\infty}^\infty S(k)e^{ik(r+r')}\,dk, \end{aligned} \tag{2.12.12}$$

where we have used (2.7.21) and (2.8.14).

Let

$$r + r' = 2a + x \qquad (x > 0). \tag{2.12.13}$$

Then the last term in (2.12.12) may be rewritten as

$$J(x) = -(1/2\pi)\int_{-\infty}^\infty S_a(k)e^{ikx}\,dk, \tag{2.12.14}$$

where $S_a(k)$ is given by (2.9.5). According to Section 2.9, $S_a(k)$ is regular in $I_+$, except possibly for simple poles on the imaginary axis. Let us assume, for simplicity, that there are only a finite number of such poles, $k = iK_n$ ($n = 1, 2, \ldots, N$), corresponding to bound states. Then, as we have seen in Section 2.9, $S_a(k)$ is bounded as $|k| \to \infty$ in $I_+$.

Let us consider a contour formed by the real axis from $-K$ to $K$ plus a semicircle $\Gamma_K$ of radius $K$ in $I_+$, where $K > \max K_n$. Then the above-mentioned properties of $S_a(k)$ imply

$$\left(\int_{-K}^K + \int_{\Gamma_K}\right) S_a(k)e^{ikx}\,dk = 2\pi\sum_{n=1}^N (b_n/K_n)\exp(-K_n x) \qquad (x > 0), \tag{2.12.15}$$

since, by (2.9.45), the residue of $S_a(k)$ at $k = iK_n$ is $-ib_n/K_n$.

We would now like to let $K \to \infty$ and to show that the semicircle at infinity does not contribute. Since $S_a(k)$ is bounded and $e^{ikx}$ is exponentially decreasing in $I_+$, there is no problem, except when we approach the real axis. As is clear from (2.12.12) and (2.12.14), the integrals we are considering actually stand for distributions, and what we want to show is that $\lim_{K\to\infty} \int_{\Gamma_K}$ vanishes (for $x > 0$) in the sense of distributions (cf. Appendix A, Section A6).

For this purpose, consider a test function $f(x)$ such that $\operatorname{supp} f(x) = [c, d]$ $(0 < c < d)$. Then

$$\int_c^d f(x)\,dx \int_{\Gamma_K} S_a(k)\,e^{ikx}\,dk$$
$$= \int_{\Gamma_K} dk\,S_a(k)[(e^{ikd}/ik)\,f(d) - (e^{ikc}/ik)\,f(c) + O(k^{-2})],$$

which tends to zero as $K \to \infty$. Thus, finally, (2.12.14) and (2.12.15) yield

$$J(x) = -\sum_{n=1}^{N} b_n \exp(-K_n x)/K_n \qquad (x > 0), \tag{2.12.16}$$

and, substituting this result in (2.12.11)–(2.12.12), we find

$$\frac{2}{\pi}\int_0^\infty \sin[kr+\eta(k)]\sin[kr'+\eta(k)]\,dk + \sum_{n=1}^{N} c_n \exp[-K_n(r+r')]$$
$$= \delta(r-r') \qquad (r > a, r' > a), \tag{2.12.17}$$

where $c_n = i \operatorname{res} S(k)|_{iK_n}$, as in (2.9.45).

If $\psi_n(r)$ denotes the wave function of the discrete spectrum associated with the $n$th bound state, assumed to be defined for all $r$, we have, in the external region,

$$\psi_n(r) = N_n \exp(-K_n r)/r \qquad (r > a), \tag{2.12.18}$$

where the normalization constant $N_n$ is determined by

$$\int \psi_m{}^*(r)\psi_n(r)\,d^3r = \delta_{mn}. \tag{2.12.19}$$

The relation (2.12.17) can then be rewritten as

$$\int_0^\infty \psi(k,r)\psi^*(k,r')\,dk + \sum_{n=1}^{N} \psi_n(r)\psi_n{}^*(r')$$
$$= \delta(r-r')/4\pi r^2 \qquad (r > a, r' > a), \tag{2.12.20}$$

where

$$c_n = 4\pi|N_n|^2. \tag{2.12.21}$$

By comparison with (2.12.6), we see that (2.12.17) represents the *resolution of the identity* in the external region. We see also that the residues of the $S$-function at the poles $iK_n$ yield the normalization constants for the corresponding bound-state wave functions [24]. The condition $c_n > 0$ [cf. (2.9.45)] follows immediately from (2.12.21).

Thus *the resolution of the identity (2.12.17) follows from the analytic properties of the S-function.* It will now be shown that the converse is also true.

For this purpose, let us assume the validity of (2.12.17); we assume, in addition, that the symmetry relation (2.8.14) is valid.

Taking into account (2.12.12), these assumptions lead to

$$\int_{-\infty}^{\infty} S_a(k) e^{ikx}\, dk = 2\pi \sum_{n=1}^{N} b_n \exp(-K_n x)/K_n \qquad (x > 0), \quad (2.12.22)$$

where $b_n$ and $c_n$ are related by (2.9.45).

Let us define

$$\Gamma(k) = i\left[\frac{S_a(k) - S(0)}{k} + \sum_{n=1}^{N} \frac{b_n}{K_n{}^2(k - iK_n)}\right], \qquad (2.12.23)$$

which, as a square integrable function [we assume that $S_a(k)$ is differentiable at the origin], belongs to the distribution space $\mathscr{D}_L'^{(1)}$ [Section 1.8(b)].

Let

$$\gamma_t = \int_{-\infty}^{\infty} \Gamma(k) e^{-ikt}\, dk. \qquad (2.12.24)$$

Then, differentiating in the sense of distributions, we find

$$\begin{aligned} g_t = \gamma_t' &= \int_{-\infty}^{\infty} S_a(k) e^{-ikt}\, dk - 2\pi S(0)\delta_t \\ &+ \sum_{n=1}^{N} \frac{b_n}{K_n{}^2} \int_{-\infty}^{\infty} \frac{k}{k - iK_n} \exp(-ikt)\, dk. \end{aligned} \qquad (2.12.25)$$

We have

$$\int_{-\infty}^{\infty} \frac{k}{k - iK_n} \exp(-ikt) = 2\pi\delta_t - 2\pi K_n \theta(-t) \exp(K_n t), \qquad (2.12.26)$$

where $\theta$ is the Heaviside step function. Thus, finally,

$$\begin{aligned} g_t = &\int_{-\infty}^{\infty} S_a(k) e^{-ikt}\, dk - 2\pi\theta(-t) \sum_{n=1}^{N} b_n \exp(K_n t)/K_n \\ &+ 2\pi\delta_t\left[\sum_{n=1}^{N} \frac{b_n}{K_n{}^2} - S(0)\right]. \end{aligned} \qquad (2.12.27)$$

Taking into account (2.12.22), we see that the right-hand side of (2.12.27) vanishes (in the sense of distributions) for $t < 0$, so that

$$g_t = \int_{-\infty}^{\infty} G_k e^{-ikt}\, dk = \int_{-\infty}^{\infty} (-ik)\Gamma(k)\, e^{-ikt}\, dk \in \mathscr{D}_+'. \qquad (2.12.28)$$

Thus we are precisely in the situation discussed in Section 1.8(f), and we can conclude that $\Gamma(k)$ verifies the dispersion relation (1.8.36), which corresponds to Van Kampen's dispersion relation (2.9.44), as can be readily verified with the help of the identity

$$\mathrm{Im}\left[\frac{1}{\pi} P \int_{-\infty}^{\infty} \frac{dk'}{(k'-k)(k'-iK_n)}\right] = -\frac{k}{k^2+K_n^2}\,. \qquad (2.12.29)$$

The analytic properties of $S(k)$ follow from these results.

We conclude that *the resolution of the identity in the external region* (*2.12.17*) *is equivalent to the analytic properties of the S-function.*[32]

It should be stressed, however, that the resolution of the identity (2.12.17) is equivalent to the completeness condition for the set of all stationary states only when we have an orthonormal set, i.e., when the definition of the wave function is extended to the internal region. If we do not want to make any assumptions about the internal region beyond those already made in Section 2.7, we can still talk about completeness of the set of stationary states *in the external region*, but (2.12.17), although it is sufficient, is no longer a *necessary* condition for completeness in this sense.

Assuming for simplicity that no bound states exist, completeness of the set (2.12.8) in the external region means that any function $f(r)$ such that $rf(r) = g(r)$ is square integrable for $r > a$; i.e.,

$$g(r) \in L^2(a, \infty) \qquad (2.12.30)$$

can be expanded in the form

$$g(r) = \int_0^{\infty} c(k) \sin[kr + \eta(k)]\, dk \qquad (r > a). \qquad (2.12.31)$$

If (2.12.17) (without the bound-state terms) is valid, such an expansion is certainly possible, with, for instance,

$$c(k) = \frac{2}{\pi} \int_a^{\infty} g(r) \sin[kr + \eta(k)]\, dr. \qquad (2.12.32)$$

[32] This was conjectured by Heisenberg [24]. The type of singularities allowed by this condition, assuming that $S(k)$ has a unique analytic continuation, was investigated by Hu [10]. Cf. also Saavedra [45].

However, the expansion, in general, is not unique; this is related to the fact that the stationary states do not form an orthonormal set over $a < r < \infty$. For instance, if the wave function is defined also in the internal region, we can continue $g(r)$ to a function $\tilde{g}(r) \in L^2(0, \infty)$ in any way we please, and replace (2.12.32) by

$$c(k) = 2^{1/2} \int_0^\infty g(r)\,\varphi(k,r)\,dr, \tag{2.12.33}$$

where $\varphi(k,r)$ is the stationary wave function, defined for $0 \leqslant r < \infty$, and normalized by (2.12.8). In this case, therefore, the expansion (2.12.31) can be made in infinitely many ways, corresponding to the existence of linear relations among the stationary states in the external region (i.e., they are not linearly independent). We can say that these states form an *overcomplete set.*

On the other hand, since the expansion coefficients are not uniquely determined,.the resolution of the identity (2.12.17) or, what amounts to the same, the analyticity of $S$, is not a necessary condition for completeness in the external region. As has been shown by Van Kampen [46], (2.12.31) can be verified even if $S(k)$ possesses no analytic continuation in $I_+(k)$.

Thus completeness in the external region is a less powerful requirement than causality or than the assumption that the resolution of the identity (2.12.17) is valid in the external region.

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CHAPTER

3

# DISPERSION RELATIONS FOR THE TOTAL SCATTERING AMPLITUDE

*... We find ourselves in what Goldberger called the Never-Never Land, where momenta become imaginary, cosines of angles become less than $-1$, and so on.*

M. GELL-MANN[1]

## 3.1. Introduction

We now go over from the individual partial waves to the total scattering amplitude. For this purpose, let us consider first a classical scalar real massless field, as in Section 2.2, and a *spherically symmetric* scatterer of radius $a$.

Let the incident wave be a monochromatic plane wave, with the $z$-axis taken along its direction of propagation,

$$\psi_{\text{inc}}(k,z,t) = e^{ik(z-ct)}. \tag{3.1.1}$$

Let $\psi(k,r,\theta,t)$ be the corresponding total wave function (which does not depend on $\varphi$ because of the assumed spherical symmetry), and let its asymptotic behavior as $r \to \infty$ along the direction $\theta$ be given by

$$\psi(k,r,\theta,t) \approx [e^{ikz}+f(k,\theta)(e^{ikr}/r)]\, e^{-ikct} \qquad (r \to \infty), \tag{3.1.2}$$

[1] *Proc. 6th Annual Rochester Conf. High-Energy Nucl. Phys.*, Sec. III, p. 33. Wiley (Interscience), New York, 1956.

where $\approx$ stands for "asymptotically equal." The quantity $f(k,\theta)$ is the total scattering amplitude in the direction corresponding to the scattering angle $\theta$.

The relation between $f(k,\theta)$ and the individual partial waves is given[2] by the well-known partial-wave expansion

$$f(k,\theta) = (1/k)\sum_{l=0}^{\infty}(2l+1)f_l(k)\,P_l(\cos\theta), \tag{3.1.3}$$

where

$$\begin{aligned} f_l(k) &= (1/2i)[S_l(k)-1] = \exp(i\eta_l)\sin\eta_l \\ &= (k/2)\int_0^{\pi} f(k,\theta)\,P_l(\cos\theta)\sin\theta\,d\theta \end{aligned} \tag{3.1.4}$$

is the $l$th *partial-wave scattering amplitude* [cf. (2.6.12)].

It follows from (3.1.4) that

$$\operatorname{Im} f_l(k) = \sin^2\eta_l = |f_l(k)|^2. \tag{3.1.5}$$

Since this is a consequence of the unitarity of the $S$-matrix (2.6.10), it is called the *unitarity condition* for partial-wave scattering amplitudes.

The symmetry relation (2.6.9), which expresses the reality of the interaction, leads to a corresponding relation for the total scattering amplitude:

$$f(-k,\theta) = f^*(k,\theta). \tag{3.1.6}$$

Now let us consider an incident wave packet

$$\psi_{\text{inc}}(z,t) = \int_{-\infty}^{\infty} A(k)\,e^{ik(z-ct)}\,dk. \tag{3.1.7}$$

The linearity of the interaction implies that the corresponding scattered wave packet is asymptotically given by

$$\psi_s(r,\theta,t) \approx (1/r)\int_{-\infty}^{\infty} A(k)f(k,\theta)\,e^{ik(r-ct)}\,dk \qquad (r\to\infty). \tag{3.1.8}$$

The energy per unit area contained in the incident wave packet is given by [cf. (2.2.4)]

$$\int_{-\infty}^{\infty} s_{z,\text{inc}}\,dt = -\int_{-\infty}^{\infty}(\partial\psi_{\text{inc}}/\partial t)(\partial\psi_{\text{inc}}/\partial z)\,dt = 2\pi c\int_{-\infty}^{\infty} k^2|A(k)|^2\,dk, \tag{3.1.9}$$

where we have made use of Parseval's theorem (1.6.2) and the reality of the field, as in (2.2.12), (2.2.13). We shall restrict ourselves to incident wave packets for which the energy per unit area is finite (the total energy is, of course, divergent), so that $kA(k)$ is square integrable.

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[2] See, for example, Messiah [1, p. 385.]

Similarly [cf. (2.2.12)], the total energy contained in the scattered wave packet follows from (3.1.8):

$$\int_{-\infty}^{\infty} dt \oint s_{r,s} r^2 \, d\Omega = 2\pi c \int_{-\infty}^{\infty} k^2 |A(k)|^2 \, dk \oint \frac{d\sigma}{d\Omega}(k,\theta) \, d\Omega$$

$$= 2\pi c \int_{-\infty}^{\infty} k^2 |A(k)|^2 \sigma_t(k) \, dk, \tag{3.1.10}$$

where

$$d\sigma/d\Omega = |f(k,\theta)|^2 \tag{3.1.11}$$

is the *differential cross section* in the direction $\theta$ and

$$\sigma_t(k) = \oint |f(k,\theta)|^2 \, d\Omega = 2\pi \int_0^{\pi} |f(k,\theta)|^2 \sin\theta \, d\theta$$

$$= \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) |f_l(k)|^2 \tag{3.1.12}$$

is the *total cross section* [cf. (2.6.11)].

Unless the interaction is very singular, it is reasonable to expect that a scatterer of finite size can only scatter a finite amount of energy out of the incident beam. We shall introduce this as an additional assumption about the interaction, namely, that (3.1.10) is finite. It follows that *the total cross section is bounded* for all $k$.

This does not necessarily imply, however, that $f(k,\theta)$ is bounded, because of the averaging over angles in going over from the differential to the total cross section. According to the optical theorem (1.10.17) [which also follows directly from (3.1.5) and (3.1.12)],

$$\operatorname{Im} f(k,0) = (k/4\pi)\,\sigma_t(k), \tag{3.1.13}$$

so that the boundedness of the total cross section implies

$$\operatorname{Im} f(k,0) = O(k) \qquad (k \to \infty). \tag{3.1.14}$$

Actually, the total cross section can approach a finite limit as $k \to \infty$ [e.g., for a hard sphere; cf. (3.3.72)], so that $\operatorname{Im} f(k,0)$ can diverge linearly with $k$.

On the other hand, it follows from (3.1.4), (3.1.5) that

$$|\operatorname{Im} f(k,\theta)| = \left| \frac{1}{k} \sum_{l=0}^{\infty} (2l+1) |f_l(k)|^2 P_l(\cos\theta) \right|$$

$$\leqslant \operatorname{Im} f(k,0) \qquad (0 \leqslant \theta \leqslant \pi), \tag{3.1.15}$$

because

$$|P_l(\cos\theta)| \leqslant P_l(1) = 1 \qquad (0 \leqslant \theta \leqslant \pi). \tag{3.1.16}$$

Thus we can extend (3.1.14) to

$$\operatorname{Im} f(k,\theta) = O(k) \qquad (k \to \infty,\ 0 \leqslant \theta \leqslant \pi). \tag{3.1.17}$$

Although this does not necessarily imply a similar bound for $\operatorname{Re} f(k,\theta)$, we shall assume that such a bound exists, i.e., that the *total* scattering amplitude cannot diverge more strongly than linearly as $k \to \infty$. This condition, together with the boundedness of $f(k,\theta)$ for finite $k$, can be expressed as follows:

$$|f(k,\theta)| \leqslant C|k+i\delta| \qquad (\text{real}\, k,\ 0 \leqslant \theta \leqslant \pi), \tag{3.1.18}$$

where $C$ and $\delta$ are positive constants. This assumption about the interaction implies, in particular, the boundedness of the total cross section. We shall see, later on, that it can be derived from other assumptions about the interaction [cf. Section 3.3(c)].

The restriction to incident wave packets such that (3.1.9) is finite, together with (3.1.18), implies that the integrand of (3.1.8) is also square integrable.

## 3.2. Dispersion Relations for Fixed Scattering Angle

The total scattering amplitude is a function of two independent variables; we have chosen $k$ and $\theta$, but, as was mentioned in Section 2.1 and will be seen later, several other choices are possible and convenient. We shall now consider the analytic properties of $f(k,\theta)$ as a function of $k$, for fixed scattering angle $\theta$. For this purpose, as in Section 2.3, we formulate a suitable causality condition.

For a classical massless field, we can build up an incident wave packet with a sharp front, located (say) at $z = ct$,

$$\psi_{\text{inc}}(z,t) = 0 \qquad (z > ct). \tag{3.2.1}$$

For this purpose, according to (3.1.7) (regarded as a function of $ct-z$) and to Titchmarsh's theorem 1.6.1, it suffices to choose $A(k)$ as a causal transform, such that (3.1.9) is finite; e.g.,

$$A(k) = A_0/(k+iK)^2 \qquad (K > 0). \tag{3.2.2}$$

Causality implies that the corresponding scattered wave packet, for an observation point at a large distance $r$ from the scatterer, in the direction $\theta$, cannot appear before a certain time $t_0$. According to Fermat's principle, the shortest path linking the incident wave front with such an observation point, via the scatterer, is the path taken by a ray that undergoes specular

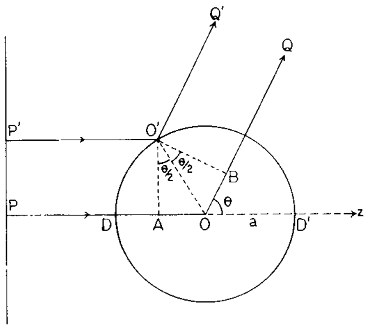


*FIG. 3.1.* The shortest path linking the wave front $PP'$ with an observation point in the direction $\theta$, via the scatterer, is the path $P'O'Q'$, corresponding to specular reflection at the surface.

reflection at the surface of the scatterer ($P'O'Q'$ in Fig. 3.1). The path difference with respect to the path $POQ$ through the center of the sphere is (cf. Fig. 3.1)

$$\overline{AO} + \overline{OB} = 2a\sin(\theta/2). \tag{3.2.3}$$

Since a wave front taking the path $POQ$ would arrive at $r$ at the time $r/c$, it follows from the causality condition (in the form III′ of Section 1.4) that the earliest possible arrival time of the scattered wave front is (for very large $r$)

$$t_0 = (1/c)[r-2a\sin(\theta/2)]. \tag{3.2.4}$$

CAUSALITY CONDITION. *If the incident wave vanishes for $t < z/c$, the scattered wave in the direction $\theta$ must vanish for $t < t_0$*, where $t_0$ is given by (3.2.4).

It is important to notice that this condition is stronger than the causality condition for each individual partial wave, i.e., for spherical multipole wave packets (Sections 2.3 and 2.6), because it implies the causal propagation of signals also *inside of the scatterer*, whereas the partial-wave condition ensures causal propagation only up to the surface of the scatterer.

In fact, if we decompose the incident wave packet (3.1.7) into partial waves [with the help of (2.2.20)], the corresponding incoming spherical multipole wave packets will all vanish at the surface of the scatterer for all times prior to the time $t = -a/c$ when the incident wave front hits the scatterer. The partial-wave causality condition formulated in Sections 2.3 and 2.6 then implies that each scattered spherical multipole wave packet also vanishes at the surface up to this time. However, this would still be consistent with the appearance of a scattered wave all over the surface at $t = -a/c$. Thus, as soon as the incident plane front touched the sphere at the point $D$ (Fig. 3.1), a scattered wave might appear at the diametrically opposite point $D'$, corresponding to an instantaneous jump over the distance $2a$. The present formulation of the causality condition excludes this possibility, telling us that we would have to wait until $t = a/c$ before observing a scattered wave at $D'$ (actually, the observation is to be made at a large distance).

The appearance of the scattered wave at different times at different points along the surface of the scatterer is only possible, of course, due to phase relations among different partial waves that allow them to interfere constructively at some points and destructively at others. Thus it is to be expected that the present causality condition entails the existence of *relations among partial waves* of different angular momenta, whereas no such relation follows from partial-wave causality conditions.

Let us rewrite (3.1.8) in the form

$$r\psi_s(r,\theta,t) \approx \int_{-\infty}^{\infty} A(k) f_a(k,\theta) \exp[-ick(t-t_0)]\, dk, \tag{3.2.5}$$

where

$$f_a(k,\theta) = e^{2ika\sin(\theta/2)} f(k,\theta). \tag{3.2.6}$$

With $A(k)$ given, for example, by (3.2.2), the integrand of (3.2.5) is square integrable and, according to the causality condition, the integral vanishes for $t-t_0<0$. It then follows from Titchmarsh's theorem 1.6.1 that $(k+iK)^{-2} \times f_a(k,\theta)$ is regular in $I_+(k)$ and square integrable over any line parallel to the real axis, for any $K>0$.

In particular, taking $K=\delta$, where $\delta$ is defined by (3.1.18), we see that the function $g_a(k,\theta)=(k+i\delta)^{-1} f_a(k,\theta)/C$ has the following properties: (i) it is regular in $I_+$; (ii) according to (3.1.18), $|g_a(k,\theta)| \leqslant 1$ on the real axis; (iii)[3] $|g_a(k,\theta)| = O(k)$ for $|k|\to\infty$ in $I_+$. It then follows from the Phragmén–Lindelöf theorem 2.5.4 that, actually, $|g_a(k,\theta)| \leqslant 1$ in $I_+$.

In conclusion, we see that $f_a(k,\theta)$ *has a regular analytic continuation in* $I_+(k)$, and

$$|f_a(k,\theta)| \leqslant C|k+i\delta| \qquad (\operatorname{Im} k \geqslant 0). \tag{3.2.7}$$

According to Section 1.7(b), these results enable us to write down a dispersion relation with two subtractions for $f_a(k,\theta)$. We shall make the subtractions at the origin, so that we get, for real $k$,

$$f_a(k,\theta) = f_a(0,\theta) + kf_a'(0,\theta) + \frac{k^2}{\pi i} \int_{-\infty}^{\infty} \frac{f_a(k',\theta) - f_a(0,\theta) - k' f_a'(0,\theta)}{k'^2(k'-k)}\, dk'.$$

The symmetry relation (3.1.6) and its derivative with respect to $k$, at $k=0$, yield

$$f_a(0,\theta) = f_a^*(0,\theta); \qquad f_a'(0,\theta) = -f_a'^*(0,\theta),$$

so that

$$\operatorname{Im} f_a(0,\theta) = \operatorname{Re} f_a'(0,\theta) = 0. \tag{3.2.8}$$

[3] Property (iii) follows easily from the dispersion relation for $(k+i\delta)^{-2} f_a(k,\theta)$, with $k \in I_+$, just as in (2.3.5), (2.3.6).

Thus, taking the real part of both sides in the above dispersion relation, we get

$$\mathrm{Re}\, f_a(k,\theta) = f_a(0,\theta) + \frac{k^2}{\pi} ⨍_{-\infty}^{\infty} \frac{\mathrm{Im}\, f_a(k',\theta) - k' \,\mathrm{Im}\, f_a'(0,\theta)}{k'^2(k'-k)}\, dk'.$$

The second term in the numerator of the integrand may be dropped, provided that we take the principal value at $k' = 0$ as well as at $k' = k$, because

$$⨍_{-\infty}^{\infty} \frac{dk'}{k'(k'-k)} = 0. \tag{3.2.9}$$

Finally, since $\mathrm{Im}\, f_a(k',\theta)$ is an odd function of $k'$ by (3.1.6), we get

$$\begin{aligned}&\mathrm{Re}[e^{2ika\sin(\theta/2)} f(k,\theta)] \\ &\quad = f(0,\theta) + \frac{2k^2}{\pi} ⨍_0^{\infty} \frac{\mathrm{Im}[e^{2ik'a\sin(\theta/2)} f(k',\theta)]}{k'(k'^2-k^2)}\, dk'.\end{aligned} \tag{3.2.10}$$

This is the dispersion relation for the scattering amplitude for a fixed scattering angle $\theta$ [1a].

For $\theta \neq 0$, the dispersion relation contains an exponential factor dependent on the radius of the scatterer, just like the dispersion relation for fixed angular momentum (2.3.11). However, this factor does not appear in the dispersion relation for the forward scattering amplitude ($\theta = 0$),

$$\mathrm{Re}\, f(k,0) = f(0,0) + \frac{2k^2}{\pi} ⨍_0^{\infty} \frac{\mathrm{Im}\, f(k',0)}{k'(k'^2-k^2)}\, dk', \tag{3.2.11}$$

which is independent of the radius of the scatterer. This is due to the fact that the radius of the scatterer does not appear in the formulation of the causality condition in the forward direction [cf. (3.2.4)]. Thus the dispersion relation for $\theta = 0$ has a more fundamental character.

According to the optical theorem (3.1.13), the relation (3.2.11) may also be rewritten as

$$\mathrm{Re}\, f(k,0) = f(0,0) + \frac{k^2}{2\pi^2} ⨍_0^{\infty} \frac{\sigma_t(k')}{k'^2-k^2}\, dk', \tag{3.2.12}$$

where $\sigma_t$ is the total cross section. The subtraction constant $f(0,\theta) = f(0,0)$ [as $k \to 0$, $f(k,\theta)$ usually becomes independent of $\theta$, i.e., only $s$-waves contribute; cf. Section 3.3], with opposite sign, is known as the *scattering length* [cf. (5.7.31)]; according to (3.1.12),

$$\sigma_t(0) = 4\pi[f(0,0)]^2. \tag{3.2.13}$$

Thus the right-hand side of (3.2.12) is expressed in terms of quantities that can be directly determined from experimental data.

The relation (3.2.12) is the correct form of the result that was heuristically inferred in Section 1.10 from the Kramers–Kronig relation [cf. (1.10.19)].

How are the above results related to those obtained in Chapter 2 for fixed angular momentum?

According to (3.1.4), the regularity of $f(k,\theta)$ in $I_+(k)$ implies that $S_l(k)$ is also regular in $I_+$. Furthermore,

$$\begin{aligned} e^{2ika}f_l(k) &= (1/2i)[S_{la}(k)-e^{2ika}] \\ &= \frac{k}{2}\int_0^\pi f_a(k,\theta)\,e^{2ika[1-\sin(\theta/2)]}P_l(\cos\theta)\sin\theta\,d\theta, \end{aligned} \tag{3.2.14}$$

where $S_{la}(k)$ is defined by (2.6.13). According to (3.2.7), the last member is $O(k^2)$ as $|k|\to\infty$ in $I_+$. Since $|S_{la}(k)|=1$ on the real axis, it follows from the Phragmén–Lindelöf theorem that $|S_{la}(k)|\leqslant 1$ in $I_+$.

Thus the analytic properties of $S_l(k)$ (Section 2.6) and the dispersion relations for fixed angular momentum follow from those for fixed scattering angle. Note, however, that the converse is not true. In fact, as has been emphasized in the above discussion, the causality condition for the present case is stronger than that employed for each individual partial wave. The results obtained for fixed angular momentum do not imply anything about the convergence of the partial-wave expansion (3.1.3).

## 3.3. Dispersion Relations for Fixed Momentum Transfer

### *(a) Introduction*

Instead of expressing the total scattering amplitude as a function of the wave number $k$ and the scattering angle $\theta$, any equivalent pair of independent variables may be chosen. The form of the dispersion relation (3.2.10) suggests that it would be advantageous to choose $k$ and $\tau$, where

$$\tau = 2k\sin(\theta/2), \tag{3.3.1}$$

so that

$$\cos\theta = 1-\tau^2/2k^2. \tag{3.3.2}$$

The variable $\tau$ has a very simple physical meaning (Fig. 3.2): it measures the magnitude of the *momentum transfer* that takes place in the collision; i.e., if $\mathbf{k}$ denotes the initial wave vector in the incident beam and $\mathbf{k}'$ is the final wave vector associated with scattering in the direction $\theta$, the momentum transfer is measured by

$$\tau = |\boldsymbol{\tau}| = |\mathbf{k}'-\mathbf{k}|. \tag{3.3.3}$$

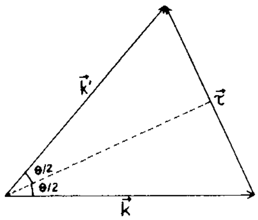


*FIG. 3.2.* The momentum transfer $\boldsymbol{\tau} = \mathbf{k}' - \mathbf{k}$.

Expressed in terms of $k$ and $\tau$, (3.2.6) would become

$$f_a(k,\tau) = e^{i\tau a} f(k,\tau), \tag{3.3.4}$$

and, for fixed $\tau$, the exponential factor is just a constant. Thus (3.2.7) leads us to expect that

$$|f(k,\tau)| \leqslant C|k + i\delta| \qquad (\operatorname{Im} k \geqslant 0 \text{ and fixed real } \tau), \tag{3.3.5}$$

and that $f(k,\tau)$ is regular in $I_+(k)$, so that it verifies dispersion relations in which the radius $a$ of the scatterer does not appear [this is also suggested by (3.2.10)]. This would be the advantage of choosing $\tau$ as a variable instead of $\theta$.

Notice, however, that neither one of the above results actually follows from Section 3.2. In fact, besides the dependence of the scattering amplitude on $k$, for fixed $\theta$, that was discussed in Section 3.2, there is, according to (3.3.2), an additional $k$-dependence, brought in through the dependence on $\cos\theta$. Thus we would need to have some knowledge about the behavior of $f(k,\cos\theta)$, also as a function of $\cos\theta$ in order to justify the heuristic inferences made above. It will be shown below that these inferences are indeed correct, but we shall follow a different method [2].

We shall define $f(k,\tau)$, for all $k$, by its partial-wave expansion [cf. (3.1.3), (3.3.2)]

$$f(k,\tau) = \sum_{l=0}^{\infty} f_l(k,\tau) = (1/k)\sum_{l=0}^{\infty}(2l+1) f_l(k)\, P_l(1-(\tau^2/2k^2)). \tag{3.3.6}$$

It will have to be shown, of course, that this expansion converges in the domain of values of $k$ under consideration.

We shall take over all the assumptions made in Sections 2.2 and 2.3 about the interaction, so that we can make use of the results derived in Chapter 2 about the analytic properties of partial-wave amplitudes. In particular, it follows from (3.3.6) and the symmetry relation (2.6.9) that

$$f(-k,\tau) = f^*(k,\tau) \qquad (\text{real } k, \tau). \tag{3.3.7}$$

For given $\tau$, not all real values of $k$ correspond to physically possible values of $\cos\theta$ ($-1 \leqslant \cos\theta \leqslant 1$). According to (3.3.2), the physical range of values of $k$ corresponding to a given $\tau$ is $|k| \geqslant |\tau|/2$. Thus we meet here for the first time an *unphysical region*, namely, the interval

$$-\tau/2 < k < \tau/2. \tag{3.3.8}$$

In this region, although $k$ and $\tau$ are real, $f(k,\tau)$ does not correspond to the scattering amplitude for a physically possible process [according to (3.3.1), $|\sin(\theta/2)| > 1$ in this region]. Thus $f(k,\tau)$ can only be defined by analytic continuation, and we shall employ (3.3.6) for this purpose.

The main results from Chapter 2 that will be required are:

$$\text{(I)}\quad f_l(k) \text{ is a meromorphic function of } k, \text{ regular in } I_+.$$

$$\text{(II)}\quad |S_{la}(k)| = |e^{2ika} S_l(k)| \leqslant 1 \qquad (\operatorname{Im} k \geqslant 0). \tag{3.3.9}$$

These results relate to the behavior of $S_l(k)$ as a function of $k$. However, as has already been mentioned in Section 3.2, they provide no information about its behavior as a function of $l$, i.e., no relations among partial waves with different angular momenta. They do not guarantee even the convergence of the partial-wave expansion (3.3.6) for any value[4] of $k$. Thus, besides the assumptions made in Chapter 2, additional assumptions about the interaction are required. We shall formulate these assumptions by a heuristic approach, trying to make them as general as possible.

### *(b) Threshold Behavior and Behavior for Large Angular Momentum*

The minimum requirements for the convergence of (3.3.6) are: first, that $f_l(k,\tau)$ be bounded for each $l$; second, that it shall approach zero sufficiently rapidly as $l \to \infty$.

Insofar as the first requirement is concerned, the only critical point is $k = 0$, because the factor $k^{-1} P_l(1-\tau^2/2k^2)$ has a pole of order $2l+1$ at this point.[5] The simplest way to compensate for this singularity is to have a zero of corresponding order in $f_l(k)$. By (3.1.4), it suffices to assume that

$$S_l(k) - 1 = O(k^{2l+1}) \qquad \text{as} \quad k \to 0 \text{ on } I_0. \tag{3.3.10}$$

This assumption is made only for physical $k$, i.e., for $k$ approaching zero along the *real* axis. However, since $S_l(k)$ is known to be meromorphic, this suffices to guarantee the same behavior as $k \to 0$ along any complex direction,[6] so that $k = 0$ is a zero of order (at least) $2l+1$, and $f_l(k,\tau)$ is regular at the origin.

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[4] For example, $S_l(k) = -1$ satisfies (I) and (II), but the corresponding series (3.3.6) would diverge for all $k$.

[5] $P_l(z)$ is a polynomial of order $l$ in $z$.

[6] Otherwise, $k = 0$ might have been an essential singularity, e.g., of the type $k^{2l+1} \exp(-ic/k)$.

The assumption (3.3.10) corresponds to a well-known law for the *threshold behavior* of phase-shifts. Its physical meaning is that, at sufficiently large wavelengths, the interaction with the scatterer is a small perturbation, and the scattering is dominated by the effect of centrifugal forces.

To see this,[7] let us introduce a quantitative measure of the effect of the interaction. Let

$$\gamma_l(k,a) = \left.\frac{r\,\partial u_l(k,r,t)/\partial r}{u_l(k,r,t)}\right|_{r=a}, \tag{3.3.11}$$

where $u_l(k,r,t)$ is the radial wave function, defined by (2.6.2), so that $\gamma_l$ represents the logarithmic derivative of this function, evaluated at $r=a$. If we define the $R$-function by analogy with (2.10.2) for $l>0$, we find that $\gamma_l$ and $R_l$ are related by

$$R_l = a/(1+\gamma_l). \tag{3.3.12}$$

In the absence of a scatterer ($S_l = 1$), we have

$$\gamma_l{}^0(k,a) = \beta j_l{}'(\beta)/j_l(\beta), \tag{3.3.13}$$

where $\beta = ka$. We can take as a measure of the distortion introduced by the scatterer the quantity

$$\delta_l(k,a) = \frac{\gamma_l(k,a)-\gamma_l{}^0(k,a)}{\gamma_l{}^0(k,a)} = \frac{\gamma_l}{\gamma_l{}^0} - 1. \tag{3.3.14}$$

The $S$-function can easily be expressed in terms of $\gamma_l$ [cf. (2.6.2)],

$$S_l(k) = -\frac{\gamma_l h_l^{(2)}(\beta) - \beta h_l'^{(2)}(\beta)}{\gamma_l h_l^{(1)}(\beta) - \beta h_l'^{(1)}(\beta)}. \tag{3.3.15}$$

Conversely, we can solve (3.3.15) to express $\gamma_l$ as a function of $S_l$ and thus conclude that it is a meromorphic function of $k$. As was found for $l=0$ in Section 2.10(b), $\gamma_l(-k) = \gamma_l(k)$, so that both $R_l$ and $\gamma_l$ are meromorphic functions of $k^2$.

It follows from (3.3.15), with the help of the Wronskian relation

$$W[h_l^{(1)}(z), h_l^{(2)}(z)] = -2iz^{-2}, \tag{3.3.16}$$

where

$$W[f(z), g(z)] = f(z)g'(z) - f'(z)g(z), \tag{3.3.17}$$

that

$$f_l(k) = \frac{1}{2i}[S_l(k)-1] = -\frac{\beta^2 j_l{}'(\beta) j_l(\beta)\,\delta_l(k,a)}{1+i\beta^2 j_l{}'(\beta)\,h_l^{(1)}(\beta)\,\delta_l(k,a)}. \tag{3.3.18}$$

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[7] For the following discussion, cf. Schiff [3, p. 107], and Messiah [1, pp. 388–393].

The behavior of $j_l(z)$ and $h_l^{(1)}(z) = j_l(z) + in_l(z)$ as $z \to 0$ follows from the expansions

$$j_l(z) = \frac{z^l}{(2l+1)!!}\left[1 + O\left(\frac{z^2}{l+(3/2)}\right)\right] \qquad (0 \leqslant |z| \lesssim 1), \tag{3.3.19}$$

$$n_l(z) = -\frac{(2l-1)!!}{z^{l+1}}\left[1 + O\left(\frac{z^2}{l-(1/2)}\right)\right] \qquad (0 \leqslant |z| \lesssim 1), \tag{3.3.20}$$

where

$$(2l+1)!! = 1 \cdot 3 \cdot 5 \cdots (2l+1) = (2l+1)!/(2^l l!). \tag{3.3.21}$$

Substituting these results in (3.3.18), we find, for $l \neq 0$,

$$f_l(k) \approx -\frac{l\delta_l(k,a)\beta^{2l+1}}{[(2l+1)!!]^2\left[1 + \dfrac{l}{2l+1}\,\delta_l(k,a)\right]} \qquad (|\beta| \ll 1, l \neq 0). \tag{3.3.22}$$

If the distortion is small, $|\delta_l(k,a)| \lesssim 1$ as $k \to 0$, the desired result (3.3.10) follows, for $l \neq 0$. For $l = 0$, $\delta_l(k,a)$ is not a good measure of distortion since, by (3.3.13),

$$\gamma_l^0(k,a) \to l \qquad \text{as} \quad k \to 0. \tag{3.3.23}$$

On the other hand, according to (2.2.23), the cross section diverges as $k \to 0$ unless (3.3.10) is satisfied for $l = 0$. In principle, this can happen,[8] i.e., we can have $S_0(0) = -1$, as was pointed out in Section 2.5. However, one would expect this to be an exceptional case, so that in general (3.3.10) would be satisfied.

It should be pointed out that, for $l \neq 0$, the threshold behavior (3.3.10) follows from (3.3.22) even if $|\delta_l(k,a)|$ is *not* small as $k \to 0$. In fact, if $|\delta_l(k,a)| \gg 1$, or even if $\delta_l \to \infty$, as it would for an impenetrable sphere, the denominator in (3.3.22) is dominated by the term in $\delta_l$, and $\delta_l$ cancels out, so that (3.3.10) is still valid.

The only way to violate (3.3.10) for $l \neq 0$ would be for the denominator to approach zero as $k \to 0$. Since $\delta_l$ is a meromorphic function of $k^2$, we would then expect to have

$$\delta_l(k,a) = -(2l+1)/l + O(k^2) \qquad \text{as} \quad k \to 0, \tag{3.3.24}$$

so that (3.3.10) would be replaced by

$$S_l(k) - 1 = O(k^{2l-1}) \qquad \text{as} \quad k \to 0 \quad (l \neq 0). \tag{3.3.25}$$

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[8] For an example, cf. (5.6.6).

Although this can happen, it is also to be regarded as an exceptional case, just like $S_0(0) = -1$, so that in general we would expect (3.3.10) to hold. This insures that each term $f_l(k, \tau)$ in the partial-wave series (3.3.6) is bounded for all finite (real) values of $k$ and $\tau$.

Let us now turn to the requirement that $f_l(k)$ should approach zero "sufficiently rapidly" as $l \to \infty$, so that the partial-wave series converges. What is to be expected on physical grounds for the behavior of $f_l(k)$ as $l \to \infty$?

If $k$ is fixed, we expect that, for sufficiently large $l$, the scattering is dominated by centrifugal forces, so that, once more, the interaction with the scatterer can be regarded as a small perturbation, and $|\delta_l(k, a)| \lesssim 1$. "Sufficiently large $l$" means not only that $l \gg 1$, but also that $l \gg \beta$. In fact, as is well known [1, 3], the $l$th term in the partial-wave decomposition (2.2.20) of the incident wave is very small for $kr \ll l$ and goes through a maximum at $kr \sim l$, so that it can be associated with an "impact parameter"

$$p_l \sim l/k. \tag{3.3.26}$$

If $p_l \gg a$, i.e., $l \gg \beta$, the corresponding incident partial wave is concentrated almost entirely outside of the scatterer, with impact parameter much larger than its radius, so that we would expect it to be almost unaffected by the scatterer. This need not be true for low $l$, due to the possibility of resonances (cf. Chapter 4), but beyond a certain value of $l$ (say $l > L$), dependent on the strength of the interaction, centrifugal forces should dominate, so that resonances no longer occur.

In conclusion, we expect to have

$$|\delta_l(k, a)| \leqslant 1 \qquad (l \geqslant N\beta,\ l \geqslant L), \tag{3.3.27}$$

where $N$ and $L$ are large numbers (independent of $k$).

On the other hand, we have,[9] under the above conditions,

$$\beta^2\, |j_l'(\beta)\, h_l^{(1)}(\beta)| \lesssim \tfrac{1}{2} \qquad (0 \leqslant \beta \ll l), \tag{3.3.28}$$

so that (3.3.18) and (3.3.27) yield

$$\begin{aligned} |f_l(k)| &\leqslant 2\beta^2\, |j_l(\beta)\, j_l'(\beta)| \\ &\leqslant \left(\frac{e\beta}{2l+1}\right)^{2l+1} \qquad (0 \leqslant \beta \leqslant l/N;\ l \geqslant L), \end{aligned} \tag{3.3.29}$$

where $e$ is the base of natural logarithms. The last result follows from Watson's inequalities [5, p. 255]. It can readily be verified by applying Stirling's formula to the factorials in (3.3.21), that (3.3.22) is compatible with (3.3.29) for large $l$.

[9] This follows from (3.3.19), (3.3.20) for $0 \leqslant \beta \lesssim 1$ and from the Debye asymptotic expansions for $\beta \gg 1$. See, for example, Jahnke and Emde [4, p. 139].

Since $N$ is large (we shall take at least $N \gtrsim 2$), it follows from (3.3.29) that

$$|f_l(k)| \leqslant (k/k_l)^{2l+1} \qquad (0 \leqslant k \leqslant k_l), \tag{3.3.30}$$

where

$$k_l = l/(Na) \qquad (l \geqslant L). \tag{3.3.31}$$

The existence, for each $l$, of an interval $0 \leqslant k \leqslant k_l$ where (3.3.30) is true follows directly from (3.3.10). However, (3.3.31) gives us, in addition, the behavior of $k_l$ for large $l$, telling us that

$$k_l \to \infty \qquad \text{as} \quad l \to \infty. \tag{3.3.32}$$

Notice that, beyond $k = k_l$, (3.3.30) remains valid, but it then contains no useful information, because (3.1.4) implies

$$|f_l(k)| \leqslant 1 \qquad (\text{real } k). \tag{3.3.33}$$

Finally, let us remark that (3.3.29) may remain valid even if (3.3.27) is not verified. This happens, for instance, in the case of an impenetrable sphere, for which $\delta_l \to \infty$. In this case, according to (2.3.7) and (2.6.2), we have

$$S_l(k) = -h_l^{(2)}(\beta)/h_l^{(1)}(\beta), \qquad f_l(k) = ij_l(\beta)/h_l^{(1)}(\beta), \tag{3.3.34}$$

and it can be shown [2] that (3.3.29) is valid, with $N = 2$, and $L$ an integer such that $(\log L)^{1/2} \gg 1$.

### *(c) Analytic Properties of $f(k, \tau)$ and Asymptotic Behavior as $k \to \infty$*

We now make use of the heuristic discussion given in the previous section to formulate the assumptions about the interaction that will be made, in addition to those of Sections 2.2 and 2.3.

ASSUMPTION I. *There exists a sequence of wave numbers $k_l$ ($l = 0, 1, 2, \ldots$) such that (3.3.30) and (3.3.32) are valid.*

This amounts to assuming the dominance of centrifugal forces over the interaction, at the threshold ($k = 0$) and in the high angular momentum limit ($l \to \infty$). Notice, however, that although we may actually expect $k_l$ to diverge *linearly* with $l$ [cf. (3.3.31)], nothing is said in Assumption I about its rate of growth: it may be arbitrarily small. Thus this is a very weak assumption, which should be valid for a large class of scatterers. However, it is already sufficient to prove the following results:

*i. For any fixed $\tau$ (real or complex), $f(k, \tau)$ is holomorphic in $I_+(k)$.*
*ii. For any fixed $k$ in $I_+$ (or $I_0$), $f(k, \tau)$ is an entire function of $\tau^2$ (i.e., it is holomorphic for all finite $\tau^2$).*

*Proof of i.* The proof is based on the following well-known theorem [6, p. 95]:

WEIERSTRASS'S THEOREM. *If each term of the series $\sum_{n=0}^{\infty} u_n(z)$ is holomorphic inside a region D, and if the series is uniformly convergent throughout every region D′ interior to D, the function $f(z) = \sum_{n=0}^{\infty} u_n(z)$ is holomorphic inside D.*

In the present case, according to (I), Section 3.3(a), $f_l(k)$ is holomorphic in $I_+$, and so is the polynomial $P_l(1-\tau^2/2k^2)$, except for the pole at the origin. However, as we have seen, it follows from Assumption I that $f_l(k,\tau)$, in (3.3.6), is regular at the origin. Thus $f_l(k,\tau)$ is holomorphic in $I_+(k)$.

To prove the uniform convergence of the partial-wave expansion, we need an inequality that follows from Assumption I. Let us consider the function

$$\varphi_l(k) = (k_l/k)^{2l+1} f_l(k)\, e^{2ika}, \tag{3.3.35}$$

which is regular in $I_+$. According to (3.3.30),

$$|\varphi_l(k)| \leqslant 1 \qquad \text{on} \quad I_0. \tag{3.3.36}$$

On the other hand, (3.3.9) implies

$$|\varphi_l(k)| < |k_l/k|^{2l+1} \to 0 \qquad \text{as} \quad |k| \to \infty \quad \text{in} \quad I_+.$$

We can therefore apply the *maximum-modulus theorem* [6, p. 165] according to which *an analytic function that is regular in a region D and on its boundary C reaches its maximum absolute value on the boundary, and not at any interior point.* We conclude that $|\varphi_l(k)| \leqslant 1$ in $I_+$, so that

$$|f_l(k)| \leqslant |k/k_l|^{2l+1}\, |e^{-2ika}| \qquad (\operatorname{Im} k \geqslant 0). \tag{3.3.37}$$

On the other hand, for any $z$ (real or complex), we have [7, p. 60]

$$|P_l(z)| \leqslant |z \pm (z^2-1)^{1/2}|^l \tag{3.3.38}$$

and

$$|z \pm (z^2-1)^{1/2}| \leqslant 2|z| + 1,$$

so that

$$|P_l(1-\tau^2/2k^2)| \leqslant (3+|\tau^2/k^2|)^l. \tag{3.3.39}$$

Let us restrict $k$ to the semicircular domain $D_K$ of $I_+$ defined by

$$D_K\colon \quad |k| \leqslant K, \qquad k \in I_+.$$

Then, according to (3.3.6), (3.3.37), and (3.3.39),

$$|f_l(k,\tau)| \leqslant u_l(\tau) = (2l+1)\,[(3K^2+|\tau|^2)^l/|k_l|^{2l+1}]\, e^{2Ka} \qquad \text{in} \quad D_K. \tag{3.3.40}$$

On the other hand,

$$\lim_{l\to\infty}[u_{l+1}(\tau)/u_l(\tau)] = (3K^2+|\tau|^2)\lim_{l\to\infty}[(1/|k_l|^2)|k_l/k_{l+1}|^{2l+1}] = 0 \tag{3.3.41}$$

by (3.3.32) (cf. Assumption I). Thus, by D'Alembert's ratio test, the partial-wave series converges absolutely and uniformly [since $u_l(\tau)$ is independent of $k$] in $D_K$. Since $K$ can be taken arbitrarily large, we conclude from Weierstrass's theorem that $f(k,\tau)$ is holomorphic in $I_+(k)$.

*Proof of ii.* Since $P_l$ is a polynomial in $\tau^2$, the function $f_l(k,\tau)$, regarded as a function of $\tau^2$, for fixed $k$ in $I_+$ (or $I_0$), is holomorphic in any domain $|\tau|^2 \leqslant T^2$. The inequality (3.3.40), with $|\tau|^2$ replaced by $T^2$, is valid in this domain, so that the partial-wave series is also absolutely and uniformly convergent with respect to $\tau^2$.

According to Hartogs' theorem [8, p. 28], a basic result in the theory of analytic functions of several complex variables, the analyticity of $f(k,\tau^2)$ in each variable separately, when the other one is kept fixed, implies its joint analyticity in both variables. Thus we can sum up the above results as follows: *$f(k,\tau^2)$ is an analytic function of both variables, regular in the topological product of the (finite) $\tau^2$-plane and the upper half of the k-plane* (i.e., the set of all pairs of points $(\tau^2,k)$ where $\tau^2$ is arbitrary and $k \in I_+$).

Let us consider next the asymptotic behavior of $f(k,\tau)$ as $|k|\to\infty$ along the real axis. For fixed $\tau$, and $|k|\to\infty$, we have $1-\tau^2/(2k^2)\to 1$, so that we expect to have

$$f(k,\tau) \to f(k,0) = O(k) \qquad \text{as} \quad |k| \to \infty \tag{3.3.42}$$

if assumption (3.1.18) is valid.

It will now be shown that this result follows from a stronger version of Assumption I, taking into account (3.3.31) and (3.3.29) for large $k$:

Assumption II. *For $k \geqslant \bar{k}$, where $\bar{k}$ can be arbitrarily large, we have*

$$|f_l(k)| \leqslant [e\beta/(2l+1)]^{2l+1} \qquad (\bar{k}a \leqslant \beta \leqslant l/N). \tag{3.3.43}$$

We could replace this by the assumption that $|\delta_l(k,a)| \leqslant 1$ under these conditions [cf. (3.3.27)], which implies (3.3.43); however, (3.3.43) is more general, since we have seen that it is valid even for an impenetrable sphere [cf. (3.3.34)], even though (3.3.27) is violated.

Note also that the assumption is being made only in the "geometrical optics" limit $k\to\infty$, since $\bar{k}$ can be taken arbitrarily large. In this limit we expect a description in terms of rays to be valid, and the "localization principle" (3.3.26) tells us that $l \geqslant N\beta$ corresponds to incident rays passing far away from the scatterer, so that the corresponding partial waves should undergo very little distortion.

Let $\tau$ be kept fixed at a real value and let us take $k \geqslant \tau/2$, so that [cf. (3.1.16)]

$$|P_l(1-(\tau^2/2k^2))| \leqslant 1.$$

Then, employing (3.3.33) for $l \leqslant [N\beta]$ and (3.3.43) for $l > [N\beta]$, where $[x]$ denotes the largest integer contained in $x$, we get from (3.3.6)

$$\begin{aligned}|f(k,\tau)| &\leqslant \frac{1}{k}\sum_{l=0}^{[N\beta]}(2l+1) + ea\sum_{l=[N\beta]+1}^{\infty}\left(\frac{e\beta}{2l+1}\right)^{2l} \\ &\leqslant \frac{(N\beta)^2}{k} + ea\sum_{l=[N\beta]}^{\infty}\left(\frac{e}{2N}\right)^{2l},\end{aligned}$$

so that, finally,

$$|f(k,\tau)| \leqslant (Na)^2 k + \frac{ea(e/2N)^{2N\beta}}{1-(e/2N)^2} \qquad (k \geqslant \bar{k},\ k \geqslant \tau/2). \tag{3.3.44}$$

According to the optical theorem (3.1.13), this implies

$$\sigma_t(k) < 4\pi(Na)^2 \quad \text{as} \quad k \to \infty, \tag{3.3.45}$$

which gives us some insight into the physical meaning of $N$.

It follows from (3.3.44) that

$$|f(k,\tau)| \leqslant C|k + i\delta| \qquad (\text{real}\,k,\ \delta > 0) \tag{3.3.46}$$

for some positive $C$ and $\delta$, as we wanted to prove. Assumption II is not the most general assumption that will lead to this result, but it is expected to hold for a wide class of scatterers. One could also postulate directly the validity of (3.3.46).

### (d) Causality Condition and Asymptotic Behavior in $I_+$

The last remaining step in the derivation of dispersion relations for fixed momentum transfer is to obtain the asymptotic behavior of $f(k,\tau)$ as $|k| \to \infty$ in $I_+$.

According to (3.3.9), each term $f_l(k,\tau)$ in (3.3.6) may be expected to blow up exponentially, like $e^{-2ika}$, as $|k| \to \infty$ in $I_+$. Nevertheless, at least for $\tau = 0$, the total amplitude, according to (3.2.7), contains no exponential factor and has at most a linear divergence. What is the physical reason for this strikingly different behavior?

As we have seen in Section 2.3, the physical origin of the exponential factor in each partial-wave amplitude is the phase advancement undergone by a spherical wave reflected at the surface of the scatterer. If such a factor were present also in the total amplitude (and, therefore, in the scattered wave), it would lead to the instantaneous appearance of a scattered wave all over the surface of the scatterer, as soon as it were hit by an incident plane wave front [cf. (3.2.1)]. This would correspond to instantaneous transmission across the diameter of the scatterer, thus violating the causality condition of Section 3.2.

According to the discussion in Section 3.2, the phases of the partial waves must be coupled by causality in such a way as to eliminate the exponential factor from the forward scattering amplitude. The heuristic argument of Section 3.3(a) suggests that the same must be true for $\tau \neq 0$.

It will now be shown that this is indeed so. For this purpose we shall apply the causality condition at a finite distance from the scatterer, in the following form:

CAUSALITY CONDITION. *For an incident wave with a plane wave front, the scattered wave cannot arrive at any point in space before the incident wave arrives there.*

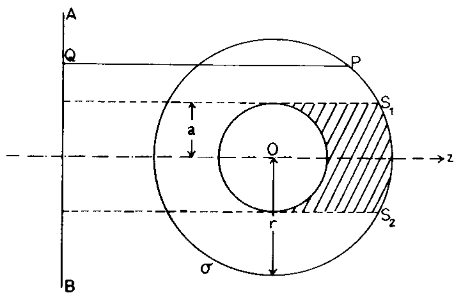


*FIG. 3.3.* For an incident wave with a plane wave front $AB$, the scattered wave cannot arrive at any point $P$ before the incident wave arrives there. If $P$ belongs to the geometrical shadow region $S_1 S_2$, no stronger condition can be formulated.

This condition is illustrated in Fig. 3.3, where $AB$ represents the incident wave front, and the observation point $P$ is at a distance $r$ from the center of the scatterer. Notice that for most points on the surface of a sphere $\sigma$ of radius $r$ centered at the scatterer, the above condition is by no means the strongest one that follows from causality. For instance, the shortest path connecting a point such as $P$ in Fig. 3.3 with the incident wave front $AB$, via the scatterer, is longer than $PQ$, so that the interval before the arrival of the scattered wave at $P$ is longer than the one implied by the above condition.

However, this condition *is* the strongest version of causality for those points on $\sigma$ that lie in the *geometrical shadow* of the scatterer (denoted by $S_1 S_2$ in Fig. 3.3). Thus it is the strongest condition that can be *uniformly* applied to all points on $\sigma$. Note that the radius of the scatterer does not appear in this formulation, just as it does not appear in the causality condition of Section 3.2 when applied to the forward direction. This will turn out to be the reason why the dispersion relations for fixed momentum transfer do not contain the radius of the scatterer. Notice also that, in contrast with the causality condition for individual partial waves, the above condition ensures the causal propagation of signals also through the scatterer.

In order to apply the causality condition, we must relate $f(k,\tau)$ with the scattered wave at a finite distance from the scatterer. Let us consider the total wave function at the point $\mathbf{R} = (R, \theta, \varphi)$ in the exterior of the scatterer, corresponding to the incident wave (3.1.1):

$$\psi(k, \mathbf{R}, t) = [e^{ikR\cos\theta} + u_s(k, R, \theta)]\, e^{-ikct} \tag{3.3.47}$$

According to Helmholtz's version of Huygens' principle [9, p. 26], the scattered wave $u_s(k, \mathbf{R})$ can be represented as follows:

$$u_s(k, \mathbf{R}) = -\oint_\sigma \left[ u_s(k, \mathbf{r}) \frac{\partial G_0}{\partial n}(k, \mathbf{r}, \mathbf{R}) - G_0(k, \mathbf{r}, \mathbf{R}) \frac{\partial u_s}{\partial n}(k, \mathbf{r}) \right] d\sigma, \tag{3.3.48}$$

where $n$ denotes the direction of the outward normal to the closed surface $\sigma$ surrounding the scatterer, and

$$G_0(k, \mathbf{r}, \mathbf{R}) = -\exp(ik|\mathbf{R}-\mathbf{r}|)/4\pi|\mathbf{R}-\mathbf{r}|. \tag{3.3.49}$$

We shall choose $\sigma$ as a sphere of radius $r = |\mathbf{r}| \geqslant a$, concentric with the scatterer (Fig. 3.3).

Now let $R \to \infty$ along the direction $\mathbf{R}/R$, so that $u_s(k, \mathbf{R}) \to f(k, \theta) e^{ikR}/R$ [cf. (3.1.2)]. Then, by (3.3.48), we find

$$f(k, \theta) = -\frac{r^2}{4\pi} \oint \left[ \frac{i\mathbf{k}\cdot\mathbf{r}}{r} u_s(k, \mathbf{r}) + \frac{\partial u_s}{\partial r}(k, \mathbf{r}) \right] e^{-i\mathbf{k}\cdot\mathbf{r}} d\Omega, \tag{3.3.50}$$

where $d\Omega$ is the element of solid angle and

$$\mathbf{k} = k\mathbf{R}/R = (k, \theta, \varphi).$$

The representation (3.3.50) holds irrespective of the radius $r$ of the sphere of integration $\sigma$. In particular, if $r \to \infty$, we may employ the well-known asymptotic formula[10]

$$e^{-i\mathbf{k}\cdot\mathbf{r}} \approx -\frac{2\pi}{ikr} [\delta(\Omega_r - \Omega_k) e^{-ikr} - \delta(\Omega_r + \Omega_k) e^{ikr}] + O(r^{-2}) \quad \text{as} \quad r \to \infty, \tag{3.3.51}$$

[10] Cf., for example, Messiah [10, p. 805].

where $\Omega_r$ and $\Omega_k$ denote the directions of $\mathbf{r}$ and $\mathbf{k}$, respectively. Using also the asymptotic form for the scattered wave,

$$u_s(k,\mathbf{r}) \approx f(k,\theta_r)e^{ikr}/r \qquad (r \to \infty),$$

we find that (3.3.50) becomes

$$f(k,\theta) = \frac{1}{2}\oint f(k,\theta_r)(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}+1)[\delta(\Omega_r-\Omega_k) - \delta(\Omega_r+\Omega_k)e^{2ikr}]\,d\Omega_r,$$

where $\hat{\mathbf{k}}$ and $\hat{\mathbf{r}}$ are unit vectors in the directions of $\mathbf{k}$ and $\mathbf{r}$, respectively. Since

$$(\hat{\mathbf{k}}\cdot\hat{\mathbf{r}}+1)\delta(\Omega_r+\Omega_k) = 0,$$

we see that (3.3.50) reduces to an identity in the limit $r \to \infty$, as it should.

Let $\mathbf{r} = (r,\theta',\varphi')$. The integration with respect to $\varphi'$ can be performed with the help of the well-known formula

$$\frac{1}{2\pi}\int_0^{2\pi} e^{iz\cos(\varphi-\varphi')}\,d\varphi' = J_0(z), \tag{3.3.52}$$

where $J_0$ denotes Bessel's function of order zero. Carrying out the integration and expressing $\theta$ in terms of $\tau$ with the help of (3.3.2), we finally get

$$\begin{aligned} f(k,\tau) = -\frac{r^2}{2}\int_0^{\pi}\Bigg\{&\Bigg[ik\cos\theta\left(1-\frac{\tau^2}{2k^2}\right)J_0(\tau r\sin\theta(1-\tau^2/4k^2)^{1/2}) \\ &+ \tau\sin\theta\left(1-\frac{\tau^2}{4k^2}\right)^{1/2} J_1\left(\tau r\sin\theta\left(1-\frac{\tau^2}{4k^2}\right)^{1/2}\right)\Bigg]u_s(k,r,\theta) \\ &+ J_0\left(\tau r\sin\theta\left(1-\frac{\tau^2}{4k^2}\right)^{1/2}\right)\frac{\partial u_s}{\partial r}(k,r,\theta)\Bigg\} \\ &\times \exp\left[-ikr\cos\theta\left(1-\frac{\tau^2}{2k^2}\right)\right]\sin\theta\,d\theta, \end{aligned} \tag{3.3.53}$$

which expresses $f(k,\tau)$ in terms of the scattered wave (and its normal derivative) over a sphere of radius $r \geqslant a$. Notice that the sign of the square root is irrelevant. One can verify directly that (3.3.53) is equivalent to (3.3.6), by substituting

$$u_s(k,r,\theta) = \tfrac{1}{2}\sum_{l=0}^{\infty}(2l+1)i^l[S_l(k)-1]h_l^{(1)}(kr)P_l(\cos\theta) \tag{3.3.54}$$

and performing the integration with the help of Gegenbauer's integral [5, p. 378].

The integral representation (3.3.53) reduces our problem to that of determining the asymptotic behavior of $u_s(k,r,\theta)$ and $\partial u_s(k,r,\theta)/\partial r$ as $|k| \to \infty$ in $I_+$. For this purpose, we shall employ the causality condition formulated above.

Let us consider an incident wave packet of the form (3.1.7):

$$\psi_{\text{inc}}(r,\theta,t) = \int_{-\infty}^{\infty} A(k)\, e^{-ik(ct-r\cos\theta)}\, dk. \tag{3.3.55}$$

The corresponding scattered wave packet can be written as

$$\psi_s(r,\theta,t) = \int_{-\infty}^{\infty} A(k)\, u_s(k,r,\theta)\, e^{-ikr\cos\theta}\, e^{-ik(ct-r\cos\theta)}\, dk. \tag{3.3.56}$$

The causality condition now implies: *if* $\psi_{\text{inc}}$ *vanishes for* $ct-r\cos\theta<0$, *so does* $\psi_s$ *(as well as* $\partial\psi_s/\partial r$*).*

Let the total incident energy per unit area be finite, so that $kA(k)$ is square integrable [cf. (3.1.9)]. Does this guarantee the square integrability of (3.3.56) and the corresponding integral for $\partial\psi_s/\partial r$? To find out, we must obtain upper bounds for $u_s(k,r,\theta)$ and $\partial u_s(k,r,\theta)/\partial r$ as $|k|\to\infty$ on $I_0$.

Such bounds can be derived from the partial-wave expansion (3.3.54) and Assumption II, in the same manner as (3.3.44), with the help of suitable inequalities and asymptotic expansions for the Hankel functions. The derivation is simplified by choosing

$$r = Na, \tag{3.3.57}$$

where $N$ is defined by (3.3.43); from now on, we shall take this value for $r$ in (3.3.53). The results are [2, Appendix B]

$$u_s(k,r,\theta) = O(k); \qquad \partial u_s(k,r,\theta)/\partial r = O(k^2) \qquad (k\to\pm\infty).$$

It follows that there exist positive constants $M$ and $\gamma$ such that

$$|u_s(k,r,\theta)| \leqslant M|k+i\gamma| \qquad \text{and} \qquad |\partial u_s(k,r,\theta)/\partial r| \leqslant M|k+i\gamma|^2$$
$$(\text{real}\, k,\ \gamma>0). \tag{3.3.58}$$

It is now possible to choose $A(k)$ in such a way that the following conditions are simultaneously fulfilled:

(i) $\psi_{\text{inc}}(r,\theta,t)=0$ for $ct-r\cos\theta<0$;
(ii) $\psi_{\text{inc}}$ and $\psi_s$ are square integrable;
(iii) The expression (3.3.56) may be differentiated under the integral sign,

$$\frac{\partial\psi_s}{\partial r}(r,\theta,t) = \int_{-\infty}^{\infty} A(k)\,\frac{\partial u_s}{\partial r}(k,r,\theta)\, e^{-ikr\cos\theta}\, e^{-ik(ct-r\cos\theta)}\, dk; \tag{3.3.59}$$

(iv) $\partial\psi_s/\partial r$ is square integrable.

The following choice satisfies all of these conditions:

$$A(k) = A_0/(k+i\gamma)^4, \tag{3.3.60}$$

where $A_0$ is a constant with appropriate dimensionality.

We can now apply Titchmarsh's theorem 1.6.1 and conclude from the causality condition that

$$u_s(k,r,\theta) \quad \text{and} \quad \partial u_s(k,r,\theta)/\partial r \quad \text{are holomorphic in } I_+(k). \tag{3.3.61}$$

Furthermore, just as in (3.2.7), it follows from (3.3.58), by applying the Phragmén–Lindelöf theorem, that

$$|u_s(k,r,\theta)\, e^{-ikr\cos\theta}| \leqslant M|k+i\gamma| \qquad (\operatorname{Im} k \geqslant 0), \tag{3.3.62}$$

$$\left|\frac{\partial u_s}{\partial r}(k,r,\theta)\, e^{-ikr\cos\theta}\right| \leqslant M|k+i\gamma|^2 \qquad (\operatorname{Im} k \geqslant 0), \tag{3.3.63}$$

which are the desired results.

It follows from (3.3.53) and the above relations, with the help of the inequalities [5, p. 49]

$$|J_0(z)| \leqslant e^{\operatorname{Im} z}, \qquad |J_1(z)| \leqslant \tfrac{1}{2}|z|\, e^{\operatorname{Im} z}, \tag{3.3.64}$$

that

$$|f(k,\tau)| \leqslant Mr^2|k+i\gamma|\,[|k+i\gamma| + |k| + |\tau^2/2k| + \tfrac{1}{2}r\tau^2(1+|\tau/2k|^2)] \\ \times \exp[\tau r(|\tau/2k| + (1+|\tau/2k|^2)^{1/2})] \qquad (\operatorname{Im} k \geqslant 0), \tag{3.3.65}$$

where $r = Na$.

For $|k| \to \infty$, all terms containing $k$ in the denominator disappear, and we get

$$f(k,\tau) = O(k^2) \qquad \text{as} \quad |k| \to \infty \quad \text{in} \quad I_+. \tag{3.3.66}$$

Taking into account (3.3.46) and the regularity of $(k+i\delta)^{-1}f(k,\tau)$ in $I_+$, we may again apply the Phragmén–Lindelöf theorem and conclude that

$$|f(k,\tau)| \leqslant C|k+i\delta| \qquad (\operatorname{Im} k \geqslant 0). \tag{3.3.67}$$

Thus *the troublesome exponential factor is indeed eliminated by the causality condition.* In conclusion, the analytic properties of $f(k,\tau)$ for $\tau \neq 0$ are similar to those of $f(k,0)$, so that it satisfies a dispersion relation of the same form [cf. (3.2.11)]:

$$\operatorname{Re} f(k,\tau) = f(0,\tau) + \frac{2k^2}{\pi} \int_0^\infty \frac{\operatorname{Im} f(k',\tau)}{k'(k'^2-k^2)}\, dk'. \tag{3.3.68}$$

This is the *dispersion relation for fixed momentum transfer.*

For $\tau \neq 0$, both $f(0,\tau)$ and $\operatorname{Im} f(k',\tau)$ for $k' < \tau/2$ are in the unphysical region; in a practical application of the dispersion relation, these quantities have to be computed by means of the partial-wave expansion (3.3.6), which relates them to the physically measurable phase shifts.

Actually, if we exclude a neighborhood of $k = 0$, the regularity of $f(k, \tau)$ in $I_+(k)$ follows directly from (3.3.53) and the causality condition, together with (3.3.58). In fact, let us consider the domain $D_{\varepsilon,K}$ of $I_+$ defined by

$$D_{\varepsilon,K}: \qquad 0 < \varepsilon \leqslant |k| \leqslant K, \qquad k \in I_+. \tag{3.3.69}$$

The coefficients of $u_s$ and $\partial u_s/\partial r$ in (3.3.53) are regular in $D_{\varepsilon,K}$, so that, according to (3.3.61), the whole integrand is regular in $D_{\varepsilon,K}$. On the other hand, if we replace $|k|$ by $\varepsilon$ in the denominators of (3.3.65) and by $K$ in the numerators, we obtain an upper bound, independent of $k$ and $\theta$, for the integrand of (3.3.53) in $D_{\varepsilon,K}$. Thus the integrand is a regular, uniformly bounded function of $k$ in $D_{\varepsilon,K}$.

We can therefore apply the following theorem [11, p. 257]:

THEOREM 3.3.1. *Let $F(t,z)$ be defined for $a \leqslant t \leqslant b$ and $z$ in a domain $D$, and such that: (i) $|F(t,z)| \leqslant M$ for $t$ in $(a,b)$ and $z \in D$; (ii) $F(t,z)$ is integrable with respect to $t$ over $(a,b)$ for each fixed $z \in D$; (iii) $F(t,z)$ is a holomorphic function of $z$ in $D$ for each fixed $t$ in $(a,b)$. Then*

$$f(z) = \int_a^b F(t,z)\,dt$$

*is a holomorphic function of $z$ in $D$.*

It follows from this theorem that $f(k, \tau)$ is regular in $D_{\varepsilon,K}$. Since $K$ may be taken arbitrarily large and $\varepsilon$ arbitrarily small, the above assertion is proved.

However, the regularity at $k = 0$ cannot be proved by this method, because of the denominators in (3.3.53). This is the reason why we had to employ the partial-wave expansion and Assumption I. Furthermore, as remarked above, the partial-wave expansion is required anyway, as it provides the sole means for evaluating $f(k, \tau)$ in the unphysical region.

On the other hand, the analyticity of $f(k, \tau)$ as a function of $\tau^2$ follows from (3.3.53) without any restrictions.

As was mentioned in connection with (3.3.34), all of the above assumptions are satisfied in the case of scattering by an impenetrable sphere, so that the corresponding scattering amplitude verifies the dispersion relation (3.3.68). The "subtraction constant" $f(0, \tau)$ can be explicitly computed in this example, by taking the limit of (3.3.6) as $k \to 0$, with $f_l(k)$ given by (3.3.34):

$$f(0,\tau) = \sum_{l=0}^{\infty} f_l(0,\tau) = a \sum_{l=0}^{\infty} (-1)^{l+1} \frac{(\tau a)^{2l}}{(2l)!} = -a\cos(\tau a). \tag{3.3.70}$$

In particular, for $\tau = 0$, this reduces to the well-known result for the scattering length: $f(0,0) = -a$ [cf. (3.2.13)].

It can also be shown that, for an impenetrable sphere,

$$f(k,\tau) \approx ika^2(J_1(\tau a)/\tau a) + O[(ka)^{1/3}] \qquad (k \to \infty). \tag{3.3.71}$$

In particular, for $\tau = 0$,

$$f(k,0) = \tfrac{1}{2}ika^2 + O[(ka)^{1/3}] \qquad (k \to \infty), \tag{3.3.72}$$

a well-known result,[11] implying that the total cross section approaches $2\pi a^2$ (i.e., twice the geometrical cross section) as $k \to \infty$. The physical origin of this effect is the diffraction peak in the forward direction ("shadow scattering"). We see from (3.3.71) that the number of subtractions in the dispersion relation (3.3.68) cannot be decreased, unless additional restrictions are imposed upon the scatterer.

## 3.4. Extension to Nonrelativistic Quantum Scattering

### *(a) Analytic Properties of the Scattered Wave*

So far we have discussed the analytic properties of the total scattering amplitude only for the case of a classical field. Let us now extend the results to nonrelativistic quantum scattering, taking over the assumptions formulated in Section 2.7.

With suitable reinterpretation, all the results of Section 3.1 remain valid (e.g., we have to speak of probability flow instead of energy flow); the factor $e^{-ikct}$ should be replaced by $e^{-iEt}$, where $E = k^2/2$, as in (2.7.3). The symmetry relation (3.1.6) represents an additional postulate (cf. Sections 2.8 and 2.9).

The total Schrödinger wave function is given by [cf. (3.3.47)]

$$\psi(k,\mathbf{R},t) = [e^{ikr\cos\theta} + u_s(k,R,\theta)]\,e^{-iEt}, \tag{3.4.1}$$

where $u_s(k,R,\theta)$ still obeys the Helmholtz wave equation outside of the scatterer, so that Huygens' principle representations (3.3.48), (3.3.50) remain valid. With the help of (3.3.52), we find

$$\begin{aligned} f(k,\theta) = -\frac{r^2}{2}\int_0^\pi &\Big\{[ik\cos\theta\cos\theta' J_0(kr\sin\theta\sin\theta') + k\sin\theta\sin\theta' \\ &\times J_1(kr\sin\theta\sin\theta')]\,u_s(k,r,\theta') + J_0(kr\sin\theta\sin\theta')\frac{\partial u_s}{\partial r}(k,r,\theta')\Big\} \\ &\times e^{-ikr\cos\theta\cos\theta'}\sin\theta'\,d\theta' \qquad (r \geqslant a). \end{aligned} \tag{3.4.2}$$

[11] See, for example, Nussenzveig [12, p. 71].

The relations (3.3.1), (3.3.2) between scattering angle and momentum transfer remain valid, so that we can still employ the representation (3.3.53).

We take over Assumptions I and II of Section 3.3(c); they can be justified by exactly the same reasoning as in the classical case. It follows, in particular, that the high-energy bounds (3.3.58) (for real $k$) remain valid.

Let us consider a wave packet of incident particles,

$$\psi_{\text{inc}}(r,\theta,t) = \int_0^\infty A(E)\, e^{ikr\cos\theta - iEt}\, dE. \tag{3.4.3}$$

By the assumed linearity of the interaction, the corresponding scattered wave packet is

$$\psi_s(r,\theta,t) = \int_0^\infty A(E)\, u_s(k,r,\theta)\, e^{-iEt}\, dE, \tag{3.4.4}$$

so that it is a linear functional of the incident wave packet,

$$\psi_s(r,\theta,t) = \int_{-\infty}^\infty g(r,\theta,t-t')\, \psi_{\text{inc}}(r,\theta,t')\, dt'. \tag{3.4.5}$$

We shall base all our analysis on a causality condition that represents the extension of the Schützer–Tiomno condition of Section 2.8 to the present case [13]:

Causality Condition. *The scattered wave packet at $(r,\theta)$ $(r \geqslant a)$ at time $t$ does not depend on the behavior of the corresponding incident wave packet at $(r,\theta)$ at later times $t' > t$.*

Thus [cf. (2.8.2)]

$$\delta\psi_s(r,\theta,t)/\delta\psi_{\text{inc}}(r,\theta,t') = 0 \qquad (t' > t). \tag{3.4.6}$$

By (3.4.5), this implies

$$g(r,\theta,\tau) = 0 \qquad (\tau < 0). \tag{3.4.7}$$

We assume next that $g(r,\theta,\tau) \in \mathscr{S}_\tau'$, i.e., that it is a temperate distribution in the time variable, so that we can define its Fourier transform,

$$g(r,\theta,\tau) = \int_{-\infty}^\infty \mathscr{G}(r,\theta,E)\, e^{-iE\tau}\, dE, \tag{3.4.8}$$

with

$$\mathscr{G}(r,\theta,E) \in \mathscr{S}_E'. \tag{3.4.9}$$

We also choose $A(E)$ in (3.4.3) in such a manner that the convolution theorem applies [cf. the discussion in Section 2.8(c)], so that (3.4.5) yields

$$\theta(E)\, A(E)\, u_s(k,r,\theta) = 2\pi \mathscr{G}(r,\theta,E)\, \theta(E)\, A(E)\, e^{ikr\cos\theta}, \tag{3.4.10}$$

where $\theta$ is the Heaviside step function. Thus

$$\mathscr{G}(r,\theta,E) = (1/2\pi)u_s(k,r,\theta)e^{-ikr\cos\theta} \qquad (E>0). \tag{3.4.11}$$

The causality condition (3.4.7) now implies[12] that $g(r,\theta,\tau) = \mathscr{F}^{-1}\mathscr{G}(r,\theta,E) \in \mathscr{D}_+'$ [cf. Section 1.8(a)]. Together with (3.4.9), this implies that $\mathscr{G}(r,\theta,E)$ fulfills conditions (a), (b), and (c) of Theorem 1.8.2. We conclude that $u_s(k,r,\theta)$ *is the boundary value (for* $E>0$*) of an analytic function, regular in* $I_+(E)$*, and*

$$|e^{-ikr\cos\theta}u_s(k,r,\theta)| = O(E^n) \qquad (\operatorname{Im} E \geqslant \varepsilon > 0;\ r \geqslant a). \tag{3.4.12}$$

The causality condition can also be applied to $\partial u_s/\partial r$, and similar considerations lead to the conclusion that $\partial u_s(k,r,\theta)/\partial r$, *regarded as a function of* $E$*, has a regular analytic continuation in* $I_+$ *and is subject to an upper bound similar to* (3.4.12):

$$|e^{-ikr\cos\theta}\,\partial u_s(k,r,\theta)/\partial r| = O(E^m) \qquad (\operatorname{Im} E \geqslant \varepsilon > 0;\ r \geqslant a). \tag{3.4.13}$$

What happens as $E$ approaches the real axis? For $E>0$, we get the physical scattered wave (3.4.11), which is regular and bounded by (3.3.58). However, for $E<0$, singularities cannot be excluded. In fact, we know from Chapter 2 that bound states give rise to simple poles for $E<0$ in the partial-wave amplitudes $f_l(k)$, and therefore also in $u_s(k,r,\theta)$ [cf. (3.3.54)]. As we have seen in (2.8.23), such poles are associated with $\delta^+$-type singularities [cf. (A11.5)] in $\mathscr{G}(r,\theta,E)$, which are compatible with (3.4.9).

The possible singularities of $\mathscr{G}(r,\theta,E)$ on the real axis are restricted by a condition that may be regarded as an addition[13] to Theorem 1.8.2:

THEOREM 3.4.1. *Under the assumptions of Theorem 1.8.2, we have, in addition to* (*a*), (*b*), *and* (*c*) *of that theorem,*

(d) *For any interval* $0 < v_1 \leqslant v \leqslant v_2$, *there exists a polynomial* $P_n(u)$ *and an integer* $m$ *such that*

$$|G(u+i\zeta v)| \leqslant P_n(u)/\zeta^m \tag{3.4.14}$$

*for all* $u$, *all* $\zeta$ *with* $0<\zeta<1$, *and all* $v$ *in the above interval.*

The polynomial boundedness in $u$ is already contained in condition (b) of Theorem 1.8.2; the new result in (3.4.14) is that $|G(u+iv)|$ can diverge at most like $v^{-m}$ as $v\to 0$ from above.

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[12] As in Section 2.8(c), Green's function is not unique; by (3.4.11), it is subject to the same ambiguity (2.8.25). What is implied by the causality condition is that *there exists a unique causal Green function, satisfying (3.4.7).*

[13] For a proof, see Streater and Wightman [14, p. 62].

To proceed further, just as in Sections 2.8(b) and 2.9(b), we need an additional mathematical assumption: we assume that $\mathscr{G}(r, \theta, E)$ *has a single-valued analytic continuation across* $E < 0$, *with only a finite number of singularities on the negative real axis.*

As in Section 2.8(b), the single-valued analytic continuation of $\mathscr{G}(r, \theta, E)$ to $I_-(E)$ that is compatible with the symmetry relation [cf. (3.1.6)]

$$\mathscr{G}(r, \theta, E-i0) = \mathscr{G}^*(r, \theta, E+i0) \qquad (E > 0) \tag{3.4.15}$$

is provided by Schwarz's reflection principle,

$$\mathscr{G}(r, \theta, E) = \mathscr{G}^*(r, \theta, E^*). \tag{3.4.16}$$

The reality of $\mathscr{G}(r, \theta, E)$ at any regular point on the negative real axis is compatible with the corresponding reality of each term in the partial-wave expansion (3.3.54) for positive imaginary $k$.

It also follows from (3.4.16) that a bound similar to (3.4.14) remains valid as $E$ approaches the negative real axis from below. Since $\mathscr{G}(r, \theta, E)$ can have only isolated singularities on $E < 0$ by the above assumption, it follows that the only possible singularities are poles of order $\leqslant m$. In view of the connection with bound states, we further assume that $m = 1$.

The above assumptions thus imply that $u_s(k, r, \theta)$ *has a regular analytic continuation in* $I_+(k)$, *except possibly for a finite number of simple poles* $iK_p$ $(p = 1, 2, \ldots, P)$ *on the positive imaginary axis, corresponding to bound states.*[14]

Let us consider the function

$$\mathscr{F}(k, r, \theta) = \prod_{p=1}^{P} \frac{(k+iK_p)}{(k-iK_p)} \frac{u_s(k, r, \theta)}{M(k+i\gamma)} e^{-ikr\cos\theta}, \tag{3.4.17}$$

where $M$ and $\gamma$ are defined by (3.3.58). This function is regular in $I_+(k)$, and $|\mathscr{F}(k, r, \theta)| \leqslant 1$ for real $k$, according to (3.3.58). Furthermore, according to (3.4.12) and (3.4.16),

$$|k|^{-1} \ln|\mathscr{F}(k, r, \theta)| \to 0 \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0. \tag{3.4.18}$$

It then follows from the Phragmén–Lindelöf theorem 2.5.4 that

$$|\mathscr{F}(k, r, \theta)| \leqslant 1 \qquad \text{for all} \quad k \in I_+. \tag{3.4.19}$$

Thus

$$|e^{-ikr\cos\theta} u_s(k, r, \theta)| = O(k) \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0 \quad (r \geqslant a). \tag{3.4.20}$$

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[14] In fact, according to (3.3.54), $u_s(k, r, \theta)$ behaves asymptotically like $\exp(-K_p r)/r$ as $r \to \infty$, for $k = iK_p$.

Similar considerations show that $\partial u_s(k,r,\theta)/\partial r$ has the same analytic properties as $u_s(k,r,\theta)$ in $I_+(k)$ and is bounded by

$$\left| e^{-ikr\cos\theta} \frac{\partial u_s}{\partial r}(k,r,\theta) \right| = O(k^2) \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0 \quad (r \geqslant a). \tag{3.4.21}$$

These results correspond to (3.3.62) and (3.3.63).

### (b) Analytic Properties of Scattering Amplitudes

Now let us apply the above results on $u_s$ and $\partial u_s/\partial r$ to derive the analytic properties of scattering amplitudes. Let us consider first the total scattering amplitude $f(k,\theta)$ for fixed scattering angle.

The coefficients of $u_s$ and $\partial u_s/\partial r$ in (3.4.2) are entire functions of $k$. The above properties of $u_s$ and $\partial u_s/\partial r$ therefore imply that the integrand of (3.4.2) is a regular, uniformly bounded analytic function of $k$ in any compact domain of the upper half-plane, excluding a neighborhood of each pole. According to Theorem 3.3.1, it follows that $f(k,\theta)$ *has a regular analytic continuation in* $I_+(k)$, *except possibly for a finite number of simple poles at* $k = iK_p$.

Furthermore, by applying (3.4.2) at $r = a$, it follows from (3.4.20) and (3.4.21) that, for sufficiently large $|k|$ in $I_+$, there exists a constant $M$ such that

$$|f(k,\theta)| \leqslant Ma^2|k|^2 \int_0^\pi [2|J_0(ka\sin\theta\sin\theta')| + |J_1(ka\sin\theta\sin\theta')|] \times |e^{ika\cos\theta'(1-\cos\theta)}| \sin\theta'\, d\theta'. \tag{3.4.22}$$

Let

$$k = k' + iK \qquad (K \geqslant 0). \tag{3.4.23}$$

Then, with the help of (3.3.64), we get

$$|f(k,\theta)| \leqslant M'a^3|k|^3 \int_0^\pi e^{\alpha\cos\theta' + \beta\sin\theta'} \sin\theta'\, d\theta', \tag{3.4.24}$$

where

$$\alpha = -Ka(1-\cos\theta); \qquad \beta = Ka\sin\theta. \tag{3.4.25}$$

We have

$$\exp(\alpha\cos\theta' + \beta\sin\theta') \leqslant \exp(\alpha^2+\beta^2)^{1/2} = \exp[2Ka\sin(\theta/2)], \tag{3.4.26}$$

so that (3.4.24) may be rewritten as

$$|e^{2ika\sin(\theta/2)} f(k,\theta)| \leqslant 2M'a^3|k|^3 \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0. \tag{3.4.27}$$

On the real axis, according to (3.1.18), $|f(k,\theta)| = O(k)$ as $|k| \to \infty$. Therefore, we can apply once more the Phragmén–Lindelöf theorem, just as we did in connection with (3.4.17), to conclude that, actually,

$$|e^{2ika\sin(\theta/2)}f(k,\theta)| = O(k) \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0. \tag{3.4.28}$$

This property is the analog of (3.2.7). In summary, $f(k,\theta)$ *has exactly the same properties as in the classical case* (*Section 3.2*), *except for the possible occurrence of a finite number of simple poles on the positive imaginary axis, corresponding to bound states.*

The analytic properties of partial-wave amplitudes follow from these results by partial-wave projection, as in (3.1.4), (3.2.14). We conclude that $S_l(k)$ *is regular in* $I_+$, *except possibly for a finite number of simple poles on the positive imaginary axis, and*

$$|S_{la}(k)| = |S_l(k)e^{2ika}| \leqslant 1 \qquad (|k| \to \infty \text{ in } I_+). \tag{3.4.29}$$

This last inequality follows from (3.2.14) and (3.4.28), again by the same method that led to (3.4.19). The results found in Chapter 2 for $s$-waves are thus extended to all values of the angular momentum.

Finally, (3.3.53) allows us to derive the analytic properties of the scattering amplitude as a function of $k$ and the momentum transfer $\tau$. For fixed momentum transfer, it follows from (3.3.53) and the above properties of $u_s$ and $\partial u_s/\partial r$, just as in Section 3.3, that $f(k,\tau)$ *has a regular analytic continuation in* $I_+(k)$, *with the possible exception of a finite number of simple poles on the positive imaginary axis.* If we exclude an arbitrarily small neighborhood of $k = 0$, these results follow directly from the representation (3.3.53) and the properties of $u_s$ and $\partial u_s/\partial r$, as remarked in Section 3.3(d). The regularity at the origin follows from the partial-wave expansion (3.3.6), with the help of Assumption I and the analytic properties of partial-wave amplitudes derived above. Alternatively, as in Section 3.3(c), we can use these properties to prove the analyticity in $I_+(k)$. Assumption II is not really necessary; it can be replaced by the more general requirement that the bounds (3.3.58) must hold.

Furthermore, as $|k| \to \infty$, it follows from (3.3.53), (3.4.20), (3.4.21), (3.3.46) and the Phragmén–Lindelöf theorem, as in Section 3.3(d) and (3.4.19), that

$$f(k,\tau) = O(k) \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0. \tag{3.4.30}$$

Thus *the troublesome exponential factor in the partial-wave amplitudes* (*3.4.29*) *is again eliminated, as a direct consequence of the causality condition.*

These results allow us to derive a dispersion relation for fixed momentum transfer. For this purpose, it is convenient to regard the scattering amplitude as a function of the energy $E = k^2/2$ and the momentum transfer $\tau$. Its analytic properties in the $E$-plane are then entirely analogous to those of the function

$S_a(E)$ discussed in Section 2.9(c) [except that it is $O(E^{-1/2})$ as $|E| \to \infty$, instead of (2.9.39)]. Therefore, we can write down at once the corresponding dispersion relation, by analogy with (2.9.41):

$$f(E,\tau) = f(0,\tau) + \frac{E}{\pi}\int_0^\infty \frac{\operatorname{Im} f(E'+i0,\tau)}{E'(E'-E)}\,dE' + E\sum_p \frac{R_p(\tau)}{E_p(E-E_p)}, \tag{3.4.31}$$

where $E_p$ are the energies of the bound states, and

$$R_p(\tau) = \operatorname{res} f(E,\tau)\Big|_{E=E_p}. \tag{3.4.32}$$

This is the dispersion relation for fixed momentum transfer. In particular, in the absence of bound states, it reduces to (3.3.68). In the unphysical region (3.3.8), the value of the amplitude can be computed by means of the partial-wave expansion (3.3.6).

Finally, let us consider the analytic behavior of $f(k,\tau)$ as a function of $\tau^2$. The dependence on $\tau^2$ is contained only in the coefficients of $u_s$ and $\partial u_s/\partial r$ in (3.3.53), which are entire functions of $\tau^2$. Therefore, according to Theorem 3.3.1, *$f(k,\tau)$ is an entire function of $\tau^2$ for any value of $k$ such that $u_s$ and $\partial u_s/\partial r$ are bounded and integrable with respect to $\theta$ over* $(0,\pi)$. Taking into account the above results, we can say that *$f(k,\tau^2)$ is an analytic function of both variables, regular in the topological product of the (finite) $\tau^2$-plane and the upper half of the $k$-plane, except possibly for a finite number of simple poles on the positive imaginary axis, corresponding to bound states.*

It is remarkable that, for physical (real) values of $k$, the analyticity in momentum transfer is in no way dependent on causality. It follows immediately from the representation (3.3.53) (and the boundedness and integrability of $u_s$ and $\partial u_s/\partial r$ with respect to $\theta$), which in its turn follows from the cutoff nature of the interaction (i.e., the absence of interaction for $r > a$). Thus *analyticity in the momentum transfer is related with the short-range character of the interaction, rather than with causality.* A similar conclusion has been reached in quantum field theory [15]. A more detailed discussion of this relationship will be given in Section 6.6.

We have seen in the present section that the causality condition (3.4.6), together with the remaining assumptions, enables us to derive the analytic properties of the scattering amplitude in both variables and, by projection, also those of the partial-wave amplitudes. It is interesting to note that the causality condition employed for a classical massless field [Section 3.3(d)] is also a particular case of the present one: in that situation, as we have seen, it is possible to build an incident wave packet that vanishes for $t < t_0$, so that, by (3.4.6), the corresponding scattered wave packet also vanishes for $t < t_0$. Thus the present treatment, suitably modified (the dispersion formula

$E = k^2/2$ is replaced by $\omega = ck$, and $\omega$ ranges from $-\infty$ to $\infty$), can be applied to both situations. Even properties such as (3.2.7), (3.4.28), which, by the argument given in Section 3.2, seemed to depend on the relativistic version of the causality condition (III′ of Section 1.4), are now seen to follow directly from the primitive form of the causality condition (III, Section 1.3), which does not depend upon the existence of a limiting velocity for the propagation of signals.

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CHAPTER

# 4

# PHYSICAL INTERPRETATION OF $S$-MATRIX SINGULARITIES

*If an agitation be communicated to the nucleus in any way, waves will be started in the medium ... The point to be here considered arises in the interpretation of the analytical expression for the waves thus generated.*

H. LAMB[1]

## 4.1. Introduction

The singularities of analytic functions play an important role in their characterization. For the physical systems that we have discussed so far, the elements of the $S$-matrix in the angular momentum representation are meromorphic functions of $k$. We have seen in Sections 2.5 and 2.9(d) that they are completely determined by the set of all their singularities (poles and essential singularity at infinity). This leads one to expect that such singularities, or at least some of them, should have a physical interpretation. The present chapter is devoted to a discussion of this problem.

In Heisenberg's original program for deriving all observable quantities from the $S$-matrix (cf. Section 1.1), the following quantities (besides those that characterize free particles) were regarded as observable: (a) the cross

[1] *Proc. London Math. Soc.* **32**(1), 208 (1900).

sections for all possible collision processes; (b) the energy levels of bound states; (c) the lifetime and decay energy of unstable states.

For the systems that we have considered, only elastic scattering is possible. The corresponding partial-wave cross section is given by (2.6.11) for both classical and quantum scattering [cf. (3.1.12)]:

$$\sigma_l(k) = (\pi/k^2)(2l+1)|1 - S_l(k)|^2 = (4\pi/k^2)(2l+1)\sin^2\eta_l, \quad (4.1.1)$$

where $\eta_l(k)$ is the scattering phase shift.

The energy levels of bound quantum states, as discussed in Section 2.8(b), are associated with poles $iK_p$ of $S_l(k)$ on the positive imaginary axis, the corresponding energy being given by (2.8.13),

$$E_p = \tfrac{1}{2}(iK_p)^2 < 0. \quad (4.1.2)$$

In addition to these *bound-state poles* in $I_+$, $S_l(k)$ may have poles in $I_-$. Do such poles also have a physical interpretation?

It was proposed by Møller [1] and by Heitler and Hu [2] that complex poles of the $S$-matrix in $I_-$ may be related to the lifetime and decay energy of unstable states. Before turning to a discussion of this problem, let us examine the possible effects of such poles on the scattering cross section.

## 4.2. Effects on the Cross Section

The simplest example of the possible effects of complex poles on the cross section is that of an *isolated low-energy resonance*. The low-energy assumption means that

$$ka \ll 1 \quad (4.2.1)$$

and this is consistent with assuming that only the $s$-wave contributes. Consider a situation in which $S_0(k)$ has a simple pole near the origin,

$$k_0 = k_0' - iK_0, \qquad 0 < k_0' \ll 1/a, \quad K_0 > 0, \quad (4.2.2)$$

located very close to the real axis and well separated from other poles,

$$K_0 \ll k_0', \qquad K_0 \ll |k_1 - k_0|, \quad (4.2.3)$$

where $k_1$ is the "nearest-neighbor" pole in the fourth quadrant (Fig. 4.1).

In a sufficiently small neighborhood of the pole, $S_0(k)$ may be approximated by the principal part of its Laurent expansion,

$$S_0(k) \approx r_0/(k-k_0). \quad (4.2.4)$$

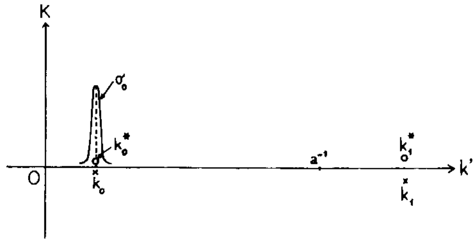


*FIG. 4.1.* At low energy, a complex pole $k_0$ located close to the real axis and far away from other poles gives rise to a Breit-Wigner type resonance in the cross section $\sigma_0$ (×, poles; O, zeros).

In this neighborhood, we say that the behavior of $S_0(k)$ is *dominated* by the pole $k_0$. Under the above conditions, it is plausible to assume that this neighborhood extends up to the real axis $k = k'$, at least over the domain

$$|k' - k_0'| \lesssim K_0. \tag{4.2.5}$$

It must be remembered, however, that $S_0(k)$ is unitary over the real axis, so that the approximation corresponding to (4.2.4) for real $k$ is

$$S_0(k') \approx \frac{k'-k_0^*}{k'-k_0} = \frac{k'-k_0'-iK_0}{k'-k_0'+iK_0}, \qquad (|k'-k_0'| \lesssim K_0). \tag{4.2.6}$$

The numerator, that has been chosen so as to ensure unitarity, renders apparent the effect of the complex conjugate zero $k_0^*$ of $S_0(k)$, located in $I_+$ (cf. Fig. 2.1).

We shall refer to (4.2.6) as the *one-pole approximation*. It can also be obtained as an approximation to the canonical product expansion (2.9.58), in which essentially only the factor associated with the pole $k_0$ contributes; the remaining factors are replaced by unity because of (4.2.3), and $e^{-2ika} \approx 1$ because of (4.2.1).

Substituting (4.2.6) in (4.1.1) for $l = 0$, we get

$$\sigma_0 \approx \frac{\pi}{k'^2}\frac{4K_0^2}{[(k'-k_0')^2 + K_0^2]} \approx \frac{\pi}{k'^2}\frac{\Gamma_0^2}{[(E'-E_0')^2 + (\Gamma_0^2/4)]} \; (|E'-E_0'| \lesssim \Gamma_0), \tag{4.2.7}$$

where $E' = k'^2/2$ is the real energy, and

$$E_0 = E_0' - i(\Gamma_0/2) = \tfrac{1}{2}k_0^2 = \tfrac{1}{2}(k_0'^2 - K_0^2) - ik_0'K_0 \tag{4.2.8}$$

is the "complex energy" associated with the pole $k_0$.

The cross section (4.2.7) has a sharp "Lorentzian" resonance peak centered at $E' = E_0'$ (resonance energy) and of half-width $\Gamma_0$. The phase shift corresponding to (4.2.6),

$$\eta_{\text{res}} \approx \tan^{-1}[K_0/(k_0'-k')] \approx \tan^{-1}[\Gamma_0/2(E_0'-E')], \tag{4.2.9}$$

is rapidly varying and changes by approximately $\pi$ across the resonance peak. Both effects are similar to those shown in Fig. 1.2 for a typical resonance.

The cross section (4.2.7) corresponds to the well-known *Breit–Wigner one-level formula*[2] for resonance scattering. In general, one must take into account also the contribution from the remaining pole factors in the canonical product expansion, as well as that of the factor $e^{-2ika}$ [especially if (4.2.1) is not satisfied]. They give rise to a slowly varying "background" contribution to the phase shift, which must be added to the resonant contribution (4.2.9):

$$\eta = \eta_{\text{res}} + \eta_{\text{pot}}, \tag{4.2.10}$$

where $\eta_{\text{pot}}$, the background contribution, is responsible for what is called "potential scattering." The contribution $-ka$ to $\eta_{\text{pot}}$ from the factor $e^{-2ika}$ corresponds to "hard-sphere scattering" [cf. (2.3.8)]. Potential scattering plays an important role outside of resonance; it can also interfere with resonance scattering,[3] distorting the resonance shape.

We see, therefore, that a pole $k_0$ sufficiently close to the real axis and far away from other poles can give rise to a resonance peak in the cross section, with resonance energy and half-width, respectively, given by the real and the imaginary part of the complex energy associated with the pole [cf. (4.2.8)]. As a pole gets farther away from the real axis, the corresponding resonance peak gets broader, and if the poles are not far apart one must take into account the interference among contributions from different poles.

A special case of the one-pole approximation is that of a pole $iK$ located close to the origin on the negative imaginary axis ($K < 0$, $|K|a \ll 1$). The "energy" $(iK)^2/2$ associated with such a pole is real and negative, just like that for a bound state [cf. (2.8.13)], but the corresponding "wave function" (2.8.12) is a purely *increasing* (rather than purely decreasing) exponential in $r$, so that it is not normalizable. Such a pole is sometimes said to correspond to an "antibound state." The corresponding one-pole approximation is obtained by keeping only one factor in the last Blaschke product of (2.9.58):

$$S_0(k') \approx (iK+k')/(iK-k') \qquad (k' \lesssim |K|).$$

Substituting in (4.1.1), we find for the cross section

$$\sigma_0(k') \approx 4\pi/(k'^2+K^2) \qquad (k'a \lesssim |K|a \ll 1). \tag{4.2.11}$$

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[2] See Breit and Wigner [3]; also Blatt and Weisskopf [4, Chapter VIII].

[3] Cf. Blatt and Weisskopf [4, Chapter VIII].

This corresponds to a very large low-energy cross section $[\sigma_0(0) = 4\pi/K^2 \gg 4\pi a^2]$. For $K \to 0$, we would have $\sigma_0(0) \to \infty$, corresponding to the fact that $S_0(0) = -1$ instead of 1 in this limit [cf. (2.3.10)].

Thus an isolated negative imaginary pole close to the origin gives rise to an anomalously large low-energy cross section. Experimentally, a cross section of the form (4.2.11) has been found in low-energy neutron–proton scattering.[4] It has been interpreted in terms of a "virtual singlet state of the deuteron": the neutron–proton interaction in the singlet state is not quite strong enough to bind them; instead of a bound-state pole, we have one corresponding to an "antibound state" near the origin.

The above discussion illustrates the main effects due to isolated poles on the scattering cross section. In particular, the connection between complex poles and resonance peaks is similar to that found in the simple example of a damped harmonic oscillator, discussed in Section 1.2. This suggests that there may also be a similar connection between complex poles and exponentially decaying "free modes of oscillation" of the system. Let us now turn our attention to this problem.

## 4.3. Complex Poles and Unstable States

Consider the $s$-wave stationary solution of Schrödinger's equation outside of the scatterer, given by (2.7.2),

$$r\psi(k,r,t) = [A(E)e^{-ikr} + B(E)e^{ikr}]\,e^{-iEt}, \qquad E = k^2/2 \quad (r > a), \tag{4.3.1}$$

in terms of which the $S$-function is given by (2.7.5),

$$S_0(k) = -B(E)/A(E). \tag{4.3.2}$$

If we can, by analytic continuation, extend the definition of $\psi$ to complex values of $k$, a pole

$$k_n = k_n' - iK_n \qquad (K_n > 0) \tag{4.3.3}$$

of $S_0(k)$ will correspond to a "purely outgoing" solution, with $A(E_n) = 0$, so that [cf. (2.8.12)]

$$\begin{aligned} r\psi(k_n,r,t) &= B(E_n)\exp(ik_n r - iE_n t) \\ &= B\exp(ik_n' r - iE_n' t)\exp(K_n r - \tfrac{1}{2}\Gamma_n t), \end{aligned} \tag{4.3.4}$$

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[4] Cf. Blatt and Weisskopf [4, Chapter II].

where $E_n = k_n^2/2$ is the "complex energy" associated with the pole, so that [cf. (4.2.8)]

$$E_n = E_n' - (i/2)\Gamma_n, \qquad E_n' = \tfrac{1}{2}(k_n'^2 - K_n^2), \quad \Gamma_n = 2k_n'K_n. \tag{4.3.5}$$

It follows that

$$|r\psi(k_n, r, t)|^2 = |B|^2 \exp(2K_n r)\exp(-\Gamma_n t). \tag{4.3.6}$$

It follows from (4.3.5) that (4.3.6) represents *exponential decay* in time if $k_n' > 0$ (pole in the fourth quadrant of the $k$-plane), and *exponential growth* if $k_n' < 0$ (pole in the third quadrant). Accordingly, it was proposed by Møller and by Heitler and Hu (cf. Section 4.1) that poles in the fourth quadrant should be associated with "decaying states" and poles in the third quadrant with "capture states." The lifetime would be given by

$$\tau_n = 1/\Gamma_n, \tag{4.3.7}$$

and the decay energy by $E_n'$. The latter, by the uncertainty relation, is only defined with an uncertainty of the order of $\Gamma_n$. The requirement that the mean decay energy be positive, $E_n' = \frac{1}{2}(k_n'^2 - K_n^2) > 0$, restricts this interpretation to poles located above the bisectors of the third and fourth quadrants; poles located below the bisectors would not be related to unstable states. In this way, the other observable quantity mentioned in Section 4.1, namely, the lifetime and decay energy of unstable states, would be directly obtained from the complex poles of the $S$-matrix.

The above interpretation, however, is greatly oversimplified and gives rise to several difficulties. The energy is an observable quantity, associated with a Hermitian operator, which cannot have complex eigenvalues. This is reflected in the fact that the "eigenfunction" (4.3.4) cannot be associated with a state vector in Hilbert space: it is not normalizable, due to its exponential increase with the distance, contained in the factor $\exp(K_n r)$.

It is true that a plane wave (eigenfunction of the momentum operator) is also not normalizable, but, in spite of this, it is an acceptable wave function, provided that it is interpreted as a limiting form of a (normalizable) wave packet.[5] In this case, however, the probability density remains finite at every point in space, whereas for (4.3.6) it would diverge as $r \to \infty$, so that the difficulty here is more serious.

The physical origin of this "exponential catastrophe" is quite simple. Taking into account (4.3.5), we can rewrite (4.3.6) in the form

$$|r\psi(k_n, r, t)|^2 = |B|^2 \exp\{-\Gamma_n[t - (r/k_n')]\}. \tag{4.3.8}$$

[5] Cf. von Neumann [5].

Thus a particle found at a distance $r$ at the time $t$ was emitted at the time $t-r/k_n'$ ($k_n'$ = decay velocity), when the density at the source was larger by a factor $\exp(\Gamma_n r/k_n')$. The exponential increase as $r \to \infty$ therefore corresponds to the increase in the probability of emission in the remote past, and to the unphysical assumption that (4.3.8) remains valid for all time, even though $|\psi|^2$ would diverge in the limit as $t \to -\infty$. Thus, to eliminate the exponential catastrophe, one must take into account the process whereby the decaying system was produced.

"Complex-energy eigenfunctions" of the type (4.3.4) were first employed in quantum mechanics in the theory of alpha-decay [6]. The lifetime and decay energy can be obtained from (4.3.5), provided that the state is long-lived ($|\Gamma_n| \ll E_n'$) and that the level width is much smaller than the level spacing, so that the level can be excited without appreciable interference with neighboring ones. These conditions are similar to those formulated in Section 4.2 for the validity of the Breit–Wigner one-level approximation.

The above discussion suggests that, in order to justify this procedure, and to find the relation between complex poles of the $S$-matrix and the transient time behavior of a system, it is necessary to take into account the excitation conditions. The decay of a system must always depend to some extent on the way in which it was excited. For long-lived states, this dependence may be very weak; the system may "lose memory" of the production process, as in Bohr's theory of the compound nucleus [7]. Nevertheless, a proper formulation of the problem must include some account of how the "decaying state" was formed.

For linear systems with a finite number of degrees of freedom (e.g., mechanical systems or electric networks), it is well known that the transient time behavior of the system is closely related to its "complex eigenfrequencies." A standard procedure[6] to treat such a system is to consider first its response to a time-harmonic excitation with frequency $\omega$. The response function usually has an analytic continuation with poles in the complex $\omega$-plane. The poles are associated with the "free modes of oscillation" of the system (cf. Section 1.2).

By decomposing the response function into a sum of terms, one associated with each pole, one finds that the response of the system to a given set of initial conditions, in the absence of external driving forces, is a superposition of its free modes of oscillation, with amplitudes determined by the initial conditions [cf. (1.2.4)]. The effect of the poles in the presence of external driving forces is obtained by convolution [cf. (1.2.16)]. In this way, one can find the general connection between excitation, decay, and "complex eigenfrequencies" (poles of the response function).

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[6] Cf. Doetsch [8], Carslaw and Jaeger [9], and Gardner and Barnes [10].

Perhaps the earliest application of the method of complex eigenfrequencies to linear systems with an infinite number of degrees of freedom was made by Thomson [11], in his treatment of the free modes of oscillation of the electromagnetic field around a perfectly conducting sphere. He determined them by the requirement that they should contain only outgoing radiation, so that they correspond, in modern language, to the poles of the $S$-matrix associated with this problem [cf. (4.3.2)]. Thomson found them to be of the form $\omega_n = \omega_n' - i\gamma_n$ ($\gamma_n > 0$), corresponding to a time dependence $\exp(-i\omega_n t)$, and he accordingly interpreted $\omega_n'$ as the frequency, and $\gamma_n$ as the damping constant, for the $n$th natural mode of oscillation.

It was subsequently remarked by Lamb [12] that Thomson's modes behave like $r^{-1}\exp\{-i\omega_n[t-(r/c)]\}$ as $r \to \infty$, so that they also give rise to an "exponential catastrophe." Lamb pointed out that the difficulty arises from the unphysical assumption that the modes have been in existence for an indefinitely long time, and that it may be overcome by taking into account the excitation conditions, of which he gave an example. Another example was treated by Love [13]. However, the dependence of the results on the excitation, for a general type of excitation, was not discussed.

A convenient way to take into account the excitation conditions is to introduce them through an initial-value problem. Through its dependence on the initial conditions, the general solution of an initial-value problem exhibits the dependence on the excitation. This will now be done for some typical cases [14, 15], leading to the physical interpretation of complex poles of the $S$-matrix by an extension of the techniques available for determining the transient behavior of linear systems with a finite number of degrees of freedom.

## 4.4. Vibrating String and Oscillator

A very simple example that brings out the main features of the method is one already treated by Lamb in a special case: the initial-value problem for a semiinfinite vibrating string coupled to a harmonic oscillator.

Let the rest position of the string coincide with the positive $x$-axis, and let $y(x,t)$ denote the transverse displacement of the string. The oscillator, constrained to move only in the $y$ direction, is located at the origin, so that $y(0,t)$ represents its displacement. Let $m$ be the mass of the oscillator and $\omega_0$ its natural frequency, and let us define

$$2\gamma = T/m > 0, \tag{4.4.1}$$

where $T$ is the tension of the string. The force exerted by the string on the oscillator is

$$T\frac{\partial y}{\partial x}(0,t),$$

so that the equation of motion of the oscillator is

$$\left(\frac{d^2}{dt^2}+\omega_0{}^2\right)y(0,t) = 2\gamma\frac{\partial y}{\partial x}(0,t). \tag{4.4.2}$$

This may be regarded as a boundary condition on the motion of the string, which must satisfy the wave equation,

$$\partial^2 y/\partial t^2 - \partial^2 y/\partial x^2 = 0 \qquad (x>0), \tag{4.4.3}$$

where the units have been so chosen that the wave velocity in the string is equal to unity.

Let the initial conditions be

$$y(x,0) = u(x), \qquad \frac{\partial y}{\partial t}(x,0) = v(x) \qquad (x \geqslant 0). \tag{4.4.4}$$

The monochromatic (stationary) solution of (4.4.2), (4.4.3) is

$$y_\omega(x,t) = A_\omega[e^{-i\omega x} - S(\omega)e^{i\omega x}]e^{-i\omega t}, \tag{4.4.5}$$

where

$$S(\omega) = \frac{\omega^2-\omega_0{}^2-2i\gamma\omega}{\omega^2-\omega_0{}^2+2i\gamma\omega} = \frac{(\omega-\omega_1{}^*)(\omega-\omega_2{}^*)}{(\omega-\omega_1)(\omega-\omega_2)}, \tag{4.4.6}$$

and

$$\omega_{1,2} = \pm(\omega_0{}^2-\gamma^2)^{1/2} - i\gamma \tag{4.4.7}$$

are the roots of the denominator, already considered in (1.2.3). We may regard $S(\omega)$ as the $S$-function for the scattering of waves in the string by the harmonic oscillator. The poles $\omega_{1,2}$ of $S(\omega)$ are located in $I_-(\omega)$ [cf. Section 1.2(d)].

To find the solution of the initial-wave problem, we write the general solution of the wave equation (4.4.3)

$$y(x,t) = f(x-t) + g(x+t), \tag{4.4.8}$$

where $f$ and $g$ are arbitrary functions, representing outgoing and incoming waves, respectively.

To satisfy the initial conditions (4.4.4), it suffices to take

$$f(x) = \tfrac{1}{2}u(x) - \frac{1}{2}\int_0^x v(x')\,dx' \qquad (x \geqslant 0), \tag{4.4.9}$$

$$g(x) = \tfrac{1}{2}u(x) + \frac{1}{2}\int_0^x v(x')\,dx' \qquad (x \geqslant 0), \tag{4.4.10}$$

where the lower limit in the integrals has been chosen as 0 for reasons of simplicity.

Since we are interested in the solution for $t > 0$, the incoming wave $g(x+t)$ is completely defined by (4.4.10). However, the outgoing wave $f(x-t)$ is defined by (4.4.9) only for $x \geqslant t$, i.e., for that portion of the outgoing wave which has not interacted with the oscillator. In this domain, the solution becomes

$$y(x,t) = \tfrac{1}{2}[u(x-t) + u(x+t)] + \frac{1}{2}\int_{x-t}^{x+t} v(x')\,dx' \qquad (x \geqslant t). \tag{4.4.11}$$

This is the well-known d'Alembert solution, representing free propagation in the string, as ought to be expected.

To determine the solution for $x < t$, we need the continuation of $f$ to negative values of its argument; $f(x-t)$ then represents the "scattered wave," i.e., the portion of the outgoing wave resulting from the interaction with the oscillator. For this purpose, we must use the "boundary condition" (4.4.2).

Substituting (4.4.8) in (4.4.2) and using the notation

$$\bar{f}(t) = f(-t), \tag{4.4.12}$$

we find

$$\bar{f}''(t) + 2\gamma\bar{f}'(t) + \omega_0{}^2\bar{f}(t) = -g''(t) + 2\gamma g'(t) - \omega_0{}^2 g(t) \qquad (t > 0). \tag{4.4.13}$$

The right-hand side is known, by (4.4.10), so this is an ordinary differential equation for the unknown function $\bar{f}(t)$.

The physical meaning of (4.4.13) becomes more transparent if we rewrite it in terms of the displacement of the oscillator,

$$y(0,t) = y_0(t) = \bar{f}(t) + g(t), \tag{4.4.14}$$

as [cf. (4.4.10)]

$$\ddot{y}_0 + 2\gamma\dot{y}_0 + \omega_0{}^2 y_0 = 4\gamma g'(t) = 2\gamma[u'(t) + v(t)]. \tag{4.4.15}$$

This is the equation of motion of a damped harmonic oscillator [cf. (1.2.1)] subject to the external driving force $4m\gamma g'(t)$. Thus *the effect of the coupling to the string on the motion of the oscillator is equivalent to a damping term (radiation damping) and an external driving force, due to the incoming waves.*

The initial conditions for the solution of (4.4.13) are, according to (4.4.4),

$$\bar{f}(0) = \tfrac{1}{2}u_0, \qquad \bar{f}'(0) = \tfrac{1}{2}[v_0 - u'(0)], \tag{4.4.16}$$

where $u_0 = u(0)$ and $v_0 = v(0)$ are, respectively, the initial displacement and the initial velocity of the oscillator.

The solution of the initial-value problem (4.4.13), (4.4.16) is a typical problem of the theory of transients in discrete systems, which can be treated by the standard Laplace transformation methods referred to in Section 4.3 [we might also apply the results obtained in Section 1.2 for the equivalent equation (1.2.1)]. The solution is found to be

$$\begin{aligned}\bar{f}(t) = &-g(t) + \sum_{j=1}^{2} a_j \int_0^t \exp[-i\omega_j(t-t')]\, g(t')\, dt' \\ &- [i/2(\omega_1+\omega_2)] \sum_{j=1}^{2} [u_0 + (iv_0/\omega_j)]\, a_j \exp(-i\omega_j t),\end{aligned} \tag{4.4.17}$$

where $\omega_1$ and $\omega_2$ are the poles of $S(\omega)$, given by (4.4.7), and

$$a_j = i \operatorname{res} S(\omega)\Big|_{\omega=\omega_j} \tag{4.4.18}$$

It follows from (4.4.8), (4.4.10), and (4.4.17) that

$$\begin{aligned} y(x,t) = {} & \tfrac{1}{2}[u(t+x) - u(t-x)] + \frac{1}{2}\int_{t-x}^{t+x} v(x')\, dx' \\ & + \tfrac{1}{2} \sum_{j=1}^{2} a_j \exp[-i\omega_j(t-x)] \\ & \times \left( \int_0^{t-x} \exp(i\omega_j x') \{u(x') + i[v(x')/\omega_j]\}\, dx' \right. \\ & \left. - [i/(\omega_1+\omega_2)][u_0 + i(v_0/\omega_j)] \right) \qquad (0 \leqslant x < t). \end{aligned} \tag{4.4.19}$$

Together with (4.4.11), this completes the general solution of the problem.

To interpret the solution, let us consider first the effect of the initial conditions for the oscillator, by assuming that the string is initially at rest and an impulse is given to the oscillator, so that

$$u(x) = 0 \quad (x \geqslant 0); \qquad v(x) = 0 \quad (x > 0); \qquad v_0 \neq 0. \tag{4.4.20}$$

Substituting in (4.4.19), we find

$$y(x,t) = \frac{v_0}{2(\omega_1+\omega_2)} \sum_{j=1}^{2} \frac{a_j}{\omega_j}\, \theta(t-x) \exp[-i\omega_j(t-x)]$$

$$= v_0\, \theta(t-x) e^{-\gamma(t-x)} \frac{\sin[(\omega_0{}^2-\gamma^2)^{1/2}(t-x)]}{(\omega_0{}^2-\gamma^2)^{1/2}}, \qquad (4.4.21)$$

where $\theta(t)$ is the Heaviside step function. Thus $y$ vanishes for $x > t$, corresponding to the domain not yet reached by the impulse propagated from the oscillator.

For $x < t$, each term of the sum in (4.4.21) is of the form of a "complex-frequency eigenfunction" associated with the corresponding pole $\omega_j$ of $S(\omega)$. However, there is no exponential catastrophe, because the step function introduces a sharp cutoff at the wave front, corresponding to the excitation at the definite instant $t = 0$.

The result (4.4.21) should be compared with (1.2.15). We may consider it as describing the analog of an emission process in which the kinetic energy initially concentrated in the oscillator is gradually transmitted to the string.

Thus we see that the terms containing $u_0$ and $v_0$ in (4.4.19) are the ordinary transients associated with the initial displacement and velocity of the oscillator, which are propagated to the string. To interpret the remaining terms, that arise from the forced motion of the oscillator, it suffices to consider the case $u_0 = v_0 = 0$. We may then rewrite (4.4.11) and (4.4.19) as

$$y(x,t) = \int_0^\infty [G_-(x,x',t) f(x') + G_+(x,x',t) g(x')]\, dx', \qquad (4.4.22)$$

where $f$ and $g$ are given by (4.4.9) and (4.4.10), and

$$G_-(x,x',t) = \delta(x-x'-t), \qquad (4.4.23)$$

$$G_+(x,x',t) = \delta(x-x'+t) - \delta(x+x'-t) + \sum_{j=1}^{2} G_j(x,x',t), \qquad (4.4.24)$$

where $\delta$ is the Dirac delta function, and

$$G_j(x,x',t) = \theta(t-x-x') a_j \exp[-i\omega_j(t-x-x')]. \qquad (4.4.25)$$

By means of (4.4.9) and (4.4.10), the initial wave function is decomposed into an initially outgoing part $f$ and an initially incoming part $g$. The subsequent propagation of $f$ and $g$ is described by $G_-$ and $G_+$, so that we shall call $G_-$ and $G_+$ the *propagators* (or Green's functions) associated with $f$ and $g$, respectively.

For the physical interpretation of $G_\pm$, it is convenient to take the (purely symbolic) initial conditions

$$u(x) = \delta(x-x_0), \qquad v(x) = 0. \qquad (4.4.26)$$

By (4.4.9), (4.4.10), and (4.4.22), we find the corresponding solution to be given by

$$y(x,t) = \tfrac{1}{2}G_-(x,x_0,t) + \tfrac{1}{2}G_+(x,x_0,t). \tag{4.4.27}$$

In this case, therefore, the initial pulse splits into two identical pulses, which propagate in opposite directions. According to (4.4.23), the initially outgoing pulse $f$ propagates freely (as it would in an unlimited string), as ought to be expected. The same is true for the initially incoming pulse $g$ before it strikes the oscillator ($t < x_0$); during this period, (4.4.24) reduces to its first term. This term vanishes for $t > x' = x_0$; the remaining terms of (4.4.24) represent the scattered wave, which consists of two parts: an inverted mirror image of the incoming pulse, such as would be produced by reflection at a fixed end of a string, together with an exponential tail, similar to (4.4.21), which is due to the excitation of the oscillator transients by the incoming pulse.

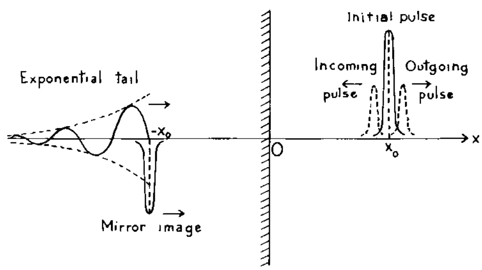


*FIG. 4.2.* An initial pulse at $x_0$ in real space splits into an incoming and an outgoing pulse; the incoming pulse gives rise in image space to a mirror image at $-x_0$, followed by an exponential tail.

These results can also be visualized by introducing an "image space," i.e., a fictitious continuation of the string for $x < 0$. The initial situation in both real and image space corresponding to (4.4.27) is represented in Fig. 4.2, which shows the splitting of the initial pulse at $x_0$, and the mirror image of the incoming pulse at $-x_0$, followed by the exponential tail. If we let the different components of this initial configuration propagate freely, in the direction indicated by the arrows in Fig. 4.2, the resulting wave function on the string is identical to (4.4.27). The advantage of this representation is that it can be extended to other situations, including cases where free propagation is not described by the wave equation (4.4.3), as will be seen later.

We see from (4.4.24) that each pole $\omega_j$ of the $S$-function gives rise to a term $G_j(x,x',t)$ in the propagator of the scattered wave. We will call $G_j$ the *propagator of the transient mode associated with the complex pole* $\omega_j$. According to (4.4.25), it may also be defined as that solution of the free wave equation which, for $t = 0$, reduces to

$$G_j(x, x', 0) = a_j \theta(-x-x') \exp[i\omega_j(x+x')]. \tag{4.4.28}$$

This is a cutoff exponential wave packet with "complex wave number" given by the pole $\omega_j$ and amplitude $a_j$ given (apart from a factor $i$) by the residue of $S$ at this pole [cf. (4.4.18)].

Thus the "complex-frequency eigenfunctions" employed in the method of complex eigenvalues (cf. Section 4.3) can be given an exact meaning: with a suitable cutoff factor, which removes the exponential catastrophe, they correspond precisely to the propagators of transient modes.

Owing to the exponential tail contained in the last term of (4.4.24), the form of the scattered wave at a given moment depends on the whole previous history, i.e., on the portion of the incoming wave packet which has reached the scatterer (oscillator) up to that moment. It is only in special cases that exponentially decaying terms like (4.4.25) will actually appear in the scattered wave. This happens in (4.4.21) and also, as illustrated by (4.4.27), in the case of excitation by a very sharp pulse. More generally, if the initial disturbance vanishes for $x > x_0$, the scattered wave will be of this form for $t - x > x_0$, i.e., after the whole incoming wave packet has stricken the oscillator.

The above treatment can be extended to other problems of wave propagation. In particular, it can be applied [14] to discuss the free oscillations of the electromagnetic field around a perfectly conducting sphere. The general solution of the initial-value problem in this case represents the decay of an arbitrary initial field outside of the sphere.

For an electric or magnetic multipole of a given order, one finds that the $S$-function has a finite number of poles, and, by partial-fraction decomposition, one can separate the contributions from the various poles. It is then found that each such contribution can be expressed in terms of the propagator of the corresponding transient mode. The expression for the propagator of a transient mode is essentially the same as (4.4.25) (the only difference arises from the fact that free propagation is modified, due to the presence of angular momentum). The transient modes are very short-lived, corresponding to poles far from the real axis. Physically, they arise from the effect of the centrifugal barrier, which can trap a wave packet, forcing it to circulate for a short time around the surface of the sphere, due to its own angular momentum.

A far-reaching generalization of these results on exponential decay has been achieved by Lax, Morawetz, and Phillips [16, 17]. They have considered the behavior for large times of solutions of the wave equation in the exterior of a smooth, bounded, impenetrable scatterer, on whose boundary the wave function is required to vanish. The only restriction on the shape of the scatterer is that it should be *star-shaped*, i.e., that it should contain some point $P_0$ such that the line segment joining $P_0$ to any other point $P$ of the scatterer lies completely in the scatterer (Fig. 4.3a). This includes any convex body as a particular case; such a body is star-shaped with respect to any of its points $P_0$.

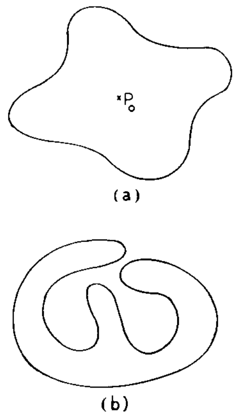


*FIG. 4.3.* The body (a) is star-shaped, whereas (b) is not.

It was shown by Lax, Morawetz, and Phillips that, *for any solution of the wave equation in the exterior of a bounded, impenetrable, star-shaped scatterer, whose initial values have finite energy and vanish outside of some bounded region, the energy remaining in any bounded region decays exponentially with the time.* From this follows, for smooth solutions, the pointwise exponential decay of the wave function itself.

The restriction to star-shaped objects helps to prevent long-lasting reverberations that might otherwise exist (cf. Fig. 4.3b). However, all that seems to be required physically is that the sojourn time for all reflected rays (in the sense of geometrical optics) within some sphere containing the obstacle is bounded, and it has been conjectured by Lax and Phillips that the exponential decay law would remain valid under these more general conditions.

## 4.5. The Transient-Mode Propagator for the Schrödinger Equation

The problem of the electromagnetic oscillations around a perfectly conducting sphere is closely related to that of an impenetrable ("hard") sphere in quantum mechanics. In fact, the $S$-function for angular momentum $l$ for the quantum-mechanical problem, given by (3.3.34), is identical with that for magnetic multipoles of order $l$ in the electromagnetic problem; one can show that it has exactly $l$ poles. The general initial-value problem in the quantum case corresponds to describing the propagation of an arbitrary initial wave packet outside of the sphere. The solution of this problem [14] can again be expressed in terms of transient-mode propagators associated with the poles of the $S$-matrix. It is found that, at $t = 0$, these propagators reduce to exactly the same form as for the wave equation [cf. (4.4.28)] in terms of "image space" (where the "mirror" now corresponds to the surface of the sphere). The only difference lies in the nature of free propagation for $t > 0$, which is now described by Schrödinger's equation rather than the wave equation.

We shall now give a general discussion of the behavior of the transient-mode propagator for Schrödinger's equation, and subsequently we shall illustrate its use by applying it to a specific example. According to the above discussion, the transient-mode propagator associated with a complex pole

$$k = k' - iK \qquad (K > 0) \tag{4.5.1}$$

of the $S$-matrix must be related to the solution $M(x, k, t)$ of the free-particle Schrödinger equation which, for $t = 0$, reduces to the cutoff exponential wave packet [cf. (4.4.28)]

$$M(x, k, 0) = \theta(-x)\, e^{ikx}. \tag{4.5.2}$$

To determine $M(x, k, t)$, let us consider Green's function $U(x, t)$ for the one-dimensional free-particle Schrödinger equation, i.e., the solution of this equation which, for $t \to 0$, satisfies

$$\lim_{t \to 0} U(x, t) = \delta(x). \tag{4.5.3}$$

In our units ($\hbar = m = 1$), $U(x, t)$ is given by [18, p. 22]

$$U(x, t) = [e^{-i\pi/4}/(2\pi t)^{1/2}] \exp(ix^2/2t). \tag{4.5.4}$$

In terms of Green's function, the solution of the one-dimensional free-particle Schrödinger equation which, for $t = 0$, takes the value $f(x)$, is given by

$$\psi(x, t) = \int_{-\infty}^{\infty} U(x - x', t) f(x')\, dx' = \int_{-\infty}^{\infty} U(x', t) f(x - x')\, dx'. \tag{4.5.5}$$

Thus, according to (4.5.2), we have

$$M(x, k, t) = \int_{-\infty}^{\infty} \theta(x' - x) \exp[ik(x - x')]\, U(x', t)\, dx'. \tag{4.5.6}$$

This function was introduced by Moshinsky [19]. According to (4.5.4), it may be rewritten as

$$M(x, k, t) = \frac{e^{-i\pi/4}}{(2\pi t)^{1/2}} \exp(ikx) \int_x^{\infty} \exp\left[i\left(\frac{x'^2}{2t} - kx'\right)\right] dx'.$$

Completing the square in the exponent of the integrand, we find

$$M(x,k,t) = \tfrac{1}{2}v(x,k,t)\operatorname{erfc}(e^{-i\pi/4}w), \tag{4.5.7}$$

where

$$v(x,k,t) = e^{i(kx-Et)}, \tag{4.5.8}$$

$$E = E' - i(\Gamma/2) = \tfrac{1}{2}k^2 = \tfrac{1}{2}(k'^2 - K^2) - ik'K, \tag{4.5.9}$$

$$w = (x-kt)/(2t)^{1/2}, \tag{4.5.10}$$

and

$$\operatorname{erfc} z = (2/\sqrt{\pi})\int_z^\infty \exp(-\zeta^2)\,d\zeta \tag{4.5.11}$$

is the error function. Note that $E$ is the "complex energy" associated with the pole (4.5.1) [cf. (4.3.5)] and $v(x,k,t)$ corresponds precisely to the "complex-energy wave function" (4.3.4).

The path of integration in (4.5.11) may be deformed in an arbitrary way, provided only that $\arg\zeta \to \alpha$ with $|\alpha| < \pi/4$ as $|\zeta| \to \infty$ along the path [$\alpha = \pi/4$ is also allowed, provided that $\operatorname{Re}(\zeta^2)$ remains bounded to the left]. The error function has the following asymptotic expansion [20, p. 298]:

$$\operatorname{erfc} z = \pi^{-1/2}z^{-1}\exp(-z^2)\left[1 + \sum_{m=1}^{n}(-1)^m[(2m-1)!!/(2z^2)^m] + R_n(z)\right], \tag{4.5.12}$$

where $(2m-1)!!$ is defined by (3.3.21), and

$$|R_n(z)| \leqslant \pi^{-1/2}2^{-n-1}(2n+1)!!\,|z|^{-2n-1} \qquad (\operatorname{Re} z \geqslant 0), \tag{4.5.13}$$

so that this expansion may be employed in the right half-plane. The asymptotic expansion of $\operatorname{erfc} z$ in the left half-plane then follows from the relation

$$\operatorname{erfc} z + \operatorname{erfc}(-z) = \operatorname{erfc}(-\infty) = 2\operatorname{erfc}(0) = 2. \tag{4.5.14}$$

The function $M(x,k,t)$ describes the propagation of the free-particle Schrödinger wave packet that, for $t = 0$, has the cutoff exponential shape (4.5.2). The propagation of this wave packet results essentially from a combination of three effects: (i) propagation without change of shape; (ii) the quantum-mechanical "spreading effect"; (iii) interference effects due to the sharp cutoff at the wave front.

Let us consider first the last of these effects. For real values of $k$, (4.5.2) may be regarded as representing a beam of particles with velocity $k$ confined to the half-space $x < 0$ by a perfectly absorbing shutter, which is suddenly removed at $t = 0$. According to classical mechanics, the behavior of the particle current at a point $x > 0$ as a function of time would be given by a step function, with a sharp rise after the time of flight $t = x/k$. For Schrödinger particles, however, the current begins to rise immediately after $t = 0$, and it approaches the classical value for $t \gg x/k$. In the neighborhood of $t = x/k$, there appear oscillations in the current that resemble the Fresnel diffraction pattern of a straight edge in optics. This effect was first investigated by Moshinsky [21], who called it "diffraction in time."

A similar effect takes place for complex $k$. The initially sharp front becomes "blurred" for $t > 0$, corresponding to the region $|w| \lesssim 1$ in (4.5.7), i.e., according to (4.5.10), to

$$|x - kt| \lesssim (2t)^{1/2} \tag{4.5.15}$$

[in ordinary units, the width of this region is $(2\hbar t/m)^{1/2}$]. In this region, complicated interference effects, of purely quantum-mechanical origin, take place. We shall only be interested in the behavior of $M(x, k, t)$ outside of this region, i.e., either ahead of or behind the wave front, but not too close to it. This means that we can take $|w| \gg 1$ in (4.5.7), and therefore we can employ the asymptotic expansion (4.5.12) of the error function.

Let $A$ and $B$ denote the regions of the complex plane above and below the second bisector, respectively, so that

$$-\pi/4 < \arg w < 3\pi/4 \quad \text{if} \quad w \in A, \qquad 3\pi/4 < \arg w < 7\pi/4 \quad \text{if} \quad w \in B. \tag{4.5.16}$$

Then we find, for $|w| \gg 1$,

$$\begin{aligned} M(x,k,t) &= M_A(x,k,t) \\ &= \frac{it}{x-kt} U(x,t) \left[ 1 - \frac{i}{2w^2} \right. \\ &\quad \left. + \cdots + \frac{(2n-1)!!}{(2iw^2)^n} + R_n(w) \right] \qquad \text{if} \quad w \in A, \end{aligned} \tag{4.5.17}$$

$$\begin{aligned} M(x,k,t) &= M_B(x,k,t) \\ &= v(x,k,t) + M_A(x,k,t) \qquad\qquad \text{if} \quad w \in B, \end{aligned} \tag{4.5.18}$$

with

$$|R_n(w)| \leqslant \sqrt{\pi}\, 2^{-n-1}(2n+1)!!\,|w|^{-2n-1} \qquad (n = 0, 1, 2, \ldots). \tag{4.5.19}$$

For $|w| \gg 1$, $M_A(x, k, t)$ differs from the free-particle Green function (4.5.4) essentially by a factor of squared modulus

$$t^2/[(x-k't)^2 + (Kt)^2], \tag{4.5.20}$$

which has a peak of width $Kt$ around the point $x = k't$. On the other hand, $M_B$ differs from $M_A$ by the additional term $v(x,k,t)$, which corresponds, as we have seen, to the "complex-energy wave function" (4.3.4). However, this term appears in the propagator only for a special class of poles, and only within a limited range of values of $x$ and $t$.

To show this, let us consider the behavior of $M(x,k,t)$ as a function of $x$, for fixed $t > 0$. We restrict ourselves to $x > 0$, because $x < 0$ will represent the image space, as in Section 4.4. For a pole $k \in B$, like the pole $k_2$ in Fig. 4.4, it is readily seen from (4.5.10) that $w \in A$ for all $x > 0$. On the other hand, for a pole $k \in A$, like $k_1$ in Fig. 4.4, we have $w \in B$ if

$$0 < x < (k' - K)t, \tag{4.5.21}$$

corresponding to the thick line segment in Fig. 4.4, and $w \in A$ if $x > (k' - K)t$. Thus it is only in the case of poles located above the second bisector, and only within the range of values of $x$ and $t$ defined by (4.5.21), that the term usually associated with a "decaying state" appears in $M(x,k,t)$.

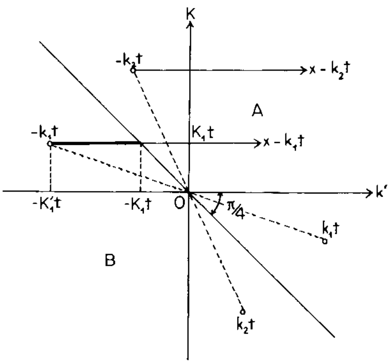


*FIG. 4.4.* For a pole $k_2 \in B$, we have $x - k_2 t \in A$. For a pole $k_1 \in A$, we have $x - k_1 t \in B$ within the range indicated by the thick line, and $x - k_1 t \in A$ outside of this range.

It is readily verified that the real part of the exponent in (4.5.8) is always negative within the range (4.5.21), so that there is no exponential catastrophe. This follows immediately from the fact that (4.5.2) is a normalizable wave packet.

There is a continuous transition between poles belonging to $A$ and those belonging to $B$, the range (4.5.21) becoming smaller and smaller as the second bisector is approached. Moreover, if we compare the order of magnitude of the two terms in (4.5.18) as a function of time, within the range (4.5.21), we find that $M_A$ always predominates over $v$ after a sufficient lapse of time. Since

$$|v(x,k,t)| = \exp[K(x - k't)], \tag{4.5.22}$$

we can associate $v$ with the propagation of the initial wave packet, without change of shape, with velocity $k'$. However, as is well known, a free-particle

Schrödinger wave packet always undergoes a broadening in the course of time. This "spreading effect" is contained in the term $M_A$.

In fact, according to (4.5.2), the width of the initial wave packet is of the order of $K^{-1}$. By the uncertainty relation, this corresponds to a momentum spread of the order of $K$, so that the wave packet must spread by an amount of the order of $Kt$ after a time $t$ [cf. (4.5.20)].

The spreading effect becomes important when this quantity becomes comparable with the initial width, i.e., for $t \gtrsim t_s$, where the "spreading time" $t_s$ is given by (cf. also Appendix E)

$$t_s \sim K^{-2}. \tag{4.5.23}$$

On the other hand, according to (4.5.22) and (4.3.7), the lifetime associated with the pole is given by

$$\tau = \Gamma^{-1} = (2k'K)^{-1}. \tag{4.5.24}$$

Thus, as a pole approaches the second bisector, $t_s$ and $\tau$ become of the same order, so that the spreading effect predominates over exponential decay within a single lifetime. There is no time, so to speak, for the exponential law to manifest itself. This clarifies the special role played by the second bisector, which was implicitly contained in Heitler and Hu's discussion (cf. Section 4.3).

The above discussion also indicates that, no matter what may be the position of the pole, the exponential decay law cannot remain valid for arbitrarily large times: it must ultimately be superseded by the decay law for a free-particle wave packet. This decay law is derived in Appendix E, where it is shown that the probability distribution at a given point decays, in general, like $t^{-3}$ for $t \to \infty$, i.e., much more slowly than exponentially (this result has a very simple physical interpretation, which is also discussed in Appendix E). An explicit example will be treated in the next section.

So far, we have studied the behavior of $M(x, k, t)$ only for $t > 0$. For $t < 0$, we have, according to (4.5.4),

$$U(x, -t) = U^*(x, t), \tag{4.5.25}$$

so that, by (4.5.6),

$$M(x, k, -t) = M^*(x, -k^*, t). \tag{4.5.26}$$

Thus poles that are symmetrical about the imaginary axis exchange their roles under time reversal. In particular, for $t < 0$, the exponential term $v(x, k, t)$ appears in the propagator only for poles located above the first bisector. It corresponds to a time-reversed emission mode, so that it may be called an absorption mode (or "capture state"; cf. Section 4.3). Such modes would appear in connection with "final-value problems," i.e, when we want to describe how a given situation was built up.

## 4.6. Application to an Explicit Model

### (a) Transient-Mode Expansion of the Propagators

In order to illustrate the application of the Schrödinger transient-mode propagator discussed in Section 4.5, we shall now consider a very simple specific model of decay and resonance scattering.[7] This is the initial-value problem for a Schrödinger particle interacting with a penetrable (partially transparent) spherical shell of radius $a$, corresponding to a "delta-function potential"

$$V(r) = (A/2a)\,\delta(r-a) \qquad (A > 0). \tag{4.6.1}$$

In our units $\hbar = m = 1$, $A$ is a dimensionless parameter that measures the opacity of the sphere. The limit $A \to \infty$ corresponds to a spherical box with impenetrable walls. It is well known that in this case there exists a series of discrete energy levels associated with stationary states of the particle within the box. For finite but large $A$, the stationary states go over into transient modes, with lifetimes that can be made arbitrarily large by suitably adjusting $A$. According to the choice of initial conditions, one can use this model to describe either the decay of a wave packet initially confined within the sphere or the scattering of a wave packet by the sphere.

We restrict ourselves to $s$-waves, and denote by the indices 1 and 2 the interior and the exterior of the sphere, respectively, with corresponding radial wave functions

$$\varphi_j(r,t) = r\psi_j(r,t) \qquad (j = 1,2). \tag{4.6.2}$$

The Schrödinger equation for $\varphi(r,t) = r\psi(r,t)$ is

$$(\partial^2/\partial r^2 + 2i\,\partial/\partial t)\,\varphi(r,t) = (A/a)\,\delta(r-a)\,\varphi(r,t). \tag{4.6.3}$$

Thus each $\varphi_j$ satisfies the free-particle equation

$$(\partial^2/\partial r^2 + 2i\,\partial/\partial t)\,\varphi_j(r,t) = 0 \qquad (j = 1,2). \qquad (4.6.4$$

The wave function must be regular at the origin and continuous at $r = a$, so that

$$\varphi_1(0,t) = 0, \tag{4.6.5}$$

$$\varphi_1(a,t) = \varphi_2(a,t). \tag{4.6.6}$$

---

[7] See Nussenzveig [15]. Cf. also Petzold [22].

On the other hand, the radial derivative of the wave function is not continuous at $r = a$; it undergoes a discontinuity that can be computed by integrating both sides of (4.6.3) from $r = a-0$ to $r = a+0$:

$$\frac{\partial \varphi_2}{\partial r}(a,t) - \frac{\partial \varphi_1}{\partial r}(a,t) = \left(\frac{A}{a}\right)\varphi_2(a,t). \tag{4.6.7}$$

The solution of (4.6.4) subject to these conditions is equivalent to that of (4.6.3). Let the initial conditions be

$$\varphi_j(r,0) = f_j(r) \qquad (j = 1,2). \tag{4.6.8}$$

We want to express the general solution of this initial-value problem in terms of propagators:

$$\varphi_j(r,t) = \int_0^a G_{j1}(r,\rho,t) f_1(\rho)\, d\rho + \int_a^\infty G_{j2}(r,\rho,t) f_2(\rho)\, d\rho, \tag{4.6.9}$$

where the propagator $G_{jk}$ describes the effect in region $j$ of the portion of the initial wave packet contained in region $k$. We also want to expand the propagators in terms of transient modes associated with the poles of the $S$-function, as discussed in Section 4.5.

The $S$-function associated with this problem may readily be obtained from the stationary solution (cf. Section 2.7)

$$\varphi_1(E,r,t) = B_1 \sin(kr) e^{-iEt},$$

$$\varphi_2(E,r,t) = B_2 [e^{-ikr} - S(k) e^{ikr}] e^{-iEt}.$$

Taking into account the boundary conditions (4.6.6) and (4.6.7), we find

$$S(k) = -Q(-2ika)/Q(2ika), \tag{4.6.10}$$

where

$$Q(z) = 1/(A+z-Ae^{-z}). \tag{4.6.11}$$

The solution of the initial-value problem can be found by an extension of the method employed in Section 4.4. For this purpose we start from the general solution of (4.6.4), given by (4.5.5):

$$\varphi_j(r,t) = \int_{-\infty}^{\infty} U(\rho,t)\chi_j(r+\rho)\, d\rho, \tag{4.6.12}$$

and note that the initial conditions (4.6.8) yield

$$\chi_1(\rho) = f_1(\rho) \qquad (0 \leqslant \rho \leqslant a), \tag{4.6.13}$$

$$\chi_2(\rho) = f_2(\rho) \qquad (a \leqslant \rho), \tag{4.6.14}$$

whereas (4.6.5) implies

$$\chi_1(-\rho) = -\chi_1(\rho). \tag{4.6.15}$$

Thus the unknown functions in (4.6.12) are $\chi_1(a+\rho)$ and $\chi_2(a-\rho)$ for $\rho > 0$. They are determined by the boundary conditions (4.6.6) and (4.6.7), which can be solved by the Laplace transformation (for the procedure, see the references on the Laplace transformation given in Section 4.3). Substituting the results in (4.6.12) and comparing with (4.6.9), one finally obtains the following expressions for the propagators:

$$G_{11}(r,\rho,t) = G_1(|r-\rho|,t) - G_1(r+\rho,t), \tag{4.6.16}$$

$$G_{12}(r,\rho,t) = G(|r-\rho|,t) - G(r+\rho,t) = G_{21}(\rho,r,t), \tag{4.6.17}$$

$$G_{22}(r,\rho,t) = U(|r-\rho|,t) + \int_0^\infty U(r+\rho-2a+\xi,t)\,R_{22}(\xi)\,d\xi, \tag{4.6.18}$$

where

$$G_1(x,t) = U(x,t) + \int_0^\infty [U(2a+x+\xi,t) + U(2a-x+\xi,t)]\,R_{11}(\xi)\,d\xi, \tag{4.6.19}$$

$$G(x,t) = \int_0^\infty U(x+\xi,t)\,R_{12}(\xi)\,d\xi, \tag{4.6.20}$$

and the functions $R_{jk}(\xi)$ are given by

$$R_{11}(\xi) = A\mathscr{L}^{-1}[Q(2ap)], \tag{4.6.21}$$

$$R_{12}(\xi) = 2a\mathscr{L}^{-1}[pQ(2ap)], \tag{4.6.22}$$

$$R_{22}(\xi) = -\mathscr{L}^{-1}[S_a(ip)], \tag{4.6.23}$$

where $\mathscr{L}^{-1}$ denotes the inverse Laplace transform, $Q$ is given by (4.6.11), and $S_a(k) = S(k)\,e^{2ika}$, with $S$ given by (4.6.10).

It is readily seen that

$$R_{22}(\xi) = -\delta(\xi) + R_{12}(\xi) - \theta(\xi-2a)\,R_{12}(\xi-2a), \tag{4.6.24}$$

where $\theta$ is Heaviside's step function. Thus the evaluation of the propagators is reduced to that of (4.6.19) and (4.6.20).

The poles of $Q(2ap)$ are related by the transformation $p = -ik$ with the poles of $S(k)$ in the $k$-plane. According to (4.6.10) and (4.6.11), the latter are the roots of the equation

$$A - 2i\beta - Ae^{2i\beta} = 0, \tag{4.6.25}$$

where

$$\beta = ka. \tag{4.6.26}$$

By applying some general methods for locating the roots of complex transcendental equations of this type [23], the following results are found.

For each value of the potential strength $A$, there exists an infinite number of simple poles, all of which are located in $I_-(k)$. For each pole $\beta_n$ in the fourth quadrant, there exists a corresponding pole

$$\beta_{-n} = -\beta_n^* \tag{4.6.27}$$

in the third quadrant, symmetrically placed with respect to the imaginary axis (cf. Section 2.4); thus it suffices to consider the poles in the fourth quadrant.

There is one pole $\beta_n$ in each strip $(n-1)\pi < \operatorname{Re}\beta < n\pi$ $(n = 1, 2, 3, \ldots)$. In the free-particle limit $A \to 0$, $\beta_n$ approaches $(n-\frac{1}{2})\pi - i\infty$. When $A$ increases, $\beta_n$ moves upwards and away from the imaginary axis, and $\beta_n \to n\pi$ when $A \to \infty$. In this limit, therefore, we get the eigenvalues for a particle within a spherical box with impenetrable walls, as ought to be expected.

For given $A$, the asymptotic behavior of the pole distribution for large $n$ is given by [cf. (5.5.29)]

$$\beta_n \approx (n-\tfrac{1}{4})\pi - (i/2)\log[(2n-\tfrac{1}{2})(\pi/A)] \qquad (n\pi \gg A). \tag{4.6.28}$$

As will be seen later (cf. Section 5.5), such "large" poles are related basically to the cutoff in the potential, and they do not reflect any other features of physical interest. Most of the physics is contained in the lower-order poles that are located closer to the origin. We shall be interested mainly in the case $A \gg 1$, in which these poles are very close to their limiting values as $A \to \infty$ on the real axis,

$$\beta_n = n\pi\left(1 - \frac{1}{A+1}\right) - i\left(\frac{n\pi}{A}\right)^2 + O\left[\left(\frac{n\pi}{A}\right)^3\right] \qquad (n\pi \ll A). \tag{4.6.29}$$

According to (4.5.24), the lifetime of the transient mode associated with (4.6.29) is

$$\tau_n \approx a^2A^2/2(n\pi)^3, \tag{4.6.30}$$

which can be made arbitrarily large by increasing the "opacity" $A$.

The physical interpretation of these long-lived transient modes is very simple. The high opacity of the potential barrier at $r = a$ gives rise to a large number of reflections before a particle is able to escape through it. This leads to sharp resonances, similar to the well-known optical interference phenomena in thin plates.[8] The transmissivity of the potential barrier for $\beta = ka \ll A$ is given by $\Theta \approx 4\beta^2/A^2$, which represents the ratio of the number of successful attempts to escape through the barrier to the total number of

[8] Cf., for example, McVoy [24].

attempts, and the lifetime (4.6.30) can be interpreted as the ratio [4, p. 389]

$$\tau_n = 2a/(k_n \Theta_n) \tag{4.6.31}$$

of the "period" $2a/k_n$ of the motion within the sphere ($k_n$ being the "velocity") to the transmissivity $\Theta_n$.

To obtain the transient-mode expansion of the propagators, according to (4.6.16)–(4.6.23), we have to find the partial fraction decomposition of $Q(2ap)$ in terms of its poles. Since there is an infinite number of poles, this corresponds to a Mittag–Leffler expansion, which may be obtained by Cauchy's method [25, p. 110]. The result is [cf. (5.7.36)]

$$Q(z) = [1/(A+1)z] + \sum_n [r_n/z_n(z-z_n)], \tag{4.6.32}$$

where $z_n = -2i\beta_n$ and $r_n$ is the residue at $z_n$,

$$r_n = z_n/(A+1+z_n). \tag{4.6.33}$$

The summation is extended over all the poles, taken in the order of increasing modulus [so that $\beta_n$ and $\beta_{-n}$ are taken together; cf. (4.6.27)]. The convergence of the series follows from the asymptotic behavior (4.6.28) of the poles.

Substituting these results in (4.6.21), (4.6.22), and noting that

$$\mathscr{L}^{-1}[1/(p-p_n)] = \theta(\xi)\exp(p_n\xi) = \theta(\xi)\exp(-ik_n\xi), \tag{4.6.34}$$

where $\theta$ is Heaviside's step function, we get

$$R_{11}(\xi) = \left[b_0 + \sum_n b_n \exp(-ik_n\xi)\right]\theta(\xi), \tag{4.6.35}$$

$$R_{12}(\xi) = \tfrac{1}{2}\delta(\xi) + \sum_n c_n\, \theta(\xi)\exp(-ik_n\xi), \tag{4.6.36}$$

where

$$\begin{aligned} b_0 &= A/[2a(A+1)]; \qquad b_n = A/[2a(A+1-2i\beta_n)]; \\ c_n &= -2i(b_n\beta_n/A) \qquad (n = \pm 1, \pm 2, \ldots). \end{aligned} \tag{4.6.37}$$

Taking into account (4.6.24), and substituting in (4.6.16)–(4.6.20), we finally obtain the transient-mode expansion of the propagators

$$\begin{aligned} G_1(x,t) = {} & U(x,t) + b_0[M(2a+x,0,t) + M(2a-x,0,t)] \\ & + \sum_n b_n[M(2a+x,k_n,t) + M(2a-x,k_n,t)], \end{aligned} \tag{4.6.38}$$

$$G(x,t) = \tfrac{1}{2}U(x,t) + \sum_n c_n M(x,k_n,t), \tag{4.6.39}$$

$$\begin{aligned} G_{22}(r,\rho,t) = {} & U(|r-\rho|,t) - \tfrac{1}{2}[U(r+\rho-2a,t) + U(r+\rho,t)] \\ & + \sum_n c_n[M(r+\rho-2a,k_n,t) - M(r+\rho,k_n,t)], \end{aligned} \tag{4.6.40}$$

where $M(x,k,t)$ is the Schrödinger transient-mode propagator, and we have made use of (4.5.6).

It is instructive to consider the behavior of $G_{11}$ in the limit as $A\to\infty$, corresponding to a spherical box with impenetrable walls. In this limit, (4.6.11) and (4.6.21) yield

$$R_{11}(\xi) = \mathscr{L}^{-1}[(1-e^{-2ap})^{-1}] = \tfrac{1}{2}\delta(\xi) + (1/2a)\left[1+\sum_n \exp(-ik_n\xi)\right], \tag{4.6.41}$$

where $k_n = n\pi/a$. Accordingly, (4.6.19) becomes

$$\begin{aligned} G_1(x,t) = {} & U(x,t) + \tfrac{1}{2}[U(2a+x,t) + U(2a-x,t)] \\ & + (1/2a)[M(2a+x,0,t) + M(2a-x,0,t)] \\ & + (1/2a)\sum_n [M(2a+x,k_n,t) + M(2a-x,k_n,t)]. \end{aligned} \tag{4.6.42}$$

On the other hand, in this limit, one can also expand $G_{11}$ in terms of the stationary states of the particle within the box,

$$G_{11}(r,\rho,t) = \frac{2}{a}\sum_{n=1}^{\infty} \sin(k_n r)\sin(k_n\rho)\exp(-\tfrac{1}{2}ik_n^2 t). \tag{4.6.43}$$

This corresponds to (4.6.16), with

$$G_1(x,t) = \frac{1}{2a} + \frac{1}{a}\sum_{n=1}^{\infty}\cos(k_n x)\exp(-\tfrac{1}{2}ik_n^2 t) = \frac{1}{2a}\,\theta\left(\frac{x}{2a}\,\middle|\, -\frac{\pi t}{2a^2}\right), \tag{4.6.44}$$

where $\theta(x|\tau)$ is Jacobi's theta function.[9] If we apply Jacobi's transformation formula

$$\theta(x|\tau) = \left(\frac{\tau}{i}\right)^{-1/2}\exp\left(-\frac{\pi i x^2}{\tau}\right)\theta\left(\frac{x}{\tau}\,\middle|\,-\frac{1}{\tau}\right), \tag{4.6.45}$$

we find

$$G_1(x,t) = U(x,t) + \sum_{n=1}^{\infty}[U(2na-x,t) + U(2na+x,t)], \tag{4.6.46}$$

which has an immediate physical interpretation: it corresponds to the result obtained by the method of images, with two "mirrors," at $r=0$ and $r=a$, giving rise to an infinite series of images.

The expansions (4.6.42), (4.6.43), and (4.6.46) are three different representations of the same propagator. In the theory of heat conduction, the

[9] Cf. Sommerfeld [26, p. 72].

transformation (4.6.45) is employed to transform a series which converges rapidly for large times into one that converges rapidly for small times. A similar result is valid here, the characteristic time interval being $T = 4a^2/\pi$, the period of the ground state.

For $t \gg T$, the dominant contribution in the stationary-state expansion (4.6.44) usually arises from the lowest terms, since contributions from large values of $n$ are rapidly oscillating and tend to cancel by destructive interference. On the other hand, for $t \ll T$, a large number of terms contributes and the convergence becomes very slow.

The opposite is true for (4.6.46): it converges rapidly for $t \ll T$, the main contribution coming from the free-particle propagator and the lowest-order reflections, as ought to be expected; the convergence becomes slow for $t \gg T$.

It can readily be shown, with the help of (4.5.17) and (4.5.18), that the transient-mode expansion (4.6.42) behaves like the stationary-state expansion for $t \gg T$, whereas for $t \ll T$, it is dominated by the free-particle propagator. Thus it combines the advantages of the stationary-state expansion with those of the expansion in terms of multiple reflections: it converges well both for small and for large times. Furthermore, unlike the other two expansions, it can still be applied for finite values of $A$.

### *(b) Application to Decay*

To describe the decay of a particle that is initially confined within the sphere, it suffices to specialize the initial conditions (4.6.8) by taking

$$f_2(r) = 0. \tag{4.6.47}$$

For an arbitrary initial wave packet $f_1(r)$ within the sphere, the behavior of the solution in regions 1 and 2 is then determined by the propagators $G_{11}$ and $G_{21}$, respectively [cf. (4.6.9)].

For times $t \ll T = 4a^2/\pi$, the "period" of the motion within the sphere, the propagators tend to be dominated by the free-particle propagator. For $t \sim T$, reflection at the surface of the sphere must be taken into account, and the behavior of the propagators becomes quite complicated.

We shall be interested only in the behavior of the propagators for $t \gg T$. Under these conditions, the asymptotic expansions (4.5.17) and (4.5.18) may be employed in (4.6.38) and (4.6.39), and we find that (4.6.16) becomes

$$\begin{aligned} G_{11}(r,\rho,t) = {} & \frac{2}{a}\sum_{A}\left[1 - \frac{1}{A+1-2i\beta_n}\right]\sin(k_n r)\sin(k_n \rho)\exp(-iE_n t) \\ & - \left(\frac{2}{\pi}\right)^{1/2}\frac{e^{i\pi/4}}{(A+1)^2}\frac{\rho r}{t^{3/2}} + O(t^{-5/2}) \qquad (t \gg T), \end{aligned} \tag{4.6.48}$$

where $\sum_A$ denotes the sum over all the poles located above the second bisector, and $E_n$ is the corresponding "complex energy," given by (4.3.5).

Each term of the series in (4.6.48) decays exponentially in time, the lifetime of the $n$th term being given by (4.6.30) for $n\pi \ll A$. On the other hand, for $t \to \infty$, $G_{11}$ is dominated by the term in $t^{-3/2}$. Thus the asymptotic decay law is in general the same as that for a free-particle wave packet, i.e., an inverse third power law, in agreement with the discussion of Section 4.5.

The propagator $G_{21}$ may be treated in a similar way. Each transient mode associated with a pole $k_n = k_n' - iK_n$ above the second bisector gives rise to an exponential wave train with a diffuse wave front at $r \approx (k_n' - K_n)t$ [cf. (4.5.21)]. For points far behind the wave front associated with the lowest mode ($r \ll k_1't$), where all the wave trains overlap, and for $t \gg r^2$, we find

$$G_{21}(r, \rho, t) \approx -2i \sum_n c_n \sin(k_n \rho) \exp(ik_n r - iE_n t)$$

$$-\left(\frac{2}{\pi}\right)^{1/2} \frac{e^{i\pi/4}}{A+1} \left[r - \frac{A}{A+1} a\right] \frac{\rho}{t^{3/2}}. \tag{4.6.49}$$

Each term of the series corresponds to a "complex-energy wave function" of the type (4.3.4). On the other hand, for $t \to \infty$, the last term of (4.6.49) predominates, and we again get an asymptotic behavior in $t^{-3/2}$.

We see that *the exponential decay law cannot be valid either for very small or for very large times*. The lower limit for its validity is set by the period of the motion of the particle within the sphere: it takes several reflections at the walls before the exponential behavior begins to appear.

The upper limit for the validity of the exponential law arises from the spreading effect. The tail of the emitted wave packet contains the "slow" components of the initial wave packet, i.e., the long-wavelength components; for wavelengths much greater than $a$, they are practically unaffected by the potential. This explains why the asymptotic decay law is similar to that for free particles.

Note that the expansion (4.6.48) goes over into (4.6.43) as $A \to \infty$. In this limit, therefore, each term in the transient mode expansion approaches the corresponding term in the stationary-state expansion for large $t$, so that the expansion coefficient gives the probability amplitude associated with the level in question; the eigenfunctions of different states are orthogonal in this limit.

In general, however, the transient modes are not even approximately orthogonal, so that one cannot ascribe an independent physical meaning to each term in the transient-mode expansion.

Transient modes occupy an intermediate position between stationary states and free-particle wave packets, sharing some of the properties of both. It is only for poles located close to the real axis, within the domain of applicability of the exponential law, that concepts taken over from the treatment of stationary states may be employed for an approximate description of the system; we then have approximate orthogonality, and the expansion coefficients may be approximately interpreted in terms of probability amplitudes. For large times, however, free-particle features predominate; the wave packets associated with different poles finally get mixed together as a result of the spreading. On the other hand, for poles located far from the real axis, free-particle features predominate for all times, and no trace of exponential decay remains.

In order to investigate the domain of validity of the exponential decay law in the most favorable conditions, let us consider the decay of a long-lived mode, e.g., the lowest mode for $A \gg 1$. To concentrate the excitation as much as possible on this mode, we choose the initial (unnormalized) wave packet

$$f_1(r) = \sin(k_1 r), \tag{4.6.50}$$

corresponding to a "complex resonance." This choice leads to higher-mode amplitudes of order $A^{-2}$ times smaller than that of the lowest mode.

According to (4.6.9), the wave function within the sphere is

$$\varphi_1(r,t) = \int_0^a G_{11}(r,\rho,t) f_1(\rho)\, d\rho. \tag{4.6.51}$$

For $t \gg T$, we may substitute $G_{11}$ by (4.6.48). It is then found that corrections to the exponential decay law are very small for $T \ll t \lesssim \tau_1$, where $\tau_1$ is the lifetime of the mode $n = 1$, given by (4.6.30).

For $t \gtrsim \tau_1$, the main correction arises from the term in $t^{-3/2}$, and one finds

$$\varphi_1(r,t) \approx \sin\left(\frac{\pi r}{a}\right) \exp\left[-\frac{i}{2}\left(\frac{\pi}{a}\right)^2 t - \frac{t}{2\tau_1}\right] - \frac{\sqrt{2}\, e^{i\pi/4}}{\pi^{3/2} A^2} \frac{a^2 r}{t^{3/2}} \qquad (t \gtrsim \tau_1). \tag{4.6.52}$$

The probability to find the particle still confined within the sphere after a time $t$ is

$$P(t) = \int_0^a |\varphi_1(r,t)|^2\, dr \Big/ \int_0^a |f_1(r)|^2\, dr, \tag{4.6.53}$$

and (4.6.52) leads to the decay law

$$P(t) \approx \exp\left(-\frac{t}{\tau_1}\right) - \left(\frac{2}{\pi}\right)^{5/2} \frac{1}{A^2} \cos\left(\frac{\pi^2 t}{2a^2} + \frac{\pi}{4}\right)\left(\frac{a^2}{t}\right)^{3/2} \exp\left(-\frac{t}{2\tau_1}\right)$$
$$+ \frac{4}{3\pi^3 A^4}\left(\frac{a^2}{t}\right)^3 \qquad (t \gtrsim \tau_1). \tag{4.6.54}$$

The deviations from the exponential decay law become important when the first term becomes comparable with the last one, i.e, when $\exp(t/\tau_1) \approx (t/\tau_1)^3 A^{10}$, yielding

$$t/\tau_1 \gtrsim 10 \log A + 3 \log\log A. \tag{4.6.55}$$

Thus the asymptotic $t^{-3}$ law predominates only after the probability that the particle still has not decayed becomes $\lesssim A^{-10}$, which is extremely small for all values of $A$ that correspond to even moderately long-lived modes. We conclude that it would be very difficult to detect deviations from the exponential decay law in this case.[9a]

### *(c) Application to Resonance Scattering*

We can also apply the general solution (4.6.9) to give a time-dependent description of the scattering of an arbitrary wave packet by the penetrable sphere. It suffices to specialize the initial conditions (4.6.8) by taking

$$f_1(r) = 0. \tag{4.6.56}$$

For an arbitrary initial wave packet $f_2(r)$ in the external region, the behavior of the solution in regions 1 and 2 is then determined by the propagators $G_{12}$ and $G_{22}$, respectively.

The most interesting case is that of resonance scattering. A convenient choice for the initial wave packet is

$$f_2(r) = \exp[-ik_0(r-a)], \qquad k_0 = k_0' - iK_0 \quad (K_0 > 0), \tag{4.6.57}$$

an exponential wave packet with mean momentum $k_0'$ and width $K_0$ in momentum space; both parameters can be varied to study their effect on the resonance. The resonant mode is taken to be the lowest one, associated with the pole $k_1$, and we assume that

$$A \gg 1, \qquad |(k_0-k_1)/k_1| \ll 1, \tag{4.6.58}$$

so that one is close to a sharp resonance.

Substituting (4.6.57) in (4.6.9), and using the expressions derived above for the propagators, one finds that the solution can be expressed in closed form in terms of the $M$ function:

$$\varphi_1(r,t) = F(r,t) - F(-r,t), \tag{4.6.59}$$

$$\varphi_2(r,t) = \varphi_{2H}(r,t) + \varphi_{2R}(r,t), \tag{4.6.60}$$

$$\varphi_{2H}(r,t) = M(a-r, k_0, t) - M(r-a, k_0, t), \tag{4.6.61}$$

$$\varphi_{2R}(r,t) = F(2a-r,t) - F(-r,t), \tag{4.6.62}$$

[9a] Similar conclusions have been reached in connection with some field-theoretical models of unstable particles; see [27] and the references quoted there.

where

$$F(r,t) = -2i\beta_0 Q(-2i\beta_0) M(a-r,k_0,t) - i\sum_n [c_n/(k_0-k_n)] M(a-r,k_n,t), \tag{4.6.63}$$

with $\beta_0 = k_0 a$ and $Q$ given by (4.6.11).

For $t \gg T$, we may employ the asymptotic expansions (4.5.17), (4.5.18). For the internal wave function (4.6.59), the quantity of interest is the amplitude of excitation of the resonant mode $k_1$ relative to the amplitudes of other modes; this is found to be given by a "gain factor"

$$\begin{aligned} g(k_0,k_1,t) &= [\exp(-iE_0 t) - \exp(-iE_1 t)]/(\beta_0 - \beta_1) \qquad (k_0 \neq k_1), \\ &= -i(k_1 t/a)\exp(-iE_1 t) \qquad (k_0 = k_1). \end{aligned} \tag{4.6.64}$$

Thus, in the case of "complex resonance," $k_0 = k_1$, the gain increases linearly with the time, to begin with, attaining its maximum value $|g|_{\max} = e^{-1}(A/\pi)^2$ after a rise time $t = 2\tau_1$; thereafter, it decreases, with a decay time also of the order of $2\tau_1$.

If we move the center of the exciting line off resonance ($k' \neq k_1'$), this gives rise to "beats" with the difference frequency, and the maximum gain decreases in proportion with the distance from exact resonance. If the center of the exciting line is kept at resonance, only its width being varied, we find

$$|g(k_0,k_1,t)| \approx (A/\pi)^2 \tau_0 [\exp(-t/2\tau_0) - \exp(-t/2\tau_1)]/(\tau_0 - \tau_1), \tag{4.6.65}$$

where $\tau_0$ is the "lifetime" associated with $E_0 = \frac{1}{2}k_0{}^2$ [cf. (4.3.5)].

It follows from (4.6.65) that, for excitation by a narrow line ($\tau_0 \gg \tau_1$), the rise time and the maximum gain are of the same order as at the complex resonance, whereas the decay time is $\sim 2\tau_0$; for excitation by a broad line ($\tau_0 \ll \tau_1$), the decay time is of the same order as at resonance, but the rise time and the maximum gain are both reduced by a factor of the order of $\tau_0/\tau_1$. In conclusion, the *rise* (*decay*) *time is the shorter* (*longer*) *of* $\tau_0$ *and* $\tau_1$; *the maximum amplitude gain is of the order of* $(\operatorname{Im}\beta_1)^{-1}$ *times the fraction of the width of the excitation that overlaps the width of the resonant mode.*

The wave function in the external region is decomposed by (4.6.60) into a "hard-sphere term" $\varphi_{2H}$, which represents the solution that would correspond to scattering by an impenetrable sphere, and a "resonance term" $\varphi_{2R}$. We take the excitation to be centered at the resonance momentum, $k_0' = k_1'$, and consider the shape of the outgoing wave packet for times $t$ greater than the lifetime $\tau_1$, but still much smaller than the "spreading time" $t_s = \min(K_1^{-2}, K_0^{-2})$; i.e.,

$$A^2 \lesssim t/a^2 \ll A^4. \tag{4.6.66}$$

The outgoing wave has a diffuse wave front, of width $\sim(2t)^{1/2}$, around $r-a \approx k_1't$ [cf. (4.5.15)]; we consider only the behavior far behind this region, i.e., for

$$\zeta = k_1't - r \gg t^{1/2}. \tag{4.6.67}$$

Under these conditions, we can employ the asymptotic expansion of the $M$-functions, and the dominant term in the outgoing wave packet is given by

$$\varphi_{2,\text{out}}(r,t) \approx \left[\exp(-K_0\zeta) + \frac{2K_1(\exp(-K_0\zeta) - \exp(-K_1\zeta))}{K_0 - K_1}\right] \times \exp[i(k_1'r - \tfrac{1}{2}k_1'^2 t)]. \tag{4.6.68}$$

The first term in the expression within square brackets represents the contribution from hard-sphere scattering. The second one originates from the resonance term, and its spatial behavior is the transmitted counterpart of the time behavior (4.6.64) within the sphere. The amplitude of the resonance term at a distance $\zeta$ behind the wave front corresponds to the amplitude of the resonant mode within the sphere at a time $t = \zeta/k_1'$.

In terms of the stationary-state expansion (2.7.8), the result (4.6.68) can be rewritten as

$$\varphi_2 = \varphi_{2,\text{in}} + \varphi_{2,\text{out}} = -\frac{1}{2\pi i}\int_{-\infty}^{\infty} \exp[-ik(r-a) - \tfrac{1}{2}ik^2 t]\frac{dk}{k-k_0} + \frac{1}{2\pi i}\int_{-\infty}^{\infty} S_a(k)\exp[ik(r-a) - \tfrac{1}{2}ik^2 t]\frac{dk}{k-k_0}, \tag{4.6.69}$$

where

$$S_a(k) = S(k)\exp(2ika) = (k-k_1^*)/(k-k_1), \tag{4.6.70}$$

corresponding to a one-level approximation [cf. (4.2.6)]. Thus, under the above conditions, only the immediate neighborhood of the resonant level gives a significant contribution.

The absolute value of the expression within square brackets in (4.6.68), as well as that of its two component terms, is plotted as a function of $\zeta$ in Fig. 4.5, in the following cases: (a) broad line ($K_0 \gg K_1$); (b) complex resonance ($K_0 = K_1$); (c) narrow line ($K_0 \ll K_1$). The amplitude of the resonance term also represents the behavior of the resonant mode within the sphere as a function of time. Probability distributions correspond to the squares of the probability amplitudes represented in Fig. 4.5.

In all cases, the outgoing wave packet consists of a peak due to direct reflection at the surface of the sphere, followed by a tail that represents the

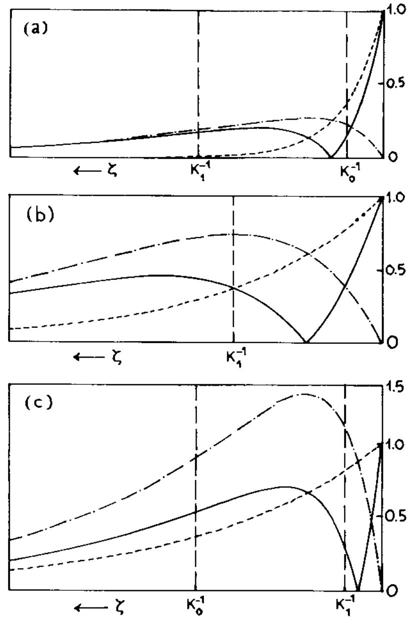


*FIG. 4.5.* Profiles of the hard-sphere term (---), the resonance term (—·—) and the total outgoing wave packet (——) as a function of the distance $\zeta$ behind the wave front. (a) $K_0 = 5K_1$ (broad line); (b) $K_0 = K_1$ (complex resonance); (c) $K_0 = K_1/5$ (narrow line) (after Nussenzveig [15]).

effect of resonance scattering. The destructive interference between the hard-sphere term and the resonance term gives rise to a dip in the surface-reflected wave, representing the part of the incident wave packet that penetrates into the scatterer to build up the resonant mode.

It is instructive to discuss the applicability of the concept of time delay, analyzed in Section 2.11(c), to the outgoing wave packet. Outside of resonance, as we have seen, (2.11.21) may actually lead to a time advance, corresponding to a "retardation" of $-2a$, due to surface reflection. On the other hand, for a sharp resonance, where the one-level approximation (4.6.70) can be applied, (2.11.22) leads to a retardation $\Delta = 2/K_1$ at resonance, corresponding to a time delay of twice the lifetime associated with the level. For an incident wave packet of width $K_0$, it has to be assumed in this case that $K_0 \ll K_1$, so that the variation of $d\eta/dk$ over the width $K_0$ can be neglected.

If we adopt the Eisenbud–Wigner interpretation of the time delay in terms of the position of the "center" of the wave packet, the results that we have found in the present example for a narrow line (Fig. 4.5c) are not in agreement with (2.11.22). The "retardation" of the outgoing wave front is $\approx -2a$, and the shape of the outgoing wave packet is appreciably different from that of the incident wave, so that a description in terms of a retardation of the "center" is not appropriate.

The reason for this discrepancy is that the momentum distribution in the incident wave packet does not fall to zero sufficiently rapidly in the wings, i.e., outside of the width $K_0$, to justify the stationary-phase argument on which the Eisenbud–Wigner interpretation is based [cf. Section 2.11(c)]. An example that would satisfy the necessary requirements is that of a Gaussian wave packet of width $K_0 \ll K_1$. Although the initial excitation within the scatterer does not vanish in this case, it can be rendered negligibly small by taking the initial position of the center of the wave packet sufficiently far from the scatterer. The strong surface reflection disappears in this case, because the internal region is excited adiabatically. The outgoing wave packet is of Gaussian shape, and the retardation of its center is given by Eisenbud's expression.

Under these conditions, however, the retardation at resonance is a small effect, much smaller than the position uncertainty of the wave packet. A retardation $\Delta = 2/K_1$ decreases the probability density at the unretarded position of the center of the Gaussian wave packet by an amount

$$1 - \exp(-K_0{}^2\Delta^2) \approx 4(K_0/K_1)^2 \ll 1. \tag{4.6.71}$$

If we try to render the retardation effect more conspicuous, by associating it with a sharper signal, the requirements for the validity of Eisenbud's expression are violated, as we have seen in the case of an initial wave packet with a sharp front. If one takes an incident wave packet of width $K_0 \gg K_1$, so that the uncertainty in the position of the wave packet is much smaller than the retardation at resonance, Eisenbud's expression can no longer be applied, because $d\eta/dk$ varies greatly over the width $K_0$. As shown in Fig. 4.5a, the outgoing wave packet then has a tail of small amplitude ($\sim K_1/K_0$) that decays with the lifetime of the level, i.e., much more slowly than the incident wave. This can also be called a time delay, but it is again a small effect.[10]

On the other hand, the definition (2.11.34) of the time delay, and the corresponding physical interpretation, discussed in Section 2.11(c), can always be employed. If we rewrite the incident wave packet in (4.6.69) in the form (2.11.15), where the normalization is given by (2.11.24), we find that, to a good approximation,

$$k^2 |A(E)|^2 \approx K_0/(8\pi^3 |k-k_0|^2), \tag{4.6.72}$$

so that (2.11.33) yields

$$\langle \Delta t \rangle \approx \frac{2K_0}{\pi} \int_0^\infty \frac{d\eta/dE}{|k-k_0|^2}\, dk. \tag{4.6.73}$$

[10] Cf. also Lee and Wick [28].

On the other hand, in the one-level approximation (4.6.70),

$$d\eta/dk = k\,d\eta/dE = -a + (K_1/|k-k_1|^2). \tag{4.6.74}$$

We consider only the case in which the center of the exciting line is at resonance, as in Fig. 4.5. Substituting (4.6.74) in (4.6.73), and noting that the main contribution to the integral arises from the neighborhood of $k = k_0' = k_1'$, we can approximate the result by

$$\langle \Delta t \rangle \approx \frac{2K_0}{\pi k_1'} \int_{-\infty}^{\infty} \left[ -a + \frac{K_1}{(k-k_1')^2 + K_1{}^2} \right] \frac{dk}{(k-k_1')^2 + K_0{}^2}, \tag{4.6.75}$$

so that the effective retardation is

$$\langle \Delta r \rangle = k_1' \langle \Delta t \rangle = -2a + [2/(K_0 + K_1)]. \tag{4.6.76}$$

Both $K_0$ and $K_1$ are $\ll a^{-1}$ under the above conditions, so that the first term of (4.6.76), representing the advance due to surface reflection, is negligible with respect to the second term. For $K_0 \ll K_1$, we find $\langle \Delta r \rangle \approx 2/K_1$, in agreement with the Eisenbud–Wigner result; however, this no longer represents the position of the center of the wave packet, but rather an average retardation. On the other hand, for $K_0 \gg K_1$, we have $\langle \Delta r \rangle \approx 2/K_0$, so that the effective retardation is then determined by the incident wave packet. Inspection of Fig. 4.5 shows that (4.6.76) gives a reasonable account of the retardation effect in all cases.

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PART

# II

# POTENTIAL SCATTERING

*A Truly General Theorem must be true not only in general, but also in each particular case.*

(quoted by G. Beck)

CHAPTER

5

# ANALYTIC PROPERTIES OF PARTIAL-WAVE AMPLITUDES

*The Schrödinger equation is a laboratory for theoretical physicists.*

M. Gell-Mann

## 5.1. Introduction

The remainder of this book will be devoted to a discussion of the analytic properties of scattering amplitudes for a specific model of an interaction: the scattering of nonrelativistic particles by a central potential $V(r)$. The purpose is twofold: to illustrate the general treatment of Part I by providing a concrete example and to show how much more detailed information may be gained by making specific assumptions about the form of the interaction.

The approach is quite different: the scattering amplitude is completely determined, in principle, once the potential has been given, via the solution of a differential or integral equation, and the problem is reduced to studying the analytic properties of the solutions of such equations. The connection between the results and the general physical properties of the interaction that was stressed in Part I becomes less apparent. On the other hand, while the methods of Part I allowed us to deal only with cutoff interactions, the present approach can be applied also to interactions with tails extending to infinity. This will enable us to deal, in particular, with a special class of potentials, known as *Yukawa-type potentials* [cf. Section 5.4(c)], that has received the greatest amount of attention because of its connection with the field theory of strongly interacting particles.

The Schrödinger equation for the stationary states of a system of two nonrelativistic spinless particles interacting through a potential that depends only on their relative position is, in the center-of-mass frame,

$$-(\hbar^2/2m)\Delta\psi + V(r)\psi = E\psi, \tag{5.1.1}$$

where $m$ is the reduced mass, $E$ is the energy, and $V$ is the potential, with $r = |\mathbf{r}|$, where $\mathbf{r}$ is the relative coordinate. As mentioned above, we need not and, in general, will not assume, unless otherwise stated, that $V$ is a cutoff potential.

In the present chapter, we deal only with the analytic properties of scattering amplitudes for fixed angular momentum. For the $l$th partial wave, with

$$\psi_l(k,\mathbf{r}) = [u_l(k,r)/r]\, Y_l^m(\theta,\varphi), \tag{5.1.2}$$

the radial wave function $u_l(k,r)$ satisfies the differential equation

$$d^2u_l/dr^2 + \{k^2 - [l(l+1)/r^2] - U(r)\}\, u_l = 0, \tag{5.1.3}$$

where

$$k^2 = 2mE/\hbar^2, \qquad U(r) = 2mV(r)/\hbar^2. \tag{5.1.4}$$

Only some of the main results will be discussed here. Additional details may be found in treatises on potential scattering, particularly in the books by De Alfaro and Regge [1] and Newton [2].

## 5.2. The Jost Function

The simplest case is that of $s$-waves, for which (5.1.3) becomes

$$d^2u/dr^2 + [k^2 - U(r)]\,u = 0. \tag{5.2.1}$$

In order to be physically acceptable, a solution of this equation, $u = \varphi(k,r)$, must satisfy[1]

$$\varphi(k,0) = 0. \tag{5.2.2}$$

[1] In fact, if $u(k,0) = c \neq 0$, the corresponding wave function $\psi = u(k,r)/r$ (cf. (5.1.2)) satisfies $(\Delta + k^2 - U(r))\psi = -4\pi c\delta(\mathbf{r})$ rather than Schrödinger's equation, i.e., there is an additional point-source term at the origin; as is well known, $\Delta(1/r) = -4\pi\delta(\mathbf{r})$.

The boundary condition (5.2.2) defines the solution up to a constant factor; it is convenient to fix this factor by the additional boundary condition

$$\varphi'(k,0) = 1, \tag{5.2.3}$$

where the prime denotes differentiation with respect to $r$. The solution $\varphi(k,r)$ of (5.2.1) that is defined by the boundary conditions (5.2.2) and (5.2.3) is known as the *regular solution.*

If the potential decreases sufficiently rapidly as $r \to \infty$ [cf. (5.3.20)], the asymptotic behavior of $\varphi(k,r)$ as $r \to \infty$ is still of the form (2.2.16),

$$\begin{aligned}\varphi(k,r) &\approx C(k)\sin[kr+\eta(k)] \\ &= i/2\, C(k)\, e^{-i\eta(k)}[e^{-ikr} - S(k)\, e^{ikr}],\end{aligned} \tag{5.2.4}$$

where $\eta(k)$ is the phase shift and

$$S(k) = e^{2i\eta(k)} \tag{5.2.5}$$

is the $S$-function for $l = 0$.

Following Jost [3], it is convenient to introduce a new solution $f(k,r)$ of (5.2.1), defined by its asymptotic behavior as $r \to \infty$,

$$f(k,r) \approx e^{-ikr} \qquad (r \to \infty), \tag{5.2.6}$$

so that the corresponding wave function behaves asymptotically like an incoming wave of unit amplitude. The origin is a sink for such a solution, so that, in contrast with (5.2.2), we have, in general, $f(k,0) \neq 0$.

As (5.2.1) does not change under the transformation $k \to -k$, it is clear that $f(-k,r)$ is also a solution, and that $f(-k,r) \approx e^{ikr}$ $(r \to \infty)$. We expect that these two solutions, in general, will be independent. To check this we evaluate the Wronskian[2] of these two functions.

The Wronskian $W[u_1,u_2]$ of two functions $u_1(r)$ and $u_2(r)$ is defined by [cf. (3.3.17)]

$$W[u_1,u_2] = u_1 u_2' - u_1' u_2. \tag{5.2.7}$$

If $u_1, u_2$ are solutions of (5.2.1), we have

$$dW/dr = u_1 u_2'' - u_1'' u_2 = 0, \tag{5.2.8}$$

so that *the Wronskian of two solutions of (5.2.1) does not depend on r*. Furthermore, $W = 0$ if and only if $u_2 = \lambda u_1$, where $\lambda$ is a constant. Thus $u_1$ *and* $u_2$ *are linearly independent if and only if*

$$W[u_1,u_2] \neq 0. \tag{5.2.9}$$

---

[2] See, for example, Morse and Feshbach [4, p. 524].

We can make use of (5.2.8) to evaluate the Wronskian of $f(k,r)$ and $f(-k,r)$ by letting $r\to\infty$ and employing (5.2.6). The result is

$$W[f(k,r),f(-k,r)] = 2ik, \tag{5.2.10}$$

so that $f(k,r)$ and $f(-k,r)$ are indeed linearly independent for $k\neq 0$. Any other solution of (5.2.1) can therefore be expressed as a linear combination of these two. In particular, we must have

$$\varphi(k,r) = A[f(k,0)f(-k,r) - f(-k,0)f(k,r)], \tag{5.2.11}$$

since the linear combination within brackets satisfies the boundary condition (5.2.2). The constant $A$ can be evaluated from the condition (5.2.3):

$$\begin{aligned}\varphi'(k,0) &= A[f(k,0)f'(-k,0) - f(-k,0)f'(k,0)]\\ &= A\,W[f(k,r),f(-k,r)] = 2ikA = 1,\end{aligned}$$

so that

$$\varphi(k,r) = (1/2ik)[f(k)f(-k,r) - f(-k)f(k,r)], \tag{5.2.12}$$

where we have introduced the notation

$$f(k) = f(k,0). \tag{5.2.13}$$

The function $f(k)$ is known as the *Jost function*. It follows immediately from (5.2.10) and (5.2.12) that

$$f(k) = W[f(k,r),\varphi(k,r)]. \tag{5.2.14}$$

Since $U(r)$ is real in (5.2.1), the complex conjugate of a solution is (for real $k$) also a solution. Together with (5.2.6), this implies

$$f(-k,r) = f^*(k,r); \qquad f(-k) = f^*(k) \qquad (\text{real}\,k). \tag{5.2.15}$$

According to (5.2.6) and (5.2.12),

$$\varphi(k,r) \approx (1/2ik)[f(k)e^{ikr} - f(-k)e^{-ikr}] \qquad (r\to\infty). \tag{5.2.16}$$

Comparing this with (5.2.4), we conclude that

$$f(k) = |f(k)|\,e^{i\eta(k)}, \tag{5.2.17}$$

so that *the phase of the Jost function is the scattering phase shift*, and

$$S(k) = f(k)/f(-k). \tag{5.2.18}$$

It follows at once that

$$S(k)S(-k) = 1, \tag{5.2.19}$$

and, taking into account (5.2.15),

$$S(-k) = S^*(k) \qquad (\text{real}\,k). \tag{5.2.20}$$

This is the symmetry relation (2.8.14), which follows here from the reality of the interaction potential.

From (5.2.16) and (5.2.17), we get

$$\varphi(k,r) \approx \frac{|f(k)|}{k} \sin(kr+\eta) \qquad (r \to \infty), \tag{5.2.21}$$

which determines the asymptotic amplitude $C(k)$ in (5.2.4).

In the absence of interaction $[U(r)=0]$, the above solutions become

$$\varphi_0(k,r) = \sin(kr)/k, \qquad f_0(k,r) = e^{-ikr}, \qquad f_0(k) = 1, \tag{5.2.22}$$

where the index zero signifies the absence of interaction.

## 5.3. Analytic Properties of the Jost Function

### *(a) The Integral Equation*

By (5.2.18), the analytic properties of $S(k)$ follow from those of the Jost function. To investigate these properties for the function $f(k,r)$, we follow the standard procedure [5, p. 351] of transforming the differential equation (5.2.1) into an equivalent integral equation that incorporates also the boundary condition (5.2.6) satisfied by $f(k,r)$, by introducing a suitable Green function.

The function $\varphi_0(k,r)$ defined by (5.2.22) is a solution of (5.2.1) with $U=0$, satisfying the boundary conditions (5.2.2) and (5.2.3). It follows that

$$\begin{aligned}
&\int_r^\infty \frac{d}{d\rho}\left[\varphi_0(k,\rho-r)\frac{df}{d\rho}(k,\rho) - \frac{d\varphi_0}{d\rho}(k,\rho-r)f(k,\rho)\right] d\rho \\
&\quad = \int_r^\infty \varphi_0(k,\rho-r)\,U(\rho)f(k,\rho)\,d\rho \\
&\quad = \lim_{\rho\to\infty}\left\{\frac{\sin[k(\rho-r)]}{k}\frac{df}{d\rho}(k,\rho)\right. \\
&\qquad \left. - \cos[k(\rho-r)]f(k,\rho)\right\} + f(k,r).
\end{aligned}$$

Making use of (5.2.6), we find that the limit of the expression in curly brackets is simply $-e^{-ikr}$, so that $f(k,r)$ satisfies the integral equation

$$\begin{aligned}
f(k,r) &= f_0(k,r) + \int_r^\infty \varphi_0(k,\rho-r)\,U(\rho)f(k,\rho)\,d\rho \\
&= e^{-ikr} + (1/k)\int_r^\infty \sin[k(\rho-r)]\,U(\rho)f(k,\rho)\,d\rho.
\end{aligned} \tag{5.3.1}$$

The function

$$g(k,r) = e^{ikr} f(k,r) \tag{5.3.2}$$

therefore satisfies the equation

$$g(k,r) = 1 + (1/k) \int_r^\infty e^{-ik(\rho-r)} \sin[k(\rho-r)]\, U(\rho) g(k,\rho)\, d\rho. \tag{5.3.3}$$

### (b) The Iteration Solution

The simplest procedure to solve (5.3.3) is the *perturbation expansion* or *solution by iteration*,[3] that leads to the *Liouville–Neumann series*,

$$g(k,r) = \sum_{n=0}^{\infty} g_n(k,r), \tag{5.3.4}$$

where

$$g_0(k,r) = 1, \tag{5.3.5}$$

$$g_n(k,r) = (1/k) \int_r^\infty e^{-ik(\rho-r)} \sin[k(\rho-r)]\, U(\rho) g_{n-1}(k,\rho)\, d\rho, \tag{5.3.6}$$

provided that the series converges. In order to investigate its domain of convergence in the complex $k$-plane [3, 6], we set

$$k = k' + iK. \tag{5.3.7}$$

To get an upper bound for each term of the series, we use the fact that $(1+|z|)\exp(-|\mathrm{Im}\, z|)|(\sin z)/z|$ is bounded both for small and for large $|z|$, so that

$$|(\sin z)/z| \leqslant C \exp(|\mathrm{Im}\, z|)/(1+|z|), \tag{5.3.8}$$

where $C$ is a constant. It follows that

$$\begin{aligned} |k^{-1} e^{-ik(\rho-r)} \sin[k(\rho-r)]| &\leqslant C \frac{(\rho-r)}{1+|k|(\rho-r)} \exp[(K+|K|)(\rho-r)] \\ &\leqslant C \frac{\rho}{1+|k|\rho} \exp[(K+|K|)(\rho-r)] \qquad (\rho \geqslant r). \end{aligned} \tag{5.3.9}$$

Substituting in (5.3.6), we find

$$|g_n(k,r)| \leqslant C e^{-\beta r} \int_r^\infty \gamma(k,\rho) |g_{n-1}(k,\rho)|\, d\rho, \tag{5.3.10}$$

[3] Cf. Courant and Hilbert [5, p. 140].

where

$$\beta = K + |K| \qquad (\beta = 0 \text{ if } K \leqslant 0, \quad \beta = 2K \text{ if } K > 0), \tag{5.3.11}$$

and

$$\gamma(k, \rho) = \frac{\rho}{1 + |k|\,\rho}\, e^{\beta\rho}\,|U(\rho)|. \tag{5.3.12}$$

It follows from (5.3.5) and (5.3.10) that

$$|g_1(k, r)| \leqslant Ce^{-\beta r} M(k, r), \tag{5.3.13}$$

where

$$M(k, r) = \int_r^\infty \gamma(k, \rho)\, d\rho. \tag{5.3.14}$$

Iterating the inequality (5.3.10), we find

$$|g_n(k, r)| \leqslant [Ce^{-\beta r} M(k, r)]^n/n!. \tag{5.3.15}$$

In fact, if this result is true for $g_{n-1}$, we find, substituting in (5.3.10),

$$\begin{aligned} |g_n(k, r)| &\leqslant [(Ce^{-\beta r})^n/(n-1)!] \int_r^\infty \gamma(k, \rho)\, M^{n-1}(k, \rho)\, d\rho \\ &= -[(Ce^{-\beta r})^n/(n-1)!] \int_r^\infty M^{n-1}(k, \rho)\, dM(k, \rho)/d\rho\, d\rho \\ &= [Ce^{-\beta r} M(k, r)]^n/n!, \end{aligned}$$

so that the result follows by complete induction.

It follows from (5.3.4) and (5.3.15) that

$$|g(k, r)| \leqslant \sum_{n=0}^{\infty} [Ce^{-\beta r} M(k, r)]^n/n! = \exp[Ce^{-\beta r} M(k, r)]. \tag{5.3.16}$$

Thus *a sufficient condition for the convergence of the Liouville–Neumann series for $g(k, r)$ is that*

$$M(k, r) = \int_r^\infty e^{\beta\rho}\, |U(\rho)|\, \frac{\rho\, d\rho}{1 + |k|\,\rho} < \infty, \tag{5.3.17}$$

where $\beta$ is given by (5.3.11).

To discuss the analytic behavior, let us consider first the half-plane $I_-(k)$, in which $\beta = 0$ [cf. (5.3.11)], excluding a neighborhood of the origin, so that $|k| \geqslant \varepsilon > 0$. In this region,

$$M(k, r) \leqslant \int_0^\infty |U(\rho)|\, \frac{\rho\, d\rho}{1 + \varepsilon\rho}, \tag{5.3.18}$$

assuming that the integral exists. Therefore, the series (5.3.16) is absolutely and uniformly convergent with respect to $k$ (and $r$) in any bounded domain within this region, for all $r \geqslant 0$. In order to apply Weierstrass's theorem [Section 3.3(c)], we have to prove that $g_n(k,r)$ is holomorphic in such a domain for all $n$.

If we consider first $g_1(k,r)$, the integrand of (5.3.6) is holomorphic and the integral, by (5.3.15), is uniformly convergent. However, since it is an improper integral, this is not sufficient to conclude that $g_1$ is holomorphic. A sufficient condition [7, p. 92] is obtained by requiring that, if we differentiate under the integral sign with respect to $k$, the resulting integral is also uniformly convergent. The differentiation introduces an additional factor $(\rho - r)$ in the integrand of (5.3.6), so that, in addition to (5.3.18), we obtain the requirement

$$\int_0^\infty |U(\rho)| \frac{\rho^2 \, d\rho}{1+\varepsilon\rho} < \infty. \tag{5.3.19}$$

It then follows by iteration that all $g_n(k,r)$ are holomorphic in the domain under consideration.

Since the boundedness of (5.3.18) is a stronger requirement on the potential near the origin, while (5.3.19) is stronger for $\rho \to \infty$, we can combine both requirements into a single one, namely, that the first moment of the potential should be finite:

$$M_1 = \int_0^\infty |U(\rho)| \rho \, d\rho < \infty. \tag{5.3.20}$$

This means, roughly, that the potential $V(r)$ should be less singular than $r^{-2}$ at the origin, and it should decrease more rapidly than $r^{-2}$ as $r \to \infty$.

We can now apply Weierstrass's theorem, and conclude:

THEOREM 5.3.1. *For all $r \geqslant 0$, $f(k,r)$ is holomorphic in $I_-(k)$ and continuous up to the real axis (except possibly at $k = 0$), provided that $M_1$ is finite.*

If $k = 0$, a convergence factor proportional to $\rho$ is missing in the denominator of (5.3.17) and (5.3.19), so that we need a stronger requirement than (5.3.20) at infinity:

$$M_2 = \int_0^\infty |U(\rho)| \rho^2 \, d\rho < \infty. \tag{5.3.21}$$

This ensures the continuity of $f(k,r)$ also at $k = 0$. For the boundedness of $f(0,r)$, it is sufficient for $M_1$ to be finite [cf. (5.3.16)].

In particular, the above statements are true for the Jost function $f(k) = f(k,0)$. It also follows that, for all $r \geqslant 0$, $f(-k,r)$ [and, in particular, $f(-k)$] is holomorphic in $I_+(k)$ and continuous down to the real axis (for $k \neq 0$), if $M_1$ is finite, and including $k = 0$ if $M_2$ is also finite.

Note, however, that the above domains of analyticity for $f(k)$ and $f(-k)$ do not overlap, so that, if all we know about the potential is that $M_1$ and $M_2$ are finite, we cannot yet say anything about the analytic continuation of the $S$-function to complex values of $k$ [cf. (5.2.18)]. To get some information concerning $S$, we must assume that the potential decreases more rapidly at infinity.

According to (5.3.17), the decrease must be at least exponential. Let us assume that

$$E_\alpha = \int_0^\infty e^{\alpha\rho} |U(\rho)| \rho \, d\rho < \infty \qquad (\alpha > 0), \tag{5.3.22}$$

so that the potential decreases more rapidly than $e^{-\alpha r}$ as $r \to \infty$. Then, in $I_+(k)$, condition (5.3.17) is satisfied for $\beta = 2K = 2\operatorname{Im} k < \alpha$ [cf. (5.3.11)], and similarly for the analog of (5.3.19), so that we get:

THEOREM 5.3.2. *If $E_\alpha$ is finite, $f(k, r)$, for $r \geqslant 0$, is holomorphic in the half-plane*

$$K = \operatorname{Im} k < \alpha/2; \tag{5.3.23}$$

*similarly, $f(-k, r)$ is holomorphic for* $\operatorname{Im} k > -\alpha/2$.

The following results are immediate consequences of this theorem, together with (5.2.18):

(*a*) *If $E_\alpha$ is finite, $S(k)$ is meromorphic in the strip*

$$-\alpha/2 < \operatorname{Im} k < \alpha/2. \tag{5.3.24}$$

(*b*) *If the potential decreases faster than any exponential for $r \to \infty$ (e.g., if it is a cutoff potential or if it has a Gaussian tail), so that $E_\alpha$ is finite for all $\alpha$, the Jost function is an entire function and $S(k)$ is meromorphic in the whole $k$-plane.*

According to (5.3.16), $|g(k, r)|$ is bounded in $I_-(k)$ if (5.3.18) is finite. It then follows from (5.3.3) and (5.3.9) that

$$|g(k, r) - 1| \leqslant B \int_r^\infty |U(\rho)| \frac{\rho \, d\rho}{1 + |k| \rho} \qquad \text{if} \quad k \in I_-, \tag{5.3.25}$$

where $B$ is a constant. If (5.3.20) holds, the integral in (5.3.25) approaches zero as $|k| \to \infty$. In fact, this integral is bounded by

$$\int_0^\infty |U(\rho)| \frac{\rho \, d\rho}{1 + |k| \rho} \leqslant \int_0^{|k|^{-1/2}} |U(\rho)| \rho \, d\rho + \frac{1}{(|k|)^{1/2}} \int_{|k|^{-1/2}}^\infty |U(\rho)| \rho \, d\rho, \tag{5.3.26}$$

and both terms go to zero as $|k| \to \infty$. It follows that [cf. (5.3.2)]

$$|e^{ikr} f(k,r) - 1| \to 0 \qquad \text{as} \quad |k| \to \infty, \quad \operatorname{Im} k \leqslant 0, \tag{5.3.27}$$

*uniformly for all* $r \geqslant 0$. In particular, the Jost function $f(k)$ approaches 1 in this limit. Note that the right-hand side of (5.3.25) also goes to zero as $r \to \infty$, so that the asymptotic behavior (5.2.6) remains valid for all $k$ in $I_-$.

### (c) Discussion of the Results

The analyticity domain that has been found for the Jost function and the corresponding requirements on the potential have a rather simple physical interpretation.

If $U(r)$ is small enough for large $r$, we can still try to define $f(k,r)$ as the solution of (5.2.1) with the asymptotic behavior (5.2.6), even for complex $k$; the other independent solution, $f(-k,r)$, behaves asymptotically like $e^{ikr}$. For $\operatorname{Im} k < 0$, $e^{-ikr}$ is exponentially decreasing and $e^{ikr}$ is exponentially increasing as $r \to \infty$. Therefore, the asymptotic condition (5.2.6) indeed defines $f(k,r)$ uniquely; any admixture with $f(-k,r)$ would blow up exponentially as $r \to \infty$ and violate (5.2.6). The result was that $f(k,r)$ can be analytically continued to $I_-(k)$ and it does have the required asymptotic behavior, as was noted at the end of Section 5.3(b).

On the other hand, for $\operatorname{Im} k > 0$, $e^{-ikr}$ is exponentially increasing, and (5.2.6) does not in general define $f(k,r)$ uniquely, because any admixture with the exponentially decreasing function $f(-k,r)$ still has the same asymptotic behavior. However, one can still try to define $f(k,r)$ in $I_+$ as that solution of (5.2.1) for which the perturbation due to the potential,

$$\Delta f(k,r) = f(k,r) - e^{-ikr}, \tag{5.3.28}$$

decreases more rapidly than $e^{ikr}$ as $r \to \infty$. This requires a sufficiently fast decrease of $U(r)$ as $r \to \infty$.

For a cutoff potential, this condition defines $f(k,r)$ uniquely in the whole of $I_+$, because $\Delta f$ vanishes identically outside the potential. If the potential has an exponential tail of the type $e^{-\alpha r}$ ($\alpha > 0$), we can estimate $\Delta f$, according to (5.3.1), by taking the first Born approximation,

$$\Delta f_1 \approx (1/k) \int_r^\infty \sin[k(\rho - r)]\, e^{-(\alpha + ik)\rho}\, d\rho = e^{-ikr - \alpha r}/\alpha(\alpha + 2ik), \tag{5.3.29}$$

where $r$ is large enough for the asymptotic behavior of $U(r)$ to be valid, and we already have to assume the validity of (5.3.23), in order that the integral be convergent. By comparing this with $e^{ikr}$, we find that (5.3.29) is more rapidly decreasing for $\operatorname{Im} k < \alpha/2$, which leads us again[4] to the analyticity domain found in Theorem 5.3.2.

The requirement from (5.3.20) that the potential be less singular than $r^{-2}$ at the origin implies that $r = 0$ is a *regular point* [7, p. 197] for the differential equation (5.2.1). For an attractive potential of the form

$$U(r) \approx -Ar^{-s} \qquad (A > 0) \tag{5.3.30}$$

as $r \to 0$, the spectrum of discrete energy levels (bound states) has no lower bound if $s > 2$, so that the particle "falls" to the center[5]; for $s < 2$, there is a lower bound [cf. Section 2.9(b)]. Similarly, if the potential has the asymptotic behavior (5.3.30) as $r \to \infty$ and $s < 2$, so that (5.3.20) is violated, there are bound states with arbitrarily small (negative) energy, so that there are infinitely many discrete levels, with an accumulation point at zero energy (a well-known instance is the Coulomb potential). On the other hand, if $s > 2$, the discrete spectrum terminates at a nonzero value, so that, if (5.3.20) is satisfied, we expect that the total number of bound states must be finite. This will be confirmed in the next section.

## 5.4. The Singularities of the $S$-Function

### (a) Poles in the Analyticity Domain

From now on, unless otherwise stated, we assume that the potential always satisfies (5.3.22), so that $S(k)$ can be analytically continued at least to the strip (5.3.24). Let us consider its behavior within this strip. By taking the complex conjugate of both sides of (5.3.1) and taking into account that the potential $V(r)$ is real, we see that

$$f^*(k, r) = f(-k^*, r), \tag{5.4.1}$$

which is the extension of (5.2.15) to complex values of $k$. It then follows from (5.2.18) that

$$S^*(k) = S(-k^*), \tag{5.4.2}$$

and (5.2.19), as an analytic relation, is also extended to complex $k$.

According to (5.2.18) and Theorem 5.3.2, *the only possible singularities of $S(k)$ in the strip (5.3.24) are poles, corresponding to the zeros of $f(-k)$.* Let

$$k_0 = k_0' + iK_0 \tag{5.4.3}$$

[4] Cf. Peieris [8].

[5] Cf. Landau and Lifshitz [9, p. 54].

be one such zero, so that [cf. (5.2.13)]

$$f(-k_0) = f(-k_0, 0) = 0. \tag{5.4.4}$$

It follows that $f(-k_0, r)$ is a solution of (5.2.1) that is regular at the origin. According to the discussion at the end of Section 5.3(b), (5.2.6) remains valid for $k \in I_-$, so that

$$f(-k_0, r) \approx \exp(-ik_0' r - K_0 r) \qquad (r \to \infty). \tag{5.4.5}$$

Thus $f(-k_0, r)$ is normalizable if $K_0 > 0$. In this case, therefore, $k_0{}^2$ is an eigenvalue of the Hermitian operator $-d^2/dr^2 + U(r)$, and, according to a well-known theorem,[6] it must be real. It follows that $k_0' = 0$ if $K_0 > 0$, and (5.4.5) then has the characteristic behavior of a *bound-state* wave function. Thus, *in the strip*

$$0 < \operatorname{Im} k < \alpha/2, \tag{5.4.6}$$

*the S-function can only have poles on the positive imaginary axis, associated with bound states.* This argument is no longer valid for $\operatorname{Im} k < 0$, because (5.4.5) is then no longer normalizable, so that $S(k)$ can have complex poles in $-\alpha/2 < \operatorname{Im} k < 0$. By (5.4.2), such poles occur in pairs: if $k_0$ is a pole, so is $-k_0{}^*$.

On the real axis, $f(k)$ cannot have any zeros for $k \neq 0$, because $f(-k)$ would then also be zero [by (5.2.15)] and $\varphi(k, r)$ would identically vanish [by (5.2.12)], which would contradict (5.2.3). The function $f(k)$ may vanish at $k = 0$, but only very exceptionally, because, by (5.2.1) and (5.2.6), this can happen only if there exists a solution of $d^2u/dr^2 - U(r)u = 0$, $u(0) = 0$, that is bounded at infinity. It can be shown that, if (5.3.20) is satisfied, a zero of $f(k)$ at $k = 0$ does not correspond to a bound state [11]. Furthermore, if $f(0) = 0$, it can be shown [12] that

$$f(k) = O(k) \qquad \text{as} \quad |k| \to 0 \quad \text{from} \quad I_-, \tag{5.4.7}$$

so that, if $f(k)$ is analytic at $k = 0$, we have a simple zero.

According to (5.4.4), each bound state corresponds to a zero of $f(k)$ on the negative imaginary axis. We can now prove the result mentioned at the end of Section 5.3(c), namely, that *if the potential satisfies (5.3.20), the total number of s-wave bound states is finite.* In fact, by Theorem 5.3.1., $f(k)$ is holomorphic in $I_-$, so that it cannot have an accumulation point of zeros at any point of the negative imaginary axis; by (5.3.27) (with $r = 0$) and (5.4.7), the zeros cannot have an accumulation point at $k = 0$ or at infinity, either; therefore, their total number $n_0$ is finite. It has been shown by Bargmann [13] that

$$n_0 \leqslant M_1. \tag{5.4.8}$$

[6] See, for example, Messiah [10, p. 172].

Let us now consider the multiplicity of the poles. We have seen in Section 2.9(a) that, for a causal cutoff interaction, all bound-state poles are simple poles. It will now be shown, by explicit evaluation of the residue of $S(k)$ at a bound-state pole, that this is also true under the assumption (5.3.20). Let $k = iK_n$ ($K_n > 0$) be a bound-state pole of $S(k)$. According to (5.2.18), the residue of $S(k)$ at this pole, if it is simple, is given by

$$\operatorname{res} S(k)\Big|_{iK_n} = -f(iK_n)/\dot{f}(-iK_n), \tag{5.4.9}$$

where we have introduced the notation

$$\dot{f} = \partial f/\partial k. \tag{5.4.10}$$

To evaluate $\dot{f}(-iK_n)$, let us take the differential equation (5.2.1) for $f$,

$$f''(k,r) + [k^2 - U(r)]f(k,r) = 0, \tag{5.4.11}$$

where

$$f' = \partial f/\partial r, \tag{5.4.12}$$

and the equation obtained by differentiation with respect to $k$,

$$\dot{f}''(k,r) + [k^2 - U(r)]\dot{f}(k,r) = -2kf(k,r). \tag{5.4.13}$$

Multiplying (5.4.11) by $\dot{f}(k,r)$ and (5.4.13) by $f(k,r)$, and subtracting one from the other, we find, for $k \in I_-$,

$$\begin{aligned}\int_0^\infty (\dot{f}f'' - f\dot{f}'')\,dr &= f(k)\dot{f}'(k,0) - \dot{f}(k)f'(k,0)\\ &= 2k\int_0^\infty f^2(k,r)\,dr \qquad (k \in I_-),\end{aligned} \tag{5.4.14}$$

where the restriction of $k$ to $I_-$ ensures the convergence of the integrals at infinity, according to (5.4.5). Taking $k = -iK_n$ and noting that $f(-iK_n) = 0$, we get

$$\dot{f}(-iK_n) = \frac{2iK_n}{f'(-iK_n,0)}\int_0^\infty f^2(-iK_n,r)\,dr. \tag{5.4.15}$$

For $k = -iK_n$, the Wronskian (5.2.10) becomes

$$f(iK_n)f'(-iK_n,0) = -2K_n. \tag{5.4.16}$$

Substituting (5.4.15) and (5.4.16) in (5.4.9), we get

$$\operatorname{res} S(k)\Big|_{iK_n} = -i\Big/\int_0^\infty f^2(-iK_n,r)\,dr. \tag{5.4.17}$$

Since the right-hand side is nonvanishing, *all bound-state poles are simple*, as we wanted to prove.

Furthermore, the normalized bound-state wave function $\psi_n(\mathbf{r})$, with

$$\int |\psi_n(\mathbf{r})|^2 d^3 r = 1, \tag{5.4.18}$$

is given by [note that $f(-iK_n, r)$ is real, by (5.4.1)]

$$\psi_n(\mathbf{r}) = N_n f(-iK_n, r)/r \approx N_n \exp(-K_n r)/r \qquad (r \to \infty), \tag{5.4.19}$$

with

$$N_n = \left[4\pi \int_0^\infty f^2(-iK_n, r)\, dr\right]^{-1/2}. \tag{5.4.20}$$

Thus (5.4.17) may be rewritten as

$$i \operatorname{res} S(k)\Big|_{iK_n} = c_n = 4\pi N_n{}^2 > 0, \tag{5.4.21}$$

showing that the relation (2.12.21) between the residues of $S(k)$ at bound-state poles and the normalization factors of bound-state wave functions [cf. (5.4.19), (2.12.18)], as well as the relation (2.9.45), are valid under the assumption that $M_1$ is finite.

### *(b) Levinson's Theorem*

We have seen that, if $M_1$ is finite, the total number $n_0$ of bound $s$-states is also finite, and it is equal to the number of zeros of $f(k)$ in $I_-$. According to a well-known theorem [14, pp. 78, 116], *if a function $f(z)$ is holomorphic within a contour $C$ and continuous up to and including $C$, and if it is not zero on $C$, the number of zeros of $f(z)$ contained within $C$ is given by*

$$n_0 = (1/2\pi i) \int_C [f'(z)/f(z)]\, dz = (1/2\pi)\Delta_C \arg f(z), \tag{5.4.22}$$

where $\Delta_C \arg f(z)$ denotes the variation of the argument of $f(z)$ around the contour $C$ [cf. (2.5.30)].

In order to apply this to $f(k)$, we have to ensure continuity also at $k = 0$, so that, in accordance with (5.3.21), we assume that $M_2$ also is finite. Furthermore, to take account of the exceptional case in which $f(0) = 0$, we take the contour $C$ as shown in Fig. 5.1, indenting it around the origin by a half-circle $\gamma$ of radius $\varepsilon$, and closing it by a half-circle of large radius $K$ ($K \to \infty$).

By (5.2.17), we have

$$\arg f(k) = \eta(k), \tag{5.4.23}$$

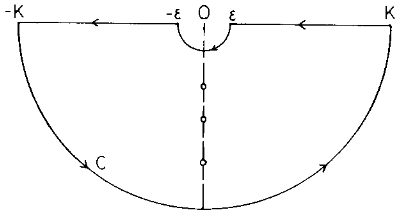



*FIG. 5.1.* The contour $C$ (O, zeros).

and it follows from (5.3.27) that

$$\lim_{|k|\to\infty} f(k) = 1 \qquad (\operatorname{Im} k \leqslant 0), \tag{5.4.24}$$

so that we may set (by convention)

$$\lim_{|k|\to\infty} \eta(k) = 0 \qquad (\operatorname{Im} k \leqslant 0). \tag{5.4.25}$$

It follows that the large half-circle gives no contribution to (5.4.22) in the limit as $K\to\infty$.

If $f(0) \neq 0$, there is also no contribution from the half-circle $\gamma$ as $\varepsilon \to 0$, and (5.4.22) yields

$$\begin{aligned} n_0 &= (1/2\pi)\lim_{\varepsilon\to 0}[\eta(\varepsilon)-\eta(\infty)+\eta(-\infty)-\eta(-\varepsilon)] \\ &= (1/\pi)[\eta(0+)-\eta(\infty)] = (1/\pi)\eta(0+) \qquad [f(0)\neq 0], \end{aligned} \tag{5.4.26}$$

where we have employed (5.4.25) and the relation $\eta(-k) = -\eta(k)$, which follows from (5.2.15). Note that, with the convention (5.4.25), the phase shift is discontinuous at the origin for $n_0 \neq 0$.

In the exceptional case $f(0) = 0$, it follows from (5.4.7) that

$$\log f(k) = \log k + O(1) \qquad \text{as} \quad |k|\to 0 \quad \text{from} \quad I_-,$$

so that

$$(1/2\pi i)\int_\gamma [f'(k)/f(k)]\, dk = (1/2\pi i)[\log k]_\varepsilon^{-\varepsilon} = -\tfrac{1}{2}. \tag{5.4.27}$$

The total number of bound states is still $n_0$, because $k = 0$ does not correspond to a bound state [cf. Section 5.4(a)].

Combining (5.4.26) and (5.4.27), we get

Levinson's Theorem [6]. *If $M_1$ and $M_2$ are finite, the zero-energy limit of the phase shift is given by*

$$\begin{aligned} \eta(0+) &= n_0\pi && \text{if} \quad f(0)\neq 0, \\ &= (n_0+\tfrac{1}{2})\pi && \text{if} \quad f(0) = 0, \end{aligned} \tag{5.4.28}$$

*where $n_0$ is the number of s-wave bound states* [and we set $\eta(\infty) = 0$ by convention].

As an illustration of Levinson's theorem, we show in Fig. 5.2 the effect of gradually increasing the strength of an attractive potential $V(r)$ so as to allow an increasing number of bound states. In each case we compare the actual wave function $u(k, r)$ at a small positive energy with the free wave function $u_0(k, r)$ at the same energy to get the phase shift [cf. (5.2.21) and (5.2.22)] and then extrapolate to zero energy.

In Fig. 5.2a, the potential is too weak to allow a bound state, and the phase shift tends to vanish at zero energy. In Fig. 5.2b, the potential is deep enough to allow a single bound state (the corresponding wave function is shown), and $u$ goes through an additional half-cycle of oscillation before "rejoining" $u_0$ as $k \to 0$, so that $\eta(0+) = \pi$. In Fig. 5.2c, the depth has increased to allow two bound states, and, correspondingly, $u$ undergoes an additional full cycle of oscillation before "rejoining" $u_0$ as $k \to 0$, leading to $\eta(0+) = 2\pi$. The ambiguity in the definition of the phase shift, which is defined only up to an additive multiple of $2\pi$, is removed by the convention (5.4.25).

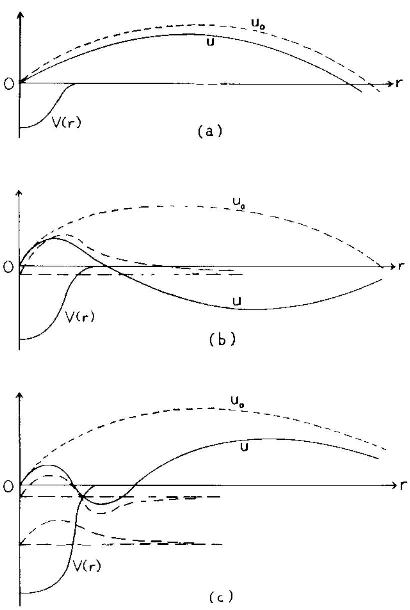



*FIG. 5.2.* Illustration of Levinson's theorem for an attractive potential $V(r)$ of increasing strength. ——, wave function $u(k,r)$ at a small positive energy; ---, wave function $u_0(k,r)$ at same energy in the absence of a potential; —·—, bound-state wave functions and energy levels. In (a), $V(r)$ is too weak to allow a bound state, $\eta(0+) = 0$; in (b), $V(r)$ is deep enough to allow a single bound state, $\eta(0+) = \pi$; in (c), $V(r)$ allows two bound states, $\eta(0+) = 2\pi$.

### (c) Yukawa-Type Potentials

If the only information we have about the potential is that it satisfies (5.3.22), we cannot assert anything about the possibility of analytically continuing $S(k)$ beyond the analyticity strip (5.3.24). We now consider a special class of potentials known as *Yukawa-type potentials*, that are of the form

$$U(r) = \int_m^\infty \sigma(\mu)(e^{-\mu r}/r)\, d\mu \qquad (m > 0). \tag{5.4.29}$$

This represents a continuous superposition of Yukawa potentials with exponential ranges $\mu^{-1}$ varying from zero up to a maximum range $m^{-1}$, distributed with weight $\sigma(\mu)$. This class of potentials has received special attention because of its relation with the nonrelativistic limit of quantum field theory for strongly interacting particles [15].

We have [cf. (5.3.20)]

$$M_1 = \int_0^\infty dr \left| \int_m^\infty \sigma(\mu)\, e^{-\mu r}\, d\mu \right| \leqslant \int_m^\infty |\sigma(\mu)| \frac{d\mu}{\mu}, \tag{5.4.30}$$

so that the finiteness of the last integral is sufficient to ensure that $M_1$ is finite.

If we set

$$\sigma(\mu) = d\rho(\mu)/d\mu \tag{5.4.31}$$

and if partial integration is allowed, (5.4.29) becomes

$$U(r) = -[\rho(m)\, e^{-mr}/r] + \int_m^\infty \rho(\mu)\, e^{-\mu r}\, d\mu,$$

so that the Yukawa-type class of potentials is essentially equivalent to the class of superpositions of exponential potentials,

$$U(r) = \int_m^\infty \rho(\mu)\, e^{-\mu r}\, d\mu. \tag{5.4.32}$$

The Yukawa potential is a particular case, with $\rho(\mu) = \text{const.}$ [corresponding to a delta function for $\sigma(\mu)$].

The class of Yukawa-type potentials is a very restrictive one. In fact, according to well-known properties of the Laplace transform [16, p. 144], it follows from (5.4.29) or (5.4.32) that $U(r)$ has an analytic continuation to complex values of $r$, regular in the half-plane $\operatorname{Re} r > 0$. In particular, no cutoff potential can belong to this class.

We see immediately from (5.4.29) and (5.4.30) that $E_\alpha$ in (5.3.22) is finite for $0 \leqslant \alpha < m$, so that, by Theorem 5.3.2, the function $f(k, r)$ is holomorphic in the half-plane

$$\operatorname{Im} k < m/2. \tag{5.4.33}$$

In this half-plane, as we have seen in Section 5.3(b), the function $g(k,r) = e^{ikr}f(k,r)$ verifies the integral equation (5.3.3), which, substituting $U(r)$ by (5.4.32), can be rewritten as

$$g(k,r) = 1 + (1/2ik)\int_m^\infty d\mu\,\rho(\mu)\,e^{-\mu r} \times \int_0^\infty (1-e^{-2ikr'})\,e^{-\mu r'}g(k,r+r')\,dr'. \tag{5.4.34}$$

Furthermore, the iteration solution (5.3.4) converges in the half-plane (5.4.33), and its first few terms, according to (5.3.5) and (5.4.34), are given by

$$\begin{aligned} g_1(k,r) &= \frac{1}{2ik}\int_m^\infty d\mu\,\rho(\mu)\,e^{-\mu r}\int_0^\infty [e^{-\mu r'} - e^{-(\mu+2ik)r'}]\,dr' \\ &= \int_m^\infty \frac{d\mu}{\mu}\frac{\rho(\mu)}{(\mu+2ik)}\,e^{-\mu r}, \end{aligned} \tag{5.4.35}$$

$$\begin{aligned} g_2(k,r) &= \frac{1}{2ik}\int_m^\infty d\mu_2\,\rho(\mu_2)\exp(-\mu_2 r)\int_m^\infty \frac{d\mu_1}{\mu_1}\frac{\rho(\mu_1)}{(\mu_1+2ik)}\exp(-\mu_1 r) \\ &\quad\times\int_0^\infty [\exp[-(\mu_1+\mu_2)r'] - \exp[-(\mu_1+\mu_2+2ik)r']]\,dr' \\ &= \int_m^\infty \frac{d\mu_1}{\mu_1}\frac{\rho(\mu_1)}{(\mu_1+2ik)}\exp(-\mu_1 r) \\ &\quad\times\int_m^\infty \frac{d\mu_2}{(\mu_1+\mu_2)}\frac{\rho(\mu_2)}{(\mu_1+\mu_2+2ik)}\exp(-\mu_2 r), \end{aligned} \tag{5.4.36}$$

so that the general term is

$$g_n(k,r) = \int_m^\infty \frac{d\mu_1}{\mu_1}\frac{\rho(\mu_1)}{(\mu_1+2ik)}\exp(-\mu_1 r)\cdots\int_m^\infty \frac{d\mu_n}{(\mu_1+\cdots+\mu_n)} \times \frac{\rho(\mu_n)\exp(-\mu_n r)}{(\mu_1+\cdots+\mu_n+2ik)}. \tag{5.4.37}$$

These results have all been obtained in the domain (5.4.33), and it is only in this domain that the integral over $r'$ in (5.4.35) converges. However, once we have obtained (5.4.37), we can employ this result for the analytic continuation of $g(k,r)$ outside of the domain (5.4.33).

If we consider first (5.4.35), we see that it is an integral of the Cauchy type, i.e., of the form

$$F(k) = \frac{1}{2\pi i}\int_L \frac{\varphi(\kappa)}{\kappa-k}\,d\kappa, \tag{5.4.38}$$

where the path $L$, in this case, extends from $im/2$ to $i\infty$. An integral of this type clearly represents a holomorphic function of $k$ in the whole plane,[7] excluding a neighborhood of the path $L$. The same is true for the general term (5.4.37).

To get an upper bound for $|g_n|$, let us exclude the neighborhood of the path $L$ shown shaded in Fig. 5.3: a sector of angular opening $2\varepsilon$, capped by an arc of a circle centered at $k = im/2$ and with radius $(m/2)\sin\varepsilon$. Outside of the shaded region, as shown in Figure 5.3, we have

$$|\mu + 2ik| = 2|k - (i\mu/2)| \geqslant \mu \sin\varepsilon. \tag{5.4.39}$$

Let us assume, for simplicity, that $\rho$ is bounded:

$$|\rho(\mu)| \leqslant C. \tag{5.4.40}$$

It then follows from (5.4.37), (5.4.39), and (5.4.40) that

$$\begin{aligned} |g_n(k,r)| &\leqslant \left(\frac{Ce^{-mr}}{\sin\varepsilon}\right)^n \int_m^\infty \frac{d\mu_1}{\mu_1{}^2} \int_m^\infty \frac{d\mu_2}{(\mu_1+\mu_2)^2} \cdots \int_m^\infty \frac{d\mu_n}{(\mu_1+\cdots+\mu_n)^2} \\ &\leqslant \frac{1}{n!}\left(\frac{Ce^{-mr}}{m\sin\varepsilon}\right)^n. \end{aligned} \tag{5.4.41}$$

It follows, as in Section 5.3(b), that the Liouville–Neumann series (5.3.4) is absolutely and uniformly convergent, the same being true for the series obtained by termwise differentiation with respect to $k$. Thus, *for all* $r \geqslant 0$, *$f(k,r)$ is holomorphic in the whole k-plane, cut along the imaginary axis from $k = im/2$ to $i\infty$.*

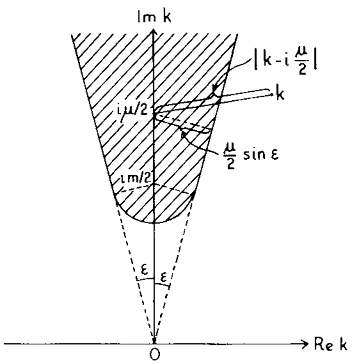



*FIG. 5.3.* The shaded region is excluded.

[7] See, for example, Hille [17, p. 182].

The restriction (5.4.40) on the weight function $\rho(\mu)$ is unnecessary. A more complete treatment, due to Martin [18, 18a], shows that it is sufficient to require

$$\int_m^\infty |\rho(\mu)|\, d\mu/\mu^2 < \infty, \tag{5.4.42}$$

which, as in (5.4.30), is equivalent to the finiteness of $M_1$. It also follows from this treatment that

$$|e^{ikr} f(k,r) - 1| \to 0 \qquad \text{as} \quad |k| \to \infty, \quad r \geqslant 0, \tag{5.4.43}$$

where $|k| \to \infty$ in any direction, outside of the shaded region in Fig. 5.3. This extends the result (5.3.27) to the whole analyticity domain. The nature of the singularity along the imaginary axis has also been investigated by Martin, who found that it corresponds in general to a branch cut, with $f(k,r)$ approaching finite limits as $k$ approaches the cut from the left or from the right, except near $k = im/2$, where $f(k,r)$ has a logarithmic singularity [8, 19].

These results can be readily verified for the first Born approximation (5.4.35), which is of the form of a Cauchy integral (5.4.38). If we formally apply (1.6.9) to (5.4.38), we infer that, as $k$ approaches a point $\kappa_0$ on $L$ from the left or from the right, respectively, the integral approaches the limits

$$F^{\pm}(\kappa_0) = \pm\, \tfrac{1}{2}\varphi(\kappa_0) + \frac{1}{2\pi i} \int_L \frac{\varphi(\kappa)}{\kappa - \kappa_0}\, d\kappa, \tag{5.4.44}$$

where the integral denotes a Cauchy principal value. Thus the discontinuity across $L$ is given by

$$F^{+}(\kappa_0) - F^{-}(\kappa_0) = \varphi(\kappa_0). \tag{5.4.45}$$

A precise form of these results is the following [20]:
*Let $\varphi(\kappa)$ satisfy a Hölder condition on $L$, i.e., let*

$$|\varphi(\kappa_2) - \varphi(\kappa_1)| \leqslant A\,|\kappa_2 - \kappa_1|^{\delta} \qquad (\delta > 0), \tag{5.4.46}$$

*for any points $\kappa_1, \kappa_2$ on $L$. Then, at any point $\kappa_0$ on $L$, except an endpoint $\hat{\kappa}$ where $\varphi(\hat{\kappa}) \neq 0$, the Cauchy integral (5.4.38) is continuous on $L$ from left and right, respectively, approaching the limits (5.4.44) as $k$ approaches $\kappa_0$ from either side. Near an endpoint $\hat{\kappa}$ with $\varphi(\hat{\kappa}) \neq 0$, we have*

$$F(k) = \frac{\varphi(\hat{\kappa})}{2\pi i} \ln\left(\frac{1}{k - \hat{\kappa}}\right) + F_0(k), \tag{5.4.46a}$$

*where $F_0(k)$ is bounded as $k \to \hat{\kappa}$.* The relations (5.4.44), which should be compared with (1.6.7), are also known as *Plemelj's formulas*. These results should also be compared with (2.9.41) and (2.8.20).

The logarithmic singularity of (5.4.35) at $k = im/2$ may be immediately checked for $r = 0$ in the case of a Yukawa potential, $U(r) = U_0 e^{-mr}/r$, for which $\rho(\mu) = U_0$ in (5.4.32). In this case we find

$$g_1(k,0) = f_1(k) = \frac{U_0}{2ik} \ln\left(\frac{m+2ik}{m}\right). \tag{5.4.47}$$

It follows from the above results, together with (5.2.18), that $S(k)$ *is meromorphic in the k-plane, cut along the imaginary axis from* $k = im/2$ *to* $i\infty$, *and from* $k = -im/2$ *to* $-i\infty$. In the upper half-plane, apart from the cut due to $f(k)$, the only possible singularities arise from the zeros of $f(-k)$, and these, by Section 5.4(a), can occur only on the imaginary axis, giving rise to simple poles of $S(k)$, associated with bound states. In the lower half-plane, in addition to the cut, there may be complex poles. Furthermore, if $|k| \to \infty$ along any direction, excluding sectors with an arbitrarily small angle $\varepsilon$ around the positive and negative imaginary axis, we have [cf. (5.4.43)]

$$\lim_{|k|\to\infty} S(k) = 1. \tag{5.4.48}$$

Translating these results into the $E$-plane, we see that, on the physical sheet [Section 2.8(b)], $S(E)$ is a holomorphic function, apart from two branch cuts, one along the positive real axis, and one along the negative real axis, from $E = -m^2/8$ to $-\infty$ (the latter is referred to as the *left-hand cut*); in addition, there may be poles on the negative real axis, corresponding to the energies of bound states. These properties, together with (5.4.48), may be employed to derive a dispersion relation for $S(E)$, by a procedure similar to that employed in the derivation of (2.9.41), but there is an additional contribution due to the left-hand cut.

A special case of (5.4.32) that is exactly soluble (for $s$-waves) is the *exponential potential*,

$$U(r) = -U_0 e^{-mr}, \tag{5.4.49}$$

for which the weight function $\rho(\mu)$ is a delta function; we take $U_0 > 0$, corresponding to an attractive potential. Substituting $U(r)$ by (5.4.49) in (5.2.1), and making the change of variable

$$x = 2\sqrt{U_0}\, e^{-mr/2}/m = \zeta e^{-mr/2}, \tag{5.4.50}$$

we find

$$d^2u/dx^2 + (1/x)\, du/dx + (1+4k^2/m^2x^2)u = 0. \tag{5.4.51}$$

Linearly independent solutions are the Bessel functions $J_{\pm\nu}(x)$, where

$$\nu = 2ik/m. \tag{5.4.52}$$

When $x \to 0$ $(r \to \infty)$, we have

$$J_\nu(x) \approx (x/2)^\nu/\Gamma(\nu+1) \qquad (x \to 0).$$

It follows that (5.2.6) is satisfied by

$$f(k,r) = (2/\zeta)^{\nu}\Gamma(\nu+1)J_{\nu}(x)$$
$$= \exp\left[-i\frac{k}{m}\log\left(\frac{U_0}{m^2}\right)\right]\Gamma\left(1+2i\frac{k}{m}\right)J_{2ik/m}\left(\frac{2}{m}\sqrt{U_0}\,e^{-mr/2}\right) \tag{5.4.53}$$

This is regular for $\operatorname{Im} k < m/2$, as it should be. The bound states $k = i\kappa_n$ ($\kappa_n > 0$) are the roots of [21]

$$J_{2\kappa_n/m}(2\sqrt{U_0}/m) = 0 \qquad (\kappa_n > 0). \tag{5.4.54}$$

The zeros of $J_{-2ik/m}(2\sqrt{U_0}/m)$ for $\operatorname{Im} k > 0$ must lie on the imaginary $k$-axis [22, p. 88], in agreement with the discussion following (5.4.5).

For $\operatorname{Im} k > m/2$, we see from (5.4.53) that $f(k,r)$ is meromorphic, its poles being those of $\Gamma(1+2ik/m)$, namely,

$$k_n = inm/2 \qquad (n = 1,2,3,\ldots). \tag{5.4.55}$$

The fact that the branch cut from $im/2$ to $i\infty$ found for Yukawa-type potentials is here reduced to a series of poles is due to the singular (deltalike) character of the weight function $\rho(\mu)$.

The $S$-function, according to (5.2.18), is also meromorphic. Its poles in $I_+$ lie on the imaginary axis, and they are of two different kinds: (i) poles due to the zeros of $f(-k)$; these correspond to the roots of (5.4.54), and they are associated with bound states; (ii) the poles of $f(k)$, given by (5.4.55); these do *not* correspond to bound states. They are known as *redundant poles*. Their existence was first pointed out by Ma [23], as an example of difficulties in the association of poles of $S$ on the imaginary axis with bound states that had been proposed by Heisenberg (cf. Section 4.1). In fact, as we have seen, such an association can only be made unambiguously within the strip (5.4.6); outside of this strip, according to the discussion in Section 5.3(c), the $S$-function can no longer be defined by the asymptotic behavior (5.2.4), because of the perturbation due to the tail of the potential. For Yukawa-type potentials, this gives rise to the cut from $im/2$ to $i\infty$, which may be regarded as a "redundant singularity"; for a potential with an exponential tail, the cut degenerates into the series of redundant poles (5.4.55).

## 5.5. Cutoff Potentials

We have seen following Theorem 5.3.2 that, for a potential decreasing faster than any exponential for $r \to \infty$ [$E_\alpha$ finite for all $\alpha \geqslant 0$ in (5.3.22)], the Jost function is an entire function and $S(k)$ is meromorphic in the whole

$k$-plane. The poles of $S(k)$ are the zeros of $f(-k)$ and, according to Section 5.4(a), $S(k)$ is holomorphic in $I_+$, apart from a finite number of simple poles on the imaginary axis, corresponding to bound states.

These properties are true, in particular, for a cutoff potential, i.e., a potential such that

$$U(r) = 0 \qquad (r > a). \tag{5.5.1}$$

It is only for such interactions that we can compare the present results with those obtained in Part I.

The restriction to cutoff potentials enables us to obtain additional information, besides (5.3.27), on the behavior of the Jost function as $|k| \to \infty$. Let $|k| \to \infty$ along any direction in $I_+$,

$$k = k' + i\kappa = |k| e^{i\theta} \qquad (|k| \to \infty,\ 0 < \theta < \pi), \tag{5.5.2}$$

and let us consider the asymptotic behavior of each term of the Liouville–Neumann series (5.3.4) for $r < a$. We then have, by (5.3.6),

$$g_n(k,r) = (1/k) \int_r^a e^{-ik(\rho-r)} \sin[k(\rho-r)]\, U(\rho) g_{n-1}(k,\rho)\, d\rho \qquad (r < a). \tag{5.5.3}$$

Let us begin with the first Born approximation,

$$g_1(k,r) = (1/2ik) \int_r^a [1 - e^{-2ik(\rho-r)}]\, U(\rho)\, d\rho. \tag{5.5.4}$$

Clearly, the second term within square brackets, which blows up exponentially in the limit (5.5.2), dominates the asymptotic behavior. We have to deal with the asymptotic behavior of a Fourier–Laplace integral, which can be determined by applying well-known methods [24]. It depends essentially on the behavior of $U(r)$ near the endpoint of the domain of integration, $r = a$. Let us assume, for simplicity, that $U(r)$ is bounded,

$$|U(r)| \leqslant U_0, \tag{5.5.5}$$

and that there is a discontinuity at the cutoff point,

$$U(a-) \neq 0. \tag{5.5.6}$$

Then the asymptotic behavior of (5.5.4) in the limit (5.5.2) is obtained by partial integration:

$$g_1(k,r) \approx -\frac{U(a-)}{4k^2} e^{-2ik(a-r)}[1 + O(k^{-1})]. \tag{5.5.7}$$

On the other hand, by (5.5.3) and (5.3.9),

$$|g_n(k,r)| \leqslant \frac{C}{|k|} \int_r^a e^{2\kappa(\rho-r)} |U(\rho)|\, |g_{n-1}(k,\rho)|\, d\rho, \tag{5.5.8}$$

so that (5.5.5) and (5.5.7) lead to

$$|g_2(k,r)| \leqslant BC|k|^{-3}e^{2\kappa(a-r)}U_0(a-r), \tag{5.5.9}$$

where $B$ is a constant. It is now easy to show, by complete induction, that

$$|g_n(k,r)| \leqslant \frac{B}{|k|^2}e^{2\kappa(a-r)}\frac{[CU_0(a-r)/|k|]^{n-1}}{(n-1)!}, \tag{5.5.10}$$

so that

$$\left|\sum_{n=2}^{\infty} g_n(k,r)\right| \leqslant \frac{B}{|k|^2}e^{2\kappa(a-r)}\left\{\exp\left[\frac{CU_0(a-r)}{|k|}\right]-1\right\}$$
$$= O(k^{-1})|g_1(k,r)|. \tag{5.5.11}$$

Finally, putting together the above results, we get,[8] for $r < a$,

$$g(k,r) = e^{ikr}f(k,r) \approx g_B(k,r)$$
$$= 1 - [U(a-)/4k^2]e^{-2ik(a-r)}[1+O(k^{-1})] \qquad (|k|\to\infty \text{ in } I_+), \tag{5.5.12}$$

where $g_B$ denotes *Born's approximation*, corresponding to the first two terms of (5.3.4). In particular, taking $r = 0$, we find

$$f(k) \approx f_B(k) = 1 - [U(a-)/4k^2]e^{-2ika}[1+O(k^{-1})] \qquad (|k|\to\infty \text{ in } I_+), \tag{5.5.13}$$

where the second term dominates along directions away from the real axis. On the other hand, according to (5.3.27),

$$f(-k) \approx 1 \qquad \text{as} \quad |k|\to\infty \quad \text{in} \quad I_+. \tag{5.5.14}$$

Combining (5.5.13) and (5.5.14) with (5.2.18), we obtain the asymptotic behavior of the $S$-function,

$$S(k) \approx 1 - [U(a-)/4k^2]e^{-2ika}[1+O(k^{-1})] \qquad (|k|\to\infty \text{ in } I_+), \tag{5.5.15}$$

and the asymptotic behavior in $I_-$ then follows from (5.2.19).

If $U(a-) = 0$, but $U'(a-) \neq 0$, the asymptotic behavior of (5.5.4) is obtained by integrating by parts once more, yielding an additional power of $k$ in the denominator of (5.5.7). More generally, if $U^{(m)}(a-)$ is the first non-vanishing derivative of $U(r)$ at $r = a-$, the asymptotic behavior of (5.5.4) is obtained by $m+1$ partial integrations [24], leading to[9]

$$S(k) \approx 1 + [U^{(m)}(a-)/(2ik)^{m+2}]e^{-2ika} \qquad (|k|\to\infty \text{ in } I_+). \tag{5.5.16}$$

[8] The obtained asymptotic behavior is not valid uniformly in $r$ up to $r = a$; a more careful derivation is given by Regge [25].

[9] Cf. Humblet [26].

In any case, it follows that

$$|S_a(k)| = |S(k)e^{2ika}| \quad \text{is bounded as} \quad |k| \to \infty, \quad \operatorname{Im} k \geqslant 0. \tag{5.5.17}$$

We can summarize all of the above properties in the following theorem:

THEOREM 5.5.1. *Let $U(r)$ be a cutoff potential with finite first moment $M_1$. Then $S(k)$ is meromorphic in the whole k-plane and holomorphic in $I_+$, except possibly for a finite number of simple poles on the positive imaginary axis [with residues satisfying (5.4.21)], associated with bound states; $S(k)$ satisfies the relations $S(-k) = S^*(k^*) = S^{-1}(k)$, and $|S_a(k)|$ remains bounded as $|k| \to \infty$, $\operatorname{Im} k \geqslant 0$.*

Thus $S(k)$ satisfies all the properties that were derived in Section 2.9 from Van Kampen's causality condition. It follows that the dispersion relation (2.9.44) is valid for cutoff potentials.

The canonical product expansion (2.9.58) was derived from the properties expressed in Theorem 5.5.1, so that its validity in the present case also follows. Furthermore, according to (5.5.16),

$$\lim_{|k| \to \infty} [\ln |S(k)|/|k|] = 2a \sin \theta \qquad (k = |k| e^{i\theta}, \ 0 < \theta < \pi), \tag{5.5.18}$$

where the limit has to be understood in the sense of the Ahlfors–Heins theorem 2.5.7. It then follows from the Ahlfors–Heins theorem that $\alpha = a$ in (2.9.58) (as had been anticipated in Section 2.9), so that

$$S(k) = \pm e^{-2ika} \prod_n \frac{[1-(k/k_n{}^*)]}{[1-(k/k_n)]}, \tag{5.5.19}$$

where the product runs over all the poles $k_n$ of $S(k)$, taken in the order of increasing modulus, and the plus sign usually holds [the minus sign applies only to the exceptional case in which $S(0) = -1$, corresponding to $f(0) = 0$].

Under the assumptions of Theorem 5.5.1, the Jost function is an entire function, so that

$$F(k^2) = f(k) f(-k) \tag{5.5.20}$$

is an entire function of $k^2$. Furthermore, if we set $k^2 = Re^{i\theta}$ and denote by $M(R)$ the maximum modulus of $F(k^2)$ on a circle of radius $R$ centered at the origin, it follows from (5.5.14) and (5.5.13) [or its generalization, analogous to (5.5.16)] that [cf. (5.5.18)]

$$\limsup_{R \to \infty} [\ln M(R)/R^{1/2}] = 2a. \tag{5.5.21}$$

Therefore, by definition [27, p. 8], $F(k^2)$ is an entire function of $k^2$ of *order* $\frac{1}{2}$ and *type* $2a$ [an entire function of order $\rho$ and type $\tau$ is, roughly, an entire function such that the growth of $M(R)$ is bounded by $\exp(\tau R^{\rho})$ as $R \to \infty$]. We can then apply a known theorem [27, p. 24], according to which an *entire function of nonintegral order has an infinite set of zeros*. We conclude that $f(k)$ has infinitely many zeros, or, equivalently, that $S(k)$ has infinitely many poles. Since the number of poles in $I_+$ is finite, we see that, *under the assumptions of Theorem 5.5.1, $S(k)$ has an infinite number of poles in $I_-$.*

The asymptotic behavior of the pole distribution for large $|k|$ follows[10] from (5.5.16). The poles of $S(k)$ are the zeros of $S(-k)$, so that, by (5.5.16), they must approach, asymptotically, the roots of

$$1 \pm [A^2/(-2i\beta)^{m+2}]\, e^{2i\beta} = 0, \tag{5.5.22}$$

where we have set

$$\beta = ka, \tag{5.5.23}$$

$$a^{m+2}\, U^{(m)}(a-) = \pm A^2, \tag{5.5.24}$$

and the plus or the minus sign must be taken according as to whether $U^{(m)}(a-)$ is positive or negative, respectively.

On account of the symmetry of the poles with respect to the imaginary axis [cf. Section 5.4(a)], it suffices to consider the roots of (5.5.22) in the fourth quadrant of the $k$-plane. Clearly, if $\beta_n$ is a root with very large modulus, $\beta_n + \pi$ will also be very close to a root, and we must have $|\mathrm{Re}\,\beta_n| \gg |\mathrm{Im}\,\beta_n|$. Let us therefore make the substitution

$$\beta_n = U_n + \varepsilon_n, \tag{5.5.25}$$

where

$$U_n = n\pi - (\pi/2)\,[(m/2) + \zeta], \tag{5.5.26}$$

$$\begin{aligned} \zeta &= 0 \quad \text{if} \quad U^{(m)}(a-) > 0, \\ &= 1 \quad \text{if} \quad U^{(m)}(a-) < 0. \end{aligned} \tag{5.5.27}$$

With this substitution, (5.5.22) becomes

$$\exp(2i\varepsilon_n) = [2(U_n + \varepsilon_n)]^{m+2}/A^2,$$

so that

$$\varepsilon_n = -(i/2)\,\{\ln[(2U_n)^{m+2}/A^2] - (m+2)\ln[1 + (\varepsilon_n/U_n)]\}. \tag{5.5.28}$$

Since $|\beta_n|$ is large, $n$ is a large integer, and we assume $0 \leqslant \mathrm{Re}\,\varepsilon_n < \pi/2$, so that $|\varepsilon_n| \ll U_n$. Thus we can expand the logarithm in (5.5.28) in a power series. In the first approximation, this leads to

$$\beta_n = U_n + \varepsilon_n \approx U_n - [(m+2)/4U_n]\ln[(2U_n)^{m+2}/A^2] - (i/2)\ln[(2U_n)^{m+2}/A^2], \tag{5.5.29}$$

which is asymptotically valid for sufficiently large $U_n$.

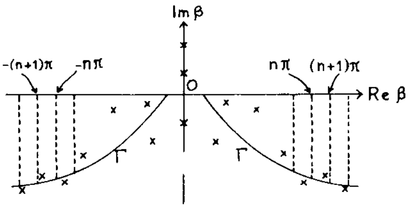
[10] Cf. Humblet [26, p. 44].

For large $n$, the distance $|\beta_{n+1}-\beta_n|$ between two consecutive poles approaches $\pi$; we have

$$\operatorname{Re}\beta_n = O(n), \qquad \operatorname{Im}\beta_n = O(\log n) \qquad (n \to \infty). \tag{5.5.30}$$

The poles tend to be distributed asymptotically on the curve $\Gamma$ defined by (5.5.22). As shown by (5.5.30) and Fig. 5.4, the two branches of this curve tend to become asymptotically parallel to the real axis.

*FIG. 5.4.* For large $n$, the poles $\beta_n$ approach asymptotically the two branches of the curve $\Gamma$, and the spacing between two consecutive poles approaches $\pi$ ($\times$, poles) [after J. Humblet, *Mém. Soc. Roy. Sci. Liège* **12**, No. 4, 46 (1952)].




It is clear from the above discussion that the asymptotic pole distribution does not depend on the behavior of the potential over the whole interval $0 \leqslant r \leqslant a$; it depends only on the way in which the potential approaches zero at $r = a$. Thus "large" poles, located far away from the origin, do not have much physical significance. They reflect the existence of a sharp cutoff, just as the branch cuts found in Section 5.4(c) reflect the existence of a Yukawa-type tail. In fact, if we cut off a Yukawa-type potential at a distance far greater than its range, we do not expect much change in the behavior of physical quantities [cf., however, Section 8.4(c)], but the branch cuts are replaced by poles with an asymptotic distribution of the type just found. This sensitivity of the analytic behavior to the tail of the potential is another example of the instability of analytic continuation discussed in Section 2.9(e).

## 5.6. An Example: Square Well or Barrier

As an illustration of the preceding results, let us consider the case of a square potential well or barrier,

$$\begin{aligned} U(r) &= U_0, \qquad r < a, \\ &= 0, \qquad r > a, \end{aligned} \tag{5.6.1}$$

where $U_0 < 0$ for a well and $U_0 > 0$ for a barrier. Let us introduce the dimensionless parameter $\beta = ka$, as in (5.5.23); the corresponding parameter for the wave number within the potential is $(\beta^2 \pm A^2)^{1/2}$, where [cf. (5.5.24)]

$$-a^2 U_0 = \pm A^2 \qquad (+\text{ for a well, } -\text{ for a barrier}) \tag{5.6.2}$$

is a measure of the well depth (barrier height).

The $S$-function can be obtained by matching (continuously and with continuous derivative) the external and the internal solution at the boundary $r = a$. For $s$-waves, this leads to

$$S(\beta) = e^{-2i\beta}\left(\frac{(\beta^2 \pm A^2)^{1/2}\cot[(\beta^2 \pm A^2)^{1/2}] + i\beta}{(\beta^2 \pm A^2)^{1/2}\cot[(\beta^2 \pm A^2)^{1/2}] - i\beta}\right). \tag{5.6.3}$$

This is a meromorphic function of $\beta$ satisfying all the conditions stated in Theorem 5.5.1.

The poles of $S(\beta)$ are the roots of the complex transcendental equation

$$(\beta^2 \pm A^2)^{1/2}\cot[(\beta^2 \pm A^2)^{1/2}] = i\beta. \tag{5.6.4}$$

The roots can be located [28] by a combination of graphical, numerical, and asymptotic methods. We confine ourselves to a discussion of the results.

As the potential strength parameter $A$ increases from 0 to $\infty$, the poles describe continuous trajectories in the $\beta$-plane. We adopt the following nomenclature for describing the poles: $a$-poles ($a$ = antibound; cf. Section 4.2) for poles on the negative imaginary axis; $b$-poles ($b$ = bound) for those on the positive imaginary axis; $c$-poles ($c$ = complex) for those outside of the imaginary axis.

### (a) The Potential Well

To keep track of the poles, we shall identify them by their limiting positions in the free-particle limit $A \to 0$, namely,

$$\lim_{A \to 0} \beta_n = n\pi - i\infty \qquad (n = 0, \pm 1, \pm 2, \ldots). \tag{5.6.5}$$

For small values of $A$, therefore, the poles are located far from the real axis. Thus, in agreement with the considerations of Sections 4.5 and 4.6, a wave packet will propagate in the presence of a very shallow well practically in the same way as it would do in free space (in spite of the infinite number of poles).

The paths described by the first few poles in the complex $\beta$-plane ($\beta = u + iv$) as a function of $A$ are shown in Fig. 5.5. The numbers beside the poles give the corresponding values of $A$. The path described by $\beta_n$ has the straight line $u = n\pi$ as vertical asymptote for $A \to 0$.

Let us consider first the behavior of $\beta_0$ (indicated by $\square$ in Fig. 5.5). It starts as an $a$-pole at $-i\infty$ for $A \to 0$ and it moves upwards along the negative

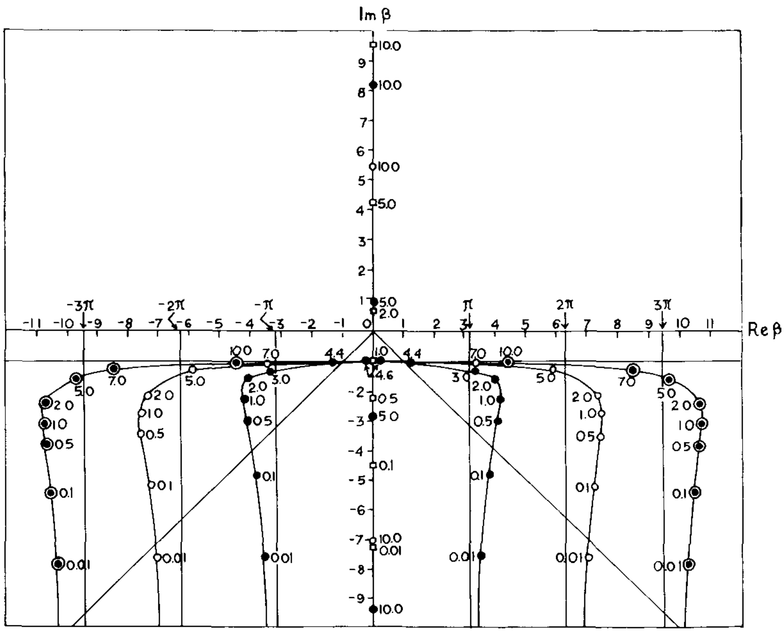



*FIG. 5.5.* The poles $\beta_n$ of $S_0(\beta)$ for a potential well (□, $n=0$; ●, $n=\pm 1$; ○, $n=\pm 2$; ◉, $n=\pm 3$). The numbers beside the poles give the corresponding values of $A$. The curves in full line are the paths described by the poles. The bisectors of the third and fourth quadrants are also indicated [after H. M. Nussenzveig, *Nucl. Phys.* **11**, 499 (1959)].

imaginary axis as $A$ increases, crossing the origin for $A = \pi/2$. For $A > \pi/2$, it becomes a $b$-pole, giving rise to the first bound state. Thereafter, it moves up from 0 to $i\infty$ as $A$ increases from $\pi/2$ to $\infty$. It always corresponds to the ground state (deepest energy level).

What happens when $\beta_0$ crosses the origin? For $A = \frac{1}{2}\pi - \varepsilon$, $|\varepsilon| \ll 1$, one finds that $\beta_0 \approx -i\pi\varepsilon/2$, and a one-pole approximation (cf. Section 4.2) is valid for $\beta$ near the origin, so that

$$S(\beta) \approx -\frac{\beta - \beta_0{}^*}{\beta - \beta_0} \approx -1 + \frac{i\pi\varepsilon}{\beta + i(\pi/2)\varepsilon}\,. \tag{5.6.6}$$

Thus $\beta_0{}^*$, which is a zero of $S(\beta)$, approaches the origin simultaneously with $\beta_0$, and annihilates the singularity there. Note that $A = \pi/2$ is one of the exceptional cases in which $S(0) = -1$.

Substituting (5.6.6) in (4.1.1), we find that the $s$-wave scattering cross-section, under the above conditions, is given by

$$\sigma(\beta) \approx 4\pi a^2/[\beta^2 + (\pi\varepsilon/2)^2], \tag{5.6.7}$$

which becomes very large as $\beta \to 0$ [for $\varepsilon \to 0$, $\sigma(0) \to \infty$]. This is an example of the situation discussed in Section 4.2, in which an $a$-pole located close to the origin gives rise to an anomalously large cross section at low energies.

Let us consider next the first pair of $c$-poles, $\beta_{\pm 1}$ (indicated by ● in Fig. 5.5), which are symmetrical about the imaginary axis. They start at $\pm\pi - i\infty$ for $A \to 0$ and they move upwards as $A$ increases, until (for $A \gtrsim 2$) they approach the straight line $v = -1$. Thereafter, they move towards the imaginary axis, where they coalesce (for $A \approx 4.6$) at the point $\beta = -i$, giving rise to a double pole. For larger values of $A$, the double pole splits into a pair of $a$-poles, which move in opposite directions: one of them moves down from $-i$ to $-i\infty$ as $A$ increases from $\sim 4.6$ to $\infty$; the other one, that moves upwards, crosses the origin for $A = 3\pi/2$, giving rise to the second bound state. Afterwards, as a $b$-pole, it moves up to $i\infty$ when $A$ increases to $\infty$.

An entirely similar process takes place with each pair of poles $\beta_{\pm n}$: after coalescing at $\beta = -i$, they move in opposite directions along the imaginary axis, and one of them gives rise to a new bound state as it crosses the origin for $A = (n+\frac{1}{2})\pi$ [note that this mechanism is in agreement with (5.4.7)]. Thus every pair of $c$-poles ultimately gives rise to an $a$-pole and a $b$-pole. We see also that there is an antibound energy level between each pair of consecutive bound-state energy levels [cf. the remarks following (2.9.46)]. This is to be expected, since an antibound level corresponds to an energy at which the regular internal solution can be smoothly joined to a purely *increasing* exponential in the external region (Section 4.2).

We can also see in Fig. 5.5 that the pole distribution, for a given value of $A$, asymptotically approaches curves similar to $\Gamma$ in Fig. 5.4.

### *(b) The Potential Barrier*

This case is much simpler, because there are no bound states. All the poles are $c$-poles, and they can be labeled by their limiting positions as $A \to 0$,

$$\lim_{A\to 0} \beta_n = (n-\tfrac{1}{2})\pi - i\infty \qquad (n = 1, 2, \ldots), \tag{5.6.8}$$

where we have restricted ourselves to the fourth quadrant.

The paths described by the first few poles as a function of $A$ are shown in Fig. 5.6. The pole $\beta_1$ starts at $\pi/2 - i\infty$ for $A \to 0$ and it moves upwards and away from the imaginary axis as $A$ increases, tending to approach the real axis for very large values of $A$. The same happens with all other poles. For $A \to \infty$, corresponding to a hard sphere, all the poles move out to infinity along the direction of the real axis, so that $S$ has no poles in this limit [cf. (2.3.8)].

For large $A$, corresponding to a high barrier, the lowest-order poles tend

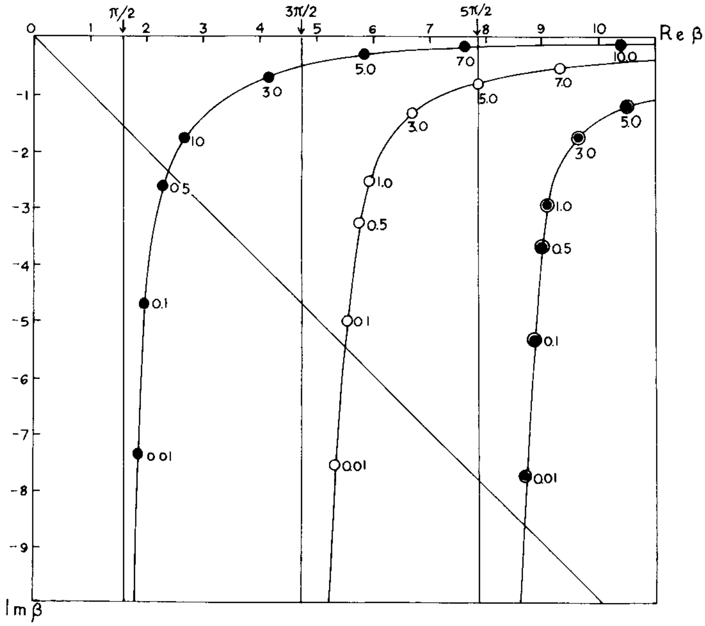



*FIG. 5.6.* The poles $\beta_n$ of $S_0(\beta)$ for a potential barrier (●, $n = 1$; ○, $n = 2$; ◉, $n = 3$). The numbers beside the poles give the corresponding values of $A$. The curves in full line are the paths described by the poles. The bisector of the fourth quadrant is also indicated [after H. M. Nussenzveig, *Nucl. Phys.* **11**, 499 (1959)].

to be located close to the real axis, and they may give rise to sharp resonances, associated with long-lived transient modes. The physical interpretation of these modes is very similar to that given for the delta-function potential in Section 4.6(a), and their lifetime can also be obtained from (4.6.31) with $k_n$ replaced by the "internal velocity" within the potential. They correspond to virtual energy levels lying above the top of the barrier.

In contrast with this, one does not get very sharp $s$-wave resonances even for a very deep potential well; the poles do not approach the real axis closer than shown in Fig. 5.5, keeping below the line $\text{Im}\,\beta = -1$. The physical reason for the short lifetime of the transient modes in this case is the large value of the "internal velocity."

## 5.7. Mittag–Leffler and Transient-Mode Expansions

### *(a) Mittag–Leffler Expansion of S(k)*

In order to extend the treatment given in Section 4.6 to an arbitrary cutoff potential, we need the partial-fraction decomposition of the $S$-function in terms of its poles.[11] We restrict ourselves to the case in which there are only simple poles [higher-order poles occur only exceptionally, and one can usually treat them as limiting cases of simple poles lying close together, by varying parameters, as in Section 5.6(a)]. We can then apply the following simple corollary of the Mittag-Leffler theorem[12]:

THEOREM 5.7.1. *Let $F(k)$ be a meromorphic function all of whose poles $k_n$ are simple and ordered in the order of increasing modulus,*

$$0 < |k_1| \leqslant |k_2| \leqslant |k_3| \leqslant \cdots, \tag{5.7.1}$$

*and let*

$$r_n = \operatorname{res} F(k)\Big|_{k=k_n}. \tag{5.7.2}$$

*If there exists an integer $M \geqslant 0$ such that*

$$\sum_n |r_n|/|k_n|^{M+1} < \infty, \tag{5.7.3}$$

*the following Mittag-Leffler expansion is valid:*

$$F(k) = \mathscr{E}(k) + \sum_n (k/k_n)^M r_n/(k-k_n), \tag{5.7.4}$$

*where $\mathscr{E}(k)$ is an entire function. Furthermore, apart from a finite number of terms that can be subtracted out, the resulting series is uniformly convergent in any finite domain of the k-plane.*

In order to apply this theorem to the $S$-function, we need an asymptotic estimate for the residue of $S(k)$ at a complex pole $k_n$ as $|k_n| \to \infty$. As in (5.4.9), we have

$$r_n = \operatorname{res} S(k)\Big|_{k_n} = -f(k_n)/\dot{f}(-k_n), \tag{5.7.5}$$

where $f(k)$ is the Jost function and $\dot{f} = \partial f/\partial k$. It follows from (5.3.27) and (5.3.25) with $r = 0$ [assuming that $U(r)$ is integrable at the origin] that

$$f(k_n) = 1 + O(k_n^{-1}) \qquad \text{as} \quad |k_n| \to \infty \quad \text{in} \quad I_-. \tag{5.7.6}$$

On the other hand, it can be shown[13] that the asymptotic expansion of $\dot{f}$ is obtained from that of $f$ by termwise differentiation with respect to $k$. It then follows, either from (5.5.13) or from the more general expression for the asymptotic behavior of $f(k)$ corresponding to (5.5.16), that

$$\dot{f}(k) = -2ia[f(k)-1][1+O(k^{-1})] \qquad \text{as} \quad |k| \to \infty \quad \text{in} \quad I_+. \tag{5.7.7}$$

[11] Cf. Humblet [26].

[12] See, for example, Hurwitz and Courant [29, p. 113].

[13] Cf. Humblet [26, Appendix].

Taking $k = -k_n$, with $f(-k_n) = 0$, we get

$$\dot{f}(-k_n) = 2ia + O(k_n^{-1}) \qquad \text{as} \quad |k_n| \to \infty \quad \text{in} \quad I_-. \tag{5.7.8}$$

Substituting (5.7.6) and (5.7.8) in (5.7.5), we finally get

$$r_n = (i/2a) + O(k_n^{-1}). \tag{5.7.9}$$

It follows from (5.5.30) and (5.7.9) that

$$\sum_n |r_n|/|k_n|^2 < \infty, \tag{5.7.10}$$

so that we can apply Theorem 5.7.1 with $M = 1$:

$$\begin{aligned} S(k) &= \mathscr{E}_1(k) + \sum_n (k/k_n) r_n/(k-k_n) \\ &= \mathscr{E}_1(k) + \sum_n r_n\{[1/(k-k_n)] + (1/k_n)\}, \end{aligned} \tag{5.7.11}$$

where $\mathscr{E}_1(k)$ is an entire function.

In the summation over the poles, the ordering (5.7.1) is to be preserved, so that the contribution from each pole $k_n$ is (except for poles on the imaginary axis) paired with that from the corresponding pole $-k_n{}^*$ [cf. Section 5.4(a)], which is associated with the residue $-r_n{}^*$ [by (5.4.2)]. It follows from (5.7.9) and (5.5.30) that

$$\begin{aligned} [r_n/(k-k_n)] - [r_n{}^*/(k+k_n{}^*)] &= i(k - \operatorname{Im} k_n)/[a(k-k_n)(k+k_n{}^*)] + O(1/|k_n|^2) \\ &= O(n^{-2}\log n) \qquad (n \to \infty), \end{aligned} \tag{5.7.12}$$

so that we can break up the series in (5.7.11) into the sum of two separately convergent series and rewrite it as

$$S(k) = \mathscr{E}(k) + \sum_n r_n/(k-k_n), \tag{5.7.13}$$

where the entire function $\mathscr{E}(k)$ differs from $\mathscr{E}_1(k)$ by a constant term.

The series that appears in this remarkably simple expansion is to be summed pairwise, so that, according to (5.7.12), it is still absolutely and uniformly convergent in any bounded domain of the $k$-plane (excluding a neighborhood of each pole).

### (b) Application to the Cross Section

According to (4.1.1), the $s$-wave scattering cross section is given by

$$\sigma(k) = (\pi/k^2)|1 - S(k)|^2 = (\pi/k^2)[2 - S(k) - S(-k)], \tag{5.7.14}$$

where we have made use of unitarity and the symmetry relation (5.2.20).

It is convenient to employ the first expression for $S(k)$ in (5.7.11). Noting that

$$\mathscr{E}_1(k) = S(0) + kP(k) = 1 + k \sum_{j=0}^{\infty} a_j k^j, \tag{5.7.15}$$

where $P(k)$ is an entire function and we have excluded the exceptional cases in which $S(0) = -1$, we get

$$\sigma(k) = -(\pi/k)[P(k) - P(-k)] - 2\pi \sum_n r_n/[k_n(k^2 - k_n{}^2)]. \tag{5.7.16}$$

According to (5.7.15), we have

$$-(1/k)[P(k) - P(-k)] = -2 \sum_{p=0}^{\infty} a_{2p+1} k^{2p} = C(E), \tag{5.7.17}$$

where $C(E)$ is an entire function of the energy $E = k^2/2$.

On the other hand, coupling together the contribution from the complex poles $k_n$ and $-k_n{}^*$ in the last term of (5.7.16), we get

$$-\frac{2r_n/k_n}{k^2 - k_n{}^2} - \frac{2r_n{}^*/k_n{}^*}{k^2 - (k_n{}^*)^2} = 2\,\mathrm{Re}\left(-\frac{2r_n/k_n}{k^2 - k_n{}^2}\right)$$

$$= \frac{A_n(E - E_n{}') + \frac{1}{2}B_n\Gamma_n}{(E - E_n{}')^2 + \frac{1}{4}\Gamma_n{}^2}, \tag{5.7.18}$$

where we have set

$$-2r_n/k_n = A_n + iB_n \tag{5.7.19}$$

and $E_n = \frac{1}{2}k_n{}^2 = E_n{}' - i\Gamma_n/2$ is the "complex energy" associated with the complex pole $k_n$ [cf. (4.3.5)].

On the other hand, if $k_n = i\kappa_n$ is a pole on the imaginary axis, the corresponding residue $r_n$ is pure imaginary [according to (5.4.2), $S(k)$ is real along the imaginary axis], and we set

$$-2r_n/i\kappa_n = 2A_n. \tag{5.7.20}$$

It follows that

$$-\frac{2r_n/i\kappa_n}{k^2 + \kappa_n{}^2} = \frac{A_n}{E + |E_n{}'|} = \frac{A_n}{E - E_n{}'}, \tag{5.7.21}$$

where $E_n{}' = -\kappa_n{}^2/2$; this may be regarded as a particular case of (5.7.18), with $\Gamma_n = 0$ [note, however, the discrepancy by a factor 2 between (5.7.19) and (5.7.20)].

Substituting (5.7.17) to (5.7.21) in (5.7.16), we finally get the cross section as a function of the energy:

$$\sigma(E) = \pi\left[C(E) + \sum_n{}' \frac{A_n(E-E_n') + \frac{1}{2}B_n\Gamma_n}{(E-E_n')^2 + \frac{1}{4}\Gamma_n^2}\right], \qquad (5.7.22)$$

where the prime in the summation sign means that the sum is restricted to complex poles $k_n$ in the fourth quadrant of the $k$-plane and poles $i\kappa_n$ on the imaginary axis [with the convention (5.7.20)].

The above expression for the $s$-wave elastic scattering cross section by a cutoff potential was obtained by Humblet, and this treatment was further extended by Humblet and Rosenfeld to the general theory of nuclear reactions [30]. The expression (5.7.22) is a many-level dispersion formula, with each complex pole of the $S$-function giving an additive contribution to the cross section.

The contribution from each complex pole is of the Breit–Wigner type [cf. (4.2.7)], except for the term $A_n(E-E_n')$ in the numerator. However, if a pole $k_n = k_n' - i\kappa_n$ is associated with an isolated narrow resonance, so that $\kappa_n \ll k_n'$, and if we can neglect all other contributions in a sufficiently small neighborhood of the pole on the real axis, $S(k)$, in this neighborhood, must be of the form $(k-k_n^*)/(k-k_n)$, by unitarity [cf. (4.2.6)], so that $r_n \approx k_n - k_n^* = -2i\kappa_n$, and (5.7.19) yields the approximations

$$A_n = 0, \qquad B_n = 4\kappa_n/k_n' = \Gamma_n/E_n'. \qquad (5.7.23)$$

The corresponding contribution to (5.7.22) is then of the Breit–Wigner form (4.2.7).

The term $A_n(E-E_n')$ introduces an asymmetry in the shape of the peak. Such asymmetries also appear in other treatments,[14] where they are ascribed to interference between resonance and potential scattering. In the present treatment, potential scattering can be defined as the contribution from the smoothly varying term $C(E)$ in (5.7.22), which is an entire function of $E$; there is here no interference term between potential and resonance scattering.

The definition of potential scattering contains many elements of arbitrariness; in other treatments, it may include the background contribution due to nonresonant pole terms, as well as "hard-sphere scattering" (cf. Section 4.2). In the present treatment, although the entire-function contribution to (5.7.22) is in principle well defined, ambiguities still remain: e.g., we could have chosen $M > 1$ in (5.7.4); each such choice of $M$ would correspond to a redistribution between "resonance" and "potential" scattering.

[14] Cf. Blatt and Weisskopf [31, p. 401].

Let us note, finally, that the contributions to the cross section from pure imaginary poles (corresponding to bound or antibound states) are, according to (5.7.21), of a form similar to (4.2.11). Here again significant contributions arise only from poles sufficiently close to the real axis.

### *(c) The Cauchy Expansion*

The above treatment suffers from the disadvantage that the form of the entire function $\mathscr{E}(k)$ in (5.7.13) or $C(E)$ in (5.7.22) remains almost completely undetermined. It is possible to employ a different procedure [8, 32], where this difficulty does not appear. For this purpose, one employs the following special form of the Mittag–Leffler theorem, due to Cauchy [14, p. 111]:

THEOREM 5.7.2. *Let $F(k)$ be a meromorphic function, all of whose poles $k_n$ are simple and satisfy (5.7.1), (5.7.2). If there exists a sequence of closed contours $C_n$ such that $C_n$ includes $k_1, \ldots, k_n$ but no other poles, that the minimum distance $R_n$ of $C_n$ from the origin tends to infinity with $n$, while the length of $C_n$ is $O(R_n)$, and if, on $C_n$,*

$$F(k) = O(R_n{}^p), \tag{5.7.24}$$

*where $p$ is a nonnegative integer, we have*

$$\begin{aligned} F(k) &= F(0) + kF'(0) + \cdots + \frac{k^p}{p!} F^{(p)}(0) + \sum_{n=1}^{\infty} \left(\frac{k}{k_n}\right)^{p+1} \frac{r_n}{k-k_n} \\ &= \sum_{q=0}^{p} \frac{F^{(q)}(0)}{q!} k^q + \sum_{n=1}^{\infty} r_n \left(\frac{1}{k-k_n} + \frac{1}{k_n} + \frac{k}{k_n{}^2} + \cdots + \frac{k^p}{k_n^{p+1}}\right), \end{aligned} \tag{5.7.25}$$

*for all values of $k$ except the poles. Furthermore, the series converges uniformly inside any closed contour that does not contain any poles.*

Let us apply this theorem to $F(k) = S_a(k) = S(k)e^{2ika}$. According to (5.5.17), $S_a(k)$ is bounded in $I_+$ as $|k| \to \infty$. In $I_-$, by (5.2.19) and (5.5.16), we have

$$S_a(k) \approx \left[e^{2ika} + \frac{U^{(m)}(a-)}{(2ik)^{m+2}}\right]^{-1} \qquad (|k| \to \infty \text{ in } I_-). \tag{5.7.26}$$

Let us choose as contours $C_n$ a sequence of squares with vertices at $\pm L_n \pm iL_n$.

To avoid the poles, that are asymptotically given by the roots of the expression within brackets in (5.7.26) [cf. (5.5.22)], we choose $L_n$ so that[15]

$$\exp(2iL_n a) = -U^{(m)}(a-)/i^m |U^{(m)}(a-)| = \mp i^{-m}. \qquad (5.7.27)$$

Then, along the vertical sides, e.g., for $k = L_n + i\kappa$, we have, by (5.7.26),

$$|S_a(k)| = \left| e^{-2\kappa a} + \frac{U^{(m)}(a-)}{[2(L_n+i\kappa)]^{m+2}} \right|^{-1} = O(L_n^{m+2}) \qquad (\kappa < 0),$$

and similarly for the other sides, so that, finally,

$$S_a(k) = O(L_n^{m+2}) \qquad \text{on} \quad C_n, \qquad (5.7.28)$$

and we can apply Theorem 5.7.2 with $p = m+2$. Let us restrict ourselves, for definiteness, to the case $m = 0$, assuming that the potential satisfies (5.5.6). Then (5.7.25) becomes

$$S_a(k) = S_a(0) + k\,S_a'(0) + (k^2/2)\,S_a''(0) + \sum_n (k/k_n)^3 [r_n \exp(2i\beta_n)/(k-k_n)], \qquad (5.7.29)$$

where $r_n$ is defined as in (5.7.5), so that [cf. (5.5.23)]

$$\operatorname{res} S_a(k)\Big|_{k_n} = r_n \exp(2i\beta_n). \qquad (5.7.30)$$

We assume, as usual, that $S(0) = 1$; also, as $k \to 0$,

$$\lim_{k\to 0}\left[\frac{S(k)-1}{2ik}\right] = \frac{S'(0)}{2i} = \lim_{k\to 0}\left[\frac{\eta(k)}{k}\right] = f(0,0) = -l, \qquad (5.7.31)$$

where $\eta(k)$ is the phase shift, $f(0,0)$ is the zero-energy limit of the forward scattering amplitude, and $l$ is the scattering length[16] [cf. (3.2.13)]. It follows that

$$S_a'(0) = 2i(a-l).$$

Furthermore, from the Taylor expansion about $k = 0$ of the relation $S(k)\,S(-k) = 1$ [cf. (5.2.19)] we get

$$S''(0) = [S'(0)]^2, \qquad (5.7.32)$$

so that (5.7.29) becomes

$$S_a(k) = 1 + 2ik(a-l) - 2k^2(a-l)^2 + \sum_n (k/k_n)^3 [r_n \exp(2i\beta_n)/(k-k_n)]. \qquad (5.7.33)$$

---

[15] According to (5.5.26), this corresponds to $L_n a = U_n + \pi/2$, i.e., to contours $C_n$ passing (asymptotically) halfway between consecutive poles.

[16] Cf. Blatt and Weisskopf [31, p. 60].

This is the Humblet–Peierls expansion.[17] In contrast with (5.7.11), the form of the entire function is completely determined, in terms of the scattering length and the radius of the scatterer.

Just as we did for (5.7.11), we can take advantage of the pairwise summation (the contributions from $k_n$ and $-k_n^*$ are paired) to simplify (5.7.33) still further. We can rewrite the last term of (5.7.33) as

$$k^2 \sum_n \frac{1}{k_n^2} r_n \exp(2i\beta_n) \left( \frac{1}{k-k_n} + \frac{1}{k_n} \right). \tag{5.7.34}$$

It follows from (5.5.22), (5.5.24) (with $m = 0$) and (5.7.9) that, as $n \to \infty$,

$$\frac{1}{k_n^2} r_n \exp(2i\beta_n) = \frac{2i}{aU(a-)} + O(k_n^{-1}). \tag{5.7.35}$$

Thus, precisely as in (5.7.12), each of the series in (5.7.34) is separately convergent, and (5.7.33) may be rewritten as

$$S_a(k) = 1 + 2ik(a-l) + Ck^2 + \sum_n (k/k_n)^2 [r_n \exp(2i\beta_n)/(k-k_n)], \tag{5.7.36}$$

where

$$C = \sum_n (r_n/k_n^3) \exp(2i\beta_n) - 2(a-l)^2. \tag{5.7.37}$$

It can readily be verified that the Mittag–Leffler expansion of $S_a(k)$ employed in Section 4.6 [cf. (4.6.32), (4.6.10), and (4.6.23)] is a special case of the Humblet–Peierls expansion (note that the delta-function potential corresponds to $m = -1$, rather than $m = 0$).

### *(d) Transient-Mode Expansions*

The above results can be employed to extend the transient-mode expansions for the Schrödinger equation, that were discussed in Section 4.6 for a special model, to more general cutoff potentials. It is clear that, in the more general case, we are only allowed to discuss explicitly the behavior of the wave function in the external region [corresponding to Section 4.6(c)].

As in (4.6.2), we set (the index 2 referring to the external region, $r > a$)

$$\begin{aligned} \varphi_2(r,t) &= r\psi_2(r,t) = \varphi_{2,\text{in}} + \varphi_{2,\text{out}} \\ &= \int_{-\infty}^{\infty} A(k) e^{-ikr-iEt} \, dk - \int_{-\infty}^{\infty} S(k) A(k) e^{ikr-iEt} \, dk. \end{aligned} \tag{5.7.38}$$

[17] There is a misprint in Peierls's paper [8, Eq. (5.16)], where the coefficient of $k^2$ appears with a + sign.

At $t=0$ we may have an arbitrary initial wave packet, contained entirely in the external region [cf. (4.6.8)]

$$\varphi_2(r,0) = \varphi_{2,\mathrm{in}}(r,0) = \int_{-\infty}^{\infty} A(k)\, e^{-ikr}\, dk = f_2(r). \tag{5.7.39}$$

The corresponding solution is, in the notation of Section 4.6,

$$\varphi_2(r,t) = \int_a^{\infty} G_{22}(r,\rho,t) f_2(\rho)\, d\rho. \tag{5.7.40}$$

The propagator $G_{22}(r,\rho,t)$ is the solution corresponding to the initial condition $f_2(r) = \delta(r-\rho)$, i.e., $A(k) = e^{ik\rho}/2\pi$; thus, by (5.7.38),

$$\begin{aligned} G_{22}(r,\rho,t) &= (1/2\pi)\int_{-\infty}^{\infty} e^{-ik(r-\rho)-iEt}\, dk + G_{22}^{(\mathrm{out})}(r,\rho,t) \\ &= U(|r-\rho|,t) + G_{22}^{(\mathrm{out})}(r,\rho,t), \end{aligned} \tag{5.7.41}$$

where $U(x,t)$ is the free-particle propagator, given by (4.5.4), and

$$\begin{aligned} G_{22}^{(\mathrm{out})}(r,\rho,t) &= -(1/2\pi)\int_{-\infty}^{\infty} S(k)\, e^{ik(r+\rho)-iEt}\, dk \\ &= -(1/2\pi)\int_{-\infty}^{\infty} S_a(k)\, e^{ik(r+\rho-2a)-iEt}\, dk. \end{aligned} \tag{5.7.42}$$

In the first expression, we can substitute $S(k)$ by its Mittag–Leffler expansion (5.7.13); in the second one, we can employ the Cauchy expansion (5.7.36) of $S_a(k)$. We then make use of the following identity:

$$-\frac{1}{2\pi i}\int_{-\infty}^{\infty} e^{ikx-iEt}\, \frac{dk}{k-k_n} = M(x,k_n,t) \qquad (\mathrm{Im}\, k_n < 0), \tag{5.7.43}$$

where $M(x,k_n,t)$ is the transient-mode propagator (4.5.7); the identity follows either by direct integration or from the fact that both sides are solutions of the free-particle Schrödinger equation that reduce, for $t=0$, to $\theta(-x)\exp(ik_n x)$ [cf. (4.5.2)]. The above substitutions then lead to the following results:

$$\begin{aligned} G_{22}^{(\mathrm{out})}(r,\rho,t) &= -\frac{1}{2\pi}\int_{-\infty}^{\infty} e^{ik(r+\rho)-iEt}\, \mathscr{E}(k)\, dk \\ &\quad + i\sum_n r_n M(r+\rho,k_n,t), \end{aligned} \tag{5.7.44}$$

$$G_{22}^{(\text{out})}(r,\rho,t) = -U(r+\rho-2a,t) - 2(a-l)\frac{\partial}{\partial r}U(r+\rho-2a,t)$$
$$+ C\frac{\partial^2}{\partial r^2}U(r+\rho-2a,t) - i\frac{\partial^2}{\partial r^2}\sum_n \frac{r_n}{k_n{}^2}\exp(2i\beta_n)$$
$$\times M(r+\rho-2a,k_n,t). \tag{5.7.45}$$

This last expression is to be understood in the sense of distribution theory; according to (4.5.2) and (4.5.3), it contains $\delta$ and its derivatives up to second order at $t=0$; for $m>0$ in (5.7.26), still higher derivatives would appear. The sums over poles in both (5.7.44) and (5.7.45) are to be carried out pairwise, in order of increasing $|k_n|$.

The transient-mode expansions (5.7.44) and (5.7.45) are two different representations for the same propagator; (5.7.45) is valid only for $m=0$ in (5.5.16) (but similar representations can be derived for any value of $m$), whereas (5.7.44) is valid without such restrictions; on the other hand, the form of the entire function $\mathscr{E}(k)$ in (5.7.44) remains undetermined, whereas (5.7.45) is completely determined. From the physical point of view, (5.7.45) corresponds to a description in terms of "hard-sphere scattering" (cf. Section 4.2): the first term of (5.7.45) is associated with total reflection at the surface $r=a$ [cf. also (4.6.61)].

As we have seen in Section 4.5, the behavior of a transient-mode propagator as a function of time depends on whether the corresponding pole $k_n$ is above or below the second bisector in the $k$-plane. For poles above the bisector—and only for such poles—we get, over a limited domain in space and time, exponentially decaying contributions. More exactly, according to (4.5.18) and (4.5.21), it follows from either (5.7.44) or (5.7.45) that *each pole $k_n = k_n' - i\kappa_n$ above the second bisector contributes to the propagator a term of the form*

$$g_n(r,\rho,t) = ir_n\exp[ik_n(r+\rho) - iE_n t], \tag{5.7.46}$$

(*where $r_n$ is the residue of $S(k)$ at $k=k_n$*), *provided that*

$$0 < r+\rho < (k_n' - \kappa_n)t, \qquad |r+\rho-k_n t| \gg (2t)^{1/2}. \tag{5.7.47}$$

According to (4.4.18) and (4.4.28), this is precisely the same cutoff "complex-energy" decaying state that was found in the special models treated in Chapter 4. The restrictions (5.7.47) ensure that the observation point is sufficiently far behind the wave front associated with the mode $k_n$, so that "diffraction in time" effects can be neglected (cf. Section 4.5).

By (4.5.17) and (4.5.18), poles located below the second bisector contribute a term proportional to the free-particle propagator, and the same term appears as an additional contribution for poles located above the second bisector. Asymptotically, if we let $t \to \infty$ at a fixed point in space, these are the dominant terms, so that the asymptotic behavior of the complete propagator, according to (5.7.41) and (5.7.45), is given by

$$G_{22}(r, \rho, t) \approx U(|r-\rho|, t) - U(r+\rho-2a, t) - 2(a-l)\frac{\partial}{\partial r} U(r+\rho-2a, t)$$
$$+ C\frac{\partial^2}{\partial r^2} U(r+\rho-2a, t) + \frac{\partial^2}{\partial r^2} \sum_n \frac{r_n}{k_n{}^2}$$
$$\times \exp(2i\beta_n)\left[\frac{tU(r+\rho-2a, t)}{r+\rho-2a-k_n t}\right] \qquad (t \to \infty), \tag{5.7.48}$$

where the sum is extended over all poles, both above and below the bisector.

Taking into account (4.5.4) and (5.7.37), one can readily show that (5.7.48) becomes

$$G_{22}(r, \rho, t) \approx -\left(\frac{2}{\pi}\right)^{1/2} e^{i\pi/4} \frac{(r-l)(\rho-l)}{t^{3/2}} + O(t^{-5/2}) \qquad (t \to \infty). \tag{5.7.49}$$

For a free-particle wave packet, according to Appendix E, (E9), one would have found the same result, with $l = 0$ [as it must be, by (5.7.31)]. Thus *the only difference between the asymptotic behavior of the propagator for $t \to \infty$ and that of the free propagator is that the origin appears displaced by an amount equal to the scattering length.*

The above discussion indicates how one can extend the treatment of transient modes given in Section 4.6 to more general cutoff potentials. In particular, in the one-level approximation, the discussion of resonant scattering and time delay given in Section 4.6(c) remains valid in this more general case.

A similar extension has been made by Rosenfeld [33], for partial waves with arbitrary values of the angular momentum. In contrast with the above analysis, which is based on the stationary-state expansion (5.7.38), Rosenfeld's treatment is based on an expansion in terms of unperturbed (free-particle) states, so that a detailed comparison of the results requires somewhat lengthy transformations. The Mittag–Leffler expansion employed in his work [33, Eq. (36)] is not convergent in the form in which it was written.[18] However, by inserting a suitable convergence factor of the form $(k/k_n)^p$, the series can be rendered convergent and the rest of Rosenfeld's analysis then remains valid, with relatively minor modifications. The results are similar to those discussed here.

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[18] For example, for $s$-waves, the residues [33, Eq. (37)] grow like $(k_n)^{1+m/2}$, where $m$ is again defined by (5.5.16).

## 5.8. Extension to Higher Angular Momenta

### (a) Analytic Properties

Most of the results so far derived for $s$-waves can be extended to higher angular momenta. The methods employed for this purpose are analogous to those employed for $s$-waves, but the detailed derivations are more laborious, due to the more complicated nature of the functions involved (spherical Bessel functions instead of simple exponentials). However, no essentially new procedures or physical insights are involved, so that we will simply list the main results, referring the reader to more specialized treatises (cf. the references quoted in Section 5.1) for additional details about the derivations[19]; besides, the more general case of complex angular momentum will be treated in Chaptér 7.

For $l > 0$, the radial equation [cf. (5.1.2), (5.1.3)] becomes

$$[d^2/dr^2 + k^2 - l(l+1)/r^2 - U(r)]\, u_l(k,r) = 0, \tag{5.8.1}$$

where $U(r)$ is given by (5.1.4). This differs from (5.2.1) by the "centrifugal potential" term $l(l+1)/r^2$.

In the absence of interaction $[U(r) = 0]$, (5.8.1) has solutions of the form

$$\varphi_l^{(0)}(k,r) = k^{-l} r j_l(kr), \tag{5.8.2}$$

$$f_l^{(0)}(k,r) = (-i)^{l+1} kr h_l^{(2)}(kr), \tag{5.8.3}$$

where $j_l$ and $h_l^{(2)}$ are spherical Bessel and Hankel functions, respectively [cf. (2.6.3)]. The solution (5.8.2) is regular at the origin, where its behavior [cf. (3.3.19)] is given by

$$\varphi_l^{(0)}(k,r) \to r^{l+1}/(2l+1)!! \qquad (r \to 0). \tag{5.8.4}$$

For $l = 0$, this is the same behavior specified by (5.2.2) and (5.2.3), and $\varphi_0^{(0)}$ reduces to $\sin(kr)/k$ [cf. (5.2.22)]. For $r \to \infty$, we have

$$\varphi_l^{(0)}(k,r) \to k^{-l-1} \sin[kr - (l\pi/2)] \qquad (r \to \infty). \tag{5.8.5}$$

On the other hand, by (2.6.4),

$$f_l^{(0)}(k,r) \to e^{-ikr} \qquad (r \to \infty), \tag{5.8.6}$$

so that it plays the same role as the free Jost solution in (5.2.22). For $r \to 0$, we have, by (3.3.19) and (3.3.20),

$$f_l^{(0)}(k,r) \to (-i)^l (2l-1)!!\,(kr)^{-l} \qquad (r \to 0). \tag{5.8.7}$$

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[19] See also Newton [12].

The Jost solution in the presence of interaction is again defined by its asymptotic behavior at infinity,

$$f_l(k,r) \approx e^{-ikr} \qquad (r \to \infty), \tag{5.8.8}$$

so that we still have [cf. (5.2.10)]

$$W[f_l(k,r), f_l(-k,r)] = 2ik. \tag{5.8.9}$$

The integral equation that generalizes (5.3.1) is [cf. (7.2.26)]

$$f_l(k,r) = f_l^{(0)}(k,r) + \int_r^\infty G_l(k;r,\rho)\, U(\rho) f_l(k,\rho)\, d\rho, \tag{5.8.10}$$

where

$$G_l(k;r,\rho) = \frac{1}{2ik}\left[f_l^{(0)}(k,r) f_l^{(0)}(-k,\rho) - f_l^{(0)}(-k,r) f_l^{(0)}(k,\rho)\right]. \tag{5.8.11}$$

For $r \to 0$, $f_l(k,r)$ will in general behave like $r^{-l}$, as in (5.8.7). The definition (5.2.13) of the Jost function is now generalized to

$$F_l(k) = \lim_{r\to 0}\left[\frac{(ikr)^l}{(2l-1)!!} f_l(k,r)\right], \tag{5.8.12}$$

so that, by (5.8.7), we have

$$F_l^{(0)}(k) = 1 \tag{5.8.13}$$

in the absence of interaction [as in (5.2.22)].

Let us define [cf. (5.2.12)]

$$\varphi_l(k,r) = \tfrac{1}{2}(i/k)^{l+1}[F_l(-k) f_l(k,r) - (-1)^l F_l(k) f_l(-k,r)]. \tag{5.8.14}$$

Then it follows from (5.8.9) that [cf. (5.2.14)]

$$F_l(k) = (ik)^l W[f_l(k,r), \varphi_l(k,r)]. \tag{5.8.15}$$

By evaluating the Wronskian at $r = 0$, taking into account that $f_l(k,r) = O(r^{-l})$ as $r \to 0$, and comparing the result with (5.8.12), we find that

$$\varphi_l(k,r) \to r^{l+1}/(2l+1)!! \qquad (r \to 0), \tag{5.8.16}$$

so that $\varphi_l(k,r)$ is the *regular solution* of the Schrödinger radial equation (5.8.1), defined by the same boundary condition (5.8.4). In the absence of interaction, (5.8.14) reduces to (5.8.2). By comparing the asymptotic behavior of (5.8.14) as $r \to \infty$, that follows from (5.8.6), with (2.6.5), we find that [cf. (5.2.18)]

$$S_l(k) = F_l(k)/F_l(-k). \tag{5.8.17}$$

The reality of the potential implies that $F_l$ again satisfies the symmetry relations (5.2.15), and, as in (5.2.17), the phase of $F_l(k)$ is the phase shift $\eta_l(k)$.

The analytic properties of $f_l(k,r)$ in the $k$-plane can be investigated in close analogy with the procedure of Section 5.3, by examining the Liouville–Neumann iteration solution of the integral equation (5.8.10). The result is that *the function $k^l f_l(k,r)$, and, therefore, in particular* [*cf.* (*5.8.12*)], *also the Jost function $F_l(k)$, has all the analytic properties stated in Theorems 5.3.1 and 5.3.2:* it is holomorphic in $I_-(k)$ if $M_1$ is finite, and it has a regular analytic continuation for $\operatorname{Im} k < \alpha/2$ if $E_\alpha$ is finite. The factor $k^l$ is required because $f_l(k,r)$ has a pole of order $l$ at $k=0$ [cf. (5.8.3)]. For potentials vanishing faster than any exponential at infinity, the Jost function is an entire function of $k$. The asymptotic behavior (5.3.27) for $|k|\to\infty$ in $I_-$ remains valid for $f_l(k,r)$ [in particular, $F_l(k)\to 1$].

For Yukawa-type potentials with maximum range $m^{-1}$ [cf. (5.4.29)], one can extend the analytic continuation above $\operatorname{Im} k = m/2$, and one again finds [34] that the Jost function is holomorphic in the whole $k$-plane, cut from $k = im/2$ to $i\infty$. Furthermore, $F_l(k)\to 1$ as $|k|\to\infty$ along any direction outside of the cut.

By the same argument as in Section 5.4(a), the zeros of $F_l(k)$ in $I_-(k)$ must be located on the imaginary axis, and they correspond to bound states. Again they are all simple. If $F_l(0)=0$ for $l>0$, however, (5.4.7) is replaced by

$$F_l(k) = O(k^2) \qquad \text{as} \quad |k|\to 0 \quad \text{from} \quad I_-, \quad l \geqslant 1, \tag{5.8.18}$$

so that, if $F_l(0)=0$ and if $F_l(k)$ is analytic at the origin, it has a double zero there. Also, in contrast with the case $l=0$, a zero of $F_l(k)$ at $k=0$ *does* correspond to a bound state of zero energy, with a normalizable wave function that is $O(r^{-l})$ as $r\to\infty$.

For a potential with finite first moment $M_1$, it also follows that the total number $n_l$ of bound states with angular momentum $l$ is finite; Bargmann's inequality (5.4.8) is extended to [13]

$$n_l \leqslant M_1/(2l+1). \tag{5.8.19}$$

This also implies that there cannot be any bound states with angular momentum larger than $L = \frac{1}{2}(M_1 - 1)$, so that the *total* number of bound states (without restrictions on the angular momentum) is finite.

Levinson's theorem [Section 5.4(b)] must also be modified. In the exceptional case $F_l(0)=0$, we now have (5.8.18), so that the integral in (5.4.27) becomes equal to $-1$; on the other hand, $k=0$ will then correspond to a zero-energy bound state, so that (5.4.28) is replaced by

$$\eta_l(0+) = n_l\pi \qquad (l \geqslant 1), \tag{5.8.20}$$

where $n_l$ is the *total* number of bound states with angular momentum $l$ [including a zero-energy bound state, if $F_l(0) = 0$].

The above properties, together with (5.8.17), enable us to determine the analytic behavior of the $S$-function. They imply the following result:

THEOREM 5.8.1. *The function $S_l(k)$ is meromorphic:*

*(a) in the strip $|\mathrm{Im}\, k| < \alpha/2$ if $E_\alpha$ is finite [cf. (5.3.22)];*

*(b) in the k-plane, cut along the imaginary axis from $im/2$ to $i\infty$ and from $-im/2$ to $-i\infty$, for Yukawa-type potentials (5.4.29);*

*(c) in the whole (finite) k-plane, if the potential decreases faster than any exponential at infinity ($E_\alpha$ finite for all $\alpha$).*

*In the analyticity domain of $S_l(k)$ in the upper half-plane [i.e., in case (a), in the strip $0 < \mathrm{Im}\, k < \alpha/2$], its poles can only be located on the positive imaginary axis; there is a finite number of such poles; they are simple and correspond to bound states.*

### *(b) Threshold Behavior and High Angular Momentum Limit*

The most interesting new results are those concerned with the threshold behavior of partial-wave scattering amplitudes ($k \to 0$), and their behavior for large angular momentum; for cutoff interactions, a physical discussion of this problem was given in Section 3.3(b).

By methods similar to those that led to (5.3.16), one can show that

$$|S_l(k) - 1| \leqslant C_l \frac{|k|^{2l+1}}{|F_l(-k)|} \int_0^\infty \left(\frac{r}{1+|k|r}\right)^{2l+2} |V(r)|\, dr, \qquad (5.8.21)$$

where $C_l$ is a constant. If $F_l(0) \neq 0$, it follows that [cf. (3.1.4)]

$$S_l(k) - 1 = 2i \exp(i\eta_l) \sin \eta_l = O(k^{2l+1}) \qquad (k \to 0), \qquad (5.8.22)$$

provided that[20]

$$\int_0^\infty r^{2l+2} |V(r)|\, dr < \infty. \qquad (5.8.23)$$

In particular, this is true for all $l$ in cases ($b$) and ($c$) of Theorem 5.8.1, so that, in these cases, we may expect all partial waves to have the usual threshold behavior (3.3.10), in agreement with the qualitative discussion given in Section 3.3(b).

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[20] Cf. Mott and Massey [35, p. 45].

In the exceptional case $F_l(0) = 0$, it follows from (5.8.18) and (5.8.21) that (5.8.22) must be replaced by

$$S_l(k) - 1 = O(k^{2l-1}) \quad (k \to 0), \qquad \text{if} \quad F_l(0) = 0 \quad (l \geqslant 1), \tag{5.8.24}$$

in agreement with (3.3.25).

Let us consider now the high angular momentum limit. Since (5.3.20) implies that the potential must decrease faster than $r^{-2}$ as $r \to \infty$, we expect that, as $l \to \infty$, the "centrifugal potential" $l(l+1)r^{-2}$ in (5.8.1) should become the dominant term, and that the effect of the potential may then be treated as a perturbation.

It was, in fact, shown by Carter[21] that, for large enough $l$, the absolute value of the phase shift is always bounded by the first Born approximation,[22] so that

$$\begin{aligned} |[S_l(k) - 1]/2i| &= |\sin \eta_l| \approx |\eta_l| \\ &\leqslant Ck \int_0^\infty r^2 |U(r)| j_l^2(kr)\, dr, \end{aligned} \tag{5.8.25}$$

where $j_l$ is the spherical Bessel function.

For a cutoff potential of radius $a$ [cf. (5.5.1)], it follows from (5.8.25), (5.3.20), and Watson's inequality [39, p. 225] that

$$|\sin \eta_l| \leqslant \frac{M_1 C}{(2l+1)} \left( \frac{eka}{2l+1} \right)^{2l+1} \qquad (0 \leqslant ka \leqslant l/2), \tag{5.8.26}$$

for sufficiently large $l$. Let $L$ be such that (5.8.25) is valid for $0 \leqslant k < \kappa$, $l \geqslant L$, and also $L \geqslant 2\kappa a$. Then it follows from (5.8.25) and (5.8.26) that, no matter how large $\kappa$ is,

$$|\sin \eta_l| \leqslant (k/\kappa)^{2l+1} \qquad (0 \leqslant k < \kappa,\ l \geqslant L), \tag{5.8.27}$$

so that the phase shifts tend to zero at least exponentially with $l$ for $l \to \infty$ [actually, like $\exp(-2l \log l)$, by (5.8.26)]. Furthermore, by comparison with (3.3.30), we find that $k_l$ cannot approach a finite limit, so that (3.3.32) is valid. Thus Assumption I of Section 3.3(c) is justified for a cutoff potential. We also expect that (5.8.26) will be valid when $l \geqslant L \sim Nka$, for sufficiently large $ka$ (where $N$ is a constant), so that Assumption II of Section 3.3(c) and (3.3.43) are also justified.

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[21] See Carter [36]. Cf. also Martin [37], and the discussion in Section 7.5.

[22] See, for example, Schiff [38, p. 165].

For Yukawa-type potentials (5.4.29), it follows from (5.8.25) that

$$|S_l(k) - 1| \leqslant \pi C \int_m^\infty d\mu\, |\sigma(\mu)| \int_0^\infty e^{-\mu r} J_{l+\frac{1}{2}}^2(kr)\, dr$$

$$= \frac{C}{k} \int_m^\infty d\mu\, |\sigma(\mu)|\, Q_l(1+(\mu^2/2k^2)), \tag{5.8.28}$$

where $Q_l$ is the Legendre function of the second kind.[23] It then follows from the asymptotic behavior of the Legendre function for large $l$ that [40]

$$|\sin \eta_l| = O(l^{-\frac{1}{2}} e^{-\chi l}) \qquad (l \to \infty), \tag{5.8.29}$$

where

$$\cosh \chi = 1 + (m^2/2k^2). \tag{5.8.30}$$

According to (5.8.29), phase shifts may be expected to decrease exponentially with $l$ for Yukawa-type potentials. In particular, if $k \gg m$, (5.8.30) yields $\chi \approx m/k$, so that

$$\exp(-\chi l) \approx \exp(-m p_l), \tag{5.8.31}$$

where $p_l \sim l/k$ is the "impact parameter" associated with the $l$th partial wave [cf. (3.3.26)]; under these conditions, therefore, the bound on the phase shifts is of the order of the value of the potential at the impact parameter.

### (c) Cutoff Potentials

The analytic properties of $S_l(k)$ in the special case of cutoff potentials follow from Theorem 5.8.1 (c). The behavior of the Jost function as $|k| \to \infty$ in $I_+$ may be obtained [26] from the Liouville–Neumann solution of (5.8.10), as in Section 5.5. It is again given by the first Born approximation,

$$F_l(k) \approx F_{lB}(k) = 1 - ik \int_0^a j_l(k\rho)\, h_l^{(2)}(k\rho)\, U(\rho)\, \rho^2\, d\rho$$

$$\approx 1 + (1/2ik) \int_0^a [1 - (-1)^l e^{-2ik\rho}]\, U(\rho)\, d\rho \qquad (|k| \to \infty \text{ in } I_+), \tag{5.8.32}$$

which differs from (5.5.4), (5.5.13) only by the factor $(-1)^l$. Accordingly, (5.5.16) is replaced by

$$S_l(k) \approx 1 + (-1)^l [U^{(m)}(a-)/(2ik)^{m+2}]\, e^{-2ika} \qquad (|k| \to \infty \text{ in } I_+). \tag{5.8.33}$$

[23] Cf. Watson [39, p. 389].

Thus *Theorem 5.5.1 remains valid for* $S_l(k)$, as well as the analytic properties and the dispersion relation given in Section 2.9. This is in agreement with the result found in Section 3.4(b). In particular, the canonical product expansion (5.5.19) (with the plus sign) holds.

Again (5.8.33) leads to the existence of an infinite number of complex poles of $S_l(k)$ in $I_-$. Since (5.8.33) differs from (5.5.16) only by the factor $(-1)^l$, it follows that the asymptotic distribution of "large" poles is still given by (5.5.29), provided that (5.5.27) is replaced by

$$\begin{aligned} \zeta &= 0 \qquad \text{if} \quad (-1)^l U^{(m)}(a-) > 0, \\ &= 1 \qquad \text{if} \quad (-1)^l U^{(m)}(a-) < 0. \end{aligned} \tag{5.8.34}$$

Due to the difference between (5.4.7) and (5.8.18), the mechanism whereby bound-state poles appear on the positive imaginary axis differs from that found for $l = 0$ [cf. Section 5.6(a)]. They arise from the confluence of a pair of $c$-poles *at the origin*, giving rise to the double zero (5.8.18); the poles move along a parabolic arc, osculating the real axis at the origin. For a square well with $l = 1$, this is explicitly illustrated in Fig. 5.7, adapted from a paper by Kaus and Pearson [41]. As a consequence of this, the appearance of a

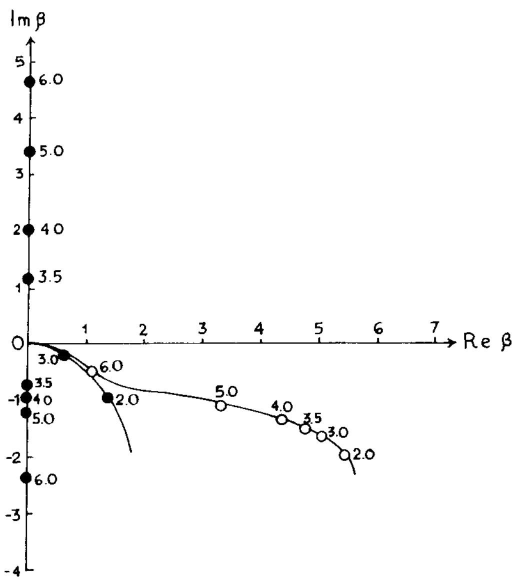


*FIG. 5.7.* The poles $\beta_n$ of $S_1(\beta)$ for a potential well (●, $n = 1$; O, $n = 2$). The numbers beside the poles give the corresponding values of $A$. The notation is the same as in Fig. 5.5 [after P. Kaus and C. J. Pearson, *Nuovo Cimento* **28**, 500 (1963)].

new bound state is preceded by a low-energy resonance,[24] due to the presence of a complex pole close to the real axis. Physically, the existence of a sharp resonance for $l \neq 0$ is due to the centrifugal barrier.

Finally, the Mittag–Leffler and transient-mode expansions of Section 5.7 can be extended to $l \neq 0$ along the lines of Rosenfeld's [33] treatment. One can reduce the problem essentially to the case $l = 0$ with the help of the multipole differential operator[25] $r^l(r^{-1}\,\partial/\partial r)^l$.

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[24] Cf. Nussenzveig [28].
[25] Cf. also Beck and Nussenzweig [42].

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CHAPTER

# 6

## ANALYTIC PROPERTIES OF THE TOTAL AMPLITUDE

*The nonphysical region is as bad and as treacherous as any point in the complex plane.*

R. JOST[1]

### 6.1. Introduction

Let us now turn our attention to the analytic properties of the total scattering amplitude. As in the previous chapter, we do not restrict our consideration to cutoff potentials; we consider also potentials with tails extending to infinity and, in particular, Yukawa-type potentials.

The Schrödinger equation for eigenstates with energy $E$ in a potential $V(r)$ reads

$$(\Delta+k^2)\psi(k,\mathbf{r}) = U(r)\psi(k,\mathbf{r}), \tag{6.1.1}$$

where $k^2 = 2mE/\hbar^2$, $U(r) = 2mV(r)/\hbar^2$. For an incident plane wave,

$$\psi_0(\mathbf{k},\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} = e^{ikr\cos\theta}, \tag{6.1.2}$$

there exists, provided that the potential falls off sufficiently fast as $r\to\infty$ (cf.

[1] *1958 Annu. Int. Conf. High-Energy Phys. CERN, Geneva, 1958*, p. 215.

Section 6.2), a solution of (6.1.1) whose asymptotic behavior differs from $\psi_0$ by an outgoing spherical wave (scattered wave). In terms of the free-space Green function (3.3.49),

$$G_0(k, \mathbf{r}, \mathbf{r}') = -\frac{\exp(ik|\mathbf{r}-\mathbf{r}'|)}{4\pi|\mathbf{r}-\mathbf{r}'|}, \tag{6.1.3}$$

this solution satisfies the well-known integral equation of scattering theory [cf. Section 6.2(a)]

$$\psi(\mathbf{k}, \mathbf{r}) = \psi_0(\mathbf{k}, \mathbf{r}) + \int \mathscr{K}(k, \mathbf{r}, \mathbf{r}')\psi(\mathbf{k}, \mathbf{r}')\, d^3 r', \tag{6.1.4}$$

with the kernel

$$\mathscr{K}(k, \mathbf{r}, \mathbf{r}') = G_0(k, \mathbf{r}, \mathbf{r}')\, U(r'). \tag{6.1.5}$$

If we now let the observation point $\mathbf{r}$ approach infinity along the direction $\hat{\mathbf{r}}$, we have

$$\begin{aligned} |\mathbf{r}-\mathbf{r}'| &= [(x-x')^2 + (y-y')^2 + (z-z')^2]^{1/2} \\ &= r[1 - 2(\mathbf{r}\cdot\mathbf{r}'/r^2) + (r'^2/r^2)]^{1/2} \\ &\approx r - \hat{\mathbf{r}}\cdot\mathbf{r}' \qquad (r \to \infty), \end{aligned} \tag{6.1.6}$$

so that

$$G_0(k, \mathbf{r}, \mathbf{r}') \approx -(1/4\pi)(e^{ikr}/r)\exp(-i\mathbf{k}'\cdot\mathbf{r}') \qquad (r \to \infty), \tag{6.1.7}$$

where

$$\mathbf{k}' = k\hat{\mathbf{r}}. \tag{6.1.8}$$

Substituting in (6.1.5), (6.1.4), we see that, if the potential falls off sufficiently fast at infinity [cf. Section 6.2(c)], we have

$$\psi(\mathbf{k}, \mathbf{r}) \approx e^{ikr\cos\theta} + f(k, \theta)\, e^{ikr}/r \qquad (r \to \infty), \tag{6.1.9}$$

where the total scattering amplitude $f(k, \theta)$ is given by

$$f(k, \theta) = -(1/4\pi)\int \exp(-i\mathbf{k}'\cdot\mathbf{r}')\, U(r')\psi(\mathbf{k}, \mathbf{r}')\, d^3 r'. \tag{6.1.10}$$

Our problem is to investigate the analytic properties of $f(k, \theta)$ as a function of both variables (or as a function of $k$ and the magnitude of the momentum transfer, as in Chapter 3). In the present section, which is intended mainly for orientation, we present some formal developments and a survey of possible approaches to this problem.

According to (6.1.10), our goal can be reached by investigating first the analytic properties of the complete wave function $\psi$, and then discussing the convergence of the integral in (6.1.10).

The integral equation (6.1.4) for $\psi$ can be written formally in operator language as

$$\psi = \psi_0 + \mathscr{K}\psi, \tag{6.1.11}$$

or

$$(I-\mathscr{K})\psi = \psi_0, \tag{6.1.12}$$

where $I$ is the identity operator. Formally, the solution can be written as

$$\psi = R\psi_0 = (I-\mathscr{K})^{-1}\psi_0, \tag{6.1.13}$$

where $R$ is called the *resolvent operator*. If we write [cf. (6.1.5)]

$$R = (I-G_0 U)^{-1} = I + GU, \tag{6.1.14}$$

the solution (6.1.13) takes the form

$$\psi = \psi_0 + GU\psi_0, \tag{6.1.15}$$

i.e., explicitly,

$$\psi(\mathbf{k},\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} + \int G(k,\mathbf{r},\mathbf{r}')\, U(r')\, e^{i\mathbf{k}\cdot\mathbf{r}'}\, d^3 r', \tag{6.1.16}$$

where $G(k,\mathbf{r},\mathbf{r}')$ is *Green's function*. The Schrödinger equation (6.1.1) implies that it satisfies the familiar equation

$$[\Delta+k^2-U(r)]\, G(k,\mathbf{r},\mathbf{r}') = \delta(\mathbf{r}-\mathbf{r}'). \tag{6.1.17}$$

In operator language, this corresponds to $(E-H)G = I$, where $H$ is the Hamiltonian operator, so that $G$ corresponds to $(E-H)^{-1}$; similarly, $G_0$ corresponds to $(E-H_0)^{-1}$. All these operators, of course, have to be defined in a suitable way to satisfy the outgoing wave boundary condition.[2]

Operating with $(I-G_0 U)$ on both sides of (6.1.14), we find that $G$ satisfies the integral equation

$$G = G_0 + G_0 UG, \tag{6.1.18}$$

i.e., explicitly,

$$G(k,\mathbf{r},\mathbf{r}') = G_0(k,\mathbf{r},\mathbf{r}') + \int G_0(k,\mathbf{r},\mathbf{r}'')\, U(r'')\, G(k,\mathbf{r}'',\mathbf{r}')\, d^3 r''. \tag{6.1.19}$$

Through (6.1.13) or (6.1.15), the analytic properties of $\psi$ can be related to those of the resolvent or of Green's function. The analytic properties of the scattering amplitude then follow from (6.1.10).

---

[2] See, for example, Newton [1]. One has to define an operator corresponding to $(E+i0-H)^{-1}$ [cf. (A4.24)].

According to (6.1.4) and (6.1.16), singularities may arise: (a) Due to lack of convergence of the integrals. In this connection, the behavior of the potential, especially as $r \to 0$ or $r \to \infty$, has to be considered. Note also that $G_0$ [cf. (6.1.3)] becomes singular as $|\mathbf{r}-\mathbf{r}'| \to 0$. (b) Due to singularities of the resolvent or Green's function. If we consider the extension to complex $k$, with $\operatorname{Im} k > 0$, we expect such singularities to appear, according to (6.1.12) and (6.1.18), when the homogeneous equations

$$\psi = \mathscr{K}\psi \qquad \text{or} \qquad G = G_0 U G = \mathscr{K} G \tag{6.1.20}$$

have nontrivial solutions. In view of the similarity between these equations, we may construct a solution $\psi(\mathbf{k}, \mathbf{r})$ of the first one by fixing the value of $\mathbf{r}'$ in a solution $G(k, \mathbf{r}, \mathbf{r}')$ of the second one. By the "outgoing wave" boundary condition, $\psi$ will behave like $e^{ikr}/r$ as $r \to \infty$, with $\operatorname{Im} k > 0$, and it may therefore be expected (cf. Section 5.4) to correspond to a bound state. Thus we expect singularities (poles on the positive imaginary axis in the $k$-plane) to develop at positions corresponding to bound states.

The simplest approach to the solution of an integral equation such as (6.1.11) is the *Liouville–Neumann iteration method*, already employed in Section 5.3(b). Iterating (6.1.11) once, we obtain

$$\psi = \psi_0 + \mathscr{K}(\psi_0 + \mathscr{K}\psi) = \psi_B + \mathscr{K}^2\psi, \tag{6.1.21}$$

where

$$\psi_B(\mathbf{k}, \mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}} + \int \mathscr{K}(k, \mathbf{r}, \mathbf{r}') e^{i\mathbf{k}\cdot\mathbf{r}'}\, d^3r' \tag{6.1.22}$$

corresponds to the well-known Born approximation, and $\mathscr{K}^2$ is the *iterated kernel*

$$\mathscr{K}^2(k, \mathbf{r}, \mathbf{r}') = \int \mathscr{K}(k, \mathbf{r}, \mathbf{r}'')\mathscr{K}(k, \mathbf{r}'', \mathbf{r}')\, d^3r''. \tag{6.1.23}$$

Repeating the procedure, we generate the complete Born series. This corresponds to the Liouville–Neumann expansion of the resolvent (6.1.13):

$$R = (I - \mathscr{K})^{-1} = I + \mathscr{K} + \mathscr{K}^2 + \mathscr{K}^3 + \cdots. \tag{6.1.24}$$

In general, this series converges only when the potential is very weak, as one would expect from a perturbation approach (if we replace $U$ by $\lambda U$, introducing a parameter $\lambda$ to measure the strength of the potential, the above series becomes a power series in $\lambda$). If the potential is attractive and if it becomes strong enough to accommodate a bound state [so that the homogeneous equations (6.1.20) have a nontrivial solution], the series fails to converge at sufficiently low energy. However, if one measures the strength of the potential by comparing it with the total energy, it can usually be regarded as "weak" for sufficiently high energies [$U$ is a small perturbation for sufficiently large $k^2$ in

(6.1.1)], so that one would expect the Born series to converge in the high-energy limit.

It was, in fact, shown by Klein and Zemach [2, 2a] that, for a suitable class of potentials (similar to those to be discussed below), the Born series becomes convergent and the first Born approximation is approached as the energy goes to infinity [cf. (5.5.13) and Section 6.4(a)]. Together with known results about the analytic properties of Green's function, this enabled them to derive dispersion relations for the total scattering amplitude for fixed momentum transfer.

Such dispersion relations were first derived by Khuri [3]. His derivation was based upon Jost and Pais's solution [4] of the scattering integral equation (6.1.11) by the *Fredholm method*. The classical Fredholm theory[3] was developed for continuous kernels. Thus it cannot be applied directly to (6.1.11), in view of the above-mentioned singularity of $\mathscr{K}(k, \mathbf{r}, \mathbf{r}')$ at $\mathbf{r} = \mathbf{r}'$ [cf. (6.1.3)]. However, the iterated kernel (6.1.23) does not have such a singularity, so that the classical Fredholm method can be applied to the iterated integral equation (6.1.21). The Fredholm solution is of the form

$$\psi(\mathbf{k}, \mathbf{r}) = \psi_B(\mathbf{k}, \mathbf{r}) + \int \frac{\Delta(k, \mathbf{r}, \mathbf{r}')}{\Delta(k)} \psi_B(\mathbf{k}, \mathbf{r}')\, d^3 r', \tag{6.1.25}$$

where (again assuming that $U$ is replaced by $\lambda U$) $\Delta(k, \mathbf{r}, \mathbf{r}')$ and the Fredholm denominator $\Delta(k)$ are power series in the potential strength $\lambda$, with coefficients given by integrals of determinants containing the iterated kernel $\mathscr{K}^2$. With suitable assumptions about the potential, these series can be shown to be uniformly and absolutely convergent, and the analytic properties of the scattering amplitude then follow from the analytic properties of the numerator and denominator series.

The classical Fredholm theory has been extended by Smithies [5, Chapter VI] to *square integrable* kernels, i.e., to kernels $\mathscr{K}(\mathbf{x}, \mathbf{y})$ such that

$$\iint |\mathscr{K}(\mathbf{x}, \mathbf{y})|^2\, d^3x\, d^3y < \infty. \tag{6.1.26}$$

The kernel $\mathscr{K}$ of (6.1.4) does not have this property. However, if we multiply both sides by $U^{1/2}(r)$, and introduce

$$\tilde{\psi}(\mathbf{k}, \mathbf{r}) = U^{1/2}(r)\psi(\mathbf{k}, \mathbf{r}), \qquad \tilde{\psi}_0(\mathbf{k}, \mathbf{r}) = U^{1/2}(r)\psi_0(\mathbf{k}, \mathbf{r}), \tag{6.1.27}$$

we find

$$\tilde{\psi} = \tilde{\psi}_0 + \tilde{\mathscr{K}}\tilde{\psi}, \tag{6.1.28}$$

[3] See, for example, Smithies [5].

where

$$\tilde{\mathscr{K}}(k,\mathbf{r},\mathbf{r}') = U^{1/2}(r)\mathscr{K}(k,\mathbf{r},\mathbf{r}')\,U^{-1/2}(r')$$
$$= U^{1/2}(r)G_0(k,\mathbf{r},\mathbf{r}')\,U^{1/2}(r') \qquad (6.1.29)$$

is square integrable for the class of potentials to be considered. The use of the square root of the potential does not cause any serious difficulties: the potential may have either sign [6, 7]. By applying Smithies's extension of the Fredholm theory to the integral equation (6.1.28), Scadron *et al.* [8] derived the analytic properties of the scattering amplitude as a function of both the energy and the momentum transfer. This approach can also be applied to potentials that are not spherically symmetric.

Similar results had previously been obtained by Hunziker [10], who applied general techniques of functional analysis in Banach space. These techniques had already been employed by Ikebe [11] in a general discussion of potential scattering. Our treatment will be based on Hunziker's approach. We restrict our consideration to spherically symmetric potentials, but much of the analysis remains valid in more general cases, as is shown in the references quoted above.[4a]

## 6.2. The Resolvent Operator in Banach Space

### (a) Functional Formulation in Banach Space

The basic idea of Hunziker's approach is to make use of the known properties of the resolvent of a class of operators known as *compact* operators in Banach space. The relevant definitions and theorems about compact operators in Banach space are given in Appendix F.

In the usual framework of quantum mechanics, one considers the solution of Schrödinger's equation as an eigenvalue problem in Hilbert space. This means that the usual scalar product of state vectors is defined, and, in agreement with the probability interpretation, we restrict ourselves to normalizable wave functions $\psi(\mathbf{r})$ (square integrable wave packets), such that $\int|\psi(\mathbf{r})|^2\,d^3r$ is finite. The plane wave (6.1.2), which does not satisfy this condition, is regarded as an "improper eigenfunction."

Hunziker reformulates the scattering problem in terms of a Banach space, which is more general than a Hilbert space, in that scalar products are not defined and the requirement of finite total probability is substituted by the requirement of a finite probability *density* $|\psi(\mathbf{r})|^2$. Specifically, let us consider

[4] See also Scadron's article in Ramakrishnan [9].

[4a] A rigorous mathematical treatment valid for a wider class of potentials may be found in the recent monograph by Simon [11a].

the Banach space $C$ of *continuous bounded functions* $\psi(\mathbf{r})$ mentioned in Appendix F, with the norm (F4).

We would like to consider the Schrödinger equation (6.1.1) as a functional equation in $C$. However, the left-hand side of (6.1.1) need not be defined for every $\psi \in C$, so that we begin by considering the subset $D_2$ of twice continuously differentiable functions for which $\Delta\psi$ is bounded. The domain of definition of $\Delta$ can be extended as follows: if $\{\psi_n\}$ is a sequence of elements $\psi_n \in D_2$, with $\psi_n \to \psi$ and $\Delta\psi_n \to \chi$, we *define* $\Delta\psi = \chi$. This extended domain of definition will be called $D$. One can show [10, Appendix I] that the elements of $D$ are continuously differentiable functions with bounded derivatives, so that their first-order partial derivatives belong to $C$.

To begin with, we make the following assumptions about the potential [cf. (5.3.20)]:

$$U(r) \in C, \tag{6.2.1}$$

$$M_1 = \int_0^\infty r|U(r)|\, dr < \infty. \tag{6.2.2}$$

Assumption (6.2.1) (which will later be relaxed) implies that $U\psi \in C$ if $\psi \in C$, so that we can regard the Schrödinger equation as a functional equation in $C$ (with $\psi$ restricted to $D$).

We will consider both real values of $k$ and complex values with $\operatorname{Im} k > 0$. For real $k > 0$, we have to add the boundary condition that $\psi$ must have the asymptotic behavior (6.1.9) as $r \to \infty$. More precisely,

$$\psi(\mathbf{k}, \mathbf{r}) = \psi_0(\mathbf{k}, \mathbf{r}) + \psi_s(\mathbf{k}, \mathbf{r}), \tag{6.2.3}$$

where the incident wave $\psi_0(\mathbf{k}, \mathbf{r})$ is given by (6.1.2) and the scattered wave $\psi_s(\mathbf{k}, \mathbf{r})$ must satisfy the Sommerfeld radiation condition[5]

$$\lim_{r\to\infty} \psi_s = 0; \qquad \lim_{r\to\infty} [r(\partial\psi_s/\partial r - ik\psi_s)] = 0. \tag{6.2.4}$$

The above boundary-value problem can now be reformulated in terms of integral equations by following the standard procedure. The above-mentioned properties of functions $\psi(\mathbf{k}, \mathbf{r}) \in D$ allow us to apply Green's theorem to a domain $V$ bounded by two concentric spheres $\sigma$ and $\Sigma$ centered at the point $\mathbf{r}$, with radii $\varepsilon$ and $R$, respectively,

$$\begin{aligned}\int_{\sigma+\Sigma} \left( G_0 \frac{\partial\psi}{\partial n} - \psi \frac{\partial G_0}{\partial n} \right) d^2 r' &= \int_V (G_0 \Delta\psi - \psi \Delta G_0)\, d^3 r' \\ &= \int_V G_0(k, \mathbf{r}, \mathbf{r}')\, U(r')\psi(k, \mathbf{r}')\, d^3 r', \end{aligned} \tag{6.2.5}$$

where $G_0$ is given by (6.1.3) and we have used (6.1.1).

---

[5] Cf. Sommerfeld [12, p. 189].

We now let $\varepsilon \to 0$ and $R \to \infty$. The volume integral approaches the last term of (6.1.4), and it is absolutely convergent. In fact, by (F4) and (6.1.3),

$$
\begin{aligned}
|\mathscr{K}\psi| &\leqslant \int |G_0 U\psi|\, d^3 r' \\
&\leqslant \|\psi\| \int_0^\infty |U(r')|\, r'^2\, dr' \int \frac{d\Omega'}{4\pi|\mathbf{r}-\mathbf{r}'|},
\end{aligned}
\tag{6.2.6}
$$

both for real $k$ and for $\operatorname{Im} k > 0$. For any function $f(|\mathbf{r}-\mathbf{r}'|)$, we have

$$
\begin{aligned}
(1/4\pi) \int f(|\mathbf{r}-\mathbf{r}'|)\, d\Omega' &= (1/2) \int_0^\pi f((r^2 - 2rr'\cos\theta' + r'^2)^{1/2}) \sin\theta'\, d\theta' \\
&= (1/2rr') \int_{|r-r'|}^{r+r'} f(\rho)\rho\, d\rho,
\end{aligned}
\tag{6.2.7}
$$

so that

$$
\int \frac{d\Omega'}{4\pi|\mathbf{r}-\mathbf{r}'|} = \min\left(\frac{1}{r}, \frac{1}{r'}\right)
\tag{6.2.8}
$$

and (6.2.6) yields

$$
|\mathscr{K}\psi| \leqslant \|\psi\| \int_0^\infty |U(r')|\, r'\, dr' = M_1 \|\psi\|.
\tag{6.2.9}
$$

As $\varepsilon \to 0$, the integral over $\sigma$ on the left-hand side of (6.2.5) approaches $\psi(k, \mathbf{r})$, as is well known [$G_0$ satisfies (6.1.17) with $U = 0$]. For $\operatorname{Im} k > 0$, we have $\int_\Sigma \to 0$ as $R \to \infty$ because $G_0$ is exponentially damped and $\psi \in C$. However, for real $k$, we have to apply Green's theorem to $\psi_s = \psi - \psi_0$, i.e., $\psi$ has to be replaced by $\psi_s$ in the first two integrals of (6.2.5) (but *not* in the third one!). It then follows from Sommerfeld's radiation condition (6.2.4) that $\int_\Sigma \to 0$ as $R \to \infty$. Finally we find

$$
\psi = \mathscr{K}\psi \qquad (\operatorname{Im} k > 0),
\tag{6.2.10}
$$

$$
\psi = \psi_0 + \mathscr{K}\psi \qquad (k \text{ real}),
\tag{6.2.11}
$$

corresponding, respectively, to the homogeneous and inhomogeneous integral equations given in Section 6.1.

### *(b) Compactness of $\mathscr{K}$*

It follows from (6.2.9) and (F5) that $\mathscr{K}(k)$ is a *bounded operator*, and that its norm satisfies

$$
\|\mathscr{K}\| \leqslant M_1, \qquad \operatorname{Im} k \geqslant 0.
\tag{6.2.12}
$$

It follows that, for a *weak potential*, satisfying

$$M_1 < 1, \tag{6.2.13}$$

the Liouville–Neumann series expansion of the resolvent (6.1.24) is convergent[6]; the inhomogeneous equation (6.2.11) may be solved by iteration (i.e., the Born perturbation expansion converges) and the homogeneous equation (6.2.10) has only the trivial solution [i.e., there are no bound states; this is consistent with (5.8.19)].

Regardless of the strength of the potential (provided that $M_1$ is finite), the operator $\mathscr{K}(k)$ is not only bounded but also *compact*, for $\operatorname{Im} k \geqslant 0$. This is true not only for potentials in the class (6.2.1) but for any measurable function $U(r)$; i.e., we have[7]:

THEOREM 6.2.1. *If $U(r)$ is measurable and if $M_1 < \infty$, $\mathscr{K}(k)$ is a compact operator in the Banach space $C$, for* $\operatorname{Im} k \geqslant 0$.

*Proof.* Let us first consider the special case in which $U(r)$ is a *bounded cutoff potential*, i.e.,

$$|U(r)| < B, \qquad U(r) = 0 \quad (r > a), \tag{6.2.14}$$

and let us consider the set $\mathscr{F}$ of all functions $\varphi$ resulting from the application of $\mathscr{K}(k)$ to functions in the unit sphere in $C$:

$$\mathscr{F} = \{\varphi | \varphi = \mathscr{K}\psi, \|\psi\| < 1\}. \tag{6.2.15}$$

By (6.2.9), we have $\|\varphi\| < M_1$, so that the functions $\varphi$ are *equibounded*. Actually, (6.2.6) and (6.2.8) yield in this case, for $r > a$,

$$|\varphi(k, \mathbf{r})| < (1/r) \int_0^a |U(r')| \, r'^2 \, dr' < Ba^3/3r = f(r), \tag{6.2.16}$$

so that all $\varphi \in \mathscr{F}$ are uniformly bounded by $f(r)$, which $\to 0$ as $r \to \infty$. Finally, since the last term in (6.1.4) is in this case similar to the potential of a mass distribution contained in a finite volume, it follows from a well-known theorem [14, p. 246] that $\varphi$ is a continuously differentiable function of $\mathbf{r}$ for all $\mathbf{r}$, and that its partial derivatives may be obtained by differentiating under the integral sign. Taking into account (6.2.14), one finds

$$|\partial\varphi/\partial x_i| < (1 + |k|)A \qquad (i = 1, 2, 3), \tag{6.2.17}$$

where $A$ depends only on $B$ and $a$. This implies, in particular, that condition (F7) (see Appendix F) is satisfied for all $\mathbf{r}$ and all $\varphi \in \mathscr{F}$; i.e., the set $\mathscr{F}$ is *equicontinuous*. We can now apply Theorem F2, and we conclude that $\mathscr{F}$ is

[6] See, for example, Yosida [13, p. 69].

[7] Cf. Ikebe [11, Lemma 4.2].

compact. Thus $\mathscr{K}(k)$ maps every bounded set into a compact set, so that it is a compact operator in $C$.

To extend the proof to potentials $U(r)$ that are not cutoff, we approximate them by a sequence $\{U_n(r)\}$ of bounded cutoff potentials, where $U_n(r)$ is defined as follows:

$$U_n(r) = \begin{cases} U(r) & \text{if} \quad r < a_n \quad \text{and} \quad |U(r)| < B_n, \\ 0 & \text{otherwise.} \end{cases} \tag{6.2.18}$$

It then follows from (6.2.9) and (F5) that

$$\begin{aligned} \|\mathscr{K} - \mathscr{K}_n\| &= \|G_0 U - G_0 U_n\| \\ &\leqslant \int_{a_n}^{\infty} |U(r')|\, r'\, dr' + \int_{E_n} |U(r')|\, r'\, dr', \end{aligned} \tag{6.2.19}$$

where $E_n$ is the set of all $r' < a_n$ such that $|U(r')| > B_n$. In view of the finiteness of $M_1$, we can choose $a_n$ and $B_n$ so large that both integrals are smaller than $1/n$. It follows that

$$\lim_{n \to \infty} \|\mathscr{K} - \mathscr{K}_n\| = 0. \tag{6.2.20}$$

Since each $\mathscr{K}_n$ is compact, we conclude from Theorem F3 that $\mathscr{K}$ is also compact, thus completing the proof of Theorem 6.2.1.

### *(c) Scattering States and Bound States*

We have seen in Section 6.2(a) that the Schrödinger equation (6.1.1), regarded as a functional equation in $C$, leads to the homogeneous equation (6.2.10) for $\operatorname{Im} k > 0$; for real $k$, together with the boundary conditions (6.2.3) and (6.2.4) as $r \to \infty$, it leads to the inhomogeneous equation (6.2.11). Let us now consider the converse of these results.

One can readily show [10, Sec. 4], for potentials of the class (6.2.1) and (6.2.2), that the solutions of (6.2.10) and (6.2.11) satisfy the Schrödinger equation (6.1.1). For cutoff potentials, the proof is again similar to that employed for solutions of Poisson's equation [cf. (6.2.17)]; this is then extended to the general case by considering a sequence of truncated potentials and applying the extended definition of $\Delta$ in the domain $D$ given in Section 6.2(a).

There remains for us to consider the asymptotic behavior of the solutions as $r \to \infty$. For this purpose, let $\psi \in C$ and let us define $\varphi = \mathscr{K}\psi$, so that $\varphi$ corresponds to $\psi$ for the homogeneous equation and to the scattered wave $\psi_s = \psi - \psi_0$ for the inhomogenous equation:

$$\varphi(\mathbf{k}, \mathbf{r}) = -\frac{1}{4\pi} \int \frac{\exp(ik|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} U(r') \psi(\mathbf{k}, \mathbf{r}')\, d^3 r'. \tag{6.2.21}$$

If we consider first the case of real $k$, we expect from (6.1.6) and (6.1.7) that $\varphi = \psi_s$ will approach [cf. (6.1.9)]

$$\varphi_\infty(\mathbf{k},\mathbf{r}) = -(1/4\pi)(e^{ikr}/r)\int \exp(-i\mathbf{k}'\cdot\mathbf{r}')\,U(r')\,\psi(\mathbf{k},\mathbf{r}')\,d^3r'$$
$$= f(k,\theta)\,e^{ikr}/r \tag{6.2.22}$$

as $r\to\infty$ in the direction $\theta$; here $\mathbf{k}'$ is defined by (6.1.8). We have

$$|f(k,\theta)| \leqslant \|\psi\|\int |U(r')|\,d^3r', \tag{6.2.23}$$

provided that the integral exists. This is tantamount to assuming [cf. (5.3.21)]

$$M_2 = \int_0^\infty |U(r)|\,r^2\,dr < \infty; \tag{6.2.24}$$

i.e., the potential must decrease more rapidly than $r^{-3}$ at infinity.[8] Assuming that this condition is verified, one can estimate the difference $|\varphi-\varphi_\infty|$ and show [10, Sec. 5; 11, Lemma 3.2] that

$$\lim_{r\to\infty}[r\,|\varphi(\mathbf{k},\mathbf{r})-\varphi_\infty(\mathbf{k},\mathbf{r})|] = 0. \tag{6.2.25}$$

Similarly, one can show that $\varphi$ satisfies the Sommerfeld radiation condition (6.2.4), if $r^2\,|U(r)|\to 0$ as $r\to\infty$.

Let us now consider the solutions of the homogeneous equation (6.2.10) for $\kappa = \operatorname{Im} k > 0$. We expect that such solutions will verify a bound of the form

$$|\psi(k,\mathbf{r})| < Ae^{-\kappa r}/r \qquad (\kappa = \operatorname{Im} k). \tag{6.2.26}$$

In fact, if we assume this bound and substitute it in (6.2.21), we find, with the help of (6.2.8),

$$|\varphi(k,\mathbf{r})| < \frac{A}{4\pi}\int \frac{\exp(-\kappa|\mathbf{r}-\mathbf{r}'|)}{|\mathbf{r}-\mathbf{r}'|}\,|U(r')|\,\frac{e^{-\kappa r'}}{r'}\,d^3r'$$
$$< Ae^{-\kappa r}\int_0^\infty |U(r')|\,r'\,dr'\int\frac{d\Omega'}{4\pi|\mathbf{r}-\mathbf{r}'|} \leqslant M_1\,A\,\frac{e^{-\kappa r}}{r}. \tag{6.2.27}$$

Thus, if $M_1<\infty$, and $\psi$ obeys a bound of the form (6.2.26), $\varphi = \mathscr{K}\psi$ also does: *all $\psi\in C$ satisfying a bound of the type (6.2.26) form a linear subspace $T$ of $C$, invariant under the operator $\mathscr{K}$.*

To show that solutions of the homogeneous equation satisfy (6.2.26), it

[8] For an heuristic discussion of this condition, cf. Landau and Lifshitz [15, p. 474].

now suffices to show that *all solutions of the homogeneous equation belong to T.* For this purpose, we can split the potential into a cutoff potential and a remainder:

$$U(r) = [1 - \theta(r-a)]\,U(r) + \theta(r-a)\,U(r) = U_a(r) + \tilde{U}(r), \quad (6.2.28)$$

where $\theta$ is the Heaviside step function. Correspondingly,

$$\mathscr{K} = G_0\,U = \mathscr{K}_a + \tilde{\mathscr{K}}. \quad (6.2.29)$$

Let $\psi \in C$ be a solution of the homogeneous equation (6.2.10),

$$\psi = (\mathscr{K}_a + \tilde{\mathscr{K}})\psi = \psi_0 + \tilde{\mathscr{K}}\psi. \quad (6.2.30)$$

Since $\psi_0 = \mathscr{K}_a\psi$ is represented by an integral of the form (6.2.21) extended over a finite volume, it is easy to show that it satisfies a bound of the form (6.2.26); i.e., $\psi_0 \in T$. In view of the invariance of $T$ under $\mathscr{K}$, it follows that we can regard (6.2.30) as a functional equation in $T$ as well as in $C$. According to (6.2.12), we can choose the cutoff radius $a$ so large that $\|\tilde{\mathscr{K}}\| < 1$. It then follows, as in (6.2.13), that (6.2.30) has a unique solution in $T$ as well as in $C$. Since $\psi$ is a solution in $C$ and $T$ is a subspace of $C$, we conclude that $\psi \in T$.

We conclude that solutions of the homogeneous equation for $\operatorname{Im} k > 0$ are square integrable solutions of Schrödinger's equation (energy eigenfunctions). Since the Hamiltonian is an Hermitian operator [the potential $U(r)$ is real], the energy eigenvalues must be real, so that the corresponding values of $k$ must lie on the positive imaginary axis. Therefore, as expected, *solutions of the homogeneous equation for* $\operatorname{Im} k > 0$ *correspond to bound states.*

It follows from Theorem 6.2.1 and the Riesz–Schauder theorem F4 that, if $U(r)$ is measurable and $M_1 < \infty$, the *Fredholm alternative* is valid for $\operatorname{Im} k \geqslant 0$; i.e., for each $k$, either the resolvent operator

$$R = (I - \mathscr{K})^{-1} \quad (6.2.31)$$

exists as a bounded operator, or else the homogeneous equation (6.2.10) has a nontrivial solution, in which case it has a finite number of linearly independent solutions. The values of $k$ for which the first alternative is valid will be called the *resolvent set* of $\mathscr{K}(k)$; the remainder set, corresponding to the $k$-eigenvalues of $\mathscr{K}$, is the *spectrum* of $\mathscr{K}$. The above discussion then shows that *the spectrum of* $\mathscr{K}(k)$ *in* $\operatorname{Im} k > 0$ *lies on the imaginary axis, and the corresponding k-eigenvalues are associated with bound states.*

We can now make use of the fact that $U(r)$ is a central potential, so that the Schrödinger equation is invariant under rotation. Let $k$ be a point in the spectrum; there is then a finite number of linearly independent eigenfunctions, so that they span a finite-dimensional subspace $F$ of $C$. However, if $\psi(k, \mathbf{r}) \in F$, so does $\psi(k, R\mathbf{r})$, where $R$ is an arbitrary rotation. This induces in $F$ a finite-dimensional representation of the rotation group,[9] so that there must exist in $F$ a finite basis of the form [cf. (5.1.2)]

$$\psi_{nlm}(k, \mathbf{r}) = \frac{u_{nl}(k, r)}{r} Y_l^m(\theta, \varphi). \tag{6.2.32}$$

The $u_{nl}(k, r)$ are radial eigenfunctions and we can therefore apply the results obtained in Chapter 5. Taking into account (5.8.19), we conclude that, *if $M_1 < \infty$, the spectrum of $\mathscr{K}(k)$ in $\operatorname{Im} k \geqslant 0$ consists of a finite number of $k$-eigenvalues, $k = i\kappa_n$, all of which lie on the imaginary axis (including possibly $k = 0$).*

For real $k \neq 0$, the inhomogeneous equation (6.2.11) has, if $M_1 < \infty$, a unique continuous and bounded solution $\psi(k, \mathbf{r})$; if $M_2 < \infty$, the asymptotic behavior of the solution as $r \to \infty$ is given by (6.1.9) and (6.1.10). Let us now apply these results to the discussion of the analytic properties of the scattering amplitude $f(k, \theta)$.

## 6.3. Analytic Properties of the Total Scattering Amplitude

### *(a) Analytic Properties of the Resolvent Operator*

From now on we restrict our consideration to central potentials satisfying the conditions

$$M_1 < \infty, \qquad M_2 < \infty. \tag{6.3.1}$$

Let us consider two points $k$ and $k'$, with $\operatorname{Im} k \geqslant 0$, $\operatorname{Im} k' \geqslant 0$. It follows from (6.1.3)–(6.1.5), (F4), and (F5) that

$$\begin{aligned} \|\mathscr{K}(k) - \mathscr{K}(k')\| &= \sup_{\|\psi\|=1} \|[\mathscr{K}(k) - \mathscr{K}(k')]\psi\| \\ &\leqslant \frac{1}{4\pi} \int \frac{d^3 r'}{|\mathbf{r} - \mathbf{r}'|} \left| e^{ik|\mathbf{r}-\mathbf{r}'|} - e^{ik'|\mathbf{r}-\mathbf{r}'|} \right| |U(r')|. \end{aligned} \tag{6.3.2}$$

We have[10]

$$|e^{ikr} - e^{ik'r}| \leqslant |k - k'|\, r \qquad (\operatorname{Im} k \geqslant 0, \operatorname{Im} k' \geqslant 0). \tag{6.3.3}$$

---

[9] See, for example, Wigner [16].

[10] This follows from the inequality $|f(z)| \leqslant 1$ in $\operatorname{Im} z \geqslant 0$ for $f(z) = (e^{iz} - 1)/z$, which in its turn follows from the fact that $f(z)$ is regular in $\operatorname{Im} z \geqslant 0$, $|f(z)| \leqslant 1$ on the real axis, and $|f(z)| \to 0$ as $|z| \to \infty$, $\operatorname{Im} z \geqslant 0$, together with the maximum-modulus theorem.

Substituting in (6.3.2), it follows that

$$\|\mathscr{K}k) - \mathscr{K}(k')\| \leqslant (1/4\pi)|k-k'| \int |U(r')|\, d^3 r' = M_2|k-k'|, \tag{6.3.4}$$

so that $\mathscr{K}(k)$ is a continuous operator function of $k$ in $\operatorname{Im} k \geqslant 0$ (cf. Appendix F). Thus the operator integral

$$\mathscr{I} = \oint_\Gamma \mathscr{K}(k)\, dk \tag{6.3.5}$$

exists over any simple rectifiable closed curve $\Gamma$ in $\operatorname{Im} k \geqslant 0$.

For any $\psi \in C$, we have

$$\begin{aligned} \mathscr{I}\psi &= \oint_\Gamma dk\, \mathscr{K}(k)\psi \\ &= -(1/4\pi) \oint_\Gamma dk \int d^3 r' (e^{ik|\mathbf{r}-\mathbf{r}'|}/|\mathbf{r}-\mathbf{r}'|)\, U(r')\psi(\mathbf{r}'). \end{aligned} \tag{6.3.6}$$

Since the integral is absolutely and uniformly convergent for all $k$ in $\operatorname{Im} k \geqslant 0$, we can interchange the order of integration, and we get

$$\mathscr{I}\psi = 0 \qquad \text{for all} \quad \psi \in C; \qquad \text{i.e.,} \quad \mathscr{I} = 0. \tag{6.3.7}$$

It then follows from Morera's theorem (cf. Appendix F) that $\mathscr{K}(k)$ is holomorphic in $\operatorname{Im} k > 0$. In conclusion, *$\mathscr{K}(k)$ is holomorphic in* $\operatorname{Im} k > 0$ *and continuous in* $\operatorname{Im} k \geqslant 0$.

We can now apply Theorem F5, provided that we exclude a neighborhood of each point in the spectrum of $\mathscr{K}(k)$ in $\operatorname{Im} k \geqslant 0$. Taking into account the results obtained in Section 6.2(c) about the spectrum of $\mathscr{K}(k)$, we conclude that *the resolvent operator $R(k)$ defined by (6.2.31) is holomorphic in* $\operatorname{Im} k > 0$, *apart from a finite number of singularities on the imaginary axis, $k = i\kappa_n$, associated with bound states; furthermore, $R(k)$ is continuous for* $\operatorname{Im} k \geqslant 0$, *excepting these singularities (and possibly also $k = 0$).*

### *(b) Analytic Properties of the Total Wave Function*

The total wave function $\psi(\mathbf{k}, \mathbf{r})$ is given by [cf. (6.1.13) and (F11)]

$$\psi(\mathbf{k}) = R(k)\psi_0(\mathbf{k}), \tag{6.3.8}$$

where [cf. (6.1.2)] $\psi_0 = e^{i\mathbf{k}\cdot\mathbf{r}} = e^{ikr\cos\theta}$. For complex $\mathbf{k}$, $\psi_0$ is no longer bounded, so that we can no longer discuss the scattering integral equation as a functional equation in the Banach space $C$ of continuous bounded functions. We therefore extend the treatment by considering the larger space $C'$ of continuous bounded functions $\psi'(\mathbf{r})$ with the modified norm [cf. (F4)]

$$\|\psi'\|_\alpha = \sup_{\mathbf{r}}[e^{-\alpha r}|\psi'(\mathbf{r})|] \qquad (r = |\mathbf{r}|), \tag{6.3.9}$$

where $\alpha$ is a fixed positive number.

The transformation

$$\psi(\mathbf{r}) = e^{-\alpha r}\psi'(\mathbf{r}) \tag{6.3.10}$$

is an *isometric mapping* between $C$ and $C'$; i.e., it preserves the norm:

$$\|\psi\| = \|\psi'\|_\alpha. \tag{6.3.11}$$

We can consider the scattering integral equation (6.1.4) as a functional equation in $C'$. By (6.3.10), the corresponding equation in $C$ is

$$\psi(\mathbf{k}, \mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r} - \alpha r} + e^{-\alpha r}\int G_0(k, \mathbf{r}, \mathbf{r}')\, U(r')\, e^{\alpha r'}\psi(\mathbf{k}, \mathbf{r}')\, d^3 r'. \tag{6.3.12}$$

All the results derived above can be applied to this equation. In particular, if the transformed potential

$$U'(r) = e^{\alpha r}\, U(r) \tag{6.3.13}$$

satisfies condition (6.2.2), i.e., if [cf. (5.3.22)]

$$E_\alpha = \int_0^\infty e^{\alpha r}\,|U(r)|\, r\, dr < \infty \qquad (\alpha > 0), \tag{6.3.14}$$

it follows that the resolvent $R(k)$ of (6.3.12) in $C$ has all the analyticity and continuity properties stated at the end of Section 6.3(a). In view of the isometry (6.3.11), the same properties are valid for the resolvent $R'(k)$ of the scattering equation in $C'$.

We have, similarly to (6.3.8),

$$\psi'(\mathbf{k}) = R'(k)\psi_0'(\mathbf{k}), \tag{6.3.15}$$

but now

$$\psi_0'(\mathbf{k}) = e^{i\mathbf{k}\cdot\mathbf{r}}$$

is a holomorphic vector function in $C'$ for complex $k$, in the domain [cf. (6.3.9)]

$$|\mathrm{Im}\,\mathbf{k}| < \alpha. \tag{6.3.16}$$

It therefore follows from (6.3.15) and from the properties of $R'(k)$ that $\psi'(\mathbf{k})$ is also a holomorphic vector function in $C'$, in the domain

$$|\mathrm{Im}\,\mathbf{k}| < \alpha, \qquad \mathrm{Im}\, k > 0, \qquad k \neq i\kappa_n, \tag{6.3.17}$$

where $i\kappa_n$ are the eigenvalues.

Let

$$\psi'(\mathbf{k}) = \psi_0'(\mathbf{k}) + \psi_s'(\mathbf{k}), \tag{6.3.18}$$

corresponding to the decomposition (6.2.3). Then $\psi_s'(\mathbf{k})$ is also a holomorphic vector function in $C'$, for $\mathbf{k}$ in the domain (6.3.17). Since convergence in the sense of $C'$ implies pointwise convergence in $\mathbf{r}$ [cf. (6.3.9) and (F2)], it follows that, regarded as an ordinary function, $\psi_s'(\mathbf{k},\mathbf{r})$ *is holomorphic in the domain (6.3.17) for each fixed* $\mathbf{r}$. Furthermore, since $\psi_s'$ corresponds to the integral in the last term of (6.3.12) and $\psi' \in C'$, we conclude, taking into account (6.3.14), that, *for each fixed* $\mathbf{k}$ *in the domain (6.3.17),* $\psi_s'(\mathbf{k},\mathbf{r})$ *is a continuous bounded function of* $\mathbf{r}$.

### (c) The Total Scattering Amplitude

We now apply the above results to discuss the analytic properties of the total scattering amplitude $f(\mathbf{k},\mathbf{k}')$. We want to use as variables the energy $E$ and the magnitude of the momentum transfer $\tau$ (cf. Sections 3.3 and 3.4), rather than $k$ and $\theta$, as in (6.1.10). For this purpose we make the change of variables

$$\begin{aligned} \mathbf{P} &= \tfrac{1}{2}(\mathbf{k}+\mathbf{k}') = P\hat{\boldsymbol{\varepsilon}} \\ \boldsymbol{\tau} &= \mathbf{k}' - \mathbf{k} = \tau\hat{\boldsymbol{\varepsilon}}', \end{aligned} \tag{6.3.19}$$

where, in units $\hbar = m = 1$,

$$E = k^2/2 = k'^2/2; \qquad \hat{\boldsymbol{\varepsilon}}^2 = \hat{\boldsymbol{\varepsilon}}'^2 = 1; \qquad \hat{\boldsymbol{\varepsilon}}\cdot\hat{\boldsymbol{\varepsilon}}' = 0, \tag{6.3.20}$$

$$P^2 = 2E - (\tau^2/4); \qquad \tau^2 = 4E(1-\cos\theta). \tag{6.3.21}$$

Substituting (6.2.3) in (6.1.10), we get

$$f(E,\tau) = f_B(\tau) + f_1(E,\tau), \tag{6.3.22}$$

where $f_B(\tau)$ is the Born approximation,

$$f_B(\tau) = -(1/4\pi)\int \exp(-i\boldsymbol{\tau}\cdot\mathbf{r})\,U(r)\,d^3r = -\int_0^\infty rU(r)[\sin(\tau r)/\tau]\,dr, \tag{6.3.23}$$

and

$$f_1(E,\tau) = -(1/4\pi)\int \exp[-i(\tau/2)\hat{\boldsymbol{\varepsilon}}'\cdot\mathbf{r} - iP\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{r}]\,U(r)\,\psi_s(\mathbf{k},\mathbf{r})\,d^3r. \tag{6.3.24}$$

According to (6.3.23) and (6.3.14), the Born approximation $f_B(\tau)$ is a holomorphic function of $\tau$ alone, in the strip

$$|\mathrm{Im}\,\tau| < \alpha. \tag{6.3.25}$$

In order to discuss the analytic properties of $f_1(E,\tau)$, it is convenient to apply a transformation similar to (6.3.10) to the scattering integral equation, setting

$$\psi^\lambda(\mathbf{r}) = \exp(i\lambda k\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{r})\,\psi(\mathbf{r}), \tag{6.3.26}$$

where $\lambda$ is a real parameter. Similarly to (6.3.12), the scattering integral equation becomes

$$\begin{aligned}\psi^\lambda(\mathbf{k},\mathbf{r}) &= \psi_0{}^\lambda(\mathbf{k},\mathbf{r}) + \int G_0{}^\lambda(k,\mathbf{r},\mathbf{r}')\,U(r')\,\psi^\lambda(\mathbf{k},\mathbf{r}')\,d^3r' \\ &= \psi_0{}^\lambda(\mathbf{k},\mathbf{r}) + \psi_s{}^\lambda(\mathbf{k},\mathbf{r}),\end{aligned} \tag{6.3.27}$$

where

$$\psi_0{}^\lambda(\mathbf{k},\mathbf{r}) = \exp[i(P+\lambda k)\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{r} - i(\tau/2)\hat{\boldsymbol{\varepsilon}}'\cdot\mathbf{r}], \tag{6.3.28}$$

$$G_0{}^\lambda(k,\mathbf{r},\mathbf{r}') = -\frac{\exp\{ik[|\mathbf{r}-\mathbf{r}'| + \lambda\hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}')]\}}{4\pi|\mathbf{r}-\mathbf{r}'|}. \tag{6.3.29}$$

We can rewrite (6.3.24) in terms of $\psi_s{}^\lambda$ as

$$f_1(E,\tau) = -(1/4\pi)\int \exp[-i(\tau/2)\hat{\boldsymbol{\varepsilon}}'\cdot\mathbf{r} - i(P+\lambda k)\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{r}]\,U(r)\,\psi_s{}^\lambda(\mathbf{k},\mathbf{r})\,d^3r. \tag{6.3.30}$$

The transformed Green function $G_0{}^\lambda$ remains bounded in $\mathrm{Im}\,k \geqslant 0$, provided that

$$-1 \leqslant \lambda \leqslant 1. \tag{6.3.31}$$

On the other hand, by (6.3.9) and (6.3.28), $\psi_0{}^\lambda$ is an element of the Banach space $C'$, provided that

$$[\mathrm{Im}(P+\lambda k)]^2 + \tfrac{1}{4}(\mathrm{Im}\,\tau)^2 < \alpha^2, \tag{6.3.32}$$

which reduces to (6.3.16) for $\lambda = 0$. It is also a regular analytic function of $E$ and $\tau$ in this domain, except for the usual branch cut along the positive real $E$ axis [arising from $k = (2E)^{1/2}$] and a branch point at $E = \tau^2/8$ [arising from $P = (2E - \tau^2/4)^{1/2}$].

All the results of the above analysis can now be applied to the transformed integral equation (6.3.27). As in Section 6.3(b), they lead to the conclusion that $\psi_s{}^\lambda(E,\tau,\mathbf{r})$ is, for fixed $\mathbf{r}$, a regular analytic function of $E$ and $\tau$ in the domain (6.3.32), apart from a branch cut along $E > 0$, a branch point at

$E=\tau^2/8$, and a finite number of singularities for $E<0$, associated with bound states. For fixed $E$ and $\tau$ in this domain, $\psi_s{}^{\lambda}(E,\tau,\mathbf{r})$ is a continuous bounded function of $\mathbf{r}$.

The integral (6.3.30) also exists in the same domain (6.3.32), and the branch point at $E=\tau^2/8$ is removed by the integration, because the integral does not depend on the direction of $\hat{\boldsymbol{\varepsilon}}$. Thus we conclude that, apart from the cut along $E>0$ and the bound-state singularities for $E<0$, $f_1(E,\tau)$ is regular in the following family of domains:

$$(\operatorname{Im}\{[2E-(\tau^2/4)]^{1/2}+\lambda(2E)^{1/2}\})^2+\tfrac{1}{4}(\operatorname{Im}\tau)^2<\alpha^2 \qquad (-1\leqslant\lambda\leqslant 1). \tag{6.3.33}$$

This requires, in the first place, $|\operatorname{Im}\tau|<2\alpha$. Points such that this condition is fulfilled and, in addition,

$$|\operatorname{Im}[2E-(\tau^2/4)]^{1/2}|\leqslant|\operatorname{Im}(2E)^{1/2}|$$

also belong to the regularity domain, because for them one can always find $\lambda$ in $[-1,1]$ such that the square bracket vanishes in (6.3.33). On the other hand, if

$$|\operatorname{Im}[2E-(\tau^2/4)]^{1/2}|>|\operatorname{Im}(2E)^{1/2}|,$$

the square bracket takes its minimum value for $\lambda=+1$ or $\lambda=-1$, yielding the condition

$$|\operatorname{Im}[2E-(\tau^2/4)]^{1/2}|-|\operatorname{Im}(2E)^{1/2}|<[\alpha^2-\tfrac{1}{4}(\operatorname{Im}\tau)^2]^{1/2}.$$

We conclude that $f_1(E,\tau)$ *is a regular analytic function of both variables in the domain*

$$\begin{gathered}|\operatorname{Im}\tau|<2\alpha,\\ |\operatorname{Im}[2E-(\tau^2/4)]^{1/2}|-|\operatorname{Im}(2E)^{1/2}|<[\alpha^2-\tfrac{1}{4}(\operatorname{Im}\tau)^2]^{1/2},\end{gathered} \tag{6.3.34}$$

*except for a branch cut along the positive real E-axis and a finite number of singularities on the negative real E-axis, associated with bound states.*

The analytic properties of the scattering amplitude for fixed energy and for fixed momentum transfer follow from these results.

### *(d) Analytic Properties in the* $\cos\theta$ *Plane*

Let the energy be fixed at a positive real value, $E>0$. The regularity domain (6.3.34) then becomes

$$\{\operatorname{Im}[2E-(\tau^2/4)]^{1/2}\}^2+\tfrac{1}{4}(\operatorname{Im}\tau)^2<\alpha^2. \tag{6.3.35}$$

According to (6.3.21), we have

$$\tau^2/4 = 2E\sin^2(\theta/2),$$

so that (6.3.35) becomes

$$(\operatorname{Im}\cos\theta/2)^2 + (\operatorname{Im}\sin\theta/2)^2 < \alpha^2/2E. \tag{6.3.36}$$

Setting

$$\theta = \xi + i\eta, \tag{6.3.37}$$

condition (6.3.36) becomes

$$\sinh^2(\eta/2) < \alpha^2/2E. \tag{6.3.38}$$

In the $\cos\theta$ plane, the boundary of the regularity domain is the curve

$$\begin{gathered}\operatorname{Re}\cos\theta = \cosh\eta_0\cos\xi, \qquad \operatorname{Im}\cos\theta = -\sinh\eta_0\sin\xi,\\ \sinh^2(\eta_0/2) = \alpha^2/2E.\end{gathered} \tag{6.3.39}$$

These are the parametric equations of an ellipse, with foci at $\cos\theta = \pm 1$, and semimajor axis

$$\cosh\eta_0 = 1 + 2\sinh^2(\eta_0/2) = 1 + (\alpha^2/E). \tag{6.3.40}$$

In quantum field theory, Lehmann [17] has also shown that the scattering amplitude is regular in the $\cos\theta$ plane within an ellipse, known as the *Lehmann ellipse*.

Note that, while $f_1(E,\cos\theta)$ is regular within the ellipse (6.3.39), the Born term, according to (6.3.25), is regular for

$$(\operatorname{Im}\sin\theta/2)^2 = \cos^2(\xi/2)\sinh^2(\eta/2) < \alpha^2/8E, \tag{6.3.41}$$

so that we find a smaller analyticity domain for the total amplitude (intersection of the two domains).

### (e) Analyticity Domain for Dispersion Relations

Instead of fixing the value of the momentum transfer and looking for the corresponding analyticity domain in the energy variable, let us look for a domain in the momentum transfer plane such that the corresponding regularity domain in the energy is the whole $E$-plane (apart from the cut and bound-state singularities), as is appropriate for deriving dispersion relations for fixed momentum transfer.

Let us set

$$(2E)^{1/2} = i(\tau/2)\sinh(u+iv), \qquad [2E-(\tau^2/4)]^{1/2} = i(\tau/2)\cosh(u+iv), \tag{6.3.42}$$

$$\tau/2 = \rho e^{i\varphi}. \tag{6.3.43}$$

In terms of these new variables, the domain (6.3.34) is described by the inequalities

$$(\rho/\alpha)\sin\varphi < 1, \tag{6.3.44}$$

$$|\cosh u\cos v\cos\varphi - \sinh u\sin v\sin\varphi| < [(\alpha/\rho)^2 - \sin^2\varphi]^{1/2} + |\sinh u\cos v\cos\varphi - \cosh u\sin v\sin\varphi|. \tag{6.3.45}$$

Squaring both sides of (6.3.45), we get

$$\cos^2 v\cos^2\varphi - \sin^2 v\sin^2\varphi = \cos^2\varphi - \sin^2 v < (\alpha/\rho)^2 - \sin^2\varphi + 2[(\alpha/\rho)^2 - \sin^2\varphi]^{1/2}|\sinh u\cos v\cos\varphi - \cosh u\sin v\sin\varphi|. \tag{6.3.46}$$

It is clear that both inequalities (6.3.44) and (6.3.46) will be satisfied, provided that

$$|\tau/2| = \rho < \alpha. \tag{6.3.47}$$

We conclude that $f_1(E,\tau)$ *is a regular analytic function of both variables in the topological product of the circle* $|\tau| < 2\alpha$ *with the whole E-plane, cut along the positive real E-axis, except possibly for a finite number of poles on the negative real E-axis, associated with bound states.*

This result was originally proved by Khuri [3] only for real values of $\tau$ with $|\tau| < 2\alpha$. We see that for any $E$, real or complex (apart from the singularities), $f_1(E,\tau)$ is also analytic in the momentum transfer within the circle $|\tau| < 2\alpha$ [for $E > 0$, this circle is contained within the ellipse (6.3.39)]. In particular, if (6.3.14) is satisfied for all $\alpha$, i.e., for potentials falling off faster than any exponential at infinity (such as Gaussian or cutoff potentials), the total scattering amplitude [cf. also (6.3.25)] is an entire analytic function of $\tau$.

For cutoff potentials, this agrees with the results obtained in Section 3.4. It was shown there that analyticity in the momentum transfer is an immediate consequence of the cutoff nature of the interaction. For potentials of (at least) exponential decrease, as expressed by condition (6.3.14), analyticity in momentum transfer also follows from the finite range of the interaction, as will be shown in Section 6.6.

## 6.4. High-Energy Behavior of the Scattering Amplitude

### *(a) Green's Function*

In order to apply the results of Section 6.3 to the derivation of dispersion relations for fixed momentum transfer, similar to those of Section 3.4, we must still find the asymptotic behavior of $f(k,\tau)$ as $|k| \to \infty$, $\operatorname{Im} k \geqslant 0$ (corresponding to the physical sheet in the $E$ variable), for fixed $\tau$. As a preliminary step, we investigate the asymptotic behavior of Green's function [2] $G(k,\mathbf{r},\mathbf{r}')$ defined in (6.1.15).

It is actually more convenient to consider the ratio

$$g(k,\mathbf{r},\mathbf{r}') = G(k,\mathbf{r},\mathbf{r}')/G_0(k,\mathbf{r},\mathbf{r}'), \tag{6.4.1}$$

which, according to (6.1.19), satisfies the integral equation

$$g(k,\mathbf{r},\mathbf{r}') = 1 + \int D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho})\, U(\rho) g(k,\boldsymbol{\rho},\mathbf{r}')\, d^3\rho, \tag{6.4.2}$$

where

$$\begin{aligned} D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho}) &= G_0(k,\mathbf{r},\boldsymbol{\rho})\, G_0(k,\boldsymbol{\rho},\mathbf{r}')/G_0(k,\mathbf{r},\mathbf{r}') \\ &= -(1/4\pi)(|\mathbf{r}-\mathbf{r}'|/|\mathbf{r}-\boldsymbol{\rho}|\,|\mathbf{r}'-\boldsymbol{\rho}|)\exp[ik(|\mathbf{r}-\boldsymbol{\rho}|+|\mathbf{r}'-\boldsymbol{\rho}|-|\mathbf{r}-\mathbf{r}'|)]. \end{aligned} \tag{6.4.3}$$

Let us consider the Born expansion (Liouville–Neumann series), corresponding to the iteration of (6.4.2):

$$g_0(k,\mathbf{r},\mathbf{r}') = 1,$$

$$g_{n+1}(k,\mathbf{r},\mathbf{r}') = \int D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho})\, U(\rho) g_n(k,\boldsymbol{\rho},\mathbf{r}')\, d^3\rho, \tag{6.4.4}$$

$$g(k,\mathbf{r},\mathbf{r}') = \sum_{n=0}^{\infty} g_n(k,\mathbf{r},\mathbf{r}'). \tag{6.4.5}$$

In order to investigate the convergence of this series, let us discuss first the term $g_1$,

$$g_1(k,\mathbf{r},\mathbf{r}') = \int D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho})\, U(\rho)\, d^3\rho. \tag{6.4.6}$$

As we are interested in uniform convergence with respect to $\mathbf{r}$ and $\mathbf{r}'$, let us introduce the norm

$$\|g_i(k,\mathbf{r},\mathbf{r}')\| = \sup_{\mathbf{r},\mathbf{r}'} |g_i(k,\mathbf{r},\mathbf{r}')|. \tag{6.4.7}$$

We consider first the case of a continuously differentiable cutoff potential $U_a(r)$, vanishing for $r > a$. It is shown in Appendix G that, in this case,

$$\lim_{|k|\to\infty} \|g_1(k,\mathbf{r},\mathbf{r}')\| = 0 \qquad (\operatorname{Im} k \geqslant 0). \tag{6.4.8}$$

For a potential $U(r)$ that is not cutoff, but that can be approximated by a cutoff potential $U_a(r)$ (in a sense to be made more precise below), we have, by (6.4.6),

$$|g_1(k,\mathbf{r},\mathbf{r}') - g_{1,a}(k,\mathbf{r},\mathbf{r}')| \leqslant \int |D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho})|\, |U(\rho) - U_a(\rho)|\, d^3\rho, \tag{6.4.9}$$

where $g_{1,a}$ denotes the corresponding approximation to $g_1$. By (6.4.3) and the triangle inequality (G4), we have

$$|D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho})| \leqslant \frac{1}{4\pi}\left(\frac{|\mathbf{r}-\boldsymbol{\rho}|+|\mathbf{r}'-\boldsymbol{\rho}|}{|\mathbf{r}-\boldsymbol{\rho}|\,|\mathbf{r}'-\boldsymbol{\rho}|}\right) = \frac{1}{4\pi}\left(\frac{1}{|\mathbf{r}-\boldsymbol{\rho}|}+\frac{1}{|\mathbf{r}'-\boldsymbol{\rho}|}\right) \qquad (\operatorname{Im} k \geqslant 0). \tag{6.4.10}$$

By the well-known expression for the potential of a spherically symmetric distribution, we have

$$\begin{aligned} I_U(r) &= (1/4\pi)\int [|U(\rho)|/|\mathbf{r}-\boldsymbol{\rho}|]\, d^3\rho \\ &= (1/r)\int_0^r |U(\rho)|\,\rho^2\, d\rho + \int_r^\infty |U(\rho)|\,\rho\, d\rho \\ &< M, \end{aligned} \tag{6.4.11}$$

where $M$ is a constant, for any potential satisfying assumptions (6.2.2) and (6.2.24); i.e.,

$$M_1 < \infty, \qquad M_2 < \infty. \tag{6.4.12}$$

Any such potential can be approximated[11] by a continuously differentiable cutoff potential $U_a(r)$ in such a way that, given $\varepsilon > 0$,

$$\int [|U(\rho) - U_a(\rho)|/|\mathbf{r}-\boldsymbol{\rho}|]\, d^3\rho < \varepsilon. \tag{6.4.13}$$

It then follows from (6.4.9) and (6.4.10) that

$$|g_1(k,\mathbf{r},\mathbf{r}') - g_{1,a}(k,\mathbf{r},\mathbf{r}')| < \varepsilon/2\pi, \tag{6.4.14}$$

so that (6.4.8) remains valid for potentials that satisfy (6.4.12).

It can readily be shown from (6.4.4) and (6.4.6) that

$$g_{n+m+1}(k,\mathbf{r},\mathbf{r}') = \int g_n(k,\mathbf{r},\boldsymbol{\rho})\, D_{\mathbf{r},\mathbf{r}'}(k,\boldsymbol{\rho})\, U(\rho)\, g_m(k,\boldsymbol{\rho},\mathbf{r}')\, d^3\rho. \tag{6.4.15}$$

According to (6.4.7), (6.4.10), and (6.4.11), this implies, for $\operatorname{Im} k \geqslant 0$,

$$\begin{aligned} \|g_{n+m+1}\| &\leqslant \|g_n\|\,\|g_m\|\,[I_U(r)+I_U(r')] \\ &\leqslant 2M\,\|g_n\|\,\|g_m\|. \end{aligned} \tag{6.4.16}$$

It follows that

$$\begin{aligned} \|g_{2n}\| &< (2M\,\|g_1\|)^n, \\ \|g_{2n+1}\| &< (2M\,\|g_1\|)^n\,\|g_1\|. \end{aligned} \tag{6.4.17}$$

[11] See, for example, Titchmarsh [18, Chapter X and Sec. 12.2].

By (6.4.8), we can always choose $K$ such that

$$2M\|g_1\| < 1 \qquad (|k| > K,\ \mathrm{Im}\,k \geqslant 0). \tag{6.4.18}$$

It then follows from (6.4.17) that, under these conditions, the Born expansion (6.4.5) is absolutely and uniformly convergent, and

$$\|g\| < (1+\|g_1\|)/(1-2M\|g_1\|) \qquad (|k| > K,\ \mathrm{Im}\,k \geqslant 0). \tag{6.4.19}$$

Thus, *for potentials satisfying (6.4.12), the Born series for Green's function $G(k,\mathbf{r},\mathbf{r}')$ is always absolutely and uniformly convergent for sufficiently large $|k|$ in the closed upper half-plane; we have*

$$\lim_{|k|\to\infty} \|g-g_0\| = 0 \qquad (\mathrm{Im}\,k \geqslant 0); \tag{6.4.20}$$

*i.e., for sufficiently large $|k|$,*

$$|G(k,\mathbf{r},\mathbf{r}') - G_0(k,\mathbf{r},\mathbf{r}')| \leqslant \delta(k)\exp(-\mathrm{Im}\,k|\mathbf{r}-\mathbf{r}'|)/4\pi|\mathbf{r}-\mathbf{r}'|, \tag{6.4.21}$$

*where*

$$\delta(k) \to 0 \qquad \text{uniformly as} \quad |k| \to \infty, \quad \mathrm{Im}\,k \geqslant 0. \tag{6.4.22}$$

For real $k$, we can apply this result also to derive the convergence of the Born expansion for the wave function in the high-energy limit. In fact, by (6.1.16), this expansion is

$$\psi(\mathbf{k},\mathbf{r}) = \sum_{n=0}^{\infty} \psi_n(\mathbf{k},\mathbf{r}), \tag{6.4.23}$$

where

$$\psi_{n+1}(\mathbf{k},\mathbf{r}) = \int G_n(k,\mathbf{r},\mathbf{r}')\,U(r')\,e^{i\mathbf{k}\cdot\mathbf{r}'}\,d^3r', \tag{6.4.24}$$

and (6.4.1) yields, for real $k$,

$$|r\,\psi_{n+1}(\mathbf{k},\mathbf{r})| \leqslant \|g_n\| \int \frac{r}{|\mathbf{r}-\mathbf{r}'|}|U(r')|\,d^3r' = 4\pi\|g_n\|\,rI_U(r). \tag{6.4.25}$$

According to (6.4.11), $r\,I_U(r)$ is bounded for any potential satisfying (6.4.12). It then follows from (6.4.25) and the preceding results that (6.4.23) is also absolutely and uniformly convergent for sufficiently large (real) $k$. Furthermore, $\psi_n = O(r^{-1})$ as $r \to \infty$ for $n \geqslant 1$, so that we can define the scattering amplitude by [cf. (6.2.22)]

$$f(\mathbf{k},\mathbf{k}') = -(1/4\pi)\int e^{-i\mathbf{k}'\cdot\mathbf{r}}\,U(r)\,\psi(\mathbf{k},\mathbf{r})\,d^3r. \tag{6.4.26}$$

Since $\|g_n\| \to 0$ as $k \to \infty$ for $n \geqslant 1$, it follows from (6.4.23) and (6.4.25) that $\psi(\mathbf{k},\mathbf{r}) \to \psi_0(\mathbf{k},\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}$, so that (6.4.26) and (6.3.23) yield

$$\lim_{k\to\infty} f(E,\tau) = f_B(\tau) \tag{6.4.27}$$

for fixed real $\tau$ and for any potential satisfying (6.4.12).

### *(b) The Scattering Amplitude*

We now want to show that, for fixed real $\tau$, (6.4.27) remains valid when $|k| \to \infty$ in $I_+$. For this purpose, we employ the integral representation (6.3.24), which provides a valid analytic continuation in $I_+(k)$. Taking into account (6.2.3) and (6.1.16), we get

$$\begin{aligned} f_1(E,\tau) &= f(E,\tau) - f_B(\tau) \\ &= -(1/4\pi)\int d^3 r \int d^3 r' \\ &\quad \times \exp[-i(\tau/2)\hat{\boldsymbol{\varepsilon}}'\cdot\mathbf{r} - iP\hat{\boldsymbol{\varepsilon}}\cdot\mathbf{r}]\, U(r)\, G(k,\mathbf{r},\mathbf{r}')\, U(r')\exp(i\mathbf{k}\cdot\mathbf{r}'). \end{aligned} \tag{6.4.28}$$

By (6.3.19)–(6.3.21), this can be rewritten as

$$\begin{aligned} f_1(E,\tau) &= -(1/4\pi)\int d^3 r \int d^3 r'\, U(r)\, G(k,\mathbf{r},\mathbf{r}')\, U(r') \\ &\quad \times \exp\{-i(\tau/2)\hat{\boldsymbol{\varepsilon}}'\cdot(\mathbf{r}+\mathbf{r}') - i[k^2-(\tau^2/4)]^{1/2}\,\hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}')\}. \end{aligned} \tag{6.4.29}$$

Writing

$$G = G_0 + \delta G = G_0 + (G - G_0), \tag{6.4.30}$$

we get a corresponding decomposition of (6.4.29),

$$f_1(E,\tau) = f_{1,B}(E,\tau) + \delta f_1(E,\tau), \tag{6.4.31}$$

where $f_{1,B}(E,\tau)$ is the *second Born approximation* and, by (6.4.21), we have, for real $\tau$,

$$\begin{aligned} |\delta f_1(E,\tau)| &\leqslant \frac{\delta(k)}{(4\pi)^2}\int d^3 r \int \frac{d^3 r'}{|\mathbf{r}-\mathbf{r}'|}\,|U(r)\,U(r')| \\ &\quad \times \exp\{\mathrm{Im}[(k^2-\tau^2/4)^{1/2}\,\hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}') - k|\mathbf{r}-\mathbf{r}'|]\}. \end{aligned} \tag{6.4.32}$$

Assuming that the potential satisfies (6.3.14) [note that this implies (6.4.12)], we certainly have, for sufficiently large $|k|$, $\mathrm{Im}\, k \geqslant 0$,

$$\begin{aligned} &\exp(\mathrm{Im}\{[k^2-(\tau^2/4)]^{1/2}\,\hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}') - k|\mathbf{r}-\mathbf{r}'|\}) \\ &\quad = \exp\{-\mathrm{Im}\, k[|\mathbf{r}-\mathbf{r}'| - \hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}')] - \hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}')\,\mathrm{Im}[(\tau^2/8k) + O(k^{-3})]\} \\ &\quad \leqslant \exp[\alpha(r+r')], \end{aligned} \tag{6.4.33}$$

so that (6.4.32) yields

$$|\delta f_1(E,\tau)| \leqslant \frac{\delta(k)}{(4\pi)^2}\int d^3r\,|U(r)|\,e^{\alpha r}\int \frac{d^3r'}{|\mathbf{r}-\mathbf{r}'|}\,|U(r')|\,e^{\alpha r'}. \qquad (6.4.34)$$

We have [cf. (6.4.11)]

$$\frac{1}{4\pi}\int \frac{d^3r'}{|\mathbf{r}-\mathbf{r}'|}\,|U(r')|\,e^{\alpha r'} = \frac{1}{r}\int_0^r e^{\alpha r'}\,|U(r')|\,r'^2\,dr' + \int_r^\infty e^{\alpha r'}\,|U(r')|\,r'\,dr', \qquad (6.4.35)$$

so that the double integral in (6.4.34) is certainly bounded when (6.3.14) is satisfied. It then follows from (6.4.34) and (6.4.22) that

$$\lim_{|k|\to\infty}\delta f_1(E,\tau) = 0 \qquad (\operatorname{Im}k \geqslant 0). \qquad (6.4.36)$$

In order to prove that (6.4.27) remains valid in $I_+$, it therefore suffices to show the vanishing as $|k|\to\infty$ of the second Born approximation, given by

$$f_{1,B}(E,\tau) = \frac{1}{(4\pi)^2}\int d^3r\int\frac{d^3r'}{|\mathbf{r}-\mathbf{r}'|}\,U(r)\,U(r')$$
$$\times\exp\left[-i\frac{\tau}{2}\hat{\boldsymbol{\varepsilon}}'\cdot(\mathbf{r}+\mathbf{r}') + ik|\mathbf{r}-\mathbf{r}'| - i\left(k^2-\frac{\tau^2}{4}\right)^{1/2}\hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}')\right]. \qquad (6.4.37)$$

This can be shown[12] by a procedure similar to that employed in Appendix G. We consider first a continuously differentiable cutoff potential. For sufficiently large $|k|$, we may employ the expansion given in the second line of (6.4.33), and it is advantageous to go over into parabolic coordinates $(\xi,\eta,\varphi)$, with focus at $\mathbf{r}$ and axis along $\hat{\boldsymbol{\varepsilon}}$, so that

$$\begin{aligned}\xi &= |\mathbf{r}-\mathbf{r}'| - \hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}'),\\ \eta &= |\mathbf{r}-\mathbf{r}'| + \hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}'),\end{aligned} \qquad (6.4.38)$$

and $\varphi$ is the azimuthal angle about $\hat{\boldsymbol{\varepsilon}}$. We then have

$$\exp\{ik[|\mathbf{r}-\mathbf{r}'| - \hat{\boldsymbol{\varepsilon}}\cdot(\mathbf{r}-\mathbf{r}')]\}\,d^3r'/|\mathbf{r}-\mathbf{r}'| \to \tfrac{1}{2}e^{ik\xi}\,d\xi\,d\eta\,d\varphi, \qquad (6.4.39)$$

and a partial integration with respect to $\xi$ gives rise to a factor $k^{-1}$, as in (G9), so that one finds $f_{1,B}(E,\tau) = O(k^{-1})$ as $|k|\to\infty$, $\operatorname{Im}k \geqslant 0$. The result can then be extended to a noncutoff potential as in (6.4.13).

In conclusion we find that *the total scattering amplitude, for fixed momentum transfer, approaches the Born approximation as* $|k|\to\infty$, $\operatorname{Im}k\geqslant 0$; i.e.,

$$\lim_{|k|\to\infty} f_1(E,\tau) = \lim_{|k|\to\infty}[f(E,\tau)-f_B(\tau)] = 0 \qquad (\operatorname{Im}k \geqslant 0), \qquad (6.4.40)$$

for any potential satisfying (6.3.14).

[12] Cf. De Alfaro and Regge [19, p. 125].

## 6.5. Dispersion Relations for Fixed Momentum Transfer

### *(a) The Dispersion Relation*

Let $V(r)$ be a potential such that (6.3.14) is satisfied, and let the momentum transfer $\tau$ be fixed at a real value, such that

$$\tau^2 < 4\alpha^2. \tag{6.5.1}$$

Then, according to Sections 6.3 and 6.4, the function

$$f_1(E,\tau) = f(E,\tau) - f_B(\tau)$$

has a regular analytic continuation in the physical sheet of the $E$-plane cut along the positive real axis, except possibly for a finite number of poles $E_1, E_2, \ldots, E_N$ on the negative real axis, associated with bound states; furthermore, by (6.4.40), $f_1(E,\tau) \to 0$ as $|E| \to \infty$.

We can therefore employ the contour of integration shown in Fig. 2.2 and derive a dispersion relation for $f_1(E,\tau)$ by the same procedure employed in the derivation of (2.9.41). The result is

$$f_1(E,\tau) = \frac{1}{\pi}\int_0^\infty [g(E',\tau)/(E'-E)]\,dE' + \sum_{p=1}^{N} [R_p(\tau)/(E-E_p)] \qquad (\operatorname{Im} k > 0), \tag{6.5.2}$$

where

$$\begin{aligned} g(E',\tau) &= (1/2i)[f(E'+i0,\tau) - f(E'-i0,\tau)] \\ &= (1/2i)[f(k',\tau) - f(-k',\tau)] \end{aligned} \tag{6.5.3}$$

and

$$R_p(\tau) = \operatorname{res} f(E,\tau)\Big|_{E=E_p}. \tag{6.5.4}$$

According to (6.3.23), the Born approximation is real for real $\tau$. For real $k$, (6.1.19) and (6.1.3) imply $G(-k,\mathbf{r},\mathbf{r}') = G^*(k,\mathbf{r},\mathbf{r}')$. It then follows from (6.4.29) that

$$f(-k,\tau) = f^*(k,\tau) \qquad (\text{real } k,\tau), \tag{6.5.5}$$

which is the symmetry relation (3.3.7). Thus (6.5.3) becomes

$$g(E',\tau) = \operatorname{Im} f(E'+i0,\tau) \qquad (E' \geqslant 0). \tag{6.5.6}$$

Letting $E$ approach the real axis from above in (6.5.2), we finally get

$$\operatorname{Re} f(E,\tau) = f_B(\tau) + \frac{1}{\pi}\int_0^\infty \frac{\operatorname{Im} f(E'+i0,\tau)}{E'-E}\,dE' + \sum_{p=1}^{N}\frac{R_p(\tau)}{E-E_p}, \tag{6.5.7}$$

which is the *dispersion relation for fixed momentum transfer*, in the form first derived by Khuri [3].

Note that the residues $R_p(\tau)$ are real. In fact, (6.5.5) together with the Schwarz reflection principle implies the reality of $f(E, \tau)$ at any regular point on the negative $E$-axis [cf. (3.4.16)].

### *(b) The Unphysical Region*

The integration over $E'$ in (6.5.7) includes the *unphysical region*

$$0 \leqslant E' < \tau^2/8, \tag{6.5.8}$$

in which there is no scattering angle $\theta$ such that [cf. (6.3.21)] $\tau^2 = 8E' \sin^2(\theta/2)$, so that $f(E', \tau)$ cannot be measured directly in this region [cf. also (3.3.8)]. In order to compare (6.5.7) with experimental results, one must therefore find some prescription for computing $\operatorname{Im} f(E', \tau)$ in the unphysical region in terms of physically accessible quantities.

For this purpose, we consider the partial-wave expansion [cf. (3.1.3) and (3.3.6)],

$$f(k, \cos\theta) = (1/k) \sum_{l=0}^{\infty} (2l+1) f_l(k) P_l(\cos\theta) \qquad (\cos\theta = 1 - (\tau^2/2k^2)). \tag{6.5.9}$$

If this expansion can be employed to compute $\operatorname{Im} f$ in the unphysical region, it provides the desired prescription, since the partial-wave amplitudes (i.e., the corresponding phase shifts) can in principle be measured.

The convergence properties of series of Legendre polynomials are somewhat analogous to those of power series, except that the circle of convergence is replaced by an ellipse of convergence. We have [20, p. 322; 21, p. 243]

THEOREM 6.5.1 (Neumann). *Let $f(z)$ be analytic on the closed segment $[-1, +1]$. Then the expansion of $f(z)$ in a series of Legendre polynomials,*

$$f(z) = \sum_{l=0}^{\infty} a_l P_l(z), \tag{6.5.10}$$

*is convergent in the interior of the largest ellipse with foci at $\pm 1$ in which $f(z)$ is regular; it is uniformly convergent in any domain lying wholly inside this ellipse; it is divergent in the exterior of the ellipse. Furthermore, the sum $R$ of the semi-axes of the ellipse of convergence is given by*

$$1/R = \overline{\lim_{l\to\infty}}\, |a_l|^{1/l}. \tag{6.5.11}$$

The result (6.5.11) is the analog of the Cauchy–Hadamard formula for the radius of convergence of a power series.

Since the Born approximation $f_B(\tau)$ is real for real $\tau$, it follows from (6.3.22) that

$$\operatorname{Im} f(E, \tau) = \operatorname{Im} f_1(E, \tau) \qquad (\tau \text{ real}), \tag{6.5.12}$$

so that it suffices to examine the convergence of the partial-wave series for $f_1(E, \tau)$ in the unphysical region. According to Section 6.3(d), $f_1(E, \cos\theta)$, for $E > 0$, is regular in the $\cos\theta$ plane within an ellipse with foci at $\pm 1$ and semi-major axis given by (6.3.40). It then follows from Theorem 6.5.1 that the partial-wave expansion for $f_1$ is convergent within the same ellipse.

In the $\tau^2$-plane, according to (6.3.21), the major axis of the ellipse extends over the interval

$$-4\alpha^2 \leqslant \tau^2 \leqslant 4\alpha^2 + 8E. \tag{6.5.13}$$

Thus, for $E \geqslant 0$, the partial-wave expansion for $f_1$ converges within the range (6.5.1) for which the dispersion relation (6.5.7) was derived. It follows, in particular, that $\operatorname{Im} f$ *may be computed by means of the partial-wave expansion in the unphysical region.*

According to the discussion at the end of Section 6.3(d), the partial-wave expansion of the total amplitude $f(E, \cos\theta)$ has a smaller domain of convergence than that for $\operatorname{Im} f(E, \cos\theta)$, due to the analytic properties of the Born term.

Consider the function [cf. (6.5.3)]

$$\begin{aligned} g(k, \tau) &= (1/2i)[f(k, \tau) - f(-k, \tau)] \\ &= (1/2i)[f_1(E + i0, \tau) - f_1(E - i0, \tau)] \qquad (\text{real } k), \end{aligned} \tag{6.5.14}$$

which is proportional to the discontinuity of $f_1(E, \tau)$ across the cut on the positive real $E$-axis. For real $\tau$, we have [cf. (6.5.5) and (6.5.6)]

$$g(k, \tau) = \operatorname{Im} f(k, \tau) \qquad (\text{real } \tau), \tag{6.5.15}$$

and it then follows from (6.5.9) and the unitarity condition (3.1.5) that $g(k, \tau)$ has the partial-wave expansion

$$g(k, \tau) = (1/k) \sum_{l=0}^{\infty} (2l+1) |f_l(k)|^2 P_l(\cos\theta) \qquad (\cos\theta = 1 - (\tau^2/2k^2)). \tag{6.5.16}$$

This function is known as the *absorptive part* [cf. Section 1.6(b)].

According to (6.5.14) and Theorem 6.5.1, this expansion is convergent within the ellipse (6.3.39) where $f_1(E, \cos\theta)$ is regular. The sum $R$ of the semiaxes of this ellipse is, by (6.3.40),

$$R = \cosh\eta_0 + \sinh\eta_0 = e^{\eta_0} = 1 + (\alpha^2/E) + \{(\alpha^2/E)[2 + (\alpha^2/E)]\}^{1/2}. \tag{6.5.17}$$

It then follows from (6.5.11) and (6.5.16) that, given any $\varepsilon > 0$ (no matter how small), there exists an integer $l_0$ such that

$$|a_l| = [(2l+1)/|k|]\,|f_l(k)|^2 < \exp[-l(\log R - \varepsilon)]$$
$$= \exp[-l(\eta_0 - \varepsilon)] \qquad (l \geqslant l_0). \tag{6.5.18}$$

Thus the partial-wave amplitudes (as well as the corresponding phase shifts) tend to zero at least exponentially with $l$ as $l \to \infty$ [and faster than any exponential if (6.3.14) holds for all $\alpha$, e.g., for a cutoff potential]. This agrees with the results of Section 5.8(b).

In particular, at high energies, when $E = k^2/2 \gg \alpha^2$, (6.5.17) yields

$$\exp \eta_0 \approx 1 + 2\alpha/k \approx 1 + \eta_0,$$

so that (6.5.18) becomes

$$|f_l(k)| = |\sin \eta_l| = O(l^{-1/2} \exp(-\alpha p_l)), \tag{6.5.19}$$

where $p_l = l/k$ is the "impact parameter" associated with the $l$th partial wave, which is meaningful under the above conditions (de Broglie wavelength much smaller than the range of the potential). This agrees with the results (5.8.29) and (5.8.31) found for Yukawa-type potentials.

Conversely, these results on the high angular momentum limit, according to Theorem 6.5.1, entail the analytic properties of the scattering amplitude in the $\cos\theta$ plane. This confirms the link established in Section 3.4(b) between the analyticity in momentum transfer and the finite range of the interaction (cf. Section 6.6).

### *(c) Connection with Causality*

Let $V(r)$ be a cutoff potential of radius $a$, so that (6.3.14) holds for all $\alpha$. According to (6.5.1), the dispersion relation for fixed momentum transfer is then valid for all $\tau$. The same result was derived in Section 3.4 from general physical assumptions about the interaction, the most important one being causality. Let us now establish the connection between the two treatments by showing that all the assumptions made in Section 3.4 are verified within the present model.

The interaction obviously satisfies the assumptions of Section 2.7, as well as the symmetry relation (3.3.7). It was shown in Section 5.8(b) that Assumptions I and II of Section 3.3(c) are also satisfied. Actually, Assumption II can be replaced by the requirement that the bounds (3.3.58) must hold, and the results of Section 6.4(a) imply the validity of even stronger bounds, leading ' (6.4.27) instead of (3.3.46) (the weaker bounds are only necessary if we wa. to include more singular interactions, such as a hard sphere).

There remains to verify the validity of the crucial assumption: the causality condition of Section 3.4(a). For this purpose we note that, according to (3.4.1), (3.4.11), and (6.1.16), we have

$$\mathscr{G}(r, \theta, E) = (1/2\pi) \int_{r' \leqslant a} G(k, \mathbf{r}, \mathbf{r}')\, U(r')\, e^{-ik(z-z')}\, d^3 r' \qquad (E > 0), \tag{6.5.20}$$

where $\mathscr{G}(r, \theta, E)$ is defined by (3.4.8), and

$$z = r\cos\theta, \qquad z' = r'\cos\theta'. \tag{6.5.21}$$

It follows from the analytic properties of the resolvent operator found in Section 6.3(a) that Green's function $G(k, \mathbf{r}, \mathbf{r}')$ has a regular analytic continuation in $I_+(k)$, except possibly for a finite number of simple poles on the positive imaginary axis, associated with bound states. Since all the limits of integration in (6.5.20) are finite, this implies [22, p. 257] that $\mathscr{G}(r, \theta, E)$ *also has a regular analytic continuation in the physical sheet of the E-plane, except possibly for a finite number of simple poles on the negative real axis, corresponding to bound states.*

If $E_p$ is a bound-state pole, we have, as $E$ approaches $E_p$ from above [cf. (A11.5)],

$$1/(E - E_p) \to 1/(E - E_p + i0) = -2i\pi\delta^+(E - E_p), \tag{6.5.22}$$

which is a temperate distribution. Thus the temperateness assumption (3.4.9) is also valid.

Let

$$k = k' + i\kappa \qquad (\kappa \geqslant 0). \tag{6.5.23}$$

Then, according to (6.4.21) and (6.1.3), we have, for sufficiently large $|k|$ in $I_+$,

$$|G(k, \mathbf{r}, \mathbf{r}')| \leqslant \frac{1 + \delta(k)}{4\pi} \frac{\exp(-\kappa|\mathbf{r} - \mathbf{r}'|)}{|\mathbf{r} - \mathbf{r}'|} \qquad (\kappa \geqslant 0), \tag{6.5.24}$$

where $\delta(k)$ satisfies (6.4.22).

Substituting in (6.5.20), we find

$$|\mathscr{G}(r, \theta, E)| < \frac{1 + \delta(k)}{8\pi^2} \int_{r' \leqslant a} U(r') \frac{\exp[-\kappa(|\mathbf{r} - \mathbf{r}'| - z + z')]}{|\mathbf{r} - \mathbf{r}'|}\, d^3 r'. \tag{6.5.25}$$

Since

$$|\mathbf{r} - \mathbf{r}'| \geqslant z - z', \tag{6.5.26}$$

we find, for $r \geqslant a$, with the help of (6.2.8),

$$|\mathscr{G}(r,\theta,E)| \leqslant \frac{1+\delta(k)}{8\pi^2}\int_{r'\leqslant a}\frac{|U(r')|}{|\mathbf{r}-\mathbf{r}'|}\,d^3r' = \frac{1+\delta(k)}{2\pi r}\int_0^r |U(r')|\,r'^2\,dr'$$

$$\leqslant \frac{1+\delta(k)}{2\pi a}\int_0^a |U(r')|\,r'^2\,dr' \qquad (r \geqslant a,\ \kappa \geqslant 0). \qquad (6.5.27)$$

It follows from (6.5.27), (6.4.22), and (6.2.2) that $|\mathscr{G}(r,\theta,E)|$ *remains bounded as* $|E| \to \infty$. Thus $\mathscr{G}(r,\theta,E)$ satisfies conditions (a)–(c) of Theorem 1.8.2. It follows that its Fourier transform $g(r,\theta,\tau)$, defined by (3.4.8), belongs to $\mathscr{D}_+' \cap \mathscr{S}'$, so that [cf. (3.4.7)] *the causality condition is satisfied.* This completes the proof that all the assumptions employed in Section 3.4 to derive the dispersion relation for fixed momentum transfer are valid in the present model.

Note that the equality sign in (6.5.26) can apply only to points located within the geometrical shadow of the scatterer. This may be regarded as a counterpart of the fact that, for a classical field [cf. Section 3.3(d)], no stronger form of the causality condition could be applied to such points.

## 6.6. Analyticity in Momentum Transfer and Finite Range of the Interaction

### *(a) Cook's Inequality*

We have seen in Section 3.4(b), for cutoff interactions, that analyticity in the momentum transfer is linked with the short-range character of the interaction. The discussion at the end of Section 6.5(b) suggests that this is also true for potentials with exponential tails; the analyticity domain in the momentum transfer plane obtained in Section 6.3(d) is determined by the inverse range of the exponential fall-off of the potential. In terms of the partial-wave amplitudes, this corresponds to the exponential fall-off with the impact parameter found in Section 6.5(b).

Omnès [23] and Kugler and Roskies [24; 24a, p. 187] have shown that the finite range of the interaction can be expressed directly as a condition on the scattering of wave packets, in the same spirit as the causality condition of Section 3.4, and that it can then be employed to derive analytic properties in the momentum transfer. To formulate the finite-range condition, the model of scattering by a potential of (at least) exponential decrease is employed, but the condition thus obtained may be regarded as a new physical assumption about the interaction, without specific reference to the potential model.

Intuitively, one might expect that the probability of scattering of a wave packet from an interaction with an exponential tail would fall off exponentially with the impact parameter. However, one must be careful in selecting the shape of the incident wave packet, due to the spreading effect. Thus, if the incident wave packet were infinitely sharply localized at $t=0$, it would spread over all space immediately thereafter, and the effect of the initial impact parameter would become completely washed out by the spreading.

The norm of the scattered wave packet for large times may be taken as a measure of the probability of scattering. In potential scattering this norm is related to the incident wave packet and to the potential by an inequality due to Cook [25].[13] Let $\psi_{\text{inc}}(\mathbf{r},t)$ be the incident wave packet and $\psi_s(\mathbf{r},t)$ the corresponding scattered wave packet, as in Section 3.4, so that the total wave function is

$$\psi(\mathbf{r},t) = \psi_{\text{inc}}(\mathbf{r},t) + \psi_s(\mathbf{r},t). \tag{6.6.1}$$

Let

$$H = H_0 + V \tag{6.6.2}$$

be the total Hamiltonian, where $H_0$ is the free-particle Hamiltonian and $V$ is the potential. We then have (taking $\hbar = 1$)

$$i\,\partial\psi/\partial t = H\psi, \tag{6.6.3}$$

$$i\,\partial\psi_{\text{inc}}/\partial t = H_0\psi_{\text{inc}}. \tag{6.6.4}$$

The norm of the scattered wave packet at time $t$ is given by

$$\|\psi_s\|^2 = \int |\psi_s(\mathbf{r},t)|^2\, d^3r. \tag{6.6.5}$$

It follows from (6.6.5), (6.6.3), and (6.6.4) that

$$\begin{aligned}\frac{\partial}{\partial t}(\|\psi_s\|^2) &= i\int [(H\psi_s + V\psi_{\text{inc}})^*\psi_s - \psi_s{}^*(H\psi_s + V\psi_{\text{inc}})]\, d^3r \\ &= 2\,\mathrm{Im}(\psi_s, V\psi_{\text{inc}}) \\ &\leqslant 2\,\|\psi_s\|\,\|V\psi_{\text{inc}}\|\end{aligned}$$

by Schwarz's inequality. Therefore we have

$$\frac{\partial}{\partial t}\|\psi_s\| \leqslant \|V\psi_{\text{inc}}\|. \tag{6.6.6}$$

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[13] Cf. also Brenig and Haag [26].

Integrating both sides over time from $-\infty$ to $t$, and remembering that the scattered wave packet vanishes as $t \to -\infty$ for the class of potentials under consideration,[14] we find

$$\|\psi_s\| \leqslant \int_{-\infty}^{t} \|V\psi_{\text{inc}}\|\, dt,$$

and, consequently, at any time $t$,

$$\|\psi_s\| \leqslant \int_{-\infty}^{\infty} \|V\psi_{\text{inc}}\|\, dt, \tag{6.6.7}$$

which is Cook's inequality. For large $t$, $\|\psi_s\|^2$ is a measure of the scattering probability, and we may employ (6.6.7) to get an upper bound for this quantity.

### (b) The Finite-Range Condition

We choose a Gaussian incident wave packet with mean momentum $\mathbf{k}$, given by the wave function

$$\tilde{\psi}_{\text{inc}}(\mathbf{p}) = (b^2/\pi)^{3/4} \exp[-(\mathbf{p}-\mathbf{k})^2(b^2/2) - i\mathbf{p}\cdot\mathbf{a}] \tag{6.6.8}$$

in momentum space [cf. (E7)]. The corresponding position probability density is [15]

$$|\psi_{\text{inc}}(r,t)|^2 = [1/\pi b^2(t)]^{3/2} \exp\{-[\mathbf{r}-\bar{\mathbf{r}}(t)]^2/b^2(t)\}, \tag{6.6.9}$$

where (taking also the particle mass $= 1$)

$$\bar{\mathbf{r}}(t) = \mathbf{a} + \mathbf{k}t, \tag{6.6.10}$$

$$b^2(t) = b^2 + (t^2/b^2). \tag{6.6.11}$$

The wave packet (6.6.9) has the width $b$ at $t = 0$, and its center moves along the line $\bar{\mathbf{r}}(t)$, corresponding to an impact parameter $a = |\mathbf{a}|$. We take

$$\mathbf{k}\cdot\mathbf{a} = 0, \tag{6.6.12}$$

so that the closest approach to the center is at $t = 0$. The width of the wave packet at time $t$ is given by $b(t)$, which increases linearly with $t$ for $t \gg t_s$, where

$$t_s = b^2 \tag{6.6.13}$$

is the spreading time [cf. (E3)].

We want to find an upper bound for the right-hand side of (6.6.7), for a potential $V(r)$ such that

$$|V(r)| < Ce^{-mr}/r \qquad (m > 0). \tag{6.6.14}$$

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[14] For a rigorous discussion of this point, cf. Cook [25].

[15] Cf. Schiff [27, p. 58].

We have

$$\|V\psi_{\text{inc}}\|^2 = \int |V(r)|^2 \, |\psi_{\text{inc}}(\mathbf{r},t)|^2 \, d^3 r, \tag{6.6.15}$$

and, according to (6.6.14), the contribution from the potential is largest near $r=0$.

For $t \gg t_s$, $|\mathbf{k}|\, t \gg |\mathbf{a}|$, it follows from (6.6.9) [cf. also (E9)] that

$$|\psi_{\text{inc}}(0,t)|^2 \approx (b/\sqrt{\pi}\, t)^3 \exp(-k^2 b^2), \tag{6.6.16}$$

so that, if the initial width $b$ of the incident wave packet is always chosen to be the same, regardless of the impact parameter $a$, the dependence of the overlap integral (6.6.15) on the impact parameter is eventually wiped out by the spreading effect. In order that the probability of scattering (as measured by $\|\psi_s\|^2$ for large times) be an exponentially decreasing function of the impact parameter, we must choose, according to (6.6.7) and (6.6.14)–(6.6.16),

$$b^2 = \lambda a; \tag{6.6.17}$$

i.e., *the width of the incident wave packet must increase like the square root of the impact parameter*. Since $b$ only increases like $\sqrt{a}$, the bulk of the incident packet still lies outside the range of the potential for sufficiently large $a$. However, the increase in width offsets the effects of spreading, and we now show that the choice (6.6.17) indeed leads to an exponential fall-off of the scattering probability with $a$.

To majorize (6.6.15), we split the integral in the following way:

$$\int d^3 r = \int_{r<\rho} d^3 r + \int_{r>\rho} d^3 r \qquad (\rho < a). \tag{6.6.18}$$

According to (6.6.9), we always have

$$|\psi_{\text{inc}}(\mathbf{r},t)|^2 \leqslant [1/\pi b^2(t)]^{3/2}. \tag{6.6.19}$$

Using this inequality, together with (6.6.14), we get

$$\int_{r>\rho} |V(r)|^2 \, |\psi_{\text{inc}}(\mathbf{r},t)|^2 \, d^3 r < (2C^2/\sqrt{\pi}\, m)[e^{-2m\rho}/b^3(t)]. \tag{6.6.20}$$

For $r<\rho$, we have, with the help of (6.6.14) and (6.6.9),

$$\begin{aligned}\int_{r<\rho} |V(r)|^2 \, |\psi_{\text{inc}}(\mathbf{r},t)|^2 \, d^3 r &< \max_{r<\rho} |\psi_{\text{inc}}(\mathbf{r},t)|^2 \cdot 2\pi C^2 \frac{(1-e^{-2m\rho})}{m} \\ &< \frac{2C^2}{\sqrt{\pi}\, m b^3(t)} \exp\left\{-\min_{r<\rho} [\mathbf{r}-\bar{\mathbf{r}}(t)]^2/b^2(t)\right\}.\end{aligned} \tag{6.6.21}$$

Since we have taken $\rho < a$, the minimum of $[\mathbf{r}-\bar{\mathbf{r}}(t)]^2$ for $r < \rho$ is reached for $\mathbf{r}/\!/\bar{\mathbf{r}}$ and $r \to \rho$, so that, finally, taking into account (6.6.10) and (6.6.12), (6.6.21) becomes

$$\int_{r<\rho} |V(r)|^2 \, |\psi_{\text{inc}}(\mathbf{r},t)|^2 \, d^3r < \frac{2C^2}{\sqrt{\pi}\, m\, b^3(t)} \exp\left\{ -\frac{[(a^2+k^2t^2)^{1/2}-\rho]^2}{b^2(t)} \right\}. \tag{6.6.22}$$

The exponent in (6.6.22) varies from $(a-\rho)^2/\lambda a$ to $k^2\lambda a$ as $t$ varies from 0 to $\pm\infty$, where we have made use of (6.6.17). If we choose

$$k\lambda \geqslant 1, \tag{6.6.23}$$

the exponent takes its smallest value for $t=0$, so that, finally, combining (6.6.20) with (6.6.22), we get from (6.6.15)

$$\|V\psi_{\text{inc}}\|^2 < \frac{2C^2}{\sqrt{\pi}\, m\, b^3(t)} \{\exp(-2m\rho) + \exp[-(a-\rho)^2/\lambda a]\}. \tag{6.6.24}$$

The best bound is obtained when the two exponentials are equal, i.e., when (remembering that $\rho < a$)

$$\rho = a\{1 + \lambda m - [(1+\lambda m)^2 - 1]^{1/2}\} = a\sigma(\lambda, m)/m. \tag{6.6.25}$$

With this choice for $\rho$, (6.6.24) yields

$$\|V\psi_{\text{inc}}\| < \frac{2C}{\pi^{1/4}\sqrt{m}\,[b(t)]^{3/2}} e^{-\sigma(\lambda,m)a}. \tag{6.6.26}$$

Substituting this result in Cook's inequality (6.6.7), and integrating over time [with $b(t)$ given by (6.6.11)], we finally get

$$\|\psi_s\| < K(C/\sqrt{m})(\lambda a)^{1/4} e^{-\sigma(\lambda,m)a}, \tag{6.6.27}$$

where $K$ is a numerical constant, $\sigma(\lambda, m)$ is defined by (6.6.25), and $\lambda$ is restricted by (6.6.23). Thus, with the choice (6.6.17), *the scattering probability decreases exponentially with $a$*, as we wanted to show. Note also from (6.6.25) that $\sigma \approx m$ for small $\lambda$, in which case we get the "intuitive" rate of decrease proportional to $e^{-ma}$.

The fact that the scattering probability for the incident wave packet (6.6.8) and (6.6.17), as measured by $\|\psi_s\|^2$ for large $t$, is proportional to $\exp(-2\sigma a)$ can now be taken as characterizing a *finite-range interaction* (of inverse range $m$), whether or not the interaction can be described by a potential field. A more precise formulation of this condition will be given below.

### *(c) Analyticity in Momentum Transfer*

To derive the implications of the finite range of the interaction on the analytic properties in the momentum transfer, we start by finding the relation between the probability of scattering and the scattering amplitude.

The incident wave packet associated with (6.6.8) is

$$\psi_{\text{inc}}(\mathbf{r},t) = \frac{1}{(2\pi)^{3/2}} \int \tilde{\psi}_{\text{inc}}(\mathbf{p}) \exp(i\mathbf{p}\cdot\mathbf{r} - \tfrac{1}{2}ip^2 t)\, d^3p. \tag{6.6.28}$$

As in (3.4.1), let

$$\psi_p(\mathbf{r},t) = [\exp(i\mathbf{p}\cdot\mathbf{r}) + u_s(\mathbf{p},\mathbf{r})]\exp(-\tfrac{1}{2}ip^2 t) \tag{6.6.29}$$

be the stationary solution corresponding to the energy $E = p^2/2$; for $r \to \infty$ in the direction $\hat{\mathbf{r}}$, we have, as in (6.1.9),

$$u_s(\mathbf{p},\mathbf{r}) \to f(p, \hat{\mathbf{p}}\cdot\hat{\mathbf{r}})\, e^{ipr}/r \qquad (r \to \infty). \tag{6.6.30}$$

The scattered wave packet that corresponds to (6.6.28) is [cf. (3.4.4)]

$$\psi_s(\mathbf{r},t) = (2\pi)^{-3/2} \int \tilde{\psi}_{\text{inc}}(\mathbf{p})\, u_s(\mathbf{p},\mathbf{r}) \exp(-\tfrac{1}{2}ip^2 t)\, d^3p. \tag{6.6.31}$$

We want to evaluate (6.6.5) in the limit of very large $t$. One readily finds by the method of stationary phase, as in Section 2.11(c), that, for the incident wave packet (6.6.8), the scattered wave packet (6.6.31), for large $t$, is concentrated[16] around the sphere $r \approx kt$. Thus the dominant contribution to the integral in (6.6.5) for large $t$ arises from large values of $r$, for which $u_s(\mathbf{p},\mathbf{r})$ in (6.6.31) may be replaced by (6.6.30). The integration over $r$ yields a delta function, leading to

$$\int |\psi_s(\mathbf{r},t)|^2\, d^3r \to (2\pi)^{-2} \int d^3p \int d^3p'\, \tilde{\psi}^*_{\text{inc}}(\mathbf{p}')\tilde{\psi}_{\text{inc}}(\mathbf{p})\,\delta(p-p')$$
$$\times \int f^*(p', \hat{\mathbf{p}}'\cdot\hat{\mathbf{r}}) f(p, \hat{\mathbf{p}}\cdot\hat{\mathbf{r}})\, d\Omega_r \qquad (t \to \infty). \tag{6.6.32}$$

In Chapter 8, Section 8.2, we derive the generalized unitarity condition (8.2.4), that enables us to perform the integration over solid angles in (6.6.32), yielding

$$\int |\psi_s(\mathbf{r},t)|^2\, d^3r \to (1/\pi) \int_0^\infty p^2\, dp \int d\Omega_p \int d\Omega_{p'}\, \tilde{\psi}^*_{\text{inc}}(p,\hat{\mathbf{p}}')\tilde{\psi}_{\text{inc}}(p,\hat{\mathbf{p}})$$
$$\times\, p\, \mathrm{Im}\, f(p, \hat{\mathbf{p}}\cdot\hat{\mathbf{p}}') \qquad (t \to \infty). \tag{6.6.33}$$

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[16] Cf. Low [28] and Goldberger and Watson [29, p. 100].

The partial-wave expansion (6.5.16) for the absorptive part, together with the addition theorem for spherical harmonics, yield

$$p \operatorname{Im} f(p, \hat{\mathbf{p}} \cdot \hat{\mathbf{p}}') = 4\pi \sum_{l=0}^{\infty} \operatorname{Im} f_l(p) \sum_{m=-l}^{l} Y_{lm}^*(\hat{\mathbf{p}})\, Y_{lm}(\hat{\mathbf{p}}'), \tag{6.6.34}$$

where $Y_{lm}$ is a normalized spherical harmonic. Substituting (6.6.34) in (6.6.33), we finally get

$$\|\psi_s\|^2 \to 4 \sum_{l=0}^{\infty} \int_0^{\infty} A_l(p) \operatorname{Im} f_l(p) p^2 \, dp, \tag{6.6.35}$$

where

$$A_l(p) = \sum_{m=-l}^{l} |\tilde{\psi}_{lm}(p)|^2, \tag{6.6.36}$$

and

$$\tilde{\psi}_{lm}(p) = \int \tilde{\psi}_{\text{inc}}(\mathbf{p})\, Y_{lm}^*(\hat{\mathbf{p}})\, d\Omega_p \tag{6.6.37}$$

is the expansion coefficient in the expansion of the incident wave packet (6.6.8) into spherical harmonics.

For large $a$, according to (6.6.8) and (6.6.17), the only significant contribution to the integral in (6.6.35) comes from the neighborhood of $p = k$. Assuming a smooth dependence of $\operatorname{Im} f_l(p)$ on $p$, we can then replace it by $\operatorname{Im} f_l(k)$ and take it out of the integral. The precise form of the finite-range condition [24] explicitly assumes it to be valid for each component of the absorptive part corresponding to a fixed energy.

To evaluate (6.6.37) for $|\mathbf{p}| = |\mathbf{k}|$, it is convenient to choose

$$\lambda = 1/k, \tag{6.6.38}$$

which corresponds to the minimum wave packet width (6.6.17) compatible with (6.6.23) [choosing the minimum possible $\lambda$ is desirable because it maximizes $\sigma$ in (6.6.25)]. The result is [24]

$$A_l(k) = 4\pi (a/\pi k)^{3/2} [(2ka)^{2l}/(2l)!]\, e^{-2ka}. \tag{6.6.39}$$

Since each term in the sum in (6.6.35) is positive, the finite-range condition is valid a fortiori if we replace the sum by a single term, leading to [cf. (6.6.27)]

$$A_l(k) \operatorname{Im} f_l(k) < C' \exp[-2\sigma(\lambda = 1/k, m)\, a], \tag{6.6.40}$$

where $C'$ is a constant (independent of $l$). To get the best possible bound for $\operatorname{Im} f_l(k)$ for large $l$, we want to choose $a$ so as to maximize $A_l(k)$. It is readily found from (6.6.39) that, for large $l$, $A_l(k)$ is maximized when

$$a \approx l/k, \tag{6.6.41}$$

in agreement with the relation (3.3.26) between the impact parameter and the partial-wave decomposition. The corresponding maximum value of $A_l$ is $A_l \approx 2a/k^2$, so that (6.6.40) finally becomes

$$\operatorname{Im} f_l(k) = |f_l(k)|^2 < C'' \exp(-2\sigma_0 l/k) \tag{6.6.42}$$

for large $l$, where, according to (6.6.25),

$$\sigma_0 = \sigma\left(\frac{1}{k}, m\right) = m\left\{1 + \frac{m}{k} - \left[\left(1 + \frac{m}{k}\right)^2 - 1\right]^{1/2}\right\}. \tag{6.6.43}$$

For large $k$, we have $\sigma_0 \approx m$, and (6.6.43) agrees with the bound (6.5.19). It follows from (6.6.42), (6.5.16), and Theorem 6.5.1 that the *absorptive part* $g(k, \cos\theta)$ *is holomorphic in* $\cos\theta$ *within an ellipse with foci at* $\pm 1$ *and semimajor axis* [cf. (6.3.39) and (6.5.17)]

$$\cosh\eta_0 = \tfrac{1}{2}[\exp(2\sigma_0/k) + \exp(-2\sigma_0/k)]. \tag{6.6.44}$$

For large $k$, this becomes

$$\cosh\eta_0 \approx 1 + (2m^2/k^2), \tag{6.6.45}$$

in precise agreement with (6.3.40). For small values of $k$, the present ellipse is contained within that found in Section 6.5(b), so that we obtain only a part of the analyticity domain. This is not surprising, in view of the many restrictive choices made in the selection of the incident wave packet.

For the total scattering amplitude (6.5.9), the bound (6.6.42) leads to analyticity within an ellipse with semimajor axis

$$\tfrac{1}{2}[\exp(\sigma_0/k) + \exp(-\sigma_0/k)], \tag{6.6.46}$$

which approaches $1 + (m^2/2k^2)$ for large $k$, corresponding to the boundary of the regularity domain (6.3.41) for the Born term. The Born term, as remarked in Section 6.5(b), is responsible for the smaller analyticity domain of the total amplitude.

According to Theorem 6.5.1, we can also reverse the above proofs, deriving the bound (6.6.42) from the analyticity domain, and going back from (6.6.42) to the short-range condition. Thus *analyticity and short-range condition are equivalent.*

Different forms of the short-range condition, related with the "interaction time" of Section 2.11(c), have also been considered [30], leading to somewhat larger analyticity domains for suitable choices of the incident wave packet. The results have also been extended to quantum field theory [31].

In conclusion, we see that *analyticity in the energy is connected with causality, whereas analyticity in the momentum transfer is connected with the finite range of the interaction.*

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CHAPTER

# 7

# REGGE POLES

*They all went off to discover the Pole, ...*
*It's a Thing you Discover, as I've been tole*

A. A. MILNE[1]

## 7.1. Introduction

The results obtained in the preceding chapter about the analyticity domain of the total scattering amplitude represent about as far as one can go for general potentials of (at least) exponential decrease at infinity, i.e., those that satisfy condition (6.3.14). The most unsatisfactory feature of these results is that the analyticity domain in the momentum transfer variable is restricted to the ellipse of Section 6.3(d).

In order to remove this restriction, we have to make more specific assumptions about the potential. One possibility is to restrict oneself to cutoff potentials, for which (6.3.14) is valid for all $\alpha$, so that the scattering amplitude is an entire function of the momentum transfer. This class of potentials will be discussed at the end of Chapter 8.

Another possibility, which will be adopted throughout most of this chapter, is to restrict our consideration to *Yukawa-type potentials* [cf. (5.4.29)],

$$U(r) = \int_m^\infty \sigma(\mu)\, e^{-\mu r}\, d\mu/r \qquad (m > 0). \tag{7.1.1}$$

[1] "Winnie-the-Pooh," Chapter VIII.

It was shown in Section 5.4(c) that, for such potentials, the partial-wave amplitudes can be analytically continued beyond the analyticity strip found for general potentials of exponential decrease into the whole $k$-plane with cuts along the imaginary axis from $im/2$ to $i\infty$ and from $-im/2$ to $-i\infty$.

It turns out, similarly, that the total scattering amplitude for Yukawa-type potentials can be analytically continued in the momentum transfer variable beyond the ellipse of Section 6.3(d) into the whole plane with suitable cuts. This is related to the fact, already remarked in Section 5.4(c), that potentials of the class (7.1.1) can be analytically continued to complex values of $r$. They have been the object of considerable interest, in view of their relation with the nonrelativistic limit of field theory.

These results enable one to derive a dispersion relation in the momentum transfer variable, which, together with the dispersion relation in the energy (for fixed momentum transfer) obtained in Section 6.5(a), leads to a double dispersion relation for the total scattering amplitude, known as the *Mandelstam representation*.

Specifically, we want to derive a dispersion relation in the momentum transfer for the absorptive part (6.5.14) which, according to (6.5.15), appears in the integrand of the dispersion relation for fixed momentum transfer (6.5.7). For this purpose we need: (i) analytic continuation in the momentum transfer beyond the ellipse of convergence of the partial-wave expansion; (ii) asymptotic behavior of the function at infinity in the momentum transfer plane.

Both (i) and (ii) follow from a transformation of the partial-wave expansion known as the *Watson transformation* [1]. It is based upon a well-known method for the summation of a series by transforming it into a contour integral. The function $\pi/\cos(\pi\lambda)$ has simple poles at the half-integers $\lambda = l+\frac{1}{2}$, with residues $(-1)^{l+1}$. It follows that

$$\sum_{l=0}^{\infty}(-1)^l f(l+\tfrac{1}{2}) = \frac{1}{2i}\int_C f(\lambda)\,\frac{d\lambda}{\cos(\pi\lambda)}, \tag{7.1.2}$$

where $C$ is the contour shown in Fig. 7.1, enclosing the positive half-integers in the clockwise sense, and $f(\lambda)$, the "interpolating function," is a regular analytic function in a neighborhood of the real positive axis, that reduces to $f(l+\frac{1}{2})$ at the half-integral points. These requirements by no means determine $f(\lambda)$ uniquely; given a function $f(\lambda)$ verifying them, there is a wide class of other functions with the same properties (e.g., $\exp[2in\pi(\lambda-\frac{1}{2})]f(\lambda)$, where $n$ is any integer).

However, the usefulness of (7.1.2) lies in the possibility of deforming the contour $C$ away from the neighborhood of the real axis. In particular, for the Watson transformation, one wants to deform it into the straight line $D$ along the imaginary axis (Fig. 7.1), so that $f(\lambda)$ must have a single-valued analytic

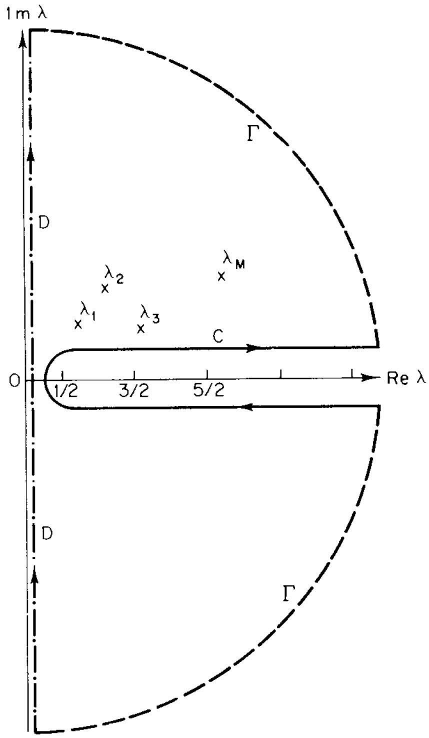


*FIG. 7.1.* Contours of integration for the Watson transformation (×, Regge poles).

continuation in the right half of the $\lambda$-plane, with growth properties as $|\lambda| \to \infty$ such that the contribution from the half-circle $\Gamma$ at infinity can be neglected. Furthermore, we want $f(\lambda)$ to have no singularities in the right half-plane other than a finite number of poles $\lambda_j$ (Fig. 7.1). As will be shown in Section 7.5(b), it turns out, for the partial-wave expansion, that these additional constraints specify $f(\lambda)$ uniquely.

Assuming the existence of $f(\lambda)$ with the above properties, the Watson transformation leads to

$$\sum_{l=0}^{\infty} (-1)^l f(l+\tfrac{1}{2}) = \frac{1}{2i} \int_{-i\infty}^{i\infty} f(\lambda) \frac{d\lambda}{\cos(\pi\lambda)} + \pi \sum_{j=1}^{M} \frac{r_j}{\cos(\pi\lambda_j)}, \tag{7.1.3}$$

where

$$r_j = \operatorname{res} f(\lambda)\Big|_{\lambda_j}. \tag{7.1.4}$$

Applying (7.1.3) to the partial-wave expansion (3.3.6) and (3.1.4), and noting that

$$(-1)^l P_l(z) = P_l(-z), \tag{7.1.5}$$

we get

$$f(k,\tau) = \sum_{l=0}^{\infty} \frac{(-1)^l}{ik}(l+\tfrac{1}{2})[S_l(k)-1]P_l(-\cos\theta)$$

$$= -\frac{1}{2k}\int_{-i\infty}^{i\infty}[S(\lambda,k)-1]P_{\lambda-\frac{1}{2}}(-\cos\theta)\frac{\lambda\,d\lambda}{\cos(\pi\lambda)}$$

$$-\frac{i\pi}{k}\sum_{j=1}^{M}\frac{\sigma_j\lambda_j}{\cos(\pi\lambda_j)}P_{\lambda_j-\frac{1}{2}}(-\cos\theta) \qquad \left(\cos\theta = 1-\frac{\tau^2}{2k^2}\right), \tag{7.1.6}$$

where

$$\sigma_j = \operatorname{res} S(\lambda,k)\Big|_{\lambda_j}. \tag{7.1.7}$$

In (7.1.6), $P_{\lambda-\frac{1}{2}}(z)$ is Legendre's function of the first kind, and $S(\lambda,k)$ is the interpolating function for $S_l(k)$, with the property

$$S(\lambda = l+\tfrac{1}{2},k) = S_l(k) \qquad (l = 0,1,2,\ldots). \tag{7.1.8}$$

The last sum in (7.1.6) is extended over all the poles $\lambda_j(k)$ of $S(\lambda,k)$ in the right half of the $\lambda$-plane, assumed to be finite in number. The application of the Watson transformation to the scattering amplitude in the present context was first discussed by Regge [2, 2a]; the poles $\lambda_j(k)$ are known as *Regge poles*.

Before attempting to justify the steps that led to (7.1.6), let us make a preliminary check that it will indeed enable us to settle both problems (i) and (ii) discussed above. The Legendre function $P_{\lambda-\frac{1}{2}}(z)$ is an entire function of $\lambda$ and a regular analytic function of $z$ in the $z$-plane cut along the real axis [3] from $-\infty$ to $-1$. Its asymptotic behavior as $|z|\to\infty$ for fixed $\lambda$, $\operatorname{Re}\lambda \geqslant 0$, is given by

$$P_{\lambda-\frac{1}{2}}(z) = O(z^{\lambda-\frac{1}{2}}) \qquad (\operatorname{Re}\lambda \geqslant 0, |z|\to\infty, |\arg z| < \pi). \tag{7.1.9}$$

If we ignore convergence questions for the moment, the above properties imply that (7.1.6) provides the desired analytic continuation of $f(k,\tau)$ in the $\tau$-plane, and that $f(k,\tau)$ for real $k$ is regular in the whole (finite) $\tau^2$-plane, cut along the real axis from $-\infty$ to 0 [actually, according to (6.3.25) and (6.5.13), the cut will not extend all the way to the origin].

Furthermore, it follows from (7.1.9) that the integral along the imaginary axis in (7.1.6), known as the *background integral*, is $O(\tau^{-1})$ as $|\tau|\to\infty$, so that the asymptotic behavior of $f(k,\tau)$ as $|\tau|\to\infty$ is dominated by the Regge pole terms,

$$f(k,\tau) = O(\tau^{2\operatorname{Re}\lambda_M - 1}) \qquad (|\tau|\to\infty), \tag{7.1.10}$$

where $\lambda_M$ is the "rightmost" Regge pole, i.e., the pole having the largest real part (Fig. 7.1).

These results imply that the function (6.5.14) verifies a dispersion relation in $\tau^2$ with a finite number of subtractions; together with (6.5.7), this leads to the Mandelstam representation.

In the absence of Regge poles, (7.1.6) would yield a representation (background integral alone) of the form

$$f(k,\cos\theta) = \int_{-\infty}^{\infty} F(k,x) P_{ix-\frac{1}{2}}(-\cos\theta)\,dx. \tag{7.1.11}$$

The functions $P_{ix-\frac{1}{2}}$ are known as *conical functions*, and an integral of this form is called a *Mehler transform* [4, p. 398]. It may be regarded as the analog of a Fourier transform for the conjugate pair of variables, scattering angle ($\theta$) and (generalized) angular momentum ($\lambda$). Thus one might say that, just as the analytic and asymptotic properties in the domain of the energy variable ($E$) are related to those in the conjugate time variable ($t$) (cf. Section 3.4), it is very natural that the analytic and asymptotic properties in $\cos\theta$ follow from those in the complex angular momentum plane [5, p. 5].

The outcome of the above heuristic discussion is that our aim of deriving a double dispersion relation will be achieved if we can show that:

A. There exists an interpolating function $S(\lambda, k)$, satisfying (7.1.8), that has a single-valued analytic continuation in the right half of the $\lambda$-plane.

B. For any real value of $k$, the only possible singularities of $S(\lambda, k)$ in the right half of the $\lambda$-plane are a finite number of Regge poles, with bounded real part, so that $f(k,\tau)$ is polynomial bounded as $|\tau| \to \infty$ [cf. (7.1.10)].

C. For real $k$ and $|\lambda| \to \infty$, $-\pi/2 \leqslant \arg\lambda \leqslant \pi/2$, $S(\lambda, k) - 1$ behaves in such a way that it is legitimate to neglect the contribution from the half-circle $\Gamma$ in Fig. 7.1, and that the convergence of the integrals in Watson's transformation is assured.

While property A is valid for more general potentials, the Yukawa-type character of the potential (with some additional restrictions to be specified later) plays an essential role in the proof of properties B and C.

In the following sections, we outline the proof of the above properties. In some cases, only the method of proof will be given, either because it is similar to methods already described before or because the details become too involved for presentation here. In such cases, the reader will be referred to the original papers. Additional details[2] may be found in the general references quoted in Section 5.1, as well as in the work by Newton [6b].

[2] For the applications of Regge poles in high-energy physics, see Refs. [6, 6a].

## 7.2. Regular and Irregular Solutions

### (a) The Regular Solution

It is natural to base the interpolation of the partial waves for complex angular momentum upon the differential equation (5.8.1) for the partial-wave radial function,

$$[d^2/dr^2 + k^2 - (\lambda^2 - \tfrac{1}{4})/r^2 - U(r)]\,u(\lambda,k,r) = 0, \tag{7.2.1}$$

where we have set

$$\lambda = l + \tfrac{1}{2}. \tag{7.2.2}$$

Assuming, as always, that the potential is less singular than $r^{-2}$ at the origin, the point $r = 0$ is a "regular" singularity for the differential equation [6c, p. 197], and it follows that (7.2.1) has two independent solutions, behaving at the origin like $r^{\lambda+1/2}$ and $r^{-\lambda+1/2}$, respectively. By analogy with (5.8.16) (though with slightly different normalization), we define the *regular solution* $\varphi(\lambda,k,r)$ by the boundary condition

$$\lim_{r\to 0} \varphi(\lambda,k,r)/r^{\lambda+1/2} = 1. \tag{7.2.3}$$

The term "regular" refers to the behavior at the origin for $\mathrm{Re}\,\lambda \geqslant 0$.

In order to discuss the properties of the regular solution, as well as those of other solutions of (7.2.1) that will be introduced later, it is convenient to transform (7.2.1), together with the boundary conditions that define the solution, into an integral equation (cf. Section 5.3). The procedure [7] for doing this is as follows.

Let $\psi(\lambda,r)$ be a solution of the differential equation

$$D(\lambda,r)\psi(\lambda,r) \equiv [d^2/dr^2 + p(\lambda,r)]\,\psi(\lambda,r) = q(\lambda,r)\psi(\lambda,r) \tag{7.2.4}$$

satisfying the boundary condition

$$\lim_{r\to r_1} [\psi(\lambda,r)/\chi(\lambda,r)] = 1, \tag{7.2.5}$$

where $\chi$ is a solution of the "free" equation (obtained by setting $q = 0$)

$$D(\lambda,r)\chi(\lambda,r) = 0. \tag{7.2.6}$$

Let $\chi_1(\lambda,r)$ and $\chi_2(\lambda,r)$ be two independent solutions of (7.2.6), so that their Wronskian [cf. (5.2.7)] is nonvanishing:

$$W[\chi_1,\chi_2] = \chi_1(d\chi_2/dr) - (d\chi_1/dr)\chi_2 \neq 0. \tag{7.2.7}$$

Consider the function

$$G(\lambda,r,r') = \chi_1(\lambda,r')\chi_2(\lambda,r) - \chi_1(\lambda,r)\chi_2(\lambda,r'). \tag{7.2.8}$$

It follows from (7.2.4) and (7.2.6) that

$$\frac{d}{dr'}\left[G(\lambda,r,r')\frac{d\psi}{dr'}(\lambda,r')-\frac{dG}{dr'}(\lambda,r,r')\psi(\lambda,r')\right] = G(\lambda,r,r')q(\lambda,r')\psi(\lambda,r').$$

Integrating both sides from $r'=r_1$ to $r'=r$, noting that $G(\lambda,r,r)=0$ and $(dG/dr')_{r'=r}=-W[\chi_1,\chi_2]$, and taking into account (7.2.5), we see that $\psi(\lambda,r)$ satisfies the integral equation

$$\psi(\lambda,r) = \chi(\lambda,r)+\frac{1}{W[\chi_1,\chi_2]}\int_{r_1}^{r} G(\lambda,r,r')q(\lambda,r')\psi(\lambda,r')\,dr'. \quad (7.2.9)$$

Let us now apply these results to (7.2.1) and (7.2.3). Taking

$$p(r) = -(\lambda^2-\tfrac{1}{4})/r^2, \qquad q(r) = U(r)-k^2,$$

and

$$\chi_1(\lambda,r) = r^{\lambda+1/2}, \qquad \chi_2(\lambda,r) = r^{-\lambda+1/2}, \qquad W[\chi_1,\chi_2] = -2\lambda, \quad (7.2.10)$$

we get

$$\varphi(\lambda,k,r) = r^{\lambda+1/2}+\frac{1}{2\lambda}\int_0^r\left[\left(\frac{r'}{r}\right)^{\lambda}-\left(\frac{r}{r'}\right)^{\lambda}\right][k^2-U(r')](rr')^{1/2}\varphi(\lambda,k,r')\,dr'. \quad (7.2.11)$$

An alternative possibility is to take $p(r)=k^2-(\lambda^2-\frac{1}{4})/r^2$, $q(r)=U(r)$, in which case (7.2.6) is the free-particle radial equation (no potential). The free-particle regular solution is

$$\varphi_0(\lambda,k,r) = \Gamma(\lambda+1)(2/k)^{\lambda}\sqrt{r}J_{\lambda}(kr), \quad (7.2.12)$$

where $\Gamma(z)$ is the gamma function. We can take

$$\chi_1 = \varphi_0(\lambda,k,r), \qquad \chi_2 = \varphi_0(-\lambda,k,r), \quad (7.2.13)$$

which have the same behavior at the origin as (7.2.10), and, consequently, also the same Wronskian. It follows that

$$\varphi(\lambda,k,r) = \varphi_0(\lambda,k,r)+\pi/[2\sin(\pi\lambda)]\int_0^r[J_{-\lambda}(kr')J_{\lambda}(kr)-J_{-\lambda}(kr)J_{\lambda}(kr')]$$
$$\times\, U(r')(rr')^{1/2}\varphi(\lambda,k,r')\,dr', \quad (7.2.14)$$

where we have employed

$$\Gamma(\lambda)\Gamma(1-\lambda) = \pi/\sin(\pi\lambda). \quad (7.2.15)$$

One can now apply the method of Section 5.3(b) to investigate the analytic properties of the regular solution as a function of $\lambda$ and $k$: one considers the iteration solution of (7.2.11) or (7.2.14), and one discusses the convergence of the Liouville–Neumann series, making use of suitable bounds for the kernel of the integral equation. The analyticity domain of the solution in the region of uniform convergence then follows from the analyticity domain of each term in the Liouville–Neumann series.

Since the integrals in (7.2.11) are taken between finite limits, it is readily seen that each term in the iteration solution is a polynomial in $k^2$ and is regular in $\operatorname{Re}\lambda \geqslant 0$, provided only that the potential satisfies (5.3.20), i.e., that its first moment $M_1$ is finite. One can say nothing about $\operatorname{Re}\lambda < 0$, because of convergence problems with the integrals at $r' = 0$, but we are not interested in the left half of the $\lambda$-plane (the analytic continuation in the left half-plane is unstable with respect to small changes in the potential).

Under the above conditions it can be shown [5, p. 18] that the iteration solution is unrestrictedly convergent. Consequently we have:

THEOREM 7.2.1. *The regular solution* $\varphi(\lambda, k, r)$ *is an entire function of* $k^2$, *holomorphic in* $\lambda$ *for* $\operatorname{Re}\lambda > 0$, *and continuous on* $\operatorname{Re}\lambda = 0$, *provided that* $M_1$ *is finite.*

The analyticity in $k^2$ also follows from a theorem due to Poincaré [8] according to which the solution of a linear differential equation with coefficients that are entire functions of a parameter, specified by initial conditions that are independent of this parameter [as (7.2.3) is independent of $k^2$], is itself an entire function of the parameter in question.

Since (7.2.1) depends only on $k^2$ and since $U(r)$ is real, the following relations are valid everywhere in the analyticity domain:

$$\varphi(\lambda, k, r) = \varphi(\lambda, -k, r) = \varphi^*(\lambda^*, k^*, r). \tag{7.2.16}$$

Finally, let us note that, for pure imaginary $\lambda$, according to Theorem 7.2.1, $\varphi(-\lambda, k, r)$ is also defined and continuous, and it follows from (7.2.3) and (7.2.10) that

$$W[\varphi(\lambda, k, r), \varphi(-\lambda, k, r)] = -2\lambda \qquad (\operatorname{Re}\lambda = 0). \tag{7.2.17}$$

### *(b) The Jost Solutions*

We consider next the analog of the Jost solution (5.2.6), defined by its behavior at infinity,

$$\lim_{r\to\infty} [e^{ikr} f(\lambda, k, r)] = 1. \tag{7.2.18}$$

The free-particle Jost solution of (7.2.1) (with $U = 0$) is

$$f_0(\lambda, k, r) = \exp[-i(\lambda + \tfrac{1}{2})\pi/2](\pi kr/2)^{1/2} H_\lambda^{(2)}(kr), \tag{7.2.19}$$

which satisfies (7.2.18) according to the well-known asymptotic behavior of Hankel functions of the second kind.

To obtain an integral equation that incorporates the above boundary condition, one possibility is to take in (7.2.4) $p = k^2$, $q = U(r) + (\lambda^2 - \frac{1}{4})/r^2$, and

$$\chi_1 = e^{-ikr}, \qquad \chi_2 = e^{ikr}, \qquad W[\chi_1, \chi_2] = 2ik, \tag{7.2.20}$$

which yields

$$f(\lambda, k, r) = e^{-ikr} + (1/k) \int_r^\infty \sin[k(r'-r)][U(r') + (\lambda^2 - \tfrac{1}{4})/r'^2] f(\lambda, k, r')\, dr'. \tag{7.2.21}$$

This should be compared with (5.3.1).

Another possibility is to take $p = k^2 - (\lambda^2 - \frac{1}{4})/r^2$, $q = U(r)$. To find a solution of the free-particle equation independent of (7.2.19), we change $k \to -k$. However, since $H_\lambda^{(2)}(kr)$ is a multivalued function of $k$ (it behaves like $k^{-\lambda}$ as $k \to 0$), we have to specify which branch of this function is considered. Since [9, p. 17]

$$H_\lambda^{(2)}(e^{-i\pi} z) = -e^{i\lambda\pi} H_\lambda^{(1)}(z), \tag{7.2.22}$$

it is convenient to define, for the general Jost solution,

$$f(\lambda, -k, r) = f(\lambda, ke^{-i\pi}, r). \tag{7.2.23}$$

In particular,

$$f_0(\lambda, -k, r) = \exp[i(\lambda + \tfrac{1}{2})\pi/2](\pi kr/2)^{1/2} H_\lambda^{(1)}(kr), \tag{7.2.24}$$

with the asymptotic behavior $f_0(\lambda, -k, r) \approx e^{ikr}$ as $r \to \infty$. Taking

$$\chi_1 = f_0(\lambda, k, r), \qquad \chi_2 = f_0(\lambda, -k, r), \qquad W[\chi_1, \chi_2] = 2ik, \tag{7.2.25}$$

we find from (7.2.9)

$$f(\lambda, k, r) = f_0(\lambda, k, r) + (i\pi/4) \int_r^\infty [H_\lambda^{(1)}(kr) H_\lambda^{(2)}(kr') - H_\lambda^{(1)}(kr') H_\lambda^{(2)}(kr)] \times U(r')(rr')^{1/2} f(\lambda, k, r')\, dr'. \tag{7.2.26}$$

For the physical values of $\lambda$ [cf. (7.1.8)], this reduces to (5.8.10). It is readily seen that the kernel of (7.2.26) is equivalent to that of (7.2.14).

The discussion of the analytic behavior of $f(\lambda, k, r)$ now proceeds in the same way as in previous cases. By investigating the iteration solution of (7.2.26), one finds [5, p. 24] that a sufficient condition for its convergence is again provided by (5.3.17). For $\operatorname{Im} k < 0$, (5.3.17) is satisfied for potentials with $M_1 < \infty$, and each term of the Liouville–Neumann series is holomorphic in $k$, so that $f(\lambda, k, r)$ is holomorphic in $I_-(k)$.

On the other hand, for a potential of (at least) exponential decrease, satisfying (5.3.22), condition (5.3.17) is satisfied for $\operatorname{Im} k \leqslant \alpha/2$. However, since

$f_0(\lambda, k, r)$ is not single valued around the origin, we must introduce a branch cut. It is convenient to take this cut along the imaginary axis [cf. Section 5.4(c)], from 0 to $i\alpha/2$. It then follows that $f(\lambda, k, r)$ is holomorphic in the cut domain, for $\operatorname{Im} k < \alpha/2$.

The analytic behavior in $\lambda$ follows most simply from the iterative solution of (7.2.21); clearly, each term in the Liouville–Neumann series is a polynomial in $\lambda^2$. One then finds that, for $k$ in the analyticity domain, $f(\lambda, k, r)$ is an entire function of $\lambda^2$. This is in agreement with Poincaré's theorem quoted at the end of Section 7.2(a), since the boundary condition (7.2.18) is independent of $\lambda$. Corresponding results are of course valid for $f(\lambda, -k, r)$. In conclusion one finds:

THEOREM 7.2.2. (*a*) *If* $M_1$ *is finite, the Jost solution* $f(\lambda, k, r)$ *is an entire function of* $\lambda^2$, *holomorphic in* $k$ *for* $\operatorname{Im} k < 0$, *and continuous on* $\operatorname{Im} k = 0$, *except at* $k = 0$; *similarly for* $f(\lambda, -k, r)$ *in* $\operatorname{Im} k > 0$. (*b*) *If* $E_\alpha$ *is finite*, $f(\lambda, k, r)$ *is an entire function of* $\lambda^2$, *holomorphic in the domain* $\operatorname{Im} k < \alpha/2$ *cut along the imaginary axis from* $k = 0$ *to* $k = i\alpha/2$; *similarly for* $f(\lambda, -k, r)$ *in* $\operatorname{Im} k > -\alpha/2$, *with a cut from* 0 *to* $-i\alpha/2$.

We obviously have

$$f(-\lambda, k, r) = f(\lambda, k, r). \tag{7.2.27}$$

On the other hand, the reality of $U(r)$ and the boundary condition (7.2.18) lead to

$$f(\lambda, -k, r) = f^*(\lambda^*, k^*, r). \tag{7.2.28}$$

Furthermore, since $f(\lambda, \pm k, r)$ behaves like $\exp(\mp ikr)$ as $r \to \infty$ [cf. (7.2.18)], we have, by (7.2.20),

$$W[f(\lambda, k, r), f(\lambda, -k, r)] = 2ik. \tag{7.2.29}$$

The nature of the branch point at $k = 0$ can be clarified by considering first the behavior of the free solution (7.2.19). Different branches of the Hankel functions are connected by the formula [9]

$$H_\lambda^{(2)}(e^{-2i\pi}z) = -H_\lambda^{(2)}(z) + 2\cos(\pi\lambda) H_\lambda^{(2)}(e^{-i\pi}z).$$

It then follows from (7.2.19) that

$$f_0(\lambda, ke^{-2i\pi}, r) = f_0(\lambda, k, r) - 2i\cos(\pi\lambda) f_0(\lambda, ke^{-i\pi}, r), \tag{7.2.30}$$

from which other determinations follow by iteration.

For potentials such that $E_\alpha$ is finite, the iteration solution of the integral equation (7.2.26) both for $f(\lambda, k, r)$ and for $f(\lambda, ke^{-i\pi}, r)$ converges within the strip $-\alpha/2 < \operatorname{Im} k < \alpha/2$ [the kernel of (7.2.26) is unchanged by the substitution $k \to ke^{-i\pi}$]. It follows that, if we make a circuit around $k = 0$ staying within this strip, the branches of $f(\lambda,k,r)$ are connected by the same relation (7.2.30) valid for the free solution:

$$f(\lambda, ke^{-2i\pi}, r) = f(\lambda, k, r) - 2i\cos(\pi\lambda) f(\lambda, ke^{-i\pi}, r). \qquad (7.2.31)$$

In other words, *the branch cut of $f(\lambda,k,r)$ from $k=0$ to $k=i\alpha/2$ is purely kinematical* (independent of the potential). Note from (7.2.31) that the branch cut disappears for physical values of $\lambda$, in which case we recover the results of Section 5.8(a).

From now on the notation $f(\lambda,k,r)$ will stand for the *principal branch* of the Jost solution, defined by the condition

$$-3\pi/2 < \arg k < \pi/2. \qquad (7.2.32)$$

## 7.3. The Jost Function and the S-Function

### (a) General Potentials

We define the *Jost function* by analogy with (5.2.14):

$$f(\lambda,k) = W[f(\lambda,k,r), \varphi(\lambda,k,r)]. \qquad (7.3.1)$$

It then follows from (7.2.12) and (7.2.19) that the free-particle Jost function is given by

$$f_0(\lambda,k) = (2/\pi)^{1/2} 2^{\lambda}\Gamma(\lambda+1) k^{-(\lambda-1/2)} \exp[-i(\lambda-\tfrac{1}{2})\pi/2]. \qquad (7.3.2)$$

One can also adopt a different normalization, by defining the function

$$F(\lambda,k) = f(\lambda,k)/f_0(\lambda,k), \qquad (7.3.3)$$

so that, in the absence of interaction ($U=0$), $F_0(\lambda,k)=1$. For physical values of $\lambda$, $F(\lambda,k)$ reduces to the function $F_l(k)$ defined by (5.8.12) and (5.8.15).

Since $f(\lambda,k,r)$ and $f(\lambda,-k,r)$ are independent solutions of (7.2.1) for $k \neq 0$ [cf. (7.2.29)], it follows that any other solution can be written as a linear combination of these two; in particular, we must have

$$\varphi(\lambda,k,r) = Af(\lambda,k,r) + Bf(\lambda,-k,r).$$

The coefficients $A$ and $B$ can be evaluated by taking the Wronskian of both sides with $f(\lambda, \pm k, r)$ and making use of (7.3.1) and (7.2.16); this leads to $B(k) = f(\lambda,k)/2ik = A(-k)$, so that

$$\varphi(\lambda,k,r) = (1/2ik)[f(\lambda,k) f(\lambda,-k,r) - f(\lambda,-k) f(\lambda,k,r)]. \qquad (7.3.4)$$

An alternative expression for $f(\lambda,k)$ can be obtained by evaluating the Wronskian (7.3.1) at $r=0$, with the help of (7.2.3). Since $f(\lambda,k,r)$ is an irregular solution, it contains terms in $r^{\lambda+1/2}$ and in $r^{-\lambda+1/2}$ as $r \to 0$; the first of these does not contribute to the Wronskian, provided that $\operatorname{Re}\lambda > 0$; for the second one, we can employ (7.2.10). In this way we find

$$f(\lambda,k) = 2\lambda \lim_{r\to 0}[r^{\lambda-1/2} f(\lambda,k,r)] \qquad (\operatorname{Re}\lambda > 0), \tag{7.3.5}$$

which, together with (7.3.3), corresponds to the previous definition (5.8.12) for physical values of $\lambda$.

Since the results of Theorems 7.2.1 and 7.2.2 can be extended by the same methods to $\partial\varphi(\lambda,k,r)/\partial r$ and $\partial f(\lambda,k,r)/\partial r$, the analytic properties of the Jost function follow immediately from (7.3.1):

THEOREM 7.3.1. *If $E_\alpha$ is finite, $f(\lambda,k)$ is holomorphic in the direct product of the half-plane* $\operatorname{Re}\lambda > 0$ *with the domain* $\operatorname{Im}k < \alpha/2$, *cut along the imaginary axis from $k = 0$ to $k = i\alpha/2$; similarly for $f(\lambda, -k)$ in the direct product of* $\operatorname{Re}\lambda > 0$ *with* $\operatorname{Im}k > -\alpha/2$, *cut from $k = 0$ to $k = -i\alpha/2$.*

Within the analyticity domain we have, from (7.2.28) and (7.3.5),

$$f(\lambda, -k) = f^*(\lambda^*, k^*). \tag{7.3.6}$$

Combining (7.3.5) with (7.2.31), we also find that

$$f(\lambda, ke^{-2i\pi}) = f(\lambda,k) - 2i\cos(\pi\lambda) f(\lambda, ke^{-i\pi}), \tag{7.3.7}$$

provided that the circuit around the origin stays entirely within the strip $-\alpha/2 < \operatorname{Im}k < \alpha/2$. This clarifies the nature of the branch point at $k = 0$, showing that the branch cuts of Theorem 7.3.1 are purely kinematical, i.e., independent of $U$, and that they disappear for physical values of $\lambda$.

To define the $S$-function, we first note from (7.2.12) that the asymptotic behavior of the free regular solution as $r \to \infty$ is given by [taking into account (7.3.2)]

$$\varphi_0(\lambda,k,r) \approx k^{-1} f_0(\lambda,k) \exp[i(\lambda-\tfrac{1}{2})\pi/2] \sin[kr-(\lambda-\tfrac{1}{2})\pi/2] \qquad (r\to\infty). \tag{7.3.8}$$

On the other hand, it follows from (7.3.4) and (7.2.18) that

$$\varphi(\lambda,k,r) \approx k^{-1} f(\lambda,k) \exp[i(\lambda-\tfrac{1}{2})\pi/2] \Big\{ \exp[ikr - i(\lambda-\tfrac{1}{2})\pi/2] - \exp[-i(\lambda-\tfrac{1}{2})\pi] \frac{f(\lambda,-k)}{f(\lambda,k)} \exp[-ikr + i(\lambda-\tfrac{1}{2})\pi/2] \Big\} \Big/ 2i \qquad (r\to\infty). \tag{7.3.9}$$

This suggests defining the *phase shift* $\eta(\lambda,k)$ by [cf. (2.6.5), (2.6.12)]

$$\varphi(\lambda,k,r) \approx k^{-1} f(\lambda,k) \exp[i(\lambda-\tfrac{1}{2})(\pi/2) - i\eta(\lambda,k)] \times \sin[kr-(\lambda-\tfrac{1}{2})(\pi/2)+\eta(\lambda,k)] \qquad (r\to\infty), \tag{7.3.10}$$

so that

$$S(\lambda,k) = \exp[2i\eta(\lambda,k)] = \exp[i(\lambda-\tfrac{1}{2})\pi]f(\lambda,k)/f(\lambda,-k). \quad (7.3.11)$$

Note that, according to (7.3.6), $\eta(\lambda,k)$ is indeed real for real $\lambda$ and $k$. By (7.3.2) and (7.3.3), we may rewrite (7.3.11) as

$$S(\lambda,k) = F(\lambda,k)/F(\lambda,-k). \quad (7.3.12)$$

For physical values of $\lambda$, this reduces to (5.8.17), so that (7.1.8) is satisfied; i.e., $S(\lambda,k)$ is an interpolating function.

The analytic properties of $S(\lambda,k)$ for potentials of at least exponential decrease at infinity follow immediately from (7.3.11) and Theorem 7.3.1:

THEOREM 7.3.2. *If $E_\alpha$ is finite, $S(\lambda,k)$ is meromorphic in the direct product of the half-plane* $\operatorname{Re}\lambda > 0$ *with the strip* $-\alpha/2 < \operatorname{Im} k < \alpha/2$, *cut along the imaginary axis from $k = -i\alpha/2$ to $k = i\alpha/2$; the poles of $S(\lambda,k)$ are the zeros of $f(\lambda,-k)$.*

Within the analyticity domain, (7.3.11) and (7.3.6) imply the extended unitarity condition [cf. (2.4.3)]

$$S(\lambda,k)\,S^*(\lambda^*,k^*) = 1. \quad (7.3.13)$$

On the other hand, (7.3.7) implies

$$S(\lambda,ke^{-i\pi}) = \frac{e^{2i\pi\lambda}}{1+e^{2i\pi\lambda}-S(\lambda,k)}, \quad (7.3.14)$$

which reduces to the symmetry relation $S_l(k)S_l(-k) = 1$ for physical $\lambda$. For a circuit around $k = 0$ within the analyticity strip, we get

$$S(\lambda,ke^{-2i\pi}) = \frac{S(\lambda,k)-1-e^{2i\pi\lambda}}{(1+e^{-2i\pi\lambda})S(\lambda,k)-e^{2i\pi\lambda}-1-e^{-2i\pi\lambda}}, \quad (7.3.15)$$

so that $S$ is single valued within the analyticity strip for physical values of $\lambda$; for other values, we have the kinematical cut from $-i\alpha/2$ to $i\alpha/2$.

Consider the function

$$Z(\lambda,k) = ik^{2\lambda}[S(\lambda,k)-e^{2i\pi\lambda}]/[S(\lambda,k)-1]. \quad (7.3.16)$$

For physical values of $\lambda$, it follows from (7.3.11) that

$$Z(\lambda = l+\tfrac{1}{2},k) = k^{2l+1}\cot\eta_l(k), \quad (7.3.17)$$

which, according to (5.8.22) and the relation $\eta_l(-k) = -\eta_l(k)$, is an even function of $k$, regular at $k = 0$. This is the basis for the so-called "effective range expansion"[3] of (7.3.17) in powers of $k^2$. The usefulness of (7.3.16) is that

[3] Cf. Mott and Massey [10, p. 50].

these properties remain valid for all $\lambda$ in the analyticity domain. In fact, it follows from (7.3.16) and (7.3.14) that

$$Z(\lambda, ke^{-i\pi}) = Z(\lambda, k), \tag{7.3.18}$$

so that $Z$ is even in $k$; furthermore, (7.3.15) leads to

$$Z(\lambda, ke^{-2i\pi}) = Z(\lambda, k), \tag{7.3.19}$$

so that $Z$ is single valued in $k$ within the analyticity strip; it does not have the kinematical cut.

From (7.3.16) we find, solving for $S$,

$$S(\lambda, k) = [Z(\lambda, k) - ik^{2\lambda}e^{2i\pi\lambda}]/[Z(\lambda, k) - ik^{2\lambda}], \tag{7.3.20}$$

which shows explicitly the nature of the branch point of $S$ at $k = 0$.

### (b) Extension of the Analyticity Domain for Yukawa-Type Potentials

For physical values of $\lambda$, we have seen in Sections 5.4(c) and 5.8(a) that the analyticity domain of the Jost functions can be extended to the whole (suitably cut) $k$-plane in the special case of Yukawa-type potentials. We now show, by a modified version of a method due to Regge [11][4], that this result is actually true for all $\lambda$ in the analyticity domain.

As was mentioned in Section 5.4(c), the class (7.1.1) of Yukawa-type potentials, besides satisfying condition (5.3.22) for $0 \leqslant \alpha < m$, is distinguished by the fact that the potential can be analytically continued to complex values of $r$, so that

$$U(r) \quad \text{is holomorphic for} \quad \operatorname{Re} r > 0. \tag{7.3.21}$$

The asymptotic behavior of $U(r)$ as $|r| \to \infty$, $\operatorname{Re} r > 0$, depends on the behavior of the weight function $\sigma(\mu)$ of (7.1.1) near the lower limit of integration [13, p. 30]; in particular, if $\sigma(\mu)$ is continuously differentiable near $\mu = m$, one finds, by partial integration,

$$U(r) = O(e^{-mr}/r^2) \qquad (|r| \to \infty, \ \operatorname{Re} r > 0).$$

In order to include more general potentials, such as the pure Yukawa potential

$$U(r) = U_0 e^{-mr}/r, \tag{7.3.22}$$

we assume instead of (7.3.22) the more general condition

$$U(r) = O(e^{-m\operatorname{Re} r}/|r|^{\gamma}) \qquad (\gamma \geqslant 1, \ |r| \to \infty, \ \operatorname{Re} r > 0). \tag{7.3.23}$$

[4] Cf. also Doetsch [12, p. 362].

Furthermore, at $r=0$, the potential must be less singular than $r^{-2}$, in order that $M_1<\infty$. Together with (7.3.21) and (7.3.23), these are the only assumptions that need to be made about the potential.

In view of (7.3.21), we can consider (7.2.1) as a differential equation in the complex domain, for $\operatorname{Re} r>0$. According to a well-known theorem [6c, p. 196], if the coefficients of this differential equation are holomorphic in a simply connected domain, any solution can also be analytically continued and is holomorphic in the same domain. In particular, the solutions $\varphi(\lambda,k,r)$ and $f(\lambda,k,r)$ defined in Section 7.2 have a regular analytic continuation in $\operatorname{Re} r>0$.

Let us set

$$r=\rho e^{i\theta} \tag{7.3.24}$$

and let us consider the behavior of the solutions along a fixed ray in the right half-plane, $\theta=\text{const}$ $(-\pi/2<\theta<\pi/2)$. Along such a ray, the differential equation (7.2.1) becomes

$$\{d^2/d\rho^2+k_\theta{}^2-[(\lambda^2-\tfrac{1}{4})/\rho^2]-U_\theta(\rho)\}\,u_\theta(\lambda,k,\rho)=0, \tag{7.3.25}$$

where

$$k_\theta = ke^{i\theta}, \tag{7.3.26}$$

$$U_\theta(\rho) = e^{2i\theta}U(\rho e^{i\theta}). \tag{7.3.27}$$

Note that (7.3.25) is of the same form as (7.2.1), but with a complex potential $U_\theta(\rho)$. According to (7.3.23), this potential satisfies

$$|U_\theta(\rho)| = |U(\rho e^{i\theta})| = O(e^{-m\rho\cos\theta}/\rho^\gamma) \qquad (\gamma\geqslant 1,\ \rho\to\infty). \tag{7.3.28}$$

Therefore, along any ray $\theta=\text{const}$ in the right half-plane, $U_\theta(\rho)$ is still a potential of (at least) exponential decrease, except at the boundary, i.e., along the imaginary axis $(\theta=\pm\pi/2)$, where $|U_{\pm\pi/2}(\rho)|=O(\rho^{-\gamma})$ as $\rho\to\infty$ [note, however, the oscillatory character of, for example, (7.3.22) along the imaginary $r$-axis.].

The results obtained in Section 7.2 [except the symmetry relations (7.2.16) and (7.2.28)] do not depend in any way on the potential being real: they remain valid for a complex potential such as $U_\theta(\rho)$, provided that it satisfies assumptions similar to those made for $U(\rho)$ in Section 7.2.

Let $\varepsilon$ be an arbitrarily small *fixed* positive number, and let $S_\varepsilon$ be the sector of the $r$-plane defined by (7.3.24), with

$$S_\varepsilon:\quad -(\pi/2)+\varepsilon\leqslant\theta\leqslant(\pi/2)-\varepsilon \qquad (\varepsilon>0). \tag{7.3.29}$$

In this sector, according to (7.3.28), $U_\theta(\rho)$ always satisfies the condition $E_\alpha<\infty$ [cf. (5.3.22)], with $\alpha\geqslant m\sin\varepsilon>0$.

It then follows from Theorem 7.2.1 that, in $S_\varepsilon$, (7.3.25) has a regular solution $\varphi_\theta(\lambda, k_\theta, \rho)$, verifying the boundary condition

$$\lim_{\rho\to 0} \varphi_\theta(\lambda, k_\theta, \rho)/\rho^{\lambda+\frac{1}{2}} = 1, \tag{7.3.30}$$

which is an entire function of $k_\theta{}^2$, holomorphic in $\lambda$ for $\operatorname{Re}\lambda > 0$. Similarly, by Theorem 7.2.2., (7.3.25) has a Jost solution $f_\theta(\lambda, k_\theta, \rho)$, defined by [cf. (7.2.18)]

$$f_\theta(\lambda, k_\theta, \rho) \approx \exp(-ik_\theta\rho) \qquad (\rho\to\infty), \tag{7.3.31}$$

which is an entire function of $\lambda^2$, holomorphic in $k_\theta$ for $\operatorname{Im} k_\theta < 0$ (actually, for $\operatorname{Im} k_\theta < \frac{1}{2}m\cos\theta$, except for a cut from $k_\theta = 0$ to $\frac{1}{2}im\cos\theta$).

On the other hand, we have seen just that the solutions $\varphi(\lambda, k, r)$ and $f(\lambda, k, r)$ of (7.2.1) have a regular analytic continuation in $S_\varepsilon$ [except at $r = 0$ for $f(\lambda, k, r)$], that must also satisfy (7.3.25) along any ray $\theta = \text{const}$. What is the relation between the analytic continuation of these functions and the solutions $\varphi_\theta$ and $f_\theta$ defined above?

In view of (7.3.28), any solution of (7.2.1) in $S_\varepsilon$, and in particular $f(\lambda, k, r)$, has the asymptotic behavior [13, p. 83]

$$f(\lambda, k, r) \approx c_-(\lambda, k, \theta)\, e^{-ikr} + c_+(\lambda, k, \theta)\, e^{ikr},$$
$$r = \rho e^{i\theta}, \qquad \rho\to\infty \quad \text{in} \quad S_\varepsilon, \tag{7.3.32}$$

where the coefficients $c_\pm$, in general, may depend on $\theta$. Thus, for $r$ real ($\theta = 0$) and $\operatorname{Im} k \leqslant 0$, we know from (7.2.21) that (7.2.18) is valid, so that

$$c_-(\lambda, k, 0) = 1; \qquad c_+(\lambda, k, 0) = 0 \qquad (\operatorname{Im} k \leqslant 0). \tag{7.3.33}$$

However, there is no reason, a priori, why these coefficients should be the same for other values of $\theta$. The asymptotic behavior of the analytic continuation of a function need not coincide with the analytic continuation of its asymptotic behavior. This corresponds to the well-known *Stokes phenomenon* [13, p. 73].

Nevertheless, we can extend (7.3.33) to other values of $\theta$ with the help of the following theorem [14, p. 65; 15]:

THEOREM 7.3.3 (Lindelöf). *Let $f(z)$ be regular in a simply connected region $G$, and let $\Gamma_1$ and $\Gamma_2$ be two branches of the boundary of $G$ that go off to infinity in the negative and positive directions relative to the region; i.e., when moving along $\Gamma_1$ the region is to the right, and along $\Gamma_2$ it is to the left. Let $f(z)$ tend to definite limits $a$ and $b$ when $|z|\to\infty$ along $\Gamma_1$ and along $\Gamma_2$, respectively. Then, if $a \neq b$, $f(z)$ is unbounded in $G$. If $a = b$, then, either $f(z)\to a$ uniformly as $|z|\to\infty$ in $G$, or $f(z)$ is unbounded in $G$.*

In order to apply this theorem to $f(\lambda, k, r)$, we have to exclude a neighborhood of $r=0$, where the Jost solution is singular. Let us therefore introduce the regions $S_\varepsilon{}^\pm$ defined by (7.3.24), with:

$$\begin{aligned} S_\varepsilon{}^+: &\qquad 0 \leqslant \theta \leqslant (\pi/2)-\varepsilon, \quad \rho \geqslant \delta, \\ S_\varepsilon{}^-: &\quad -(\pi/2)+\varepsilon \leqslant \theta \leqslant 0, \qquad \rho \geqslant \delta, \end{aligned} \qquad (\varepsilon > 0, \delta > 0). \tag{7.3.34}$$

Consider the function

$$F_+(\lambda,k,r) = e^{-ikr}f(\lambda,k,r),$$

which is regular in $S_\varepsilon{}^\pm$. According to (7.3.32),

$$F_+(\lambda,k,r=\rho e^{i\theta}) \approx c_-(\lambda,k,\theta)e^{-2ikr} + c_+(\lambda,k,\theta) \quad (\rho\to\infty \text{ in } S_\varepsilon{}^\pm). \tag{7.3.35}$$

Let

$$k = k' + i\kappa. \tag{7.3.36}$$

Then we have

$$|\exp(-2ik\rho e^{i\theta})| = \exp[2\rho(\kappa\cos\theta + k'\sin\theta)],$$

so that the first term of (7.3.35) tends to zero as $\rho\to\infty$ in $S_\varepsilon{}^+$ if $k'\leqslant 0$, $\kappa<0$, and in $S_\varepsilon{}^-$ if $k'\geqslant 0$, $\kappa<0$; i.e.,

$$\lim_{\rho\to\infty} F_+(\lambda,k,r) = c_+(\lambda,k,\theta) \qquad (\kappa<0, \text{ in } S_\varepsilon{}^+ \text{ if } k'\leqslant 0, \text{ in } S_\varepsilon{}^- \text{ if } k'\geqslant 0). \tag{7.3.37}$$

We can now apply Theorem 7.3.3. to $F_+(\lambda,k,r)$, taking $G=S_\varepsilon{}^+$ if $k'\leqslant 0$ and $G=S_\varepsilon{}^-$ if $k'\geqslant 0$. Taking into account (7.3.33), we conclude that

$$c_+(\lambda,k,\theta) = 0 \qquad (\kappa<0, \text{ in } S_\varepsilon{}^+ \text{ if } k'\leqslant 0, \text{ in } S_\varepsilon{}^- \text{ if } k'\geqslant 0). \tag{7.3.38}$$

The function

$$F_-(\lambda,k,r) = e^{ikr}f(\lambda,k,r) \tag{7.3.39}$$

therefore satisfies, under the same conditions stated in (7.3.38),

$$\lim_{\rho\to\infty} F_-(\lambda,k,r) = c_-(\lambda,k,\theta), \tag{7.3.40}$$

and, again by applying Theorem 7.3.3 to $F_-(\lambda,k,r)$, with the help of (7.3.33), we conclude that

$$c_-(\lambda,k,\theta) = 1 \qquad (\kappa<0, \text{ in } S_\varepsilon{}^+ \text{ if } k'\leqslant 0, \text{ in } S_\varepsilon{}^- \text{ if } k'\geqslant 0). \tag{7.3.41}$$

It follows from (7.3.32), (7.3.38), and (7.3.41) that

$$f(\lambda,k,r) \approx e^{-ikr} \qquad \begin{aligned}&(\rho\to\infty, \kappa<0, \text{ in } S_\varepsilon{}^+ \text{ if } k'\leqslant 0,\\ &\text{in } S_\varepsilon{}^- \text{ if } k'\geqslant 0).\end{aligned} \tag{7.3.42}$$

Comparing this with (7.3.31), we conclude that

$$f(\lambda, k, \rho e^{i\theta}) = f_\theta(\lambda, ke^{i\theta}, \rho) \qquad (\kappa < 0, \text{ in } 0 \leqslant \theta \leqslant (\pi/2)-\varepsilon \text{ if } k' \leqslant 0, \text{ in } -(\pi/2)+\varepsilon \leqslant \theta \leqslant 0 \text{ if } k' \geqslant 0). \quad (7.3.43)$$

The relationship between the function $\varphi_\theta(\lambda, k_\theta, \rho)$ defined by (7.3.30) and the analytic continuation to complex $r$ of $\varphi(\lambda, k, r)$ is obtained in a similar way, by noting that [cf. (7.2.3)]

$$\lim_{\rho\to 0} \varphi(\lambda, k, \rho e^{i\theta})/\rho^{\lambda+\frac{1}{2}} = \exp[i(\lambda+\tfrac{1}{2})\theta],$$

from which it follows that

$$\varphi(\lambda, k, \rho e^{i\theta}) = \exp[i(\lambda+\tfrac{1}{2})\theta]\,\varphi_\theta(\lambda, ke^{i\theta}, \rho). \qquad (7.3.44)$$

If we now define the Jost function associated with (7.3.25) by

$$f_\theta(\lambda, k_\theta) = W[f_\theta(\lambda, k_\theta, \rho), \varphi_\theta(\lambda, k_\theta, \rho)], \qquad (7.3.45)$$

it follows from (7.3.43) and (7.3.44), together with corresponding relations for the derivatives with respect to $r$ [cf. also (7.3.5)], that

$$f(\lambda, k) = \exp[i(\lambda-\tfrac{1}{2})\theta] f_\theta(\lambda, ke^{i\theta}) \qquad (\kappa < 0, \text{ in } 0 \leqslant \theta \leqslant (\pi/2)-\varepsilon \text{ if } k' \leqslant 0, \text{ in } -(\pi/2)+\varepsilon \leqslant \theta \leqslant 0 \text{ if } k' \geqslant 0). \quad (7.3.46)$$

However, according to Theorem 7.3.1, $f_\theta(\lambda, ke^{i\theta})$ is holomorphic for

$$\operatorname{Im} k_\theta = \operatorname{Im}(ke^{i\theta}) \leqslant 0, \qquad (7.3.47)$$

regardless of whether $\kappa = \operatorname{Im} k$ is $<0$ or $\geqslant 0$. Therefore, we can employ the relation (7.3.46) to *define* the analytic continuation of $f(\lambda, k)$ to $\operatorname{Im} k \geqslant 0$. According to (7.3.46) and (7.3.47), the extended analyticity domain is the union of all the domains

$$\operatorname{Im} k \geqslant 0, \qquad \operatorname{Im}(ke^{i\theta}) \leqslant 0, \qquad (0 \leqslant \theta \leqslant (\pi/2)-\varepsilon \text{ if } \operatorname{Re} k \leqslant 0, \; -(\pi/2)+\varepsilon \leqslant \theta \leqslant 0 \text{ if } \operatorname{Re} k \geqslant 0). \quad (7.3.48)$$

The union of all these domains is the upper half of the $k$-plane, excluding a sector of angular opening $2\varepsilon$ centered around the imaginary axis (cf. Fig. 5.3). Since $\varepsilon$ is arbitrarily small, we obtain the upper half-plane cut along the positive imaginary axis from 0 to $i\infty$.

Combining this result with (7.3.11) and Theorems 7.3.1 and 7.3.2 we finally obtain[5]

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[5] The derivations of Theorem 7.3.4 given by Regge and collaborators [5, p. 49; 11] do not take sufficient account of the singularity of $f(\lambda, k, r)$ at $r = 0$ and of the problems involved in letting $\varepsilon \to 0$ in (7.3.29) [cf. (7.3.28)]. The relation (7.3.46) differs from that given in these references.

THEOREM 7.3.4. *Let $U(r)$ be a potential satisfying (7.3.21) and (7.3.23), with $M_1 < \infty$. Then $f(\lambda, k)$ is holomorphic in the direct product of the half-plane* $\operatorname{Re}\lambda > 0$ *with the k-plane, cut along the imaginary axis from $k = 0$ to $i\infty$; similarly for $f(\lambda, -k)$, in the direct product of* $\operatorname{Re}\lambda > 0$ *with the k-plane, cut from $k = 0$ to $-i\infty$. The function $S(\lambda, k)$, which satisfies (7.1.8), is meromorphic in the direct product of* $\operatorname{Re}\lambda > 0$ *with the two (disjoint) half-planes* $\operatorname{Re}k > 0$ *and* $\operatorname{Re}k < 0$, *and its poles are the zeros of $f(\lambda, -k)$. The cut from $k = -im/2$ to $k = im/2$ is purely kinematical, and it is absent for physical values of $\lambda$.*

The results given in Sections 5.4(c) and 5.8(a) for physical values of $\lambda$ are particular cases of this theorem. The theorem shows that the interpolating function $S(\lambda, k)$ satisfies condition A of Section 7.1.

## 7.4. Properties of the Pole Distribution

### *(a) General Potentials*

In order to verify condition B of Section 7.1, we now investigate the distribution of Regge poles, starting with general properties valid for all potentials of at least exponential decrease.

The poles of $S(\lambda, k)$ are the zeros of $f(\lambda, -k)$. Note that, by (7.3.6), if $(\lambda, k)$ is a pole, so is $(\lambda^*, -k^*)$. According to (7.3.4), we have, at a pole of $S(\lambda, k)$,

$$\begin{aligned} \varphi(\lambda, k, r) &= (1/2ik) f(\lambda, k) f(\lambda, -k, r) \\ &\approx (1/2ik) f(\lambda, k)\, e^{ikr} \qquad (r \to \infty). \end{aligned} \tag{7.4.1}$$

Note that $\lambda$ and $k$ are assumed to be within the analyticity domain of $S(\lambda, k)$, given by Theorem 7.3.2 in the general case, and by Theorem 7.3.4 in the special case of Yukawa-type potentials.

Consider now the differential equation (7.2.1) for $\varphi$, together with its complex conjugate,

$$\{d^2/dr^2 + k^{*2} - [(\lambda^{*2} - \tfrac{1}{4})/r^2] - U(r)\}\, \varphi^*(\lambda, k, r) = 0. \tag{7.4.2}$$

Multiplying (7.4.2) by $\varphi(\lambda, k, r)$ and subtracting the result from (7.2.1) for $\varphi$ multiplied by $\varphi^*$, we get

$$\left[(k^{*2} - k^2) - \frac{(\lambda^{*2} - \lambda^2)}{r^2}\right] |\varphi(\lambda, k, r)|^2 = \frac{d}{dr}\left[\frac{d\varphi}{dr}(\lambda, k, r)\, \varphi^*(\lambda, k, r) - \frac{d\varphi^*}{dr}(\lambda, k, r)\, \varphi(\lambda, k, r)\right].$$

Integrating both terms over $r$ from 0 to $\infty$, and noting that the bracketed expression on the right-hand side vanishes as $r \to 0$ by (7.2.3), in the domain $\operatorname{Re}\lambda > 0$ under consideration, we get

$$\operatorname{Re}k \operatorname{Im}k \int_0^\infty |\varphi(\lambda,k,r)|^2\,dr - \operatorname{Re}\lambda \operatorname{Im}\lambda \int_0^\infty [|\varphi(\lambda,k,r)|^2/r^2]\,dr$$
$$= \frac{i}{4}\lim_{r\to\infty}\left[\frac{d\varphi}{dr}(\lambda,k,r)\,\varphi^*(\lambda,k,r) - \frac{d\varphi^*}{dr}(\lambda,k,r)\,\varphi(\lambda,k,r)\right] \qquad (\operatorname{Re}\lambda > 0). \tag{7.4.3}$$

Several results follow from (7.4.3). In the first place, at a pole of $S(\lambda,k)$ with $\operatorname{Im}k > 0$, the right-hand side of (7.4.1) is exponentially decreasing as $r \to \infty$, so that (7.4.3) becomes

$$\operatorname{Re}k \operatorname{Im}k \int_0^\infty |\varphi(\lambda,k,r)|^2\,dr$$
$$= \operatorname{Re}\lambda \operatorname{Im}\lambda \int_0^\infty [|\varphi(\lambda,k,r)|^2/r^2]\,dr \qquad (\operatorname{Im}k > 0,\ \operatorname{Re}\lambda > 0). \tag{7.4.4}$$

Thus $\operatorname{Re}k$ and $\operatorname{Im}\lambda$, if $\neq 0$, must have the same sign.

If $\operatorname{Im}\lambda = 0$, (7.4.4) implies $\operatorname{Re}k = 0$; i.e., *for all real $\lambda$, the poles of $S(\lambda,k)$ in $I_+(k)$ can only be located on the imaginary axis* (corresponding to bound states for physical $\lambda$). This is an extension of the result previously obtained for physical $\lambda$ (cf. Theorem 5.8.1).

If $\operatorname{Im}k = 0$, we have to go back to (7.4.3), which, taking into account (7.3.4), becomes

$$\operatorname{Re}\lambda \operatorname{Im}\lambda \int_0^\infty [|\varphi(\lambda,k,r)|^2/r^2]\,dr$$
$$= (1/8k)[|f(\lambda,k)|^2 - |f(\lambda,-k)|^2] \qquad (\operatorname{Im}k = 0,\ \operatorname{Re}\lambda > 0). \tag{7.4.5}$$

At a pole of $S(\lambda,k)$, the last term in square brackets in (7.4.5) vanishes, and we see that $\operatorname{Im}\lambda$ must have the same sign as $k$.

Finally, if $\operatorname{Re}k = 0$, $\operatorname{Im}k > 0$, it follows from (7.4.4) that $\operatorname{Im}\lambda = 0$, unless $\operatorname{Re}\lambda = 0$. However, we can exclude $\operatorname{Re}\lambda = 0$, except possibly at $\lambda = 0$. In fact, if $k = iK$, $\lambda = iL$ is a pole of $S(\lambda,k)$ with $\operatorname{Re}k = \operatorname{Re}\lambda = 0$, we have $f(iL, -iK) = 0$. It then follows from (7.3.6) that $f(-iL, -iK) = 0$ as well. On the other hand, for purely imaginary $\lambda$, $\varphi(-\lambda,k,r)$ is defined, and we can define, by analogy with (7.3.1), taking into account (7.2.27),

$$f(-\lambda,k) = W[f(\lambda,k,r),\ \varphi(-\lambda,k,r)] \qquad (\operatorname{Re}\lambda = 0). \tag{7.4.6}$$

It then follows, as for (7.3.4), that

$$\varphi(-\lambda,k,r) = (1/2ik)[f(-\lambda,k)f(\lambda,-k,r)-f(-\lambda,-k)f(\lambda,k,r)] \quad (\operatorname{Re}\lambda = 0). \tag{7.4.7}$$

Substituting (7.3.4) and (7.4.7) in (7.2.17), and taking into account (7.2.29), we get the identity

$$f(\lambda,-k)f(-\lambda,k)-f(\lambda,k)f(-\lambda,-k) = 4i\lambda k \qquad (\operatorname{Re}\lambda = 0). \tag{7.4.8}$$

Applying this to the case under consideration, in which $k = iK\ (K > 0)$, $\lambda = iL$ is a pole, we conclude that $4KL = 0$; i.e., $L = 0$. Therefore, poles with $\operatorname{Re}k = 0$ and $\operatorname{Im}k > 0$ must have $\operatorname{Im}\lambda = 0$.

The above results may be summed up as follows:

THEOREM 7.4.1. *The poles of* $S(\lambda,k)$ *in the analyticity domain for* $\operatorname{Im}k \geqslant 0$, $\operatorname{Re}\lambda \geqslant 0$, *can only be located in the following regions:*

$$\text{(I)}\quad \operatorname{Im}k \geqslant 0,\quad \operatorname{Re}k > 0,\quad \operatorname{Im}\lambda > 0,\quad \operatorname{Re}\lambda \geqslant 0, \tag{7.4.9}$$

$$\text{(II)}\quad \operatorname{Im}k \geqslant 0,\quad \operatorname{Re}k < 0,\quad \operatorname{Im}\lambda < 0,\quad \operatorname{Re}\lambda \geqslant 0, \tag{7.4.10}$$

$$\text{(III)}\quad \operatorname{Re}k = 0,\quad \operatorname{Im}k > 0,\quad \operatorname{Im}\lambda = 0,\quad \operatorname{Re}\lambda \geqslant 0. \tag{7.4.11}$$

For $\operatorname{Im}k = 0$, these results have a simple physical interpretation. In this case, according to (7.4.1), $\varphi(\lambda,k,r)$ is a stationary solution of the Schrödinger equation, regular at the origin, which is purely outgoing (incoming) for $k > 0$ ($k < 0$). Therefore, the effective potential must act as a source for $k > 0$ (as a sink for $k < 0$). The imaginary part of the effective potential arises from the centrifugal term $(\lambda^2 - \frac{1}{4})/r^2$, so that $\operatorname{Im}\lambda$ must have the same sign as $k$. Actually, the derivation of (7.4.3) is analogous to that of the continuity equation.

Finally, let us note that, by (7.3.11), we may rewrite (7.4.5) as

$$\operatorname{Re}\lambda \operatorname{Im}\lambda \int_0^\infty [|\varphi(\lambda,k,r)|^2/r^2]\,dr = [|f(\lambda,-k)|^2/8k][|e^{-i\pi\lambda}S(\lambda,k)|^2 - 1] \qquad (\operatorname{Im}k = 0,\ \operatorname{Re}\lambda > 0). \tag{7.4.12}$$

Therefore, taking $k > 0$, for definiteness, it follows that

$$|S(\lambda,k)| < \exp(\pi|\operatorname{Im}\lambda|) \qquad (k > 0,\ \operatorname{Re}\lambda > 0,\ \operatorname{Im}\lambda < 0), \tag{7.4.13}$$

which provides a bound for $S(\lambda,k)$ in the pole-free region $k > 0$, $\operatorname{Im}\lambda < 0$.

### *(b) Yukawa-Type Potentials*

We now discuss the additional restrictions on the Regge pole distribution that are found in the special case of Yukawa-type potentials. We restrict ourselves to physical values of $k$, $k > 0$, although the results can be extended to complex $k$ [16, 17].

The assumptions about the potential are the same as in Theorem 7.3.4. It follows that we can consider the analytic continuation of the regular solution for complex $r$ along any ray in the right half-plane. In particular, along the imaginary $r$-axis, let

$$\varphi(\lambda, k, i\rho) = \chi(\lambda, k, \rho). \tag{7.4.14}$$

According to (7.3.25) with $\theta = \pi/2$, the function $\chi$ satisfies the equation

$$\{d^2/d\rho^2 - k^2 - [(\lambda^2 - \tfrac{1}{4})/\rho^2] + U(i\rho)\}\,\chi(\lambda, k, \rho) = 0. \tag{7.4.15}$$

At a Regge pole $\lambda$, we have,[6] according to (7.4.1),

$$\chi(\lambda, k, \rho) = O(e^{-k\rho}) \qquad (\rho \to \infty). \tag{7.4.16}$$

It then follows, by repeated partial integrations, that

$$\int_0^\infty \chi^* \, d^2\chi/d\rho^2 \, d\rho = -\int_0^\infty |d\chi/d\rho|^2 \, d\rho = -\int_0^\infty |(d\chi/d\rho) - (\chi/2\rho)|^2 \, d\rho$$
$$-\tfrac{1}{4}\int_0^\infty |\chi|^2/\rho^2 \, d\rho, \tag{7.4.17}$$

where the contributions from the integrated parts vanish at $\rho = 0$ and at $\rho \to \infty$ because of (7.2.3) (with $\operatorname{Re}\lambda > 0$) and (7.4.16).

Multiplying (7.4.15) by $\chi^*(\lambda, k, \rho)$, integrating with respect to $\rho$ from 0 to $\infty$, and taking into account (7.4.17) we get, at a Regge pole $\lambda$,

$$\lambda^2 \int_0^\infty |\chi|^2/\rho^2 \, d\rho = \int_0^\infty U(i\rho)|\chi|^2 \, d\rho - k^2 \int_0^\infty |\chi|^2 \, d\rho$$
$$- \int_0^\infty |(d\chi/d\rho) - (\chi/2\rho)|^2 \, d\rho. \tag{7.4.18}$$

The restrictions on the pole distribution that follow from (7.4.18) depend[7] on the value of $\gamma$ in (7.3.28).

If we assume that

$$|U(i\rho)| < C/\rho^2, \tag{7.4.19}$$

where $C$ is a constant, it follows from (7.4.18) that

$$\operatorname{Re}(\lambda^2) = (\operatorname{Re}\lambda)^2 - (\operatorname{Im}\lambda)^2 < C, \qquad \operatorname{Im}(\lambda^2) = 2\operatorname{Re}\lambda \operatorname{Im}\lambda < C,$$

[6] For $\gamma > 1$ in (7.3.28), the asymptotic behavior (7.4.1) remains valid along the imaginary $r$-axis (cf. Newton [18, p. 332]). For $\gamma = 1$, the bound (7.3.28) along the imaginary $r$-axis corresponds to the Coulomb potential. However, in view of the oscillatory nature of the potential in this case [cf. the remarks following (7.3.28)], the result (7.4.16) remains valid (cf. Mott and Massey [10, p. 23]).

[7] The present derivation follows Martin [19, p.55].

and, in the first quadrant of the $\lambda$-plane [cf. (7.4.9)], this leads to

$$0 \leqslant \operatorname{Re} \lambda < [(\sqrt{2}+1)C/2]^{1/2}. \tag{7.4.20}$$

On the other hand, if we assume that

$$|U(i\rho)| < C/\rho, \tag{7.4.21}$$

which includes the case of the pure Yukawa potential (7.3.22), and if we make use of Schwarz's inequality

$$\int_0^\infty |\chi|^2/\rho \, d\rho \leqslant \left(\int_0^\infty |\chi^2|/\rho^2 \, d\rho\right)^{1/2} \left(\int_0^\infty |\chi|^2 \, d\rho\right)^{1/2}, \tag{7.4.22}$$

it follows from (7.4.18) that

$$\operatorname{Re}(\lambda^2) < C\zeta - k^2\zeta^2, \qquad \operatorname{Im}(\lambda^2) < C\zeta, \tag{7.4.23}$$

where we have set

$$\zeta = \left(\int_0^\infty |\chi|^2 \, d\rho\right)^{1/2} \bigg/ \left(\int_0^\infty |\chi|^2/\rho^2 \, d\rho\right)^{1/2}. \tag{7.4.24}$$

Eliminating $\operatorname{Im} \lambda$ between the inequalities (7.4.23), in the first quadrant of the $\lambda$-plane, we find

$$\begin{aligned}(\operatorname{Re} \lambda)^2 &< \frac{\zeta}{2}\left[(C-k^2\zeta)+((C-k^2\zeta)^2+C^2)^{1/2}\right] \\ &= \frac{C^2}{2k^2 + (2/\zeta)\{[C^2+(C-k^2\zeta)^2]^{1/2}-C\}} \\ &< \frac{C^2}{2k^2},\end{aligned}$$

so that

$$0 \leqslant \operatorname{Re} \lambda < C/(\sqrt{2}k). \tag{7.4.25}$$

From (7.4.20) and (7.4.25) we see that, in both cases (7.4.19) and (7.4.21), the real part of Regge poles in the first quadrant of the $\lambda$-plane is bounded from above, except possibly for $k \to 0$ [when $k \to 0$, the parameter $\zeta$ in (7.4.24) may become unbounded], so that *there is a half-plane* $\operatorname{Re} \lambda > \Lambda$ *which is free of poles*. In the case of (7.4.25), the whole right half-plane $\operatorname{Re} \lambda > 0$ is free of poles in the limit as $k \to \infty$.

A more detailed analysis of the pole distribution, based on estimates of the asymptotic behavior of the Jost solutions, employing an extension of Martin's techniques described below in Section 7.5 and in Appendix H, has been carried out by Bessis [17, 20]. His assumptions about the potential are similar to

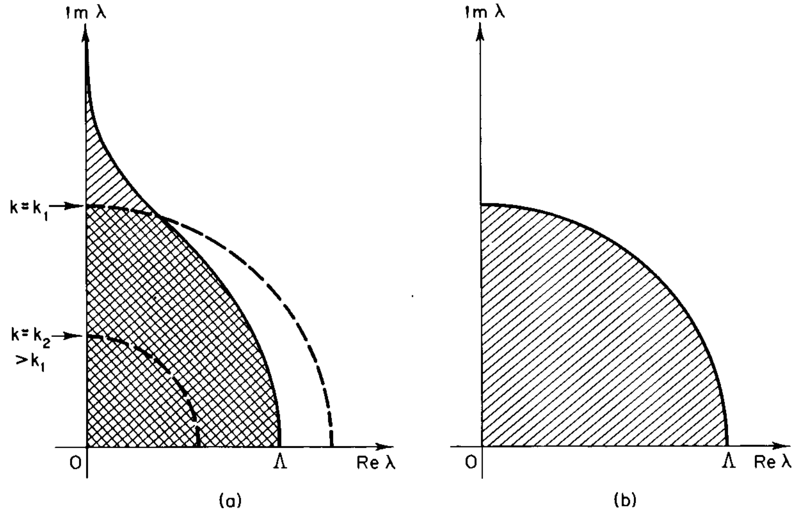


*FIG. 7.2.* For potentials satisfying (7.3.21), (7.4.26), and (7.4.27), all Regge poles in the right half-plane, for all values of $k$, must be located within the shaded regions: (a) corresponds to (7.4.29), $1 < \gamma \leqslant 2$, and (b) corresponds to (7.4.30), $\gamma > 2$. In (a), the poles must also be contained in the circle $|\lambda|^{\gamma-1}k^{2-\gamma} \leqslant D(\gamma)$, so that, for $k = k_1$, they are restricted to the cross-hatched region; $k = k_2 > k_1$ is also shown. The radius of the circle shrinks to zero as $k \to \infty$ (after D. Bessis [17, 20]).

those of Theorem 7.3.4: it is assumed that the potential satisfies (7.3.21) and that, in the right half of the $r$-plane,

$$|U(\rho e^{i\theta})| < K/\rho^{\delta} \qquad (\rho \leqslant 1,\ \delta < 2), \tag{7.4.26}$$

$$|U(\rho e^{i\theta})| < Ke^{-m\rho\cos\theta}/\rho^{\gamma} \qquad (\rho > 1, \gamma \geqslant 1) \tag{7.4.27}$$

corresponding to the finiteness of $M_1$ and to (7.3.23).

Let

$$\lambda = |\lambda|e^{i\psi}. \tag{7.4.28}$$

Then, according to Bessis's results, the Regge poles in the first quadrant of the $\lambda$-plane can only be located in the following regions:

$$0 \leqslant \operatorname{Re}\lambda \leqslant \Lambda(\cos\psi)^{\gamma-1} \qquad \text{if} \quad 1 \leqslant \gamma < 2, \tag{7.4.29}$$

$$|\lambda| \leqslant \Lambda \qquad \text{if} \quad \gamma > 2, \tag{7.4.30}$$

where $\Lambda = \Lambda(U)$ is a constant depending only on the potential, i.e., *independent of $k$*, so that these bounds remain valid for $k \to +0$. The shape of these regions is shown[8] in Fig. 7.2. Furthermore, for $1 < \gamma < 2$, it is found that the poles

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[8] Although (7.4.29) remains valid for $0 < \gamma < 1$, it leads to a domain in which $\operatorname{Re}\lambda$ is unbounded, contrary to what is shown in Fig. 2 of Bessis [20].

must also be contained within the domain $|\lambda|^{\gamma-1} \leqslant D(\gamma)/k^{2-\gamma}$, where $D(\gamma)$ is a constant, so that they have to be within the intersection of this circle with the region (7.4.29) shown in Fig. 7.2a. Since the radius of the circle shrinks to zero as $k \to \infty$, this result agrees with that obtained from (7.4.25).

In any case, *the real part of all Regge poles is bounded for all $k$*, $0 \leqslant k < \infty$. This is one of the results that we wanted to prove, contained in condition B of Section 7.1. Since $S(\lambda, k)$ is meromorphic for $\operatorname{Re}\lambda > 0$, it cannot have an accumulation point of poles at finite distance. Therefore, in order to prove the remainder of condition B, it suffices to show that the poles cannot cluster along the imaginary $\lambda$-axis. This will follow from the asymptotic behavior of $S(\lambda, k)$ as $\lambda \to i\infty$, discussed in the next section.

## 7.5. Asymptotic Behavior of $S(\lambda, k)$ as $|\lambda| \to \infty$

### *(a) Asymptotic Behavior along the Real Axis*

To find the asymptotic behavior of $S(\lambda, k)$ as $|\lambda| \to \infty$ in the right half-plane, it will be sufficient to consider this behavior along two directions: the real axis and the negative imaginary $\lambda$-axis. With the help of the Phragmén–Lindelöf theorem, one may then bound the asymptotic behavior in the whole lower right quadrant of the $\lambda$-plane, and the extended unitarity condition (7.3.13) enables us to extend the bound to the upper right quadrant. As in Section 7.4(b), we restrict the discussion throughout to physical values of $k$, $k > 0$, although complex values of $k$ have also been considered.

We begin with the asymptotic behavior along the real $\lambda$-axis, $\lambda \to \infty$. For physical values of $\lambda$, it was already mentioned in Section 5.8(b) that the phase shifts in this case are bounded by Born's approximation, leading to (5.8.29). We now give the proof of this result for any $\lambda$.

Following Martin [21], we consider the "physical solution" $\psi(\lambda, k, r)$ of (7.2.1), which is regular at $r = 0$ and has the asymptotic behavior [cf. (2.6.2) and (2.6.4)]

$$\psi(\lambda, k, r) \approx \exp[i\eta(\lambda, k)] \sin[kr - (\lambda - \tfrac{1}{2})(\pi/2) + \eta(\lambda, k)] \qquad (r \to \infty). \tag{7.5.1}$$

By comparison with (7.3.10) and (7.3.11), we see that

$$\psi(\lambda, k, r) = k \exp[i(\pi/2)(\lambda - \tfrac{1}{2})]\, \varphi(\lambda, k, r)/f(\lambda, -k). \tag{7.5.2}$$

For free particles ($U = 0$, $\eta = 0$), we have

$$\psi_0(\lambda, k, r) = (\pi k r/2)^{1/2} J_\lambda(kr). \tag{7.5.3}$$

To obtain an integral equation for $\psi(\lambda, k, r)$, we consider the Green function

$$G(\lambda, k, r, r') = -(i\pi/2)(rr')^{1/2} J_\lambda(kr_<) H_\lambda^{(1)}(kr_>), \qquad [r_< = \min(r, r'),$$
$$r_> = \max(r, r')], \tag{7.5.4}$$

which generalizes the well-known expression[9] appearing in the multipole expansion of the free-space Green function. It satisfies the free radial equation for $r \neq r'$; it is regular at the origin and purely outgoing at infinity; furthermore,

$$(dG/dr)_{r=r'+0} - (dG/dr)_{r=r'-0} = (i\pi kr/2)\, W[H_\lambda^{(1)}(kr), J_\lambda(kr)] = 1,$$

so that [cf. (A4.4)]

$$\{(d^2/dr^2) + k^2 - [(\lambda^2 - \tfrac{1}{4})/r^2]\}\, G(\lambda, k, r, r') = \delta(r - r'). \tag{7.5.5}$$

Since the "scattered wave" $\psi_s = \psi - \psi_0$ must be regular at the origin and purely outgoing at infinity, we therefore get the integral equation

$$\psi(\lambda, k, r) = \psi_0(\lambda, k, r) + \int_0^\infty G(\lambda, k, r, r')\, U(r')\, \psi(\lambda, k, r')\, dr'. \tag{7.5.6}$$

Comparing the asymptotic form of (7.5.6) as $r \to \infty$ with (7.5.1), we get

$$\begin{aligned}(1/2i)[S(\lambda, k) - 1] &= \exp[i\eta(\lambda, k)] \sin \eta(\lambda, k) \\ &= -(\pi/2k)^{1/2} \int_0^\infty \sqrt{r}\, J_\lambda(kr)\, U(r)\, \psi(\lambda, k, r)\, dr.\end{aligned} \tag{7.5.7}$$

For sufficiently large $\lambda$, the following inequality is valid [21; 5, p. 190]:

$$|J_\lambda(kr_<) H_\lambda^{(1)}(kr_>)| < 1/\sqrt{\lambda}. \tag{7.5.8}$$

It then follows from (7.5.6) and (7.5.4) that

$$|\psi(\lambda, k, r)| < |\psi_0(\lambda, k, r)| + \frac{\pi}{2}\left(\frac{r}{\lambda}\right)^{1/2} \int_0^\infty \sqrt{r'}\, |U(r')|\, |\psi(\lambda, k, r')|\, dr'. \tag{7.5.9}$$

Multiplying both sides by $\sqrt{r}\,|U(r)|$ and integrating from $r = 0$ to $\infty$, we get

$$\int_0^\infty \sqrt{r}\, |U(r)\, \psi(\lambda, k, r)|\, dr < \frac{\int_0^\infty \sqrt{r}\, |U(r)\, \psi_0(\lambda, k, r)|\, dr}{1 - (\pi M_1/2\sqrt{\lambda})}, \tag{7.5.10}$$

where $M_1$ is given by (6.2.2), and $\lambda$ is assumed to be sufficiently large for the denominator to be positive.

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[9] See, for example, Jackson [22, p. 541].

Substituting (7.5.10) back in (7.5.9), and substituting the result in (7.5.7), we get, with the help of (7.5.3),

$$\frac{1}{2}|S(\lambda,k)-1| = |\sin\eta(\lambda,k)| < \frac{\pi}{2}\int_0^\infty r|U(r)|J_\lambda{}^2(kr)\,dr$$
$$+\frac{\pi}{2}\cdot\frac{(\pi/2\sqrt{\lambda})(\int_0^\infty r|U(r)J_\lambda(kr)|\,dr)^2}{1-(\pi M_1/2\sqrt{\lambda})}. \tag{7.5.11}$$

By Schwarz's inequality,

$$\left(\int_0^\infty r|U(r)J_\lambda(kr)|\,dr\right)^2 \leqslant \int_0^\infty r|U(r)|\,dr\int_0^\infty r|U(r)|\,J_\lambda{}^2(kr)\,dr, \tag{7.5.12}$$

so that, taking into account (6.2.2), the result (7.5.11) becomes

$$\frac{1}{2}|S(\lambda,k)-1| = |\sin\eta(\lambda,k)| < \frac{(\pi/2)\int_0^\infty r|U(r)|J_\lambda{}^2(kr)\,dr}{1-(\pi M_1/2\sqrt{\lambda})}. \tag{7.5.13}$$

The numerator, according to (7.5.7) and (7.5.3), is Born's approximation for the potential $|U(r)|$. Therefore, *for large enough* $\lambda$ *and* $M_1<\infty$, *the absolute value of the phase shift is always bounded by Born's approximation*, which is the result mentioned in Section 5.8(b).

On the other hand, if [cf. (7.3.23)]

$$|U(r)| < Ke^{-mr}/r, \tag{7.5.14}$$

it follows from (7.5.13), as in (5.8.28), that

$$|S(\lambda,k)-1| < \frac{(K/k)\,Q_{\lambda-\frac{1}{2}}[1+(m^2/2k^2)]}{1-(\pi K/2m\sqrt{\lambda})} \tag{7.5.15}$$

for large enough $\lambda$, where $Q_{\lambda-\frac{1}{2}}$ is the Legendre function of the second kind. We have [23, p. 136, Eq. (45)]

$$Q_{\lambda-\frac{1}{2}}(\cosh\chi) = \sqrt{\pi}\,\frac{\Gamma(\lambda+\frac{1}{2})}{\Gamma(\lambda+1)}\,e^{-\chi(\lambda+\frac{1}{2})}F(\tfrac{1}{2},\,\lambda+\tfrac{1}{2};\,\lambda+1;\,e^{-2\chi}), \tag{7.5.16}$$

where $F$ denotes the hypergeometric function. On the other hand, we have [24, p. 61; 24a]

$$|F(\tfrac{1}{2},\,\lambda+\tfrac{1}{2};\,\lambda+1;\,\zeta)| \leqslant |1-\zeta|^{-\frac{1}{2}} \qquad (\operatorname{Re}\lambda \geqslant -\tfrac{3}{4},\,|\zeta|<1). \tag{7.5.17}$$

Combining these results with (7.5.15), we obtain, as in (5.8.29),

$$|S(\lambda,k)-1| = O(\lambda^{-\frac{1}{2}}e^{-\chi\lambda}) \qquad (\lambda\to\infty), \tag{7.5.18}$$

where

$$\cosh\chi = 1+(m^2/2k^2). \tag{7.5.19}$$

### (b) Behavior in Other Directions

As remarked at the beginning of Section 7.5(a), it is now sufficient to consider the asymptotic behavior of $S(\lambda,k)$ along the imaginary $\lambda$-axis. It is not difficult to show [5, p. 86], with the help of (7.5.18) and (7.4.13), that $S(\lambda,k)\to 1$ as $|\lambda|\to\infty$, $|\arg\lambda|<\pi/2$. However, in order to perform the Watson transformation [cf. (7.1.6)], we need the asymptotic behavior also for $|\arg\lambda|=\pi/2$.

This is the most difficult step in the derivation of the Watson transformation. Several methods have been proposed to deal with this problem, including the variable phase approach [25], Langer's method [26], and the Smithies theory of Fredholm integral equations [27]. Here we give an outline of Martin's approach [19, p. 58; 21], which is an extension of his method for real $\lambda$, described in Section 7.5(a).

We again restrict ourselves to $k>0$. It is convenient to employ, instead of (7.5.6), a Volterra integral equation for a function $\tilde{\psi}(\lambda,k,r)$ that still reduces to (7.5.3) in the absence of a potential. The integral equation that defines $\tilde{\psi}$ is analogous to (7.2.14); since the kernels of (7.2.14) and (7.2.26) are equivalent, it may be written as

$$\tilde{\psi}(\lambda,k,r) = \psi_0(\lambda,k,r)+\int_0^r G(\lambda,k,r,r')\,U(r')\,\tilde{\psi}(\lambda,k,r')\,dr', \quad (7.5.20)$$

where $\psi_0$ is given by (7.5.3) and

$$G(\lambda,k,r,r') = (i\pi/2)(rr')^{1/2}[J_\lambda(kr)\,H_\lambda^{(1)}(kr')-J_\lambda(kr')\,H_\lambda^{(1)}(kr)]. \quad (7.5.21)$$

Letting $r\to\infty$ and employing the asymptotic form of the cylindrical functions, we find that (7.5.20) yields

$$\tilde{\psi}(\lambda,k,r) \approx [1+iB(\lambda,k)]\sin[kr-(\lambda-\tfrac{1}{2})(\pi/2)] - A(\lambda,k) \times \exp\{i[kr-(\lambda-\tfrac{1}{2})(\pi/2)]\} \qquad (r\to\infty), \quad (7.5.22)$$

where

$$A(\lambda,k) = (\pi/2k)^{1/2}\int_0^\infty \sqrt{r}\,J_\lambda(kr)\,U(r)\,\tilde{\psi}(\lambda,k,r)\,dr, \quad (7.5.23)$$

$$B(\lambda,k) = (\pi/2k)^{1/2}\int_0^\infty \sqrt{r}\,H_\lambda^{(1)}(kr)\,U(r)\,\tilde{\psi}(\lambda,k,r)\,dr. \quad (7.5.24)$$

Comparing (7.5.23) with (7.5.1), we get

$$(1/2i)[S(\lambda,k)-1] = -A(\lambda,k)/[1+iB(\lambda,k)]. \quad (7.5.25)$$

We also see that $\psi$ and $\tilde{\psi}$ are related by

$$\psi(\lambda,k,r)=\frac{\tilde{\psi}(\lambda,k,r)}{1+iB(\lambda,k)}=\frac{\tilde{\psi}(\lambda,k,r)}{1+i(\pi/2k)^{1/2}\int_0^\infty\sqrt{r}\,H_\lambda^{(1)}(kr)\,U(r)\,\tilde{\psi}(\lambda,k,r)\,dr}\tag{7.5.26}$$

so that (7.5.7) and (7.5.25) are equivalent.

We now set

$$\lambda=i\nu\qquad(\nu>0).\tag{7.5.27}$$

By analogy with (7.5.6) to (7.5.9), we look for bounds for $\psi_0(i\nu,k,r)$ and $G(i\nu,k,r,r')$ in (7.5.20) that become small for large $\nu$, and, furthermore, such that the bound for $G$ is *separable* in $r$ and $r'$. With such bounds, (7.5.20) gives rise to a separable inequality, like (7.5.9), that can be integrated to provide an inequality for $|\tilde{\psi}(i\nu,k,r)|$.

To achieve this aim, according to (7.5.3) and (7.5.21), we need bounds for $H_{i\nu}^{(1,2)}(kr)$ [which imply a bound also for $J_{i\nu}(kr)$] that become small for large $\nu$. For real $r$, this is not possible, because $H_{i\nu}^{(1)}(kr)$ then diverges for large $\nu$.[10] However, restricting ourselves to Yukawa-type potentials, we can take advantage of the fact that $U(r)$, as well as $\tilde{\psi}(i\nu,k,r)$ [cf. (7.5.26), (7.5.2), and Section 7.3(b)], can be analytically continued to $\operatorname{Re} r>0$. This enables us to shift the path of integration in the complex $r$-plane in (7.5.20), (7.5.23), and (7.5.24). The problem is then reduced to finding a path $\Gamma(\nu)$ in the half-plane $\operatorname{Re} r>0$, extending from $r=0$ to $|r|\to\infty$, along which $|H_{i\nu}^{(1)}(kr)|$ and $|H_{i\nu}^{(2)}(kr)|$ are small for large $\nu$.

The solution of this problem is given in Appendix H. The path $\Gamma(\nu)$ runs from $r=0$ to $r=i\nu/k$ along the positive imaginary axis, and from $r=i\nu/k$ to $r=(i\pi\nu/2k)+\infty$ along the curve defined by (H16) and (H17); it is represented by the curve in thick line in Fig. H1. Along this path, $H_{i\nu}^{(1)}$ and $H_{i\nu}^{(2)}$ are bounded by (H18) and (H19).

Let us now consider the integral equation (7.5.20) for $\lambda=i\nu$ and complex $r$, such that $kr$ lies on the path $\Gamma(\nu)$ and the integration in (7.5.20) is performed along this path. It then follows from (7.5.21) and (H18) that

$$|G(i\nu,k,r,r')|<\frac{2C^2|r|^{1/2}|r'|^{1/2}}{|\nu^2+(kr)^2|^{1/4}|\nu^2+(kr')^2|^{1/4}}\qquad[(kr,kr')\in\Gamma(\nu)],\tag{7.5.28}$$

which is small for large $\nu$ and separable in $r$ and $r'$; furthermore, by (7.5.3) and (H18),

$$|\psi_0(i\nu,k,r)|<C|kr|^{1/2}/|\nu^2+(kr)^2|^{1/4}\qquad[kr\in\Gamma(\nu)].\tag{7.5.29}$$

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[10] See, for example, Nussenzveig [28, Appendix A].

Note that (H19) enables us to improve these bounds near $kr=i\nu$, $kr'=i\nu$; however, this will not be necessary.

Substituting (7.5.28) and (7.5.29) in (7.5.20), and setting

$$\Psi(i\nu,k,r) = \{[\nu^2+(kr)^2]^{1/4}/C(kr)^{1/2}\}\,\tilde{\psi}(i\nu,k,r), \tag{7.5.30}$$

we get

$$|\Psi(i\nu,k,r)| < 1 + 2C^2\int_0^r \frac{|r'U(r')|}{|\nu^2+(kr')^2|^{1/2}}|\Psi(i\nu,k,r')|\,dr' \qquad [kr\in\Gamma(\nu)]. \tag{7.5.31}$$

By applying a simple extension of a lemma due to Titchmarsh,[11] it follows from (7.5.31) that

$$|\Psi(i\nu,k,r)| < \exp\left[2C^2\int_0^r \frac{|r'U(r')|}{|\nu^2+(kr')^2|^{1/2}}dr'\right] \qquad [kr\in\Gamma(\nu)]; \tag{7.5.32}$$

i.e., by (7.5.30),

$$\begin{aligned}|\tilde{\psi}(i\nu,k,r)| &< \frac{C|kr|^{1/2}}{|\nu^2+(kr)^2|^{1/4}}\exp\left[2C^2\int_0^r \frac{|r'U(r')|}{|\nu^2+(kr')^2|^{1/2}}dr'\right]\\ &< \frac{C|kr|^{1/2}}{|\nu^2+(kr)^2|^{1/4}}\exp[2C^2 I(\nu,k)] \qquad [kr\in\Gamma(\nu)],\end{aligned} \tag{7.5.33}$$

where

$$\begin{aligned}I(\nu,k) &= \int_0^{\infty+(i\pi\nu/2k)} \frac{|rU(r)|}{|\nu^2+(kr)^2|^{1/2}}dr\\ &= \frac{1}{k^2}\int_{\Gamma(\nu)} \frac{|\rho U(\rho/k)|}{|\nu^2+\rho^2|^{1/2}}d\rho\\ &= \frac{\nu}{k^2}\int_{\Gamma(1)} \frac{|tU(\nu t/k)|}{|1+t^2|^{1/2}}dt.\end{aligned} \tag{7.5.34}$$

In the last integral of (7.5.34), we have made use of the scaling property of the path $\Gamma(\nu)$ mentioned in Appendix H; namely, that its parametric equations depend only on the variable $kr/\nu = t$; $\Gamma(1)$ denotes the ($\nu$-independent) path obtained from (H16) and (H17) by setting $\nu=1$.

Now let us go back to (7.5.23) and (7.5.24), and let us shift the path of integration in these integrals so that $kr$ lies along $\Gamma(\nu)$. This is possible because $|\sqrt{r}J_{i\nu}(kr)|$, $|\sqrt{r}H_{i\nu}^{(1)}(kr)|$, and $|\tilde{\psi}(i\nu,k,r)|$ all remain bounded along a path from $r=\infty$ to $r=\infty+(i\pi\nu/2k)$ [cf. (7.5.22)], whereas $U(r)$ is exponentially

[11] See Titchmarsh [29, p. 97]. The result can also be derived by iterating (7.5.31) repeatedly.

decreasing for $\operatorname{Re} r \to \infty$; thus, the contribution from this path vanishes. Substituting (7.5.33) and (H18) in the integrals along $\Gamma(\nu)$, we get

$$\left\{\begin{matrix} |A(i\nu,k)| \\ |B(i\nu,k)| \end{matrix}\right\} < C^2 \exp[2C^2 I(\nu,k)] \cdot \frac{1}{k^2} \int_{\Gamma(\nu)} \frac{|\rho U(\rho/k)|}{|\nu^2+\rho^2|^{1/2}}\, d\rho;$$

i.e., by (7.5.34),

$$|A(i\nu,k)| < C^2 I(\nu,k) \exp[2C^2 I(\nu,k)], \tag{7.5.35}$$

$$|B(i\nu,k)| < C^2 I(\nu,k) \exp[2C^2 I(\nu,k)]. \tag{7.5.36}$$

The problem is thus reduced to investigating the behavior of $I(\nu,k)$ when $\nu \to \infty$. The simplest case to treat is that in which we assume the following bound for the potential:

$$|U(r)| < K/|r|^{\gamma} \qquad (1 < \gamma < 2,\ \operatorname{Re} r \geqslant 0). \tag{7.5.37}$$

It then follows from (7.5.34) that

$$I(\nu,k) < \frac{K}{k^{2-\gamma}\nu^{\gamma-1}} \int_{\Gamma(1)} \frac{dt}{|t|^{\gamma-1}|1+t^2|^{1/2}}. \tag{7.5.38}$$

Since the integral is convergent, it follows that $I(\nu,k) \to 0$ as $\nu \to \infty$, and (7.5.25), (7.5.35), and (7.5.36) yield

$$S(i\nu,k) - 1 = O(\nu^{1-\gamma}) \qquad (1 < \gamma < 2,\ \nu \to \infty), \tag{7.5.39}$$

so that $S$ approaches 1 also along the positive imaginary axis.

The restriction to $\gamma < 2$ in (7.5.37) arises from the requirement that $M_1 < \infty$ on the behavior of $U(r)$ near $r = 0$, and from employing a single bound both for small and for large $|r|$. If we separate these two bounds, by assuming that the potential satisfies (7.4.26) and (7.4.27), it is readily seen, by splitting the integral in (7.5.34) accordingly, that [20]

$$I(\nu,k) < K'/\nu \qquad (\gamma > 2), \tag{7.5.40}$$

where $K'$ is a constant depending only on the potential. Thus, in this case also, $S(i\nu,k) - 1 \to 0$ as $\nu \to \infty$.

The most difficult case to treat is that of a pure Yukawa potential, in which $\gamma = 1$ in (7.4.27). In this case one finds only that $I(\nu,k)$ is bounded, and this does not even ensure the boundedness of $S(i\nu,k) - 1$, because the denominator in (7.5.25) might vanish. This case requires a more refined treatment, in which one makes use of the oscillatory character of $H_{i\nu}^{(1,2)}(kr)$ for pure imaginary $kr$ between 0 and $i\nu$ [cf. (H1) and (H2)], instead of relying only on the rougher bound (H18). For the details of this treatment the reader is referred to the original paper [21]; here we just quote the result, namely, that

$$S(i\nu,k) - 1 = O(\nu^{-1/2}) \qquad (\nu \to \infty, \quad \text{pure Yukawa potential}). \tag{7.5.41}$$

In all these cases, therefore, we have

$$\lim_{v\to\infty} S(iv,k) = 1 \qquad (k>0). \tag{7.5.42}$$

The extended unitarity condition (7.3.13) then implies also

$$\lim_{v\to\infty} S(-iv,k) = 1 \qquad (k>0). \tag{7.5.43}$$

Let us now consider the function $S(\lambda,k)$, $k>0$, in the lower right quadrant of the $\lambda$-plane, $-\pi/2\leqslant \arg\lambda \leqslant 0$. According to Theorems 7.3.4 and 7.4.1, $S(\lambda,k)$ is a holomorphic function of $\lambda$ in this quadrant. By (7.5.18) and (7.5.43), $S(\lambda,k)\to 1$ along the boundaries, i.e., for $\lambda\to\infty$ and for $\lambda\to -i\infty$. Finally, according to (7.4.13), $|S(\lambda,k)|<e^{\pi|\mathrm{Im}\lambda|}$ in this quadrant, so that $S(\lambda,k)$ is of order[12] at most equal to one. We can therefore apply the following consequence of the Phragmén–Lindelöf principle [30, p. 29]:

THEOREM 7.5.1. *Let $f(z)$ be regular and of order at most $\rho<\pi/\beta$ in the sector $\theta_1\leqslant \arg z\leqslant \theta_1+\beta$, $|z|\geqslant l$, and let $f(z)$ be bounded on the frontier of this sector. Suppose that $f(z)$ tends to $a$ and $b$ as $|z|\to\infty$ along* $\arg z=\theta_1$ *and* $\arg z=\theta_1+\beta$, *respectively. Then $a=b$ and $f(z)\to a$ uniformly as $|z|\to\infty$ in the sector.*

Applying Theorem 7.5.1 to $S(\lambda,k)$, with $\theta_1=-\pi/2=-\beta$, we conclude that $S(\lambda,k)\to 1$ as $|\lambda|\to\infty$, $-\pi/2\leqslant\arg\lambda\leqslant 0$. By the extended unitarity condition (7.3.13), the same is true for $0\leqslant\arg\lambda\leqslant\pi/2$. For a pure Yukawa potential, we can even improve this bound, by applying Theorem 7.5.1 to the function $\lambda^{1/2}e^{\chi\lambda}[S(\lambda,k)-1]$, which, according to (7.5.18) and (7.5.41), is bounded for $\lambda\to\infty$ and for $\lambda\to -i\infty$. We then conclude, in the same way as above, that the Born approximation bound (7.5.18) actually holds for $-\pi/2\leqslant\arg\lambda\leqslant\pi/2$. We have therefore:

THEOREM 7.5.2. *Let $U(r)$ be a potential holomorphic for* $\mathrm{Re}\, r>0$ *and bounded by (7.4.26) and (7.4.27), with $\gamma>1$. Then we have, for $k>0$,*

$$\lim_{|\lambda|\to\infty} S(\lambda,k) = 1 \qquad (-\pi/2\leqslant\arg\lambda\leqslant\pi/2). \tag{7.5.44}$$

*For a pure Yukawa potential (7.3.22), we have*

$$S(\lambda,k)-1 = O(\lambda^{-1/2}e^{-\chi\lambda}) \qquad (|\lambda|\to\infty,\ -\pi/2\leqslant\arg\lambda\leqslant\pi/2), \tag{7.5.45}$$

*where $\chi$ is defined by (7.5.19).*

It follows from (7.5.42) that the poles of $S(\lambda,k)$ in the upper right quadrant cannot cluster around $\lambda\to i\infty$; since there cannot be any accumulation point

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[12] A function $f(\lambda)$ is of order $\rho$ if $f(\lambda)=O[\exp(|\lambda|^{\rho+\varepsilon})]$ as $|\lambda|\to\infty$ for every $\varepsilon>0$, no matter how small, but not for any $\varepsilon<0$.

of poles at finite distance, the total number of poles in the right half-plane must be finite. Therefore, combining Theorem 7.5.2 with the results of Section 7.4, we get

THEOREM 7.5.3. *Under the assumptions of Theorem 7.5.2, the total number of poles of* $S(\lambda,k)$ *in* $\operatorname{Re}\lambda \geqslant 0$ *is finite. The poles can only be located in the upper right quadrant, and their real part is bounded for all* $k$, $0 \leqslant k < \infty$.

Theorems 7.3.4, 7.5.2, and 7.5.3 complete the derivation of properties A–C of Section 7.1. We can now also settle the problem of the uniqueness of the "interpolating" function $S(\lambda,k)$ mentioned in Section 7.1, with the help of the following theorem [31, p. 186]:

THEOREM 7.5.4 (Carlson's theorem). *If* $f(z)$ *is regular and is* $O(e^{c|z|})$, $c<\pi$, *for* $\operatorname{Re}z \geqslant 0$, *and* $f(z)=0$ *for* $z=0,1,2,\ldots$, *then* $f(z)=0$ *identically.*

Let $S(\lambda,k)$ and $S'(\lambda,k)$ be two interpolating functions verifying all the properties contained in Theorems 7.3.4, 7.5.2, and 7.5.3. It follows that, for a given value of $k$, both $S$ and $S'$ have only a finite number of poles in $\operatorname{Re}\lambda \geqslant 0$; let $(\lambda_1,\ldots,\lambda_N)$ be the poles of $S$ and $(\lambda_1{}',\ldots,\lambda_M{}')$ be those of $S'$ (each counted a number of times equal to its multiplicity). The function

$$f(z) = \prod_{i=1}^{N} (\lambda-\lambda_i) \prod_{j=1}^{M} (\lambda-\lambda_j{}')\,[S(\lambda,k)-S'(\lambda,k)], \tag{7.5.46}$$

where $z=\lambda-\frac{1}{2}$, is then regular for $\operatorname{Re}z \geqslant 0$; it is bounded by a power of $z$ as $|z|\to\infty$, because both $S$ and $S'$ must satisfy (7.5.44), and it vanishes at $z=0,1,2,\ldots$ by (7.1.8). Therefore, by Carlson's theorem 7.5.4 it vanishes identically. Thus we have:

THEOREM 7.5.5 (Uniqueness). *There exists only one function* $S(\lambda,k)$ *with the properties given in Theorems 7.3.4, 7.5.2, and 7.5.3.*

Note that other interpolations, e.g., $S(\lambda,k)+\sigma\cos(\pi\lambda)$, violate (7.5.44) as well as the assumptions of Carlson's theorem.

## 7.6. Watson Transformation and Analytic Continuation in $\cos\theta$.

We can now go back to the discussion in Section 7.1 about the legitimacy of the Watson transform expression (7.1.6) for the total scattering amplitude. We restrict ourselves, to begin with, to physical values of $k$ and $\theta$, such that

$$k>0, \qquad 0<\varepsilon \leqslant \theta \leqslant \pi-\varepsilon. \tag{7.6.1}$$

The justification of (7.1.6) hinges upon showing that

$$\int_{\Gamma} [S(\lambda,k)-1]P_{\lambda-\frac{1}{2}}(-\cos\theta)\frac{\lambda\, d\lambda}{\cos(\pi\lambda)} = 0, \tag{7.6.2}$$

where $\Gamma$ is the half-circle at infinity, shown in Fig. 7.1. The asymptotic behavior of $P_{\lambda-\frac{1}{2}}(-\cos\theta)$ as $|\lambda|\to\infty$, for $\theta$ in the domain (7.6.1), is given by [9, p. 71]

$$P_{\lambda-\frac{1}{2}}(-\cos\theta) = (2/\pi\lambda\sin\theta)^{\frac{1}{2}}\cos[\lambda(\pi-\theta)-(\pi/4)]\,[1+O(\lambda^{-1})],$$
$$(|\lambda|\to\infty, 0<\varepsilon\leqslant\theta\leqslant\pi-\varepsilon). \tag{7.6.3}$$

Since $\cos[\lambda(\pi-\theta)-(\pi/4)]/\cos(\pi\lambda)$ tends to zero at least like $\exp(-\varepsilon|\mathrm{Im}\,\lambda|)$ along $\Gamma$, the validity of (7.6.2) for Yukawa-type potentials then follows from Theorem 7.5.2.

The remaining requirements for the validity of (7.1.6) follow from Theorems 7.3.4 and 7.5.3. Thus, in the domain (7.6.1), we have completed the derivation of the *Watson transform* (7.1.6),

$$f(k,\cos\theta) = \frac{1}{2k}\int_{-\infty}^{\infty}[S(i\nu,k)-1]P_{i\nu-\frac{1}{2}}(-\cos\theta)\frac{\nu\, d\nu}{\cosh(\pi\nu)}$$
$$-\frac{i\pi}{k}\sum_{j=1}^{M}\frac{\sigma_j\lambda_j}{\cos(\pi\lambda_j)}P_{\lambda_j-\frac{1}{2}}(-\cos\theta), \tag{7.6.4}$$

where $\sigma_j$ is defined by (7.1.7) and $\lambda_j(k)$ $(j=1,2,\ldots,M)$ are the Regge poles of $S(\lambda,k)$ in the right half of the $\lambda$-plane. We adopt the convention of ordering them by increasing real part, so that $\lambda_M$ always denotes the pole with the largest real part (rightmost pole).

Having shown the equivalence between (7.6.4) and the partial-wave expansion for physical values of $k$ and $\cos\theta$, we can now proceed with the program formulated in Section 7.1, by employing the Watson transform to *define* the analytic continuation of $f(k,\cos\theta)$ in the $\cos\theta$ plane beyond the domain of convergence of the partial-wave expansion. Since $P_{\lambda-\frac{1}{2}}(z)$, for fixed $\lambda$, is a holomorphic function of $z$ in the $z$-plane cut along the real axis from $-\infty$ to $-1$, it follows from (7.6.4) that $f(k,\cos\theta)$ is holomorphic in any domain of the $\cos\theta$ plane, cut from $\cos\theta=1$ to $\infty$, in which the integral is uniformly convergent.

It follows from the relation

$$\pi\tan(\pi\lambda)P_{\lambda-\frac{1}{2}}(z) = Q_{-\lambda-\frac{1}{2}}(z) - Q_{\lambda-\frac{1}{2}}(z), \tag{7.6.5}$$

together with (7.5.16) and (7.5.17), that, for real or complex $\theta$ in the strip $0<\mathrm{Re}\,\theta<\pi$, we have

$$|P_{i\nu-\frac{1}{2}}(-\cos\theta)| \leqslant \frac{C\exp|\nu(\pi-\mathrm{Re}\,\theta)|}{|\nu\sin\theta|^{\frac{1}{2}}} \qquad (\nu\to\pm\infty, |\nu\sin\theta|\gg 1), \tag{7.6.6}$$

where $C$ is a constant. On the other hand, in the neighborhood of $\theta=\pi$, for $|\nu\sin\theta|\lesssim 1$, we have [3, Vol. II, p. 243]

$$P_{i\nu-1/2}(-\cos\theta) = J_0[2i\nu\cos(\theta/2)] + O(\nu^{-2}) \qquad (\nu\to\pm\infty, |\nu(\pi-\theta)|\lesssim 1), \tag{7.6.7}$$

which remains bounded and of order unity in this neighborhood.

It follows from (7.6.6) and (7.6.7) that

$$|P_{i\nu-1/2}(-\cos\theta)/\cosh(\pi\nu)| = O(e^{-\varepsilon|\nu|}) \qquad (\nu\to\pm\infty, 0<\varepsilon\leqslant \operatorname{Re}\theta\leqslant\pi). \tag{7.6.8}$$

Together with (7.5.42) and (7.5.43), this implies that the integral in (7.6.4) is indeed uniformly convergent in the strip $0<\varepsilon\leqslant\operatorname{Re}\theta\leqslant\pi$, for arbitrarily small $\varepsilon$. In the plane $\cos\theta=z=x+iy$, the line $\operatorname{Re}\theta=\varepsilon$ corresponds to the right branch of the hyperbola

$$(x^2/\cos^2\varepsilon) - (y^2/\sin^2\varepsilon) = 1,$$

so that the above strip corresponds to the whole $\cos\theta$ plane, cut along the real axis from $\cos\theta=1$ to $\infty$. The analytic continuation of $f(k,\cos\theta)$ defined by (7.6.4) is therefore holomorphic in the same domain. Actually, according to Section 6.3(c) and (6.3.41) (with $\xi=0$, corresponding to $\cos\theta$ real and $\geqslant 1$), there is no cut between $\cos\theta=1$ and $\cos\theta=1+m^2/4E$, where $m$ is the inverse range of the potential.

It also follows from (7.6.6) that

$$\int_{-\infty}^{\infty}[S(i\nu,k)-1]P_{i\nu-1/2}(-\cos\theta)\frac{\nu\,d\nu}{\cosh(\pi\nu)}$$
$$= O(|\cos\theta|^{-1/2}) \qquad (|\cos\theta|\to\infty, 0<\varepsilon\leqslant\operatorname{Re}\theta\leqslant\pi), \tag{7.6.9}$$

so that the background integral indeed goes to zero as $|\cos\theta|\to\infty$ in the cut plane (cf. Section 7.1). It then follows from (7.1.9) and (7.6.4) that, in agreement with (7.1.10),

$$f(k,\cos\theta) = O(|\cos\theta|^{\operatorname{Re}\lambda_M-1/2}) \qquad (|\cos\theta|\to\infty, 0<\varepsilon\leqslant\operatorname{Re}\theta\leqslant\pi), \tag{7.6.10}$$

where $\lambda_M$ is the rightmost Regge pole. According to Theorem 7.5.3, the real part of $\lambda_M(k)$ is bounded for all $k$, so that $f(k,\cos\theta)$ is bounded by a finite ($k$-independent) power of $|\cos\theta|$ as $|\cos\theta|\to\infty$. If the potential satisfies (7.4.21), so that (7.4.25) holds (e.g., for a pure Yukawa potential), the right half of the $\lambda$-plane is entirely free of poles for sufficiently large $k$, in which case the asymptotic behavior of the scattering amplitude is dominated by (7.6.9).

Along the cut, i.e., for $\operatorname{Re}\theta=0$, we see from (7.6.8) that one cannot guarantee the convergence of the background integral in (7.6.4). Indeed, in the case of a pure Yukawa potential, one finds [cf. (8.1.15)] that there is a pole on the cut.

We can summarize the results as follows:

THEOREM 7.6.1. *For Yukawa-type potentials (under the assumptions of Theorem 7.5.2), $f(k, \cos\theta)$, for $k > 0$, is holomorphic in the whole $\cos\theta$ plane, cut along the real axis from $\cos\theta = 1 + m^2/4E$ to $\infty$, where $m$ is the inverse range of the potential. For $|\cos\theta| \to \infty$ in the cut plane, the asymptotic behavior of $f(k, \cos\theta)$ is given by (7.6.10), where $\lambda_M$ is the rightmost Regge pole, or by (7.6.9) if there are no Regge poles in the right half of the $\lambda$-plane. In any case, $f(k, \cos\theta)$ is bounded by a fixed ($k$-independent) power of $|\cos\theta|$.*

In Section 8.1 these results will be employed to derive the Mandelstam representation; before doing this, however, let us discuss some further properties of Regge poles.

## 7.7. Regge Poles

### (a) Regge Trajectories

As we have seen in Section 7.3, the poles of $S(\lambda, k)$ are the zeros of $f(\lambda, -k)$. We want to investigate the behavior of the Regge poles $\lambda_j(k)$ in the $\lambda$-plane as a function of $k$.

According to the *implicit function theorem* [32, p. 39], if $g(\lambda, k)$ is a holomorphic function of two complex variables $\lambda$ and $k$ in a domain $\mathscr{D}(\lambda, k)$, and if

$$g(\lambda_0, k_0) = 0, \qquad \frac{\partial g}{\partial \lambda}(\lambda_0, k_0) \neq 0 \tag{7.7.1}$$

at a point $(\lambda_0, k_0) \in \mathscr{D}$, then there exists a neighborhood $N$ of $(\lambda_0, k_0)$ such that, for each $k$ in $N$, there exists in $N$ a unique solution $\lambda(k)$ of the equation $g(\lambda, k) = 0$, taking the value $\lambda_0$ for $k = k_0$, and such that $\lambda$ is a holomorphic function of $k$ in $N$.

Applying this theorem [33] to the function $g(\lambda, k) = f(\lambda, -k)$, which, according to Theorem 7.3.4, is holomorphic in the direct product of $\mathrm{Re}\,\lambda > 0$ with the $k$-plane, cut along the negative imaginary axis from $k = 0$ to $-i\infty$ (for Yukawa-type potentials), we conclude that *a Regge pole $\lambda = \lambda(k)$ defines $\lambda$ as a holomorphic function of $k$, except on the negative imaginary $k$-axis and at isolated critical points, where*

$$f(\lambda, -k) = \partial f(\lambda, -k)/\partial\lambda = 0. \tag{7.7.2}$$

This is precisely the condition for a multiple root, so that such a critical point would correspond to a branching point for the function $\lambda(k)$.

Of special interest is the curve described by a Regge pole $\lambda(k)$ as the energy

$E = k^2/2$ ranges over real values from $-\infty$ to $\infty$, i.e., the map in the $\lambda$-plane of the positive imaginary $k$-axis, from $i\infty$ to 0, followed by the positive real $k$-axis, from 0 to $\infty$. This curve is called a *Regge trajectory*. Actually, we are only interested in that portion of a Regge trajectory lying in the right half of the $\lambda$-plane.

According to (7.4.11), for $E < 0$ ($k = i\kappa, \kappa > 0$), a Regge trajectory in the right half of the $\lambda$-plane must lie along the real $\lambda$-axis. By (7.2.3) and (7.4.1), at a Regge pole $\lambda(k)$ with $\operatorname{Re}\lambda > 0$, $\operatorname{Im} k > 0$, the regular solution $\varphi(\lambda, k, r)$ is regular at $r = 0$ and exponentially decreasing at infinity, so that, *if a Regge trajectory goes through a physical value, $\lambda = l + \frac{1}{2}$, for $E < 0$, this corresponds to a bound state of angular momentum $l$ and energy $E$.*

To find out more about the behavior of Regge poles as a function of energy, let us differentiate the radial equation (7.2.1) for the regular solution $\varphi(\lambda, k, r)$, taken at the position of a Regge pole $\lambda = \lambda(E)$, with respect to the energy:

$$\frac{\partial^2}{\partial r^2}\left(\frac{\partial\varphi}{\partial E}\right) = \left(-2E + \frac{\lambda^2 - \frac{1}{4}}{r^2} + U\right)\frac{\partial\varphi}{\partial E} - 2\varphi + \frac{\varphi}{r^2}\frac{d\lambda^2}{dE}$$

$$= \frac{1}{\varphi}\frac{\partial^2\varphi}{\partial r^2}\frac{\partial\varphi}{\partial E} - 2\varphi + \frac{\varphi}{r^2}\frac{d\lambda^2}{dE},$$

so that

$$\frac{\partial}{\partial r} W\left[\varphi, \frac{\partial\varphi}{\partial E}\right] = \left(\frac{1}{r^2}\frac{d\lambda^2}{dE} - 2\right)\varphi^2, \tag{7.7.3}$$

where $W$ is the Wronskian (5.2.7). Integrating both sides with respect to $r$ from 0 to $\infty$, and noting that at a Regge pole with $\operatorname{Re}\lambda > 0$ and $\operatorname{Im} k > 0$ the integral of the left-hand side vanishes according to the above remarks (the Wronskian vanishes at both ends), we get

$$\frac{d\lambda^2}{dE} = \frac{2\int_0^\infty \varphi^2(\lambda, E, r)\,dr}{\int_0^\infty \varphi^2(\lambda, E, r)\,dr/r^2} = \frac{2}{\langle r^{-2}\rangle} \qquad (\operatorname{Im} k > 0,\ \operatorname{Re}\lambda > 0), \tag{7.7.4}$$

where the last term defines a kind of "expectation value." We may compare this with the relation $dl^2/dE = 2I$ in classical mechanics, where $l$ is the angular momentum and $I = r^2$ is the moment of inertia for a particle with mass $m = 1$.

In particular, for $k = i\kappa$, $\kappa > 0$, it follows from (7.2.16), since $\lambda$ must be real, that $\varphi(\lambda, i\kappa, r) = \varphi^*(\lambda, i\kappa, r)$, so that $\varphi$ is real. Therefore, according to (7.7.4),

$$d\lambda^2/dE > 0 \qquad (k = i\kappa,\ \kappa > 0,\ \lambda > 0). \tag{7.7.5}$$

It follows that, for $E < 0$, a Regge pole on the positive real $\lambda$-axis must move *to the right* as $|E|$ decreases, generating a bound state each time it passes through a physical value of $\lambda$. This corresponds to the intuitive fact that the binding energy of a bound state decreases as its angular momentum increases, due to the repulsive centrifugal barrier.

For $E > 0$ ($k > 0$), according to (7.4.9), Regge poles can no longer be located on the positive real $\lambda$-axis: poles in the right half of the $\lambda$-plane must be complex, and they can only be located in the first quadrant, within the shaded regions of Fig. 7.2, with their real part bounded for all $k > 0$. In particular, if the potential satisfies (7.4.21) (e.g., for a pure Yukawa potential), the excursion of a Regge pole into the right half-plane is limited by (7.4.25), which shows that no poles can remain for $k \to \infty$. In this case, therefore, Regge trajectories in the first quadrant must eventually turn back for large enough energies, going over to the left half of the $\lambda$-plane (the behavior of trajectories in the left half-plane is very unstable with respect to small perturbations of the potential, and it will not be discussed here).

The behavior of $\lambda(k)$ in the neighborhood of the threshold point $k = 0$ requires a separate investigation, because $k = 0$ is, in general, a branch point of $f(\lambda, -k)$ [cf. (7.3.7)], so that $\lambda(k)$ is not holomorphic at $k = 0$. The threshold behavior can be obtained by discussing the roots of the denominator of (7.3.20) near $k = 0$. Here we only quote the results [6b, Chapter 9]. Let $\lambda_0$ be the position reached by a Regge pole at $E = 0$. In the right half-plane, according to the above discussion, $\lambda_0$ must be on the real axis, so that either $\lambda_0 > 0$ (Newton [6b, Chapter 9] calls such poles "$C$-type") or $\lambda_0 = 0$ ("0-type" poles according to Newton).

If $\lambda_0 > 0$, it is found that the Regge trajectory leaves the real axis towards the first quadrant at right angles if $\lambda_0 = \frac{1}{2}$, making an obtuse angle with the real axis (i.e., towards the left) if $0 < \lambda_0 < \frac{1}{2}$, and making an acute angle (towards the right) if $\frac{1}{2} < \lambda_0 < 1$. If $\lambda_0 > 1$, the trajectory leaves the real axis tangentially towards the right, osculating it more and more closely the larger $\lambda_0$ is [its slope $\gamma$ in the latter case being given by $\tan\gamma = O(E^{\lambda_0 - 1})$ as $E \to 0+$].

As $E \to 0+$, there are infinitely many pole trajectories (of 0-type) that [33a] approach $\lambda_0 = 0$ along parabolic arcs osculating the imaginary $\lambda$-axis; for $E \to 0-$, they come in from the left half-plane, and they return to the left half-plane as $E \to \infty$ if the potential satisfies (7.4.21). Thus these 0-type poles have no connection at all with bound states. Note that $\lambda = 0$ is ***not*** an accumulation point of poles of $S(\lambda, k)$; at any fixed energy, there is only a finite number of poles in a neighborhood of $\lambda = 0$.

The trajectory of the rightmost Regge pole for an attractive pure Yukawa potential (7.3.22) with inverse range $m = 1$ for various values of the potential strength $U_0$, obtained by numerical computation [34], is shown in Fig. 7.3. The greater the strength of the attractive potential, the larger is the number of bound states it can accommodate and, therefore, the larger is the excursion of the pole to the right along the real axis and into the first quadrant before

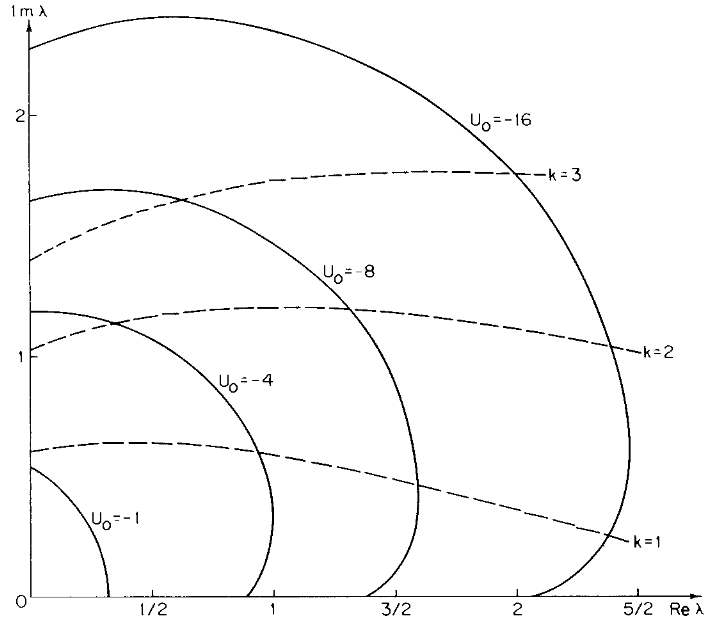


*FIG. 7.3.* Trajectories of the rightmost Regge pole for the Yukawa potential $U(r) = -U_0 e^{-r}/r$ for various values of $U_0$. ---, lines of constant $k$ [after C. Lovelace and D. Masson, *Nuovo Cimento* **26,** 472 (1962)].

turning back. The results given above about the threshold behavior of the trajectories and the angle at which they leave the real axis are well illustrated by the trajectories shown in Fig. 7.3.

### *(b) Physical Interpretation*

The Regge poles $\lambda(k)$ are obtained by solving with respect to $\lambda$ the equation

$$f(\lambda, -k) = 0. \tag{7.7.6}$$

If $\partial f/\partial k \neq 0$, it follows from the implicit function theorem that we can also solve this equation with respect to $k$, defining the poles in the $k$-plane $k(\lambda)$ as holomorphic functions of $\lambda$ for $\operatorname{Re}\lambda > 0$ and $k$ not on the negative imaginary axis. We may regard (7.7.6) as defining a *singularity surface* in the 4-dimensional space of the two complex variables $(k, \lambda)$; poles in the $\lambda$-plane and poles in the $k$-plane correspond to different sections of this same surface. Just as we defined Regge trajectories as pole paths $\lambda(E)$ in the $\lambda$-plane when $E$ ranges between $-\infty$ and $\infty$, we may define pole paths $k(\lambda)$ in the $k$ (or $E$) plane when $\lambda$ ranges from 0 to $\infty$. For physical values of $\lambda$, the poles in the $k$-plane are just those the physical interpretation of which was discussed in Chapter 4. We can make use of this correspondence to transcribe the results of that discussion to the physical interpretation of Regge poles.

Consider a Regge trajectory $\lambda_j(E)$ that lies very close to the real axis near a physical value of $\lambda$, say $\lambda = l+\frac{1}{2}$, when $E$ is in the neighborhood of some energy $E_0 > 0$ (an example would be the trajectory $U_0 = -8$ in Fig. 7.3, near $l = 1$). Thus

$$\lambda_j(E_0) = l + \tfrac{1}{2} + \mu_0 + i\nu_0 \qquad (|\mu_0| \ll 1, 0 < \nu_0 \ll 1). \tag{7.7.7}$$

Since $\lambda_j(E)$ is holomorphic, we can expand it in a power series about $E_0$; in a sufficiently small neighborhood of $E_0$, we have

$$\lambda_j(E) \approx \lambda_j(E_0) + (E-E_0)\,\lambda'_{j0}, \tag{7.7.8}$$

where

$$\lambda'_{j0} = d\lambda_j/dE\Big|_{E=E_0} \approx \operatorname{Re}\lambda'_{j0}; \tag{7.7.9}$$

i.e., we expect $\lambda'_{j0}$ to be almost purely real. This follows from the fact that $\lambda_j(E_0)$ is almost real and from (7.7.4) (which can be applied for $E = E_0 + i\varepsilon$, $\varepsilon > 0$ arbitrarily small), where $\varphi$ should be almost real for almost real $\lambda_j$ and $E$.

If we now set

$$\lambda_j(E) = l + \tfrac{1}{2}$$

in (7.7.8), we can find the "corresponding" pole of $S(l+\frac{1}{2}, E) = S_l(E)$ in the second sheet of the $E$-plane. From (7.7.7) to (7.7.9), we get, making the above substitution and solving for $E$,

$$E = E_0{}' - i(\Gamma_0/2) \approx E_0 - (\mu_0/\lambda'_{j0}) - i(\nu_0/\lambda'_{j0}). \tag{7.7.10}$$

For sufficiently small $\Gamma_0$, we know from (4.2.8) that this pole must be associated with a *resonance* of half-width $\Gamma_0$ around the energy $E_0{}'$ for the $l$th partial wave.

This can be verified [35] by projecting the $l$th partial wave out of the Watson transform representation (7.6.4) of the total scattering amplitude, with the help of (3.1.4),

$$f_l(k) = (S_l(k)-1)/2i = (k/2)\int_0^\pi f(k,\cos\theta)P_l(\cos\theta)\sin\theta\,d\theta. \tag{7.7.11}$$

If $\lambda_j(k)$ is very close to $l+\frac{1}{2}$ in (7.6.4), the denominator $\cos(\pi\lambda_j)$ is very small, and the contribution from the pole $\lambda_j$ may be expected to dominate the amplitude in (7.7.11). This contribution is

$$f_l^{(j)}(k) = -\frac{i\pi}{2}\frac{\sigma_j\lambda_j}{\cos(\pi\lambda_j)}\int_0^\pi P_{\lambda_j-\frac{1}{2}}(-\cos\theta)P_l(\cos\theta)\sin\theta\,d\theta. \tag{7.7.12}$$

We have [23, p. 144, Eq. (14), p. 170, Eqs. (7) and (13)]

$$\int_{-1}^{1} P_\nu(-x)P_l(x)\,dx = \frac{2}{\pi}\cdot\frac{\sin(\pi\nu)}{(\nu-l)(\nu+l+1)}$$

$$(\nu+l+1\neq 0,\ \mathrm{Re}\,\nu>0,\ l=0,1,2,\ldots),\quad (7.7.13)$$

so that

$$f_l^{(j)}(k) = i\sigma_j\lambda_j/(\lambda_j-l-\tfrac{1}{2})(\lambda_j+l+\tfrac{1}{2}),$$

and, substituting $\lambda_j$ by (7.7.8), we find, in the neighborhood of $E_0$,

$$f_l^{(j)}(E) \approx i\sigma_j/\{2\lambda'_{j0}[E-E_0'+i(\Gamma_0/2)]\},\qquad (7.7.14)$$

where $E$ is now the physical energy, and $E_0'$ and $\Gamma_0$ are defined by (7.7.10). We see that (7.7.14) indeed corresponds to a Breit–Wigner type formula (Section 4.2) for a resonance of half-width $\Gamma_0$ centered at $E_0'$. The results of Section 7.7(a) on the angles at which Regge trajectories leave the real axis can now readily be interpreted in terms of the effects of the centrifugal barrier on resonance widths near threshold [cf. the discussion near the end of Section 5.8(c)].

In view of the fact that $\lambda'_{j0}$ is almost real [cf. (7.7.9)], it follows from (7.7.10) that the resonance half-width associated with a Regge pole of imaginary part $\nu_0$ very close to the real axis is given by

$$\Gamma_0 \approx 2\nu_0/\lambda'_{j0}.\qquad (7.7.15)$$

This result has a simple physical interpretation [2a] in terms of the classical mechanics analogy discussed following (7.7.4). Classically, a resonance with a given angular momentum $l$ corresponds to a scattering process in which the orbit of the particle winds around the scattering center several times, leading to a long lifetime for the interaction. The lifetime $\tau_0$ associated with the resonance is given by [cf. (4.3.7)] $\tau_0 = 1/\Gamma_0$.

On the other hand, as we have seen following (7.7.9), we can apply (7.7.4), where $\varphi$ is almost real, to get

$$\lambda'_{j0} \approx (\lambda_{j0}\langle r^{-2}\rangle)^{-1} \approx r^2/l,\qquad (7.7.16)$$

where $r$ is the radius of the classical orbit and $l$ is the classical angular momentum, $l=r^2\dot\theta$, $\dot\theta$ being the angular velocity [note that (7.7.16) implies $\lambda'_{j0}>0$, so that $\Gamma_0>0$ in (7.7.15), as it should be]. Substituting these results in (7.7.15), we get

$$\nu_0 \sim 1/(2\tau_0\dot\theta) = 1/(2\,\Delta\theta),\qquad (7.7.17)$$

where $\Delta\theta$ is the total angle described by the particle around the scattering center during the lifetime of the resonance ($\Delta\theta$ should be $\gg 2\pi$ for a narrow resonance). We therefore expect that the angular part of the wave function will contain an exponential decay factor

$$\exp[-\theta/(2\Delta\theta)] \sim \exp(-\theta \operatorname{Im} \lambda_j), \tag{7.7.18}$$

so that *the imaginary part of a Regge pole represents an angular damping factor for the wave function.*

In fact, for large $|\lambda_j|$, (7.6.3) shows that the Regge pole contributions in (7.6.4) contain angular damping factors of the form (7.7.18). The "complex angular momentum eigenfunction" $P_{\lambda_j - 1/2}(-\cos\theta)$ is not, of course, an acceptable wave function (it is singular at $\theta = 0$ and it is not single-valued), any more than the "complex-energy eigenfunctions" of Section 4.3 were (the singularity there was in the radial function, at $r \to \infty$). One may establish a parallel between the two ways of interpreting resonant states, in terms of poles in the energy plane (time decay) or in the angular momentum plane (angular decay); angular momentum and angle are in some respects conjugate variables, as energy and time are (cf. Section 7.1). The treatment of some model problems involving cutoff potentials [36, p. 203] allows one to justify the above interpretation in terms of "complex angular momentum eigenfunctions" and to resolve the singularity at $\theta = 0$, just as the method described in Chapter 4 enabled us to properly interpret "complex energy eigenfunctions" and to resolve the singularity at $r \to \infty$. We will describe some of the results in Section 8.4(c).

Finally, let us note that a single Regge trajectory can describe several bound states (as it goes through physical values of $\lambda$ along the real axis for $E < 0$) and resonances (if it stays close to the real axis and physical values of $\lambda$ for $E > 0$). In this way, bound states and resonances can be grouped into families, obeying the selection rule $\Delta l = 1$ for consecutive members of a family lying on the same Regge trajectory. This grouping of bound states and resonances into families lying along common Regge trajectories has been considerably developed in high-energy physics, although the situation there is different in several respects [6, 37].

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CHAPTER

# 8

# THE MANDELSTAM REPRESENTATION

*In this way, by using dispersion relations and conservation of probability, suitably extended into the Never-Never Land, one may hope to generate the entire S-matrix from more or less fundamental principles.*

M. GELL-MANN[1]

## 8.1. Derivation of the Mandelstam Representation

### (a) Dispersion Relation for the Absorptive Part

We now derive the Mandelstam representation for Yukawa-type potentials. For this purpose, as explained in Section 7.1, we have to obtain a dispersion relation in the momentum transfer for the absorptive part $g(k, \tau)$ defined by (6.5.14), with $k > 0$ [as it appears in the integrand of (6.5.2)].

The analytic properties in $\cos\theta$ of $f(k, \cos\theta)$ for $k > 0$ are given by Theorem 7.6.1. According to (6.5.14), we need also the properties of $f(-k, \cos\theta)$, which is equal to $f^*(k, \cos\theta)$ for physical $k$ and $\theta$ [cf. (6.5.5)]. However, since the complex conjugate of an analytic function is not an analytic function, we cannot employ complex conjugation to define the analytic continuation of $f(-k, \cos\theta)$.

[1] *Proc. 6th Annual Rochester Conf. High-Energy Nucl. Phys.*, Sec. III, p. 34. Wiley (Interscience), New York, 1956.

For physical $k$ and $\theta$, we can employ the partial wave expansion in (7.1.6):

$$f(-k,\cos\theta) = \sum_{l=0}^{\infty} [(-1)^l/-ik](l+\tfrac{1}{2})[S_l^*(k)-1]P_l(-\cos\theta). \quad (8.1.1)$$

Let us consider the function [cf. (7.3.13)]

$$S^*(\lambda^*,k) = 1/S(\lambda,k) \qquad (k>0). \quad (8.1.2)$$

For Yukawa-type potentials, according to the results derived in Chapter 7, this function has the following properties: (i) It reduces to $S_l^*(k)$ for physical values of $\lambda$ [cf. (7.1.8)]. (ii) It is meromorphic in $\mathrm{Re}\,\lambda > 0$, by (8.1.2) and Theorem 7.3.4. (iii) Its poles in the $\lambda$-plane are the complex conjugates of the poles of $S(\lambda,k)$, so that, according to Theorem 7.5.3 and (7.6.4), they are given by $\lambda_j^*(k)$ $(j=1,2,\ldots,M)$, and their real part is bounded for all $k \geqslant 0$. (iv) By (7.5.44) and (8.1.2), we have

$$\lim_{|\lambda|\to\infty} S^*(\lambda^*,k) = 1 \qquad (k>0,\ -\pi/2 \leqslant \arg\lambda \leqslant \pi/2).$$

Therefore, we can apply to (8.1.1) the Watson transformation (7.1.3), and we find, similarly to (7.6.4),

$$\begin{aligned} f(-k,\cos\theta) &= \frac{1}{2k}\int_{-\infty}^{\infty} [S^*(iv,k)-1]P_{iv-1/2}(-\cos\theta)\frac{v\,dv}{\cosh(\pi v)} \\ &\quad + \frac{i\pi}{k}\sum_{j=1}^{M}\frac{\sigma_j^*\lambda_j^*}{\cos(\pi\lambda_j^*)}P_{\lambda_j^*-1/2}(-\cos\theta), \end{aligned} \quad (8.1.3)$$

where we have made use of the symmetry property $P_{\lambda-1/2}(z) = P_{-\lambda-1/2}(z)$. Note that (8.1.3) indeed reduces to $f^*(k,\cos\theta)$ for physical values of $k$ and $\theta$; for such values we could have obtained it directly from (7.6.4), by complex conjugation.

By the same reasoning as in Section 7.6, it follows that, for $k>0$, the analytic continuation of $f(-k,\cos\theta)$, defined by (8.1.3), is holomorphic in the whole $\cos\theta$ plane, cut along the real axis from $\cos\theta = 1+m^2/4E$ to $\infty$, and it obeys the same bounds as $f(k,\cos\theta)$ for $|\cos\theta|\to\infty$.

Combining these results with Theorem 7.6.1, we see that the absorptive part (6.5.14) has the same analytic properties, except that the Born term drops out in the difference, so that, according to (6.3.40), the absorptive part has no cut between $\cos\theta = 1$ and $\cos\theta = 1+m^2/E$.

From now on we adopt the conventional notation

$$s = k^2, \qquad t = -\tau^2. \quad (8.1.4)$$

We refer to $s$ as the energy, although $s = 2E$ in the units of (6.3.20) (alternatively, we could adopt units in which the mass of the particle is $\frac{1}{2}$). According to (6.3.21),

$$t = -2s(1-\cos\theta). \quad (8.1.5)$$

The above results then imply that, *for $s > 0$, the absorptive part $g(s,t)$ is holomorphic in the t-plane cut along the real axis, from $t = 4m^2$ to $\infty$, and it is bounded by a fixed integral power of t, say $t^L$, as $|t| \to \infty$ in the cut plane.* Here, according to (7.6.10),

$$L + \tfrac{1}{2} \geqslant \max_{0 \leqslant s < \infty} \operatorname{Re} \lambda_M(s). \tag{8.1.6}$$

It follows, in particular, that $L \geqslant l_b$, where $l_b$ is the largest angular momentum for which a bound state occurs.

Actually [see the remarks following (7.6.10)], we have derived the asymptotic behavior (7.6.10) with the exclusion of an angular sector of arbitrarily small opening around the cut. With somewhat more restrictive assumptions on the potential, it can be shown [cf. Section 8.1(b)] that this bound remains valid also along the cut, and that the discontinuity across the cut is finite for finite $t$.

With the help of a contour of integration similar to that shown in Fig. 2.2, we can then derive a dispersion relation in $t$ with $L+1$ subtractions (cf. Section 1.7) for the absorptive part:

$$g(s,t) = \sum_{l=0}^{L} g_l(s)(t^l/l!) + (t^{L+1}/\pi) \int_{4m^2}^{\infty} \rho(s,t')/[t'^{\,L+1}(t'-t)]\,dt' \qquad (s > 0), \tag{8.1.7}$$

where

$$g_l(s) = \left[\frac{\partial^l}{\partial t^l} g(s,t)\right]_{t=0} \tag{8.1.8}$$

and

$$\rho(s,t) = (1/2i)[g(s,t+i0) - g(s,t-i0)] \tag{8.1.9}$$

is the discontinuity of $g(s,t)$ across the cut. The function $\rho(s,t)$ is known as the *double spectral function*, and the functions $g_l(s)$ are called *single spectral functions.*

### *(b) The Mandelstam Representation*

We now go back to the dispersion relation (6.5.2) for fixed momentum transfer, which we rewrite in terms of the variables $s$ and $t$:

$$f(s,t) = f_B(t) + \sum_{p=1}^{N} [\Gamma_p(t)/(s-s_p)] + (1/\pi) \int_0^{\infty} [g(s',t)/(s'-s)]\,ds' \\ [\operatorname{Im} \sqrt{s} > 0,\ -4m^2 < t \leqslant 0]. \tag{8.1.10}$$

Here, $s_p < 0$ correspond to the bound-state energies; each bound state $s_p$, as we have seen in Section 6.2(c), is associated with a pole in a given partial-wave amplitude with angular momentum $l_p$, so that the associated residue $\Gamma_p(t)$ is proportional to the corresponding Legendre polynomial evaluated at $s = s_p$,

$$\Gamma_p(t) = c_p P_{l_p}(1+(t/2s_p)), \tag{8.1.11}$$

where $c_p$ is a constant. Thus $\Gamma_p(t)$ is a polynomial in $t$ of degree $l_p$. As was noted following (8.1.6),

$$\max l_p = l_b \leqslant L. \tag{8.1.12}$$

Substituting $U(r)$ by (7.1.1) in (6.3.23), we see that the Born approximation $f_B(t)$ can be written as

$$f_B(t) = -\int_m^\infty \frac{\sigma(\mu)\,d\mu}{\mu^2+\tau^2} = -\int_m^\infty \frac{\sigma(\mu)\,d\mu}{\mu^2 - t}. \tag{8.1.13}$$

Since (8.1.13) is a Cauchy-type integral, we see from the discussion following (5.4.38) and (5.4.45) that, if $\sigma(\mu)$ satisfies a Hölder condition, $f_B(t)$ is holomorphic in the $t$-plane cut along the real axis from $t = m^2$ to $\infty$. This corresponds to the cut in the $\cos\theta$ plane referred to in Theorem 7.6.1. Furthermore, in view of (5.4.30), we must have $\sigma(\mu) \to 0$ as $\mu \to \infty$, so that[2]

$$f_B(t) \to 0 \qquad \text{as} \quad |t| \to \infty. \tag{8.1.14}$$

For the pure Yukawa potential (7.3.22), $\sigma(\mu)$ is a delta function, and $f_B(t)$ becomes

$$f_B(t) = U_0/(t - m^2) \qquad \text{(pure Yukawa)}, \tag{8.1.15}$$

so that in this case $t = m^2$ is a pole rather than a branch point (cf. Section 7.6).

If we now substitute $g(s',t)$ in the integrand of (8.1.10) by (8.1.7), we get

$$\begin{aligned} f(s,t) &= f_B(t) + \sum_{p=1}^{N} \frac{\Gamma_p(t)}{s - s_p} + \frac{1}{\pi} \sum_{l=0}^{L} \frac{t^l}{l!} \int_0^\infty \frac{g_l(s')}{s'-s}\,ds' \\ &\quad + \frac{t^{L+1}}{\pi^2} \int_0^\infty \frac{ds'}{s'-s} \int_{4m^2}^\infty \frac{dt'}{(t'-t)} \frac{\rho(s',t')}{t'^{L+1}}. \end{aligned} \tag{8.1.16}$$

Having derived (8.1.16) in the domain $-4m^2 < t \leqslant 0$, Im $\sqrt{s} > 0$ (physical sheet of the $s$-plane), we would now like to employ it to define the analytic continuation of $f(s,t)$ as a function of two complex variables, regular (apart from the bound-state poles) in the direct product of the $s$-plane, cut along the positive real axis, with the $t$-plane, cut along the real axis from $m^2$ to $\infty$ (from $4m^2$ to $\infty$ apart from the Born term). For this purpose, however, we have to investigate the convergence properties of the various terms in (8.1.16).

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[2] Cf. Widder [1].

The Born term and the bound-state contributions have obvious analytic properties in $s$ and $t$, that have already been discussed. The single spectral functions appearing in the one-dimensional integrals are derivatives of $g$ with respect to $t$ evaluated at $t = 0$ [cf. (8.1.8)], and, according to (6.5.14) and Section 6.3(e), $g(s,t)$ is holomorphic within the circle $|t| < 4m^2$ [where the partial-wave expansion (6.5.16) converges uniformly], and $g \to 0$ as $|s| \to \infty$ by (6.4.40). Thus the one-dimensional integrals in (8.1.16) are convergent and holomorphic in the cut $s$-plane.[3]

There remains to discuss the convergence of the repeated integral involving the double spectral function. We would like to show that it is in fact convergent as a double integral, regardless of the order of integration, so that we can interpret it as a two-dimensional Cauchy integral, representing an analytic function of two complex variables regular in the direct product of the cut $s$ and $t$ planes [3, pp. 229, 237]. For this purpose, we want to find an upper bound for the double spectral function $\rho(s,t)$.

This problem has been investigated by Bessis [4], whose main results are outlined below; the reader is referred to the original paper for details of the derivations. In view of (8.1.9) and (6.5.3), the problem amounts to finding upper bounds for $f(s,t)$ along the cuts in both variables. We cannot employ theRegge representations (7.6.4) and (8.1.3), because it is difficult to bound the Regge pole contributions as well as the function $S(\lambda,k)$ in the background integral.

The first step is to obtain a Regge representation for $f(s,t)$ in terms of an integral along a path $D$ which avoids the region in the $\lambda$-plane where $S(\lambda,k)$ can become unbounded. The path $D$ is shown in Fig. 8.1: it runs parallel to the imaginary axis from $\varepsilon - i\infty$ $(0 < \varepsilon < \frac{1}{2})$ to $P^*$, then along the arc of circle $P^*LP$ of radius $L$ centered at the origin [where $L+1$ is the number of subtractions in (8.1.7)], and from $P$ to $\varepsilon + i\infty$ parallel to the imaginary axis. The shaded region where all Regge poles must be contained (cf. Fig. 7.2) lies entirely within $D$, as follows from (8.1.6) and from the discussion in Section 7.4(b). We take $L > 1$, as can always be done.

Since the path $D$ passes halfway between the physical values $\lambda = L - \frac{1}{2}$ and $\lambda = L + \frac{1}{2}$, it follows that, if we perform the Watson transformation as in Section 7.6, but deforming the contour $C$ of Fig. 7.1 into the path $D$ of Fig. 8.1, instead of deforming it into the imaginary axis, we get, instead of (7.1.6),

$$f(k,t) = -\frac{1}{2k}\int_D [S(\lambda,k)-1]P_{\lambda-\frac{1}{2}}\left(-1-\frac{t}{2s}\right)\frac{\lambda\,d\lambda}{\cos(\pi\lambda)}$$

$$+\frac{1}{ik}\sum_{l=0}^{L-1}\left(l+\frac{1}{2}\right)[S_l(k)-1]P_l\left(1+\frac{t}{2s}\right) \qquad (k>0). \quad (8.1.17)$$

[3] Cf. Martin [2].

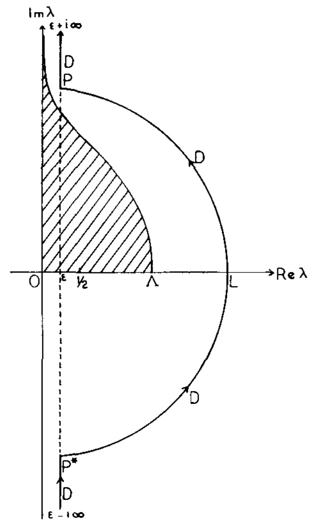


*FIG. 8.1.* The path $D$. Cross-hatched area is region where all Regge poles must be contained (after D. Bessis [4]).

According to the discussion given in the derivation of (8.1.3), a similar procedure, applied to (8.1.1), leads to

$$f(-k,t) = \frac{1}{2k}\int_D [S^*(\lambda^*,k)-1]\,P_{\lambda-\frac{1}{2}}\left(-1-\frac{t}{2s}\right)\frac{\lambda\,d\lambda}{\cos(\pi\lambda)}$$
$$-\frac{1}{ik}\sum_{l=0}^{L-1}\left(l+\frac{1}{2}\right)[S_l^*(k)-1]\,P_l\left(1+\frac{t}{2s}\right) \qquad (k>0). \tag{8.1.18}$$

It follows from (6.5.14), (8.1.17), and (8.1.18) that

$$g(s,t) = -\frac{i}{4k}\int_D [S(\lambda,k)-1]\,[S^*(\lambda^*,k)-1]\,P_{\lambda-\frac{1}{2}}\left(-1-\frac{t}{2s}\right)\frac{\lambda\,d\lambda}{\cos(\pi\lambda)}$$
$$+\frac{1}{k}\sum_{l=0}^{L-1}(2l+1)\,\mathrm{Im}\,f_l(k)\,P_l\left(1+\frac{t}{2s}\right), \tag{8.1.19}$$

where we have employed the extended unitarity condition (7.3.13) and $\mathrm{Im}\,f_l(k)$ is given by (3.1.4) and (3.1.5).

The discontinuity of the Legendre function across the cut from $-\infty$ to $-1$ is given by [5, p. 140, Eq. (10)]

$$P_{\lambda-\frac{1}{2}}(-x-i0) - P_{\lambda-\frac{1}{2}}(-x+i0) = 2i\cos(\pi\lambda)\,P_{\lambda-\frac{1}{2}}(x) \qquad (x>1), \tag{8.1.20}$$

so that (8.1.9) and (8.1.19) finally yield the Regge representation for the double spectral function,

$$\rho(s,t) = -\frac{i}{4k}\int_D [S(\lambda,k)-1][S^*(\lambda^*,k)-1]P_{\lambda-\frac{1}{2}}\left(1+\frac{t}{2s}\right)\lambda\,d\lambda \qquad (s>0,\ t>4m^2). \tag{8.1.21}$$

Upper bounds for the integrand on $D$ follow from (7.5.25) and from the inequalities (7.5.35) and (7.5.36), which remain valid when $i\nu$ is replaced by $\lambda$, with a suitable redefinition of the path of integration in (7.5.34). Assuming that the potential satisfies (7.4.26) and (7.4.27), one finds, similarly to (7.5.38) and (7.5.40),

$$|S(\lambda,k)-1| < C/(\varepsilon^{2-\gamma}|\lambda|^{\gamma-1}) \qquad \text{if} \quad 1<\gamma<2, \tag{8.1.22}$$

$$|S(\lambda,k)-1| < C/|\lambda| \qquad \text{if} \quad \gamma>2, \tag{8.1.23}$$

where $C$ is a constant depending only on the potential. The dependence on $\varepsilon = \operatorname{Re}\lambda$ in (8.1.22) along the vertical portions of the path is similar to that in (7.4.29). Using the bound [cf. (7.6.5), (7.5.16), (7.5.17)]

$$|P_{\varepsilon-\frac{1}{2}+i\nu}(\cosh\alpha)| \leqslant 3\cosh(\alpha\varepsilon)/(\nu\sinh\alpha)^{\frac{1}{2}} \qquad (\nu>1,\ -\tfrac{1}{2}<\varepsilon<\tfrac{1}{2},\ \alpha\geqslant 0), \tag{8.1.24}$$

we find that, if

$$|S(\lambda,k)-1| < C(\varepsilon)/|\lambda|^{\chi},$$

the contribution to (8.1.21) from the vertical portions of the path is bounded by

$$C(\varepsilon)\frac{[1+(t/2s)]^{\varepsilon}}{(st)^{\frac{1}{4}}[1+(t/4s)]^{\frac{1}{4}}}\int_1^{\infty}\frac{d\nu}{\nu^{2\chi-\frac{1}{2}}}. \tag{8.1.25}$$

The integral converges if $\chi > \frac{3}{4}$, i.e., by (8.1.22) and (8.1.23), if $\gamma > \frac{7}{4}$.

Taking into account also the contribution from the arc of circle, one finds [4] that, in all cases,

$$|\rho(s,t)| \leqslant C\{[1/(st)^{\frac{1}{4}}]+[(2+t)^L/s^{\frac{1}{2}}]\} \qquad (\gamma>\tfrac{7}{4}), \tag{8.1.26}$$

where the first contribution within square brackets arises from the rectilinear portions of $D$ and the second one is due to the arc of circle.

It follows from (8.1.26) that the repeated integral in (8.1.16) in fact converges as a double integral, defining a function of two complex variables holomorphic in the direct product of the two cut planes, so that we obtain:

Theorem 8.1.1. *Let $U(r)$ be a potential holomorphic for* $\operatorname{Re} r>0$ *and bounded by (7.4.26) and (7.4.27), with $\gamma>\frac{7}{4}$. Then $f(s,t)$ is an analytic function of two complex variables, regular in the direct product of the s-plane cut along the positive real axis with the t-plane cut along the real axis from $m^2$ to $\infty$, apart* *from a finite number of bound-state poles on the negative s-axis. It verifies the double dispersion relation*

$$f(s,t) = f_B(t) + \sum_{p=1}^{N} \frac{\Gamma_p(t)}{s-s_p} + \frac{1}{\pi}\sum_{l=0}^{L} \frac{t^l}{l!}\int_0^\infty \frac{g_l(s')}{s'-s}\,ds'$$

$$+\frac{t^{L+1}}{\pi^2}\int_0^\infty ds' \int_{4m^2}^\infty dt' \frac{\rho(s',t')}{t'^{L+1}(s'-s)(t'-t)}, \tag{8.1.27}$$

*where $f_B(t)$ is the Born approximation (8.1.13), and the residues $\Gamma_p(t)$ are of the form (8.1.11).*

The representation (8.1.27) is known as the *Mandelstam representation.* The corresponding representation in high-energy physics was first proposed by Mandelstam [6]. For scattering by Yukawa-type potentials, it was obtained (although the derivations were incomplete in several respects) by Blankenbecler *et al.* [7] by the Fredholm method, and by Klein[4] in terms of the Born expansion.

The restriction to $\gamma > \frac{7}{4}$ in Theorem 8.1.1 arose from the requirement of finiteness of the upper bound (8.1.25) to the double spectral function. Bessis [4] has given some indications about how one might extend the result to $1 \leqslant \gamma \leqslant \frac{7}{4}$ (which would include the pure Yukawa potential), by improving the bounds for $S(\lambda,k)-1$ in (8.1.21). In this case, however, one can no longer expect that $\rho(s,t)$ will be a continuous function: in general, it will have to be interpreted as a distribution.

## 8.2. The Unitarity Condition

The unitarity condition (3.1.5) has been expressed so far only in terms of the partial-wave amplitudes. We now find its expression in terms of the total scattering amplitude,[5] by generalizing the well-known property expressing the conservation of flux for the probability current. Let $\psi_{\mathbf{k}}(\mathbf{r})$ and $\psi_{\mathbf{k}'}(\mathbf{r})$ be the scattering wave functions associated with incident plane waves in the directions $\hat{\mathbf{k}}$ and $\hat{\mathbf{k}}'$, respectively, corresponding to the same energy $k^2 = s$. Their asymptotic behavior as $r \to \infty$ in the direction $\hat{\mathbf{r}}$ is given by [cf. (6.1.9)]

$$\psi_{\mathbf{k}}(\mathbf{r}) \approx e^{i\mathbf{k}\cdot\mathbf{r}} + f(\mathbf{k},\mathbf{k}'')\,e^{ikr}/r \qquad (r \to \infty), \tag{8.2.1}$$

$$\psi_{\mathbf{k}'}(\mathbf{r}) \approx e^{i\mathbf{k}'\cdot\mathbf{r}} + f(\mathbf{k}',\mathbf{k}'')\,e^{ikr}/r \qquad (r \to \infty), \tag{8.2.2}$$

where $\mathbf{k}'' = k\hat{\mathbf{r}}$ [cf. (6.1.8)].

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[4] See Klein [8]. The analytic properties of each term in the Born series had previously been derived by Bowcock and Martin [9].

[5] Cf. Glauber [10] and Messiah [11, p. 863].

Since both $\psi_{\mathbf{k}}$ and $\psi_{\mathbf{k}'}$ satisfy the Schrödinger equation (6.1.1), it follows that

$$\psi_{\mathbf{k}} \Delta \psi^*_{\mathbf{k}'} - \psi^*_{\mathbf{k}'} \Delta \psi_{\mathbf{k}} = 0,$$

so that, applying Green's second identity to a spherical volume of radius $r$ centered at the origin,

$$\int [\psi_{\mathbf{k}}(\partial \psi^*_{\mathbf{k}'}/\partial r) - \psi^*_{\mathbf{k}'}(\partial \psi_{\mathbf{k}}/\partial r)]\, r^2 \, d\Omega_r = 0, \tag{8.2.3}$$

where $d\Omega_r$ denotes an element of solid angle in the direction $\hat{\mathbf{r}} = \hat{\mathbf{k}}''$. If we now let $r \to \infty$ and substitute $\psi_{\mathbf{k}}$ and $\psi_{\mathbf{k}'}$ by their asymptotic expressions (8.2.1) and (8.2.2), making use of (3.3.51), we find that (8.2.3) reduces to

$$\operatorname{Im} f(\mathbf{k}, \mathbf{k}') = (k/4\pi) \int f^*(\mathbf{k}', \mathbf{k}'') f(\mathbf{k}, \mathbf{k}'')\, d\Omega_{k''}, \tag{8.2.4}$$

where the integration is over all directions $\hat{\mathbf{k}}''$.

The relation (8.2.4) is the unitarity condition for the total scattering amplitude. In the forward direction $\mathbf{k} = \mathbf{k}'$, according to (3.1.12), it reduces to the optical theorem (3.1.13). Taking the $z$-axis along the direction $\hat{\mathbf{k}}$ in (8.2.4), and taking into account (6.5.3) and (6.5.5), we may rewrite (8.2.4) as

$$g(k, \cos\theta) = (k/4\pi) \int_0^{2\pi} d\varphi' \int_0^{\pi} f(-k, \cos\zeta) f(k, \cos\theta') \sin\theta' \, d\theta', \tag{8.2.5}$$

where (taking the $xz$-plane so that $\hat{\mathbf{k}}'$ lies in it)

$$\cos\zeta = \cos\theta \cos\theta' + \sin\theta \sin\theta' \cos\varphi'. \tag{8.2.6}$$

If we substitute each $f$ in (8.2.5) by the partial-wave expansion (6.5.9) and perform the angular integrations, with the help of the addition theorem for spherical harmonics

$$P_l(\cos\zeta) = P_l(\cos\theta) P_l(\cos\theta') + 2 \sum_{m=1}^{l} (-1)^m P_l^m(\cos\theta) P_l^{-m}(\cos\theta') \cos(m\varphi'), \tag{8.2.7}$$

as well as the orthonormality relation

$$(-1)^m (l + \tfrac{1}{2}) \int_{-1}^{1} P_l^m(x) P_{l'}^{-m}(x)\, dx = \delta_{l,l'}, \tag{8.2.8}$$

we find that (8.2.5) reduces to (6.5.16).

The unitarity condition has been derived for physical values of the variables; i.e., $k \geqslant 0$ and $-1 \leqslant \cos\theta \leqslant 1$. However, for real $k$ and [cf. (8.1.5)]

$$\cos\theta = 1 + (t/2s), \tag{8.2.9}$$

both sides of (8.2.5), according to Theorem 8.1.1, can be analytically continued into the cut $t$-plane, so that (8.2.5) remains valid by analytic continuation.

The unitarity condition can also be expressed in terms of momentum-transfer variables. In (8.2.5), let

$$u = \cos\theta' = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}}'', \qquad v = \cos\zeta = \hat{\mathbf{k}}'\cdot\hat{\mathbf{k}}'', \qquad w = \cos\theta = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}}', \tag{8.2.10}$$

so that (8.2.5) may be rewritten as

$$g(s,w) = (\sqrt{s}/4\pi)\int_0^{2\pi} d\varphi' \int_{-1}^{1} f(s-i0,v)f(s+i0,u)\,du. \tag{8.2.11}$$

We change the integration variables from $(u,\varphi')$ to $(u,v)$. By (8.2.10) and (8.2.6), the Jacobian of the transformation is

$$\begin{aligned}|\partial(u,v)/\partial(u,\varphi')| &= |\partial v/\partial\varphi'| = \sin\theta\sin\theta'|\sin\varphi'|\\ &= [(1-u^2)(1-w^2)-(v-uw)^2]^{1/2}\\ &= [\Delta(u,v,w)]^{1/2},\end{aligned} \tag{8.2.12}$$

where

$$\begin{aligned}\Delta(u,v,w) = 1-u^2-v^2-w^2+2uvw &= \begin{vmatrix}1 & w & u\\ w & 1 & v\\ u & v & 1\end{vmatrix}\\ &= \begin{vmatrix}\hat{\mathbf{k}}^2 & \hat{\mathbf{k}}\cdot\hat{\mathbf{k}}' & \hat{\mathbf{k}}\cdot\hat{\mathbf{k}}''\\ \hat{\mathbf{k}}'\cdot\hat{\mathbf{k}} & \hat{\mathbf{k}}'^2 & \hat{\mathbf{k}}'\cdot\hat{\mathbf{k}}''\\ \hat{\mathbf{k}}''\cdot\hat{\mathbf{k}} & \hat{\mathbf{k}}''\cdot\hat{\mathbf{k}}' & \hat{\mathbf{k}}''^2\end{vmatrix}\\ &= \Gamma(\hat{\mathbf{k}},\hat{\mathbf{k}}',\hat{\mathbf{k}}'')\\ &= |\hat{\mathbf{k}}\cdot(\hat{\mathbf{k}}'\times\hat{\mathbf{k}}'')|^2\end{aligned} \tag{8.2.13}$$

is the Gram determinant [12, p. 34; 13, p. 105] of the vectors $\hat{\mathbf{k}}, \hat{\mathbf{k}}', \hat{\mathbf{k}}''$.

According to (8.2.6), each pair of values $(u,v)$ is taken twice when $\varphi'$ ranges from 0 to $2\pi$ (once for $\varphi'$ and once again for $2\pi-\varphi'$), so that, taking into account (8.2.12), the unitarity condition (8.2.11) becomes

$$g(s,w) = \frac{\sqrt{s}}{2\pi}\iint f(s-i0,v)f(s+i0,u)\frac{\theta(\Delta)\,du\,dv}{[\Delta(u,v,w)]^{1/2}}, \tag{8.2.14}$$

where the Heaviside step function $\theta(\Delta)$ ($=1$ for $\Delta>0$, $=0$ for $\Delta<0$) expresses the fact that, in accordance with (8.2.13), the integration is extended over the range of the variables $(u,v)$ where $\Delta>0$. The variables (8.2.10) may be expressed in terms of momentum transfer variables by [cf. (8.2.9)]

$$u = 1+(t'/2s), \qquad v = 1+(t''/2s), \qquad w = 1+(t/2s). \tag{8.2.15}$$

## 8.3. Determination of the Scattering Amplitude from Mandelstam's Representation and Unitarity

### (a) Mandelstam's Iteration Method

Mandelstam's representation (8.1.27) expresses the analytic properties of the scattering amplitude arising from causality (analyticity in $s$ for fixed $t$) and from the analytic character of the interaction [cf. (7.3.21)] as well as its finite range properties (analyticity in $t$ for fixed $s$). Together with unitarity (Section 8.3), this seems to exhaust all the properties of the scattering amplitude that follow from general physical principles verified by the interaction.

One is then led to the question whether this information is also sufficient to enable us to completely determine the scattering amplitude. Specifically: given the Mandelstam representation and the unitarity condition, can one construct the scattering amplitude from these properties alone?

To "give" the Mandelstam representation implies giving the subtraction term $f_B(t)$ in (8.1.27). Since Born's approximation $f_B(t)$ is essentially the Fourier transform of the potential [cf. (6.3.23)], this is clearly equivalent to giving the potential. It would thus be possible, in principle, to substitute it in Schrödinger's equation and then solve it to find the scattering amplitude.

The whole point of the above question, however, is that, if the answer is affirmative, one can bypass Schrödinger's equation: Mandelstam's representation, together with unitarity, would form a complete dynamical scheme, embodying all the information contained in Schrödinger's equation, insofar as the scattering amplitude is concerned.

The suggestion that the $S$-matrix may be completely determined by dispersion relations and unitarity in quantum field theory was made by Gell-Mann [14].[6] A specific program to do this in terms of double dispersion relations, together with an iteration scheme for constructing the scattering amplitude, was proposed by Mandelstam [6]. Mandelstam's program in the case of potential scattering by Yukawa-type potentials was implemented by Blankenbecler *et al.* [7].

Let us consider first, for simplicity, the case in which no subtractions are necessary in the dispersion relation (8.1.7) for the absorptive part and no bound states are present; according to (7.6.9) and (7.6.10), the first assumption is always valid for large enough energy, if the potential satisfies (7.4.21). The Mandelstam representation (8.1.27) becomes

$$f(s,t) = -\int_m^\infty \frac{\sigma(\mu)\,d\mu}{\mu^2 - t} + \frac{1}{\pi^2}\int_0^\infty \frac{ds'}{s'-s}\int_{4m^2}^\infty \frac{dt'}{t'-t}\,\rho(s',t'), \qquad (8.3.1)$$

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[6] Cf. the quotation at the beginning of the present chapter.

where we have employed (8.1.13). On the other hand, according to (8.1.7), the unitarity condition in the form (8.2.11) becomes

$$g(s,t) = \frac{1}{\pi}\int_{4m^2}^{\infty} \rho(s,t')\frac{dt'}{t'-t} = \frac{\sqrt{s}}{4\pi}\int_0^{2\pi} d\varphi' \int_{-1}^{1} f(s-i0,v)f(s+i0,u)\,du, \tag{8.3.2}$$

where, according to (8.2.6), (8.2.9), and (8.2.10),

$$v = uw + [(1-u^2)(1-w^2)]^{1/2}\cos\varphi' \qquad (w = 1+t/2s). \tag{8.3.3}$$

Substituting $f$ in the right-hand side of (8.3.2) by the Mandelstam representation (8.3.1), we get

$$\begin{aligned}
16s^{3/2}\int_{4m^2}^{\infty}\rho(s,t')\frac{dt'}{t'-t} &= \int_m^{\infty} d\mu_1\,\sigma(\mu_1)\int_m^{\infty} d\mu_2\,\sigma(\mu_2)\,I\left(1+\frac{\mu_1{}^2}{2s}, 1+\frac{\mu_2{}^2}{2s}, t\right)\\
&\quad - \frac{2}{\pi^2}P\int_0^{\infty}\frac{ds_1}{s_1-s}\int_{4m^2}^{\infty} dt_1\,\rho(s_1,t_1)\\
&\quad \times \int_m^{\infty} d\mu_1\,\sigma(\mu_1)\,I\left(1+\frac{t_1}{2s}, 1+\frac{\mu_1{}^2}{2s}, t\right)\\
&\quad + \frac{1}{\pi^4}\int_0^{\infty}\frac{ds_1}{s_1-s+i0}\int_0^{\infty}\frac{ds_2}{s_2-s-i0}\int_{4m^2}^{\infty} dt_1\\
&\quad \times \int_{4m^2}^{\infty} dt_2\,\rho(s_1,t_1)\rho(s_2,t_2)\,I\left(1+\frac{t_1}{2s}, 1+\frac{t_2}{2s}, t\right),
\end{aligned} \tag{8.3.4}$$

where

$$I(q_1,q_2,t) = \int_0^{2\pi} d\varphi' \int_{-1}^{1}\frac{du}{(q_1-u)(q_2-v)}, \tag{8.3.5}$$

and we have employed [cf. (A11.5)]

$$[1/(s_1-s+i0)] + [1/(s_1-s-i0)] = 2P/(s_1-s),$$

where $P$ denotes the Cauchy principal value.

We now want to find a dispersion relation in $t$ for the function $I(q_1,q_2,t)$, so that we may substitute it in (8.3.4) and then identify the discontinuities of the left- and the right-hand sides across the cut. This problem is tackled in

Appendix I, and the result is given by (I12). Substituting (I12) in (8.3.4) and identifying the discontinuities across the cut in the $t$-plane, we are led to

$$\begin{aligned}
\theta(t-4m^2)\rho(s,t) &= \int_m^\infty d\mu_1\,\sigma(\mu_1)\int_m^\infty d\mu_2\,\sigma(\mu_2)K(s,t;\mu_1{}^2,\mu_2{}^2)\\
&\quad-\frac{2}{\pi^2}P\int_0^\infty\frac{ds_1}{s_1-s}\int_{4m^2}^\infty dt_1\,\rho(s_1,t_1)\\
&\quad\times\int_m^\infty d\mu_1\,\sigma(\mu_1)\,K(s,t;t_1,\mu_1{}^2)\\
&\quad+\frac{1}{\pi^4}\int_0^\infty\frac{ds_1}{(s_1-s+i0)}\int_0^\infty\frac{ds_2}{(s_2-s-i0)}\int_{4m^2}^\infty dt_1\\
&\quad\times\int_{4m^2}^\infty dt_2\,\rho(s_1,t_1)\rho(s_2,t_2)K(s,t;t_1,t_2)\\
&= \rho_1(s,t)+\rho_2(s,t)+\rho_3(s,t)\qquad(s\geqslant 0,\ t\geqslant 0),
\end{aligned}\tag{8.3.6}$$

where $K(s,t;t_1,t_2)$ is defined by (I13).

According to (8.1.13), giving the Born approximation $f_B(t)$ is equivalent to giving the weight function $\sigma(\mu)$ of the Yukawa superposition (7.1.1), so that (8.3.6) becomes an integral equation for the double spectral function $\rho(s,t)$. By (8.3.1), the solution of this integral equation completely determines the scattering amplitude in the no-subtraction case.

The integral equation (8.3.6) can now be solved by *Mandelstam's iteration method*. The key to the solution is the Heaviside step function in (I13), which tells us that $K(s,t;t_1,t_2)\neq 0$ if and only if

$$\begin{aligned}
t &> t_1+t_2+\frac{t_1t_2}{2s}+\frac{(t_1t_2)^{1/2}}{2s}\left[16s^2+4s(t_1+t_2)+t_1t_2\right]^{1/2}\\
&= \left[\left(t_1+\frac{t_1t_2}{4s}\right)^{1/2}+\left(t_2+\frac{t_1t_2}{4s}\right)^{1/2}\right]^2,
\end{aligned}\tag{8.3.7}$$

or, equivalently [cf. (I11)], if the expression under the square root in the denominator of (I13) is positive. Given $t_1$ and $t_2$, the right-hand side of (8.3.7) is a monotonically decreasing function of $s$, which is asymptotic to the hyperbola $t=t_1t_2/s$ for $s\to 0$ and to the straight line $t=(\sqrt{t_1}+\sqrt{t_2})^2$ for $s\to\infty$. Thus a *necessary* condition to have $K(s,t;t_1,t_2)\neq 0$ is

$$\sqrt{t}>\sqrt{t_1}+\sqrt{t_2}.\tag{8.3.8}$$

Note also from (8.3.7) that, if $K(s,t;t_{10},t_{20})=0$, we have a fortiori $K(s,t;t_1,t_2)=0$ if $t_1>t_{10}$ and $t_2>t_{20}$, so that if (8.3.8) is not satisfied for the minimum values of $t_1$ and $t_2$ in one of the terms in (8.3.6), the contribution from that term vanishes.

It follows that, in (8.3.6),

$$\rho_1 = 0 \quad (t < 4m^2), \qquad \rho_2 = 0 \quad (t < 9m^2), \qquad \rho_3 = 0 \quad (t < 16m^2). \tag{8.3.9}$$

For $t < 4m^2$, both sides of (8.3.6) vanish. In fact, substituting $t_1$ and $t_2$ by their minimum possible values $m^2$ in (8.3.7), we see that

$$\rho(s,t) \neq 0 \qquad \text{iff} \quad t > m^2[4+(m^2/s)], \tag{8.3.10}$$

corresponding to the region above the curve $\Gamma_1$ in Fig. 8.2.

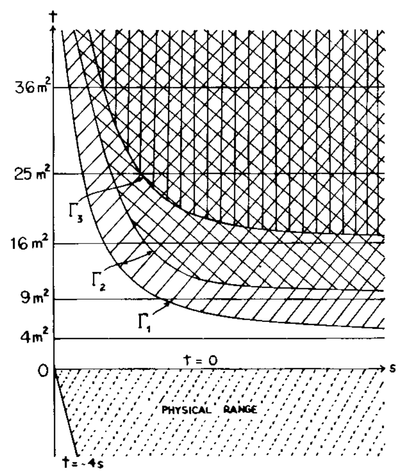


*FIG. 8.2.* Decomposition of the $(s,t)$ plane in Mandelstam's iteration method. The terms $\rho_1, \rho_2, \rho_3$ in (8.3.6) differ from zero only above the curves $\Gamma_1, \Gamma_2, \Gamma_3$, respectively. In the singly hatched region, the double spectral function is given exactly by the second Born approximation. The physical range is also shown.

If $4m^2 < t < 9m^2$, only $\rho_1$ is $\neq 0$ in (8.3.6), so that $\rho = \rho_1$ in this strip of the $(s,t)$-plane. Since $\rho_1$ depends only on $\sigma(\mu)$, it is a known term, and it is readily seen[7] that the corresponding contribution to $f(s,t)$ in (8.3.1) is nothing but the second Born approximation (6.4.37). If we take $\sigma(\mu) = \lambda\sigma_0(\mu)$, introducing a parameter $\lambda$ to measure the strength of the potential, we have $\rho_1 = O(\lambda^2)$.

In the second strip, $9m^2 < t < 16m^2$, according to (8.3.9), only $\rho_1$ and $\rho_2$ contribute to $\rho$. Actually, according to (8.3.7), $\rho_2 \neq 0$ if and only if

$$t > m^2[(4+(m^2/s))^{1/2} + (1+(m^2/s))^{1/2}]^2, \tag{8.3.11}$$

corresponding to the region above the curve $\Gamma_2$ in Fig. 8.2. To compute $\rho_2$ in this strip, we need the value of $\rho(s_1, t_1)$ in the second integral in (8.3.6). However,

[7] Cf., for example, Goldberger and Watson [15, p. 605].

for $t < 16m^2$ and $t_2 = \mu_1{}^2 \geqslant m^2$, condition (8.3.8) cannot be satisfied unless $t_1 < 9m^2$, so that the integration over $t_1$ in $\rho_2$ effectively extends only up to $9m^2$. Thus, to compute $\rho$ in this strip, we only need the value of $\rho$ in the preceding strip, where it is already known. We therefore find the exact solution in the second strip, containing terms in $\lambda^2$ and $\lambda^3$.

In the third strip, $16m^2 < t < 25m^2$, all three terms in (8.3.6) contribute, but the contribution from $\rho_3$ is nonvanishing only for

$$t > 16m^2[1+(m^2/s)], \tag{8.3.12}$$

corresponding to the region above the curve $\Gamma_3$ in Fig. 8.2. Note that this also implies that $\rho(s,t)$ is given *exactly* by the second Born approximation throughout the singly hatched region in Fig. 8.2. It again follows from (8.3.8) that, to compute $\rho_2$ in this strip, we only need the values of $\rho(s_1,t_1)$ for $t_1 < 16m^2$, i.e., in the first and in the second strip; similarly, in the computation of $\rho_3$, the effective upper limits of integration in (8.3.6) are $t_1 = 9m^2$, $t_2 = 9m^2$, so that $\rho_3$ in the third strip depends only on the values of $\rho$ in the first strip. Thus we can find the exact solution in the third strip, containing terms in $\lambda^2$, $\lambda^3$, and $\lambda^4$.

In the same way, we see from (8.3.8) that, for $nm < \sqrt{t} < (n+1)m$, $n \geqslant 5$, the contribution from $\rho_2$ in (8.3.6) depends only on the value of $\rho$ for $\sqrt{t_1} < nm$, $\sqrt{t_2} < (n-1)m$. We can therefore proceed by iteration to obtain the exact solution in any strip. At any point in the finite $(s,t)$-plane, the double spectral function $\rho(s,t)$ is a polynomial in $\lambda$ of finite degree; the degree increases with the order of the strip.

If we subdivide the domain of integration in (8.3.1) into strips in the same way, we can in principle substitute the solution found for $\rho$ in each strip and perform the integration strip by strip. This leads to an expression for the scattering amplitude $f(s,t)$ as the limit of a sequence of polynomials of increasing degree in $\lambda$. Since we obtain the exact solution in each strip, this sequence necessarily converges if our initial assumption that the unsubtracted Mandelstam representation (8.3.1) converges is valid.

The curves (such as $\Gamma_1$, $\Gamma_2$, $\Gamma_3$ in Fig. 8.2) that constitute the boundaries of regions where successive contributions to the double spectral function arise are known as *Landau curves* [16]. The stepwise structure of $\rho(s,t)$, with additional contributions starting at these boundaries, appears in the absorptive part (8.1.7) as a series of superimposed branch cuts, starting at $t = 4m^2$, with additional branch points asymptotically approaching (for large $s$) the successive strip boundaries $t = (nm)^2$ $(n = 3,4,5,\ldots)$. If we interpret the Yukawa potential in terms of the field-theoretical picture of meson exchange [17], these successive branch points can be associated with the thresholds corresponding to the exchange of $n$ mesons. It can readily be shown [8, 9] that the $n$th term in the Born series has a cut starting at $t = (nm)^2$.

### (b) Subtractions and Partial-Wave Projection

The analysis in Section 8.3(a) was based on the assumed validity of the unsubtracted Mandelstam representation without bound states (8.3.1). Although this assumption may hold for a repulsive or a weakly attractive potential, it certainly breaks down for sufficiently strong attractive potentials. Let us now reconsider the problem of determining $f(s,t)$ from Mandelstam's representation [given $f_B(t)$] and unitarity in the general case of (8.1.27). Besides the double spectral function $\rho(s,t)$, we then have, as additional unknowns, the single spectral functions $g_l(s)$ and the bound state energies $s_p$ and residues $\Gamma_p(t)$.

If we insert (8.1.27) instead of (8.3.1) in the right-hand side of the unitarity condition (8.3.2), we see that the right-hand side of (8.3.4) is replaced by a sum of sixteen terms, corresponding to all possible double products formed from the four terms of (8.1.27). However, we are interested only in the discontinuity of the right-hand side across the cut in the $t$-plane [cf. (8.3.6)], so that any term which is a holomorphic function of $t$ [e.g., the product of two polynomials in $t$, corresponding to the second and third terms of (8.1.27)] does not contribute.

To see what happens to the remaining contributions, it suffices to consider the analog of the last term in (8.3.4). The only difference here lies in the replacement

$$I\left(1+\frac{t_1}{2s}, 1+\frac{t_2}{2s}, t\right) \to \left(\frac{2s}{t_1}\right)^{L+1}\left(\frac{2s}{t_2}\right)^{L+1} \int \frac{(u-1)^{L+1}(v-1)^{L+1}}{[1+(t_1/2s)-u][1+(t_2/2s)-v]}\, d\Omega. \tag{8.3.13}$$

In the numerator, we can write

$$u-1 = [u-1-(t_1/2s)] + (t_1/2s), \qquad v-1 = [v-1-(t_2/2s)] + (t_2/2s),$$

and we can then make use of the binomial expansion to expand the numerator in powers of $u-1-(t_1/2s)$ and $v-1-(t_2/2s)$, splitting the integral into the sum of the corresponding integrals. In any integral containing a power greater than zero of either one of the two terms, at least one of the two factors in the denominator cancels out. If only the factor $1+(t_1/2s)-u$ is left in the denominator, we perform the integration over solid angles by taking $\theta'$ in (8.2.10) as the polar angle ($z$-axis along $\hat{\mathbf{k}}$), with the help of (8.2.6); if only $1+(t_2/2s)-v$ is left in the denominator, we take $\zeta$ in (8.2.10) as the polar angle ($z$-axis along $\hat{\mathbf{k}}'$), expressing $\cos\theta'$ in terms of $\theta$ and $\zeta$ by means of a relation analogous to (8.2.6). We then see that all these terms give rise to polynomials in $\cos\theta$, and they therefore do not contribute to the discontinuity across the cut in the $t$-plane.

The only surviving term in the numerator is that which contains the zeroth powers of both factors in the denominator, i.e., $(t_1/2s)^{L+1}(t_2/2s)^{L+1}$; substituting this in (8.3.13) and comparing it with (8.3.5), we see that the contribution of this term to (8.3.6) is unchanged. Similar arguments apply to the other terms so that, finally, we see that *(8.3.6) still holds and Mandelstam's iteration method for the construction of $\rho(s,t)$ remains valid and unchanged, even when there are subtractions and bound states.*

Once $\rho(s,t)$ is determined, we are still left with the problem of evaluating the single spectral functions, the bound-state poles, and their residues. This problem turns out to be closely related with the properties of partial-wave amplitudes, which may be obtained from the total amplitude by partial-wave projection, with the help of (3.1.4):

$$f_l(s) = (1/2i)[S_l(s)-1] = \exp(i\eta_l)\sin\eta_l$$
$$= (\sqrt{s}/2)\int_0^\pi f(s,\cos\theta)P_l(\cos\theta)\sin\theta\,d\theta. \quad (8.3.14)$$

Substituting (8.1.27), with $t = 2s(\cos\theta - 1)$, in the right-hand side of (8.3.14), we are led to the evaluation of integrals of the type

$$I_{n,l}(x) = \int_{-1}^{1} [(z-1)^n/(z-x)]P_l(z)\,dz \qquad (x>1). \quad (8.3.15)$$

To evaluate this integral we write

$$\frac{(z-1)^n}{z-x} = \frac{[(z-x)+(x-1)]^n}{z-x} = \sum_{j=0}^{n-1}\binom{n}{j}(z-x)^{n-j-1}(x-1)^j + \frac{(x-1)^n}{z-x}.$$

Taking into account Neumann's formula [18, p. 320]

$$\int_{-1}^{1} P_l(z)\frac{dz}{z-x} = -2Q_l(x), \quad (8.3.16)$$

where $Q_l$ is the Legendre function of the second kind, we get

$$I_{n,l}(x) = -2(x-1)^n Q_l(x) + \Pi_{n-1-l}(x), \quad (8.3.17)$$

where $\Pi_{n-1-l}(x)$ denotes a polynomial of degree $n-1-l$ in $x$ (if $l \geqslant n$, this term is not present).

According to (8.1.11) and (8.1.12), the second and third terms of (8.1.27) do not contribute for $l \geqslant L+1$, so that, taking into account (8.1.13), we finally get

$$\frac{f_l(s)}{\sqrt{s}} = \theta_{l,L}\sum_p \frac{c_{p,l}}{s-s_{p,l}} + \frac{1}{\pi}\sum_{n=l}^{L} s^n \int_0^\infty \frac{\gamma_n(s')}{s'-s}\,ds'$$
$$- \int_m^\infty d\mu\,\frac{\sigma(\mu)}{2s}\,Q_l\left(1+\frac{\mu^2}{2s}\right) + \frac{1}{\pi^2}\int_0^\infty \frac{ds'}{2s(s'-s)}\int_{4m^2}^\infty dt'\rho(s',t')$$
$$\times\left[Q_l\left(1+\frac{t'}{2s}\right) - \left(\frac{2s}{t'}\right)^{L+1}\Pi_{L-l}\left(\frac{t'}{2s}\right)\right], \quad (8.3.18)$$

where $s_{p,l}$ are the energies of the bound states with angular momentum $l$, and $\theta_{l,L}=1$ for $0 \leqslant l \leqslant L$, $\theta_{l,L}=0$ for $l \geqslant L+1$.

For $l \geqslant L+1$, the first two terms, as well as the last term within square brackets, are not present, and the result may be rewritten as

$$f_l(s) = (1/2\pi\sqrt{s}) \int_0^\infty Q_l[1+(t/2s)]\,h(s,t)\,dt \qquad (l \geqslant L+1), \quad (8.3.19)$$

where

$$h(s,t) = -\frac{2\pi}{\sqrt{t}}\,\theta(t-m^2)\,\sigma(\sqrt{t}) + \frac{1}{\pi}\,\theta(t-4m^2) \int_0^\infty \rho(s',t)\,\frac{ds'}{s'-s}. \quad (8.3.20)$$

It follows that *all partial-wave amplitudes with $l \geqslant L+1$ are already determined by the double spectral function*; they do not depend on the remaining unknowns.

The function $h(s,t)$ has a very simple interpretation: it gives the discontinuity of the scattering amplitude across the cut in the $t$-plane:

$$h(s,t) = (1/2i)[f(s,t+i0)-f(s,t-i0)]; \qquad (8.3.21)$$

this follows immediately from the Mandelstam representation (8.1.27). The function $h(s,t)$ would appear in writing down a dispersion relation in the momentum transfer (for fixed energy) for $f(s,t)$; such a dispersion relation follows from the Mandelstam representation [(8.3.21) should be compared with (6.5.3)].

We have now reduced the problem of completely determining the scattering amplitude to the determination of the partial-wave amplitudes with $0 \leqslant l \leqslant L$ [in fact, this determines the bound-state poles and residues, and the partial-wave expansion may be employed to evaluate the single spectral functions (8.1.8)]. Can we determine these lower partial-wave amplitudes by analytic continuation in the angular momentum?

Let us make the substitution $l=\lambda+\frac{1}{2}$ in (8.3.19) and (8.3.14), defining

$$(1/2i)[\tilde{S}(\lambda,s)-1] = (1/2\pi\sqrt{s}) \int_0^\infty Q_{\lambda-1/2}[1+(t/2s)]\,h(s,t)\,dt. \quad (8.3.22)$$

The function $Q_{\lambda-1/2}(z)$ is holomorphic in $\operatorname{Re}\lambda > 0$. Its asymptotic behavior for $|z| \gg 1$ is given by [19, p. 66]

$$Q_{\lambda-1/2}(z) = \frac{\sqrt{\pi}\,\Gamma(\lambda+\frac{1}{2})}{2^{\lambda+1/2}\,\Gamma(\lambda+1)}\,\frac{1}{z^{\lambda+1/2}}\,[1+O(z^{-2})] \qquad (|z| \gg 1). \quad (8.3.23)$$

On the other hand, according to (8.3.21) and Theorem 8.1.1, $h(s,t)=O(t^L)$ as $t\to\infty$ [cf. also (8.1.26)]. It then follows from (8.2.23) that the integral in (8.3.22) converges and represents a holomorphic function of $\lambda$, in the half-plane

$$\operatorname{Re}\lambda > L+\tfrac{1}{2}. \qquad (8.3.24)$$

For $|\lambda| \to \infty$ in this half-plane, we have, according to (7.5.16) and (7.5.17),

$$|Q_{\lambda-\frac{1}{2}}(\cosh\chi)| \lesssim (\pi/|\lambda|)^{\frac{1}{2}} |e^{-\chi(\lambda+\frac{1}{2})}/(1-e^{-2\chi})^{\frac{1}{2}}| \qquad (|\lambda| \to \infty), \tag{8.3.25}$$

so that, according to (8.3.22), we have, in the half-plane (8.3.24),

$$|\tilde{S}(\lambda,s)-1| \to 0 \qquad (|\lambda| \to \infty). \tag{8.3.26}$$

The function $\tilde{S}(\lambda,s)-1$ is therefore holomorphic in the half-plane (8.3.24) and it verifies (8.3.26) in this half-plane; for $\lambda = l+\frac{1}{2}$, $l = L+1, L+2, \ldots$, according to (8.3.19), it reduces to $S_l(s)-1$. It then follows from Carlson's theorem 7.5.4, as in Theorem 7.5.5, that

$$\tilde{S}(\lambda,s) = S(\lambda,s) \qquad (\operatorname{Re}\lambda > L+\tfrac{1}{2}), \tag{8.3.27}$$

where $S(\lambda,s)$ is the Regge interpolation defined in Chapter 7. We may thus rewrite (8.3.22) as

$$S(\lambda,k) - 1 = (i/\pi k)\int_0^\infty Q_{\lambda-\frac{1}{2}}[1+(t/2k^2)]\,h(s,t)\,dt \qquad (\operatorname{Re}\lambda > L+\tfrac{1}{2}). \tag{8.3.28}$$

This explicit representation of the $S$-function is known as the *Froissart–Gribov projection* [20].

We know from Chapter 7 that $S(\lambda,k)$ has a unique analytic continuation down to $\operatorname{Re}\lambda = 0$. It is therefore possible, at least in principle, to analytically continue (8.3.28) along the real axis, and in this way to determine the partial-wave amplitudes $f_l$ for $l = 0, 1, \ldots, L$. As we have seen, this in turn determines the remaining unknowns $g_l(s)$, $s_p$, and $\Gamma_p(t)$ in (8.1.27), thus completing the determination of the total scattering amplitude. A direct procedure for the determination of these unknowns by analytic continuation in the angular momentum, without going through the partial-wave amplitudes, has been given by Bessis [20a].

We conclude that, for Yukawa-type potentials, *Mandelstam's representation and unitarity indeed enable us, in principle, to completely determine the total scattering amplitude, without going through Schrödinger's equation.* This very remarkable result may be regarded as the most complete realization, to date, of Heisenberg's original program of constructing a "pure $S$-matrix theory" (cf. Section 1.1).[7a]

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[7a] In high-energy physics it has been shown [20b], for pion-pion scattering, that there exists an infinite class of functions satisfying the Mandelstam representation, crossing symmetry, elastic unitarity, and consistent with the inelastic unitarity constraints (inequalities) above the inelastic threshold; the corresponding double spectral functions differ by the contributions due to inelastic processes.

### (c) The N/D Method

Although we have just seen that the lower partial waves can in principle be determined from the Froissart–Gribov projection by analytic continuation in the angular momentum, it is useful to discuss another method for constructing them which may be more amenable to practical calculations.

From (8.3.18) we can rederive the analytic properties of partial-wave amplitudes that have already been given in Theorem 5.8.1. As is obvious from (8.3.16), the function $Q_l(x)$ has a cut along the real axis, from $x=-1$ to $+1$, and the discontinuity across the cut is given by

$$(1/2i)[Q_l(x+i0)-Q_l(x-i0)] = -(\pi/2)P_l(x)\theta(1-x^2), \qquad (8.3.29)$$

where $\theta$ is the Heaviside step function. It then follows from (8.3.18) that, besides the possible bound-state poles, the singularities of $f_l(s)$ are: (a) the usual kinematical "right-hand cut" from $s=0$ to $\infty$; (b) arising from the terms (8.3.19) involving $Q_l$, a "left-hand cut" from $s=-m^2/4$ to $-\infty$, with the discontinuity across the cut following from (8.3.29),

$$\begin{aligned} f_l(s+i0)-f_l(s-i0) &= (1/2\sqrt{|s|})\int_0^{-4s} P_l(1+(t/2s))\,h(s,t)\,dt \\ &= \alpha_l(s) \qquad (-\infty < s \leqslant -m^2/4), \end{aligned} \qquad (8.3.30)$$

where $\alpha_l(s)$, according to (8.3.20), may be regarded as a known function.

According to (8.3.14) and (5.8.17), we may express $f_l(s)$ in terms of the Jost function $F_l(k)$ as[8]

$$\frac{f_l(s)}{\sqrt{s}} = \frac{(1/2ik)[F_l(k)-F_l(-k)]}{F_l(-k)} = \frac{N_l(s)}{D_l(s)}. \qquad (8.3.31)$$

According to Section 5.8(a), the denominator function $D_l(s)=F_l(-k)$ is holomorphic in $I_+(k)$, which corresponds to the physical sheet of the $s$-plane, so that $D_l(s)$ *has only the right-hand cut*, from $s=0$ to $\infty$. Furthermore, still by Section 5.8(a),

$$D_l(s)\to 1 \qquad (|s|\to\infty). \qquad (8.3.32)$$

On the other hand, the numerator function $N_l(s)=[F_l(k)-F_l(-k)]/2ik$ depends only on $k^2=s$ so that it has no right-hand cut: $N_l(s)$ *has only the left-hand cut*, from $-\infty$ to $-m^2/4$. Since both $F_l(k)$ and $F_l(-k)$ approach 1 as $|k|\to\infty$,

$$N_l(s)\to 0 \qquad (|s|\to\infty). \qquad (8.3.33)$$

[8] The $N/D$ method was first suggested by Chew and Mandelstam [21]. Cf. Noyes and Wong [22].

Applying (7.3.3) and (7.3.6) for physical values of $\lambda$, we see that

$$F_l(k) = F_l^*(-k^*),$$

so that $D_l(s)$ *is real for* $s<0$ *and* $N_l(s)$ *is real for* $s>0$.

It follows from the above properties, together with (8.3.30) and (8.3.31), that

$$\frac{N_l(s+i0)-N_l(s-i0)}{2i\,D_l(s)} = \frac{\operatorname{Im} N_l(s)}{D_l(s)} = -\frac{\alpha_l(s)}{2\,|s|^{1/2}} \qquad (-\infty < s \leqslant -m^2/4). \tag{8.3.34}$$

On the other hand, (8.3.31) and the unitarity condition (3.1.5) imply

$$\begin{aligned}\operatorname{Im}[1/f_l(s)] = -\operatorname{Im} f_l/|f_l|^2 &= -1 \\ &= \operatorname{Im} D_l(s)/\sqrt{s}\,N_l(s) \\ &= [D_l(s+i0)-D_l(s-i0)]/2i\sqrt{s}\,N_l(s) \qquad (s \geqslant 0).\end{aligned} \tag{8.3.35}$$

From (8.3.32) to (8.3.35), together with the analytic properties of $N_l$ and $D_l$, we get the following dispersion relations for these functions:

$$N_l(s) = -(1/2\pi)\int_{-\infty}^{-m^2/4} [\alpha_l(s')\,D_l(s')/|s'|^{1/2}(s'-s)]\,ds', \tag{8.3.36}$$

$$D_l(s) = 1-(1/\pi)\int_0^{\infty} [\sqrt{s'}\,N_l(s')/(s'-s)]\,ds'. \tag{8.3.37}$$

Substituting $N_l$ by (8.3.36) in (8.3.37) and interchanging the order of integration, we find

$$D_l(-s) = 1-(1/2\pi)\int_{m^2/4}^{\infty} [\alpha_l(-s')\,D_l(-s')/\sqrt{s'}\,(\sqrt{s'}+\sqrt{s})]\,ds'. \tag{8.3.38}$$

With the substitution $s' = 1/x^2$, this becomes a nonsingular Fredholm integral equation, which has a unique solution. With the help of (8.3.36) and (8.3.31), the solution completely determines the partial-wave amplitudes. In particular, according to (8.3.31), the zeros of the denominator function determine the bound-state energies.

An exactly soluble example that can be treated to illustrate these results is that of the exponential potential (5.4.49) for $l=0$. In this case, the left-hand cut degenerates into the series of poles (5.4.55), and (8.3.38) reduces to a difference equation which can be solved by standard techniques. The result [7] agrees with the exact solution [cf. (5.4.53)] and the bound states are given by (5.4.54).

If we had not imposed the additional constraints (8.3.32) and (8.3.33), the $N/D$ representation (8.3.31) would still have allowed a considerable amount of arbitrariness in the determination of the numerator and denominator functions (note that additional common factors cancel out in the ratio $N/D$). We could have made additional subtractions, involving arbitrary subtraction constants, e.g., to exhibit explicitly the correct threshold behavior (5.8.22) as $s \to 0$. In practical calculations, where approximations must be made, such arbitrary subtraction constants may introduce spurious singularities. These arbitrary elements that may give rise to difficulties in approximate calculations correspond to the CDD ambiguities [23, 24] already discussed in Section 2.5.

Some numerical tests of Mandelstam's iteration method have been carried out. Burke and Tate [25] considered the Yukawa-type potential (7.1.1) defined by $\sigma(\mu) = -10\mu/9\pi$ for $2 \leqslant \mu \leqslant 20$, $\sigma(\mu) = 0$ otherwise. They then performed the Mandelstam iteration numerically until they reached a value of $t$ sufficiently large that the asymptotic behavior

$$g(s,t) \approx \beta(s)\, t^{\lambda_M(s)-\frac{1}{2}} \tag{8.3.39}$$

for the absorptive part [cf. (7.6.10)] had developed. From (8.3.39) they computed the trajectory $\lambda_M(s)$ of the rightmost Regge pole; the result was compared with that obtained from the numerical solution of the Schrödinger equation. They found agreement within 1%. However, the numerical accuracy was not sufficient to obtain the second Regge trajectory. A similar calculation for a different potential was performed by Bali [26], who also evaluated the residue function $\beta(s)$ in (8.3.39). Collins and Johnson [27] combined the Mandelstam iteration with the $N/D$ method to compute Regge trajectories and bound-state positions for somewhat stronger attractive Yukawa-type potentials.

## 8.4. Cutoff Potentials

### *(a) Asymptotic Behavior in the Momentum Transfer*

Besides Yukawa-type potentials, there exists another class of potentials for which the analytic behavior of the scattering amplitude in the whole momentum transfer plane is known, namely, cutoff potentials (cf. Section 7.1). In this case, the scattering amplitude is an entire function of the momentum transfer, represented, for all values of $t$, by the partial-wave expansion (6.5.9),

$$f(s,t) = (1/\sqrt{s}) \sum_{l=0}^{\infty} (2l+1) f_l(s) P_l(1+(t/2s)). \tag{8.4.1}$$

The dispersion relation for fixed momentum transfer (8.1.10) is valid for all $t$. Thus, in order to find a double dispersion representation for $f(s,t)$, we need only to determine the asymptotic behavior of the absorptive part $g(s,t)$ in (8.1.10) as $|t| \to \infty$. According to (6.5.16),

$$g(s,t) = (1/\sqrt{s}) \sum_{l=0}^{\infty} (2l+1)|f_l(s)|^2 P_l(1+(t/2s)). \tag{8.4.2}$$

The asymptotic behavior of $P_l$ as $|t| \to \infty$ follows from [28, p. 189]

$$P_l(z) \approx (2l-1)!!z^l/l! \qquad (|z| \to \infty,\ l = 0,1,2,\ldots), \tag{8.4.3}$$

where $(2l-1)!! = 1 \cdot 3 \cdot 5 \cdots (2l-1)$. Thus (8.4.2) behaves like a power series in $t$ as $|t| \to \infty$. As is well known in the theory of entire functions, the asymptotic behavior of $g(s,t)$ as $|t| \to \infty$ is therefore determined by the asymptotic behavior of the partial waves as $l \to \infty$.

According to Sections 5.8(b) and 7.5, the asymptotic behavior of the partial-wave amplitudes as $l \to \infty$ is bounded by Born's approximation,

$$f_l(k) \approx -k \int_0^a r^2 j_l^2(kr)\, U(r)\, dr. \tag{8.4.4}$$

For $l \gg (ka)^2$, we can substitute $j_l$ by (3.3.19). If $U^{(m)}(a-)$ denotes the first nonvanishing derivative of $U(r)$ at $r = a-0$, we then find from (8.4.4), by repeated partial integration,

$$|f_l(k)| \approx \frac{a^{m+2}|U^{(m)}(a-)|(ka)^{2l+1}}{[(2l+1)!!]^2(2l+3)(2l+4)\cdots(2l+3+m)} \qquad (l \to \infty). \tag{8.4.5}$$

Substituting (8.4.3) and (8.4.5) in (8.4.2), we find that, up to a polynomial in $t$, the asymptotic behavior of $g(s,t)$ as $|t| \to \infty$ is the same as that of the function

$$\gamma(k,\tau) = ka^{2m+6}|U^{(m)}(a-)|^2 \times \sum_{l=0}^{\infty} (-1)^l \frac{(l!)^2}{[(2l+1)!]^3} \frac{(2a^2k\tau)^{2l}}{(2l+3)^2(2l+4)^2\cdots(2l+3+m)^2}, \tag{8.4.6}$$

where $k$ and $\tau$ are given by (8.1.4).

We can now apply (7.1.2) to rewrite (8.4.6) as a contour integral in the $\lambda$-plane,

$$\gamma(k,\tau) = \frac{ka^{2m+6}}{2i}|U^{(m)}(a-)|^2 \times \int_C \frac{[\Gamma(\lambda+\frac{1}{2})]^2}{[\Gamma(2\lambda+1)]^3} \frac{(2a^2k\tau)^{2\lambda-1}}{(2\lambda+2)^2\cdots(2\lambda+2+m)^2} \frac{d\lambda}{\cos(\pi\lambda)}, \tag{8.4.7}$$

where $C$ is the contour of Fig. 7.1. The asymptotic behavior of (8.4.7) as

$|\tau|\to\infty$ can now be determined [29] by the saddle-point method. The saddle points are located at $(a/2)(\pm ik\tau)^{1/2}$. The result is that $\gamma(k,\tau)$, and therefore also the absorptive part, behaves asymptotically like

$$g(s,t) \approx \frac{ka}{2^{5/2}}|U^{(m)}(a-)|^2$$

$$\times\left\{\frac{\exp[2a(ik\tau)^{1/2}]}{(ik\tau)^{m+5/2}}+\frac{\exp[2a(-ik\tau)^{1/2}]}{(-ik\tau)^{m+5/2}}\right\} \qquad (|\tau|\to\infty). \quad (8.4.8)$$

We have therefore an energy-dependent essential singularity at infinity in the $\tau$-plane. The first term within curly brackets dominates the behavior in the lower half of the $\tau$-plane and the second one dominates in the upper half-plane. Along the real axis, both terms are of the same order and we have oscillations with exponentially increasing amplitude.

One might have expected, at first, on the basis of (3.2.6), (3.2.7), and (3.3.1), to find that $f(k,\tau)=O(e^{-i\tau a})$ as $|\tau|\to\infty$, $\operatorname{Im}\tau\geqslant 0$. However, this holds for the asymptotic behavior in a fixed direction $\theta$, $\tau=2k\sin(\theta/2)$, corresponding to $|\tau|\to\infty$, $|k|\to\infty$ simultaneously. When $|\tau|\to\infty$ with $k$ fixed, the behavior of $f(k,\tau)$ as $\tau\to\infty$ along the real axis is dominated by its imaginary part $g(k,\tau)$, i.e., by (8.4.8); when $|\tau|\to\infty$ in the upper half-plane, in directions away from the real axis, one does find that $f(k,\tau)=O(e^{-i\tau a})$. At any rate, we see that the radius of the scatterer, which did not appear in the asymptotic behavior in $k$ for fixed $\tau$, reappears in the asymptotic behavior for $|\tau|\to\infty$.

In order to derive a dispersion relation in $t$ for $g(s,t)$, one must first remove the essential singularity by introducing a compensating exponential factor, as in (3.2.6). The function

$$G(s,t) = \exp[-2a(st)^{1/4}]g(s,t) \qquad (8.4.9)$$

is holomorphic in the $t$-plane cut along the real axis from $-\infty$ to 0 ($-\pi\leqslant\arg t<\pi$), and, by (8.4.8), it tends to zero as $|t|\to\infty$. Taking into account (6.5.15), it follows that

$$G(s,t) = -\frac{1}{\pi}\int_{-\infty}^{0}\exp[-a(-4st')^{1/4}]\sin[a(-4st')^{1/4}]$$

$$\times \operatorname{Im} f(s,t')\frac{dt'}{t'-t} \qquad (\operatorname{Im} t\neq 0). \qquad (8.4.10)$$

In substituting (8.4.9) and (8.4.10) in the dispersion relation for fixed momentum transfer (8.1.10), it must be remembered that (8.4.8) is valid for fixed $s$ and $|t|\to\infty$, whereas the limit for $s\to\infty$, $|t|\to\infty$ depends on the manner in which both variables approach infinity. We must therefore interpret the integral in (8.1.10) as the limit of an integral with finite upper limit of integration, leading to

$$f(s,t) = f_B(t) + \sum_{p=1}^{N} \frac{\Gamma_p(t)}{s-s_p} - \frac{1}{\pi^2} \lim_{S\to\infty} \int_0^S \frac{\exp[2a(s't)^{1/4}]}{s'-s} ds'$$

$$\times \int_{-\infty}^{0} \exp[-a(-4s't')^{1/4}] \sin[a(-4s't')^{1/4}] \frac{\operatorname{Im} f(s',t')}{t'-t} dt' . \tag{8.4.11}$$

This double dispersion representation is not so useful as Mandelstam's representation (8.1.27), because interchange of the order of integration after proceeding to the limit $S\to\infty$ is not allowed, due to the nonuniformity of the asymptotic behavior in $t$ as $s\to\infty$.

### *(b) Mandelstam's Program*

If we replace $f$ by (8.4.11) in the right-hand side of the unitarity condition (8.2.5), we get, according to (6.5.15), an integral equation for $\operatorname{Im} f(s,t)$. However, the structure of this equation is quite different from that found for Yukawa-type potentials, and Mandelstam's iteration method cannot be applied.

It is nevertheless possible to reformulate Mandelstam's program in the following way: the analytic properties of $f(s,t)$ are contained in the dispersion relation for fixed momentum transfer (8.1.10), valid for all $t$, together with the analyticity in $t$, which entails the validity for all $t$ of the partial-wave expansion (8.4.1). Given these analytic properties, as well as the unitarity condition, can they be employed for the complete determination of the scattering amplitude?

In the present case, in contrast with that of Yukawa-type potentials, it is possible to project the partial waves directly out of the dispersion relation for fixed momentum transfer (8.1.10), with the help of (8.3.14),

$$f_l(k) = (1/2i)[S_l(k)-1] = (k/2) \int_0^\pi f(k,\tau = 2k\sin\theta/2) P_l(\cos\theta) \sin\theta \, d\theta, \tag{8.4.12}$$

where we substitute $f(k,\tau)$ by (8.1.10), with the absorptive part $g$ replaced by its partial-wave expansion (6.5.16). The reason why this is not possible for Yukawa-type potentials is that in (8.4.12) the variable $k$, and therefore also $\tau$, is unrestricted, whereas the partial-wave expansion of the absorptive part for Yukawa-type potentials converges only for $\tau^2 < 4m^2$ [cf. (6.5.1), (6.5.13)]. In the present case, however, the partial-wave expansion converges everywhere.

In carrying out the above-indicated procedure, we have to evaluate the integral

$$C_{l',l}(x) = (l'+\tfrac{1}{2}) \int_0^\pi P_{l'}(1-2x\sin^2\theta/2)\,P_l(\cos\theta)\sin\theta\,d\theta, \tag{8.4.13}$$

where $x = k^2/k'^2$. It can readily be shown [29, Appendix B] that

$$\begin{aligned} C_{l',l}(x) &= 0 && \text{if } l' < l, \\ &= x^l && \text{if } l' = l, \\ &= x^l(1-x)\sum_{s=0}^{m-1}(-1)^s(l;m;s)x^s && \text{if } l' = l+m, \\ &&& m = 1,2,\ldots, \end{aligned} \tag{8.4.14}$$

where

$$(l;m;s) = \frac{(2l+2m+1)}{m(2l+m+1)}\,\frac{(2l+m+s+1)!}{s!(m-s-1)!(2l+s+1)!}. \tag{8.4.15}$$

Let $i\rho_{ln}$ be the residue of $S_l(k)$ at the pole $i\kappa_{ln}$, corresponding to the $n$th bound state of angular momentum $l$ [$\rho_{ln}$ is real; cf. (2.10.30)]. Then, in (8.1.10) [cf. (8.1.11)],

$$\Gamma_{ln}(\tau) = (2l+1)\rho_{ln}P_l(1+(\tau^2/2\kappa_{ln}^2)). \tag{8.4.16}$$

With the help of (8.4.14), the partial-wave projection (8.4.12) out of (8.1.10), (6.5.16) then leads to

$$\begin{aligned} \operatorname{Re} f_l(k) &= f_{lB}(k) + (-1)^l k^{2l+1}\sum_n \frac{\rho_{ln}}{\kappa_{ln}^{2l}(k^2+\kappa_{ln}^2)} \\ &+ \frac{k^{2l+1}}{\pi}\int_{-\infty}^{\infty}\frac{\operatorname{Im} f_l(k')}{k'^{2l+1}(k'-k)}\,dk' \\ &+ (-1)^l k^{2l+1}\sum_{m\geqslant 1}\sum_n \frac{\rho_{l+m,n}}{\kappa_{l+m,n}^{2l+2}}\sum_{s=0}^{m-1}(l;m;s)\frac{k^{2s}}{\kappa_{l+m,n}^{2s}} \\ &+ \frac{k^{2l+1}}{\pi}\sum_{m=1}^{\infty}\sum_{s=0}^{m-1}(-1)^s(l;m;s)k^{2s}\int_{-\infty}^{\infty}\frac{\operatorname{Im} f_{l+m}(k')}{k'^{2l+2s+2}}\,dk', \end{aligned} \tag{8.4.17}$$

where the sums over $m$ and $n$ in the fourth term are both finite sums, and

$$f_{lB}(k) = -k\int_0^a U(r)j_l^2(kr)r^2\,dr \tag{8.4.18}$$

is Born's approximation [cf. (7.5.13)].

Written explicitly, the first few terms in the first few equations of (8.4.17) are (assuming, for simplicity, that no bound states are present)

$$\begin{aligned}
\operatorname{Re} f_0(k) &= f_{0B}(k) + \frac{k}{\pi} \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_0(k')}{k'(k'-k)}\, dk' + \frac{k}{\pi}\left[3 \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_1(k')}{k'^2}\, dk' \right.\\
&\quad \left. + 5 \int_{-\infty}^{\infty} \left(1 - 2\frac{k^2}{k'^2}\right) \frac{\operatorname{Im} f_2(k')}{k'^2}\, dk' + \cdots \right],\\
\operatorname{Re} f_1(k) &= f_{1B}(k) + \frac{k^3}{\pi} \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_1(k')}{k'^3(k'-k)}\, dk' + \frac{k^3}{\pi}\left[5 \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_2(k')}{k'^4}\, dk' \right.\\
&\quad \left. + 7 \int_{-\infty}^{\infty} \left(2 - 3\frac{k^2}{k'^2}\right) \frac{\operatorname{Im} f_3(k')}{k'^4}\, dk' + \cdots \right]. \qquad (8.4.19)
\end{aligned}$$

This is an infinite system of *coupled* partial-wave dispersion relations. It was first considered in the relativistic case by MacDowell [30], and it was also mentioned by Goldberger [31, p. 53] in the nonrelativistic case.

In contrast with dispersion relations involving a single partial wave, such as (2.9.44), the relations (8.4.17) exhibit explicitly the coupling among different partial waves, due to causality, arising from the fact that they all correspond to scattering by the same scatterer (cf. the discussion in Section 3.2). In (8.4.17), the real part of each partial-wave amplitude is coupled with the imaginary part of the same amplitude and with that of all *subsequent* partial waves, so that the system of equations has a "triangular" structure. The coupling to higher-order partial waves appears in each equation through a series of polynomials in $k^2$.

In addition to the system of equations, the partial-wave amplitudes must satisfy the unitarity condition (3.1.5),

$$\operatorname{Im} f_l(k) = |f_l(k)|^2. \qquad (8.4.20)$$

Note that (8.4.17) and (8.4.20) automatically lead to the correct threshold behavior (5.8.22) of the partial-wave amplitudes,

$$\operatorname{Re} f_l(k) = O(k^{2l+1}), \qquad \operatorname{Im} f_l(k) = O(k^{4l+2}) \qquad (k \to 0). \qquad (8.4.21)$$

The question now arises whether (8.4.17) and (8.4.20) are sufficient to determine the scattering amplitude uniquely, or whether there exist additional solutions, besides the physical one. If such solutions exist, Mandelstam's program cannot be carried out in this form unless supplementary conditions are given to select the physical solution.

As we have seen in Section 2.5, a dispersion relation involving a single unitary partial wave has an infinite set of solutions, corresponding to the CDD ambiguities. In the original work by Castillejo *et al.* [23] these ambiguities were expressed in terms of the poles and residues of the $R$-function (2.10.18), which is related to the $S$-function by (2.10.3), but one may equally well, as was done in Section 2.5, take the poles of $S$ as the arbitrary parameters, employing the canonical product representation (5.5.19).

The infinite system (8.4.17), however, is much more restrictive than a dispersion relation for a single partial wave, because it contains the coupling among different partial waves. As we have seen in Sections 3.3(d) and 3.4(b), it is precisely due to this coupling, i.e., due to the phase relations among different partial waves arising from causality, that the exponential factor $e^{-2ika}$ that dominates the behavior of each partial wave as $|k| \to \infty$ in $I_+$ is eliminated from the total amplitude for fixed momentum transfer.

If one attempted to modify a single partial wave, say, $f_0(k)$, by changing the position of a finite number of poles of the $S$-function [according to Section 2.5, this would yield an equally acceptable solution $f_0'(k)$ of the dispersion relation for a single partial wave], the difference $f_0'(k) - f_0(k)$ would still be dominated by $e^{-2ika}$ as $|k| \to \infty$ in $I_+$, so that it would violate (6.4.40) and the dispersion relation for fixed momentum transfer (8.1.10), from which (8.4.17) was derived.

However, as was shown by Gasiorowicz and Ruderman [32], ambiguities involving a single partial wave are still present. One of the simplest examples is that of a square potential well. In this case, the physical $s$-wave solution is given by (5.6.3), which corresponds to (2.10.27), with the $R$-function given by

$$R_0(k^2) = \tan(k_1 a)/k_1, \tag{8.4.22}$$

where $k_1 = (k^2 - U_0)^{1/2}$ is the wave number within the potential. Let the well depth be such that there are no bound states, i.e., $A = a(-U_0)^{1/2} < \pi/2$. If, without changing any other partial wave, we substitute $R_0$ in (2.10.27) by

$$R_0'(k^2) = \tan(k_1 b)/k_1, \tag{8.4.23}$$

where

$$a < b < \pi/[2(-U_0)^{1/2}], \tag{8.4.24}$$

the new scattering amplitude $f'(s,t)$ has exactly the same analytic properties in $s$ and $t$ as the physical amplitude $f(s,t)$, it still satisfies the unitarity condition, and, as may readily be verified,

$$\begin{aligned} f'(s,t) - f(s,t) &= (1/k)[f_0'(k) - f_0(k)] \\ &= (1/2ik)[S_0'(k) - S_0(k)] \\ &= O(k^{-1}) \qquad (|k| \to \infty, \operatorname{Im} k \geqslant 0), \end{aligned} \tag{8.4.25}$$

so that (6.4.40) and the dispersion relation for fixed momentum transfer (8.1.10) remain valid for the nonphysical solution $f'(s,t)$.

It is true that $S_0'(k)$ would not satisfy (5.5.15); the corresponding phase shift would give rise to oscillatory behavior as $k \to \infty$, violating (5.4.25). Other examples of nonphysical solutions were also found to violate Levinson's theorem[9] of Section 5.4(b). However, Levinson's theorem is an extra requirement, not contained in Mandelstam's program as formulated above.

We conclude that, in contrast with Yukawa-type potentials, Mandelstam's program cannot be carried out in the present case without specifying supplementary conditions to select the physical solution. Equivalently, we can say that analyticity of the scattering amplitude in $s$ and $t$, together with unitarity, do not exhaust the physical content of the Schrödinger equation in the case of cutoff potentials. The fact that Mandelstam's program may be carried out, at least in principle, for Yukawa-type potentials, is probably due to the specially "smooth" character of these potentials, reflected in their analyticity in configuration space [cf. (7.3.21)].

How is the possibility of modifying a single partial wave to be reconciled with the uniqueness theorem 7.5.5? The answer, of course, is that the uniqueness theorem does not apply in the present case. In the above example of the square potential well, the function $S(\lambda, k)$ behaves like $\exp(\pi|\lambda|)$ as $\lambda \to -i\infty$, so that the assumptions of Carlson's theorem 7.5.4 are violated.

In spite of the above difficulties, one may still try to determine the physical solution of the infinite system (8.4.17). The triangular structure of this system suggests trying to solve it "backwards," starting at some very high value of $l$ to find the asymptotic behavior of the solution for large $l$, and then going back step by step to lower values of $l$. For each $l$, one then has to solve a dispersion relation involving a single unknown partial wave, with a known inhomogeneous term. Since $f_l$ should approach the Born approximation $f_{lB}$ as $l \to \infty$ [cf. Section 7.5(a)], one could take as the first approximation, for sufficiently large $l$,

$$\operatorname{Re} f_l \approx f_{lB}, \qquad \operatorname{Im} f_l = 0,$$

which would effectively reduce (8.4.17) to a finite system. One would encounter CDD ambiguities at each step, and supplementary conditions would be required to select the physical solution.

On the other hand, if the ambiguities are expressed in terms of the poles of the $S$-function, as in Section 2.5, the positions of the poles are not completely arbitrary. This follows from the fact that the infinite system (8.4.17) automatically leads to the correct threshold behavior (8.4.21) of the partial-wave amplitudes. According to (2.5.42), (2.9.59), and (5.5.19), the following "sum rule" must be satisfied for $l = 0$:

$$\sum_n (1/k_n)^{\cdot} = ia + \tfrac{1}{2} S_0'(0), \tag{8.4.26}$$

[9] The relation between Levinson's theorem and CDD ambiguities in the relativistic case is discussed, for example, by Frautschi [33, p. 29].

where the summation is extended over all the poles of $S_0(k)$, both in $I_+(k)$ and in $I_-(k)$, taken in order of increasing modulus, and $S_0' = dS_0/dk$.

By considering the integral

$$I = \int_C \log S_l(k) \frac{dk}{k^{2p+2}} \qquad (1 \leqslant p \leqslant l), \tag{8.4.27}$$

where $C$ is a contour of the type shown in Fig. 5.1, but taken in the upper half-plane, and by taking into account the threshold behavior (8.4.21), one finds [29] that, for each $l \geqslant 1$, the following $l+1$ sum rules must be verified by the poles:

$$\sum_n (1/k_n) = ia, \tag{8.4.28}$$

$$\sum_n (1/k_n^{2p+1}) = 0 \qquad (p = 1,2,\ldots,l-1), \tag{8.4.29}$$

$$\sum_n (1/k_n^{2l+1}) = \tfrac{1}{2} \lim_{k\to 0} [S_l'(k)/k^{2l}]. \tag{8.4.30}$$

In the special case of scattering by a hard sphere, the backwards method of solution, with the help of these sum rules, leads uniquely to the exact solution. In this case, by partial-wave projection out of (3.3.68), with the inhomogeneous term given by (3.3.70), one finds, in the place of (8.4.17), the infinite system of equations

$$\begin{aligned} \operatorname{Re} f_l(k) = {} & -k \frac{d}{dk} [ka j_l^2(ka)] + \frac{k^{2l+3}}{\pi} \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_l(k')}{k'^{2l+3}(k'-k)} dk' \\ & + \frac{k^{2l+3}}{\pi} \sum_{m=1}^{\infty} \sum_{s=0}^{m-1} (-1)^s (l;m;s) k^{2s} \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_{l+m}(k')}{k'^{2l+2s+4}} dk'. \end{aligned} \tag{8.4.31}$$

Let us assume that the exact solution of (8.4.31) is given for $l > l_0$, where $l_0$ may be arbitrarily large. Substituting the solution for $l > l_0$ in the right-hand side of the equation for $l = l_0$, one gets an equation of the form

$$\operatorname{Re} f_l(k) = \frac{k^{2l+3}}{\pi} \int_{-\infty}^{\infty} \frac{\operatorname{Im} f_l(k')}{k'^{2l+3}(k'-k)} dk' + F_l(k), \tag{8.4.32}$$

where $F_l(k)$ is known.

It can be shown [29] that the general solution of (8.4.32) is given by (5.5.19) (with plus sign), where the poles $k_n$ are located in the lower half-plane and must fulfill the sum rules given above. Furthermore, the asymptotic behavior of $S_l(k)$ for $|k| \to \infty$, derived from (8.4.32), implies that the total number of poles is finite. They are therefore the roots of an algebraic equation.

It is well known that the coefficients of an algebraic equation of degree $m$ can be expressed in terms of the sums of the first $m$ powers of the roots. However, they can also be expressed [34] in terms of the sums of the first $m$ odd powers. It follows from this result that the poles $k_n$ in the present case are uniquely determined by the sum rules (8.4.27)–(8.4.30), leading to a unique solution of (8.4.32). We may then substitute the result in the equation (8.4.31) for $l = l_0 - 1$, yielding another equation of the form (8.4.32). The same procedure can therefore be applied to all remaining equations of the system.

Thus, in the special case of a hard sphere, if the exact solution is given for angular momenta larger than some (arbitrarily high) value, the remaining finite system of coupled partial-wave dispersion relations can be explicitly solved with the help of the sum rules, and the solution is unique. This is probably due to the specially simple structure of the $S$-matrix in this case: there is a finite number of poles for each partial wave, whereas the number of poles is infinite in the general case of a cutoff potential.

### *(c) Regge Poles*

Additional insight into the physical interpretation of Regge poles and into their role in the analytic properties of scattering amplitudes for cutoff potentials is provided by an investigation of some simple models, namely, scattering by a hard sphere [35] or by a square well [36].

Complex angular momentum and the Watson transformation were originally introduced in connection with just these models, in order to treat short-wavelength scattering by a sphere [37]. When the wavelength is much shorter than the radius of the sphere, i.e., for $ka \gg 1$, we may employ the concept of the "impact parameter" $p_l \sim l/k$ associated with the $l$th partial wave [cf. (3.3.26)], and all partial waves with $p_l \lesssim a$ undergo substantial distortion by the scatterer, so that the number of terms one has to keep in the partial-wave series to get a good approximation is of the order $ka \gg 1$. Watson's transformation was introduced in order to convert the partial-wave series into a rapidly converging series for $ka \gg 1$.

The standard Watson transformation for scattering by a hard sphere is illustrated in Fig. 8.3. The partial-wave series for the total wave function [corresponding to the incident plane wave (3.1.1)]

$$\psi(k,r,\theta) = \sum_{l=0}^{\infty} (l+\tfrac{1}{2}) i^l [h_l^{(2)}(kr) + S_l(k) h_l^{(1)}(kr)] P_l(\cos\theta), \quad (8.4.33)$$

where $S_l$ is given by (3.3.34), is first written as a contour integral along $C$, with the help of (7.1.2). The path of integration is then deformed away from the

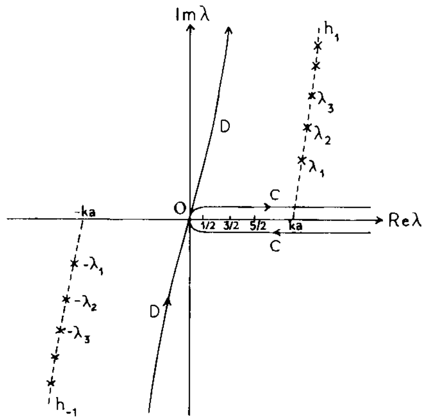


*FIG. 8.3.* The standard Watson transformation for a hard sphere (×, Regge poles).

real axis into another path $D$, giving rise to a series of residues at the Regge poles located in the first quadrant. It turns out that, in this quadrant, one cannot deform the path all the way up to the positive imaginary axis, as in Fig. 7.1. However, for observation points within the shadow region of the sphere, one can get a path $D$ which is symmetric about the origin, as shown in Fig. 8.3. Since the integrand is an odd function of $\lambda$, it follows that the contribution from the background integral vanishes identically, and the wave function is reduced to a pure residue series.

The Regge poles, according to (3.3.34), are the zeros of $H_\lambda^{(1)}(ka)$ in the $\lambda$-plane. There is an infinite number of poles, symmetric about the origin, lying along the curves $h_1$ and $h_{-1}$ in the first and third quadrants, as shown in Fig. 8.3.

Physically, the residue series represents a series of surface waves, excited by incident rays that are tangential to the sphere at $T_1$ and $T_2$ (Fig. 8.4). These waves travel around the sphere, shedding radiation in tangential directions at each point along their path. This radiation damping leads to an exponential decay of the surface wave amplitudes with the angles $\delta_\pm$ described along the surface from their respective launching points. A point $P$ in the shadow is reached by rays coming from $T_1'$ and $T_2'$, the intercepts of the tangents from $P$ to the sphere (Fig. 8.4). This physical interpretation in terms of "diffracted rays" is in agreement with the geometrical theory of diffraction [38].

The angular damping constant for a surface wave associated with the Regge pole $\lambda_n$ is given by $\mathrm{Im}\,\lambda_n$ [the corresponding contributions, with reference to Fig. 8.4, are proportional to $\exp(i\lambda_n\delta_\pm)$]. This is in complete agreement with (7.7.18). In the "deep shadow region," i.e., the shaded region in Fig. 8.4, the surface waves are strongly damped, so that only the poles closest to the real axis give an appreciable contribution (by the same token, one can neglect contributions corresponding to additional turns around the sphere). Thus, in

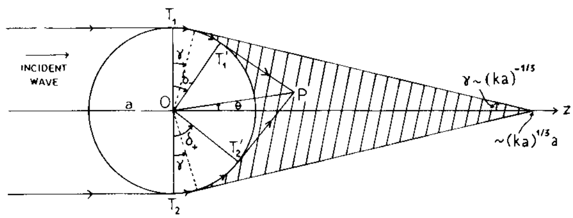


*FIG. 8.4.* Diffracted rays $T_1T_1'P$ and $T_2T_2'P$ reaching a point $P$ in the deep shadow region (the shaded region). The angle $\gamma$ is greatly exaggerated in this figure.

contrast with the original partial-wave series, the residue series is rapidly convergent in the deep shadow region. This is no longer true outside of the deep shadow region but, by suitably modifying Watson's transformation, one can still achieve rapid convergence in the other regions [35]; in particular, the dominant contributions (representing the incident and reflected waves) in the lit regions arise from the background integral.

A square potential well of depth $V_0$ corresponds, in optical terms, to a transparent sphere with refractive index $N$, where

$$N = [1+(2mV_0/\hbar^2 k^2)]^{1/2}. \tag{8.4.34}$$

The Regge trajectories for this potential have been investigated by several authors [39].

The distribution of Regge poles for $ka \gg 1$ and $N \gg 1$, corresponding to a deep well, is shown [36] in Fig. 8.5. The poles fall into two sharply differentiated classes. Class-II poles (indicated by $\times$ in Fig. 8.5) are located very close to the hard-sphere Regge poles of Fig. 8.3. There is an infinite number of Class-II poles, with unbounded real parts, in the first quadrant. As $k \to 0$, they move towards the origin, so that they have "0-type" trajectories [cf. Section 7.7(a)].

Class-I poles (indicated by $\circ$ in Fig. 8.5) are located near the real axis. Their Regge trajectories are of "$C$-type" (Section 7.7) and they resemble those for Yukawa-type potentials, except for not "turning back" as $k \to \infty$; instead, they proceed to infinity in the first quadrant. For a sufficiently deep well, they give rise to bound states at negative energies; at finite energy, the number of Class-I poles in the first quadrant is finite.

The physical interpretation of Class-I poles can be obtained by considering the corresponding poles [cf. Section 7.7(b)] in the $k$-plane, discussed in Section 5.6. The poles farthest to the right, between $ka$ and $Nka$ in Fig. 8.5, are closest to the real axis: they correspond to narrow resonances at high angular momenta. With a deep well surrounded by a high centrifugal barrier, such sharp resonances appear below the top of the barrier. The poles between 0 and $ka$

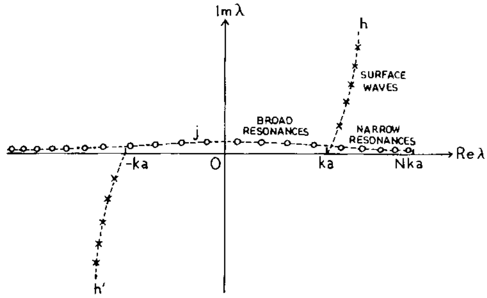


*FIG. 8.5.* Regge poles for a square potential well with $ka \gg 1$, $N \gg 1$ (O, Class-I poles; ×, Class-II poles).

in Fig. 8.5 are associated with broader resonances lying above the top of the centrifugal barrier. On the other hand, Class-II poles are associated with surface waves similar to those described for a hard sphere.

We see, therefore, that Class-I poles are associated with the "interior" of the potential, i.e., with the region $0 \leqslant r < a$; their behavior resembles that of Regge poles for Yukawa-type potentials, and they may also be interpreted in terms of families of bound states and resonances. Class-II poles represent the surface effects arising from the sharp cutoff in the potential: they are insensitive to the behavior of the potential for $r < a$ and they are almost completely determined by the geometrical shape (radius of curvature) of the surface.

The existence of an infinite number of Class-II poles in the first quadrant at any finite energy is related to the essential singularity at infinity in the momentum transfer plane that was found in Section 8.4(a), leading to the basic differences in the analytic behavior of the scattering amplitude for cutoff potentials as compared with Yukawa-type potentials. Thus the drastic change in analytic behavior that takes place when the tail of a Yukawa-type potential is cut off, no matter at how large a distance (cf. Section 5.5), may be interpreted as arising from the effect of surface waves: a potential with an infinitely sharp cutoff can support surface waves at arbitrarily high energy.

It is not only in the deep shadow region (Fig. 8.4) that Regge pole contributions are dominant. In the scattering by transparent spheres of moderately large size, a very striking effect known as the glory occurs near the backward direction. There is a strong enhancement in the backscattering, and the differential cross section for near-backward angles undergoes very large and rapid fluctuations. It can be shown [36; 40, p. 203] that, even though near-backward scattering corresponds to a lit region, the scattering amplitude is dominated by Regge pole contributions; these are the contributions that give rise to the glory.

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APPENDIX

# A

# DISTRIBUTION THEORY

In this appendix, we shall collect, for the reader's convenience, the main results from distribution theory that are used in the text, particularly in Section 1.8. Most of the results will be quoted without proof, but in cases where the proof is short and easy it will usually be given. The omitted proofs and much more complete treatments of the subject may be found in Refs. [1–11].

## A1. Introduction

Physicists are often quite happy to call "$\delta$-function" an object with the following properties:

$$\delta(t) = \begin{cases} 0 & (t \neq 0), \\ \infty & (t = 0), \end{cases} \tag{A1.1}$$

$$\int_{-\infty}^{\infty} \delta(t)\, dt = 1. \tag{A1.2}$$

Mathematically, these properties are contradictory. If a function vanishes almost everywhere, so does its (Riemann or Lebesgue) integral.

A more satisfactory approach is to consider $\delta$ as the limit in some sense of a sequence of functions that become more and more sharply peaked at the origin; e.g.,

$$\delta(t) = \lim_{n\to\infty} (n/\pi)^{1/2} \exp(-nt^2), \tag{A1.3}$$

$$\delta(t) = \lim_{\varepsilon\to 0} \varepsilon/[\pi(t^2+\varepsilon^2)]. \tag{A1.4}$$

This approach has been taken by Temple and by Lighthill [2]. However, as shown by the above examples, many different sequences lead to the same result, so that we really have to deal with an equivalence class of sequences. The equivalence property that characterizes them is that they must all lead to

$$\int_{-\infty}^{\infty} \delta(t)\varphi(t)dt = (\delta,\varphi) = \varphi(0) \qquad \text{(A1.5)}$$

for any "sufficiently well-behaved" function $\varphi(t)$. Here, $(\delta,\varphi)$ is the familiar scalar product notation.

Whenever $\delta$-functions appear, what is ultimately meant is that their scalar product with some function shall be taken. Thus $\delta$ plays the role of a *linear functional* rather than a function; i.e., it associates a number to each sufficiently well-behaved function of $t$ (and this is obviously a linear correspondence), rather than associating a number to each value of $t$.

We shall here follow Schwartz's original procedure and define distributions as linear functionals operating on some set of functions, which will be called *test functions*. We first have to agree on what set of test functions to take, i.e., on the meaning of "sufficiently well-behaved." This is to a large extent a matter of choice, and different sets of test functions are suited to treat different classes of problems. If we never had to deal with objects more singular than $\delta$, continuous test functions might suffice. As a rule, the more singular the class of distributions that one wants to discuss, the more regular has to be the corresponding set of test functions.

We start by discussing a class of distributions sufficiently broad for most applications, which are called *Schwartz distributions*. Later, we shall discuss some more restrictive classes. Unless otherwise specified, we deal only with distributions in a single real variable, and with complex-valued test functions of this real variable.

## A2. The Space $\mathscr{D}$ and Schwartz Distributions

To get a very broad class of distributions, we employ a very restrictive class of test functions $\varphi(t)$. We shall use the notations

$$\varphi^{(p)}(t) = D^p\varphi(t) = d^p\varphi/dt^p \qquad (p = 0,1,2,\ldots).$$

We say that $\varphi$ is of class $C^n$ (or $\varphi \in C^n$) if $\varphi^{(p)}(t)$ exists and is continuous for $0 \leqslant p \leqslant n$. If $\varphi$ is continuous, it is said to be $C^0$. If $\varphi$ is infinitely differentiable, it is said to be $C^\infty$.

The *support* $K$ of a function $\varphi(t)$ is the smallest *closed* set outside of which $\varphi(t) = 0$. We shall also write $K = \operatorname{supp}\varphi$. An example is shown in Fig. A1,

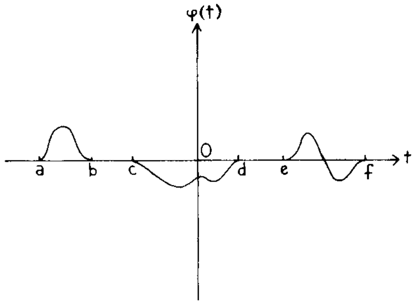


*FIG. A1.* In this example, $\operatorname{supp}\varphi = [a,b]+[c,d]+[e,f]$.

where $[t_0, t_1]$ denotes the closed interval $t_0 \leqslant t \leqslant t_1$. If $\operatorname{supp}\varphi$ is bounded, we say that $\varphi$ has *compact support* ("compact" here means bounded and closed).

We now define the set $\mathscr{D}$ of test functions that will be considered: *$\mathscr{D}$ is the set of all $C^\infty$ functions with compact support.* It is immediate that $\mathscr{D}$ is a linear vector space. Note that, if $\varphi \in \mathscr{D}$, $\operatorname{supp}\varphi$ is bounded, but the bound can be arbitrarily large: there exists no *common* bound for the supports of all test functions in $\mathscr{D}$.

An example of a test function $\varphi \in \mathscr{D}$, with $\operatorname{supp}\varphi = [-a, a]$, is

$$\varphi(t,a) = \begin{cases} 0 & (|t| \geqslant a), \\ C\exp[-a^2/(a^2-t^2)] & (|t| \leqslant a), \end{cases} \tag{A2.1}$$

where $C$ is a constant. Note that $\varphi(t,a)$ and all its derivatives vanish at $t = \pm a$. An idea of the behavior of $\varphi(t,a)$ is given by Fig. A2.

If we choose the normalization constant $C$ in such a way that

$$\int_{-\infty}^{\infty} \varphi(t,a)\, dt = 1,$$

the function $\varphi(t,a)$ will behave like $\delta(t)$ as $a \to 0$ [cf. (A1.3) and (A1.4)]. This enables one to show that any continuous function $f(t)$ with compact support $K$ can be *uniformly* approximated arbitrarily closely by a test function $\varphi \in \mathscr{D}$ and such that $\operatorname{supp}\varphi$ contains $K$ but is arbitrarily close to $K$. To do this, one takes

$$\varphi_a(t) = \int_{-\infty}^{\infty} f(\tau)\,\varphi(t-\tau, a)\, d\tau, \tag{A2.2}$$

where $\varphi(t,a)$ is normalized in the above indicated way, so that it has $\delta$-like behavior as $a \to 0$. For sufficiently small $a$, $\varphi_a(t)$ satisfies the required conditions. At the same time, (A2.2) provides arbitrarily many examples of other test functions in $\mathscr{D}$.

It will be necessary to introduce a concept of *convergence in $\mathscr{D}$*; i.e., given a sequence of functions $\varphi_n \in \mathscr{D}$, we must define what is meant by $\varphi_n \to \varphi$ in the sense of $\mathscr{D}$. Since we then have $\varphi_n - \varphi \to 0$, it suffices to define convergence to zero in $\mathscr{D}$.

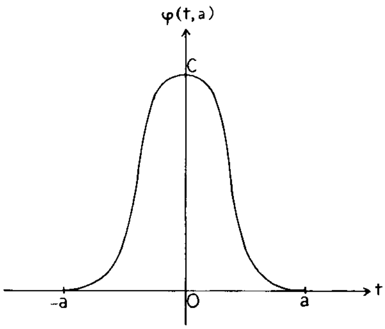


*FIG. A2.* The function $\varphi(t,a)$.

DEFINITION. Let $\{\varphi_n(t)\}$ $(n=1,2,3,\ldots)$ be a sequence of test functions $\in \mathscr{D}$. We say that $\varphi_n(t) \to 0$ (in the sense of $\mathscr{D}$) when: (a) there exists a *common* interval $K$ outside of which *all* $\varphi_n$ vanish; (b) $\varphi_n^{(p)}(t) \to 0$ *uniformly in* $K$ *for all* $p$ $(p=0,1,2,\ldots)$. This implies

$$\max_{t\in K} |\varphi_n^{(p)}(t)| \to 0. \tag{A2.3}$$

This is an extremely restrictive concept of convergence (it is "$C^\infty$ convergence"). Note, however, that "uniformly" refers to each order $p$ separately; the convergence need not be uniform independently of the order; i.e., we have uniformity in $t$, not in $p$.

*Examples.* Let $\varphi(t,a)$ be given by (A2.1). Then $\varphi_n = (1/n)\varphi(t,a) \to 0$ in $\mathscr{D}$, but $\hat{\varphi}_n = (1/n)\varphi(t/n,a)$ does *not* $\to 0$ in $\mathscr{D}$, because, although condition (b) is satisfied, (a) is not: $\operatorname{supp} \hat{\varphi}_n = [-na, na]$ is not bounded for all $n$.

It can be shown that the space $\mathscr{D}$ is *complete*; i.e., if $\varphi_n(t) \in \mathscr{D}$ and[1] $\varphi_n(t) \to \varphi(t)$, then $\varphi(t) \in \mathscr{D}$. This is related to the fact that the limit of a uniformly convergent sequence of continuous functions is a continuous function.

We can now give the definition of Schwartz distributions:

DEFINITION. *A distribution $T$ is a continuous linear functional on $\mathscr{D}$.*

"Linear functional" means that, to each $\varphi \in \mathscr{D}$, $T$ associates a complex number $(T,\varphi)$, in such a way that

$$(T, \varphi_1 + \varphi_2) = (T,\varphi_1) + (T,\varphi_2), \tag{A2.4}$$

$$(T, \lambda\varphi) = \lambda(T,\varphi) \qquad \text{for any complex number } \lambda. \tag{A2.5}$$

"Continuous" means that

$$\varphi_n \to 0 \Rightarrow (T,\varphi_n) \to 0, \tag{A2.6}$$

[1] Henceforth we shall omit, for brevity, the qualification "in the sense of convergence in $\mathscr{D}$."

so that

$$\varphi_n \to \varphi \Rightarrow (T, \varphi_n) \to (T, \varphi).$$

*Examples*

i. Let $f(t)$ be a *locally integrable* function, i.e., a function that is integrable (in the Lebesgue sense) over any bounded interval. Then we can associate to $f$ a distribution $T_f$ by

$$(T_f, \varphi) = \int_{-\infty}^{\infty} f(t)\varphi(t)\,dt. \tag{A2.7}$$

The integral exists because $f$ is locally integrable and $\varphi$ has compact support; it is clearly a linear functional. It is also continuous because, if $\varphi_n \to 0$, there exists an interval $[a, b]$ such that $\operatorname{supp}\varphi_n \subset [a, b]$ for all $n$, so that

$$|(T_f, \varphi_n)| \leqslant \max_{t \in [a,b]} |\varphi_n| \int_a^b |f(t)|\,dt \to 0.$$

Henceforth, when we define a distribution, we shall not verify in each case that it is a continuous linear functional; we shall be satisfied with verifying that the definition makes sense.

If $f_1 = f_2$ almost everywhere, $(T_{f_1} - T_{f_2}, \varphi) = 0$ for any $\varphi \in \mathscr{D}$ by (A2.7), so that $T_{f_1} = T_{f_2}$. Conversely, it can be shown that

$$T_{f_1} = T_{f_2} \Rightarrow f_1 = f_2 \qquad \text{almost everywhere.} \tag{A2.8}$$

Thus two locally integrable functions define the same distribution if and only if they are almost everywhere equal. The distribution is associated with a whole class of locally integrable functions equal to one another almost everywhere. In this sense, distributions can be regarded as a generalization of the concept of locally integrable function, which explains why they are also known as *generalized functions*.

Since we can modify the values taken by $f$ over an arbitrary set of measure zero (e.g., at a denumerable set of points) without changing the associated distribution $T_f$, it cannot make sense to speak of the value of a distribution at a given point. However, as will be seen later, it makes sense to speak about its value in the neighborhood of a point.

Keeping in mind the above remarks about equality almost everywhere, it is often convenient to identify $T_f$ with $f$ and to write

$$(f, \varphi) = \int_{-\infty}^{\infty} f(t)\varphi(t)\,dt. \tag{A2.9}$$

ii. The $\delta$-function is defined by

$$(\delta, \varphi) = \varphi(0). \tag{A2.10}$$

This is an example of a distribution which does not correspond to a locally integrable function.

iii. The Cauchy principal value of $1/t$ is defined by

$$(P(1/t),\varphi) = P\int_{-\infty}^{\infty}[\varphi(t)/t]\,dt$$

$$= \lim_{\varepsilon\to 0}\left[\int_{-\infty}^{-\varepsilon}[\varphi(t)/t]\,dt+\int_{\varepsilon}^{\infty}[\varphi(t)/t]\,dt\right]. \tag{A2.11}$$

Note that $1/t$ is not locally integrable and does not define a distribution; for this reason, $P(1/t)$ is called a *regularization* of $1/t$.

## A3. Operations with Distributions

The sum of distributions and the product by a complex number $\lambda$ are defined by

$$(T_1+T_2,\varphi) = (T_1,\varphi)+(T_2,\varphi), \tag{A3.1}$$

$$(\lambda T,\varphi) = \lambda(T,\varphi). \tag{A3.2}$$

With these definitions, the distributions over $\mathscr{D}$ also form a vector space, the space $\mathscr{D}'$ of all Schwartz distributions.

An operation $O$ on distributions will often be defined by a relation of the form

$$(OT,\varphi) = (T,O'\varphi), \tag{A3.3}$$

where $O'$ is an operation on the test functions $\varphi$. The choice of a definition is usually dictated by a kind of "correspondence principle": in the particular case that $T$ is a locally integrable function $f$, $OT$ must reduce to $Of$.

We shall usually verify only that the right-hand side of (A3.3) makes sense; in particular, we must have $\varphi\in\mathscr{D} \Rightarrow O'\varphi\in\mathscr{D}$.

### *Translation*

For a locally integrable function $f(t)$,

$$\tau_a f(t) = f(t-a) \tag{A3.4}$$

is called the *translation of $f(t)$ by $a$*. We have

$$\int_{-\infty}^{\infty} f(t-a)\varphi(t)\,dt = \int_{-\infty}^{\infty} f(t)\varphi(t+a)\,dt.$$

This leads us to define the *translation by a of a distribution T*, denoted as $\tau_a T = T_{(a)} = T_{t-a}$, by

$$(T_{(a)}, \varphi(t)) = (T, \varphi(t+a)). \tag{A3.5}$$

In particular,

$$(\delta_{(a)}, \varphi) = (\delta, \varphi(t+a)) = \varphi(a), \tag{A3.6}$$

which is usually written as

$$\int_{-\infty}^{\infty} \delta(t-a)\varphi(t)\,dt = \varphi(a). \tag{A3.7}$$

***Reflection***

The reflection $\check{f}(t)$ of a locally integrable function $f(t)$ with respect to the origin is defined by

$$\check{f}(t) = f(-t) \tag{A3.8}$$

and we have

$$(\check{f}, \varphi) = \int_{-\infty}^{\infty} f(-t)\varphi(t)\,dt = \int_{-\infty}^{\infty} f(t)\varphi(-t)\,dt = (f, \check{\varphi}),$$

so that we shall define the *reflection $\check{T}$ of a distribution T* by

$$(\check{T}, \varphi) = (T, \check{\varphi}). \tag{A3.9}$$

If $\check{T} = T$, we say that $T$ is *even*; e.g., $T = \delta$ is even. If $\check{T} = -T$, we say that $T$ is *odd*; e.g., $T = P(1/t)$ is odd.

## A4. Differentiation of Distributions

For a locally integrable function $f(t)$, we have

$$\begin{aligned}(f'(t), \varphi) &= \int_{-\infty}^{\infty} f'(t)\varphi(t)\,dt \\ &= [f(t)\varphi(t)]_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f(t)\varphi'(t)\,dt \\ &= -(f, \varphi'),\end{aligned}$$

where the integrated term vanishes because $\varphi = 0$ outside of some finite interval. This leads us to define *the derivative $T'$ of a distribution T* by

$$(T', \varphi) = -(T, \varphi'). \tag{A4.1}$$

Clearly, the right-hand side is defined for any $\varphi \in \mathscr{D}$ and it is a linear functional. It is also continuous, because $\varphi_n \to 0 \Rightarrow \varphi_n' \to 0$, according to (A2.3).

More generally, the derivative of order $p$ is given by

$$(T^{(p)}, \varphi) = (D^p T, \varphi) = (-1)^p (T, \varphi^{(p)}). \tag{A4.2}$$

Thus any distribution has derivatives of arbitrarily high orders, which are also distributions.

*Examples*

i. Let us consider the Heaviside step function

$$\theta(t) = \begin{cases} 0 & (t < 0), \\ 1 & (t > 0), \end{cases} \tag{A4.3}$$

which is locally integrable. Its derivative as a distribution is given by

$$(\theta', \varphi) = -(\theta, \varphi') = -\int_0^\infty \varphi'(t)\, dt = \varphi(0),$$

so that

$$\theta' = \delta. \tag{A4.4}$$

Similarly, if $\varepsilon(t)$ denotes the sign function,

$$\varepsilon(t) = \begin{cases} -1 & (t < 0), \\ 1 & (t > 0), \end{cases} \tag{A4.5}$$

we have

$$\varepsilon' = 2\delta. \tag{A4.6}$$

More generally, if a locally integrable function $f(t)$ has a jump discontinuity $f(t_0+0) - f(t_0-0) = C$, its derivative (in the sense of distributions) will contain a term $C\delta_{t-t_0}$.

Similarly,

$$(\delta', \varphi) = -(\delta, \varphi') = -\varphi'(0), \tag{A4.7}$$

$$(\delta^{(p)}, \varphi) = (-1)^p \varphi^{(p)}(0). \tag{A4.8}$$

ii. Let us compute the derivatives of the locally integrable function $\ln|t|$.

We have

$$
\begin{aligned}
((\ln|t|)', \varphi) &= -(\ln|t|, \varphi') \\
&= -\lim_{\varepsilon\to 0}\left[\left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty}\right)\ln|t|\,\varphi'(t)\,dt\right] \\
&= -\lim_{\varepsilon\to 0}\left\{[\ln|t|\,\varphi(t)]_{-\infty}^{-\varepsilon} + [\ln|t|\,\varphi(t)]_{\varepsilon}^{\infty}\right. \\
&\qquad \left. - \left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty}\right)[\varphi(t)/t]\,dt\right\} \\
&= \lim_{\varepsilon\to 0}\{\ln\varepsilon[\varphi(\varepsilon)-\varphi(-\varepsilon)]\} + P\int_{-\infty}^{\infty}[\varphi(t)/t]\,dt \\
&= P\int_{-\infty}^{\infty}[\varphi(t)/t]\,dt,
\end{aligned}
$$

so that

$$(\ln|t|)' = P(1/t). \tag{A4.9}$$

Similarly,

$$
\begin{aligned}
\left(\left(P\frac{1}{t}\right)', \varphi\right) &= -\left(P\frac{1}{t}, \varphi'\right) \\
&= -\lim_{\varepsilon\to 0}\left\{\left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty}\right)\frac{\varphi'(t)}{t}\,dt\right\} \\
&= \lim_{\varepsilon\to 0}\left\{\frac{\varphi(\varepsilon)+\varphi(-\varepsilon)}{\varepsilon} - \left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty}\right)\frac{\varphi(t)}{t^2}\,dt\right\} \\
&= -\lim_{\varepsilon\to 0}\left\{\int_{\varepsilon}^{\infty}[\varphi(t) + \varphi(-t)]\frac{dt}{t^2} - 2\varphi(0)\int_{\varepsilon}^{\infty}\frac{dt}{t^2} + O(\varepsilon)\right\},
\end{aligned}
$$

so that

$$(P(1/t))' = -Pf(1/t^2), \tag{A4.10}$$

where

$$\left(Pf\frac{1}{t^2}, \varphi\right) = \lim_{\varepsilon\to 0}\int_{\varepsilon}^{\infty}[\varphi(t)+\varphi(-t)-2\varphi(0)]\frac{dt}{t^2}. \tag{A4.11}$$

The symbol $Pf$ stands for *pseudofunction.* The right-hand side of (A4.11) also corresponds to what Hadamard called the *finite part* of the divergent integral

$$\int_{-\infty}^{\infty}\varphi(t)\frac{dt}{t^2}$$

[the subtraction of the term in $\varphi(0)$ amounts to subtracting out the divergent part]. Note that (A4.11) can also be rewritten in terms of a Cauchy principal value, as

$$\left(Pf\frac{1}{t^2},\varphi\right) = \lim_{\varepsilon\to 0}\left\{\left(\int_{-\infty}^{-\varepsilon} + \int_{\varepsilon}^{\infty}\right)[\varphi(t)-\varphi(0)]\frac{dt}{t^2}\right\}$$

$$= P\int_{-\infty}^{\infty}[\varphi(t)-\varphi(0)]\frac{dt}{t^2}. \tag{A4.12}$$

By successive differentiation, one is led to the general result

$$D^n P(1/t) = (-1)^n n!\,Pf(1/t^{n+1}), \tag{A4.13}$$

where

$$\left(Pf\frac{1}{t^n},\varphi\right) = P\int_{-\infty}^{\infty}\left[\varphi(t)-\varphi(0)-t\varphi'(0)-\cdots-\frac{t^{n-2}}{(n-2)!}\varphi^{(n-2)}(0)\right]\frac{dt}{t^n}. \tag{A4.14}$$

The Cauchy principal value $P(1/t)$ is a particular case, for $n=1$, and we have

$$[Pf(t^{-n})]' = -nPf(t^{-n-1}). \tag{A4.15}$$

iii. *Convergence of distributions.* Let $\{T_n\}$ be a sequence of distributions in $\mathscr{D}'$ $(T_n \in \mathscr{D}')$. We say that

$$\lim_{n\to\infty} T_n = T \tag{A4.16}$$

when, for *any* $\varphi \in \mathscr{D}$,

$$\lim_{n\to\infty} (T_n,\varphi) = (T,\varphi). \tag{A4.17}$$

It can then be shown that $T \in \mathscr{D}'$; i.e., the space $\mathscr{D}'$ is closed under convergence.

Similarly, we can define a series of distributions (by considering the sequence of partial sums) and limits such as $\lim_{u\to a} T_t(u)$, where $T_t(u)$ is a distribution in $t$ dependent on the parameter $u$.

The operations of going to the limit and of differentiation can always be interchanged. Thus (A4.16) implies

$$\lim_{n\to\infty} T_n^{(p)} = T^{(p)}, \tag{A4.18}$$

and any convergent series of distributions can be differentiated term by term any number of times.

For example, the "sawtooth" function $f(t)$ shown in Fig. A3 has the (convergent) Fourier expansion

$$f(t) = \tfrac{1}{2} + \sum_{n=-\infty}^{\infty}{}' e^{2in\pi t}/2in\pi, \tag{A4.19}$$

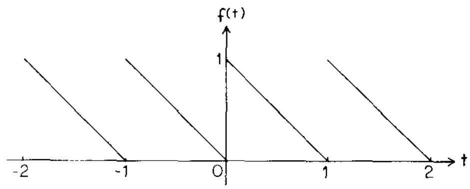


*FIG. A3.* The periodic "sawtooth" function $f(t)$.

where the prime in the summation sign indicates the exclusion of the term $n = 0$. The function $f(t)$ has a derivative (in the ordinary sense) equal to $-1$ everywhere, except at the integers, where $f(t)$ has a jump discontinuity of value unity. Thus, by the result given after (A4.6), we get, differentiating both sides of (A4.19) in the sense of distributions,

$$f'(t) = \sum_{n=-\infty}^{\infty}{}' e^{2in\pi t} = -1 + \sum_{l=-\infty}^{\infty} \delta_{(l)},$$

which can be rewritten as

$$\sum_{n=-\infty}^{\infty} e^{2in\pi t} = \sum_{l=-\infty}^{\infty} \delta(t-l). \tag{A4.20}$$

The right-hand side, consisting of a delta function at all integral points, is known as "periodic delta function." Applying both sides to a test function $\varphi(t)$, we get

$$\sum_{l=-\infty}^{\infty} \varphi(l) = \sum_{n=-\infty}^{\infty} \int_{-\infty}^{\infty} \varphi(t) e^{2in\pi t}\, dt, \tag{A4.21}$$

which is the well-known *Poisson summation formula.*

iv. Let us consider the function

$$\ln(t+iu) = \ln|t+iu| + i\arg(t+iu),$$

where

$$0 \leqslant \arg(t+iu) < 2\pi. \tag{A4.22}$$

Letting $u$ approach zero from above, we get the function

$$\lim_{u\to 0+} \ln(t+iu) = \ln(t+i0) = \ln|t| + i\pi\theta(-t). \tag{A4.23}$$

Differentiating with respect to $t$ and taking into account (A4.4) and (A4.9), we get

$$\lim_{u\to 0+} 1/(t+iu) = 1/(t+i0) = P(1/t) - i\pi\delta. \tag{A4.24}$$

Taking the $n$th derivative with respect to $t$, we find, with the help of (A4.13),

$$1/(t+i0)^{n+1} = Pf(1/t^{n+1}) - [(-1)^n/n!]\, i\pi\delta^{(n)}. \tag{A4.25}$$

## A5. Product of Distributions

The product of two locally integrable functions $f(t)$ and $g(t)$ need not be locally integrable (e.g., $f = g = |t|^{-1/2}$). Thus one cannot expect to be able to define the product of two arbitrary distributions $S$ and $T$. As a rule, the more singular one of them is, the more regular the other one must be in order that the product may be defined. We shall consider only the case in which $S$ is an infinitely differentiable function $\alpha(t)$, while $T$ can be an arbitrary distribution in $\mathscr{D}'$.

If $T$ is a locally integrable function, we have

$$(\alpha T, \varphi) = \int \alpha(t) T(t) \varphi(t) \, dt = (T, \alpha\varphi),$$

so that we are led to define, for any $T \in \mathscr{D}'$,

$$(\alpha T, \varphi) = (T, \alpha\varphi). \tag{A5.1}$$

Note that, since $\alpha \in C^\infty$, we have $\alpha\varphi \in \mathscr{D}$ for any $\varphi \in \mathscr{D}$, so that the right-hand side of (A5.1) is well defined.

*Examples.*

$$(tP(1/t), \varphi) = (P(1/t), t\varphi) = P\int_{-\infty}^{\infty} (1/t)\, t\varphi \, dt = \int_{-\infty}^{\infty} \varphi \, dt,$$

so that

$$tP(1/t) = 1. \tag{A5.2}$$

Similarly, taking into account (A4.14) and the relation $(t^n\varphi)^{(k)} = 0$ at $t = 0$ for $k < n$, we get

$$t^n Pf(1/t^n) = 1. \tag{A5.3}$$

The assumption that $\alpha \in C^\infty$ may be replaced by weaker ones if we restrict the class of distributions $T$ to which (A5.1) is to be applied. Thus

$$(\alpha\delta, \varphi) = (\delta, \alpha\varphi) = \alpha(0)\varphi(0);$$

i.e.,

$$\alpha(t)\delta = \alpha(0)\delta, \tag{A5.4}$$

and here it suffices that $\alpha(t)$ be continuous ($\alpha \in C^0$). Similarly,

$$(\alpha\delta', \varphi) = (\delta', \alpha\varphi) = -(\alpha\varphi)'(0) = -\alpha(0)\varphi'(0) - \alpha'(0)\varphi(0);$$

i.e.,

$$\alpha(t)\delta' = \alpha(0)\delta' - \alpha'(0)\delta. \tag{A5.5}$$

The requirement $\alpha \in C^1$ is sufficient in this case.

In particular,

$$t\delta = 0, \qquad t\delta' = -\delta, \qquad t^2\delta' = 0. \tag{A5.6}$$

Conversely, it can be shown that

$$tT = 0 \Rightarrow T = C\delta, \tag{A5.7}$$

where $C$ is an arbitrary constant.

More generally, the following results are valid:

$$t\delta^{(p)} = -p\delta^{(p-1)}, \tag{A5.8}$$

$$\begin{aligned} t^m\delta^{(p)} &= (-1)^m p(p-1)\cdots(p-m+1)\delta^{(p-m)} && (m \leqslant p), \\ &= 0 && (m > p), \end{aligned} \tag{A5.9}$$

$$t^nT = 0 \Leftrightarrow T = \sum_{p=0}^{n-1} C_p\delta^{(p)}, \tag{A5.10}$$

where $C_p$ are arbitrary constants. From (A5.3) and (A5.10) we can also obtain the general solution of the inhomogeneous equation $t^nT = 1$:

$$t^nT = 1 \Leftrightarrow T = Pf(1/t^n) + \sum_{p=0}^{n-1} C_p\delta^{(p)}. \tag{A5.11}$$

Finally, combining (A4.1) and (A5.1), we get the ordinary formula for the derivative of a product:

$$(\alpha T)' = \alpha' T + \alpha T'. \tag{A5.12}$$

## A6. Support of a Distribution

As we have seen in Section A2, one cannot speak about the value of a distribution at a given point. However, one can define the value of a distribution $T$ in a neighborhood $\Omega$ of a point $t_0$; it suffices, of course, to consider the value $T = 0$:

*$T = 0$ in an open set $\Omega$ when $(T, \varphi) = 0$ for all $\varphi \in \mathscr{D}$ such that* $\operatorname{supp}\varphi \subset \Omega$ (is contained in $\Omega$).

*Example.* $\delta = 0$ in any open set that does not contain the origin. Thus $\delta = 0$ in $-\infty < t < 0$ and in $0 < t < \infty$.

It can be shown that, *if $\Omega = \bigcup_i \Omega_i$ is the union of a (finite or countable) number of open sets $\Omega_i$, then $T = 0$ in each $\Omega_i \Rightarrow T = 0$ in $\Omega$.* Thus, if a distribution vanishes in a neighborhood of every point, it also vanishes globally; i.e., $(T, \varphi) = 0$ for any $\varphi \in \mathscr{D}$.

If $T \neq 0$ in *every* neighborhood of $t_0$, we say that $t_0$ is an *essential point* of $T$; e.g., $t = 0$ is an essential point of $\delta$.

*The support of a distribution T is the set of all its essential points.* It is also *the complement of the largest open set in which* $T = 0$. It is always a closed set.

*Example.* $\operatorname{supp} \delta$ is the point $t = 0$. The same is true for $\delta^{(m)}$. In fact, one can prove that *any distribution with support at a point a is a finite linear combination of δ and its derivatives taken at this point:*

$$T = \sum_{n=0}^{N} c_n \delta_{t-a}^{(n)}. \tag{A6.1}$$

The reason for the name "essential point" is that $\varphi$ can be modified in an arbitrary way outside of $\operatorname{supp} T$ without changing $(T, \varphi)$. This can be used to extend distributions with compact support to test functions $\varphi$ not having compact support, i.e., not belonging to $\mathscr{D}$. What matters is only that $\operatorname{supp} T \cap \operatorname{supp} \varphi$ be compact, which is always true when $\operatorname{supp} T$ is compact.

To carry out the extension, one constructs a function $\alpha \in \mathscr{D}$ such that $\alpha = 1$ in a set of points containing $\operatorname{supp} T$ (it can be shown that this is always possible) and one defines

$$(T, \varphi) = (T, \alpha\varphi). \tag{A6.2}$$

Let $\mathscr{E}$ be *the space of all $C^\infty$ functions*, without any support restrictions. Then, if $\varphi \in \mathscr{E}$, it follows that $\alpha\varphi \in \mathscr{D}$, so that the right-hand side of (A6.2) is well defined. Thus any distribution $T \in \mathscr{D}'$ with compact support can be extended to a distribution in $\mathscr{E}'$, the space of distributions corresponding to $\mathscr{E}$. In fact, it can be shown that $\mathscr{E}'$ *is the space of distributions with compact support.*

## A7. Direct Product

So far, we have been considering test functions and distributions in a single real variable $t$. However, there is no difficulty in extending the above concepts to more than one variable. Thus, in the case of two variables, we say that $\varphi(t,u) \in C^\infty$ if $D^{m+n}\varphi = \partial^{m+n}\varphi/\partial t^m \partial u^n$ exists for all $m, n$. We define $\mathscr{D}_{t,u}$ as the space of all $C^\infty$ test functions with compact support, and $\mathscr{D}'_{t,u}$, the space of Schwartz distributions in two variables, as the space of all continuous linear functionals on $\mathscr{D}_{t,u}$, where continuity is defined by an obvious extension of (A2.6).

The space $\mathscr{D}_{t,u}$ contains as a subspace $\mathscr{D}_t \otimes \mathscr{D}_u$, the *direct product* of the spaces $\mathscr{D}_t$ and $\mathscr{D}_u$, which is defined as the set of all $\varphi(t,u)$ which are of the form

$$\varphi(t,u) = \varphi_1(t)\varphi_2(u). \tag{A7.1}$$

The direct product of two locally integrable functions coincides with their ordinary product:

$$f(t) \otimes g(u) = f(t)g(u). \tag{A7.2}$$

It follows that

$$\begin{aligned}(f(t) \otimes g(u), \varphi_1(t)\varphi_2(u)) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t)g(u)\varphi_1(t)\varphi_2(u)\,dt\,du \\ &= \int_{-\infty}^{\infty} f(t)\varphi_1(t)\,dt \int_{-\infty}^{\infty} g(u)\varphi_2(u)\,du \\ &= (f, \varphi_1)(g, \varphi_2).\end{aligned} \tag{A7.3}$$

If we now consider a test function $\varphi(t,u) \in \mathscr{D}_{t,u}$ that does *not* belong to $\mathscr{D}_t \otimes \mathscr{D}_u$, we have

$$\begin{aligned}(f(t) \otimes g(u), \varphi(t,u)) &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} f(t)g(u)\varphi(t,u)\,dt\,du \\ &= \int_{-\infty}^{\infty} dt\, f(t) \int_{-\infty}^{\infty} du\; g(u)\varphi(t,u) \\ &= \int_{-\infty}^{\infty} du\; g(u) \int_{-\infty}^{\infty} dt\, f(t)\varphi(t,u) \\ &= \big(f(t), (g(u), \varphi(t,u))\big) \\ &= \big(g(u), (f(t), \varphi(t,u))\big),\end{aligned} \tag{A7.4}$$

where the interchanges of the order of integration are justified by the fact that the integrand is locally integrable and the integral is actually extended over a compact domain (Fubini's theorem).

The above results can now be generalized to distributions, as expressed by the following theorem: *Let $S_t \in \mathscr{D}_t'$ and $T_u \in \mathscr{D}_u'$; then there exists one and only one distribution $S_t \otimes T_u \in \mathscr{D}'_{t,u}$ such that, for $\varphi(t,u) = \varphi_1(t)\varphi_2(u) \in \mathscr{D}_t \otimes \mathscr{D}_u$,*

$$(S_t \otimes T_u, \varphi_1(t)\varphi_2(u)) = (S, \varphi_1)(T, \varphi_2). \tag{A7.5}$$

*This distribution is called the direct product of $S_t$ and $T_u$*, and it is defined by

$$\begin{aligned}(S_t \otimes T_u, \varphi(t,u)) &= \big(S_t, (T_u, \varphi(\underline{t}, u))\big) \\ &= \big(T_u, (S_t, \varphi(t, \underline{u}))\big),\end{aligned} \tag{A7.6}$$

for any $\varphi \in \mathscr{D}_{t,u}$.

In (A7.6), the underlined variables are first kept fixed, so that, for example, $\varphi(\underline{t},u)$ is regarded as a test function in $u$ dependent on the parameter $t$. It can then be shown that $(T_u, \varphi(\underline{t},u))$ is a test function in $\mathscr{D}_t$, so that the right-hand side of (A7.6) is well defined.

*Example*

$$\delta_t \otimes \delta_u = \delta_{t,u}. \tag{A7.7}$$

In fact, we have

$$\begin{aligned}(\delta_t \otimes \delta_u, \varphi_1(t)\varphi_2(u)) &= (\delta, \varphi_1)(\delta, \varphi_2)\\ &= \varphi_1(0)\varphi_2(0)\\ &= (\delta_{t,u}, \varphi_1(t)\varphi_2(u)).\end{aligned}$$

The right-hand side of (A7.7) is often written as $\delta(t)\delta(u)$.

*The support of the direct product is the direct product of the supports:*

$$\operatorname{supp}(S_t \otimes T_u) = \operatorname{supp} S_t \otimes \operatorname{supp} T_u, \tag{A7.8}$$

where the right-hand side is the set of all *pairs* of points $(t_1, u_1)$, with $t_1 \in \operatorname{supp} S_t$, $u_1 \in \operatorname{supp} T_u$. This result is illustrated in Fig. A4.

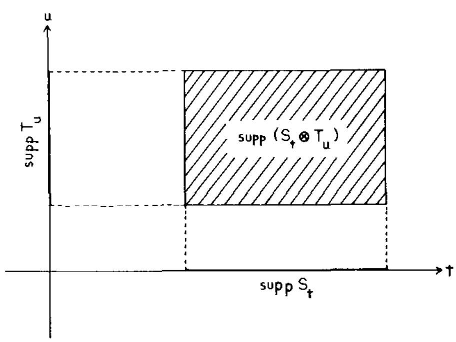


*FIG. A4.* The support of the direct product $S_t \otimes T_u$ is the shaded area.

Finally, it may be noted that the direct product is always commutative and associative.

## A8. Convolution

The *convolution product* $h(t) = f(t) * g(t)$ of two locally integrable functions is defined by

$$h(t) = \int_{-\infty}^{\infty} f(t-u)g(u)\,du = \int_{-\infty}^{\infty} f(u)g(t-u)\,du, \tag{A8.1}$$

which is also a locally integrable function.

It follows that

$$(h(t), \varphi(t)) = \int_{-\infty}^{\infty} dt \int_{-\infty}^{\infty} du\, f(t-u) g(u) \varphi(t).$$

Making the change of variables $t-u=\sigma$, $u=\tau$, this becomes

$$(h, \varphi) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\sigma) g(\tau) \varphi(\sigma+\tau)\, d\sigma\, d\tau$$
$$= (f(\sigma) \otimes g(\tau), \varphi(\sigma+\tau)).$$

This leads us to define the convolution product of two distributions $S_t$ and $T_t$, both belonging to $\mathscr{D}'$, by

$$(S * T, \varphi) = (S_\sigma \otimes T_\tau, \varphi(\sigma+\tau)). \tag{A8.2}$$

However, the right-hand side need by no means exist for arbitrary $S, T$. In fact, unless $\varphi \equiv 0$, $\varphi(\sigma+\tau)$ is *never* a function with compact support; e.g., if $\operatorname{supp} \varphi(t) = [a, b]$, then $\operatorname{supp} \varphi(\sigma+\tau)$ is a *band* in the $(\sigma, \tau)$-plane, parallel to the second bisector, drawn through the endpoints of the interval $[a, b]$, as shown in Fig. A5 (in the more general case shown in Fig. A1, it is a finite set of such bands, but it is always contained in a band of finite width).

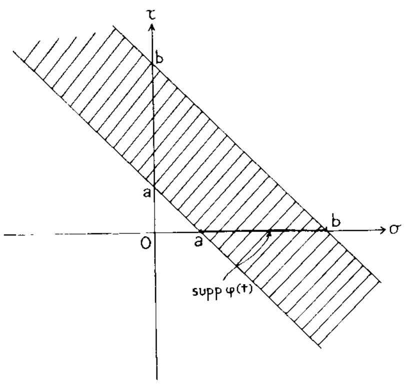


*FIG. A5.* If $\operatorname{supp} \varphi(t)$ is the interval $[a, b]$, the support of $\varphi(\sigma+\tau)$ is the shaded band $a \leqslant \sigma+\tau \leqslant b$, which is unbounded.

On the other hand, $\operatorname{supp}(S_\sigma \otimes T_\tau) = \operatorname{supp} S_\sigma \otimes \operatorname{supp} T_\tau$. *The right-hand side of (A8.2) exists if the intersection of* $\operatorname{supp} \varphi(\sigma+\tau)$ *with* $\operatorname{supp} S_\sigma \otimes \operatorname{supp} T_\tau$ *is bounded for any* $\varphi \in \mathscr{D}$. In fact, as we have seen at the end of Section A6, we can then modify $\varphi(\sigma+\tau)$ outside of the intersection of the supports and substitute it by a test function belonging to $\mathscr{D}_{\sigma,\tau}$. Thus the convolution product $S * T$ exists if the following *support condition* is verified:

$$(\operatorname{supp} S_\sigma \otimes \operatorname{supp} T_\tau) \cap \operatorname{supp} \varphi(\sigma+\tau) \quad \text{is bounded for any} \quad \varphi \in \mathscr{D}, \tag{A8.3}$$

and we then say that $S$ and $T$ are *convolutionable*. Graphically, this means that the intersection of $\operatorname{supp} S_\sigma \otimes \operatorname{supp} T_\tau$ with *any* band of finite width parallel to the second bisector in the $(\sigma,\tau)$-plane must be bounded. In particular, the following conditions are sufficient, as illustrated graphically in the accompanying figures:

a. *Either $S$ or $T$ has compact support* ($S$ *or* $T \in \mathscr{E}'$) (*Fig. A6*).

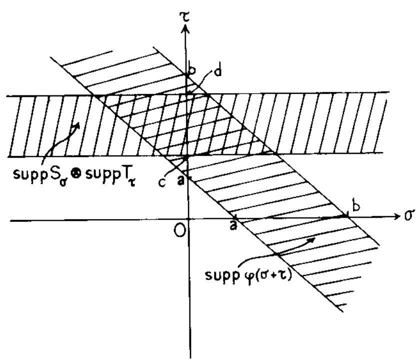


*FIG. A6.* If $\operatorname{supp} S$ is unbounded, but $\operatorname{supp} T = [c, d]$ is bounded, $S$ and $T$ are convolutionable: for any $\varphi \in \mathscr{D}$, with $\operatorname{supp} \varphi = [a, b]$, the intersection of supports defined in (A8.3) (cross-hatched region) is bounded.

b. *Both* supp $S$ *and* supp $T$ *are bounded on the left* (*or on the right*) (*Fig. A7*).

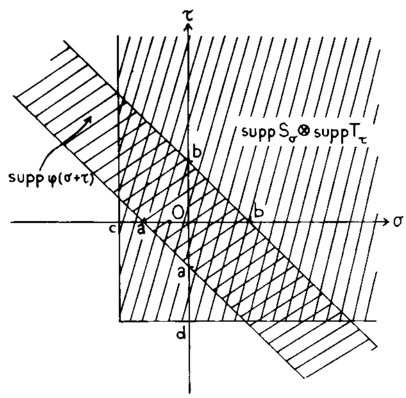


*FIG. A7.* If both $\operatorname{supp} S \subset [c, \infty)$ and $\operatorname{supp} T \subset [d, \infty)$ are bounded on the left, $S$ and $T$ are convolutionable: for any $\varphi \in \mathscr{D}$, with $\operatorname{supp} \varphi = [a, b]$, the intersection of supports defined in (A8.3) (cross-hatched region) is bounded.

In particular, $S * T$ exists if $S \in \mathscr{D}_+'$ and $T \in \mathscr{D}_+'$, where $\mathscr{D}_+'$ is the space of all distributions in $\mathscr{D}'$ having their support in $[0, \infty)$.

*Convolution of a distribution $T$ with a $C^\infty$ function $\alpha(t)$*: Let us assume that the support condition is satisfied (e.g., $T \in \mathscr{D}'$, $\alpha \in \mathscr{D}$, or $T \in \mathscr{E}'$, $\alpha \in \mathscr{E}$), so that $T * \alpha$ exists. It can then be shown that $T * \alpha$ is *also a $C^\infty$ function*, given by

$$h(t) = T * \alpha = (T_\tau, \alpha(t-\tau)), \tag{A8.4}$$

which should be compared with (A8.1). The function $h(t)$ is sometimes called the *regularization of T by* $\alpha$. An example is provided by (A2.2).

*Examples and Properties of the Convolution Product*

i. We have

$$\delta * T = T. \tag{A8.5}$$

*Proof.*

$$\begin{aligned}(\delta * T, \varphi) &= (\delta_\sigma \otimes T_\tau, \varphi(\sigma+\tau)) \\ &= \big(T_\tau, (\delta_\sigma, \varphi(\sigma+\tau))\big) \\ &= (T_\tau, \varphi(\tau)) \\ &= (T, \varphi).\end{aligned}$$

In particular, for a locally integrable function $f(t)$, (A8.5) corresponds to the formula

$$\int_{-\infty}^{\infty} \delta(t-\tau) f(\tau)\, d\tau = f(t),$$

which is an extension of (A3.7).

ii. We have

$$\delta' * T = T' \tag{A8.6}$$

and, more generally,

$$\delta^{(m)} * T = T^{(m)}. \tag{A8.7}$$

The proof is similar to that of i.

iii. We have

$$\delta_{(a)} * T = T_{(a)}, \tag{A8.8}$$

where $T_{(a)}$ is the translation of $T$ by $a$ [cf. (A3.5)].

In particular,

$$\delta_{(a)} * \delta_{(b)} = \delta_{(a+b)}, \tag{A8.9}$$

which is often written as

$$\int_{-\infty}^{\infty} \delta(t-\tau-a)\delta(\tau-b)\, d\tau = \delta(t-a-b).$$

iv. We have

$$(S * T, \varphi) = (S, \check{T} * \varphi) = (T, \check{S} * \varphi). \tag{A8.10}$$

In fact, by (A8.4) and (A3.9),

$$(S, \check{T} * \varphi) = \big(S_\sigma, (\check{T}_\tau, \varphi(\underline{\sigma} - \tau))\big) = \big(S_\sigma, (T_\tau, \varphi(\sigma + \tau))\big).$$

v. *To differentiate a convolution product, it suffices to differentiate one of the factors*:

$$(S * T)' = S' * T = S * T'. \tag{A8.11}$$

*Proof.*

$$\begin{aligned} ((S * T)', \varphi) &= -(S * T, \varphi') \\ &= -\big(S_\sigma, (T_\tau, \varphi'(\underline{\sigma} + \tau))\big) \\ &= \big(S_\sigma, (T_\tau', \varphi(\underline{\sigma} + \tau))\big) \\ &= (S * T', \varphi). \end{aligned}$$

Note that (A8.6) is a particular case of this result.

The convolution product is commutative, but not necessarily associative. However, it can be shown to be associative if two of the three factors have bounded support, or if all three supports are bounded on the left (or on the right).

### *Support of the Convolution Product*

*If* $\operatorname{supp} S \subset A$ *and* $\operatorname{supp} T \subset B$, *then* $\operatorname{supp}(S * T) \subset A + B$, *where* $A + B$ *is the set of all* $\sigma + \tau$, *with* $\sigma \in A$, $\tau \in B$.

In fact, if $\operatorname{supp} \varphi$ is not contained in $A + B$, we have $(S * T, \varphi) = (S_\sigma \otimes T_\tau, \varphi(\sigma + \tau)) = 0$, because $\varphi(\sigma + \tau) = 0$ for all $\sigma \in A$, $\tau \in B$.

It follows that, if both $S$ and $T$ have bounded support, so does $S * T$:

$$S \in \mathscr{E}', T \in \mathscr{E}' \Rightarrow S * T \in \mathscr{E}'. \tag{A8.12}$$

Also, if $\operatorname{supp} S \subset [a, \infty)$ and $\operatorname{supp} T \subset [b, \infty)$, then $\operatorname{supp} S * T \subset [a + b, \infty)$. In fact, one can prove even more: *if the left extremity of* $\operatorname{supp} S$ *is* $a$ *and that of* $\operatorname{supp} T$ *is* $b$, *then the left extremity of* $\operatorname{supp} S * T$ *is* $a + b$. This generalizes a theorem due to Titchmarsh. An illustration is provided by (A8.9).

## A9. Fourier Transforms and the Space $\mathscr{S}$

The Fourier transform of a locally integrable function $f(t)$ is defined by

$$\mathscr{F} f(t) = \hat{f}(\omega) = \int_{-\infty}^{\infty} f(t) e^{i\omega t}\, dt. \tag{A9.1}$$

Assuming that it exists and that the Fourier inversion theorem applies, we have, at a point where $f(t)$ is continuous,

$$f(t) = \mathscr{F}^{-1}\tilde{f}(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{f}(\omega)\, e^{-i\omega t}\, d\omega, \tag{A9.2}$$

where $\mathscr{F}^{-1}$ denotes the inverse Fourier transform. Thus

$$\mathscr{F}^{-1}\mathscr{F} = \mathscr{F}\mathscr{F}^{-1} = 1. \tag{A9.3}$$

If we now regard $\tilde{f}(\omega)$ as a distribution, and let it operate on a test function $\varphi(\omega) \in \mathscr{D}_\omega$, we must have

$$\begin{aligned}
(\mathscr{F}f(t), \varphi) &= (\tilde{f}(\omega), \varphi(\omega)) \\
&= \int_{-\infty}^{\infty} d\omega\, \varphi(\omega) \int_{-\infty}^{\infty} dt\, f(t)\, e^{i\omega t} \\
&= \int_{-\infty}^{\infty} dt\, f(t) \int_{-\infty}^{\infty} d\omega\, \varphi(\omega)\, e^{i\omega t} \\
&= (f(t), \tilde{\varphi}(t)) \\
&= (f(t), \mathscr{F}\varphi).
\end{aligned}$$

We may therefore try to define the Fourier transform $\tilde{T}_\omega = \mathscr{F}T_t$ of a distribution $T_t \in \mathscr{D}'$ by

$$(\mathscr{F}T, \varphi) = (\tilde{T}_\omega, \varphi(\omega)) = (T_t, \tilde{\varphi}(t)) = (T, \mathscr{F}\varphi). \tag{A9.4}$$

However, in order that this definition shall make sense for any $T \in \mathscr{D}'$, we must require that

$$\varphi \in \mathscr{D} \Rightarrow \mathscr{F}\varphi \in \mathscr{D},$$

which is *never* true (unless $\varphi \equiv 0$). In fact, since $\varphi \in \mathscr{D}$, $\operatorname{supp}\varphi$ is contained in some finite interval $[a, b]$, so that

$$\mathscr{F}\varphi = \int_a^b e^{i\omega t}\, \varphi(\omega)\, d\omega. \tag{A9.5}$$

The right-hand side of (A9.5) is an entire analytic function of $t$; therefore, it cannot vanish over any interval, no matter how small, without vanishing identically. Thus $\mathscr{F}\varphi$ cannot belong to $\mathscr{D}$.

This shows that $\mathscr{D}$ is not a suitable space in which to define the Fourier transform. We need a space such that, whenever $\varphi$ belongs to it, so does $\mathscr{F}\varphi$. We shall now see that there is a space with this property, the space $\mathscr{S}$ of "rapidly decreasing functions."

*A rapidly decreasing function is a function $\varphi(t) \in C^\infty$ which, together with all its derivatives, decreases faster than any inverse power of $t$ as $|t| \to \infty$:*

$$\lim_{|t| \to \infty} |t^p \varphi^{(m)}(t)| = 0 \qquad (p = 0,1,2,\ldots; m = 0,1,2,\ldots). \tag{A9.6}$$

We denote by $\mathscr{S}$ *the space of all rapidly decreasing functions.*

*Examples.* Any $\varphi \in \mathscr{D}$ is a rapidly decreasing function, but there are many such functions which do not belong to $\mathscr{D}$; e.g.,

$$\varphi(t) = \exp(-t^2) \in \mathscr{S}.$$

However, $\exp(-t)$ does not belong to $\mathscr{S}$, because of its behavior as $t \to -\infty$. Similarly, $\exp(-|t|) \notin \mathscr{S}$, because it is not $C^\infty$.

*Convergence in $\mathscr{S}$: A sequence $\{\varphi_n(t)\}$ of test functions belonging to $\mathscr{S}$ is said to converge to zero in $\mathscr{S}$ when*

$$t^p \varphi_n^{(m)}(t) \to 0 \qquad \text{as} \quad n \to \infty \tag{A9.7}$$

*uniformly in $t$ over the whole real axis, for all $p \geqslant 0$, $m \geqslant 0$.* Note that this is weaker than the concept of convergence in $\mathscr{D}$, because there is no support condition. Thus $\mathscr{D} \subset \mathscr{S}$.

Let us consider a test function $\varphi(t) \in \mathscr{S}$ and its Fourier transform

$$\tilde{\varphi}(\omega) = \mathscr{F}\varphi = \int_{-\infty}^{\infty} \varphi(t)\, e^{i\omega t}\, dt. \tag{A9.8}$$

Integrating by parts $p$ times and noting that the integrated term always vanishes, by (A9.6), we get

$$\tilde{\varphi}(\omega) = [1/(-i\omega)^p] \int_{-\infty}^{\infty} \varphi^{(p)}(t)\, e^{i\omega t}\, dt,$$

which can be rewritten as

$$\mathscr{F}[\varphi^{(p)}(t)] = (-i\omega)^p \tilde{\varphi}(\omega) = (-i\omega)^p \mathscr{F}[\varphi(t)]. \tag{A9.9}$$

Similarly, we can differentiate (A9.8) under the integral sign any number of times:

$$\tilde{\varphi}^{(m)}(\omega) = \int_{-\infty}^{\infty} (it)^m \varphi(t)\, e^{i\omega t}\, dt;$$

i.e.,

$$\tilde{\varphi}^{(m)}(\omega) = [\mathscr{F}\varphi(t)]^{(m)} = \mathscr{F}[(it)^m \varphi(t)]. \tag{A9.10}$$

Combining (A9.9) and (A9.10), we get

$$(-i\omega)^p \tilde{\varphi}^{(m)}(\omega) = \frac{1}{-i\omega} \int_{-\infty}^{\infty} \frac{d^{p+1}}{dt^{p+1}} [(it)^m \varphi(t)]\, e^{i\omega t}\, dt.$$

The integral is bounded, by (A9.6), so that the right-hand side tends to zero as $|\omega| \to \infty$. Thus $\tilde{\varphi}(\omega)$ also satisfies condition (A9.6). Also, $\varphi(t) \in C^\infty \Rightarrow \tilde{\varphi}(\omega) \in C^\infty$. We conclude that *the Fourier transform of a rapidly decreasing function is also a rapidly decreasing function*:

$$\varphi \in \mathscr{S} \Leftrightarrow \mathscr{F}\varphi \in \mathscr{S}. \tag{A9.11}$$

The space $\mathscr{S}$ is therefore mapped onto itself by the Fourier transformation.

*Example.*

$$\mathscr{F}[\exp(-t^2)] = \sqrt{\pi}\exp(-\omega^2/4).$$

## A10. Temperate Distributions and Their Fourier Transforms

DEFINITION. *A temperature distribution is a continuous linear functional on $\mathscr{S}$* ("continuous" with respect to convergence in $\mathscr{S}$). The space of temperate distributions is denoted by $\mathscr{S}'$.

Since $\mathscr{D} \subset \mathscr{S}$, every temperate distribution is also a distribution in $\mathscr{D}'$, but the converse is not true; i.e., $\mathscr{S}' \subset \mathscr{D}'$, but $\mathscr{D}' \not\subset \mathscr{S}'$ [cf. Examples c and d below].

### *Examples of Temperate Distributions*

a. Any function $f(t) \in L$, i.e., such that

$$\int_{-\infty}^{\infty} |f(t)|\, dt < \infty. \tag{A10.1}$$

In fact, if $f \in L$, so does $f\varphi$ for any bounded function $\varphi(t)$. Since any $\varphi \in \mathscr{S}$ is bounded, $(f, \varphi)$ exists for any $\varphi \in \mathscr{S}$.

b. Any function $f(t)$ of polynomial growth, i.e., such that

$$|f(t)| \leqslant A|t|^k \qquad (|t| \to \infty). \tag{A10.2}$$

In fact, by (A9.6), $|\varphi(t)| \leqslant B|t|^{-k-2}$ for $|t| \to \infty$ and any $\varphi \in \mathscr{S}$, so that $|f\varphi| = O(t^{-2})$ ($|t| \to \infty$), and consequently $(f, \varphi)$ exists. In particular, $f(t) = C$ (constant) is a temperate distribution.

Thus functions of "slow" growth are temperate distributions. This is the reason for the name "temperate."

c. Functions of exponential growth, such as $\exp(\pm t)$, $\exp(t^2)$, do *not* correspond to temperate distributions, although they are locally integrable, and therefore define distributions in $\mathscr{D}'$.

d. Let

$$T = \sum_{n=0}^{\infty} \exp(n^2)\,\delta(t-n).$$

Then $T \in \mathscr{D}'$, since, for any $\varphi \in \mathscr{D}$,

$$(T,\varphi) = \sum_{n=0}^{\infty} \exp(n^2)\,\varphi(n)$$

is in effect only a finite sum ($\varphi$ has compact support). However, $T \notin \mathscr{S}'$, because, for example,

$$(T, \exp(-t^2)) = \sum_{n=0}^{\infty} 1 \to \infty.$$

e. *Every distribution T with bounded support is temperate.* In fact, we have seen at the end of Section A6 that $(T,\varphi)$ is then defined for any $\varphi \in \mathscr{E}$ (i.e., $\varphi \in C^\infty$, with no restrictions on the behavior at infinity), and thus also for $\varphi \in \mathscr{S}$. We see that $\mathscr{E}' \subset \mathscr{S}' \subset \mathscr{D}'$.

In particular, $\delta$ and its derivatives are temperate distributions.

f. For any $n \geqslant 1$, $Pf(t^{-n})$ is a temperate distribution.

g. *If T is a temperate distribution, so are its derivatives to all orders.* In fact, they can be defined by (A4.2), since $\varphi \in \mathscr{S} \Rightarrow \varphi^{(p)} \in \mathscr{S}$.

Combining b and g, we see that $T = f^{(m)}(t)$, where $f(t)$ is a function of polynomial growth and the derivatives are taken in the sense of distributions, is a temperate distribution. It is remarkable that the converse is also true. More precisely, we have:

*Every temperate distribution $T \in \mathscr{S}'$ is a finite order derivative (in the sense of distributions) of some continuous function $f(t)$ of polynomial growth:*

$$T = f^{(m)}(t) \qquad (m \geqslant 0), \tag{A10.3}$$

where $f(t) \in C^0$ and satisfies condition (A10.2).

*Examples.*

i. $\delta_t = f''(t)$, where $f(t) = 0$ for $t < 0$ and $f(t) = t$ for $t > 0$; similarly for $\delta_t^{(m)}$.

ii. According to (A4.9), $P(1/t) = f''(t)$, where $f(t) = t\ln|t| - t \in C^0$; similarly for $Pf(t^{-n})$.

It can also be shown that a similar property is valid for any distribution $T \in \mathscr{D}'$, but only *locally*. This means that, if $K$ is a finite closed interval and $\mathscr{D}_K$ is the space of test functions in $\mathscr{D}$ with support in $K$, and if $T$ is defined over a neighborhood of $K$, then $(T,\varphi) = (f^{(n)},\varphi)$ for every $\varphi \in \mathscr{D}_K$, where $n \geqslant 0$ and $f \in C^0$. An example is provided by $T = \sum_{n=0}^{\infty} \delta^{(n)}(t-n)$. If $\mathscr{D}_K$ does not extend beyond the integer $N$, only the first $N$ terms contribute to $(T,\varphi)$ for any $\varphi \in \mathscr{D}_K$, and each term is a finite-order derivative of a continuous function (cf. Example i). However, the result is not valid over $\mathscr{D}$, because the order of the derivative increases with the size of $\mathscr{D}_K$.

### *Fourier Transform*

We can now employ (A9.4) to define the Fourier transform $\tilde{T}_\omega$ of any temperate distribution $T_t \in \mathscr{S}'$:

$$(\mathscr{F}T, \varphi) = (\tilde{T}_\omega, \varphi(\omega)) = (T_t, \tilde{\varphi}(t)) = (T, \mathscr{F}\varphi). \tag{A10.4}$$

In fact, $\varphi \in \mathscr{S} \Rightarrow \tilde{\varphi} \in \mathscr{S}$, so that the right-hand side is defined for any $\varphi \in \mathscr{S}$, and it can be shown to be a continuous linear functional. We conclude that:

*Every temperate distribution $T$ has a Fourier transform $\tilde{T}$, which is also a temperate distribution:*

$$T \in \mathscr{S}' \Rightarrow \mathscr{F}T = \tilde{T} \in \mathscr{S}'.$$

This is the most important property of temperate distributions.

Similarly, we can define the inverse Fourier transform by

$$(\mathscr{F}^{-1}T, \varphi) = (T, \mathscr{F}^{-1}\varphi), \tag{A10.5}$$

where $\mathscr{F}^{-1}\varphi$ is defined as in (A9.2). Since the inversion theorem (A9.3) holds for any $\varphi \in \mathscr{S}$, we get

$$(\mathscr{F}(\mathscr{F}^{-1}T), \varphi) = (\mathscr{F}^{-1}T, \mathscr{F}\varphi) = (T, \mathscr{F}^{-1}\mathscr{F}\varphi) = (T, \varphi),$$

and similarly for $\mathscr{F}^{-1}\mathscr{F}$, so that *the Fourier inversion theorem (A9.3) is valid for any temperate distribution.*

### *Properties of the Fourier Transform*

The properties (A9.9) and (A9.10) of the Fourier transform, which are valid for any $\varphi \in \mathscr{S}$, can be immediately extended to temperate distributions:

$$\mathscr{F}(T_t^{(p)}) = (-i\omega)^p \mathscr{F}T_t, \tag{A10.6}$$

$$(\mathscr{F}T_t)^{(m)} = \mathscr{F}[(it)^m T_t]. \tag{A10.7}$$

To prove (A10.6), for instance, we have

$$\begin{aligned}(\mathscr{F}(T_t^{(p)}), \varphi(\omega)) &= (T_t^{(p)}, \mathscr{F}\varphi(\omega)) \\ &= (-1)^p(T_t, \tilde{\varphi}^{(p)}(t)) \\ &= (-1)^p(T_t, \mathscr{F}[(i\omega)^p\varphi(\omega)]) \\ &= (-i\omega)^p(\mathscr{F}T_t, \varphi(\omega)).\end{aligned}$$

Similarly, we have

$$\mathscr{F}^{-1}(T_\omega^{(p)}) = (it)^p \mathscr{F}^{-1} T_\omega, \tag{A10.8}$$

$$(\mathscr{F}^{-1} T_\omega)^{(m)} = \mathscr{F}^{-1}[(-i\omega)^m T_\omega]. \tag{A10.9}$$

Now let $U = \mathscr{F}T$ and $\psi = \mathscr{F}\varphi$. Then, by (A9.8) and (A3.9), we have

$$\check{\psi}(\omega) = \psi(-\omega) = \int_{-\infty}^{\infty} \varphi(t) e^{-i\omega t}\, dt = \int_{-\infty}^{\infty} \varphi(-t) e^{i\omega t}\, dt = \mathscr{F}\check{\varphi},$$

$$\begin{aligned}(\mathscr{F}\check{T}, \varphi) &= (\check{T}, \mathscr{F}\varphi) = (\check{T}, \psi) = (T, \check{\psi}) = (T, \mathscr{F}\check{\varphi}) \\ &= (\mathscr{F}T, \check{\varphi}) = (U, \check{\varphi}) = (\check{U}, \varphi),\end{aligned}$$

so that

$$\mathscr{F}T = U \Rightarrow \mathscr{F}\check{T} = \check{U}. \tag{A10.10}$$

Thus *the Fourier transform of an even (odd) distribution is an even (odd) distribution.*

### *Fourier Transform in $\mathscr{E}'$*

If $T_t \in \mathscr{E}'$ (i.e., if it is a distribution with bounded support), its Fourier transform is an ordinary function, which can be calculated by a formula similar to (A9.1):

$$\mathscr{F}T_t = \tilde{T}(\omega) = (T_t, e^{i\omega t}). \tag{A10.11}$$

In fact, the right-hand side exists, because $e^{i\omega t} \in \mathscr{E}$. Furthermore, with the help of (A7.6), we get

$$\begin{aligned}(\mathscr{F}T_t, \varphi(\omega)) &= (T_t, \mathscr{F}\varphi) \\ &= (T_t, (\varphi(\omega), e^{i\omega t})) \\ &= (T_t \otimes \varphi(\omega), e^{i\omega t}) \\ &= (\varphi(\omega), (T_t, e^{i\omega t})).\end{aligned}$$

Similarly, if $\tilde{T}_\omega \in \mathscr{E}'$, its inverse Fourier transform is given by

$$\mathscr{F}^{-1}\tilde{T}_\omega = (1/2\pi)(\tilde{T}_\omega, e^{-i\omega t}). \tag{A10.12}$$

In the particular case of a function with bounded support, we have seen in (A9.5) that the Fourier transform is an entire function. It can be shown that the same is true for (A10.11): $\tilde{T}(\omega)$ is an entire analytic function of $\omega$.

*Examples.* Applying (A10.11), we find

$$\mathscr{F}\delta = 1, \tag{A10.13}$$

$$\mathscr{F}[\delta^{(m)}] = (-i\omega)^m, \tag{A10.14}$$

$$\mathscr{F}[\delta_{(a)}] = e^{i\omega a}. \tag{A10.15}$$

Similarly, applying (A10.12) to $\tilde{T}_\omega = \delta_\omega$, we find

$$\mathscr{F}^{-1}\delta_\omega = 1/2\pi,$$

so that, by (A9.3),

$$\mathscr{F}1 = 2\pi\delta, \tag{A10.16}$$

and, by (A10.7), this implies

$$\mathscr{F}[(it)^m] = 2\pi\delta^{(m)}. \tag{A10.17}$$

### The Convolution Theorem

Let $S$ and $T$ be two distributions in $\mathscr{E}'$. Then, according to (A8.12), their convolution product $S * T$ exists and also belongs to $\mathscr{E}'$, so that we can compute its Fourier transform by (A10.11):

$$\begin{aligned}\mathscr{F}(S * T) &= (S * T, e^{i\omega t})\\ &= (S_\sigma \otimes T_\tau, e^{i\omega(\sigma+\tau)})\\ &= (S_\sigma, e^{i\omega\sigma})(T_\tau, e^{i\omega\tau})\\ &= \mathscr{F}S \cdot \mathscr{F}T,\end{aligned}$$

where we have employed (A8.2) and (A7.5). Thus, finally,

$$\mathscr{F}(S * T) = \mathscr{F}S \cdot \mathscr{F}T. \tag{A10.18}$$

Similarly, with the help of (A10.12), we get

$$\mathscr{F}^{-1}(\tilde{S} * \tilde{T}) = 2\pi\mathscr{F}^{-1}\tilde{S} \cdot \mathscr{F}^{-1}\tilde{T}, \tag{A10.19}$$

where it is assumed that $\tilde{S} \in \mathscr{E}'$, $\tilde{T} \in \mathscr{E}'$. Note that, under the above conditions, the right-hand sides of (A10.18) and (A10.19) are $C^\infty$ functions (in fact, they are entire analytic functions, as we have just seen).

The above results are particular cases of the *convolution theorem*, according to which convolution is transformed into multiplication by the Fourier transformation. This theorem is valid under much more general conditions. In particular, it can be shown to be valid in the following cases:

a. *If $S$ is any temperate distribution ($S \in \mathscr{S}'$) and $T$ has bounded support* ($T \in \mathscr{E}'$). The product of distributions on the right-hand side of (A10.18) always exists in this case, because $\mathscr{F}T$ is a $C^\infty$ function [cf. (A5.1)].

b. *If $S$ is any temperate distribution ($S \in \mathscr{S}'$) and $T$ is a "rapidly decreasing distribution"* ($T \in \mathscr{O}_C'$).

A "rapidly decreasing distribution" is one that "decreases at infinity faster than any inverse power," i.e., such that $(1+t^2)^n T$ is a *bounded distribution* for any $n$. A distribution $U$ is bounded when $(U, \varphi)$ is bounded for any $\varphi \in \mathscr{D}$. The space of rapidly decreasing distributions is denoted by $\mathscr{O}_C'$. Note that $\mathscr{E}' \subset \mathscr{O}_C'$.

It can be shown that the Fourier transform $\tilde{T}_\omega$ of a rapidly decreasing distribution $T_t \in \mathscr{O}_C'$ is a *"slowly increasing function"*: $\tilde{T}_\omega \in \mathscr{O}_M$. *We say that $\tilde{T}_\omega \in \mathscr{O}_M$ when $\tilde{T}_\omega \in \mathscr{E}$ (i.e., is $C^\infty$) and $D^{(n)}\tilde{T}_\omega$ is bounded by a polynomial for any $n$* (the degree of the polynomial may depend on $n$).

Note that the right-hand side of (A10.18) always exists under the present conditions. In fact ,since $\tilde{S}_\omega \in \mathscr{S}'$ and $\tilde{T}_\omega \in \mathscr{O}_M$, the product $\tilde{S}_\omega \tilde{T}_\omega$ can be defined by analogy with (A5.1):

$$(\tilde{S}_\omega \tilde{T}_\omega, \varphi(\omega)) = (\tilde{S}_\omega, \tilde{T}_\omega \varphi(\omega)).$$

In fact, $\varphi(\omega) \in \mathscr{S} \Rightarrow \tilde{T}_\omega \varphi(\omega) \in \mathscr{S}$ for any $\tilde{T}_\omega \in \mathscr{O}_M$.

## A11. Fourier Transform of $P(1/t)$ and Related Distributions

As we have seen in (A5.11), the general solution of the inhomogeneous equation

$$tT = 1 \tag{A11.1}$$

is $T = P(1/t) + C\delta$. However, since $P(1/t)$ is odd and $\delta$ is even under reflection [cf. (A3.9)], there is only one *odd* solution, namely, $T = P(1/t)$. We want to compute the Fourier transform $\tilde{T}$ of this distribution.

Taking the Fourier transform of both sides of (A11.1), we find, by (A10.7) and (A10.16),

$$\tilde{T}' = 2\pi i \delta_\omega. \tag{A11.2}$$

Since the reflection character is preserved under the Fourier transformation [cf. (A10.10)], it follows that $\tilde{T} = \mathscr{F}P(1/t)$ is the unique odd solution of the differential equation (A11.2). Taking into account (A4.6), this finally leads to

$$\mathscr{F}P(1/t) = i\pi\,\varepsilon(\omega). \tag{A11.3}$$

This relation is usually written as [cf. (1.6.18)]

$$\varepsilon(\omega) = \frac{1}{\pi i} \fint_{-\infty}^{\infty} e^{i\omega t}\, \frac{dt}{t}. \tag{A11.4}$$

Let us define the distributions

$$\begin{aligned}\delta^+ &= \tfrac{1}{2}\delta - (1/2i\pi)P(1/t) = -(1/2i\pi)[1/(t+i0)],\\ \delta^- &= \tfrac{1}{2}\delta + (1/2i\pi)P(1/t) = (1/2i\pi)[1/(t-i0)]\end{aligned} \tag{A11.5}$$

[cf. (A4.24)]. Then, according to (A10.13) and (A11.3), we have

$$\begin{aligned}\mathscr{F}\delta^+ &= \tfrac{1}{2}[1-\varepsilon(\omega)] = \theta(-\omega) = \check{\theta}(\omega),\\ \mathscr{F}\delta^- &= \tfrac{1}{2}[1+\varepsilon(\omega)] = \theta(\omega).\end{aligned} \tag{A11.6}$$

These results are usually written as

$$\delta^+ = \mathscr{F}^{-1}\check{\theta}(\omega) = (1/2\pi)\int_{-\infty}^{0} e^{-i\omega t}\,d\omega = (1/2\pi)\int_{0}^{\infty} e^{i\omega t}\,d\omega, \tag{A11.7}$$

and similarly for $\delta^-$.

Similarly, we find

$$\mathscr{F}^{-1}P(1/\omega) = -(i/2)\,\varepsilon(t), \tag{A11.8}$$

$$\mathscr{F}^{-1}\delta_\omega{}^{\pm} = \theta(\pm t)/2\pi. \tag{A11.9}$$

Finally, (A4.13), (A10.6), and (A11.3) imply

$$\mathscr{F}Pf(1/t^n) = [i^n\pi/(n-1)!]\,\omega^{n-1}\,\varepsilon(\omega). \tag{A11.10}$$

In particular, for $n=2$, we have

$$\mathscr{F}Pf(1/t^2) = -\pi\,|\omega|.$$

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APPENDIX

# B

# PASSIVITY AND CAUSALITY

The example discussed in Section 1.2 provides a good illustration of the relation between the passive character of a system and its causal behavior. Let us again consider the equation of motion (1.2.1) of the damped harmonic oscillator and let us assume that the oscillator was at rest at $t \to -\infty$, so that (1.2.12) is the complete solution.

The energy balance at time $t$ is then obtained by multiplying both sides of (1.2.1) by $\dot{x}$ and integrating from $-\infty$ to $t$:

$$\int_{-\infty}^{t} \mathscr{F}(t')\dot{x}(t')\,dt' = E(t) + 2m\gamma \int_{-\infty}^{t} \dot{x}^2\,dt', \tag{B1}$$

where

$$E(t) = \tfrac{1}{2}m\dot{x}^2(t) + \tfrac{1}{2}m\omega_0{}^2 x^2(t) \tag{B2}$$

is the energy of the oscillator at time $t$ [note that $E(-\infty) = 0$ due to our assumption].

The left-hand side of (B1) represents the total energy supplied by the driving force up to the time $t$. The second term on the right is the energy dissipation due to the damping force. The dissipative character of this term ($\gamma > 0$) implies that, for any driving force $\mathscr{F}(t)$,

$$\int_{-\infty}^{t} \mathscr{F}(t')\dot{x}(t')\,dt' \geqslant 0. \tag{B3}$$

This is the characteristic property of a *passive system*: it can only absorb, and not generate, energy. The "passivity condition" (B3) was first formulated by Meixner [1] (see also König and Meixner [2]).

It will now be shown that *any linear passive system* (*in the above sense*) *is causal* [3,4]. In fact, let $\dot{x}(t)$ correspond to a driving force $\mathscr{F}(t)$ that vanishes for $t<0$, and let us consider any other driving force $\mathscr{F}_1(t)$, associated with $x_1(t)$, so that (B3) is also valid for $\mathscr{F}_1$, $\dot{x}_1$. Then, since the system is linear, the driving force $\mathscr{F}_1(t)+\lambda\mathscr{F}(t)$ [$=\mathscr{F}_1(t)$ for $t<0$] is associated with $\dot{x}_1(t)+\lambda\dot{x}(t)$ for any real constant $\lambda$, and the passivity condition implies, for $t<0$,

$$\int_{-\infty}^{t} \mathscr{F}_1(t')\dot{x}_1(t')\,dt' + \lambda \int_{-\infty}^{t} \mathscr{F}_1(t')\dot{x}(t')\,dt' \geqslant 0. \tag{B4}$$

Since the first integral is $\geqslant 0$ and $\lambda$ is any real number, this is only possible if the second integral vanishes for $t<0$. As $\mathscr{F}_1(t)$ is arbitrary, this implies that $\dot{x}(t)=0$ for $t<0$. Since the system is at rest at $t\to-\infty$, it also follows that $x(t)=0$ for $t<0$. Thus

$$\mathscr{F}(t) = 0 \quad \text{for} \quad t<0 \quad \Rightarrow \quad x(t)=0 \quad \text{for} \quad t<0, \tag{B5}$$

and, according to (1.2.12), this leads to (1.2.14), so that the system is causal.

## References


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APPENDIX

# C

## PROPERTIES OF HERGLOTZ FUNCTIONS

A *Herglotz function*[1] $F(z)$ is a function with the following properties:

$$F(z) \quad \text{holomorphic in} \quad I_+, \tag{C1}$$

$$\operatorname{Im} F(z) \geqslant 0 \quad \text{in} \quad I_+. \tag{C2}$$

It is remarkable that just these properties are sufficient to obtain a general representation for such a function [3, 4]:

THEOREM C1. *Necessary and sufficient condition for $F(z)$ to be a Herglotz function is that there exists a bounded nondecreasing real function $\beta(t)$ such that*

$$F(z) = Az + C + \int_{-\infty}^{\infty} \frac{1+tz}{t-z}\, d\beta(t) \qquad (\operatorname{Im} z > 0), \tag{C3}$$

*where $A$ and $C$ are real constants, and $A \geqslant 0$. Furthermore,*

$$F(z)/z \to A \qquad \text{as} \quad |z| \to \infty \qquad (0 < \varepsilon \leqslant \arg z \leqslant \pi - \varepsilon). \tag{C4}$$

The integral in (C3) is a Stieltjes integral.[2] The function $\beta(t)$ need not be continuous. However, as a bounded monotonic function, it can only have

[1] Herglotz [1] considered functions with these properties in the unit circle and proved the analog of theorem C1 for them. The extension to a half-plane, which can be derived from Herglotz's result by conformal mapping, was given by Nevanlinna [2].

[2] See, for example, Smirnov [5].

discontinuities of the first kind, i.e., finite jumps (which must be positive, because $\beta$ is nondecreasing):

$$\beta_n = \beta(t_n+0) - \beta(t_n-0) > 0. \tag{C5}$$

The set of discontinuity points $t_n$ can be infinite, but it must be denumerable. Furthermore, since $\beta$ is bounded, we must have

$$\sum_n \beta_n < \infty. \tag{C6}$$

It follows from (C3) that

$$\operatorname{Im} F(x+iy) = y\left[A + \int_{-\infty}^{\infty} \frac{(1+t^2)}{(t-x)^2+y^2}\, d\beta(t)\right], \tag{C7}$$

so that (C2) is obviously satisfied. The proof of the remainder of Theorem C1 can be found in the references.

At any point where $\beta(t)$ is differentiable, we have

$$d\beta(t) = \beta'(t)\,dt, \tag{C8}$$

and, if we let $z$ in (C3) approach the real axis from above, we find, with the help of (A4.24),

$$\operatorname{Im} F(t+i0) = \pi(1+t^2)\beta'(t). \tag{C9}$$

More generally, the following "inversion formula" holds [4, p. 126]:

$$\tfrac{1}{2}[\psi(x+0)+\psi(x-0)] - \tfrac{1}{2}[\psi(c+0)+\psi(c-0)] = (1/\pi)\int_c^x \operatorname{Im} F(t+i0)\,dt, \tag{C10}$$

where

$$\psi(x) = \int_0^x (1+t^2)\,d\beta(t). \tag{C11}$$

The function $\beta(t)$ can be decomposed into the sum of a *continuous* nondecreasing function $\tilde{\beta}(t)$ and a "jump function" representing the sum of the jumps of $\beta(t)$ at all its points of discontinuity. The latter can be differentiated in the sense of distributions, giving rise to a series of delta functions at the points of discontinuity (cf. Section A4, Examples i and iii). Thus (C3) can be rewritten as

$$F(z) = Az + C + \sum_n \beta_n\left(\frac{1+t_n z}{t_n - z}\right) + \int_{-\infty}^{\infty} \frac{1+tz}{t-z}\, d\tilde{\beta}(t) \qquad (\operatorname{Im} z > 0), \tag{C12}$$

where $\beta_n$ is given by (C5) and $\tilde{\beta}(t)$ is a bounded nondecreasing *continuous* function.

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APPENDIX

# D

# PROPERTIES OF $R$-FUNCTIONS

A function $R(z)$ is called an *R-function* [1] if it has the following properties:

$$R(z) \quad \text{is meromorphic,} \tag{D1}$$

$$\operatorname{Im} R(z) \operatorname{Im} z \geqslant 0; \tag{D2}$$

i.e., $\operatorname{Im} R(z) \geqslant 0$ in $I_+$ and $\operatorname{Im} R(z) \leqslant 0$ in $I_-$. Although $R(z) = C$, where $C$ is a real constant, satisfies both of these conditions, it is not usually considered to be an $R$-function.

THEOREM D1. *An R-function is real on the real axis and only on the real axis;* i.e., if $z = x + iy$,

$$\operatorname{Im} R = 0 \quad (y = 0), \qquad y \operatorname{Im} R > 0 \quad (y \neq 0). \tag{D3}$$

This follows from the well-known theorem according to which the imaginary (or real) part of an analytic function cannot have a maximum or a minimum value at an interior point of a region of analyticity, so that the value $\operatorname{Im} R = 0$, which is a minimum for $I_+$ and a maximum for $I_-$, is attained on their common boundary, and only there ($R = \text{const}$ has already been excluded).

Since $R(z)$ is real on the real axis, it follows from Schwarz's reflection principle, as in (2.8.16), that

$$R(z^*) = [R(z)]^*, \tag{D4}$$

so that it suffices to consider the behavior of $R(z)$ in $I_+$.

THEOREM D2. *We have*

$$dR(x)/dx > 0 \tag{D5}$$

*at every point $x$ of the real axis where $R$ is regular.*

In fact, $R(z)$ can be expanded in a power series about the regular point $x$:

$$R(z) = c_0 + c_n(z-x)^n + \cdots, \tag{D6}$$

where (by Theorem D1) all the coefficients are real and we assume $c_n \neq 0$.
Let

$$z - x = \rho e^{i\varphi}. \tag{D7}$$

Then, in a sufficiently small neighborhood of the point $x$,

$$\operatorname{Im} R(z) \approx c_n \rho^n \sin(n\varphi), \tag{D8}$$

and this has to be $>0$ in $I_+$ and $<0$ in $I_-$. This is only possible if $n=1$ and $c_n = c_1 > 0$, which is equivalent to (D5).

THEOREM D3. *All the zeros and poles of an R-function lie on the real axis; they interlace; they are all simple and the residues at the poles are negative.*

The proof of this theorem is based on the relation (2.5.30). According to (D3), we have

$$\begin{aligned} 0 < \arg R(z) < \pi \qquad (z \in I_+), \\ -\pi < \arg R(z) < 0 \qquad (z \in I_-), \end{aligned} \tag{D9}$$

so that

$$|\Delta_C \arg R(z)| < 2\pi \tag{D10}$$

for any contour $C$ located entirely in $I_+$ or $I_-$, and

$$|\Delta_C \arg R(z)| \leqslant 2\pi \tag{D11}$$

for any contour $C$ that intersects the real axis.

According to (2.5.30), (D10) implies that $R(z)$ has no zeros or poles outside of the real axis, and (D11) implies that all zeros or poles on the real axis are simple. Furthermore, there must be a zero between any pair of poles and a pole between any pair of zeros, since otherwise we would have $|\Delta_C \arg R| = 4\pi$ for a suitable contour round the pair. Thus poles and zeros are interlaced.

To show that the residues at the poles must be negative, it suffices to remark that (D8), with $n = -1$, is valid in the neighborhood of a pole; by (D3), it follows that the residue $c_{-1} < 0$.

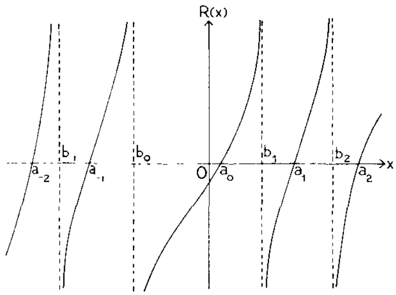



*FIG. D1.* Behavior of a typical $R$-function along the real axis.

The behavior of a typical $R$-function on the real axis, according to the above properties, is illustrated in Fig. D1. It is reminiscent of the graph of the tangent function, and indeed, as can easily be verified,

$$\alpha \tan[\beta(z-x_0)] \qquad (\alpha\beta > 0) \tag{D12}$$

is an $R$-function. Another example is

$$(\alpha z+\beta)/(\gamma z+\delta) \qquad (\alpha\delta-\beta\gamma > 0). \tag{D13}$$

The above results show that an $R$-function satisfies conditions (C1) and (C2), and it is therefore a particular case of a Herglotz function, characterized by being real on the real axis and meromorphic.

At this point we can restate Čebotarev's theorem 2.5.5 as a theorem about $R$-functions:

Čebotarev's Theorem. *The only entire R-functions are the linear functions*

$$R(z) = \alpha z + \beta \qquad (\alpha > 0, \beta \text{ real}). \tag{D14}$$

Theorem D4. *Necessary and sufficient condition for* $R(z)$ *to be an R-function is that it can be represented in the form*

$$R(z) = c\left(\frac{z-a_0}{z-b_0}\right)\prod_{n=-\infty}^{\infty}{}' \frac{[1-(z/a_n)]}{[1-(z/b_n)]}, \tag{D15}$$

*where* $b_n < a_n < b_{n+1}$ $(n = 0, \pm 1, \pm 2, \ldots)$, $b_0 < 0 < b_1$ ($a_0$ *can be either* $>0$ *or* $<0$), $c > 0$, *and the prime in the infinite product*[1] *indicates that* $n \neq 0$.

To prove the sufficiency of the condition, let us discuss the convergence of the infinite product. Writing it in the form

$$\prod_{n=-\infty}^{-1} (1+u_n) \prod_{n=1}^{\infty} (1+u_n),$$

[1] The product may be finite; the derivation remains valid (and becomes much simpler) in this case.

we find, for example, for the second product, just as in (2.9.48),

$$\sum_{n=1}^{\infty} |u_n| = |z| \sum_{n=1}^{\infty} |1-(z/b_n)|^{-1}[(1/b_n)-(1/a_n)]$$
$$\leqslant |z|[\min|1-(z/b_n)|]^{-1}\, 1/b_1,$$

so that, by Theorem 2.5.1, the infinite product is absolutely and uniformly convergent in any bounded closed set not containing any of the points $b_n$, and it therefore represents a regular analytic function in such a set. Thus (D15) defines a meromorphic function $R(z)$ with the poles $b_n$.

On the other hand, we have

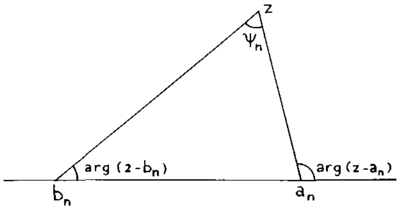
$$\arg R(z) = \sum_{n=-\infty}^{\infty} [\arg(z-a_n)-\arg(z-b_n)] = \sum_{n=-\infty}^{\infty} \psi_n, \qquad \text{(D16)}$$

where $\psi_n$ is the angle subtended by the interval $(b_n, a_n)$ as seen from the point $z$ (cf. Fig. D2). Thus

$$0 \leqslant \arg R(z) \leqslant \pi \qquad (z \in I_+), \qquad \text{(D17)}$$

so that $R(z)$ is an $R$-function.

*FIG. D2.* Geometrical interpretation of the angle $\psi_n$.




To prove that the condition is necessary, let us assume that $R(z)$ is an $R$-function. Then, by Theorem D3, its zeros $a_n$ and poles $b_n$ lie on the real axis and interlace each other. Let us consider the product

$$P(z) = \left(\frac{z-a_0}{z-b_0}\right) \prod_{n=-\infty}^{\infty}{}' \frac{[1-(z/a_n)]}{[1-(z/b_n)]}$$

extended over all the zeros and poles arranged in the above order. According to the derivation just given, $P(z)$ is an $R$-function, and

$$Q(z) = R(z)/P(z)$$

is an entire function without zeros, such that

$$|\arg Q(z)| = |\arg R - \arg P| \leqslant 2\pi$$

for all $z$. It follows, just as in the proof of Čebotarev's theorem 2.5.5, that $Q(z) = c$ is a constant, which must be $>0$ in order that (D17) be satisfied.

THEOREM D5. *Necessary and sufficient condition for $R(z)$ to be an R-function is that it can be represented in the form*[2]

$$R(z) = \alpha z + \beta + \sum_{n=-\infty}^{\infty} \gamma_n{}^2\left(\frac{1}{b_n - z} - \frac{1}{b_n}\right), \tag{D18}$$

*where $\alpha \geqslant 0$, $\beta$, $b_n$, and $\gamma_n$ are all real and such that*

$$\sum_{n=-\infty}^{\infty} \gamma_n{}^2/b_n{}^2 < \infty. \tag{D19}$$

The sufficiency of the condition is obvious, since the convergence of (D19) implies the uniform convergence of (D18) in any bounded closed set not containing the points $b_n$, so that $R(z)$ is a meromorphic function with the poles $b_n$, and

$$\operatorname{Im} R(z) \operatorname{Im} z = (\alpha + \sum_{n=-\infty}^{\infty} \gamma_n{}^2/|b_n - z|^2) \times (\operatorname{Im} z)^2 \geqslant 0. \tag{D20}$$

That the condition is necessary follows as a particular case of the representation (C12) for Herglotz functions. Just as in Section 2.9(a), we can apply (C9) at any point of the real axis where $R(z)$ is regular, and, since $R(z)$ is real at any such point, we conclude that $\tilde{\beta}'(t) = 0$, so that the last term of (C12) is absent. The remaining sum over the poles can be transformed just like (2.9.17), leading immediately to the result (D18).

The necessity of the condition can also be proved directly. In fact, it follows from Theorem D4 that

$$R_N(z) = c\left(\frac{z - a_0}{z - b_0}\right) \prod_{n=-N}^{N}{}' \frac{[1 - (z/a_n)]}{[1 - (z/b_n)]} \tag{D21}$$

tends uniformly to $R(z)$ as $N \to \infty$. By partial-fraction expansion,

$$R_N(z) = A_N \prod_{n=-N}^{N} \left(\frac{a_n - z}{b_n - z}\right) = \sum_{n=-N}^{N} \frac{\gamma_{n,N}^2}{b_n - z} + A_N, \tag{D22}$$

where $A_N > 0$ and

$$\gamma_{n,N}^2 = A_N(a_n - b_n) \prod_{m \neq n} \left(\frac{a_m - b_n}{b_m - b_n}\right) > 0. \tag{D23}$$

---

[2] Again, the series may reduce to a finite sum.

It follows that

$$\lim_{N\to\infty} \gamma_{n,N}^2 = \gamma_n{}^2 = -\operatorname{res} R(z)\Big|_{z=b_n} > 0. \tag{D24}$$

On the other hand, (D22) implies

$$R_N{}'(0) = \sum_{n=-N}^{N} \gamma_{n,N}^2/b_n{}^2 \geqslant \sum_{n=-M}^{M} \gamma_{n,N}^2/b_n{}^2 \qquad (N \geqslant M). \tag{D25}$$

Fixing $M$ and letting $N\to\infty$, we get

$$\sum_{n=-M}^{M} \gamma_n{}^2/b_n{}^2 \leqslant R'(0), \tag{D26}$$

which implies (D19) and consequently the convergence of the series appearing in (D18).

Now consider, for fixed $M$, the difference

$$\begin{aligned} D_{NM}(z) &= R_N(z) - \sum_{n=-M}^{M} \gamma_{n,N}^2\left(\frac{1}{b_n - z} - \frac{1}{b_n}\right) \\ &= \sum_{M<|n|\leqslant N} \frac{\gamma_{n,N}^2}{b_n - z} + B_{NM} \qquad (N > M). \end{aligned} \tag{D27}$$

Just as in (D20), we have $\operatorname{Im} D_{NM}(z) \geqslant 0$ in $I_+$, so that, letting $N\to\infty$, we get

$$\operatorname{Im}\left[R(z) - \sum_{n=-M}^{M} \gamma_n{}^2\left(\frac{1}{b_n - z} - \frac{1}{b_n}\right)\right] \geqslant 0 \qquad \text{in} \quad I_+.$$

It follows that

$$R(z) - \sum_{n=-\infty}^{\infty} \gamma_n{}^2\left(\frac{1}{b_n - z} - \frac{1}{b_n}\right)$$

is an entire $R$-function, and therefore, according to Čebotarev's theorem, it is of the form $\alpha z + \beta$, with $\alpha > 0$ and real $\beta$.

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APPENDIX

# E

## ASYMPTOTIC TIME BEHAVIOR OF FREE SCHRÖDINGER WAVE PACKETS

The time evolution of a free-particle Schrödinger wave packet from an initial state $\psi(\mathbf{x},0)$ is given by [1, p. 22]

$$\psi(\mathbf{x},t) = \int U(\mathbf{x}-\mathbf{x}',t)\,\psi(\mathbf{x}',0)\,d^3x', \tag{E1}$$

where the Green function $U(\mathbf{x},t)$ is the three-dimensional generalization of (4.5.4),

$$U(\mathbf{x},t) = e^{-3i\pi/4}(m/2\pi\hbar t)^{3/2}\exp(im\mathbf{x}^2/2\hbar t), \tag{E2}$$

and we have switched back to conventional units for the purposes of this appendix.

We assume that the initial wave packet has a width $\sim\Delta x$ along each coordinate axis, so that it occupies a volume $\sim(\Delta x)^3$. The corresponding momentum uncertainty is $\Delta p \sim \hbar/\Delta x$, and this leads to a spread in width after a time $t$ given by $t\,\Delta p/m \sim \hbar t/(m\,\Delta x)$. The "spreading time" $t_s$ is defined as the time after which the spread in width becomes comparable with the initial width, so that

$$t_s \sim (m/\hbar)(\Delta x)^2. \tag{E3}$$

We are interested in the asymptotic behavior of the wave function for $t \gg t_s$. According to the above argument, we expect that the width of the packet increases linearly with time for such large times, so that the volume occupied by the wave packet increases with $t^3$. Probability conservation then requires that the wave function at a fixed point should decrease asymptotically like $t^{-3/2}$.

According to (E1) and (E2),

$$\psi(\mathbf{x},t) = e^{-3i\pi/4}(m/2\pi\hbar t)^{3/2}\exp(im\mathbf{x}^2/2\hbar t) \times \int \exp(im\mathbf{x}'^2/2\hbar t)\exp[-(im/\hbar t)\mathbf{x}\cdot\mathbf{x}']\,\psi(\mathbf{x}',0)\,d^3x'. \qquad \text{(E4)}$$

Without loss of generality, we can assume that the origin of coordinates is taken at the center of mass of the initial wave packet, so that $\psi(\mathbf{x}',0)$ in (E4) is appreciable only for $|\mathbf{x}'| \lesssim \Delta x$. It then follows from (E3) that

$$(m/2\hbar t)\mathbf{x}'^2 \ll 1 \qquad (t \gg t_s) \qquad \text{(E5)}$$

within the important part of the domain of integration in (E4). We can then expand the first exponential in the integrand in a power series

$$\exp(im\mathbf{x}'^2/2\hbar t) = 1 + (im\mathbf{x}'^2/2\hbar t) + \cdots. \qquad \text{(E6)}$$

Let

$$\tilde{\varphi}(\mathbf{p}) = [1/(2\pi\hbar)^{3/2}]\int \psi(\mathbf{x}',0)\exp[-(i/\hbar)\mathbf{p}\cdot\mathbf{x}']\,d^3x' \qquad \text{(E7)}$$

be the wave function in the momentum representation. Then, taking the Laplacian with respect to $\mathbf{p}$,

$$\hbar^2\,\Delta_{\mathbf{p}}\tilde{\varphi}(p) = -[1/(2\pi\hbar)^{3/2}]\int \mathbf{x}'^2\,\psi(\mathbf{x}',0)\exp[-(i/\hbar)\mathbf{p}\cdot\mathbf{x}']\,d^3x', \qquad \text{(E8)}$$

assuming that $\psi$ is sufficiently smooth and falls off sufficiently rapidly for the integral to exist.

Substituting (E6)–(E8) in (E4), we find the desired asymptotic expansion:

$$\psi(\mathbf{x},t) = e^{-3i\pi/4}(m/t)^{3/2}\exp(im\mathbf{x}^2/2\hbar t) \times [\tilde{\varphi}(m\mathbf{x}/t) - (im/2\hbar t)\hbar^2\,\Delta_{\mathbf{p}}\tilde{\varphi}(m\mathbf{x}/t) + \cdots] \qquad (t \gg t_s). \qquad \text{(E9)}$$

Clearly, according to (E3), this is an expansion in powers of $t_s/t$. The result can also be derived by applying the method of stationary phase to the momentum representation [2], but the above derivation is simpler and more transparent.

Note that the relevant momentum component in (E9) is that corresponding to the velocity $\mathbf{x}/t$, the classical velocity associated with free particles moving from the origin to $\mathbf{x}$ in the time $t$. In particular, for very large $t$, only the "very slow" components of the initial wave packet are relevant.

## References

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APPENDIX

# F

# COMPACT OPERATORS IN BANACH SPACE

The results given below are discussed in many books on functional analysis. We quote them without proof; for the derivation, the reader is referred to the literature [1–8].

A linear vector space $X$ is said to be *normed* if to every vector $u \in X$ is associated a real number $\|u\|$, called the *norm* of $u$, with the properties: (a) $\|u\| \geqslant 0$, and $\|u\| = 0$ if and only if $u = 0$; (b) $\|\lambda u\| = |\lambda|\,\|u\|$ for any scalar (complex number) $\lambda$; (c) $\|u+v\| \leqslant \|u\| + \|v\|$ (triangle inequality).

The introduction of a norm allows us to consider $X$ as a metric space, by introducing a *distance* $\|u-v\|$ between two vectors $u$ and $v$, and to define a corresponding topology, by the introduction of the concept of convergence. A sequence $\{u_n\}$ is said to *converge* to $u$ (this is called *convergence in norm* or *strong convergence*), denoted by

$$u_n \to u \qquad (n \to \infty) \qquad \text{or} \qquad \lim_{n\to\infty} u_n = u, \tag{F1}$$

when

$$\lim_{n\to\infty} \|u_n - u\| = 0. \tag{F2}$$

By the Cauchy convergence criterion, it follows that

$$\lim_{n,m\to\infty} \|u_n - u_m\| = 0. \tag{F3}$$

A sequence $\{u_n\}$ such that (F3) is satisfied is called a *Cauchy sequence.* A normed linear space $X$ is said to be *complete* if every Cauchy sequence $\{u_n\}$ of elements $u_n \in X$ converges to an element $u \in X$, i.e., if (F3) implies (F1), with $u \in X$.

A *Banach space* is a *complete normed linear vector space.* An example[1] is the space $C$ of all continuous bounded functions $\psi(\mathbf{x})$ considered in Section 6.2, with

$$\|\psi\| = \sup_{\mathbf{x}} |\psi(\mathbf{x})|, \tag{F4}$$

where sup denotes the least upper bound. According to (F2), convergence in norm then means *uniform convergence throughout all space* (this is why the limit of a sequence of continuous functions in $C$ is continuous).

A linear operator $\mathscr{K}$ is said to be *bounded* when

$$\|\mathscr{K}\| = \sup_{\psi \neq 0} \|\mathscr{K}\psi\|/\|\psi\| = \sup_{\|\psi\|=1} \|\mathscr{K}\psi\| < \infty \tag{F5}$$

and $\|\mathscr{K}\|$ is then called the *norm* of $\mathscr{K}$. A sequence of operators $\{\mathscr{K}_n\}$ *converges in norm* to $\mathscr{K}$ when

$$\lim_{n\to\infty} \|\mathscr{K}_n - \mathscr{K}\| = 0. \tag{F6}$$

We are interested in the solution of functional equations such as (6.2.10) and (6.2.11). In finite-dimensional linear vector spaces, the solutions to such problems have properties that are closely analogous to those of the solutions of finite systems of linear equations in a finite number of unknowns.[2] In infinite-dimensional spaces, this need not be so, unless the operator $\mathscr{K}$ appearing in the functional equations has special properties.

A class of operators in many respects analogous to operators in finite-dimensional spaces is that of *compact* (also called *completely continuous*) operators. *An operator $\mathscr{K}$ is called compact if it maps every bounded set of vectors into a compact one.* A set $S$ is called *compact* when every subset of $S$ with infinitely many elements contains a subsequence converging to an element of $S$ [1, p. 34; 2, p. 55; 4, p. 62]. Thus, by the Bolzano–Weierstrass theorem, any bounded and closed infinite set of points of the real line (or, more generally, of a finite-dimensional metric space) is compact. On the other hand, the whole real line is not compact, as we see by taking, for example, the subset consisting of the integers.

In a metric space, any compact set is bounded [1, p. 39], but the converse is not generally valid in infinite-dimensional spaces. For instance, in the space $C$ that was defined above, the unit sphere, i.e., the set of all $\psi(\mathbf{x})$ such that $\|\psi(\mathbf{x})\| \leqslant 1$, is bounded, but not compact. In fact, if we consider a sequence of functions $\psi_n \in C$ with compact supports and $\|\psi_n\| = 1$, such that their supports are all disjoint, we have $\|\psi_n(\mathbf{x}) - \psi_m(\mathbf{x})\| = 1$ for $n \neq m$, so that we cannot extract a convergent subsequence.

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[1] Cf. Dunford and Schwartz [8, p. 261].

[2] Cf. Kato [6, Chapter 1].

An important criterion for compactness in the space $C$ is provided by a theorem due to Ascoli and Arzelà. Let $\mathscr{F}$ be a set of functions $\varphi(x)$ belonging to a normed space $X$, and defined for $x \in E$, where $E$ is also a normed space. The set $\mathscr{F}$ is said to be *equicontinuous at a point* $x_0 \in E$ if, for every $\varepsilon > 0$, there exists a neighborhood $\|x-x_0\| < \delta(\varepsilon)$ such that, *for all* $\varphi \in \mathscr{F}$,

$$\|\varphi(x)-\varphi(x_0)\| < \varepsilon \qquad \text{for} \quad \|x-x_0\| < \delta(\varepsilon). \tag{F7}$$

The important point here is that $\delta$ depends only on $\varepsilon$, and not on $\varphi$; i.e., the neighborhood is the same for all functions in the set $\mathscr{F}$. The set $\mathscr{F}$ is said to be *equicontinuous* if it is equicontinuous at every point $x \in E$.

We then have the following result[3]:

THEOREM F1 (Ascoli–Arzelà). *Let $\mathscr{F}$ be an equicontinuous set of functions $\varphi(x)$ from a separable[4] normed space $E$ into a Banach space $X$. If a sequence $\{\varphi_n(x)\}$ in $\mathscr{F}$ is such that, for each $x \in E$, the closure[5] $\overline{\{\varphi_n(x)\}}$ is compact, then $\{\varphi_n(x)\}$ contains a convergent subsequence that converges to a continuous function. Furthermore, the convergence is uniform on every compact subspace of $E$.*

Since the three-dimensional Euclidean space is separable [1, p. 32], we may apply this result, in particular, to an equicontinuous set $\mathscr{F}$ of functions $\varphi(\mathbf{x}) \in C$. By the Bolzano–Weierstrass theorem, the condition that $\overline{\{\varphi_n(x)\}}$ be compact for every infinite sequence belonging to $\mathscr{F}$ will be satisfied if the set $\mathscr{F}$ is *equibounded*, i.e., if

$$\sup_{\varphi \in \mathscr{F}} \sup_{\mathbf{x}} |\varphi(\mathbf{x})| < \infty. \tag{F8}$$

However, we still cannot conclude from this that $\mathscr{F}$ is compact, because Theorem F1 asserts only that the convergence is uniform on every compact subspace, whereas convergence in $C$, according to (F4), means uniform convergence throughout all space. To ensure this last condition, it is sufficient to add the requirement that *all* $\varphi(\mathbf{x}) \in \mathscr{F}$ be uniformly bounded by a function $f(\mathbf{x})$ which $\to 0$ as $|\mathbf{x}| \to \infty$; together with uniform convergence on every compact subspace, this leads to uniform convergence throughout all space. Thus we have:

THEOREM F2. *Let $\mathscr{F}$ be an equicontinuous, equibounded set of functions $\varphi(\mathbf{x}) \in C$, such that all $\varphi \in \mathscr{F}$ are uniformly bounded by a function $f(\mathbf{x})$, with*

$$\lim_{|\mathbf{x}| \to \infty} f(\mathbf{x}) = 0.$$

[3] The usual formulation of the Ascoli–Arzelà theorem [2, p. 137; 8, p. 266] is for functions defined in a *compact* space. The present formulation is given by Hu [9, p. 202].

[4] A separable space is one that contains a countable everywhere dense set.

[5] The closure $\bar{S}$ of a set $S$ is the smallest closed set that contains $S$ (it contains all the limit points of $S$).

*Then the set $\mathscr{F}$ is compact.*

A compact operator $\mathscr{K}$ is bounded; it is also continuous, in the sense that $u_n \to u$ implies $\mathscr{K}u_n \to \mathscr{K}u$. However, the converse need not be true: the identity operator in an infinite-dimensional space is bounded and continuous, but it is not compact, because it maps the unit sphere into itself.

Let $\{\mathscr{K}_n\}$ be a sequence of compact operators in Banach space converging in norm to $\mathscr{K}$, i.e., such that (F6) is verified. Then $\mathscr{K}$ is also compact; i.e., we have [1, p. 130]:

THEOREM F3. *The limit in norm of a sequence of compact operators in Banach space is also compact, i.e.,*

$$\mathscr{K}_n \text{ compact and } \|\mathscr{K}_n - \mathscr{K}\| \to 0 \Rightarrow \mathscr{K} \text{ compact.}$$

Let us now consider the functional equation

$$\lambda\psi - \mathscr{K}\psi = (\lambda I - \mathscr{K})\psi = \psi_0, \tag{F9}$$

where $I$ is the identity operator, $\lambda$ is a complex number, and $\psi$ and $\psi_0$ are vectors in a Banach space $X$.

In a finite-dimensional vector space, (F9) corresponds to a finite system of linear equations (for the components of the vector in a given basis). It is then well known that there are only two possibilities for a given $\lambda$: *either* (a) the equation has a unique solution, $\psi = (\lambda I - \mathscr{K})^{-1}\psi_0$ ($\lambda$ is then called a *regular value*); *or* (b) the homogeneous equation $\mathscr{K}\psi = \lambda\psi$ has nontrivial solutions. In case (b), $\lambda$ is called an *eigenvalue* of $\mathscr{K}$; it corresponds to a root of the determinant of the system, so that the number of eigenvalues is at most equal to the dimension of the space. For each eigenvalue, the homogeneous equation has a finite number of linearly independent solutions (called the *multiplicity* of the corresponding eigenvalue).

For an infinite-dimensional space, the situation is in general more complicated. However, the analogy between compact operators and operators in finite-dimensional spaces [1, p. 139; 3, p. 222] allows one to obtain an extension of the above result in the special case of compact operators.

The *resolvent set* $\rho(\mathscr{K})$ of an operator $\mathscr{K}$ is the set of complex numbers $\lambda$ for which $(\lambda I - \mathscr{K})^{-1}$ exists as a bounded operator. All complex numbers $\lambda$ that are not in $\rho(\mathscr{K})$ form a set $\sigma(\mathscr{K})$ called the *spectrum* of $\mathscr{K}$. The operator-valued function

$$R(\lambda, \mathscr{K}) = (\lambda I - \mathscr{K})^{-1}, \tag{F10}$$

defined on $\rho(\mathscr{K})$, is called the *resolvent* of $\mathscr{K}$. In general, the spectrum consists of several components (point spectrum, continuous spectrum, residual spectrum [5, p. 209]). However, when $\mathscr{K}$ is a compact operator, we have the following basic theorem [3, p. 219; 2, p. 319; 4, p. 281; 5, p. 283; 6, p. 185; 8, p. 579]:

THEOREM F4 (Riesz–Schauder). *Let $\mathscr{K}$ be a compact operator in a Banach space $X$. Then (i) its spectrum consists of an at most denumerable set of points, with no point of accumulation in the complex plane except possibly $\lambda = 0$; (ii) every $\lambda \neq 0$ in the spectrum is an eigenvalue of $\mathscr{K}$ with finite multiplicity.*

Since every complex number $\lambda \neq 0$ either belongs to $\rho(\mathscr{K})$ or is an isolated eigenvalue with finite multiplicity, it follows that, just as for finite-dimensional spaces, the *Fredholm alternative* is valid for compact operators: *either the functional equation (F9) has the unique solution in $X$*

$$\psi = R(\lambda, \mathscr{K})\psi_0 \tag{F11}$$

*for any $\psi_0 \in X$, or else the homogeneous equation*

$$\mathscr{K}\psi = \lambda\psi \tag{F12}$$

*has a nontrivial solution; in the latter case, the number of linearly independent solutions of the homogeneous equation is finite.*

In the applications of the above results made in Chapter 6, $\psi$ and $\mathscr{K}$ depend on the complex parameter $k$, and the concepts of vector-valued and operator-valued analytic functions of a parameter arise. Since the concept of convergence is defined for vectors in a Banach space, one can introduce the concepts of derivative of a vector-valued function and of holomorphic vector functions and extend many results of analytic function theory to this domain [7, p. 92; 8, p. 224; 6, p. 8; 2, Chapter IX].

Let $X$ be a Banach space and let $\psi(k) \in X$ be a vector-valued function defined in a domain $D$ of the complex $k$-plane. If

$$\lim_{h\to 0}\left\| \frac{\psi(k_0+h)-\psi(k_0)}{h} - \psi'(k_0)\right\| = 0 \tag{F13}$$

at a point $k_0 \in D$, we say that $\psi(k)$ is *differentiable* at $k_0$ and $\psi'(k_0)$ is its *derivative*. If $\psi(k)$ is defined and differentiable everywhere in $D$, it is said to be *holomorphic* in $D$. Similarly, if $\mathscr{K}(k)$ is a *bounded* operator-valued function in a Banach space $X$, defined in a domain $D$ of the complex $k$-plane, convergence can again be defined by (F6). We can therefore define the derivative $\mathscr{K}'(k)$ and *holomorphic operator functions* (since the set of all bounded operators in a Banach space is itself a Banach space [6, p. 150], we can also regard $\mathscr{K}(k)$ as a vector in this space).

We have

$$(d/dk)[\mathscr{K}(k)\psi(k)] = \mathscr{K}'(k)\psi(k) + \mathscr{K}(k)\psi'(k), \tag{F14}$$

whenever the products are meaningful and the derivatives exist.

We can also, in the usual manner, define the integral of a holomorphic vector or operator function along a path in the complex plane. If $\mathscr{K}(k)$ is holomorphic in a domain $D$ of the $k$-plane and $\Gamma$ is a simple closed rectifiable curve in $D$, the interior of $\Gamma$ being also in $D$, we have (*Cauchy's theorem*)

$$\oint_\Gamma \mathscr{K}(k)\,dk = 0, \tag{F15}$$

where the right-hand side is the zero operator in $X$. Conversely, if (F15) is true for any circuit $\Gamma$ in $D$, $\mathscr{K}(k)$ is holomorphic in $D$ (*Morera's theorem*).

If $\mathscr{K}(k)^{-1}$ and $\mathscr{K}'(k)$ exist, the inverse operator $\mathscr{K}(k)^{-1}$ is also differentiable, and we have [6, p. 32]

$$(d/dk)[\mathscr{K}(k)^{-1}] = -\mathscr{K}(k)^{-1}\mathscr{K}'(k)\mathscr{K}(k)^{-1}. \tag{F16}$$

If $\mathscr{K}(k)$ is holomorphic in a neighborhood of $k = k_0$ and if $\lambda_0 \in \rho(\mathscr{K}(k_0))$, so that $R(\lambda_0, \mathscr{K}(k_0)) = [\lambda_0 I - \mathscr{K}(k_0)]^{-1}$ exists, one would expect this to remain true within sufficiently small neighborhoods of $\lambda_0$ and $k_0$, and, furthermore, by (F16), the resolvent should also be holomorphic in $k$ in a neighborhood of $k_0$. Indeed,

$$\Delta\mathscr{K}(k) = \mathscr{K}(k) - \mathscr{K}(k_0) \tag{F17}$$

has arbitrarily small norm for $k$ sufficiently close to $k_0$, so that the Neumann series [cf. (6.1.24)]

$$\begin{aligned} R(\lambda, \mathscr{K}(k)) &= [\lambda I - \mathscr{K}(k)]^{-1} \\ &= [\lambda I - \mathscr{K}(k_0) - \Delta\mathscr{K}]^{-1} \\ &= R(\lambda, \mathscr{K}(k_0)) \sum_{n=0}^{\infty} [\Delta\mathscr{K} R(\lambda, \mathscr{K}(k_0))]^n \end{aligned} \tag{F18}$$

should be absolutely and uniformly convergent for sufficiently small $|k - k_0|$. In fact, we have the result [8, p. 585; 6, p. 367]:

THEOREM F5. *Let $\mathscr{K}(k)$ be a holomorphic operator function defined for*

$$|k - k_0| < \gamma \qquad (\gamma > 0),$$

*and let $N(\lambda)$ be an open set with closure*

$$\bar{N}(\lambda) \subset \rho(\mathscr{K}(k_0)).$$

*Then there exists a $\delta > 0$ such that, if $|k - k_0| < \delta$, we have*

$$\bar{N}(\lambda) \subset \rho(\mathscr{K}(k))$$

*and $R(\lambda, \mathscr{K}(k))$ is a holomorphic function of $k$ for each $\lambda \in N(\lambda)$.*

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APPENDIX

# G

## ASYMPTOTIC BEHAVIOR OF GREEN'S FUNCTION

In order to prove (6.4.8), we have to show that, given $\varepsilon > 0$, we can find $K$, independent of $\mathbf{r}, \mathbf{r}'$, such that

$$|g_1(k, \mathbf{r}, \mathbf{r}')| < \varepsilon, \qquad |k| > K, \quad \operatorname{Im} k \geqslant 0. \tag{G1}$$

The assumptions on the potential are

$$|U(r)| \leqslant U_0 \delta_a(r), \qquad |\nabla U(r)| \leqslant U_1 \delta_a(r), \tag{G2}$$

where $U_0$ and $U_1$ are constants, and

$$\delta_a(r) = 1 \quad \text{if} \quad r < a, \qquad \delta_a(r) = 0 \quad \text{if} \quad r \geqslant a. \tag{G3}$$

In view of the triangle inequality

$$|\mathbf{r} - \mathbf{r}'| \leqslant |\mathbf{r} - \boldsymbol{\rho}| + |\mathbf{r}' - \boldsymbol{\rho}|, \tag{G4}$$

it follows from (6.4.3) that

$$D_{\mathbf{r},\mathbf{r}'}(k, \boldsymbol{\rho}) \to 0 \quad \text{as} \quad r \to \infty, \quad r' \to \infty,$$
$$\text{for all } k, \quad \operatorname{Im} k \geqslant 0.$$

Therefore, by (6.4.6) and (G2), the result (G1) is always valid for sufficiently large $r$ and $r'$, say for $r \geqslant R$, $r' \geqslant R$, where $R$ depends only on $\varepsilon$. Thus we may restrict ourselves to the case in which at least one of $r$ and $r'$ is less than $R$; for definiteness, assume it is $r'$,

$$r' < R. \tag{G5}$$

Let

$$|\mathbf{r}-\boldsymbol{\rho}| = r_1, \qquad |\mathbf{r}'-\boldsymbol{\rho}| = r_2, \qquad |\mathbf{r}-\mathbf{r}'| = 2c, \tag{G6}$$

and let us adopt a set of prolate spheroidal coordinates [1, p. 149] $(\xi,\eta,\varphi)$, where

$$\begin{aligned} \xi &= \tfrac{1}{2}(r_1+r_2), & c \leqslant \xi < \infty, \\ \eta &= (1/2c)(r_1-r_2), & -1 \leqslant \eta \leqslant 1, \end{aligned} \tag{G7}$$

and $\varphi$ is the azimuth about the axis $\mathbf{r}-\mathbf{r}'$ (Fig. G1). We have

$$d\xi\, d\eta\, d\varphi = [D(\xi,\eta,\varphi)/D(x,y,z)]\, dx\, dy\, dz = dx\, dy\, dz/r_1 r_2,$$

so that (6.4.6) becomes

$$g_1(k,\mathbf{r},\mathbf{r}') = -ce^{-2ikc}\int_{-1}^{1} d\eta \int_c^{\infty} e^{2ik\xi}\, U(\xi,\eta)\, d\xi. \tag{G8}$$

Integrating by parts with respect to $\xi$, we get

$$g_1(k,\mathbf{r},\mathbf{r}') = g_1{}'(k,\mathbf{r},\mathbf{r}') + g_1''(k,\mathbf{r},\mathbf{r}'), \tag{G9}$$

where

$$g_1{}'(k,\mathbf{r},\mathbf{r}') = (c/2ik)\int_{-1}^{1} U(c,\eta)\, d\eta, \tag{G10}$$

$$g_1''(k,\mathbf{r},\mathbf{r}') = (c/2ik)\, e^{-2ikc}\int_{-1}^{1} d\eta \int_c^{\infty} e^{2ik\xi}(\partial U/\partial\xi)\, d\xi. \tag{G11}$$

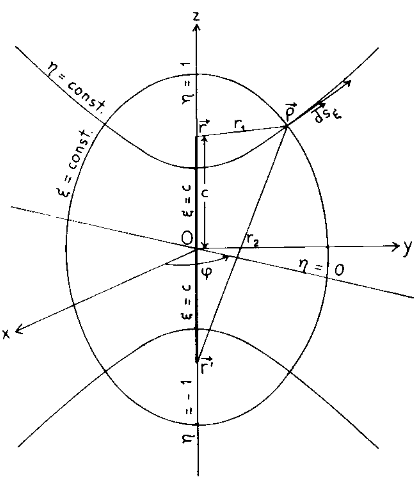


*FIG. G1.* Prolate spheroidal coordinates.

In (G10), the integration is along the straight line joining $\mathbf{r}'$ to $\mathbf{r}$, and $c\,d\eta$ is the line element along this straight line, so that (G2) yields

$$|g_1'(k,\mathbf{r},\mathbf{r}')| \leqslant (1/2|k|) \cdot 2aU_0 = aU_0/|k|. \tag{G12}$$

We have

$$(\partial U/\partial \xi)\, d\xi = \nabla U \cdot d\mathbf{s}_\xi,$$

where $d\mathbf{s}_\xi$ is an element of arc along a hyperbola $\eta = \text{const}$ (Fig. G1). Thus (G2) and (G11) imply

$$|g_1''(k,\mathbf{r},\mathbf{r}')| \leqslant (cU_1/2|k|) \int_{-1}^{1} d\eta \int_{\xi=c}^{\infty} \delta_a(\rho)\, ds_\xi \qquad (\operatorname{Im} k \geqslant 0). \tag{G13}$$

According to (G4) and (G6),

$$2c(1-\eta) = 2c - r_1 + r_2 \leqslant 2r_2 = 2|\mathbf{r}'-\boldsymbol{\rho}| \leqslant 2(r'+\rho),$$

so that, by (G5), we have, in the integral (G13),

$$\eta \geqslant 1 - [(R+a)/c]. \tag{G14}$$

On the other hand, the integral over $ds_\xi$ is the length of a hyperbolic arc $\eta = \text{const}$ (starting on the straight line joining $\mathbf{r}'$ to $\mathbf{r}$) that is contained within the sphere $\rho = a$; this length is certainly smaller than $2\pi a$. It then follows from (G13) and (G14) that

$$\begin{aligned} |g_1''(k,\mathbf{r},\mathbf{r}')| &\leqslant (cU_1/2|k|) \cdot 2\pi a \int_{1-(R+a)/c}^{1} d\eta \\ &= (\pi/|k|)\, a(R+a)\, U_1 \qquad (\operatorname{Im} k \geqslant 0). \end{aligned} \tag{G15}$$

Combining (G9), (G12), and (G15), we see that (G1) is satisfied in all cases.

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APPENDIX

# H

## THE PATH $\Gamma(\nu)$

We want to find a path $\Gamma(\nu)$ in the half-plane $\operatorname{Re} z > 0$, extending from $z = 0$ to $|z| \to \infty$, along which both $|H_{i\nu}^{(1)}(z)|$ and $|H_{i\nu}^{(2)}(z)|$ are small for large $\nu$, $\nu > 0$.

For small $z$, $|z| \ll \sqrt{\nu}$, and large $\nu$, we find from the power series expansions that (apart from constants) $|H_{i\nu}^{(1,2)}(z)|$ is bounded by $\nu^{-1/2} \exp[\pm \nu(\pi/2 - \arg z)]$, so that the path $\Gamma(\nu)$ should start along the positive imaginary axis, $z = iy$. From the asymptotic expansions of Debye type [1, p. 142], we get

$$H_{i\nu}^{(1)}(iy) \approx -2i\left(\frac{2}{\pi}\right)^{1/2} \frac{\sin[\nu \cosh^{-1}(\nu/y) - (\nu^2 - y^2)^{1/2} + (\pi/4)]}{(\nu^2 - y^2)^{1/4}}, \tag{H1}$$

$$H_{i\nu}^{(2)}(iy) \approx \left(\frac{2}{\pi}\right)^{1/2} \frac{\exp\{i[\nu \cosh^{-1}(\nu/y) - (\nu^2 - y^2)^{1/2} + (\pi/4)]\}}{(\nu^2 - y^2)^{1/4}}, \tag{H2}$$

for $y < \nu$ and $\nu - y \gg \nu^{1/3}$, whereas

$$|H_{i\nu}^{(1,2)}(iy)| = O(\nu^{-1/3}), \qquad |y - \nu| \lesssim \nu^{1/3}. \tag{H3}$$

Thus we can take the path $\Gamma(\nu)$ along the positive imaginary axis from $z = 0$ to $z = i\nu$.

On the other hand, for $|z| \to \infty$, we can use the Hankel-type expansions

$$H_{i\nu}^{(1,2)}(z) \approx (2/\pi z)^{1/2} \exp\{\pm i[z - i\nu(\pi/2) - (\pi/4)]\} \qquad (|z| \to \infty), \tag{H4}$$

from which we conclude that $\Gamma(\nu)$ should have the straight line $z = i\nu\pi/2$ as an asymptote.

Note from (H4) that $|H_{iv}^{(1,2)}(z)|$ changes from exponentially increasing to exponentially decreasing from one side to the other of the line $\Gamma(v)$, so that $\Gamma(v)$ corresponds to a *neutral or anti-Stokes line* [2; 3, p. 32], where the asymptotic character of the function changes from dominant to subdominant.

To find the shape of $\Gamma(v)$ between $z = iv$ and $z = iv\pi/2 + \infty$, we can employ [4] the Sommerfeld integral representation of the Hankel functions,

$$H_{iv}^{(1)}(z) = (1/\pi) \int_{-\eta+i\infty}^{\eta-i\infty} \exp[\varphi(v,z,t)]\, dt, \tag{H5}$$

$$H_{iv}^{(2)}(z) = (1/\pi) \int_{\eta-i\infty}^{2\pi-\eta+i\infty} \exp[\varphi(v,z,t)]\, dt, \tag{H6}$$

where

$$\varphi(v,z,t) = iz\cos t + v[(\pi/2)-t], \tag{H7}$$

$$-\eta < \arg z < \pi - \eta, \qquad 0 \leqslant \eta \leqslant \pi. \tag{H8}$$

In order that both $|H_{iv}^{(1)}(z)|$ and $|H_{iv}^{(2)}(z)|$ be small, we want to have $\operatorname{Re}\varphi \leqslant 0$ along both paths of integration in (H5) and (H6). It is not possible for $\operatorname{Re}\varphi$ to be strictly negative along both paths because

$$\varphi(v,z,\pi-t) = -\varphi(v,z,t), \tag{H9}$$

so that, if we had $\operatorname{Re}\varphi < 0$ along the path $(-\eta+i\infty, \eta-i\infty)$ in (H5), we would have $\operatorname{Re}\varphi > 0$ along the path $(\pi+\eta-i\infty, \pi-\eta+i\infty)$, which necessarily intersects the path $(\eta-i\infty, 2\pi-\eta+i\infty)$ in (H6). However, we can have $\operatorname{Re}\varphi \leqslant 0$ along both paths, provided that $\operatorname{Re}\varphi = 0$ at the above-mentioned point of intersection. In order that $\operatorname{Re}\varphi < 0$ on both sides of the point $\operatorname{Re}\varphi = 0$ along each path (rather than changing sign), this point must be a *saddle point*. Thus we must have a point $\bar{t}$ that is a saddle point,

$$d\varphi(v,z,\bar{t})/dt = 0, \tag{H10}$$

and such that, at the same time,

$$\operatorname{Re}\varphi(v,z,\bar{t}) = 0. \tag{H11}$$

Let

$$z = x + iy, \qquad \bar{t} = u + iv. \tag{H12}$$

Then, by (H7), the real and imaginary parts of (H10) yield

$$x\sin u\cosh v - y\cos u\sinh v = 0, \tag{H13}$$

$$x\cos u\sinh v + y\sin u\cosh v = v, \tag{H14}$$

and (H11) yields

$$x\sin u\sinh v - y\cos u\cosh v = -v[(\pi/2)-u]. \tag{H15}$$

Eliminating $\nu$ among these equations, we get

$$\sin u \cos u (x^2+y^2) = \nu^2 [(\pi/2)-u],$$
$$y^2 \cos^2 u - x^2 \sin^2 u = \nu^2 [(\pi/2)-u]^2,$$

and, solving with respect to $x^2$ and $y^2$, we finally get the parametric equations of the path $\Gamma(\nu)$,

$$x^2 = \nu^2 [(\pi/2)-u][\cot u - (\pi/2)+u], \tag{H16}$$
$$y^2 = \nu^2 [(\pi/2)-u][\tan u + (\pi/2)-u]. \tag{H17}$$

As $u$ ranges between 0 and $\pi/2$, the point $z = x+iy$ ranges from $i\nu\pi/2+\infty$ to $i\nu$, in agreement with the previous discussion. The shape of the whole path $\Gamma(\nu)$ from $z=0$ to $z=i\nu\pi/2+\infty$ is shown by the curve in thick line in Fig. H1. Note from (H16) and (H17) that it depends only on the variable $z/\nu$; i.e., $\nu$ has the effect simply of a scaling factor.

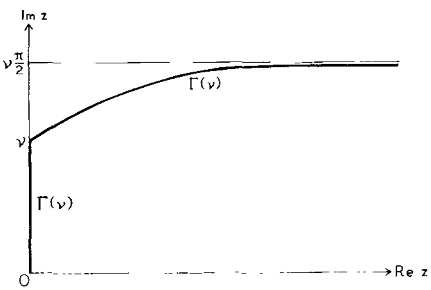


*FIG. H1.* The path $\Gamma(\nu)$ in the $z$-plane along which the bounds (H18) and (H19) are valid.

We have seen that the above conditions are necessary, but they can also be shown [4] to be sufficient, i.e., when $z$ lies on $\Gamma(\nu)$, one can choose paths of integration in (H5) and (H6), going through the saddle points $\bar{t}$ and $\pi-\bar{t}$, respectively, along which $\mathrm{Re}\,\varphi \leqslant 0$. The behavior of the integrals can then be estimated and bounded with the help of the steepest descent method.

The result is that

$$(\pi/2)^{1/2} |H_{i\nu}^{(1,2)}(z)| < C/|\nu^2+z^2|^{1/4} \qquad (z \in \Gamma(\nu)), \tag{H18}$$

where $C$ is a purely numerical constant.[1] In the neighborhood of $z = i\nu$, for $|z-i\nu| \lesssim \nu^{1/3}$, this bound can be replaced by

$$(\pi/2)^{1/2} |H_{i\nu}^{(1,2)}(z)| < C/\nu^{1/3}, \qquad |z-i\nu| \lesssim \nu^{1/3}, \quad z \in \Gamma(\nu). \tag{H19}$$

Note that these results agree with the asymptotic estimates from (H1) to (H4). Since (H18) and (H19) tend to zero for $\nu \to \infty$, our goal has been achieved.

[1] According to Bessis [5], it can be shown that, for $\nu \geqslant 1$, one has $C < 23$.

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APPENDIX

# I

# DISPERSION RELATION FOR THE BASIC MANDELSTAM INTEGRAL

According to (8.2.4) and (8.2.10), the integral (8.3.5) may be rewritten as

$$I(q_1,q_2,t) = \int \frac{d\Omega_{k''}}{(q_1-\hat{\mathbf{k}}\cdot\hat{\mathbf{k}}'')(q_2-\hat{\mathbf{k}}'\cdot\hat{\mathbf{k}}'')} \qquad (q_1 > 1, q_2 > 1). \tag{I1}$$

We combine the two denominators into one by the Feynman–Schwinger [1] parametrization method, with the help of the identity

$$\frac{1}{ab} = \int_0^\infty \frac{dx}{(ax+b)^2}. \tag{I2}$$

This leads to

$$I(q_1,q_2,t) = \int_0^\infty dx \int \frac{d\Omega_{k''}}{[q_1x+q_2-\hat{\mathbf{k}}''\cdot(\hat{\mathbf{k}}x+\hat{\mathbf{k}}')]^2}. \tag{I3}$$

To perform the angular integration, we take spherical coordinates, with the $z$-axis along the direction of $\hat{\mathbf{k}}x+\hat{\mathbf{k}}'$, so that

$$\hat{\mathbf{k}}x + \hat{\mathbf{k}}' = \lambda\hat{\mathbf{z}}, \qquad \lambda^2 = x^2 + 2wx + 1, \tag{I4}$$

where, by (8.2.9) and (8.2.10),

$$w = \hat{\mathbf{k}}\cdot\hat{\mathbf{k}}' = 1 + (t/2s). \tag{I5}$$

The angular integration can now immediately be performed, yielding

$$I(q_1,q_2,t) = 4\pi\int_0^\infty \frac{dx}{(q_1x+q_2)^2-\lambda^2} = 2\pi\int_0^\infty \frac{dx}{x(y-w)}, \tag{I6}$$

where

$$y = \tfrac{1}{2}(q_1{}^2-1)x + q_1 q_2 + \tfrac{1}{2}(q_2{}^2-1)x^{-1}. \tag{I7}$$

As $x$ varies from 0 to $\infty$, $y$ first decreases monotonically from $\infty$ to $y_{\min}$, and then it increases monotonically from $y_{\min}$ to $\infty$, where

$$y_{\min} = q_1 q_2 + [(q_1{}^2-1)(q_2{}^2-1)]^{1/2} \tag{I8}$$

is the minimum value of $y$, attained for $x = x_{\min}$, where

$$x_{\min} = [(q_2{}^2-1)/(q_1{}^2-1)]^{1/2}. \tag{I9}$$

To change variables in (I6) from $x$ to $y$, we split the integral into two parts, from 0 to $x_{\min}$ and from $x_{\min}$ to $\infty$. We have

$$dx/x = \mp dy/[(y-q_1q_2)^2-(q_1{}^2-1)(q_2{}^2-1)]^{1/2}, \tag{I10}$$

where the upper (lower) sign is valid for $x < x_{\min}$ ($x > x_{\min}$), so that (I6) becomes

$$\begin{aligned} I(q_1,q_2,t) &= 4\pi \int_{y_{\min}}^{\infty} \frac{dy}{(y-w)\,[(y-q_1q_2)^2-(q_1{}^2-1)(q_2{}^2-1)]^{1/2}} \\ &= 4\pi \int_0^{\infty} \frac{\theta(y-y_{\min})dy}{(y-w)\{(y-y_{\min})\,[y-y_{\min}+2((q_1{}^2-1)(q_2{}^2-1))^{1/2}]\}^{1/2}} \\ &= 4\pi \int_0^{\infty} \frac{\theta(-\Delta)dy}{(y-w)[-\Delta(q_1,q_2,y)]^{1/2}}, \end{aligned} \tag{I11}$$

where the notation is the same as in (8.2.12) to (8.2.14).

Setting $y = 1+(t'/2s)$ and taking into account (I5), it follows that

$$I\left(1+\frac{t_1}{2s}, 1+\frac{t_2}{2s}, t\right) = 16s^{3/2} \int_0^{\infty} K(s,t';t_1,t_2)\frac{dt'}{t'-t}, \tag{I12}$$

where

$$\begin{aligned} &K(s,t';t_1,t_2) \\ &= \frac{\pi}{2} \frac{\theta[t'-t_1-t_2-t_1t_2/2s-((t_1t_2)^{1/2}/2s)(16s^2+4s(t_1+t_2)+t_1t_2)^{1/2}]}{\{s[t'-(\sqrt{t_1}+\sqrt{t_2})^2]\,[t'-(\sqrt{t_1}-\sqrt{t_2})^2]-t'\,t_1t_2\}^{1/2}}. \end{aligned} \tag{I13}$$

The result (I12) is the sought-for dispersion relation in the momentum transfer variable. It has been assumed throughout that $t$ is any real or complex value not on the cut.

## Reference

1. R. P. Feynman, *Phys. Rev.* **76**, 769 (1949); J. Schwinger, *Phys. Rev.* **76**, 790 (1949).

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# AUTHOR INDEX

Numbers in parentheses are reference numbers and indicate that an author's work is referred to, although his name is not cited in the text. Numbers in italics show the page on which the complete reference is listed.

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